CPCL

ENQUIRY DOCUMENT - MR

(Document No : B649-095-VV-MR-6009-04-RFQ)

इंजीनियर्स इंडिया लिमिटेड
(भारत सरकार का उपक्रम)
ENGINEERS INDIA LIMITED
(A Govt. of India Undertaking)

हर एक काम देश के नाम

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GROUP-II / III LOBS PROJECT

BIDDING DOCUMENT
FOR

PRESSURE VESSEL-CS/LAS+SS CLAD

(BIDDING DOCUMENT NO.: MKP/B649-095-VV-MR-6009/04)

PREPARED AND ISSUED BY:

इंजीनियर्स इंडिया लिमिटेड
(भारत सरकार का उपक्रम)
ENGINEERS INDIA LIMITED
(A Govt. of India Undertaking)

इंजीनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

REQUEST FOR QUOTATION (RFQ) COVER

ON

SINGLE STAGE COMPOSITE BID SYSTEM

(Important Instructions to Bidder)

RFQ No.: MKPB649-095-VV-MR-6009/04

ITEM: PRESSURE VESSEL-CS/LAS+SS CLAD

A. CHECK- LIST FOR MANDATORY SUBMISSION OF DOCUMENTS IN COMPOSITE BID:

(Bidder to submit these cover pages duly filled in, signed and stamped along with the bid)

S. NO.	DESCRIPTION	STATUS
1.	Agreed Terms & Conditions (ATC) duly filled)	☐
2.	Prices filled in BoQ Format of the Price Schedule (without any condition/ deviation/ clarification).	☐
3.	Form-2: Self Certification by Bidder Under PPP-MII Order, 2017	☐
4.	Form-IA / II as applicable (regarding restrictions on procurement from a Bidder of a country which shares a land border with India) - **Annexure-III to ATC**.	☐
5.	Annexure-A (Self Declaration of Conflict of Interest / Alternate Bids / Multiple Bids) - **Annexure-IV to ATC**	☐

FAILURE TO SUBMIT ANY/ALL OF THE ABOVE DOCUMENTS SHALL RENDER THE BID TO BE SUMMARILY REJECTED

B. CHECKLIST FOR SUBMISSION OF DOCUMENTS AS APPLICABLE

S. NO.	DESCRIPTION	STATUS
1.	Agreed Terms & Conditions (ATC) duly filled	☐
2.	Form-2: Self Certification by Bidder Under PPP-MII Order, 2017	☐
3.	Form-IA / II as applicable (regarding restrictions on procurement from a Bidder of a country which shares a land border with India) - **Annexure-III to ATC**.	☐
4.	Annexure-A (Self Declaration of Conflict of Interest / Alternate Bids / Multiple Bids) - **Annexure-IV to ATC**	☐
5.	Certified documentary evidence to avail MSE purchase preference	☐
6.	Addendum / Amendment (if any) duly signed & stamped	☐
7.	Priced Copy of 2 years O&M spares in Price Schedule	☐
8.	Signed and Stamped Copy of Unit Rates for Addition/Deletion in line with Annexure-I to BDS	☐
9.	Prices filled in BoQ Format of the Price Schedule (without any condition/ deviation/ clarification).	☐

Bidder's Signature & Stamp

C. BIDDER MUST READ FOLLOWING INSTRUCTIONS ON COMPOSITE BIDDING CAREFULLY AND COMPLY:

a) Bidder to note that no correspondence shall be entertained after the due date for bid submission.

b) Bidder shall quote prices for only those items which the bidder can supply material as per specifications and for which bidder is approved by EIL/ OWNER.

c) **No technical details of any nature shall be specified in the bid.** If the bid stipulates any technical deviations, it shall be summarily rejected.

Bidder shall note that enquiry is issued on **ZERO DEVIATION BASIS**. Bidder shall comply with all technical specification & commercial terms and other conditions of the enquiry document without any deviations/ clarifications. **Any deviation stipulated by bidder shall result in summary rejection of the bid.**

d) The submission of prices by the bidder shall be construed that bidder has confirmed compliance with all technical specifications of the quoted items as well as commercial terms & conditions of enquiry document.

e) Other than the details and documents specifically required in the bid as per enquiry document, bidder shall not submit any other detail/ document in the bid.

f) Bidder should ensure submission of complete information/ documentation in the first instance itself. Evaluation of bids shall be completed based on the details/ documents so furnished in bids without seeking any subsequent additional information/ clarification/ documents.

Bidder's Signature & Stamp

MASTER INDEX

NAME OF ITEM : PRESSURE VESSEL-CS/LAS+SS CLAD

BIDDING DOCUMENT NO. : MKP/B649-095-VV-MR-6009/04

COMMERCIAL SECTION	
S. NO.	**DESCRIPTION**
1.	COVER PAGE
2.	MASTER INDEX
3.	REQUEST FOR QUOTATION (RFQ)
4.	BID DATA SHEET(BDS) ALONG WITH ANNEXURE-I TO BDS
5.	INSTRUCTION TO BIDDERS (ITB)
6.	APPENDICES TO ITB
	i) APPENDIX-I TO ITB – INSTRUCTIONS TO BIDDERS FOR E-TENDERING
	ii) APPENDIX-II TO ITB – TERMS & CONDITIONS FOR REVERSE AUCTION (**NOT APPLICABLE**)
	iii) APPENDIX-III TO ITB: PREFERENCE TO MICRO AND SMALL ENTERPRISES (MSEs)
	iv) APPENDIX-IV TO ITB: PUBLIC PROCUREMENT (POLICY FOR PREFERENCE TO MAKE IN INDIA) (PPP-MII)
	v) APPENDIX-V TO ITB: METHODOLOGY OF CONCURRENT APPLICATION OF PPP-2012 (MSE) POLICY & PPP-2017 (MII) POLICY
	vi) APPENDIX-VI TO ITB: POLICY FOR PROVIDING PREFERENCE TO DOMESTICALLY MANUFACTURED IRON & STEEL PRODUCTS (DMI&SP) **– NOT APPLICABLE**
	vii) APPENDIX-VII TO ITB: PROCUREMENT FROM BIDDERS FROM A COUNTRY OR COUNTRIES SHARING LAND BORDER WITH INDIA
	viii) APPENDIX-VIII TO ITB: STARTUP INDIA POLICY **-NOT APPLICABLE**
	ix) APPENDIX-IX TO ITB: FOREIGN TRADE POLICY ON IMPORTS & EXPORTS TO DEMOCRATIC PEOPLES REPUBLIC OF KOREA
7.	AGREED TERMS AND CONDITIONS
8.	SPECIAL PURCHASE CONDITIONS
9.	ANNEXURES TO SPC
	i) ANNEXURE-I: TERMS & CONDITIONS FOR INDIAN SOURCED COMPONENTS / SERVICES BY FOREIGN BIDDER – **NOT APPLICABLE**
	ii) ANNEXURE-II: TERMS & CONDITIONS FOR SUPERVISION OF ERECTION, TESTING & COMMISSIONING / TRAINING AT SITE - **NOT APPLICABLE**
	iii) ANNEXURE-III: TERMS & CONDITIONS FOR SITE WORK – **NOT APPLICABLE**
	iv) ANNEXURE-IV: TERMS & CONDITIONS FOR ANNUAL MAINTENANCE CONTRACT – **NOT APPLICABLE**
	v) ANNEXURE-V: TERMS & CONDITIONS FOR TRAINING AT VENDOR'S/OEM'S WORKS / TRAINING FACILITY – **NOT APPLICABLE**
	vi) ANNEXURE-VI: TAXES AND DUTIES
	vii) ANNEXURE-VII: LIST OF INTERNATIONAL SEAPORTS / AIRPORTS
10.	GENERAL PURCHASE CONDITIONS
11.	PREAMBLE, BoQ & OTHER PRICE FORMATS (IF APPLICABLE)
TECHNICAL SECTION	
MR No. B649-095-VV-MR-6009	

इंजीनियर्स इंडिया लिमिटेड
(भारत सरकार का उपक्रम)
ENGINEERS INDIA LIMITED
(A Govt. of India Undertaking)
Regd. Office: Engineers India Bhawan, 1 Bhikaiji Cama Place,
New Delhi - 110 066, India

REQUEST FOR QUOTATION (RFQ)- LIMITED-INDIGENOUS
[BIDDING THROUGH CPCL eProcurement Portal]
COVERING LETTER

Date: 27.08.2026

SL NO.	SALIENT FEATURES		DETAILS
a)	Client	:	M/s CHENNAI PETROLEUM CORPORATION LIMITED
b)	Project	:	GROUP-II/III LOBS PROJECT AT MANALI REFINERY, CHENNAI
c)	Item	:	PRESSURE VESSEL-CS/LAS+SS CLAD
d)	EIL RFQ No.	:	MKP/B649-095-VV-MR-6009/04
e)	MR Category	:	I (Other than Bulk)
f)	Pre-Bid meeting	:	Applicable
g)	Pre-Bid Meeting Details and Venue	:	At 11:00 Hrs. IST on 03.09.2026 Meeting ID: 471 000 009 027 522 Passcode: RB9oG96J PRE-BID MEETING PRESSURE VESSEL-CS/LAS+SS CLAD FOR GROUP-II/III LOBS PROJECT AT MANALI REFINERY, CHENNAI | Meeting-Join | Microsoft Teams
h)	Last date of receipt for pre-bid queries	:	**Cut-off date for sending queries: One day before Pre-Bid Meeting** Any Clarification after cut-off date for submission of Query may not be entertained. The consolidated reply shall be uploaded on CPCL eProcurement portal. ***Bidder to note that this is a Nil deviation tender, and any clarification should be cleared by the bidder before bid submission. Any deviation in their bid shall be liable for rejection.***
i)	Earnest Money Deposit (EMD)	:	Not Applicable
j)	BID DUE DATE (Last date and time of submission of e-Bids)	:	Up to 12:00 Hrs. IST on 17.09.2026
k)	Composite BID OPENING Opening of Unpriced and Priced e-Bids	:	At 14:00 Hrs. IST on 18.09.2026
l)	Contact Person for any query/ clarification	:	Ms. Arya Subash, Dy Manager (SCM-C&P) Phone no. 0124 289 1348 E-mail ID: subash.arya@eil.co.in Mr. Mitesh Patel, AGM (SCM-C&P),

SL NO.	SALIENT FEATURES		DETAILS
			Phone no. 0124 289 1341 E-mail ID: mitesh.patel@eil.co.in Mr. Jaydip Purkayastha, GM (SCM-C&P), Phone no. 0124 289 1287 E-mail ID: jaydip@eil.co.in
m)	**e-Tendering portal for submission of e-bids**	**:**	The complete **Enquiry Document along with Technical Part and subsequent Addendums/Corrigendum's, if any,** can be viewed/downloaded from the following website: https://cpcletenders.nic.in. A link will be available on EIL tender website http://tenders.eil.co.in.
n)	**Delivery Period/ Completion Period**	**:**	**SUPPLY:** Within **9 Months** on FOT Project Site on freight pre-paid & door delivery basis from the date of issue of LOA. The date of receipt of material at site shall be considered as the date of delivery.
o)	**DEVIATIONS NOT PERMITTED** Bids are invited on **"Zero Deviation Basis".** Accordingly offer is to be submitted in complete compliance to RFQ Terms & Conditions without any deviation. **Offer with deviation shall be rejected without raising any TQ/CQ. No TQ/CQ shall be issued in composite bidding. Only minor clarifications, which do not lead to any price implication may be sought from the bidder.** Hence, Bidder shall ensure submission of complete bid without any deviation as per requirement of Enquiry document in first instance itself.		
p)	This enquiry is issued on **Limited Bidding basis for "captioned item(s) mentioned at sl. No. c above" only**. A separate intimation to the EIL suppliers eligible for this enquiry shall be sent through mail subsequent to issue of enquiry in e-Tendering Portal. **Only those suppliers who are intimated of publication of the enquiry through EIL e-mail will be eligible to submit their bids. Bids received from any other supplier shall not be considered for evaluation and shall be rejected.** In case a bidder does not register on the e-procurement Portal and as a consequence, does not submit their bid on e-Tendering Portal, it shall be deemed that he is not interested in bidding against this enquiry and no further correspondence will be entertained.		

If any of the dates indicated at Sl. No. **(k)** above happens to be a declared/closed holiday in EIL-Gurugram, the next working day shall be considered.

Dear Sir/Madam,

1. Chennai Petroleum Corporation Limited (CPCL) operates a 10.5 Million Metric Tons Per Annum (MMTPA) Refinery at Chennai, Tamil Nadu, which is one of the most complex refineries in India with fuel, lube, wax and petrochemical feedstock production facilities. CPCL is setting-up "LOBS" Project along with associated facilities, at its MANALI Refinery, CHENNAI in the State of TAMILNADU.

2. The Group-II/ III LOBS Project comprises following facilities:-

   - New Catalytic Dewaxing Unit (CDWU)
   - Revamp of exiting OHCU Unit
   - Augmentation of Utilities & Offsite facilities

3. Engineers India Limited (EIL) has been engaged by CPCL as Engineering, Procurement, and Construction Management (EPCM) Consultant to execute LOBS Project.

4. Bids are required to be submitted only through Govt. of India CPCL eProcurement Portal on or before the Bid submission date and time. Bidders are required to register themselves at https://cpcletenders.nic.in. No registration fee would be charged from the bidders.

5. E-bids under **single stage single bid system (single packet bid/composite bid)** are requested for the captioned item based on the requirement specified in this MR and in total compliance to technical specifications, scope and terms & conditions of RFQ documents/attachments.

6. Bidder can download the RFQ Documents from e-tendering website of CPCL eProcurement portal (https://cpcletenders.nic.in). All amendments, time extension, clarifications etc. will be uploaded in e-Tendering Portal only. Bidders should regularly visit the above website to keep themselves updated. No extension in the bid due date / time shall be considered on account of delay in receipt of any document by mail.

7. Provisions mentioned in Instruction to Bidders with regard to uploading of bidding document, replies/clarifications to bidder queries, techno-commercially amendments etc., on EIL website shall stands modified to the extent that these shall be uploaded on e-Tendering portal only.

8. Bidders to refer E-Tendering methodology enclosed with this RFQ. Various links such as “Help for Contractor”, “Information about DSC”, “FAQ”, “Resources Required”, “Bidders Manual Kit” etc. are available on home page of https://cpcletenders.nic.in facilitating vendors to participate in the bidding process. Bidder are advised to download & utilize the available information/documents under these links for activities like Registration in CPPP, obtaining User ID & Password, uploading & submission of e-bids etc. Bidders are advised in their own interest to carefully go through Instructions for E-tendering and other related document available against various help links so as to ensure that bid are uploaded in E-tendering website well before the closing date and time of bid submission.

9. e-Tendering Portal mandates that the bidders are to be registered on the portal before any enquiry can be issued to them. In order to expedite issue of enquiries, the information of enquiry is being issued through EIL Tender Portal, however, document shall be available on e-Tendering Portal. The name of bidder whose NIC ID details are submitted/available to EIL shall be issued the enquiry and enquiry to the balance bidders can only be issued once they will register themselves on e-Tendering portal and provide their registration details to EIL.

   Therefore, all those vendors who have still not registered on the e-tendering Portal are required to register on the same (immediately after issue of enquiry) and inform the undersigned the registration details for facilitating issue of enquiry to them on the e-tendering Portal failing which it will not be possible for them to upload their bids.

   All those bidders who have not complied with the above registration requirements will not be issued this enquiry on e-tendering portal and will not be eligible to bid. Request for extension in due date of submission of bids due to non-registration or delayed registration in e-tendering portal shall not be entertained.

10. Bidders are required to submit their bids complete in all respects in accordance with Instructions to Bidders (ITB) only through e-tendering website of CPCL eProcurement portal (https://cpcletenders.nic.in) on or before the bid-submission due date & time as specified above. Bidder to carefully read and understand the instructions for e-tendering as indicated in ITB to upload their bid on e-tendering website.

    Physical Bids sent through Fax/E-mail/Courier/Post will not be accepted. However, any specific document required to be submitted in physical form as per RFQ shall also be submitted in physical form.

    The vendor registration on e-tendering Portal is a very user-friendly process. However, in case of any doubt, the vendor may contact the undersigned.

    Bidder(s) are advised to upload their bid as soon as they are permitted to upload the bid. In the event of failure in bidder’s connectivity with e-tendering Portal during the last few hours on account of any problem and they are likely to miss the deadline for bid submission in such a case request for extension in bid due date will not be entertained.

11. Please submit your Acknowledgement against the RFQ on EIL’s website http://tenders.eil.co.in within the due date & time, with reason(s) of not participating in the bidding process in case of

regret/negative acknowledgment. In case there is no response, the bidder shall be liable for suitable action including review of their enlistment with EIL for the subject item.

12. Submission of authentic documents is the prime responsibility of the bidder. Wherever EIL has concern or apprehension regarding the authenticity/ correctness of any document, EIL reserves a right of getting the document cross verified from the document issuing authority.

13. Bidders are mandatorily required to indicate their name on each page of excel file / pdf file of Un-priced copy of Price Schedule while uploading the same.

14. **EIL enlisted bidder shall quote only for those Items for which they are registered with EIL as on the date of issuance of RFQ. Unless otherwise specified, Items quoted by EIL registered bidder other than the approved range with EIL shall not be considered for evaluation/award. Separate email communication shall be sent to the eligible enlisted bidders subsequent to issuance of enquiry in CPCL eProcurement Portal. Only the bids received from these eligible enlisted bidders shall be accepted and considered for further evaluation. Bids received from all other bidders shall be rejected.**

15. **DELETED**

16. Bidder to note that the Inspection shall be carried out in line with Inspection Philosophy mentioned in RFQ Documents.

17. **Wherever the term – "Black-listing / Holiday Listing" is appearing anywhere in the Bidding Document including formats, declaration etc, the same stands replaced by – "Black-listing/ Holiday Listing/ Suspension/ Banning List".**

18. **The offer should be valid for a period of 2 months from final Bid Due date .**

19. RFQ Document is non-transferrable.

20. Subsequent to the submission of bid, bidders are not allowed to change the price or substance of the bid i.e. scope of work, specifications, delivery schedule, completion period etc.

21. Bids received after the due date and time shall be rejected.

22. Consortium / Unincorporated Joint Venture Bids are not acceptable.

23. EIL reserves the right to assess Bidder's capabilities and capacity to execute the work using in-house information. Further in case Purchase Order awarded for the tendered work have to be terminated due to supplier's default and separate enquiry have to be floated to get the remaining / complete work executed, such defaulting supplier will not be considered for enquiries issued for the refloated works of the same project.

24. In case any bidder is found to be involved in cartel formation, his bid will not be considered for evaluation/placement of order. Such bidder will be debarred from bidding in future.

25. Canvassing in any form by the Bidder or by any other Bidder on their behalf may lead to disqualification of their Bid.

26. The bidder shall bear all costs associated with the preparation (including site visit, if required and other expenses incurred during bidding process) and submission of its bid, and the Purchaser/EIL shall in no case be responsible or liable for such costs regardless of the conduct or outcome of the bidding process.

27. EIL reserves the right to reject any or all Bids received and to annul the Bidding Process at its discretion without assigning any reason, whatsoever.

28. "Engineers India Ltd including its officers and representatives, shall under no circumstances solicit or request any payment, monetary consideration from bidders other than those specifically stipulated in the published tender/enquiry document and its amendments (if any).

Bidders are cautioned that, should they receive any phone calls, messages, emails, or personal approaches requesting payments other than specified in Tender Document, such communications are fraudulent and do not originate from Engineers India Ltd.

Engineers India Ltd shall not be responsible for any financial loss, penalty, or damage caused due to actions taken by Bidders/Contractors on the basis of such communications."

29. For detailed specifications, terms and conditions and other details, refer complete Bidding Document.

***Please specify EIL RFQ No. in all correspondence**

This limited invitation for e-bid is open only to suppliers to whom this RFQ is issued.

THIS IS NOT AN ORDER

Very truly yours,

(Mitesh Patel)
AGM (SCM-C&P)
Engineers India Limited

<table>
<tr><th>S.NO.</th><th>REF. CL. / SUBJECT</th><th>CONTENT</th></tr>
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RFQ No: MKP/B649-095-VV-MR-6009/04</td></tr>
<tr><td>2.</td><td>ITB 2.1</td><td>Enquiry Basis is: Domestic Limited Basis</td></tr>
<tr><td>3.</td><td>EIL vis-à-vis Supplier Commitments</td><td>
<table>
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<tr><td>PR</td><td>Within 2 weeks from FOA date</td></tr>
<tr><td>Nozzle Orientation</td><td>With PR</td></tr>
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<tr><td>FE report</td><td>Within 12 weeks</td></tr>
<tr><td>Balance Drawings as per VDR</td><td>Within 12 weeks</td></tr>
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Note: Time extension due to non-availability of drawing and/or details shall be considered only if the progress of work is commensurate to the drawing front released.
</td></tr>
<tr><td>4.</td><td>Payment Terms</td><td>Payment Terms shall be as per following:
A. Supply:
- 5% against approval of drawings, wherever required (identified in PO/ PR) at least in Code-2 and against submission of ABG of equivalent amount.
- 10% against identification of raw materials (*) at Supplier’s works (to be identified during kick off meeting, if not specified in enquiry) and against submission of ABG of equivalent amount.
- 70% against despatch documents (along with copy of LR/GR and IRN) through Bank or directly to CPCL/ CONSULTANT together with full taxes, duties and transportation charges.
- 5% on receipt of Final/As built Drawings/Documents/Data/Manual in requisite number of copies/sets/CDs as per VDR specified in</td></tr>
</table>

S.NO.	REF. CL. / SUBJECT	CONTENT
		Purchase Requisition. - 10% on receipt of all material at site. **(*) Major Raw Material for payment against receipt & identification of raw materials:** Plates & Forgings **For other terms, refer Special Purchase Conditions and its Annexure(s).**
5.	**GPC Cl. No. 12.3(A)**	The provisions of GPC Cl. No. 12.3(A) are modified to the following extent: The words "Such discount shall be given by the Vendor by equivalent reduction in the invoice value before presentation of documents to the Bank/CPCL for payment. Should the Vendor fail to deliver the Material(s) or to make such adjustment, the discount may be recovered by any other means" *stands replaced by* "Vendor will be raising the invoice for the full value and CPCL will deduct applicable Price Discount on basis price only. No credit notes will be given for Price Discount as it is not a supply".
6.	**ITB 5.4**	The detail of Contact Persons **for site visit**: Mr. Alok Singh, General Manager (SCM-C&P) Engineers India Limited, Tower-2, 2nd Floor, R&D Complex Sector-16, Gurugram Haryana-122001 Telephone: +91 (0) -124-289-2505 E-mail : ***alok.singh@eil.co.in***
7.	**ITB 6.2,**	The detail of Contact Persons for any query/ clarification/ communication/ **Postal address including original EMD (Bid Security) submission** is: Mr. Jaydip Purkayastha, General Manager (SCM-C&P) Engineers India Limited, Tower-1, 1st Floor, R&D Complex Sector-16, Gurugram Haryana-122001 Telephone: +91 (0) -124-289-1348 / 1341 /1287 E-mail : subash.arya@eil.co.in / mitesh.patel@eil.co.in / jaydip@eil.co.in

S.NO.	REF. CL. / SUBJECT	CONTENT
8.	**ITB 11.1 (a)**	List of Document required in the COMPOSITE BID: a) **Agreed Terms & Conditions (ATC) duly filled** b) **Formats under Annexure-I to ATC:** - Form-2: Self Certification by Bidder Under PPP-MII Order, 2017 c) **Formats under Annexure-III to ATC:** - Form-IA / II as applicable (regarding restrictions on procurement from a Bidder of a country which shares a land border with India) d) **Formats under Annexure-IV to ATC:** - Annexure-A (Self Declaration of Conflict of Interest / Alternate Bids / Multiple Bids) e) Certified documentary evidence to avail MSE purchase preference. f) Addendum / Amendment (if any) duly signed & stamped g) Any other document required as per MR / RFQ Documents h) **Signed and Stamped Copy of Unit Rates for Addition/Deletion in line with Annexure-I to BDS** i) Prices duly filled in the BoQ Format j) Priced Copy of 2 years O&M spares in BoQ Format
9.	**STOCKIST/ TRADERS**	**Not Applicable**
10.	**ITB 12.0**	**Integrity Pact: Not Applicable**
11.	**ITB 13.2.5, 13.3.2**	Not Applicable
12.	**ITB 16.0**	Bid shall remain valid for a period of **60 Days** from bid (Unpriced/techno-commercial) opening date.
13.	**ITB 32.1**	Purchase preference as admissible under the prevailing procurement policy for MSEs is **applicable** to this Enquiry Document.
14.	**ITB 32.2**	Policy for Public Procurement (Preference to Make in India), order 2017 is **Applicable** in this Enquiry Document.
15.	**ITB 30.2**	**Reverse Auction: Not Applicable**
16.	**ITB 32.3**	Policy for Providing Preference to domestically manufactured iron & steel products **(DMI&SP)** in government procurement- revised, 2025 is **Not Applicable** for this Enquiry.
17.	**ITB 32.4**	Policy for Domestically Manufactured Electronic Products **(DMEP)** is **Not Applicable** for this Enquiry Document.
18.	**ITB 32.5**	**Startup India Policy: Not Applicable**
19.	**Evaluation and Ordering Philosophy**	Evaluation and Ordering shall be done on **Item Wise** basis.
20.	**---**	**Commercial Loading of Offers: Not Applicable**

S.NO.	REF. CL. / SUBJECT	CONTENT
21.	**SPC 14.0**	Following changes has been made: i) The words *"and Sl. No. 5 of Bid Data Sheet"* stands added at the end of Cl. 14.1 of SPC. ii) Cl. No. 14.4 stands deleted.
22.	**SPC 17.0**	**Repeat Order: Not Applicable**
23.	**SPC 11.0**	**Clause no. 11. 1 is Not Applicable**
24.	**ANNEXURE-VI TO SPC TAXES & DUTIES**	MOOWR Scheme: **Not Applicable** for this Enquiry
25.	**Warranty / Guarantee Period / Defect Liability Period GPC 1.1(d)**	Applicable as per GPC 1.1 (d)-ii
26.	**QUALIFICATION UNDER MAKE IN INDIA POLICY**	**Not Applicable**.

	ADDITION DELETION RATES B649-095-VV-MR-6009, RevA, Dated 22/07/2026		
Sr. No.	**Description**	**Qty**	**Unit Rate (Rs/Kg)**
6.001	Change in support dimension SA 387 GR 22 CL.2	1 Kg	410
6.002	Change in shell dimension SA 387 GR 22 CL.2 (IMPACT TESTED) + 347 WOL	1 Kg	800
6.003	Change in size of nozzle flange SA 182 F22 CL.3 (IMPACT TESTED) + 347 WOL	1 Kg	670
6.004	Change in size of nozzle neck SA 335 P22 (IMPACT TESTED) + 347 WOL	1 Kg	660
6.005	Change in shell dimension SA 516 GR 70 (IMPACT TESTED) + 304L CLAD (NACE COMPLIANT)	1 Kg	460
6.006	Change in size of nozzle flange SA 266 GR 4 (IMPACT TESTED) + 304L WOL (NACE COMPLIANT)	1 Kg	
6.009	Change in support dimension SA 516 GR 70	1 Kg	240
6.010	Change in support dimension IS 2062 E250BR/BO	1 Kg	121
6.011	Change in pipe fitting SA 234 WP 22 CL.3 (IMPACT TESTED) + SS 347 WOL	1 Kg	1030
6.012	Change in pipe fitting SS 347	1 Kg	1170
6.013	Change in pipe fitting SA 234 WPB (IMPACT TESTED) + SS 304L WOL (NACE COMPLIANT)	1 Kg	730
6.014	Change in pipe fitting SA 403 WP 304L (NACE COMPLIANT)	1 Kg	910
6.015	Change in pipe SA 312 TP 304L (NACE COMPLIANT)	1 Kg	540
6.016	Change in plate dimension SS 321 (NACE COMPLIANT)	1 Kg	610
6.017	Change in pipe SA 106 GR B	1 Kg	160
6.018	Change in plate dimension SA 240 GR 304L (NACE COMPLIANT)	1 Kg	540
6.019	Change in plate SS 316	1 Kg	580
6.020	Change in Plate dimension SS 347	1 Kg	640

the above rates are excluding GST

INSTRUCTIONS
TO
BIDDERS (ITB)
(MR CPPP PORTAL)

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Appendix-VIII	:	Startup India Policy

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A. GENERAL

1.0 SCOPE OF BID

1.1 In connection with the Request for Quotation (RFQ), Engineers India Limited on behalf of CPCL invites e-Bid(s) under single part single envelope system, for the scope of supply and services covered under the Bidding Document (hereinafter for the purpose of these instructions collectively referred to as the "ITEM" as specified in Bid Data Sheet (BDS)).

1.2 Throughout the Bidding Document:

(a) **CLIENT / OWNER:** Client / Owner means Project Owner M/s. Chennai Petroleum Corporation Limited (**CPCL**), a company incorporated in India having its registered office at New No: 536, Anna Salai, Teynampet, Chennai, Tamil Nadu – 600018.

(b) **CONSULTANT:** Consultant means EPCM Consultant, M/s Engineers India Limited (**EIL**), a company incorporated in India having its registered office at Engineers India Bhawan, 1, Bhikaiji Cama Place, RK Puram, New Delhi-110066 and shall include its successors and assignees.

(c) except where the context requires otherwise, words indicating the singular also include the plural and words indicating the plural also include the singular; and word written in capital or small letter or where the first letter capital have the same meaning; and

(d) "Day" means calendar day.

(e) The term 'Bid Due Date' shall mean the 'Final Bid Due Date'.

(f) the term "bid"/ "tender" / "offer" / "proposal" and their derivatives ("RFQ/Bidding/Enquiry" documents, "bidder/tenderer", "Vendor/Supplier/Seller", "Bid/tender", "bidding/tendering", "Contract/Purchase Order" etc.) are synonymous, and day means calendar day. Singular also means plural and vice versa.

(g) **INCOTERM 2020:** INCOTERM 2020 means the terms which shall be applicable for procurement of equipment/material under the subject RFQ / Enquiry document.

(h) Material Requisition (MR) means the Technical Part of RFQ Document.

(i) E-Tendering website (or e-procurement CPP portal / CPPP/ E-tendering system/ e-Procurement Portal/ e-tendering portal) means online portal on which Bidders are required to submit their bid. For this RFQ, e tendering website is https://cpcletenders.nic.in.

1.3 The successful bidder / "supplier" shall complete the Scope of RFQ within the Time Period/Completion Period stated in RFQ document.

2.0 ELIGIBLE BIDDERS

2.1 An Eligible bidder may be a firm or a company, who, have been issued/intimated for issuance of RFQ Document on Limited Competitive Basis.

The bidding is open only to the bidders to whom Bidding Document has been issued on e-tendering portal or intimation for issuance of Bidding Document has been sent through EIL tender portal.

2.2 If a supplier / an agency is placed on **Black-listing / Holiday Listing /Suspension / Banning List/ Negative List** of EIL or Owner or MoPNG after issue of enquiry but before Composite bid opening, their bid will not be considered for further evaluation.

If a supplier/ an agency is placed on **Black-listing/ Holiday Listing/ Suspension/ Banning List/ Negative List** EIL or Owner or MoPNG after opening of Composite bids and such a supplier/ agency happens to be the lowest bidder, the next lowest bidder shall be considered as L1.

2.3 In case of Limited enquiry, if the RFQ Document is/was issued inadvertently to such Bidder/downloaded from website by such Bidder, then Bid submitted by such Bidder shall not be considered for evaluation/ award.

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2.4 Department of Expenditure (DoE) vide O.M. No. 7/10/2021-PPD(1) dated 23.02.2023 has issued the revised guidelines for Restrictions on Procurement from Bidders from Country(ies) sharing Border with India. These guidelines are available on the website of DoE (https://doe.gov.in). In view of the above, Bidder to submit signed and stamped copy of Form-IA (on Company's Letterhead) attached with **Appendix-VII of this ITB** for compliance.

2.5 Ministry of Petroleum & Natural Gas (MoPNG) vide File no. FP-20013/2/2017-FP-PNG-Part(4)(E-41432) dated 26.04.2022, have issued a directive that revised Public Procurement (Preference to Make in India) (PPP-MII) Order, 2017 issued by DPIIT and as amended from time to time shall be applicable to all the Public Sector Undertakings and their wholly owned subsidiaries under MoP&NG, Joint Ventures that have 51% of more equity by one or more Public Sector Undertakings under MoP&NG, with effect from 01.04.2022.

Under this policy, buyer reserves the right to allow Suppliers or Service providers in a tender and purchase preference as admissible under the prevailing policy, subject to complying with the requirements/ conditions defined herein and submitting documents required to support the same.

Further, following documents and its amendment thereof shall be applicable:

i. PPP-MII Policy O.M. No P-45021/2/2017-PP (BE-II)-Part(4)Vol.II Dated 19.07.2024 issued by DPIIT,
ii. MoPNG O.M. File No. FP-20013/2/2017-FP-PNG-Part(4)(E-41432) dated 26.04.2022 and 26.03.2024.
iii. File No. W-43/4/2019-IPHW-Meity dated 07.09.2020 and S.O. No 4591(E) dated 12.10.2023 issued by Meity.
iv. File No.1(10)/2017-CLES dated 06.12.2019 issued by Meity.
v. File No FP-18-10/2017-IP dated 21.10.2024 issued by Ministry of Telecommunication

Refer **Appendix-IV to ITB** for details.

2.6 **REJECTION CRITERIA:**

Bid is liable for rejection in the following circumstances:

a) Submits the tender late i.e. after due date and time
b) Unsolicited bids.
c) Stipulates the validity period less than what is stated in the Tender Documents
d) Stipulates his own conditions and does not agree to withdraw the deviations, rendering his bid unacceptable
e) Does not disclose the full names and addresses of all his partners or Directors as applicable wherever called for in the tender.
f) Does not fill in and submit the required annexure, specifications, etc. as specified in the tender.
g) Does not submit bid in the prescribed format making it impossible to evaluate the bid
h) Indulges in tampering of tender documents
i) Does not conform to any tender condition which stipulates non-conformance of tender conditions as rejection criteria.
j) Failure of bidder to submit tenders without taking cognizance of corrigendum / amendment issued.
k) Failure of bidder to submit reply to Technical / Commercial Query (TQ/CQ) within the given dead line.
l) In case the bidder alters / modifies the bid suo-moto after opening the bids within the validity period.
m) Bidder who are under liquidation, court receivership or similar proceedings. Bidder shall submit a self-certificate stating that they are not under liquidation, court receivership or similar

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proceedings. Failure to do so or if the bidder is under liquidation, court receivership or similar proceedings, their bids shall not be considered

n) A bidder Who is on Holiday/ Suspension /Banning List / Negative list of CPCL or MoPNG or Holiday/ Negative list of EIL on due date of submission of bid / during the process of evaluation of the bids, the offers of such bidders shall not be considered for bid opening/evaluation/Award.

3.0 FRAUDULENT PRACTICES

Relevant clause as per Special Purchase Conditions (SPC) shall be applicable.

B. CONTENTS OF BIDDING DOCUMENT

4.0 BIDDING DOCUMENT

4.1 The RFQ document consisting of Commercial & Technical Section has been hosted on e-procurement website (i.e. CPCL e-tendering Portal https://cpcletenders.nic.in). However, a link shall be provided in EIL website http://tenders.eil.co.in.

4.2 The Bidding Document shall in general consist of the following and should be read in conjunction with any amendment issued in accordance with Clause No. **7.0** of ITB.

i. Request for Quotation (RFQ)
ii. Bid Data Sheet (BDS)
iii. Instructions to Bidders (ITB) and its Appendices
iv. Agreed Terms and Conditions (ATC) and it's Annexures
v. Special Purchase Conditions (SPC) and its Annexure
vi. General Purchase Conditions (GPC)
vii. Format for Performance Bank Guarantee (PBG) and Advance Bank Guarantee (ABG)
viii. Additional Documents Mentioned in RFQ / BDS
ix. Price Schedule / Schedule of Prices (SOP/BoQ) / Schedule of Rates (SOR)
x. Material Requisition (MR)

4.3 Although all the details presented in this Bidding Document have been compiled with reasonable care, the bidder is expected to examine the Bidding Document, including all instructions, forms, terms, specifications and drawings in the Bidding Document and it includes all the documents as per Table of Contents. In case of any ambiguity or incompleteness, the bidder must bring it to the attention of EIL immediately and prior to submission of Bid.

4.4 The bidder is expected to examine the RFQ document downloaded from the e-Procurement Portal, bidding guidelines receipt from EIL or available on EIL website, all instructions, formats, terms, specifications and drawings etc., enclosed in the RFQ documents. The invitation for bid (bidding) together with all its attachment thereto, shall be considered to be read, understood and accepted by the bidder in co-ordination to each other. Failure to furnish all information required by the RFQ documents or submission of a bid not substantially responsive to the RFQ documents in every respect will be at bidder's risk and may result in the rejection of the Bid.

4.5 RFQ document once issued in case of limited competitive bidding is non-transferable. However, in case of Open Competitive Bidding, since the RFQ Document is available for free downloading on the portal, any bidder may submit the Bid and the same shall be evaluated in line with the provisions of the Bidding Document.

4.6 **Additional requirements for Category-I MRs (Refer Material Requisition for MR Category):**

4.6.1 Bidder to note that no correspondence shall be entertained after the due date for bid submission.

4.6.2 Bidder shall quote prices for only those items which the bidder can supply material as per specifications and for which bidder is approved by EIL.

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4.6.3 No technical details of any nature shall be specified in the bid unless otherwise specifically sought as per the Bidding Document. If the bid stipulates any technical deviations, it shall be summarily rejected.

4.6.4 Bidder shall note that enquiry is issued on zero deviation basis. Bidder shall comply with all technical specifications & commercial terms and other conditions of the enquiry document without any deviations/ clarifications. Any deviation stipulated by bidder shall result in summary rejection of the bid.

4.6.5 The submission of prices by the bidder shall be construed that bidder has confirmed compliance with all technical specifications of the quoted items as well as commercial terms & conditions of enquiry document.

4.6.6 Other than the details & documents specifically required in the bid as per enquiry document, bidder shall not submit any other detail / document in the bid.

4.6.7 Bidders in their own interest should ensure submission of complete information/ documentation in the first instance itself. Buyer reserves the right to evaluate the bids based on the details/ documents so furnished in bids without seeking any subsequent additional information/ clarification/ documents.

4.6.8 Bidders to refer the requisite format provided in MR (if sought in MR) for providing the Technical information which are essential for PR preparation.

- Bidder is requested to fill the above format and submit the same without fail in their Composite Bid. Further, Bidder to note that post opening of Composite, no query will be raised either due to non-submission or incomplete submission.
- Therefore, it is requested for Bidder to ensure submission of the above Technical information complete in all respects in the Composite Bid so that the Purchase Requisition (PR) can be issued to the successful Vendor without any delay past issuance of LOA/PO
- Bidders to note that any delays in issuance of PR post award by EIL, due to either non-submission or incomplete submission of the requisite Technical Information as sought above in Composite Bid, shall be to Vendor's account.

5.0 SITE VISIT

5.1 For Scope involving Site Work, bidder, if required, may visit and examine the Project Site and its surroundings and obtain for itself on its own responsibility all information that may be necessary for preparing the bid.

5.2 Bidder is advised to visit and examine the site (at his own cost), its surroundings and familiarise themselves of the existing facilities & environment, and collect all other information which he may require for preparing and submitting the bid and entering into the Contract. Claims and objections due to ignorance of existing conditions or inadequacy of information will not be considered after submission of the bid and during implementation. Bidder shall provide at least two days in advance the details of their representatives along with details of laptop, mobile or any other electronic device or any equipment if any, being carried so as to make arrangement for gate passes. Owner's decision on this regard shall be final and binding upon the bidder.

5.3 Bidder and any of its personnel will be granted permission by EIL/Client to enter upon its premises and lands/ Project Site for the purpose of such visit, but only upon the express condition that the bidder and its personnel will relieve and indemnify EIL/Client and its personnel from and against all liability in respect thereof, and will be responsible for death or personal injury, loss of or damage to property, and any other loss, damage, costs, and expenses incurred as a result of the inspection.

5.4 For site visit, Bidder may contact the persons in EIL/CPCL as per details provided in Bid Data Sheet (BDS).

6.0 CLARIFICATION OF BIDDING DOCUMENT & PRE-BID MEETING

6.1 The enquiry is issued on "***Zero Deviation Bidding***" basis wherein no post bid correspondence of any nature may be entertained. Bidders in their own interest are advised to take part in the pre-bid meeting

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seriously by issuing their genuine queries and also by attending the pre-bid meeting by their competent personnel. This will help bidder not only to cut short of the evaluation time leading to requirement of shorter bid validity but also enable them to submit their best prices based on the clear techno-commercial scope.

6.2 Bidder shall upload the pre-bid queries on above referred e-procurement portal within the cut off dare specified in the portal. Post cut off date bidder shall not be able to upload their queries on the portal. EIL/Owner reserves the right not to entertain the queries raised through email / any other mode. Reply to Pre-Bid Queries shall be hosted on E-Tendering Portal.

6.3 Pre-bid meeting shall be attended by competent representative(s) of the bidder.

6.4 Replies to the pre-bid queries will be uploaded in e-procurement portal**.** After pre-bid meeting, no further queries will be entertained from the bidders. The response of pre bid queries shall not form part of the Bidding Document unless issued as an Addendum/ Amendment.

6.5 The bidders may carry their qualification documents in case they desire them to be reviewed by EIL officials during the pre-bid meeting.

6.6 **Any modification to the RFQ document, which may become necessary as a result of the pre-bid queries / pre bid discussion, shall be intimated to all bidders through the issue of an Addendum/Amendment.** Record notes of pre-bid meeting/reply to pre-bid queries shall not be considered as part of RFQ.

6.7 Technical/Commercial queries (TQ/CQ) shall normally not be issued once the bid has been opened. However, wherever CQ/ TQ are unavoidable, the same shall be raised as deemed necessary and the cut-off date given for CQ/ TQ replies shall be adhered to. Offers shall be evaluated based on the information available upto cut-off date for CQ/ TQ replies. Bidder shall not submit any Price against TQ/CQ raised unless specifically sought in writing.

7.0 AMENDMENT OF BIDDING DOCUMENT

7.1 At any time prior to the deadline for submission of bids, EIL may, for any reason whether at his own initiative or in response to the clarification requested by the prospective bidder(s), issue amendment in the form of Amendment/ Addendum.

7.2 All Amendments, (if any) will be uploaded in e-procurement portal. Generally, intimation of uploading of documents in website is sent to the bidders (all bidders in case of Limited bidding and to bidders who have submitted acknowledgement on tender portal in case of NIT) in their email id. It is the responsibility of the bidder to regularly check the website and provide current and authenticated email id. Any amendment thus issued shall become part of Bidding Document. Bidder shall submit a copy of the Addendum duly signed and stamped in token of his acceptance. Any revision, addendum, corrigendum, time extension, etc. will be hosted on these websites only. Bidders are requested to visit the websites regularly to keep themselves updated. Bidders have to take into consideration of all the addendum(s) / corrigendum (s) issued/web hosted, before submitting the bid and shall consider its impact in his bid.

7.3 To give bidders reasonable time in which to take an addendum into account in preparing their bids, EIL may extend the deadline for the submission of bids, pursuant to Clause No. **19.2** of ITB.

7.4 In exceptional circumstances, EIL may also for any reason issue amendment subsequent to opening of un-priced bids. Any amendment thus issued shall become part of Bidding Document. Bidder shall follow the instructions issued along with Amendment with regard to submission of impact on quoted price / revised price, if any.

8.0 ACKNOWLEDGEMENT & CONFIRMATION

8.1 Within 10 days of downloading of Bidding Document, bidder shall acknowledge and confirm his intention to bid for the requested item/work on EIL website http://tenders.eil.co.in.

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8.2 In case of limited competitive bidding, Bidders should respond to this enquiry either by submitting their bids or by explaining the reason for non-submission of the offer. In case there is no response either way continuously, the bidder shall be liable for suitable action including review of their enlistment with EIL for that particular item.

C. PREPARATION OF BIDS

9.0 COST OF BIDDING

9.1 Bidder shall bear all direct and indirect costs associated with the preparation and submission of its Bid (including clarification meetings, and site visit, if any), and EIL shall not be responsible or liable for those costs, regardless of the conduct or outcome of the bidding process.

10.0 LANGUAGE OF BID

10.1 The Bid prepared by the bidder, all correspondence/drawings and documents relating to the bid exchanged by the bidder with the Owner/Consultant shall be in English Language only. In case, any printed literature furnished by bidder written in another language so long as accompanied by an English translation, in which case, for the purpose of evaluation and interpretation of the bid, the English translation shall govern.

10.2 In the event of submission of any document/ certificate by the Bidder in a language other than English, the English translation of the same shall be submitted duly authenticated by either of the following:

a) Local chamber of commerce
b) Any recognized/ authorized translator in India

11.0 DOCUMENTS COMPRISING THE BID

No Physical Bids / Offers shall be permitted (other than the documents which are specifically mentioned to be submitted in physical form for which scanned copy already uploaded along with the bid). The offers/bids submitted online through e-procurement portal as mentioned in clause 1.2(i) above shall only be considered as bid for evaluation and ordering. Bidders are required to upload the Bid/offer along with all supporting documents including Priced bid on the **E-Tendering website only.** Electronic Bids (e-bid) shall be submitted on e-tendering Portal as per available functionalities.

Bidders to upload the **Composite bid (both unpriced and priced bid)** strictly at designated place on e-tendering Portal. Offer prices shall not be mentioned at any place other than the designated field(s) provided on e-tendering Portal.

Notes:

1. Deviation to terms & conditions, presumptions etc. shall not be stipulated in bid. In case of any conditions stipulated in bid, the bids of such bidders shall be summarily rejected and shall not be considered for further evaluation. Evaluation shall be carried out excluding such bidder(s).
2. Other than details / documents specifically spelt out in the RFQ document, bidder shall not furnish any other details/comments/documents/clarifications elsewhere in the e- bid. In case of any deviations/comments/clarifications to RFQ terms and conditions stipulated in the bid, the bid shall be liable for rejection.

11.1 **TECHNO-COMMERCIAL/ COMPOSITE BID (through e-tender portal)**

(a) List of Document required in the COMPOSITE BID is as per BDS/RFQ

(b) Bidder shall fill-in the prices in the BoQ format and upload the same on e-tendering Portal.

Note:

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i) Prices for Two Years O&M spares, Overtime rates for Supervision Services, any-other optional rates/charges sought separately as per instructions provided in 'Preamble to Submission of Prices ' in excel format shall be filled in the BoQ format in e-tendering portal.

ii) The prices as submitted in the price part only shall be considered for evaluation and ordering, if any. If prices are not found in the format of Price Schedule/BoQ and even if mentioned elsewhere, the bid shall be rejected

12.0 INTEGRITY PACT (applicability shall be as per BDS)

12.1 Bidder shall submit, the scanned copy of the Integrity Pact (along with Cover Letter) duly signed and stamped in token of compliance in Part-I of the Bid (**enclosed with RFQ**). **Non-submission of Integrity pact in the bid shall be a ground of disqualification. However, in case of non-submission of Integrity Pact in the bid, an opportunity shall be given in the CQ to submit the Integrity Pact.**

12.2 For detail of Independent External Monitor(s) (IEMs), appointed by CPCL, in terms of Integrity Pact (IP) refer Annexure-V to ATC.

12.3 No deviations / modifications in IP shall be acceptable. Bidder shall submit the same format of signed integrity pact (as provided in the bidding document duly signed by the buyer), duly filled-in, signed & stamped by the authorised signatory of bidder.

12.4 Integrity Pact shall be signed by the authorised signatory of the bid subject to following:

12.4.1 In case of Proprietorship firm IP may be signed by the Proprietor

12.4.2 In case of Partnership firm, IP shall be signed by at all the Partners of the firm.

12.4.3 In case of consortium bids (if permitted), Integrity Pact shall be signed by all consortium members

12.4.4 In Case of sub-contracting by the main/principle contractor, the main/Principal Contractor shall take the responsibility of the adoption of Integrity Pact by the Sub-contractor (of main sub-contractor).

13.0 CONFLICT OF INTEREST/ ALTERNATIVE OFFERS / MULTIPLE BIDS

13.1 Bidders having a conflict of interest shall not be eligible to participate in the bidding process. During evaluation, any bidder found to have conflict of interest shall be disqualified. Where an Enquiry has evaluation basis on Part(s)/Group(s)/Item(s), the conditions of Conflict of Interest/ Alternative Offers/ Multiple Bids shall be applicable on individual basis for each Part/Group/Item.

13.2 The bidder shall be considered to have a conflict of interest in the bidding process and execution of the resultant contract in the following situations:

13.2.1 If, Director(s) (in Board of Directors) of Bidder is a relative of any Director (in Board of Director) of Owner/ EIL Or the bidder is a firm in which any Director (in Board of Director) of Owner/EIL or their relative is a partner.

13.2.2 If, the bidder (or his allied firm) provided services for the need assessment/ procurement planning of the bidding process in which it is participating.

Notes:

a. "Allied firm" ('affiliates'/ 'affiliated firm', 'sister concern', 'associated firm', or 'related party') of a bidder/ contractor (Principal firm, including Joint Venture Company) is a firm/ concern (including Joint Venture Company) that comes within the sphere of effective control/ influence of the principal firm, wherein the Principal Firm –

i) being a proprietary firm, owns the Allied Firm,

ii) being a partnership firm, has common (all or majority of) partners, or any one of its partners has profit share of 20% or more, in the Allied Firm

iii) has common Management (say majority of director) with the Allied firm;

iv) its partners or directors have a majority interest in the management of the Allied Firm;

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v) has a controlling voice by owning substantial (20% or more) shares in the Allied Firm;

vi) directly or indirectly controls or is controlled by or is under common control, by way of any agreement/ MoU or otherwise with the Allied Firm,

vii) has the Allied Firm as its successor/ subsidiary or vice-a-versa;

viii) has common offices/ manufacturing facilities with the Allied Firm.

b. Inter-alia services for "need assessment" preparation of - feasibility/ cost estimates (except Budgetary Quote)/ Detailed Project Report (DPR), design/ technical specifications, terms of reference (ToR)/ Activity Schedule/ schedule of requirements or the Tender Document etc.

c. Further, in case of a holding company having more than one independently manufacturing units, or more than one unit having common business ownership/management, only one offer should be submitted. Similar restrictions would apply to closely related sister companies. Bidders must proactively declare such sister/ common business/ management units in same/ similar line of business.

13.2.3 If, a Principal and their agent participate in the tender process: A Principal can authorize only one agent, and an agent should not represent or quote on behalf of more than one Principal. However, this shall not debar more than one Authorised distributor (with/ or without the OEM) from quoting equipment manufactured by an Original Equipment Manufacturer (OEM) in procurements under a Proprietary Article Certificate.

Note: "Agent" is a person, or a legal entity employed to act for/ represent another (called the Principal) in dealings with a third person or legal entity. The Agent is a representative participating in the Tender Process or the execution of a Contract for and on behalf of its principals.

13.2.4 If, a bidder participates in more than one bid in the tender process: Participation in any capacity by a Bidder (including the participation of a Bidder as a partner/ JV member or sub-contractor in another bid or vice-versa) in more than one bid shall result in the disqualification of all bids in which he is a party. However, this does not limit the participation of an entity as a sub-contractor in more than one bid if he is not bidding independently in his own name or as a member of a JV.

13.2.5 If, the bidder(s) who are selling agents/ authorized distributors/ authorized dealers/ authorized supply houses/Stockist/ Traders/ Resellers of the domestic manufacturer(s) and domestic manufacturer(s) also submit bids for the same item(s)/Group(s)/Part(s).

Where both the Manufacturer and one or more of its selling agents/ authorized distributors/ authorized dealers/ authorized supply houses/Stockist/ Traders/ Resellers submit bids for the same item(s)/Group(s)/Part(s) only the bid submitted by the manufacturer shall be considered for evaluation. This rule applies irrespective of whether the manufacturing shops, plants, mills, or works offered are identical or distinct. Bids from one or more selling agents/ authorized distributors/ authorized dealers/ authorized supply houses/ Stockist/ Traders/ Resellers of the same manufacturer for the same item(s)/Group(s)/Part(s) shall be acceptable only if the Manufacturer does not participate in the bidding process on their own.

In case, if complete offer of selling agents/ authorized distributors/ authorized dealers/ authorized supply houses/Stockist/ Traders/ Resellers is rejected on above grounds, Earnest Money Deposit (EMD), as applicable, shall be returned.

Note: Above clause 13.2.5, shall be applicable only for procurement of the items covered under Domestically manufactured iron & steel products Policy.

Refer BDS for applicability of this clause.

13.3 Submission of Declaration on conflict of interest:

13.3.1 The Bidder shall declare any actual, potential, or perceived conflict of interest in tender process as per the prescribed format enclosed as **Annexure-IV to ATC**.

13.3.2 Additionally, selling agents/ authorized distributors/ authorized dealers/ authorized supply houses/Stockist/ Traders/ Resellers (**as per clause no 13.2.5 above**) shall upfront submit the details

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of manufacturers in Composite bids itself as per **Annexure-IV to ATC**. **Refer BDS for applicability of this clause.**

13.3.3 Wherever bidders do not submit the declaration in the offer citing that they are not aware of the other bidders, such a declaration can be submitted through CQ. In case bidder(s) do not submit the declaration even after CQ, their bid shall be rejected.

13.4 A Bidder found to have a conflict of interest, EIL/Owner may, at its sole discretion, seek further clarification, mitigation measures, disqualify or reject the bid. Failure to disclose a conflict of interest, or submission of a false declaration, shall render the Bidder liable to disqualification/ rejection of bid and other actions as specified in Bidding Document.

14.0 BID PRICES AND DISCOUNTS

14.1 The prices must be quoted strictly in the format of Price Schedule/BoQ provided in the Bidding Document. Bidder shall not quote any discount separately for any item or on overall price. Conditional discount, if offered, shall not be considered for evaluation.

14.2 Rates/ prices quoted by the bidder shall remain **FIRM, FIXED AND VALID** till the completion of the scope of supply/Works (as applicable) and will not be subject to variation on any account except as otherwise specifically provided in the Bidding Documents.

14.3 The bidder shall quote the prices after careful analysis of cost involved for the performance of complete work considering all parts of the RFQ documents. In case, any activity though specifically not covered but is required to complete the work as per scope of work, scope of supply, specifications, standards, drawings, GPC, SPC or any other part of RFQ Document, the prices quoted shall be deemed to be inclusive of full and final cost incurred for such activity(ies).

14.4 It shall be the responsibility of the Bidder to duly enquire of their own and comply with all applicable laws, rules, regulations, orders and formalities applicable to Goods and Services Tax (GST), Custom Duty etc. on the manufacture, sale and/or supply of any material / services to Owner / EIL. It is clearly understood that Owner/ EIL will not have any additional liability towards payment of applicable Taxes & Duties as a result of Bidder's wrong assessment / interpretation of applicable taxes & duties.

14.5 **Price basis w.r.t inclusion/exclusion of taxes**

14.5.1 In case of indigenous bidders, the quoted price shall be deemed to be inclusive of all taxes and duties except GST (i.e. IGST or CGST and SGST/UTGST applicable in case of interstate supply or intra state supply respectively and GST compensation cess, if applicable).

14.5.2 In case of foreign bidders, bidders shall submit their prices on either FOB international sea port of exit basis or CFR destination port basis (INCOTERM 2020), as specified in the price schedule. Quoted prices shall be inclusive of all applicable taxes and duties up to the respective delivery point.

14.5.3 Transportation charges upto respective project site, if sought separately, shall be exclusive of GST.

14.5.4 Site work prices, if applicable, shall be exclusive of GST but inclusive of all other taxes and duties. All necessary registrations, if required, for carrying out the site activities shall be done by the bidder and costs towards the same shall be included in quoted site work prices.

14.5.5 Prices of per diem supervision charges, hazop study, 3D modelling, training/AMC, if applicable, shall be exclusive of GST. However, GST shall not be applicable on training provided at foreign bidder's work/training facilities in their country.

14.6 All duties, taxes, and other levies payable shall be as per Special Purchase Conditions (SPC).

14.7 It shall be the responsibility of the Bidder to duly enquire on their own and comply with all applicable laws, rules, regulations, orders and formalities applicable to Goods and Services Tax (GST), Customs Duty etc. on the manufacture, sale and/or supply of any material / services to Owner / EIL.

14.8 **The Bidder shall keep the Owner / EIL indemnified from and against any and all claims, demands, prosecutions, penalties, damages, demurrages and/or other levies whatsoever** made

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or levied by any Court, Tribunal or the Customs or other Authorities with respect to any alleged breach, evasion or infraction of such duties, taxes, charges or levies or any breach or infraction of such laws, rules, regulations, orders or formalities concerning the same and from the consequence thereof.

14.9 Foreign Exchange rate variation shall not be provided.

15.0 CURRENCIES OF BID AND PAYMENT

15.1 Currency of bid and Payment shall be as specified in Special Purchase Conditions/ Price Schedule.

16.0 PERIOD OF VALIDITY OF BIDS

16.1 Bids shall remain valid for the period specified in the BDS after the bid submission deadline date prescribed by EIL/Owner in accordance with ITB. A bid with shorter validity period shall be rejected by EIL/Owner as non-responsive. Bidders shall not be entitled during the above period, without the consent of EIL/Owner in writing, to revoke or cancel their Bid or to vary the Bid given or any term thereof. In case of bidders revoking or cancelling their Bid or varying any terms in regard thereof without the consent of EIL/Owner in writing, Owner/EIL shall forfeit Bid security, submitted along with the bid and shall reject their bids. Such Bidder may also be put on **Black-listing/ Holiday Listing/ Suspension/ Banning List/ Negative List**.

16.2 In exceptional circumstances, prior to the expiration of the bid validity period, EIL may request bidders to extend the period of validity of their bids without any price implication and change in terms & conditions. The request and the responses shall be made in writing. A bidder granting the request shall not be permitted to modify its bid.

A bidder may refuse the request after which their offer will not be considered for further evaluation. Bid security shall not be forfeited in case a bidder refuses validity extension.

17.0 SPECIAL INSTRUCTIONS TO FOREIGN BIDDERS

17.1 The bidder must ascertain and confirm along with supporting documents in the bid, if any Customs Duty exemption / waiver is applicable to the products being supplied by him under any multi-lateral / bi-lateral trade agreement between India and bidder's country.

17.2 The bidder shall be liable to provide all documentation to ensure availing of the exemption / waiver. In case, the bidder defaults on this due to any reason, whatsoever, he shall be liable to bear the incremental Customs Duty applicable, if any.

17.3 Any Customs Duty applicability on account of any change in the bi-lateral / multi-lateral agreement shall be to bidder's account.

17.4 Documentation to be furnished for availing the exemption / waiver of Customs Duty shall be specifically listed in the Letter of Credit also as the pre-requisite for release of payment against shipping documents and this documentation shall necessarily form a part of shipping documents.

18.0 FORMAT AND SIGNING OF BID

18.1 The e-bid shall be uploaded duly digitally signed (e-signed) using the digital signature of a person duly authorized to sign on behalf of the bidder.

18.2 Bidders to upload their e-bids (comprising of Un-price and Price part) strictly at the designated place in the e-tendering portal. Non-Compliance to the same may lead to rejection of their Bids.

18.3 DIGITAL SIGNATORY

18.3.1 In case of e-tendering, login and submission of bids by the bidder on the e-tendering portal is through requisite & valid digital signature certificate.

~~18.3.2 e-Bid Signatory who is using his digital signature for uploading / signing the e-bid, shall also carry the Power of Attorney.~~

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18.3.3 All documents as submitted in the e-Procurement Portal using digital signature at any time during the Bidding process shall be deemed considered signed by the POA holder only, irrespective of POA holder's signature on all pages or not.

D. SUBMISSION AND OPENING OF BIDS

19.0 DEADLINE FOR SUBMISSION OF BIDS

19.1 Bidders shall submit their bids electronically in the e-tendering portal, within the date and time **specified in the RFQ Covering Letter** or any extension thereof. Bidders submitting bids electronically shall follow the electronic bid submission procedures of e-tendering Portal.

No Physical bid shall be permitted other than submission of the documents categorically asked for in hard copies as part of e-bids submitted online through the e-tendering Portal. The bids submitted online through e-tendering Portal (https://cpcletenders.nic.in) shall only be considered for evaluation and ordering. The e-bid must be complete in all respects, leaving no scope for ambiguity. Bidder shall be fully responsible for the bid submitted in e-tendering portal and no relief or consideration can be given for errors and omissions.

19.2 EIL/Owner may, at its discretion, extend the deadline for the submission of bids by amending the Bidding Documents in accordance with Clause No. **7.0** of ITB, in which case all rights and obligations of EIL/Owner and bidders previously subject to the deadline shall thereafter be subject to the deadline as extended.

19.3 Bidder should avoid the last hour rush to the website wherein bidder is likely to miss the deadline for bid submission. Due date extension request due to above reason shall not be entertained.

20.0 LATE BIDS

20.1 E-tendering system shall close immediately after the deadline for submission of bid. Any bids being submitted in physical form shall not be considered for opening / evaluation / award and will be returned to such bidders.

20.2 Unsolicited bids or bids being submitted to place/website other than one specifically stipulated in the RFQ documents shall not be considered for opening/evaluation/award.

21.0 WITHDRAWAL, SUBSTITUTION, AND MODIFICATION OF BIDS

21.1 The bidder may modify, re-submit or withdraw its e-bid after the bid submission, but, before the due date and time for submission of Bid following the electronic bid submission procedures of e-tendering portal**.**

21.2 No bid shall be withdrawn, substituted, or modified in the interval between the deadline for submission of bids and the expiration of the period of bid validity or any extension thereof. Withdrawal of a bid during this interval shall result in the forfeiture of bidder's EMD/Bid security.

21.3 In case bidder fail to accept the order or withdraw the bid after bid opening, punitive action shall be taken against them as per the EIL/Owner Policy.

22.0 BID OPENING

22.1 The Composite/ Single Packet Bid (unpriced & priced) shall be opened online through e-tendering portal, at the specified date and time given in the RFQ document or on the extended due date, as applicable. Bidders can view the details by logging in E-Tendering Portal.

Note: Online opening as above, will be deemed as opened in public and there will be no separate public opening. Hence, bidders will not be called during opening of the bids.

E. EVALUATION AND COMPARISON OF BIDS

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23.0 CONFIDENTIALITY AND COMMUNICATION PROTOCOL

23.1 A bidder may seek clarification regarding the Bidding Document provisions, bidding process and / or rejection of his bid. EIL shall respond to such requests within a reasonable time. However, such information relating to the evaluation of bids and recommendation of award shall not be disclosed to any other persons not officially concerned with the bidding process.

23.2 Any attempt by a bidder to influence EIL in the evaluation of the bids or award decisions shall render their Bid liable for rejection.

23.3 Notwithstanding Clause No. 23.2 of ITB, from the time of bid opening to the time of placement of Order, if a bidder wishes to contact EIL on any matter related to the bidding process, it shall done so in writing only.

23.4 Bidder shall strictly follow the communication protocol as prescribed in Enquiry. In case the queries/ representation is not addressed to the persons specified in the communication protocol stipulated in the Bidding document, EIL/Owner may not give any cognizance to the same and EIL/Owner shall not be liable to reply them.

23.5 Only a bidder who has participated in the concerned procurement process i.e. prequalification or bidding, as the case may be, can make such representation in respect of the procurement process in which they have participated concerning their bid only.

23.6 Any representation pertaining to Award stage by a bidder whose Composite bid is opened and is directly affected by such award shall have to submit such representation within 10 days of the issue of LOA.

23.7 Following decisions of the EIL/Owner shall not be subject to review

i. Determination of the need for procurement;
ii. Selection of the mode of procurement or bidding system;
iii. Choice of selection procedure;
iv. Provisions limiting participation of bidders in the procurement process;
v. The decision to enter into negotiations with the L1 bidder;
vi. Cancellation of the procurement process except where it is intended to subsequently re-tender the same requirements
vii. Issues related to ambiguity in contract terms after award of contract.
viii. Complaints against specifications except under the premise that they are either vague or too specific so as to limit competition may be permissible.

24.0 ZERO DEVIATION

24.1 Bidders to note that enquiry is issued on Zero deviation basis, accordingly all clarification should be sought by the bidder before bid submission. Bids shall be submitted as "**Zero deviation Basis**". Accordingly offer is to be submitted in complete compliance to Terms & Conditions of Bidding Document without any deviation. Offer with deviation be liable for rejection without any further correspondence.

In view of this Bidder shall ensure submission of complete bid without any deviation as per requirement of Bidding Document in first instance itself.

25.0 DETERMINATION OF RESPONSIVENESS

25.1 EIL's determination of a bid's responsiveness is to be based on the contents of the bid itself, as defined in Clause 11.0 of ITB.

25.2 A substantially responsive bid is one that meets the requirements of the Bidding Documents without deviation.

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25.3 A substantially responsive Bid is one which conforms to all the terms & conditions, specifications etc. of the Tender Document without any material deviation or reservation or omission, for this purpose Purchaser defines the foregoing terms below:-

a) 'Deviation' is departure from the requirement specified in the Tender Documents.

b) 'Reservation' is the setting of limiting condition(s) or withholding from complete acceptance of the requirement in the Bidding Document.

c) 'Omission' is the failure to submit part or all of the information or documentation required in the Tender Document.

d) A material deviation, reservation or omission is one that, if accepted would,

i) Affect in any substantial way the scope, quality or performance of the job as specified in the Tender Document.

ii) Limit, in any substantial way, inconsistent with the Tender Document, the Owner/EIL's rights or the Bidder's obligation under the proposed Contract.

iii) If rectified, would unfairly affect the competitive position of other bidders presenting substantially responsive bids.

25.4 The Owner/EIL shall examine all aspects of the Bid to confirm that all requirements have been met without any material deviation, reservation or omission.

25.5 Bidder should not be under liquidation, court receivership or similar proceedings. Bidder shall submit confirmation in Agreed Terms and Conditions in this regard.

In case Bidder(s) affirms that present litigations/arbitrations have impact on their obligations to perform the Contract or doesn't provide the affirmations as above, their bids shall be rejected. Bidder shall submit confirmation in Agreed Terms and Conditions in this regard.

Note: In case it comes to the notice of EIL/Owner that the bidder has given wrong declaration in this regard, the same shall be dealt as 'Fraudulent Practices' and action shall be initiated as per the procedure for action in case of corrupt / Fraudulent / collusive / coercive practices.

25.6 **Bidder shall not be allowed to submit any Price Implication or Revised Price after submission of e- Bid. In case Exceptions and Deviations submitted by Bidder along with Bid are not considered acceptable and no Amendment is issued incorporating such changes, then such Bid(s) shall be considered as non-responsive and liable for rejection.**

26.0 CLARIFICATION OF BIDS

26.1 Bidders should ensure that the Bid submitted is substantially responsive Bid in the first instance itself. Evaluation may be completed based on the content of the Bid itself without seeking any subsequent additional information, which may result in rejection of Bid. However, EIL may, at its discretion, may request bidder to submit the necessary information or documentation, within a reasonable period of time rectify omission in the bid related to documentation requirements. Requesting information or documentation on such account shall not be related to any aspect of the price of the Bid. EIL's request for clarification and the response shall be in writing.

26.2 No change, including any voluntary increase or decrease, in the prices or substance of the bid shall be sought, offered, or permitted. Bidder shall not be allowed to submit any price implication or revised price after submission of Bid unless the same is called for by EIL in writing.

26.3 Any clarification submitted by a bidder that is not in response to a request by EIL/Owner shall not be considered. Failure of the bidder to comply with the request may result in the rejection of its Bid.

26.4 If a bidder does not provide clarifications of its bid by the date and time set in EIL/Owner's request for clarification, its bid shall be evaluated with available information which may result in rejection of their bid.

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27.0 EVALUATION OF COMPOSITE BIDS

27.1 EIL will determine whether each bid is accompanied with requisite documents;

i) Fulfils the technical, financial and other requirements as specified in the Bidding Document;
ii) Bid's responsiveness in accordance with Clause No. 25.0 of ITB and clarifications thereafter, if any;
iii) Any other relevant factor, if any that EIL/Owner deemed necessary or prudent to be taken into consideration.

27.2 EIL/Owner reserves the right to use in-house information for assessment of capability of bidder and their performance on jobs completed/ in progress for evaluation purpose.

27.3 Bidders who are enlisted with EIL are required to quote as per their capabilities registered with EIL as on the date of issuance of RFQ (applicable in case of limited enquiries).

27.4 In view of the enquiries being issued on zero deviation basis, Technical or commercial queries shall not be raised. However, EIL reserves the right to raise queries in exception situations.

27.5 In view of zero deviation bidding, in case of MRs (group/ basket/ block/ itemwise) / Lumpsum Enquiry, during Composite bid evaluation if a bidder has not quoted any part/component/ freight etc, the bid of such a bidder for the corresponding group/ basket/ block/ item (or the entire MR/ lumpsum Enquiry, in case there is only one group/ block) shall stand rejected.

It is clarified here that for grouped/ basket/ block, if any item is not quoted, the offer shall be rejected for that group/ basket/ block.

In such an event, the lowest bidder shall be determined out of the remaining bidders.

28.0 CORRECTION OF ARITHMETICAL ERRORS

28.1 The prices quoted by bidders shall be checked for arithmetic correction, if any, based on rate and amount filled by the bidder in the SOR/price schedule formats. If some discrepancies are found between the rate / amount given in figures/words, the total amount shall be corrected as per the following procedure, which shall be binding upon the bidder:

i. When there is a discrepancy between the rate in figures and in words for an item, the rate which corresponds to the amount shall be taken as correct.

ii. When the rate quoted by the bidder in figures and words tallies but the amount is incorrect, the rate quoted by the tenderer shall be taken as correct.

iii. When it is not possible to ascertain the correct rate as detailed above, the rate quoted for the item in words shall be adopted as the quoted rate.

28.2 Bidder(s) confirms acceptance to arithmetical errors. Failure to accept the correction in accordance with Clause No. 28.1 of ITB, shall result in the rejection of the Bid.

29.0 CONVERSION TO SINGLE CURRENCY

29.1 NOT APPLICABLE

30.0 EVALUATION AND COMPARISON OF PRICE BIDS

30.1 EIL shall compare the price bids based on the evaluation methodology and loading criteria, if any, as indicated in the below table to determine the lowest evaluated bid.

1.	FOT Dispatch Point Price excluding GST but including all other taxes and duties: (FOT Dispatch Point Price, Cost of mandatory spares identified in the Material Requisition, commissioning Spares and special tools & tackles but excluding costs of Spares for two years operation & maintenance)	As quoted by the bidder

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2.	Transportation Charges excluding GST but including all other taxes and duties	As quoted by the bidder
3.	GST on Supply & Freight	As Applicable
4.	Total Price for Supplies on FOT Site basis including GST	**SUM OF SR. NOS. (1+2+3)**
5.	Price for Site Work/ Supervision/ Training/ 3-D Modelling/ Hazop Study/ AMC (if applicable)	
	(a) Site work (as per MR) excluding GST but including all other taxes and duties	As quoted by the bidder
	(b) Supervision of Erection, Testing, Commissioning, as applicable (for no. of days to be considered for evaluation as per Enquiry Document) (excluding GST but including all other taxes and duties)	As quoted by the bidder
	(c) Training (as per MR) **(for no. of days to be considered / Lump sum for evaluation as per Enquiry Document)** (excluding GST but including all other taxes and duties)	As quoted by the bidder
	(d) 3-D Modelling/ Hazop Study/ AMC etc.	As quoted by the bidder
6	GST on Site work/ Supervision/ Training/ 3-D Modelling/ Hazop Study/ AMC Price	As Applicable
7.	Total of Site work / Supervision / Training / 3-D Modeling / Hazop Study / AMC Price including GST	**SUM OF SR. NOS. (5+6)**
8.	Total Landed cost at site	**SUM OF SR. NOS. (4+7)**
9.	Less Input Tax Credit: **100%** of GST (as quoted for Supply, Freight, Site work, supervision, Training, 3-D Modelling, Hazop Study, AMC Price- as applicable)	As applicable
10.	**Total Evaluated Price considering 100% Input Tax Credit / Total Landed cost for price comparison**	**SR. NO. 8 MINUS (-) SR. NO. 9**

30.2 **REVERSE AUCTION**

Refer BDS for Applicability

30.3 Conditional discount, if offered, shall not be considered for evaluation.

30.4 **Un-conditional discount:** Any unsolicited reduction in price offered by a bidder within the bid validity by way of discount or revised prices, subsequent to the bid due date, shall not be taken into account for comparison. However, such reduction in price shall be taken into account for ordering if such bidder happens to be recommended as per the originally quoted prices.

30.5 **Unsolicited price increase:** In the event of any unsolicited price increase sought by any bidder, subsequent to the bid due date, the bid of such a bidder shall be rejected, in case the bidder does not agree to withdraw the price increase. In case, price increase is sought by the L1 bidder after priced bid opening and the bidder does not agree to withdraw the price increase, the order shall not be placed with price increase and the enquiry shall be re-floated.

Wherever, decision is taken to reject a bid-appropriate action will be taken against the bidder as per EIL/Owner guidelines.

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30.6 Any implication on account of wrong quotation of taxes and duties shall be to Bidder's account. However, the payment towards such taxes and duties quoted by the bidder shall be restricted to lower of taxes and duties quoted by the bidder or the documentary evidence provided by the bidder.

30.7 The prices as submitted in the BoQ Format of CPP Portal only shall be considered for evaluation and ordering, if any.

30.8 **TIE AMONG LOWEST EVALUATED PRICES**

30.8.1 In an extreme situation, when there is a 'TIE' based on the evaluated price (after loadings, if any), following provision shall apply:

i. In case quantity can be equally divided, the same shall be divided equally amongst the bidders.

ii. In case quantity cannot be equally divided / does not make commercial sense to divide, discount may be taken from all the lowest bidders and ordered on the lowest bidder (after discount).

iii. In case there is still a tie, after obtaining price discount, the bidder having the maximum order may be ordered the full quantity.

30.9 **SHIFTING OF ORDER**

30.9.1 In order to avoid splitting of order on number of bidders, particularly when order value is small, shifting shall be resorted to so that time and effort on ordering, expediting, inspection etc. is saved in dealing with extra suppliers. For the purpose of shifting, cost of order management per order is considered as Rs. 50,000/-.

30.9.2 Based on lowest evaluated item wise prices the shifting shall be carried out provided the following conditions are met

a) Shifting shall be done only if shifting cost of each order is upto Rs.50000/-.

b) Shifting shall not be resorted to if the value of the order is more than 25% of the total order value of the recommendation. However, this ceiling will not be applicable where total order value of recommendation is upto Rs. 5.0 Lakh.

c) In case the order value on L1 bidder is more than Rs 2,00,000/- then shifting shall not be resorted

d) With above shifting the selected bidder for award will be deemed to be L1 bidder.

30.10 Preference to MSEs, if applicable as per BDS, shall be carried out to distribute the item / quantity as per clause no. **32.1** below.

30.11 Public Procurement (Preference to Make in India) (PPP-MII), if applicable as per BDS, shall be carried out to distribute the item / quantity as per clause no. **32.2** below.

30.12 Shifting shall not be resorted to, where MSE is the recommended bidder even if the items are being shifted to another MSE bidder.

31.0 EIL/OWNER'S RIGHT TO ACCEPT ANY BID, AND TO REJECT ANY OR ALL BIDS

31.1 EIL/Owner reserves the right to accept or reject any bid, and to annul the bidding process and reject all bids at any time prior to award of the order without thereby incurring any liability to the affected bidder or bidders or any obligations to inform the affected bidder or bidders on the ground of such action by EIL/Owner .

32.0 POLICIES ISSUED BY GOVERNMENT INCLUDING PURCHASE PREFERENCE POLICIES

32.1 **Preference to Micro and Small Enterprises (MSEs) (Applicable for Procurement of Goods & Services only):**

- For applicability Refer Bid Data Sheet (BDS)
- Refer **Appendix – III to ITB** for details.

32.2 **Public Procurement (Preference to Make in India), Order 2017:**

- For applicability Refer Bid Data Sheet (BDS)

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- Refer **Appendix – IV to ITB** for details.

For Concurrent Applicability of PPP-MII & MSE Policy refer Appendix-V to ITB.

32.3 **Policy for providing Preference to Domestically Manufactured Iron & Steel Products (DMI&SP) in Government Procurement:**

- For applicability Refer Bid Data Sheet (BDS)
- Refer **Appendix – VI to ITB** for details.

32.4 **Policy for Purchase Preference to Domestically Manufactured Electronic and Telecom Products:**

- For applicability Refer Bid Data Sheet (BDS)
- Refer **Appendix – IV to ITB** for details.

32.5 **Startup India Policy:**

- For applicability Refer Bid Data Sheet (BDS)
- Refer **Appendix – VIII to ITB** for details.

32.6 **Foreign Trade Policy on Imports & Exports to Democratic Peoples Republic of Korea:**

- This policy shall be **applicable.**
- Refer **Appendix – IX to ITB** for details.

F. AWARD OF CONTRACT

33.0 AWARD CRITERIA

33.1 Pursuant to clause no. 30.0 and Clause no. 31.0 of ITB, the Owner shall place Purchase Order to the bidder(s) who has (have) been determined to be the lowest successful bidder(s) for item(s) / Group(s) of MR as per the item wise / Groupwise ordering philosophy defined in Bidding Document. However, if in the opinion of Owner, the total price or certain **item rates quoted by the lowest evaluated bidder are considered high OR where there is a suspicion of cartel formation OR in case of proprietary / OEM items OR where limited bids are available for the item(s)**, EIL/Owner may invite such bidder for price negotiation. Lowest quoted bidder shall attend such negotiation meetings and if requested by Owner shall provide the analysis of rates/break-up of amount quoted by him for any or all items of Price Schedule Format to demonstrate the reasonability. As a result of negotiation, bidder may offer rebate on his earlier quoted Price(s).

33.2 In case of MRs containing supplies as well site work/ services, a single order, covering the supply part as well site work/ services, shall be placed. ***Split ordering is not permitted.***

34.0 NOTIFICATION OF AWARD

34.1 Prior to the expiration of the period of bid validity, Owner shall notify the successful bidder, in writing, that its bid has been accepted. The notification letter [hereinafter and in the General Purchase Conditions called the "**Letter of Acceptance (LOA)/ Purchase Order (PO)**"] shall specify the sum that the Owner will pay the Supplier in consideration of the execution and completion of the Supply and Services. The notification letter will constitute the formation of the contract.

34.2 The Delivery Schedule shall commence from the date of issue of LOA/PO, whichever is earlier.

34.3 LOA/PO shall contain price, delivery and other salient terms of bid and RFQ Document.

34.4 Bidder will be required to acknowledge receipt of the same by returning "Copy of the Contract" duly signed and stamped by the bidder as a token of acknowledgement to Owner and EIL.

35.0 CONTRACT CUM PERFORMANCE BANK GUARANTEE (CPBG)

35.1 The supplier shall furnish the SFMS based Contract cum Performance Bank Guarantee for the required value, period and in the proforma, inline with bidding document within the stipulated period indicated in SPC.

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35.2 Failure of the successful bidder to submit the CPBG shall constitute sufficient grounds for the annulment / termination / cancelation of the award.

36.0 WAIVER OR TRANSFER OF THE AGREEMENT

36.1 The successful bidder shall not waive the Agreement or transfer it to third parties, whether in part or in whole, nor waive any interest that is included in the Agreement without the prior written permission of the Purchaser.

37.0 CARTEL FORMATION

37.1 Based on the documents submitted by Bidder(s) in the Composite bid if a suspension of cartel formation is observed, bid(s) will not be considered for evaluation /placement of order. Such bidder will also be debarred from bidding in future and EMD shall be forfeited.

38.0 UNIQUE DOCUMENT IDENTIFICATION NUMBER

The Institute of Chartered Accountant of India at its 379th council meeting dated 17-18th December, 2018 made generation of Unique Document Identification Number (UDIN) mandatory for every signature of full time practicing chartered accountant in phased manner for following services:

a) All certificates with effect from 1st February, 2019.

b) GST and Income Tax Audit with effect from 1st April, 2019.

c) All Audit and Assurance Functions with effect from 1st July, 2019.

In view of the above, bidder shall ensure that any certificate/ reports issued/ attested by an practicing chartered accountant in India and submitted in the bid shall mandatorily include the UDIN number. Certificate / reports issued/ attested without UDIN number of practicing chartered accountant in India shall not be considered for evaluation. The requirement of including UDIN shall not be applicable for certificate/ reports issued/ attested by a chartered accountant practising in Foreign Country.

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E-TENDERING METHODOLOGY

1.1 GENERAL

Chennai Petroleum Corporation Ltd. has developed a secured and user friendly system which will enable Vendors / Bidders to Search, View, Download tenders directly from Chennai Petroleum Corporation Ltd. secured website and also enables them to participate & submit On-line Bids on the e-tendering site https://cpcletenders.nic.in directly from the website in secured and transparent manner maintaining confidentiality and security throughout the tender evaluation process and award.

Bidders are requested to read following conditions in conjunction with various conditions, wherever applicable appearing with this bid invitation for e-Tendering.

1.1.1 The subject tender is an e-tender and available on CPCL website https://cpcletenders.nic.in . The bidding documents are and shall remain the exclusive property of OWNER without any right of the Bidder to use them for any purpose except bidding and for use by successful Bidder with reference to the work. All interested bidders are requested to register themselves with the portal indicated above and enroll their digital certificate with the 'user id' for participation in the tender.

1.2 Digital Signature is a unique digital code which can be transmitted electronically and primarily identifies a unique sender. The objective of digital signature is to guarantee that the individual sending the message is who he or she really claims to be just like the written signature. The Controller of Certifying Authorities of India (CCA) has authorized certain trusted Certifying Authorities (CA) who in turn allot on a regular basis Digital Certificates, Documents which are signed digitally are legally valid documents as per the Indian IT Act (2000).

1.3 In order to bid for CPCL e-tenders all the vendors are required to obtain a legally valid Digital Certificate as per Indian IT Act from the licensed Certifying Authorities (CA) operating under the Root Certifying Authority of India (RCAI), Controller of Certifying Authorities (CCA) of India. The Digital Certificate is issued by CA in the name of a person authorized for filing Bids / Offers on behalf of his Company. A Vendor / Bidder can submit their Bids / Offers On-line only after digitally signing the bid / documents with the above allotted Digital Signatures.

1.4 Bidders have to procure Digital Certificate on their own from any of the Certifying Authorities in India. (For detail bidders are advised to visit https://www.cpcletenders.nic.in

1.4.1 Bidders already possessing the digital signature issued from authorized CAs can use the same in this tender. Further, the bidder should ensure that the email address given in the Registration Form is valid & active as all the communications will be made through this email

1.4.2 The bidder shall submit their offer through e-tendering site as mentioned above, following the steps in the e-tendering portal. Physical Bids sent through Fax / E-mail / Courier / Post will not be acceptable.

Bidder(s) are advised to quote strictly as per terms and conditions of the tender documents for ebidding and not to stipulate any deviations / exceptions. Bidder must ensure that the RTGS/NEFT Transfer/ Bank Guarantee (BG) against EMD are received by OWNER before techno-commercial bid opening schedule (as indicated in the NIT).

1.4.3 On no account will any person, who has downloaded the bidding documents using digital signature, part with possession thereof or copy or disclose the provisions thereof or any of them or disclose or take copies of tracings or of any drawing, plan or route forming part thereof, it being understood that the information therein is confidential, and that the bidding documents are being downloaded in strictest confidence.

1.4.4 The details as called for in the bidding documents shall be filled and completed by the Bidders in all respect and shall be submitted with requisite information and Annexure.

1.4.5 If the space in any Performa of bidding document is insufficient, additional pages shall be separately added. These shall be page numbered and shall also carry the bidding document number before uploading the offer.

1.4.6 Bidder shall clearly indicate their legal constitution and the person signing the bid shall state his capacity as also the source of his ability to bind the Bidder. The Power of Attorney or Authorization or other document constituting adequate proof of the ability of the signatory to bind the Bidder, shall be annexed to the bid. OWNER may reject outright any bid unsupported by adequate proof of the signatory's authority.

1.4.7 Bidders/Agencies shall submit their eligibility and qualification documents, Technical bid, Financial part etc in the standard formats prescribed in the Tender Documents, uploaded in eprocurement website. The bidder shall upload all the scanned copies of all the relevant certificates, documents etc in support of their eligibility criteria/technical bids in the eprocurement website. The bidder shall digitally sign on the statements, documents, certificates uploaded by him, owning responsibility for their correctness/authenticity. If any of the documents furnished by the bidder is found to be false/ fabricated, the bidder is liable for black listing, forfeiture of EMD & cancellation of work.

1.4.8 A pre-bid meeting as per the schedule indicated in the Notice Inviting Tender will be held. Apart from any technical queries and any other queries regarding the conditions of tender, bidders/ their representatives shall be familiarized about the OWNER e-tendering site & how to submit their offer online, getting digital signatures etc. Bidders are requested to clarify all the technical queries related to tender during the Pre-bid meeting and as far as possible avoid any deviations/Queries in their offer. Bidders to ensure compliance of all provisions of the bidding document.

1.4.9 Resource Requirement:

Vendors / Bidders must use any computer having Windows XP or higher of Window operating System and an internet web browser version internet explorer V6.0 or higher recommended.

1.4.10 Help Desk:

Bidders are requested to go through the “Bidders Manual Kit” available in the homepage of the e tendering portal i.e. https://iocletenders.gov.in to have a clear understanding of the steps to be followed for bid submission. The Name and Contact Numbers of the persons from our service

Provider NIC (National informatics Centre) for troubleshooting in e-Tendering Process are:

Contact Us

For any technical related queries please call at 24 x 7 Help Desk Number

0120-4001 002

0120-4001 005

0120-6277 787

International Bidders are requested to prefix 91 as country code

EMail Support

A) For any Issues or Clarifications relating to the published tenders,
bidders are requested to contact the respective Tender Inviting Authority

Technical - support-eproc(at)nic(dot)in

APPENDIX-II TO ITB

TERMS & CONDITIONS FOR REVERSE AUCTION

NOT APPLICABLE

APPENDIX-III TO ITB

PREFERENCE TO MICRO AND SMALL ENTERPRISES (MSEs)

PREFERENCE TO MICRO AND SMALL ENTERPRISES (MSES)
(APPLICABLE FOR PROCUREMENT OF GOODS & SERVICES ONLY)

1. Ministry of Micro, Small & Medium Enterprises notified the Public Procurement Policy (PPP), 2012 for facilitating the promotion and development of micro and small enterprises in Government Procurement. Subsequently, Micro, Small & Medium Enterprises vide circular no. 21(8)/2018-MA dated 09.11.2018 has issued an amendment to the said policy.

   Further, Ministry of MSMEs (MoMSME) vide **Gazette notification No. S.O. 4926(E) dated 18.10.2022** have issued the following amendment in sub-paragraph (5) of paragraph 8 of MoMSME S.O. No. 2119 (E) dated 26.06.2020:

   "In case of an upward change in terms of investment in Plant and Machinery or Equipment or Turnover or both, and consequent re-classification, an enterprise shall continue to avail of all non- tax benefits of the category (Micro or Small or Medium) it was in before the re-classification, for a period of three years from the date of such upward change".

   As per the press notification dated 19.10.2022 issued by Press Information Bureau, **Non-tax benefits include benefits of various schemes of the Government, including Public Procurement Policy, Delayed Payments, etc.**

2. For claiming price preference against enquiries in which provision of price preference is to be extended to the MSE Bidders, the MSE certificate should be valid as on date of un-priced bid opening.

3. The owner reserves the right to allow Micro & Small enterprises as well as MSEs Owned by SC/ST entrepreneur and MSEs Owned by Women, purchase preference as admissible under the prevailing procurement policy for MSEs.

4. The quantity against the item(s) of MR may be split to enable ordering of 25% quantity against the item(s) of MR to MSEs bidder within the price range of L1 bidder's evaluated price + 15% subject to their matching L1 bidder's price, in line with purchase preference as admissible under the prevailing procurement policy for MSEs. Accordingly, the quoted prices against various items of MR shall remain valid in case of splitting of quantities of the items in view of the same.

   Out of this 25% from MSEs, 4% is to be from MSEs owned by SC/ ST entrepreneurs. However, in event of failure of MSEs owned by SC/ ST entrepreneurs to participate in the bidding process or meet the tender requirements and L1 price, this 4% will be met from other MSEs.

   Additionally, out of this 25% from MSEs, 3% is to be from MSEs owned by Women. However, in event of failure of MSEs owned by Women to participate in the bidding process or meet the tender requirements and L1 price, this 3% will be met from other MSEs.

   While granting purchase preference as above, procurement of goods through MRs may include certain small work, or some services, which are incidental or consequential to supply of such goods such as transportation, insurance, installation, commissioning, training & maintenance.

   Clarification:

   - In case where quantity against a line item cannot be split (i.e., minimum 25% to MSEs) or items with single quantity or in grouped item, the complete line item/ group shall be awarded on MSE bidder within the price range of L1 bidder's evaluated price + 15% subject to their matching L1 bidder's price.
   - In case, quantity(ies) against an item/ group cannot be split among MSEs owned by SC/ ST entrepreneurs, MSEs owned by Women and non- SC/ ST or non-women MSE entrepreneurs, first opportunity shall be given to MSEs owned by Women than to MSEs owned by SC/ ST entrepreneurs and lastly to other MSEs, within the price range of L1 non-MSE bidder's evaluated price + 15%.

Important Note:

In case where tender quantity can be split and MSE vendor is already getting order for more than 25% of the tender value, no additional purchase preference is required to be given in that tender.

In case MSE vendor is already getting order for less than 25% of the tender quantity, purchase preference to this and other MSE vendors (together) shall be given only up to the differential quantity to make total as 25% to MSE vendor.

5. MSE bidders shall submit the following:

   i) Bidder claiming to be a Micro or Small Enterprises (MSEs) must be registered on Udyam Registration portal having valid “Udyam Registration Certificate”.

   ii) Ministry of MSME vide Gazette Notification no. 1364(E) dated 21.03.2025 has notified the revised criteria for classification of micro, small and medium enterprises as under:

   - In a Micro Enterprise, the investment in Plant and Machinery or Equipment does not exceed two crore and fifty lakh rupees and turnover does not exceed ten crore rupees;
   - In a Small Enterprise, the investment in Plant and Machinery or Equipment does not exceed twenty-five crore rupees and turnover does not exceed one hundred crore rupees;
   - In a Medium Enterprise, the investment in Plant and Machinery or Equipment does not exceed one hundred twenty-five crore rupees and turnover does not exceed five hundred crore rupees.

iii) The Udyam registration certificate shall be verified from GoI Udyam registration portal (present web address is https://udyamregistration.gov.in). The status of the MSE Bidder as on the date of Unpriced Bid opening shall be considered for benefits defined in the Bidding Document under the Purchase Preference Policy for MSEs, 2012.

iv) In order for MSE owned by SC/ST Entrepreneurs to qualify for benefits available under Public Procurement Policy for MSEs order 2012, the MSE shall additionally submit a relevant SC/ ST certificate in the name of SC/ ST proprietor or partner(s) holding minimum 51% shares in case of partnership or person(s) holding minimum 51% shares in limited companies, issued by any of the following:

- District Magistrate/ Additional District Magistrate/ Collector/ Deputy Commissioner/ Deputy Collector/ 1st Class Stipendiary Magistrate/ City Magistrate/ Sub-Divisional Magistrate/ Taluka Magistrate/ Executive Magistrate/ Extra Assistant Commissioner (not below the rank of 1st Class Stipendiary Magistrate).
- Chief Presidency Magistrate/ Additional Chief Presidency Magistrate/ Presidency Magistrate.
- Revenue Officers not below the rank of Tehsildar
- Sub-Divisional Officer of the area where the candidate and/ or his family normally resides.
- Administrator/ Secretary to Administrator/ Development Officer (Lakshadweep Islands).

Definition of MSEs owned by SC/ST Entrepreneurs, is as follows:

- In case of proprietary MSE, proprietor should be SC/ST.
- In case of partnership MSE, the SC/ST partner(s) should be holding at least 51% shares in the unit.
- In case of Limited companies, at least 51% share should be held by SC/ST shareholder(s).
- In case of Public Limited Companies, at least 51% share should be held by SC/ST entrepreneurs at any given point of time.

v) Micro & Small Enterprises shall be considered as owned by Women Entrepreneurs as per the below definition:

- In case of proprietary MSE, Women should be SC/ST.
- In case of partnership MSE, the Women partner(s) should be holding at least 51% shares in the unit.
- In case of Limited companies, at least 51% share should be held by Women shareholder(s)

In order for MSE owned by Women Entrepreneurs to qualify for benefits available under Public Procurement Policy for MSEs order 2012 to Women Entrepreneurs, the MSE shall additionally submit the following:

- In case of a proprietorship firm, the name and address of proprietor, and certified copy of Bank Account Details.
- In case bidder is a partnership firm, certified copy of the partnership deed.
- In case of company (whether private or public), certified copy of the `Certificate of
- Incorporation' together with certified copy of Memorandum/Articles of Association.

vi) To avail the benefits available to MSEs under the bidding document, MSEs shall provide the necessary valid documentary evidence, i.e. Udyam Registration Certificate. MSE bidders shall submit MSE documents, as above.

vii) No verification of bidder's status as Micro or Small Enterprises from the financial statement of bidders shall be required.

viii) In case a MSE bidder has submitted Udyam Registration Certificate, the same shall be accepted provided the details of the same could be verified from the Government Udyam Registration portal, as applicable.

ix) In case, a MSE bidder doesn't submit the MSE document or submits invalid document or, such bidders shall not be considered for providing the benefits under MSE Policy.

6. In case at any stage, if it is found the bidder has claimed / availed the benefits by providing false/forged details, their bid shall be rejected and bidder shall be placed on suspension/ banning. However, if this is found after order placement, bidder shall also be liable for encashment of PBG. Bidder's organization is responsible for any and all actions of their employees. Any claims seeking to pin point/shift the blame on any employee and consequently attempting to absolve the bidder's organization will not be entertained.

7. Owner/EIL is committed to promotion and development of micro and small enterprises. Accordingly, bidders are also encouraged to promote the same by considering MSE sub-suppliers/ sub-contractors to the extent possible under the Orders/ Contracts awarded on them Owner/EIL. All efforts shall be made by bidder to encourage procurement thru MSEs (holding valid certificate) through their Sub orders/ Sub contract procurement. The successful bidder shall provide the following only for the purpose of reporting.

i) Monthly Reports for purchase of material/services/ sub-contracts (from MSE vendors) shall be submitted by bidder to .....(Name of CPSU).

i) The report shall be signed by authorised person of contractor (Successful Bidder).

ii) Report shall be submitted in the following format.

Sl. No.	Sub-Supplier/ Subcontractor	Work Order no.& date	Item Description	Amount (In Rs)	Category of MSE Gen/ SC/ ST	MSME Status Micro/ Small with code	Remarks

APPENDIX-IV TO ITB

POLICY TO PROVIDE PURCHASE PREFERENCE UNDER PUBLIC PROCUREMENT (PREFERENCE TO MAKE IN INDIA) (PPP-MII) ORDER, 2017 & POLICY FOR PURCHASE PREFERENCE TO DOMESTICALLY MANUFACTURED ELECTRONIC & TELECOM PRODUCTS

POLICY TO PROVIDE PURCHASE PREFERENCE UNDER PUBLIC PROCUREMENT (PREFERENCE TO MAKE IN INDIA) (PPP-MII) ORDER, 2017

1. MoP&NG vide File no. FP-20013/2/2017-FP-PNG-Part(4)(E-41432) dated 26.04.2022 (enclosed herwith), have issued a directive that **Public Procurement (Preference to Make in India) (PPP-MII) Order, 2017 issued by DPIIT** and as amended from time to time shall be applicable to all the Public Sector Undertakings and their wholly owned subsidiaries under MoP&NG, Joint Ventures that have 51% of more equity by one or more Public Sector Undertakings under MoP&NG.

1.1 EVALUATION AND AWARD PHILOSOPHY

1.1.1 Bidder is hereby instructed that relevant documents and information pertaining to MSE and PPP-MII, which are necessary for availing the relevant benefits during evaluation, must be updated/ uploaded in the designated folder on e-Tendering Portal.

1.1.2 The 'Class-I local supplier'/ 'Class-II local supplier' shall be required to indicate percentage of local content & details of the location(s) of the local value addition, as per formats enclosed with Annexure-I to ATC.

1.1.3 While evaluating the Bids, for price matching opportunities, and/ or distribution of quantities/ Parts (wherever applicable) amongst Bidders, purchase preference shall be given as per following order of precedence:

a. Purchase preference to MSE Bidders over Class-I local supplier under PPP-MSE, 2012
b. Purchase preference to 'Class-I local supplier' over 'Class-II local supplier' as well as 'Non-local supplier'

1.1.4 'Purchase preference' means the opportunity given to 'MSE bidder' or 'Class-I local supplier' for matching the Price of L1 Bidder, despite its Price being not the lowest valid Price Bid (i.e. not L1) but because it is within the margin of purchase preference prescribed as follows:

a. As per PPP-MSE policy 2012 : 15% of L1 price
b. As per PPP-MII policy 2017 : 20% of L1 price

1.1.5 In case where quantity against a line item cannot be split or items with single quantity or in grouped item, the complete line item/ group shall be awarded on PPP-MSE/ PPP-MII compliant bidder within the price range of L1 bidder's evaluated price (as defined above), subject to their matching L1 bidder's price.

1.1.7 Purchase preference shall be as per **Appendix-V of ITB.**

POLICY FOR PURCHASE PREFERENCE TO DOMESTICALLY MANUFACTURED ELECTRONIC AND TELECOM PRODUCTS

2.1 MEiTY on the preference to domestically manufactured Electronic Products have issued the following notifications (copies available on website https://dpiit.gov.in/ministryelectronics-and-information-technology):

a) Notification F. no. W-43/4/2019-IPHW-Meity dated 07.09.2020 and S. 0. No 4591 (E) dated 12.10.2023 on purchase preference in domestically manufactured Electronic Products.

b) Notification no. 1(10)2017-CLES dated 06.12.2019 on purchase preference in domestically manufactured / produced Cyber Security Products.

c) Notification no. W-43/2/2025-IPHW dated 27.03.2025 on purchase preference in Cellular mobiles Phones

2.2 In furtherance of the Public Procurement (Preference to Make in India) Order 2017 [PPP-MII Order 2017] notified vide O.M. No P-45021/2/2017-PP (BE-ll)-Part(4)Vol.ll Dated 19.07.2024, and in supersession of the Electronic Product Notification Nos.33(1)/2017- IPHW dated 14.09.2017 and 33(5)/2017-IPHW dated 01.08.2018 issued thereunder, the Ministry of Electronics and Information Technology (MeitY) hereby notifies that preference shall be provided by all procuring entities to locally manufactured Electronic Products as per the aforesaid Order dated 19.07.2024, as amended from time to time.

Also, Ministry of Electronics and Information Technology (MeitY) vide Order No.1 (1 0)/2017-CLES dated 06.12.2019 had issued revised Public Procurement (Preference to Make in India) for cyber security products.

2.3 The mechanism for calculation of local content has been prescribed for each notified Electronic Product in Notification F. no. W-43/4/2019-IPHW-Meity dated 07.09.2020 and S.O. 4591 (E) dated 12.10.2023, in accordance with paragraph 5 of the DPIIT PPP-MII Order 2017 No. P-45021 /2/2017-PP (BE-ll)-Part(4)Vol.ll Dated 19.07.2024, as amended from time to time.

2.4 The Ministry of Communications (Department of Telecommunications), in supersession of its previous notifications No. 18-10/2017-IP dated 29.08.2018 and 31.08.2021, hereby issued Notifications no. F. No. 18-10/2017-IP dated 21.10.2024 on the preference to domestically manufactured Telecom Products (copy available on website https://dpiit.gov.in/departmenttelecommunications).

This Notification shall be applicable in respect of the procurement made by all attached or subordinate offices or autonomous bodies under the Government of India including Government Companies as defined in the Companies Act, and/or the States and Local Bodies making procurement under all Central Schemes/ Central Sector Schemes where the Scheme is fully or partially funded by the Government of India. This Notification shall also be applicable in respect of projects wherein funding of goods, services or works is by the Digital Bharat Nidhi to the extent of having being awarded through bids.

2.5 List & definitions of Telecom products, services & works, Percentage of Procurement for which preference to domestically manufactured Telecom Products is to be provided and Percentage of local content in terms of Bill of Material (BOM) required for the products to qualify as domestically manufactured Telecom Products and conditions for inputs to be qualified as Local Content & Maximum ceiling for Design as local content out of the total Local content are defined in above notification.

2.6 Above Notifications with respect to Electronics Products, Cyber Security Products and telecom Products shall be followed in Procurement of Goods, Works and Services contracts [including System integrator (SI)/ EPC / Turnkey Contacts] as main item(s) or sub-item(s) (in part of SI/EPC/Turnkey Contracts) to be procured from suppliers being out of the notified Electronic Products including Cyber Security Products or Telecom products. Accordingly, the items, notified as having sufficient local capacity and competition, shall mandatorily source from Class-I local Suppliers.

Annexure-2

FORM – 2

SELF CERTIFICATION BY BIDDER UNDER PPP-MII ORDER, 2017

To

M/s _____ <Insert name of Client>

NAME OF WORK / ITEM:

TENDER /RFQ NO. :

Dear Sir,

We, M/s _____ *<Name of Bidder>* hereby confirm that we meet the mandatory minimum requirement of Local Content (LC) for **Class-I local supplier / Class-II local supplier** *(strike out which is not applicable)* under Public Procurement (Preference to Make in India), Order 2017 (PPP-MII, 2017) and the work / Item shall have **LC of** _____**%.**

The details of the location(s) at which the local value addition is/shall be made, is/ areas under:

.....
.....
.....

We confirm compliance of Public Procurement (Preference to Make in India) (PPP-MII) Order 2017, vide Order No P-45021/2/2017-PP(BE-II)-Part-4)Vol.II, Notifications with respect to Electronics Products [File No. W-43/4/2019-IPHW-Meity dated 07.09.2020 and S.O. No 4591(E) dated 12.10.2023 issued by Meity], Cyber Security Products [File No.1(10)/2017-CLES dated 06.12.2019 issued by Meity] and Telecom Products [File No FP-18-10/2017-IP dated 21.10.2024 issued by Ministry of Telecommunication]. {Strike out which is not applicable}.

We also confirm that, the imported items sourced locally from resellers/distributors are excluded from calculation of local content. The cost of such locally sourced imported items (inclusive of taxes) along with break up on license/royalties paid/technical expertise etc. sourced from outside India shall be submitted at the time of execution as per Form-2A.

We further confirm that in case we fail to meet the minimum local content/domestic value addition, the same shall be treated as false information and Owner reserves the right to take appropriate action for debarment, Penalty after award etc. as per provision of the PPP-MII Order, 2017 and its subsequent Amendments and as more specifically mentioned in the Bidding/ RFQ document.

We also confirm that we are not debarred by any other procuring entity (GOI/PSU/PSE etc.)
or
Our debarment period is completed as on _____ <insert date>, and as such we are eligible to avail purchase preference under this Order.
(## Strike out which is not applicable)

_____
Signature of Bidder

Place:

Name :

Date:

Designation

Seal :

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FORM-2A

DETAILS OF IMPORTED COMPONENTS SOURCED FROM RESELLERS/DISTRIBUTORS EXCLUDED UNDER LOCAL CONTENT

RFQ/TENDER No:

ITEM/WORKS DESCRIPTION:

S. No	Cost component	Cost in INR (Refer Note-2)
a.	Locally-sourced imported items from resellers/distributors (inclusive of taxes)	i. ii. iii. iv.
b.	License/Royalties paid/ Technical Expertise cost etc. for the items sourced from outside India i. License ii. Royalties Paid iii. Technical Expertise Cost iv. other	i. ii. iii. iv.
DETAILS OF ITEMS SOLD BY BIDDER AS RESELLER		
c.	Details of Items	OEM Certificate for country of origin
	i. ii. iii. iv.	i. ii. iii. iv.

Note –

1. For contracts involving supply of multiple items, weighted average of all items to be taken while calculating the local content.
2. As regards cases where currency for above is other than Indian Rupee, exchange rate prevailing on the Bid Due Date shall be considered for the calculation of Local Content.
3. Bidder to enclose OEM Certificate for country of origin as specified in S. No (c) above.

	Signature of Bidder
Place:	Name :
Date:	Designation
	Seal :

ANNEXURE-3

FORM – 3

(FOR CLASS-I LOCAL SUPPLIER/ CLASS-II LOCAL SUPPLIER HAVING VALUE OF PROCUREMENT MORE THAN INR 10 CRORE)

CERTIFICATE UNDER PPP-MII ORDER, 2017 TOWARDS MANDATORY MINIMUM LOCAL CONTENT

To
M/s _____ <Insert name of Client/Owner>

NAME OF WORK / ITEM:

TENDER /RFQ NO.:

Dear Sir,

We, _____, the Statutory Auditor or Cost Auditor of the Company (in case of Companies) or are a practicing Cost Accountant or practicing Chartered Accountant (in respect of supplier other than companies) of M/s. _____ *<Name of Bidder>*, hereby certify that as per definition specified in above policy, M/s. _____ *<Name of Bidder>* is a Class-I Local Supplier / Class-II Local Supplier <as applicable>

We further confirm that M/s _____ *<Name of Bidder>* meet the mandatory minimum Local content requirement specified for Class-I Local supplier/ Class-II Local supplier <as applicable> under Policy for Public Procurement (Preference to Make in India), Order 2017 (PPP-MII) for their offer for the abovementioned Work/ Item and shall have local content of _____% .

The details of the location(s) at which the local value addition is/shall be made, is/ areas under:

.....
.....

We confirm compliance of Public Procurement (Preference to Make in India) (PPP-MII) Order 2017, vide Order No P-45021/2/2017-PP(BE-II)-Part-4)Vol.II, Notifications with respect to Electronics Products [File No. W-43/4/2019-IPHW-Meity dated 07.09.2020 and S.O. No 4591(E) dated 12.10.2023 issued by Meity], Cyber Security Products [File No.1(10)/2017-CLES dated 06.12.2019 issued by Meity] and Telecom Products [File No FP-18-10/2017-IP dated 21.10.2024 issued by Ministry of Telecommunication]. ***{Strike out which is not applicable}.***

We also confirm that, the imported items sourced locally from resellers/distributors are excluded from calculation of local content. The cost of such locally sourced imported items (inclusive of taxes) along with break up on license/royalties paid/technical expertise etc. sourced from outside India is enclosed as Form-3A.

We also confirm that M/s _____ <Name of Bidder> is not debarred by any other procuring entity (GOI/PSU/PSE etc.).
OR
M/s _____<Name of Bidder>'s debarment period is completed as on _____<insert date>, and as such M/s _____ <Name of Bidder> is eligible to availed purchase preference under this Order.
(Strike out which is not applicable)

_____
Signature of certificate issuing authority

Place:

Date:

Name :

Name of Audit Firm:

Designation

Seal :

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FORM-3A

DETAILS OF IMPORTED COMPONENTS SOURCED FROM RESELLERS/DISTRIBUTORS EXCLUDED UNDER LOCAL CONTENT

RFQ/TENDER No:

ITEM/WORKS DESCRIPTION:

S. No	Cost component	Cost in INR (Refer Note-2)
a.	Locally-sourced imported items from resellers/distributors (inclusive of taxes)	i. ii. iii.
b.	License/Royalties paid/ Technical Expertise cost etc. for the items sourced from outside India i. License ii. Royalties Paid iii. Technical Expertise Cost iv. other	i. ii. iii. iv.
DETAILS OF ITEMS SOLD BY BIDDER AS RESELLER		
c.	Details of items	OEM Certificate for country of origin
	i. ii. iii.	i. ii. iii.

Note –

1. For contracts involving supply of multiple items, weighted average of all items to be taken while calculating the local content.
2. As regards cases where currency for above is other than Indian Rupee, exchange rate prevailing on the Bid Due Date shall be considered for the calculation of Local Content.
3. Bidder to enclose OEM Certificate for country of origin as specified in S. No (c) above.
4. The above formats are indicative, the statutory auditor/Cost auditor/Chartered Accountant/ Cost Accountant can modify the format without changing the Intent of certification.

_____

Signature of certificate issuing authority

Place:

Name:

Date:

Name of Audit Firm:

Designation

Seal:

Form -2B

Format for Self-Certification regarding Local Content (LC) for Telecom Product, Services or Works to be submitted on non-judicial stamp paper of the value Rs. 100/-

Date:

I_ _____S/o, D/o, W/o, Resident of _____ do hereby solemnly affirm and declare as under:

That I agree to abide by the terms and conditions of Department of Telecommunications, Government of India issued vide Notification No:_____ dated_____.

That the information furnished hereinafter is correct to the best of my knowledge and belief and I undertake to produce relevant records before the procuring entity or any other authority so nominated by the Department of Telecommunications, Government of India for the purpose of assessing the LC.

That the LC for all inputs which constitute the said Telecom Product/Services/Works has been verified by me and I am responsible for the correctness of the claims made therein.

That in the event of the LC of the Telecom Product/Services/Works mentioned herein is found to be incorrect and not meeting the prescribed LC norms, based on the assessment of an authority so nominated by the Department of Telecommunications, Government of India, I and my Statutory auditor or cost auditor (if applicable) will be liable for actions as specified in Clause (9) of the DPIIT PPP-MII Order dated 19.07.2024 for all incorrect/false facts and figures.

I agree to maintain detailed breakup / information (separately for each product) to substantiate my claim for LC in the Company's record for a period of 2 years and shall make this available for verification to any authority. I shall also maintain records of local content pertaining to items bought from other domestic manufacturers / traders.

[Please provide following information]

i	Name and details of the local supplier - Registered Office, Manufacturing unit location, nature of legal entity
Ii	Date of certificate issuance - Date on which this certificate is issued
iii	Telecom Product/Services/Works - Telecom Product/Services/Works for which the certificate is produced
iv	Procuring agency - Procuring agency to whom the certificate is furnished
v	Percentage of Local Content claimed - Percentage of LC claimed
vi	Manufacturer unit details - Name and contact details of the unit of the manufacturer
vii	Sale Price of the product - Sale Price of the product
viii	Ex-Factory Price of the product - Ex-Factory Price of the product
ix	Freight, insurance and handling - Freight, insurance and handling charges
x	Total Bill of Material - Total Bill of Material

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xi	Local inputs cost - List and total cost value of inputs used for manufacture of the Telecom product/Services/Works
xii	Locally sourced inputs - List and total cost of inputs which are locally sourced. Please attach LC certificates from local suppliers, if the input is not produced in-house
xiii	Imported inputs cost - List and cost of inputs which are imported, directly or indirectly

I hereby certify that, having read all the provisions of the above order and principles / basis of calculations, the local content calculation does not include the following:

a. Imported items sourced locally from resellers/distributors.

b. The license fees / royalties paid/ technical charges paid out of India

c. Procurement / supply of repackaged / refurbished/rebranded imported products

I hereby also certify to the best of my knowledge and belief that all the particulars furnished above are correct and complete. I agree to comply with the terms and conditions of the DPIIT PPP-MII order dated 19.07.2024 and DoT PPPMII Notification dated.....

I understand that any incorrect declaration regarding the local content or failure to substantiate the claim of LC will result in penalties as specified in Clause (9) of the DPIIT PPP-MII Order dated 19.07.2024.

I further certify and take personal responsibility that I have applied my mind to the calculations and principles of LC as specified in this order and I shall, having declared the LC shall not seek recourse to change it on any ground. Any changes made by me on any grounds in a bid in LC after bid submission shall make my bid non-responsive and I shall hold myself liable for civil/criminal action arising out of any such change. I understand and agree that any such post bid change in LC content shall also be a valid ground for blacklisting of the firm from future contracts/bids.

Signature:
Name:
Designation: CEO/MD
Address:
Email Address:
Mobile No.:
Place:
Date:

Form-2C

Format for Self-Declaration regarding local supplier' for Cyber Security Products

This is to Certify that the organization_____ registered as_____with registration number_____ do hereby solemnly affirm and declare as under:

That we agree to abide by the terms and conditions of the Notification issued by Ministry of Electronics and Information Technology (MeitY), Government of India no: _____dated_____.

That the information furnished hereinafter is correct and we undertake to produce relevant records before the procuring entity or any other authority so nominated by the Ministry of Electronics and Information Technology (MeitY), Government of India for the purpose of establishing ownership of the Intellectual Property Rights(IPR), legal existence and revenue accrual, local content for the cyber security products nominated by the aforesaid organization.

That all IPR which constitute the said cyber security product has been verified by us and we are responsible for the correctness of the claims made therein and we take complete responsibility of the same.

We agree to maintain all information regarding our claim(s) for IPR ownership, legal existence and revenue accrual, local content in the organization's record throughout the existence of the product and shall make this available for verification to any authorities specified by Government of India.

In case of any change in the IPR ownership, legal existence and revenue accrual, local content we will be accountable to inform Ministry of Electronics and Information Technology, Government of India within one week or before applying for any public procurement or before referring this order for taking any advantage which so ever occurs first.

That in the event of the cyber security product mentioned herein is found to be incorrect and not meeting the prescribed norms, based on the assessment of an authority so nominated by the MeitY, Government of India and we will be liable as under clause 7 (f) of Public Procurement (Preference to Make in India) Order 2019 for cyber security product.

1. Name and details of the organization nominating product under this order (Registered Office, Manufacturing unit location, nature of legal entity)

2. Entity Registration Certificate number
   a. Type of registration

3. Date on which this certificate is issued

4. Percentage of Royalty/License fee to be paid to other entity with respect to estimated cost of the product

5. Name and contact details of the unit of the manufacturer

For and on behalf of_____ (Name of firm/entity)

Authorize signatory (To be duly authorized by the Board of Directors)

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FORM – 2D

CERTIFICATE UNDER DOMESTICALLY MANUFACTURED/PRODUCED CYBER SECUIRTY PRODUCTS

(HAVING VALUE OF PROCUREMENT MORE THAN INR 10 CRORE)

To
M/s _____ <Insert name of Client/Owner>

NAME OF WORK / ITEM:

TENDER /RFQ NO.:

Dear Sir,

We, _____, the Statutory Auditor or Cost Auditor of the Company (in case of Companies) or are a practicing Cost Accountant or practicing Chartered Accountant (in respect of supplier other than companies) of M/s. _____ *<Name of Bidder>*, hereby certify offered product(s) by M/s _____ *<Name of Bidder>* meets the definition of 'local supplier' of domestically manufactured/produced Cyber Security Products as per para 4 of Public Procurement (Preference to Make in India) Order 2019 for Cyber Security Products.

_____
Signature of certificate issuing authority

Place:

Name :

Date:

Name of Audit Firm:

Designation

Seal :

No. P-45021/2/2017-PP (BE-II)-Part(4)Vol.II
Government of India
Ministry of Commerce and Industry
Department for Promotion of Industry and Internal Trade
(Public Procurement Section)

Vanijya Bhawan, New Delhi
Dated: 19 July,2024

To

All Central Ministries/Departments/CPSUs/All concerned

ORDER

Subject: Public Procurement (Preference to Make in India), Order 2017– Revision; regarding.

Department for Promotion of Industry and Internal Trade, in partial modification [Paras 2, 3, 5, 10 & 13] of Order No.P-45021/2/2017-B.E.-II dated 15.6.2017 as amended by Order No.P-45021/2/2017-B.E.-II dated 28.05.2018, Order No.P-45021/2/2017-B.E.-II dated 29.05.2019, Order No.P-45021/2/2017-B.E.-II dated 04.06.2020 and Order No.P-45021/2/2017-B.E.-II dated 16.09.2020 hereby issues the revised 'Public Procurement (Preference to Make in India), Order 2017" dated 19.07.2024 effective with immediate effect.

Whereas it is the policy of the Government of India to encourage 'Make in India' and promote manufacturing and production of goods and services in India with a view to enhancing income and employment, and

Whereas procurement by the Government is substantial in amount and can contribute towards this policy objective, and

Whereas local content can be increased through partnerships, cooperation with local companies, establishing production units in India or Joint Ventures (JV) with Indian suppliers, increasing the participation of local employees in services and training them,

Now therefore the following Order is issued:

1. This Order is issued pursuant to Rule 153 (iii) of the General Financial Rules 2017.
2. **Definitions**: For the purposes of this Order:
   'Local content' means the amount of value added in India which shall, unless otherwise prescribed by the Nodal Ministry, be the total value of the item procured (excluding net domestic indirect taxes) minus the value of imported content in the item (including all customs duties) as a proportion of the total value, in percent.

Explanatory notes for calculation of local content given above

a. Imported items sourced locally from resellers/distributors shall be excluded from calculation of local content.
b. The license fees/royalties paid/ technical charges paid out of India shall be excluded from local content calculation.

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c. Procurement/Supply of repackaged/refurbished/rebranded imported products as understood commonly shall be treated as reselling of imported products and shall be excluded from calculation of local content. The definition of repackaged/refurbished/rebranded imported products is as follows;

'Refurbishing' means repair or reconditioning of an imported product does not amount to manufacture because no new goods come into existence.

'Repackaging' means repacking of imported goods from bulk pack to smaller packs would not ordinarily amount to manufacture of a new item.

'Rebranding' means relabeling or renaming or change in symbol or logo/makes or corporate image of a company/organization/ firm for an imported product would amount to rebranding.

d. To ensure that imported items sourced locally from resellers/distributors are excluded from calculation of local content, procuring entities to obtain from bidders, the cost of such locally-sourced imported items (Inclusive of taxes) along with break-up on license/royalties paid/technical expertise cost etc. sourced from outside India. For items sold by bidder as reseller, OEM certificate for country of origin to be submitted.

e. For contracts involving supply of multiple items, weighted average of all items to be taken while calculating the local content.

'*Class-I local supplier*' means a supplier or service provider, whose goods, services or works offered for procurement, meets the minimum local content as prescribed for 'Class-I local supplier' under this Order.

'*Class-II local supplier*' means a supplier or service provider, whose goods, services or works offered for procurement, meets the minimum local content as prescribed for 'Class-II local supplier' but less than that prescribed for 'Class-I local supplier' under this Order.

'*Non - Local supplier*' means a supplier or service provider, whose goods, services or works offered for procurement, has local content less than that prescribed for 'Class-II local supplier' under this Order.

'*L1*' means the lowest tender or lowest bid or the lowest quotation received in a tender, bidding process or other procurement solicitation as adjudged in the evaluation process as per the tender or other procurement solicitation.

'*Margin of purchase preference*' means the maximum extent to which the price quoted by a "Class-I local supplier" may be above the L1 for the purpose of purchase preference.

'*Nodal Ministry*' means the Ministry or Department identified pursuant to this order in respect of a particular item of goods or services or works.

'*Procuring entity*' means a Ministry or department or attached or subordinate office of, or autonomous body controlled by, the Government of India and includes Government companies as defined in the Companies Act.

'Works' means all works as per Rule 130 of GFR- 2017, and will also include *'turnkey works'*.

2A. *Special treatment for items covered under PLI Scheme*

The manufacturers manufacturing an item under PLI scheme shall be treated as deemed Class II local supplier for that item unless they have minimum local content equal to or higher than that notified for Class-I local supplier for that item, provided the manufacturer has received incentive from the concerned PLI Ministry for the Item. The above shall be applicable for the specific time period only, as notified by concerned PLI Ministry.

3. Eligibility of 'Class-I local supplier'/ 'Class-II local supplier'/ 'Non-local suppliers' for different types of procurement

(a) In procurement of all goods, services or works in respect of which the Nodal Ministry / Department has communicated that there is sufficient local capacity and local competition, only 'Class-I local supplier', as defined under the Order, shall be eligible to bid irrespective of purchase value.

(b) Only 'Class-I local supplier' and 'Class-II local supplier', as defined under the Order, shall be eligible to bid in procurement undertaken by procuring entities, except when Global tender enquiry has been issued. In global tender enquiries, 'Non-local suppliers' shall also be eligible to bid along with 'Class-I local suppliers' and 'Class-II local suppliers'. In procurement of all goods, services or works, not covered by sub-para 3(a) above, and with estimated value of purchases less than Rs. 200 Crore, in accordance with Rule 161(iv) of GFR, 2017, Global tender enquiry shall not be issued except with the approval of competent authority as designated by Department of Expenditure.

(c) For the purpose of this Order, works includes Engineering, Procurement and Construction (EPC) contracts and services include System Integrator (SI) contracts.

3.1 Mandatory sourcing of items, with sufficient local capacity and competition, from Class-I local suppliers in SI/EPC/Turnkey Contracts/Service Tenders

a. The items, notified as having sufficient local capacity and competition, shall mandatory be sourced from Class-I local suppliers in SI/EPC/Turnkey Contracts/ Services tenders. This provision will be applicable only for those items which have been notified by the Nodal Ministry as Class I i.e. having sufficient local capacity and competition, with specific HSN codes."

b. Notwithstanding above, if in any project, it is considered that it is not practically feasible to source such items from Class I local suppliers, it may take relaxation from such stipulation with the approval of Secretary of the administrative Ministry/ Department concerned or with the approval of the Competent Authority specified by the Administrative Ministry/Department, on case-specific basis.

3A. Purchase Preference

(a) Subject to the provisions of this Order and to any specific instructions issued by the Nodal Ministry or in pursuance of this Order, purchase preference shall be given to 'Class-I local supplier' in procurement undertaken by procuring entities in the manner specified here under.

(b) In the procurement of goods or works, which are covered by para 3(b)

above and which are divisible in nature, the 'Class-I local supplier' shall get purchase preference over 'Class-II local supplier' as well as 'Non-local supplier', as per following procedure:

i. Among all qualified bids, the lowest bid will be termed as L1. If L1 is 'Class-I local supplier', the contract for full quantity will be awarded to L1.

ii. If L1 bid is not a 'Class-I local supplier', 50% of the order quantity shall be awarded to L1. Thereafter, the lowest bidder among the 'Class-I local supplier' will be invited to match the L1 price for the remaining 50% quantity subject to the Class-I local supplier's quoted price falling within the margin of purchase preference, and contract for that quantity shall be awarded to such 'Class-I local supplier' subject to matching the L1 price. In case such lowest eligible 'Class-I local supplier' fails to match the L1 price or accepts less than the offered quantity, the next higher 'Class-I local supplier' within the margin of purchase preference shall be invited to match the L1 price for remaining quantity and so on, and contract shall be awarded accordingly. In case some quantity is still left uncovered on Class-I local suppliers, then such balance quantity may also be ordered on the L1 bidder.

(c) In the procurement of goods or works, which are covered by para 3(b) above and which are not divisible in nature, and in procurement of services where the bid is evaluated on price alone, the 'Class-I local supplier' shall get purchase preference over 'Class-II local supplier' as well as 'Non-local supplier', as per following procedure:

i. Among all qualified bids, the lowest bid will be termed as L1. If L1 is Class -I local supplier', the contract will be awarded to L1.

ii. If L1 is not 'Class-I local supplier', the lowest bidder among the 'Class-I local supplier', will be invited to match the L1 price subject to Class-I local supplier's quoted price falling within the margin of purchase preference, and the contract shall be awarded to such 'Class-I local supplier' subject to matching the L1 price.

iii. In case such lowest eligible 'Class-I local supplier' fails to match the L1 price, the 'Class-I local supplier' with the next higher bid within the margin of purchase preference shall be invited to match the L1 price and so on and contract shall be awarded accordingly. In case none of the 'Class-I local supplier' within the margin of purchase preference matches the L1 price, the contract may be awarded to the L1 bidder.

(d) "Class-II local supplier" will not get purchase preference in any procurement, undertaken by procuring entities.

3B. Applicability in tenders where contract is to be awarded to multiple bidders- In tenders where contract is awarded to multiple bidders subject to matching of L1 rates or otherwise, the 'Class-I local supplier' shall get purchase preference over 'Class-II local supplier' as well as 'Non-local supplier', as per following procedure:

a. In case there is sufficient local capacity and competition for the item to be procured, as notified by the nodal Ministry, only Class I local suppliers shall be eligible to bid. As such, the multiple suppliers, who would be awarded the contract, should be all and only 'Class I Local suppliers'.

b. In other cases, 'Class II local suppliers' and 'Non local suppliers' may also participate in the bidding process along with 'Class I Local suppliers' as per provisions of this Order.

c. If 'Class I Local suppliers' qualify for award of contract for at least

50% of the tendered quantity in any tender, the contract may be awarded to all the qualified bidders as per award criteria stipulated in the bid documents. However, in case 'Class I Local suppliers' do not qualify for award of contract for at least 50% of the tendered quantity, purchase preference should be given to the 'Class I local supplier' over 'Class II local suppliers'/ 'Non local suppliers' provided that their quoted rate falls within 20% margin of purchase preference of the highest quoted bidder considered for award of contract so as to ensure that the 'Class I Local suppliers' taken in totality are considered for award of contract for at least 50% of the tendered quantity.

d. First purchase preference has to be given to the lowest quoting 'Class-I local supplier', whose quoted rates fall within 20% margin of purchase preference, subject to its meeting the prescribed criteria for award of contract as also the constraint of maximum quantity that can be sourced from any single supplier. If the lowest quoting 'Class-I local supplier', does not qualify for purchase preference because of aforesaid constraints or does not accept the offered quantity, an opportunity may be given to next higher 'Class-I local supplier', falling within 20% margin of purchase preference, and so on.

e. To avoid any ambiguity during bid evaluation process, the procuring entities may stipulate its own tender specific criteria for award of contract amongst different bidders including the procedure for purchase preference to 'Class-I local supplier' within the broad policy guidelines stipulated in sub- paras above.

4. **Exemption of small purchases:** Notwithstanding anything contained in paragraph 3, procurement where the estimated value to be procured is less than Rs. 5 lakhs shall be exempt from this Order. However, it shall be ensured by procuring entities that procurement is not split for the purpose of avoiding the provisions of this Order.

4A. Exemption in sourcing of spares and consumables of closed systems:

Procurement of spare parts, consumables for closed systems and Maintenance/ Service contracts with Original Equipment Manufacturer/Original Equipment Supplier/Original Part Manufacturer shall be exempted from this Order.

5. **Minimum local content**: The 'local content' requirement to categorize a supplier as 'Class-I local supplier' is minimum 50%. For 'Class-II local supplier', the 'local content' requirement is minimum 20%. Nodal Ministry/ Department may prescribe only a higher percentage of minimum local content requirement to categorize a supplier as 'Class-I local supplier'/ 'Class- II local supplier'. For the items, for which Nodal Ministry/ Department has not prescribed higher minimum local content notification under the Order, it shall be 50% and 20% for 'Class-I local supplier'/ 'Class-II local supplier' respectively.

6. **Margin of Purchase Preference**: The margin of purchase preference shall be 20%.

7. **Requirement for specification in advance**: The minimum local content, the margin of purchase preference and the procedure for preference to Make in India shall be specified in the notice inviting tenders or other form of procurement solicitation and shall not be varied during a particular procurement transaction.

8. **Government E-marketplace**: In respect of procurement through the Government E-marketplace (GeM) shall, as far as possible, specifically mark the items which meet the minimum local content while registering the item for

display, and shall, wherever feasible, make provision for automated comparison with purchase preference and without purchase preference and for obtaining consent of the local supplier in those cases where purchase preference is to be exercised.

9. **Verification of local content:**

a. The 'Class-I local supplier'/ 'Class-II local supplier' at the time of tender, bidding or solicitation shall be required to indicate percentage of local content and provide self-certification that the item offered meets the local content requirement for 'Class-I local supplier'/ 'Class-II local supplier', as the case may be. They shall also give details of the location(s) at which the local value addition is made.

b. In cases of procurement for a value in excess of Rs. 10 crores, the 'Class-I local supplier'/ 'Class-II local supplier' shall be required to provide a certificate from the statutory auditor or cost auditor of the company (in the case of companies) or from a practicing cost accountant or practicing chartered accountant (in respect of suppliers other than companies) giving the percentage of local content.

c. The bidder shall give self-certification for local content in the quoted item (goods/works/services) at the time of tendering. However, at the time of execution of the project, for all contracts above INR 10 Crore, the contractor/ supplier shall be required to give local content certification duly certified by cost/ chartered accountant in practice. For cases where it is not possible to provide certification by Cost/Chartered Accountant at the time of execution of project, the supplier shall be permitted to provide the certificate for local content from Cost/ Chartered Accountant after completion of the contract, within time limit acceptable to the procuring entity. In case the contractor/ supplier does not meet the stipulated local content requirement and the category of the supplier changes from Class-I to Class-II/ Non-local or from Class-II to Non-local, a penalty upto 10% of the contract value may be imposed. However, contract once awarded shall not be terminated on this account.

d. Decisions on complaints relating to implementation of this Order shall be taken by the competent authority which is empowered to look into procurement-related complaints relating to the procuring entity.

e. Nodal Ministries may constitute committees with internal and external experts for independent verification of self-declarations and auditor's/ accountant's certificates on random basis and in the case of complaints.

f. Nodal Ministries and procuring entities may prescribe fees for such complaints.

g. False declarations will be in breach of the Code of Integrity under Rule 175(1)(i)(h) of the General Financial Rules for which a bidder or its successors can be debarred for up to two years as per Rule 151 (iii) of the General Financial Rules along with such other actions as may be permissible under law.

h. A supplier who has been debarred by any procuring entity for violation of this Order shall not be eligible for preference under this Order for procurement by any other procuring entity for the duration of the debarment. The debarment for such other procuring entities shall take effect prospectively from the date on which it comes to the notice of other procurement entities, in the manner prescribed under paragraph 9

i below.

l. The Department of Expenditure shall issue suitable instructions for the effective and smooth operation of this process, so that:
   i. The fact and duration of debarment for violation of this Order by any procuring entity are promptly brought to the notice of the Member-Convenor of the Standing Committee and the Department of Expenditure through the concerned Ministry /Department or in some other manner;
   ii. On a periodical basis such cases are consolidated and a centralized list or decentralized lists of such suppliers with the period of debarment is maintained and displayed on website(s);
   iii. In respect of procuring entities other than the one which has carried out the debarment, the debarment takes effect prospectively from the date of uploading on the website(s) in the such a manner that ongoing procurement are not disrupted.

10. **Specifications in Tenders and other procurement solicitations:**
   a. Every procuring entity shall ensure that the eligibility conditions in respect of previous experience fixed in any tender or solicitation do not require proof of supply in other countries or proof of exports.
   b. Procuring entities shall endeavour to see that eligibility conditions, including on matters like turnover, production capability and financial strength do not result in unreasonable exclusion of 'Class-I local supplier'/ 'Class-II local supplier' who would otherwise be eligible, beyond what is essential for ensuring quality or creditworthiness of the supplier.
   c. Procuring entities shall, within 2 months of the issue of this Order review all existing eligibility norms and conditions with reference to sub-paragraphs 'a' and 'b' above.
   d. **Reciprocity Clause**
      i. When a Nodal Ministry/Department identifies that Indian suppliers of an item are not allowed to participate and/ or compete in procurement by any foreign government, due to restrictive tender conditions which have direct or indirect effect of barring Indian companies such as registration in the procuring country, execution of projects of specific value in the procuring country etc., it shall provide such details to all its procuring entities including CMDs/CEOs of PSEs/PSUs, State Governments and other procurement agencies under their administrative control and GeM for appropriate reciprocal action.
      ii. Entities of countries which have been identified by the nodal Ministry/Department as not allowing Indian companies to participate in their Government procurement for any item related to that nodal Ministry shall not be allowed to participate in Government procurement in India for all items related to that nodal Ministry/ Department, except for the list of items published by the Ministry/ Department permitting their participation.
      iii. The stipulation in (ii) above shall be part of all tenders invited by the Central Government procuring entities stated in (i) above. All purchases on GeM shall also necessarily have the above provisions for items identified by nodal Ministry/ Department.
      iv. State Governments should be encouraged to incorporate similar provisions in their respective tenders.
      v. The term 'entity' of a country shall have the same meaning as under the FDI Policy of DPIIT as amended from time to time.
   e. Specifying foreign certifications/ unreasonable technical specifications/

brands/ models in the bid document is restrictive and discriminatory practice against local suppliers. If foreign certification is required to be stipulated because of non-availability of Indian Standards and/or for any other reason, the same shall be done only after written approval of Secretary of the Department concerned or any other Authority having been designated such power by the Secretary of the Department concerned.

f. "All administrative Ministries/Departments *whose procurement exceeds Rs. 1000 Crore per annum* shall notify/update their procurement projections every year, including those of the PSEs/PSUs, for the next 5 years on their respective website."

10A. Action for non-compliance of the Provisions of the Order: In case restrictive or discriminatory conditions against domestic suppliers are included in bid documents, an inquiry shall be conducted by the Administrative Department undertaking the procurement (including procurement by any entity under its administrative control) to fix responsibility for the same. Thereafter, appropriate action, administrative or otherwise, shall be taken against erring officials of procurement entities under relevant provisions. Intimation on all such actions shall be sent to the Standing Committee.

11. **Assessment of supply base by Nodal Ministries**: The Nodal Ministry shall keep in view the domestic manufacturing / supply base and assess the available capacity and the extent of local competition while identifying items and prescribing the higher minimum local content or the manner of its calculation, with a view to avoiding cost increase from the operation of this Order.

12. **Increase in minimum local content**: The Nodal Ministry may annually review the local content requirements with a view to increasing them, subject to availability of sufficient local competition with adequate quality.

13. **Manufacture under license/ technology collaboration agreements with phased indigenization**: While notifying the minimum local content, Nodal Ministries may make special provisions for exempting suppliers from meeting the stipulated local content if the product is being manufactured in India under a license from a foreign manufacturer who holds intellectual property rights and where there is a technology collaboration agreement / transfer of technology agreement for indigenous manufacture of a product developed abroad with clear phasing of increase in local content.

13A. In procurement of all goods, services or works in respect of which there is substantial quantity of public procurement and for which the nodal ministry has not notified that there is sufficient local capacity and local competition, the concerned nodal ministry shall notify an upper threshold value of procurement beyond which foreign companies shall enter into a joint venture with an Indian company to participate in the tender. Procuring entities, while procuring such items beyond the notified threshold value, shall prescribe in their respective tenders that foreign companies may enter into a joint venture with an Indian company to participate in the tender. The procuring Ministries/Departments shall also make special provisions for exempting such joint ventures from meeting the stipulated minimum local content requirement, which shall be increased in a phased manner.

14. **Powers to grant exemption and to reduce minimum local content**: The administrative Department undertaking the procurement (including

procurement by any entity under its administrative control), with the approval of their Minister-in-charge, may by written order, for reasons to be recorded in writing,

a. reduce the minimum local content below the prescribed level; or
b. reduce the margin of purchase preference below 20%; or
c. exempt any particular item or supplying entities from the operation of this Order or any part of the Order.

The Administrative Department, while seeking exemption under this para, shall certify that such an item(s) has not been notified by Nodal Ministry/ Department concerned under para 3 (a) of the Order.

A copy of every such order shall be provided to the Standing Committee and concerned Nodal Ministry / Department. The Nodal Ministry / Department concerned will continue to have the power to vary its notification on Minimum Local Content.

15. **Directions to Government companies**: In respect of Government companies and other procuring entities not governed by the General Financial Rules, the administrative Ministry or Department shall issue policy directions requiring compliance with this Order.
16. **Standing Committee**: A standing committee is hereby constituted with the following membership:
Secretary, Department for Promotion of Industry and Internal Trade - Chairman
Secretary, Commerce—Member
Secretary, Ministry of Electronics and Information Technology—Member Joint Secretary (Public Procurement), Department of Expenditure—Member Joint Secretary (DPIIT)—Member-Convenor

The Secretary of the Department concerned with a particular item shall be a member in respect of issues relating to such item. The Chairman of the Committee may co-opt technical experts as relevant to any issue or class of issues under its consideration.

17. **Functions of the Standing Committee**: The Standting Committee shall meet as often as necessary, but not less than once in six months. The Committee

a. shall oversee the implementation of this order and issues arising therefrom, and make recommendations to Nodal Ministries and procuring entities.
b. shall annually assess and periodically monitor compliance with this Order
c. shall identify Nodal Ministries and the allocation of items among them for issue of notifications on minimum local content
d. may require furnishing of details or returns regarding compliance with this Order and related matters
e. may, during the annual review or otherwise, assess issues, if any, where it is felt that the manner of implementation of the order results in any restrictive practices, cartelization or increase in public expenditure and suggest remedial measures
f. may examine cases covered by paragraph 13 above relating to manufacture under license/ technology transfer agreements with a view to satisfying itself that adequate mechanisms exist for enforcement of such agreements and for attaining the underlying objective of progressive indigenization

g. may consider any other issue relating to this Order which may arise.

18. **Removal of difficulties**: Ministries /Departments and the Boards of Directors of Government companies may issue such clarifications and instructions as may be necessary for the removal of any difficulties arising in the implementation of this Order.

19. **Ministries having existing policies**: Where any Ministry or Department has its own policy for preference to local content approved by the Cabinet after 1st January 2015, such policies will prevail over the provisions of this Order. All other existing orders on preference to local content shall be reviewed by the Nodal Ministries and revised as needed to conform to this Order, within two months of the issue of this Order.

20. **Transitional provision**: This Order shall not apply to any tender or procurement for which notice inviting tender or other form of procurement solicitation has been issued before the issue of this Order.

19/7/24

(Himani Pande)
Additional Secretary to the Government of India
Tel: 011-23038888
E-mail: ashp.dpiit@gov.in

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File No: FP-20013/2/2017-FP-PNG-Part (4) (E-41432)
Government of India
Ministry of Petroleum and Natural Gas
(Flagship Programme Cell)
****

Shastri Bhawan, New Delhi
Dated: 26th March, 2024

To

1. Chairman, IOCL
2. C&MD, ONGC/ BPCL/ HPCL/ OIL/ GAIL/ EIL/ Balmer Lawrie
3. Managing Director, CPCL/ NRL/ MRPL/ BCPL/ OVL
4. DG, DGH
5. DG, PPAC
6. Secretary, OIDB
7. ED, OISD
8. ED, CHT
9. Director, RGIPT/ IIPE
10. Secretary, PNGRB
11. CEO & MD, ISPRL

Subject: Public Procurement (Preference to Make in India) Order, 2017-reg.

Sir/ Madam,

I am directed to refer to this Ministry's letter of even number dated 26.04.2022 on the above mentioned subject.

2. In this regard, it is stated that the sector specific relaxation granted under para 2(b) of aforementioned letter, has been reviewed in the Ministry and it has been decided that scope of this relaxation shall be limited to the items (list enclosed) to be installed/ operated in flammable environment of oil and gas process industry.

3. This issues with the approval of Hon'ble Minister, P&NG.

Yours faithfully

Kala 26/3/24
(Kala)
Under Secretary to the Govt. of India
Tel.: 011-23381029

Encl.: as above

Copy to:

a. PS to Hon'ble Minister, P&NG
b. PS to Hon'ble MoS, P&NG
c. PPS/ PS to Secretary, P&NG
d. PPS/PS to AS&FA/ AS, MoPNG
e. PPS/ PS to JS (E & BR)/ JS(GP)/ JS (M& OR)/ JS (IC)/ JS (G)/ JS (IFD)/ DDG, MoPNG
f. Technical Director, NIC: with a request to upload the letter on MoPNG website

Copy for information to,

Secretary, DPIIT

File No: FP-20013/2/2017-FP-PNG-Part (4) (E-41432)
Government of India
Ministry of Petroleum and Natural Gas
(Flagship Programme Cell)
****

Shastri Bhawan, New Delhi
Dated 26th April, 2022

To,

1. Chairman, IOCL
2. C&MD, BPCL/ HPCL/ ONGC/ OIL/ GAIL/ EIL/ Balmer Lawrie
3. Managing Director, MRPL/NRL/CPCL// BCPL/ OVL
4. DG, DGH
5. DG, PPAC
6. Secretary, OIDB
7. ED, PCRA
8. ED, OISD
9. ED, CHT
10. Director, RGIPT
11. Secretary, PNGRB
12. CEO & MD, ISPRL

Sub: Public Procurement (Preference to Make in India) (PPP-MII) Order, 2017-reg.

Sir/Madam,

I am directed to refer to this Ministry's letter of even number dated 23.02.2022 regarding Policy to Provide Purchase Preference (linked with local content) (PP-LC) in all Public Sector Undertakings under the Ministry of Petroleum and Natural Gas (MoP&NG) and to say that Public Procurement (Preference to Make in India) Order, 2017 issued by DPIIT and as amended time to time shall be applicable to all the Public Sector Undertakings and their wholly owned subsidiaries under MoP&NG; Joint Ventures that have 51% or more equity by one or more Public Sector Undertakings under MoP&NG; attached and subordinate offices of MoPNG w.e.f. 01.04.2022.

2. Moreover, as per para 14 of the PPP-MII Order, the following modifications in the order shall be applicable on the procuring entities under this Ministry:

a. Limit for exemption of small purchase under para 4 of the PPP-MII Order, 2017 shall be Rs. 1 crore.

b. Local value addition through services such as transportation, insurance, installation, commissioning, training and after sales services support like AMC/ CMC etc. shall continue to be considered in local content calculation.

c. HP-HT operations in upstream oil and gas business activities shall be exempted from applicability of the Order.

3. This issues with the approval of Hon'ble Minister, Petroleum and Natural Gas.

Yours faithfully

26/04/2022

(Santanu Dhar)
Under Secretary to the Govt. of India
Tel.: 011-23388652

Copy to:

a. PS to Minister, PNG
b. PPS/ PS to Secretary/ AS&FA/ Sr. Economic Advisor, MoPNG
c. PPS/ PS to AS (E)/ JS(R)/ JS (M& GP)/ OSD (IC)/ JS (G)/ JS (IFD)/ DDG (ED), MoPNG
d. PPS/PS to Dir.(BR)/Dir.(E-II)/Dir.(E-I)/DS(GP)/DS(Mkt.)/DS(LPG)/DS(Admn.)/DS(RTI)/ DS (Gen) MoPNG

Copy for information to:

Secretary, DPIIT

Items specifically to be used in flammable environment of oil and gas process industry in which value addition through services such as transportation, insurance installation, commissioning, training and after sales service support like AMC/CMC shall be continued to be conserved wile computing Local Content:

Sl.	Items Category of procurement
1	**Instrumentation items** like DCS/ ESD/ PLC, Emission Monitoring System, Machine Monitoring System, Condition Monitoring System, Fire Detection & Alarm System, Gas & Liquid Metering Systems, Mass Flow Meter, Process Analysers, Ultrasonic flowmeter, Anti surge & Governor Control system, Master control station for MOVs, Actuators, Transmitters, Radar gauges, multi-spot temperature sensors, Batch Controller unit, Additive blocks, Hydrocarbon detectors, Field instruments like Radar Gauges, Nucleonic gauges, Servo level gauging, Hydrastep, SMART Positioner in control valves, Multi point Reactor Thermocouple, Flame scanner, Viscosity cum Density meter, High Pressure/Temperature Special type valves, IS Test Equipment, Flare flow meter, Tank Farm Management System, Loading Automation System, Gas Detection System, Corrosion Monitoring System, Wireless Instrumentation, Surge Relief Valve Skid, Meter Provers, Pipeline Leak detection system (Negative Pressure wave type) etc
2	Equipment involved in drilling, completion, testing and production of oil and gas wells, Electronic carousel, Safety Relief valves, Compressors, turbines and blowers, heat ejectors, exchangers, condensers where SITC and / Expert Supervisory Services are involved, Equipment for which Life Cycle cost evaluation is done considering AMC/CAMC etc.
3	Electrical Equipment like Flameproof Plant Communication system, GIS (Gas Insulated Switchgear), Numerical Relays, Flameproof and/or explosion proof CCTV, Synchronous Machines, TETRA System etc
4	Laboratory and R&D Equipment like Gas Chromatographs, Spectrometers, Analytical equipment, Automatic liquid sampler, Dry colorimetric (tape) detectors; Mercury Free PVT Equipment; HT-HP Corrosion cell; Wheel Test Machine; Atmospheric and HT-HP- Consistometer; Ultra Sonic Cement Analyzer / HT-Ultra Sonic Cement Analyzer; Compressive strength tester; Stirred fluid loss apparatus; Total Sulfur analyzer; Colony counter; Laminar flow systems; Microscope with digital camera; Anaerobic Chamber; RockEval; TOC analyser; Simdist analyser; Cold Finger test apparatus; Microcoulometer; Atomic Absorption Spectrometer; High Performance Ion Chromatograph; Permeameter; Helium Porosimeter; Laser Scattering Particle Size Distribution Analyser; Microscope high magnification; Flame Photometer; U.V.- Visible Spectrophotometers; EP Lube tester; Differential sticking tester; Electrical stability Meter; Stemi-2000 microscope/ equivalent; Core Gama Logger; Energy Dispersive Spectrometer; Thin Section Machine; X-Ray Diffractometer; Wettability tester; Ambient Resistivity System; Flouroscope; Spin Drop Tensiometer; Fluid Eval; Mercury Inclusion Porosity meter; Tri-Axial test machine; Smoke point detector; Vitrinite Reflectance VRo Microscope; Curing Chamber/ HT curing chamber; Source Rock Pyrolyser with Sulphur/ without sulphur; NGA Gas Analyzer Flash Point Apparatus/ automatic; Compressive Strength Analyser; HT-HP Curing Chamber; Crush resistance test equipment; Manual Dry bath pore point Apparatus; Bench Type Dissolved Oxygen meter; Bench Type GRAIN Moisture meter; Rotational Viscometer; Capillary pressure instrument; Core Plugging Machine; Core Trimming machine; HP-HT filter press; BENCH TOP DENSITY METER; HP-HT ROLLER OVEN with AGING CELLS; Crude Oil Analyser; Static Gel Strength analyser; Oilwell cement Mechanical properties analyser; Poroperm; Acoustic velocity system; Auto Saturator; Porosimeter cum permeameter; Rock Testing System; Auto Core Saturator; Auto Imbibiometer for reverse permeability; Proppant Conductivity Tester etc

5	Pipeline Intrusion and Detection System (PIDS), SCADA & APPS system; Rim seal protection system, Chemical Treatment program which involves supply of Process chemicals, Dosing & monitoring the parameters etc
6	**Geophysical Services** Seismic Data Acquisition System; VSP Data Acquisition System; Field Processing Unit; Geophysical equipment (non-Seismic including Gravity Magnetic, MT, EM etc.); Seismic Data Processing Software ; Seismic Data Interpretation Software ; Hydrocarbon Reservoir Software ; Seismic Data Archival Software ; Petro -Physical Software
7	**Geophysical Equipment** Field Processing Unit, Global Navigation Satellite System, Specialised Geological Lab Equipment, Seismic Data Acquisition System, Seismic Data Processing/ Imaging Software, Tape Drive Unit, VSP Data Acquisition System, VSP Processing Software; High-End Servers for G&G Applications, HPCC Solution; Virtualization and Container Software
8	**Logging Services** Logging unit, equipment and tools rated for hostile HPHT environment (Temperature > 300°F & Pressure > 10.000 psi).

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रजिस्ट्री सं. डी.एल.- 33004/99 REGD. No. D. L.-33004/99

भारत का राजपत्र
The Gazette of India

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असाधारण
EXTRAORDINARY

भाग I—खण्ड 1
PART I—Section 1

प्राधिकार से प्रकाशित
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इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय

(आई.पी.एच.डब्ल्यू.)

अधिसूचना

नई दिल्ली, 7 सितंबर, 2020

विषय: सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश 2017 के अनुक्रम में इलेक्ट्रॉनिक उत्पादों को अधिसूचित करना

संदर्भ : उद्योग और आंतरिक व्यापार संवर्धन विभाग (डीपीआईआईटी) के दिनांक 28.05.2018, 29.05.2019 और 04.06.2020 के आदेशों के जरिए यथासंशोधित आदेश संख्या पी-45021/2/2017-बीई-II दिनांक 15.06.2017

फा. सं. डब्ल्यू-43/4/2019-आईपीएचडब्ल्यू-एमईआईटीवाई.—सरकार ने 'मेक इन इंडिया' को प्रोत्साहित करने और आय और रोजगार बढ़ाने के उद्देश्य से भारत में वस्तुओं और सेवाओं के विनिर्माण और उत्पादन को बढ़ावा देने के लिए उद्योग और आंतरिक व्यापार संवर्धन विभाग (डीपीआईआईटी) के दिनांक 28.05.2018, 29.05.2019 और 04.06.2020 के आदेशों के जरिए यथासंशोधित आदेश संख्या पी-45021/2/2017-बीई-II, दिनांक 15.06.2017 के तहत सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश, 2017 जारी किया है।

2. उपर्युक्त संदर्भ के जरिए अधिसूचित सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश, 2017 [पीपीपी-एमआईआई आदेश 2017] के अनुक्रम में और इसके तहत जारी की गई दिनांक 14.09.2017 की इलेक्ट्रॉनिक उत्पाद अधिसूचना संख्या 33(1)/2017-आईपीएचडब्ल्यू और दिनांक 01.08.2018 की अधिसूचना संख्या 33(5)/2017-आईपीएचडब्ल्यू के अधिक्रमण में इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय (एमईआईटीवाई) एतदद्वारा अधिसूचित करता है कि समय-समय पर संशोधित पूर्वोक्त आदेश के अनुसार स्थानीय रूप से विनिर्मित इलेक्ट्रॉनिक उत्पादों को सभी क्रय संस्थाओं द्वारा वरीयता प्रदान की जाएगी।

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3. इस अधिसूचना के उद्देश्य से:

क. श्रेणी-I स्थानीय आपूर्तिकर्ता, श्रेणी-II स्थानीय आपूर्तिकर्ता और गैर-स्थानीय आपूर्तिकर्ता की परिभाषा समय-समय पर यथासंशोधित डीपीआईआईटी पीपीपी-एमआईआई आदेश, 2017 संख्या पी-45021/2/2017-पीपी (बीई-II), दिनांक 04.06.2020 के पैरा 2 के अनुसार होगी। समय-समय पर यथासंशोधित डीपीआईआईटी पीपीपी-एमआईआई आदेश, 2017 संख्या पी-45021/2/2017-पीपी (बीई-II), दिनांक 04.06.2020 के पैरा 5 के अनुसार इस अधिसूचना में अधिसूचित प्रत्येक इलेक्ट्रॉनिक उत्पाद के लिए स्थानीय सामग्री की गणना के लिए तंत्र निर्धारित किया गया है।

ख. समय-समय पर यथासंशोधित डीपीआईआईटी पीपीपी-एमआईआई आदेश, 2017 संख्या पी-45021/2/2017-पीपी (बीई-II), दिनांक 04.06.2020 के पैरा 3क क्रय के प्रतिशत के लिए संदर्भित किया जाएगा जिसके लिए अधिमानतः घरेलू स्तर पर विनिर्मित इलेक्ट्रॉनिक उत्पादों (मूल्य के संदर्भ में) को वरियता दी जानी है।

4. सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश 2017 के तहत निम्नलिखित इलेक्ट्रॉनिक उत्पाद अधिसूचित किए गए हैं:

4.1 डेस्कटॉप पर्सनल कंप्यूटर (पीसी)

(क) परिभाषा:

इस अधिसूचना के प्रयोजन से एक डेस्कटॉप पीसी में आवश्यक रूप से एक सीपीयू, मेमोरी, हार्ड डिस्क ड्राइव, की-बोर्ड, माउस और एक अलग या एकीकृत डिस्प्ले यूनिट शामिल होगी और इसे स्वतंत्र रूप से कार्य करने में सक्षर्थ होना चाहिए।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र:

डेस्कटॉप पीसी की सामग्री का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बी.ओ.एम के प्रमुख इनपुट/डेस्कटॉप पीसी के विनिर्माण के लिए चरण	**घरेलू बीओएम के रूप में वर्गीकृत किए जाने वाले इनपुट के लिए आवश्यक मूल्यवर्धन/स्थानीय सामग्री**
1	**2**
मुख्य बोर्ड/मदरबोर्ड और सीपीयू/जीपीयू	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण, जिसमें सेमीकंडक्टर्स** और सीपीयू/जीपीयू** का मूल्य शामिल है और बेयर पीसीबी का मूल्य इसमें शामिल नहीं है। हालाँकि, सीपीयू के कुल मूल्य का भार डेस्कटॉप पीसी के कुल बीओएम के 30% से अधिक नहीं होगा।
मेमोरी मॉड्यूल	बेयर पीसीबी के मूल्य को छोड़कर, आयातित/घरेलू स्तर पर विनिर्मित बेयर पीसीबी पर आयातित/घरेलू स्तर पर विनिर्मित मेमोरी चिप्स** और कलपुर्जों से घरेलू पीसीबी असेंबली* एवं परीक्षण।
हार्ड डिस्क ड्राइव/सॉलिड स्टेट ड्राइव (एसएसडी)	बेयर पीसीबी के मूल्य को छोड़कर आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* एवं फाइनल उत्पाद असेम्बली और परीक्षण।
एलसीडी/एलईडी मॉनिटर	आयातित/घरेलू रूप से असेम्बल किए गए एलसीडी/एलईडी पैनल वाले कलपुर्जों से एलसीडी/एलईडी मॉनीटर की घरेलू स्तर पर असेम्बली और परीक्षण और घरेलू पीसीबी असेंबली*, घरेलू प्लास्टिक मोल्डिंग और बेयर पीसीबी के मूल्य को छोड़कर धातुई कलपूर्जों की स्टैम्पिंग।

डीवीडी ड्राइव	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू असेंबली और परीक्षण।
कैबिनेट + एमएसपीएस	घरेलू रूप से विनिर्मित कैबिनेट और घरेलू पीसीबी असेम्बली* और आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से एसएमपीएस की फाइनल असेम्बली और परीक्षण, इस शर्त के अध्यधीन कि "एसएमपीएस" की असेम्बली में उपयोग किए जाने वाले घरेलू विनिर्मित कलपुर्जे और संघटक न्यूनतम 20% होंगे ("एसएमपीएस" के विनिर्माण में उपयोग किए जाने वाले कलपुर्जों और संघटकों के कुल मूल्य के)।
की-बोर्ड और माउस	घरेलू पीसीबी असेम्बली* और आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से फाइनल असेम्बली और परीक्षण।
बेयर पीसीबी	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू रूप से विनिर्मित।
(i) फाइनल असेंबली/परीक्षण और (ii) डिजाइन/विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

**यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करते हुए बेयर पीसीबी पर सीपीयू (एस)/जीपीयू/प्रोसेसर (एस) कलपुर्जों/संघटकों के प्रिंटेड सर्किट बोर्ड की असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

***भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

4.2 थिन क्लाइंट

(क) परिभाषा:

इस अधिसूचना के प्रयोजन से किसी थिन क्लाइंट (टीसी) में अनिवार्य रूप से सीपीयू, मेमोरी, डोम, की-बोर्ड, माउस और एक अलग या एकीकृत डिस्प्ले यूनिट शामिल होते हैं और यह स्वतंत्र रूप से कार्य करने में सक्षम होना चाहिए।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र:

थिन क्लाइंट की सामग्री का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम के प्रमुख इनपुट/थिन क्लाइंट के विनिर्माण के लिए चरण	**घरेलू बीओएम के रूप में वर्गीकृत किए जाने वाले इनपुट के लिए आवश्यक मूल्यवर्धन/स्थानीय सामग्री**
1	2
मुख्य बोर्ड/मदरबोर्ड और सीपीयू	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण, जिसमें सेमीकंडक्टर्स** और सीपीयू** का मूल्य शामिल है और बेयर पीसीबी का मूल्य इसमें शामिल नहीं है। हालाँकि, सीपीयू के कुल मूल्य का भार थिन क्लाइंट के कुल बीओएम के 30% से अधिक नहीं होगा।

मेमोरी मॉड्यूल/डोम	बेयर पीसीबी के मूल्य को छोड़कर आयातित/घरेलू स्तर पर विनिर्मित बेयर पीसीबी पर आयातित/घरेलू स्तर पर विनिर्मित मेमोरी चिप्स** और कलपुर्जों से घरेलू पीसीबी असेंबली* और परीक्षण।
एसएमपीएस/पावर अडैप्टर	घरेलू रूप से पीसीबी असेंबलीं * और आयातित/ घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से एसएमपीएस / पावर अडैप्टर की फाइनल असेंबली और टेस्टिंग, इस शर्त के अधीन है कि "एसएमपीएस" की असेंबली में प्रयुक्त घरेलू विनिर्मित पुर्जे और कलपुर्जे न्यूनतम 20% होंगे ("एसएमपीएस") के विनिर्माण में उपयोग किए जाने वाले कलपुर्जों और संघटकों के कुल मूल्य के)।
एलसीडी/एलईडी मॉनिटर	आयातित/घरेलू रूप से असेंबली किए गए एलसीडी/एलईडी पैनल वाले कलपुर्जों से एलसीडी/एलईडी मॉनीटर की घरेलू स्तर पर असेम्बली और परीक्षण और घरेलू पीसीबी असेंबली*, घरेलू प्लास्टिक मोल्डिंग और बेयर पीसीबी के मूल्य को छोड़कर धातुई कलपूर्जों की स्टैम्पिंग।
कैबिनेट	घरेलू तौर पर विनिर्मित।
की-बोर्ड और माउस	घरेलू पीसीबी असेम्बली* और आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से फाइनल असेम्बली और परीक्षण।
बेयर पीसीबी	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू रूप से विनिर्मित ।
(i) फाइनल असेंबली / परीक्षण और (ii) डिजाइन / विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

**यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करते हुए बेयर पीसीबी पर सीपीयू (एस)/जीपीयू/प्रोसेसर (एस) कलपुर्जों/संघटकों के प्रिंटेड सर्किट बोर्ड की असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

***भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

4.3 लैपटॉप पर्सनल कंप्यूटर (पीसी)

(क) परिभाषा :

इस अधिसूचना के प्रयोजन से किसी लैपटॉप पीसी (जिसे आमतौर पर लैपटॉप/नोटबुक/नेटबुक/अल्ट्राबुक आदि के रूप में बाजार में जाना जाता है) में आवश्यक रूप से सीपीयू, मेमोरी, हार्ड डिस्क ड्राइव, कीबोर्ड, टचपैड और/या ट्रैकपॉइंट एकीकृत प्रदर्शन इकाई, एकीकृत बैटरी शामिल होंगे और यह स्वतंत्र रूप से प्रचालित होने में सक्षम होना चाहिए।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र:

लैपटॉप पीसी का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट व्यक्तिगत रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम प्रमुख इनपुट लैपटॉप के विनिर्माण के लिए चरण	**घरेलू बीओएम के रूप में वर्गीकृत किए जाने वाले इनपुट के लिए आवश्यक मूल्यवर्धन/ स्थानीय सामग्री**
1	2
मुख्य बोर्ड/मदरबोर्ड और सीपीयू/जीपीयू	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण, जिसमें सेमीकंडक्टर्स** और सीपीयू/जीपीयू** का मूल्य शामिल है और बेयर पीसीबी का मूल्य इसमें शामिल नहीं है। हालाँकि, सीपीयू के कुल मूल्य का भार लैपटॉप के कुल बीओएम के 35% से अधिक नहीं हो
पावर मॉड्यूल	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण ।
मेमोरी मॉड्यूल	बेयर पीसीबी के मूल्य को छोड़कर, आयातित/घरेलू स्तर पर विनिर्मित बेयर पीसीबी पर आयातित/घरेलू स्तर पर विनिर्मित मेमोरी चिप्स** और कलपुर्जों से घरेलू पीसीबी असेंबली* और परीक्षण ।
हार्ड डिस्क ड्राइव/सॉलिड स्टेट ड्राइव (एसएसडी)	बेयर पीसीबी के मूल्य को छोड़कर, आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और फाइनल उत्पाद असेम्बली और परीक्षण।
डिस्प्ले पैनल (एलसीडी, एलईडी, आदि)	बैक कवर और बेज़ेल में घरेलू असेंबली और बैक कवर और बेज़ेल के मूल्य को छोड़कर, आयातित/घरेलू स्तर पर डिस्प्ले पैनल, एलईडी बैक लाइट, एंटेना से फाइनल परीक्षण।
डीवीडी ड्राइव	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू असेंबली और परीक्षण ।
पावर अडैप्टर	घरेलू पीसीबी असेंबली के साथ आयातित/घरेलू स्तर पर विनिर्मित पूर्जों और संघटकों से फाइनल घरेलू असेंबली* और परीक्षण, इस शर्त के अध्यधीन है कि "पावर अडैप्टर" की असेंबली में इस्तेमाल होने वाले घरेलू स्तर पर विनिर्मित पूर्जों और संघटकों का मूल्य न्यूनतम 40% होगा ("पावर अडैप्टर" के विनिर्माण में उपयोग किए जाने वाले कलपूर्जों और संघटकों के कुल मूल्य का) ।
बेयर पीसीबी	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू रूप से विनिर्मित।
कीबोर्ड/टचपैड और/या ट्रैकपॉइंट	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण ।
बैटरी	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण ।
बॉटम कवर, बैक कवर, कीबोर्ड फेस बेजेल और डिस्प्ले फेस बेजल और हिंगेस सहित लैपटॉप के कैबिनेट/चेसिस	आयातित/घरेलू इनपुट का उपयोग करके इंजेक्शन मोल्डिंग / स्टांपिंग के माध्यम से घरेलू स्तर पर विनिर्मित ।
(i) फाइनल असेंबली/परीक्षण और (ii) डिजाइन/विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

**यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करते हुए बेयर पीसीबी पर सीपीयू (एस)/जीपीयू/प्रोसेसर (एस) कलपुर्जों/संघटकों के प्रिंटेड सर्किट बोर्ड की असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

***भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

4.4 कंप्यूटर मॉनिटर

(क) परिभाषा :

इस अधिसूचना के प्रयोजन से, एक कंप्यूटर मॉनीटर (जिसे आमतौर पर बाजार में मॉनीटर के रूप में जाना जाता है) में एक एलसीडी/एलईडी डिस्प्ले असेंबली, लॉजिक कार्ड, पॉवर सप्लाई अडैप्टर और कैबिनेट होता है जो इसे डेस्कटॉप पीसी या लैपटॉप पीसी या टैबलेट पीसी से कनेक्ट करने के लिए डिस्प्ले पोर्ट्स से युक्त होता है।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र :

कंप्यूटर मॉनिटर का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम के प्रमुख इनपुट/कंप्यूटर मॉनिटर के विनिर्माण के लिए चरण	**घरेलू बीओएम के रूप में वर्गीकृत किए जाने वाले इनपुट के लिए आवश्यक मूल्यवर्धन/स्थानीय सामग्री**
1	2
लॉजिक कार्ड/माइक्रो कंट्रोलर	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण, जिसमें सेमीकंडक्टर्स का मूल्य शामिल है और बेयर पीसीबी का मूल्य शामिल नहीं है।
एलसीडी/एलईडी पैनल	कैबिनेट/चेसिस के मूल्य को छोड़कर, मॉनिटर कैबिनेट/चेसिस में आयातित/घरेलू रूप से एकत्रित एलसीडी/एलईडी पैनल *** का घरेलू एकीकरण और परीक्षण । हालाँकि, एलसीडी/एलईडी पैनल के कुल मूल्य का भार कंप्यूटर मॉनीटर के कुल बीओएम के 35% से अधिक नहीं होगा।
केबल्स	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्मित और परीक्षण किया गया।
पावर अडैप्टर	घरेलू पीसीबी असेंबली के साथ आयातित/ घरेलू स्तर पर विनिर्मित पूर्जों और संघटकों से फाइनल घरेलू असेंबली* और परीक्षण, इस शर्त के अध्यधीन है कि "पावर अडैप्टर" की असेंबली में इस्तेमाल होने वाले घरेलू स्तर पर विनिर्मित पूर्जों और संघटकों का मूल्य न्यूनतम 40% होगा ("पावर अडैप्टर" के विनिर्माण में उपयोग किए जाने वाले कलपूर्जों और संघटकों के कुल मूल्य का)।
बेयर पीसीबी	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू रूप से विनिर्मित।
कैबिनेट/चेसिस	मॉनिटर बेज़ेल, बैक कवर, स्टैंड, बेस और चेसिस जिसमें प्लास्टिक मोल्डिंग और धातु के हिस्सों की स्टैम्पिंग को आयातित/घरेलू इनपुट का उपयोग करके घरेलू स्तर पर किया जाता है।

(i) फाइनल असेंबली/परीक्षण और (ii) डिजाइन/विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

**यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करते हुए बेयर पीसीबी पर सीपीयू (एस)/जीपीयू/प्रोसेसर (एस) कलपुर्जों/संघटकों के प्रिंटेड सर्किट बोर्ड की असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

***भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

**** इसकी समीक्षा तब की जाएगी जब भारत में कंप्यूटर मॉनिटर्स (प्रत्येक आकार और प्रकार के) का एलसीडी / एलईडी पैनल विनिर्माण संचालन में हो।*

4.5 टैबलेट पर्सनल कंप्यूटर (पीसी)

क. परिभाषा :

इस अधिसूचना के प्रयोजन से, एक टैबलेट पीसी में आवश्यक रूप से एक एकीकृत मदरबोर्ड जिसमें सीपीयू/प्रोसेसर, मेमोरी और पावर मॉड्यूल और बोर्ड होंगे शामिल होगा; डिस्प्ले पैनल (टच पैनल + एलसीडी/एलईडी मॉड्यूल) और एकीकृत बैटरी होंगे और यह स्वतंत्र रूप से संचालित करने में सक्षम होना चाहिए।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र :

टैबलेट पीसी का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम के प्रमुख इनपुट/टैबलेट पीसी के विनिर्माण के लिए चरण	**घरेलू बीओएम के रूप में वर्गीकृत किए जाने वाले इनपुट के लिए आवश्यक मूल्यवर्धन/स्थानीय सामग्री**
1	2
ऑन बोर्ड सीपीयू/जीपीयू/प्रोसेसर के साथ इंटीग्रेटेड मदरबोर्ड	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण, जिसमें सेमीकंडक्टर्स** और प्रोसेसर/ सीपीयू/जीपीयू** का मूल्य शामिल है और बेयर पीसीबी का मूल्य इसमें शामिल नहीं है। हालाँकि, सीपीयू के कुल मूल्य का भार टैबलेट पीसी के कुल बीओएम के 35% से अधिक नहीं होगा।
मेमोरी और पावर मॉड्यूल	बेयर पीसीबी के मूल्य को छोड़कर, आयातित/ घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण ।
डिस्प्ले पैनल (टच पैनल + एलसीडी/एलईडी मॉड्यूल)	आयातित/घरेलू रूप से विनिर्मित टच पैनल, एलसीडी/एलईडी मॉड्यूल या संयोजन से घरेलू असेंबली और परीक्षण, इस शर्त के अध्यधीन है कि डिस्प्ले पैनल का बैकलाइट असेंबली और परीक्षण घरेलू स्तर पर किया जाएगा।
पावर अडैप्टर	घरेलू पीसीबी असेंबली के साथ, आयातित/घरेलू स्तर पर विनिर्मित पूर्जों और संघटकों से फाइनल घरेलू असेंबली और परीक्षण, इस शर्त के अध्यधीन है कि "पावर अडैप्टर" की असेंबली में इस्तेमाल होने वाले घरेलू स्तर पर विनिर्मित

	कलपूर्जों और संघटकों का मूल्य न्यूनतम 40% होगा ("पावर अडैप्टर" के विनिर्माण में उपयोग किए जाने वाले कलपूर्जों और संघटकों के कुल मूल्य का)।
बेयर पीसीबी	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू रूप से विनिर्मित।
केसिंग	घरेलू स्तर पर विनिर्मित केसिंग।
बैटरी	आयातित/घरेलू स्तर पर विनिर्मित कलपूर्जों और संघटकों से घरेलू असेंबली और परीक्षण।
ऐक्सेसरी (कैमरा, स्पीकर, वाईफाई एंटीना, आदि)	आयातित/घरेलू स्तर पर कलपूर्जों और संघटकों से घरेलू असेंबली और परीक्षण।
(i) फाइनल असेंबली/परीक्षण और (ii) डिजाइन/विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

**यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करते हुए बेयर पीसीबी पर सीपीयू (एस)/जीपीयू/प्रोसेसर(एस) कलपुर्जों/संघटकों के प्रिंटेड सर्किट बोर्ड की असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

***भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

4.6 डॉट मैट्रिक्स प्रिंटर

(क) परिभाषा :

डॉट मैट्रिक्स प्रिंटर एक प्रकार का प्रभावी प्रिंटर है, जो 0.2 मिमी से 0.3 मिमी के व्यास के धातु पिन द्वारा कागज पर डॉट बनाता है, जो सोलनॉइड सिद्धांत के आधार पर इलेक्ट्रोमैग्नेट द्वारा संचालित होता है और डॉट मैट्रिक्स प्रिंटहेड केन क्षैतिज और ऊर्ध्वाधर रिजोलूशन द्वारा अपेक्षित कैरेक्टर का मैट्रिक्स उतपन्न होता है। डॉट मैट्रिक्स प्रिंटर पिन के यांत्रिक दबाव के आधार पर कार्बन प्रतियां और कार्बन रहित प्रतियां बना सकता है।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र :

डॉट मैट्रिक्स प्रिंटर का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग- अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम के प्रमुख इनपुट/डॉट मैट्रिक्स प्रिंटर के विनिर्माण के लिए चरण	**घरेलू बीओएम के रूप में वर्गीकृत किए जाने वाले इनपुट के लिए आवश्यक मूल्यवर्धन/ स्थानीय सामग्री**
1	**2**
मुख्य पीसीबी	आयातित/घरेलू स्तर पर विनिर्मित कलपूर्जों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण, इस शर्त के अधीन है कि "मुख्य पीसीबी" की असेंबली में प्रयुक्त घरेलू स्तर पर विनिर्मित कलपूर्जों और संघटकों का मूल्य न्यूनतम 15% होगा (मुख्य पीसीबी के विनिर्माण में प्रयुक्त कलपूर्जों और संघटकों के कुल मूल्य का), बेयर पीसीबी के मूल्य को छोड़कर।
बेयर पीसीबी	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू रूप से विनिर्मित।

एसएमपीएस	आयातित/घरेलू स्तर पर विनिर्मित कलपूर्जों और संघटकों से घरेलू असेंबली और परीक्षण, इस शर्त के अध्यधीन है कि "एसएमपीएस" की असेंबली में इस्तेमाल होने वाले घरेलू स्तर पर विनिर्मित कल-पूर्जों और संघटकों का मूल्य न्यूनतम 20% होगा ("एसएमपीएस" के विनिर्माण में उपयोग किए जाने वाले कलपूर्जों और संघटकों के कुल मूल्य का)।
कैरिज मोटर्स और पेपर फीड मोटर्स	उप-असेंबली के रूप में आयात किया गया और मुख्य प्रिंटर तंत्र के साथ घरेलू रूप से परीक्षण किया गया।
फ्रंट कंट्रोल पैनल	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू असेंबली और परीक्षण।
होम पोजीशन / पेपर एंड सेंसर	आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से घरेलू असेंबली और परीक्षण।
मुख्य प्रिंटर कैबिनेट और अन्य छोटे प्लास्टिक घटक	प्रिंटर कैबिनेट और अन्य भागों के घरेलू मोल्डिंग ।
प्रिंटर मेकेनिज़्म असेंबली	घरेलू स्तर पर विनिर्मित रबर प्लेट्स, छोटे रबर भागों, शीट धातु के घटकों, प्लास्टिक गियर और टण्ड स्टील शाफ्ट के साथ अन्य प्लास्टिक भागों और ऊपर वर्णित सेंसर और मोटर्स का उपयोग करके घरेलू असेंबली।
प्रिंट हेड्स और इंटरकनेक्टिंग केबल्स	उप- असेंबली के रूप में आयात किया गया और मुख्य प्रिंटर तंत्र के साथ घरेलू रूप से परीक्षण किया गया।
(i) फाइनल असेंबली/परीक्षण और (ii) डिजाइन/विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

** यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करते हुए बेयर पीसीबी पर कल पूर्जों/ संघटकों के प्रिंटेड सर्किट बोर्ड असेंबली (पीसीबीए) अनिवार्य रूप से भारत में किया जाना चाहिए।*

4.7 स्मार्ट कार्ड

(क) परिभाषा :

इस अधिसूचना के प्रयोजन से, स्मार्ट कार्ड आमतौर पर एक क्रेडिट कार्ड के आकार का प्लास्टिक कार्ड होता है, जिसमें एक एकीकृत सर्किट (आईसी) होता है। इस आईसी में एक माइक्रोप्रोसेसर और मेमोरी होती है। स्मार्ट कार्ड संपर्क, संपर्क रहित या दोहरे इंटरफ़ेस (संपर्क और संपर्क रहित दोनों) पर आधारित हो सकते हैं। स्मार्ट कार्ड के कुछ अनुप्रयोगों में पहचान पत्र, बैंकिंग कार्ड, स्वास्थ्य कार्ड, वाहन पंजीकरण कार्ड आदि हैं।

I (ख) कांटैक्ट स्मार्ट कार्ड के लिए स्थानीय सामग्री की गणना के लिए तंत्र

कांटैक्ट स्मार्ट कार्ड का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम के प्रमुख इनपुट/कॉन्टैक्ट स्मार्ट कार्ड के विनिर्माण के लिए चरण	घरेलू बीओएम के रूप में वर्गीकृत किए जाने वाले इनपुट के लिए आवश्यक मूल्यवर्धन/स्थानीय सामग्री
1	2
प्लास्टिक कार्ड बॉडी	आयातित/घरेलू रूप से विनिर्मित, कलपुर्जों और संघटकों का उपयोग करके शीट कटिंग और पंचिंग, प्रिंटिंग, लेमिनेशन और परीक्षण सहित घरेलू विनिर्माण ।
आईसी चिप मॉड्यूल	आयातित/घरेलू रूप से विनिर्मित कच्चे माल, कलपुर्जों और संघटकों का उपयोग करते हुए, आईसी चिप मॉड्यूल* की घरेलू असेंबली, पैकेजिंग और परीक्षण।
प्लास्टिक कार्ड पर आईसी चिप मॉड्यूल की मिलिंग और एंबेडिंग	प्लास्टिक कार्ड पर आईसी चिप मॉड्यूल की मिलिंग और एंबेडिंग घरेलू स्तर पर करना।
(i) फाइनल असेंबली और टेस्टिंग (ii) डिजाइन / विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

** भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

II (ख) कॉन्टैक्टलेस स्मार्ट कार्ड के लिए स्थानीय सामग्री की गणना के लिए क्रियाविधि

कॉन्टैक्टलेस स्मार्ट कार्ड का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम में प्रमुख इनपुट/कॉन्टैक्टलेस स्मार्ट कार्ड के विनिर्माण के लिए चरण	इनपुट को घरेलू बीओएम के तौर पर वर्गीकृत किए जाने के लिए आवश्यक मूल्यवर्धन/स्थानीय सामग्री
1	2
प्लास्टिक कार्ड बॉडी	आयातित/घरेलू रूप से विनिर्मित कच्चे माल, कलपुर्जों और संघटकों का उपयोग करके शीट कटिंग और पंचिंग, प्रिंटिंग, लेमिनेशन और परीक्षण सहित घरेलू विनिर्माण ।
कार्ड इनले (एंटीना)	घरेलू स्तर पर विनिर्मित आयातित कच्चे माल, कलपुर्जों और संघटकों से घरेलू असेंबली और परीक्षण।
आईसी चिप मॉड्यूल	आयातित/घरेलू रूप से विनिर्मित कच्चे माल, कलपुर्जों और संघटकों का उपयोग करके आईसी चिप* मॉड्यूल की घरेलू असेंबली, पैकेजिंग और परीक्षण।
प्लास्टिक कार्ड पर आईसी चिप मॉड्यूल की मिलिंग और एंबेडिंग	प्लास्टिक कार्ड पर आईसी चिप मॉड्यूल की मिलिंग और एंबेडिंग घरेलू स्तर पर करना ।
(i) फाइनल असेंबली और टेस्टिंग (ii) डिजाइन / विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

** भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

4.8 एलईडी उत्पाद

(क) परिभाषा :

इस अधिसूचना के प्रयोजन से, एलईडी उत्पाद वे हैं जिनका कार्य एलईडी द्वारा उत्पादित प्रकाश का उपयोग करना और निम्नलिखित क्षेत्रों में इसका उपयोग बढ़ाना है:- (i) रोशनी, (ii) टू एलईडी टीवी सहित ऑप्टिकल डिस्प्ले, (iii) बैकलाइटिंग, (iv)) सिग्नलिंग और इंडिकेशन और (v) परिवहन।

(ख) स्थानीय सामग्री की गणना के लिए क्रियाविधि :

एलईडी उत्पादों का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें: हालाँकि, (घ) हीट सिंक या थर्मल मैनेजमेंट सॉल्यूशंस, (ङ) सेकेंडरी ऑप्टिक्स और (च) सिस्टम फिक्सेटर और फिटिंग के कुल मूल्य का भार एलईडी उत्पाद के कुल बीओएम से 20% से अधिक नहीं होगा:

बीओएम के प्रमुख इनपुट/एलईडी उत्पादों के विनिर्माण के लिए चरण	**इनपुट को घरेलू बीओएम के तौर पर वर्गीकृत किए जाने वाले के लिए आवश्यक मूल्यवर्धन**
1	2
एलईडी एमिटर	आयातित/घरेलू रूप से विनिर्मित बेयर एलईडी डाई से पैकेजिंग, इस शर्त के अधीन है कि बेयर एलईडी डाई को आयातित/घरेलू विनिर्मित आदानों का उपयोग करके घरेलू रूप से विनिर्मित किया जाएगा।
ड्राइविंग इलेक्ट्रॉनिक्स	घरेलू असेंबली* और आयातित/घरेलू स्तर पर विनिर्मित कलपूर्जों और संघटकों से परीक्षण, इस शर्त के अध्यधीन है कि "ड्राइविंग इलेक्ट्रॉनिक्स " की असेंबली में इस्तेमाल होने वाले घरेलू स्तर पर विनिर्मित कल-पूर्जों और संघटकों (बेयर पीसीबी का मूल्य छोड़कर) का मूल्य न्यूनतम 30% होगा ("ड्राइविंग इलेक्ट्रॉनिक्स" के विनिर्माण में उपयोग किए जाने वाले कलपूर्जों और संघटकों के कुल मूल्य का)।
एमसीपीसीबी सहित बेयर पीसीबी	घरेलू रूप से आयातित/घरेलू स्तर पर विनिर्मित इनपुट का उपयोग करके विनिर्मित।
हीट सिंक या थर्मल प्रबंधन समाधान	घरेलू रूप से आयातित/घरेलू स्तर पर विनिर्मित इनपुट का उपयोग करके विनिर्मित।
माध्यमिक प्रकाशिकी	घरेलू रूप से आयातित/घरेलू स्तर पर विनिर्मित इनपुट का उपयोग करके विनिर्मित।
सिस्टम फिक्सर और फिटिंग	घरेलू तौर पर विनिर्मित।
फाइनल असेंबली/परीक्षण	समय-समय पर अधिसूचित भारतीय मानकों के अनुसार घरेलू रूप से असेंबली/परीक्षण किए गए।

4.9 <u>बायोमेट्रिक एक्सेस कंट्रोल/ऑथेंटिकेशन डिवाइस</u>

(क) परिभाषा :

इस अधिसूचना के प्रयोजन से, बायोमेट्रिक एक्सेस कंट्रोल/ऑथेंटिकेशन डिवाइस में *अन्य बातों के साथ-साथ* एक फिंगर प्रिंट सेंसर/आइरिस सेंसर, नियंत्रक मॉड्यूल और बिजली की आपूर्ति शामिल होगी। इसमें डिस्प्ले यूनिट हो सकती है या नहीं भी। बायोमेट्रिक एक्सेस कंट्रोल/ऑथेंटिकेशन डिवाइस के कुछ एप्लिकेशन फिजिकल एक्सेस कंट्रोल, टाइम और अटेंडेंस कंट्रोल आदि हैं।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र :

बायोमेट्रिक एक्सेस कंट्रोल/ऑथेंटिकेशन डिवाइस का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम के प्रमुख इनपुट बायोमेट्रिक एक्सेस कंट्रोल/ऑथेंटिकेशन डिवाइस के विनिर्माण के लिए चरण	**इनपुट को घरेलू बीओएम के तौर पर वर्गीकृत किए जाने वाले के लिए आवश्यक मूल्यवर्धन**
1	2
फिंगर प्रिंट सेंसर/आइरिस सेंसर	घरेलू स्तर पर विनिर्मित जैसा कि अधिसूचित है (पैराग्राफ 4.10 और 4.11 को देखें)।
मुख्य पीसीबी (नियंत्रक मॉड्यूल)	घरेलू असेंबली और आयातित/घरेलू स्तर पर विनिर्मित कलपुर्जों और संघटकों से परीक्षण, बेयर पीसीबी के मूल्य को छोड़कर।
बेअर पीसीबी	आयातित/घरेलू स्तर पर विविनिर्मित इनपुट से घरेलू स्तर पर विविनिर्मित ।
बिजली की आपूर्ति/बैटरी (यदि अलग हो)	आयातित/घरेलू स्तर पर विनिर्मित कलपूर्जों और संघटकों से घरेलू असेंबली और परीक्षण।
डिस्प्ले यूनिट	आयातित/घरेलू स्तर पर विनिर्मित कलपूर्जों और संघटकों से घरेलू असेंबली और परीक्षण।
वैकल्पिक विशेषताएं जैसे कि कैमरा, कीबोर्ड, आरएफआईडी, स्मार्ट कार्ड रीडर, जीपीआरएस मॉड्यूल, वाई-फाई, ब्लू टूथ आदि।	आयातित/घरेलू स्तर पर विनिर्मित कलपूर्जों और संघटकों से घरेलू असेंबली और परीक्षण।
प्लास्टिक हाउसिंग	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्मित।
युएसबी केबल्स	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्मित।
(i) फाइनल असेंबली और परीक्षण (ii) डिजाइन/विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

4.10 बायोमेट्रिक फिंगर प्रिंट सेंसर

(क) परिभाषा :

इस अधिसूचना के उद्देश्य से, बॉयोमीट्रिक फिंगर प्रिंट सेंसर में एक नियंत्रक मॉड्यूल, सीएमओएस सेंसर और ऑप्टिक्स शामिल हैं। बायोमेट्रिक फिंगर प्रिंट सेंसर के अनुप्रयोग व्यक्तिगत पहचान और सत्यापन आदि हैं।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र:

बायोमेट्रिक फिंगर प्रिंट सेंसर की घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम के प्रमुख मुख्य इनपुट/बॉयोमीट्रिक फिंगर प्रिंट सेंसर के निर्माण के लिए चरण	इनपुट को घरेलू बीओएम के तौर पर वर्गीकृत किए जाने वाले के लिए आवश्यक मूल्यवर्धन
1	2
मुख्य पीसीबी (नियंत्रक मॉड्यूल)	बेअर पीसीबी के मूल्य को छोड़कर आयातित / घरेलू स्तर पर विनिर्मित कलपूर्जों और संघटकों का उपयोग करके घरेलू असेंबली और परीक्षण ।
ऑप्टिक्स	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्मित ।
सीएमओएस सेंसर	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से उपयोग करके सीएमओएस सेंसर की घरेलू असेंबली, पैकेजिंग और परीक्षण*।
बेअर पीसीबी	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्माण।
प्लास्टिक हाउसिंग	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्माण।
यूएसबी केबल्स	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्माण।
(i) फाइनल असेंबली और टेस्टिंग (ii) डिजाइन/विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

** भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

4.11 बायोमेट्रिक आइरिस सेंसर

(क) परिभाषा :

इस अधिसूचना के उद्देश्य से, बॉयोमीट्रिक आइरिस सेंसर में एक नियंत्रक मॉड्यूल, सीएमओएस सेंसर और ऑप्टिक्स शामिल हैं। बायोमेट्रिक आइरिस सेंसर के आवेदन व्यक्तिगत पहचान और सत्यापन आदि हैं।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र:

बायोमेट्रिक आइरिस सेंसर का का घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम के प्रमुख इनपुट/बॉयोमीट्रिक फिंगर प्रिंट सेंसर के विनिर्माण के लिए चरण	इनपुट को घरेलू बीओएम के तौर पर वर्गीकृत किए जाने वाले के लिए आवश्यक मूल्यवर्धन
1	2
मुख्य पीसीबी (नियंत्रक मॉड्यूल)	बेअर पीसीबी के मूल्य को छोड़कर आयातित/ घरेलू स्तर पर विनिर्मित भागों और संघटकों का उपयोग करके घरेलू असेंबली और परीक्षण ।
ऑप्टिक्स	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्मित ।

सीएमओएस सेंसर	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से उपयोग करके सीएमओएस सेंसर की घरेलू असेंबली, पैकेजिंग और परीक्षण *
बेअर पीसीबी	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्माण।
प्लास्टिक हाउसिंग	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्माण ।
यूएसबी केबल्स	आयातित/घरेलू स्तर पर विनिर्मित इनपुट से घरेलू स्तर पर विनिर्माण ।
(i) फाइनल असेंबली और परीक्षण (ii) डिजाइन/विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

* *भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी ।*

4.12 सर्वर

(क) परिभाषा :

इस अधिसूचना के प्रयोजन से, एक सर्वर में आवश्यक रूप से एक मदर बोर्ड, सीपीयू, मेमोरी (रैम), हार्ड डिस्क ड्राइव (एचडीडी)/सॉलिड स्टेट स्टोरेज ड्राइव (एसएसएसडी), पावर सप्लाई यूनिट (एसएमपीएस), चेसिस, कनेक्टिंग केबल्स और फर्मवेयर और ओएस शामिल होंगे।

(ख) स्थानीय सामग्री की गणना के लिए तंत्र :

सर्वर की घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करें:

बीओएम के प्रमुख इनपुट/सर्वर के विनिर्माण के लिए चरण	**इनपुट को घरेलू बीओएम के तौर पर वर्गीकृत किए जाने वाले के लिए आवश्यक मूल्यवर्धन**
1	**2**
सर्वर बोर्ड/मदर बोर्ड और सीपीयू(एस)/जीपीयू (एस)	सेमीकंडक्टर्स** और सीपीयू (एस)/जीपीयू (एस)को शामिल करके** और बेअर पीसीबी के मूल्य को छोड़कर आयातित/घरेलू स्तर पर विनिर्मित भागों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण
मेमोरी मॉड्यूल	पीसीबी असेंबली और बेअर पीसीबी के मूल्य को छोड़कर, आयातित/घरेलू स्तर पर विनिर्मित बेअर पीसीबी आयातित/घरेलू स्तर पर विनिर्मित मेमोरी चिप्स** और कलपुर्जों/संघटको से घरेलू परीक्षण।
हार्ड डिस्क ड्राइव/सॉलिड स्टेट स्टोरेज ड्राइव (एसएसडी)	आयातित/घरेलू स्तर पर विनिर्मित भागों और संघटकों से घरेलू असेंबली और परीक्षण।
कैबिनेट + एसएमपीएस	आयातित/घरेलू स्तर पर विनिर्मित भागों और संघटकों से घरेलू रूप से घरेलू स्तर पर विनिर्मित कैबिनेट और एसएमपीएस का घरेलू असेंबली और परीक्षण, इस शर्त के

	अध्यधीन कि "एसएमपीएस" की असेंबली में उपयोग किए गए घरेलू स्तर पर विनिर्मित भागों और संघटकों का मूल्य न्यूनतम 25% होगा ("एसएमपीएस " के विनिर्माण में उपयोग किए जाने वाले भागों और संघटकों का कुल मूल्य)
बेअर पीसीबी	आयातित/घरेलू विनिर्मित इनपुट से घरेलू स्तर पर विनिर्मित
सहायक उपकरण (पावर केबल, कनेक्टर्स,आदि)	आयातित/घरेलू स्तर पर विनिर्मित भागों और संघटकों से घरेलू असेंबली और परीक्षण।
(i) फाइनल असेंबली और टेस्टिंग (ii) डिजाइन और विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

** यह आवश्यक है कि सीपीयू (एस) /जीपीयू (एस) /प्रोसेसर (एस) के प्रिंटेड सर्किट बोर्ड असेंबली (पीसीबीए), एसएमटी प्रक्रिया का उपयोग कर बेअर पीसीबी पर भागों/संघटकों को अनिवार्य रूप से भारत में किया जाना चाहिए।*

*** भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

4.13 सेल्युलर मोबाइल फ़ोन

(क) परिभाषा :

इस अधिसूचना के प्रयोजन के लिए, एक सेलुलर मोबाइल फोन (फीचर फोन या स्मार्ट फोन) में मुख्य रूप से मुद्रित सर्किट बोर्ड (पीसीबी), बैटरी पैक, डिस्प्ले यूनिट, कुंजी पैड/टच पैनल, चार्जर/एडाप्टर, माइक्रोफोन और रिसीवर वाइब्रेटर मोटर/रिंगर और मैकेनिक्स शामिल होंगे। ,

(ख) स्थानीय सामग्री की गणना के लिए तंत्र :

सेल्युलर मोबाइल फ़ोनों के घरेलू बीओएम में निम्न तालिका के कॉलम1 में विनिर्दिष्ट मुख्य इनपुट की लागत का योग होगा, बशर्ते इनपुट व्यक्तिगत रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्य अतिरिक्त आवश्यकता को पूरा करते हैं:

बीओएम के प्रमुख इनपुट/सेलुलर मोबाइल फोन के निर्माण के लिए चरण	**इनपुट को घरेलू बीओएम के तौर पर वर्गीकृत किए जाने वाले के लिए आवश्यक मूल्यवर्धन /स्थानीय सामग्री**
1	2
मुख्य पीसीबी*	प्रोसेसर/सेमीकंडक्टर** बीओएम(यानी मेन पीसीबी पर सेमीकंडक्टर चिप्स और मॉड्यूल),को शामिल करके और बेअर पीसीबी के मूल्य को छोड़कर ,आयातित / घरेलू स्तर पर विनिर्मित भागों और संघटकों से घरेलू असेंबलीऔर परीक्षण।
बेयर पीसीबी	घरेलू रूप से आयात सामग्री का विनिर्माण/घरेलू स्तर पर विनिर्मित इनपुट।
बैटरी पैक	आयात सामग्री की घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
चार्जर/एडाप्टर	आयात सामग्री से घरेलू असेंबली और परीक्षण/ घरेलू स्तर पर विनिर्मित इनपुट।
वायर्ड हेडसेट	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
यांत्रिकी***	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
डाई कट कलपुर्ज़े***	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
माइक्रोफोन और रिसीवर	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
की-पैड	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।

यूएसबी केबल	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
कैमरा मॉड्यूल	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
कनेक्टर्स	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
डिसप्ले यूनिट	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
टच पैनल/कवर ग्लास असेंबली	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
वाइब्रेटर मोटर/रिंगर	आयात सामग्री से घरेलू असेंबली और परीक्षण/घरेलू स्तर पर विनिर्मित इनपुट।
(i) अंतिम असेंबली और परीक्षण (ii) डिजाइन और विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

**यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करते हुए बेयर पीसीबी पर संसाधक/संघटक की प्रिंटेड सर्किट बोर्ड असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

*** भारत में सेमीकंडक्टर फैब प्रचालनरत होने पर इसकी समीक्षा की जाएगी।*

**** अनुबंध का संदर्भ लें।*

5. अधिसूचना तुरंत प्रभाव से लागू है। संशोधित अधिसूचना जारी होने तक यह अधिसूचना मान्य रहेगी।

6. सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश 2017 के तहत कोई भी इलेक्ट्रॉनिक उत्पाद अधिसूचना का भूतलक्षी प्रभाव नहीं होगा।

7. उपरोक्त इलेक्ट्रॉनिक उत्पादों की क्रय के लिए दिनांक 04.06.2020 के सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश 2017 में उल्लिखित प्रावधानों के अनुसार क्रय वरीयता प्रदान की जाएगी।

8. यदि भारत सरकार द्वारा परियोजना या योजना पूरी तरह से या आंशिक रूप से वित्त पोषित होती है, तो राज्यों और स्थानीय निकायों द्वारा बनाए गए इलेक्ट्रॉनिक उत्पादों की क्रय के लिए अधिसूचना सभी केंद्रीय योजनाओं (सीएस)/केंद्रीय क्षेत्र की योजनाओं (सीएसएस) पर भी लागू होगी।

9. स्थानीय सामग्री/घरेलू मूल्यवर्धन की गणना करने हेतु प्रक्रिया

9.1 घरेलू विनिर्माताओं (डीओएम-बीओएम) से प्राप्त सामग्री के बिल की गणना उपलब्ध आंकड़ों के आधार पर निम्नलिखित में से किसी एक के आधार पर की जा सकती है। इनमें से प्रत्येक गणना के सुसंगत परिणाम प्राप्त होने चाहिए।

क. सभी इनपुटों की लागतों का योग जो उस उत्पाद को बनाने के लिए आवश्यक है (इनपुटों की क्रय पर लगाए गए शुल्क और करों सहित, सिवाय उनको छोड़कर जिनके लिए क्रेडिट/सेट-ऑफ का लाभ लिया जा सकता है) और जिनका आयात प्रत्यक्ष रूप से या घरेलू व्यापारी या माध्यस्थ के माध्यम से नहीं किया गया है।

ख. उत्पाद के फैक्ट्री बाह्य मूल्य से उत्पाद के उत्पादन में इनपुटों के रूप में उपयोग की गई सामग्री के आयातित बिल (प्रत्यक्ष या अप्रत्यक्ष रूप से) को घटाकर (इनपुटों की क्रय पर लगाए गए शुल्क और करों सहित, सिवाय उनको छोड़कर जिनके लिए क्रेडिट/सेट-ऑफ का लाभ लिया जा सकता है) और उसमें से वारंटी लागत को घटाकर।

ग. बाजार मूल्य से उत्पादन पश्चात मालभाडा, बीमा और अन्य हथालन लागतों को घटाकर, उसमें से कर पश्चात लाभ, वारंटी लागत को घटाया जाता है, उसमें से उत्पादों के उत्पादन (इनपुटों की क्रय पर लगाए गए शुल्क और करों सहित, सिवाय उनको छोड़कर जिनके लिए क्रेडिट/सेट-ऑफ लिया जा सकता है) में इनपुटों के रूप में उपयोग की गई सामग्री के आयातित बिल के योग को घटाया जाता है और बिक्री और विपणन खर्च को घटाया जाता है।

9.2 सामग्री के कुल बिल (कुल-बीओएम) की गणना उपलब्ध आंकड़ों पर आधारित निम्नलिखित में से किसी एक के आधार पर की जा सकती है। इनमें से प्रत्येक गणना के सुसंगत परिणाम प्राप्त होने चाहिए।

क. सभी इनपुटों की लागतों का योग जो उस उत्पाद को बनाने के लिए आवश्यक है (इनपुटों की क्रय पर लगाए गए शुल्क और करों सहित, सिवाय उनको छोड़कर जिनके लिए क्रेडिट/सेट-ऑफ का लाभ लिया जा सकता है)।

ख. उत्पाद के फैक्टरी बाह्य मूल्य से कर पश्चात लाभ और वारंटी लागतों को घटाया जाता है।

ग. बाजार मूल्य से उत्पादन पश्चात मालभाडा, बीमा और अन्य हथालन लागतों को घटाकर उसमें से कर पश्चात लाभ, वारंटी लागत, बिक्री और विपणन खर्च घटाया जाता है।

9.3 निम्नलिखित फॉर्मूला के अनुसार प्रस्तुत जानकारी के आधार पर घरेलू मूल्य-वृद्धि के प्रतिशत की गणना की जा सकती है:

स्थानीय सामग्री/घरेलू मूल्य-वृद्धि का प्रतिशत $= \frac{\text{डीओएम} - \text{बीओएम}}{\text{कुल} - \text{बीओएम}} \times 100$

यह सिफ़ारिश की जाती है कि आकलन करने वाली प्रत्येक एजेंसी को उपरोक्त सूत्रों में से कम से कम दो का उपयोग करके स्थानीय सामग्री/घरेलू मूल्य-वृद्धि की गणना करनी चाहिए ताकि इस संबंध में आकलन को मान्य किया जा सके और यह सुनिश्चित किया जा सके कि दावाकृत घरेलू मूल्य सुसंगत है।

10. स्थानीय सामग्री/घरेलू मूल्यवृद्धि का सत्यापन

क. निविदा, बोली प्रक्रिया या सिफ़ारिश के समय स्थानीय आपूर्तिकर्ता स्व-प्रमाणन प्रदान करेगा कि प्रसतावित की गई वस्तुएं न्यूनतम स्थानीय सामग्री की आवश्यकताओं को पूरा करती हैं और उस स्थान (स्थानों) का विवरण देगा, जिस पर स्थानीय मूल्य वृद्धि की गई है।

ख.10 करोड़ रुपए से अधिक मूल्य की क्रय के मामलों में, स्थानीय आपूर्तिकर्ता स्थानीय सामग्री का प्रतिशत देते हुए कंपनी के सांविधिक लेखा परीक्षक (कंपनियों के मामले में) या लागत लेखापरीक्षक या किसी पेशेवर लागत लेखाकार या पेशेवर चार्टेड लेखाकार (कंपनियों के अलावा अन्य आपूर्तिकर्ताओं के संबंध में) से एक प्रमाणपत्र उपलब्ध कराएगा।

ग. यदि किसी इलेक्ट्रॉनिक उत्पाद के संबंध में स्थानीय सामग्री/घरेलू मूल्य वृद्धि के संबंध में बोलीकर्ता के दावे के खिलाफ क्रय एजेंसी या संबंधित मंत्रालय/विभाग द्वारा शिकायत प्राप्त की जाती है, तो उसे एसटीक्यूसी के पास भेजा जाएगा।

घ. एसटीक्यूसी को संदर्भित कोई भी शिकायत 4 सप्ताह के भीतर निपटा दी जाएगी। बोलीकर्ता को एसटीक्यूसी हेतु किसी इलेक्ट्रॉनिक उत्पाद में दावा की गई घरेलू मूल्य वृद्धि के समर्थन में आवश्यक दस्तावेज प्रस्तुत करने की आवश्यकता होगी। यदि बोलीकर्ता के द्वारा कोई जानकारी प्रस्तुत नहीं की जाती है, तो ऐसी प्रयोगशालाएं दावे की वास्तविकता को सिद्ध करने के लिए अगली आवश्यक कार्रवाई कर सकती हैं ।

ड़. 2 लाख रुपए या खरीदे जा रहे घरेलू रूप से विनिर्मित इलेक्ट्रॉनिक उत्पादों की कीमत का 1% शिकायत शुल्क (5 लाख रुपए की अधिकतम सीमा के अध्यधीन), जो भी अधिक हो, का डिमांड ड्राफ्ट द्वारा भुगतान किया जाएगा, जो एसटीक्यूसी में जमा जमा होगा। यदि शिकायत गलत पाई जाती है, तो शिकायत शुल्क जब्त कर लिया जाएगा। यदि, शिकायत को बरकरार रखा जाता है और काफी हद तक सही पाया जाता है तो शिकायतकर्ता को जमा शुल्क बिना किसी ब्याज के वापस कर दिया जाएगा।

च. सामान्य वित्तीय नियमों के विनियम 175(1) (i) (h) के तहत अखंडता संहिता के उल्लंघन में झूठी घोषणाएं की जाएगी तो इसके लिए एक बोलीकर्ता या उसके उत्तराधिकारियों पर सामान्य वित्तीय नियमों के विनियम 151(iii) के अनुसार दो साल तक के लिए रोक लगा दी जाएगी, इसके साथ-साथ कानून के तहत स्वीकार्य होने वाली अन्य कार्रवाई भी की जा सकती है।

11. इलेक्ट्रॉनिक उत्पाद अधिसूचना के कार्यान्वयन की निगरानी के लिए नोडल मंत्रालय एमईआईटीवाई होगा।

12. यदि प्रश्न उठता है कि क्या खरीदी जा रही कोई वस्तु एक इलेक्ट्रॉनिक उत्पाद है, जिसे सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश 2017 के तहत शामिल किया जाना है, तो इस मामले को स्पष्टीकरण के लिए इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय को संदर्भित किया जाएगा।

सौरभ गौड़, संयुक्त सचिव

अनुबंध

यान्त्रिकी और डाई कट कलपुर्ज़े की सूची

क. यान्त्रिकी (मैकेनिक्स)

1. बैटरी कवर (एचएस 39209999)
2. फ्रंट कवर (एचएस 39209999)
3. फ्रंट कवर (जिंक कास्टिंग के साथ) (एचएस 39209999)
4. मिडल कवर (एचएस 39209999)
5. जीएसएम एंटीना/किसी भी प्रौद्योगिकी का एंटीना (एचएस 39209999)
6. साइड की (एचएस 85389000)
7. मेन लेंस (एचएस 39209999)
8. कैमरा लेंस (एचएस 39209999)
9. पेंच (एचएस 73181500)
10. माइक्रो रबर केस (एचएस 40169990)
11. सेंसर रबड़ केस/सीलबंद गैसकेटों सहित सीलबंद गैसकेट/रबड़ केस यानि एसबीआर, ईपीडीएम, सीआर, सीएस, सिलिकॉन और सभी अन्य विशेष रबड़ या संयोजन/रबड़ के संयोजन (एचएस 40169990)

    11.1 पीयू केस /सीलबंद गैसकेट (एचएस 39269091) - सीलबंदी गैसकेट/केस की तरह पॉलीयुरेथेन फोम की अन्य सामग्री।

    11.2 सीलबंदी गास्केट/पीई, पीपी,ईपीएस, पीसीसे केस और सभी अन्य विशेष बहुलक या संयोजन/बहुलकों का संयोजन(एच एस 39269099)

12. सिम सॉकेट/अन्य मैकेनिकल (यान्त्रिकी) मदें (धातु) (एचएस 73269099)
13. सिम सॉकेट/अन्य मैकेनिकल (यान्त्रिकी) मदें (प्लास्टिक) (एचएस 39269099)
14. बैक कवर (एचएस 39209999)

ख. डाई कट कलपुर्ज़े

1.सुचालक कपड़ा (एचएस 39269099)

2. ऊष्मा उत्सर्जक स्टीकर बैटरी कवर (एचएस 39199090)

3. स्टीकर-बैटरी स्लॉट (एचएस 39199090)

4. मैन लेंस के लिए प्रोटेक्टिव फिल्म (एचएस 39199090)

5. एलसीडी एफपीसी के लिए माइलर (एचएस 39199090)

6. एलसीडी सुचालक फोम (एचएस 39269099)

7. फिल्म-फ्रंट फ्लैश (एचएस 39199090)

8. एलसीडी फोम (एचएस 39269099)

9. बीटी फोम (एचएस 39269099)

MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

(IPHW DIVISION)

NOTIFICATION

New Delhi, the 7th September, 2020

Subject : Public Procurement (Preference to Make in India) Order 2017-Notifying Electronic Products in furtherance of the Order

Reference : Department for Promotion of Industry and Internal Trade (DPIIT) Order No. P-45021/2/2017-B.E.-II dated 15.06.2017, as amended by Orders dated 28.05.2018, 29.05.2019 and 04.06.2020

F. No. W-43/4/2019-IPHW-MeitY.—The Government has issued Public Procurement (Preference to Make in India) Order 2017 vide the Department for Promotion of Industry and Internal Trade (DPIIT) Order No. P-45021/2/2017-B.E.-II dated 15.06.2017, as amended by Orders dated 28.05.2018, 29.05.2019 and 04.06.2020, to encourage 'Make in India' and promote manufacturing and production of goods and services in India with a view to enhancing income and employment.

2. In furtherance of the Public Procurement (Preference to Make in India) Order 2017 [PPP-MII Order 2017] notified vide reference cited above, and in supersession of the Electronic Product Notification Nos.33(1)/2017-IPHW dated 14.09.2017 and 33(5)/2017-IPHW dated 01.08.2018 issued thereunder, the Ministry of Electronics and Information Technology (MeitY) hereby notifies that preference shall be provided by all procuring entities to locally manufactured Electronic Products as per the aforesaid Order, as amended from time to time.

3. For the purpose of this notification:

a. The definition of Class-I local supplier, Class-II local supplier and Non-local supplier shall be as per paragraph 2 of the DPIIT PPP-MII Order 2017 No.P-45021/2/2017-PP(BE-II) dated 04.06.2020, as amended from time to time. The mechanism for calculation of local content has been prescribed for each notified Electronic Product in this notification, in accordance with paragraph 5 of the DPIIT PPP-MII Order 2017 No.P-45021/2/2017-PP(BE-II) dated 04.06.2020, as amended from time to time.

b. Paragraph 3A of the DPIIT PPP-MII Order 2017 No.P-45021/2/2017-PP(BE-II) dated 04.06.2020, as amended from time to time, shall be referred for percentage of procurement for which preference to domestically manufactured Electronic Products is to be provided (in value terms).

4. Following Electronic Products are notified under the Public Procurement (Preference to Make in India) Order 2017:

4.1 <u>Desktop Personal Computers (PCs)</u>

(A) Definition:

For the purpose of this Notification, a Desktop PC shall necessarily consist of a CPU, Memory, Hard disk drive, Keyboard, Mouse and a separate or integrated display unit and should be able to operate independently.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of Desktop PC would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/stages for manufacture of Desktop PC	**Value addition/local content required for the input to be classified as domestic BOM**
1	**2**
Main Board / Motherboard and CPU / GPU	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components, including the value of Semiconductors** and CPU/ GPU** and excluding

	the value of bare PCB. However, the weightage of total value of CPU shall not exceed 30% of the total BOM of the Desktop PC.
Memory Module	Domestic PCB Assembly* and testing from imported / domestically manufactured memory chips** and parts / components on imported/ domestically manufactured bare PCB, excluding the value of bare PCB.
Hard Disk Drive / Solid State Drive (SSD)	Domestic PCB Assembly* and final product assembly and testing from imported / domestically manufactured parts and components, excluding the value of bare PCB.
LCD / LED Monitor	Domestic assembly and testing of LCD / LED Monitor from parts consisting of imported/domestically assembled LCD / LED Panel and Domestic PCB Assembly*, Domestic plastic moulding and Domestic stamping of metal parts, excluding the value of bare PCB.
DVD Drive	Domestic assembly and testing from imported / domestically manufactured parts and components.
Cabinet + SMPS	Domestically manufactured Cabinet and Domestic PCB Assembly* and the final assembly and testing of SMPS from imported/ domestically manufactured parts and components, subject to the condition that the domestically manufactured parts and components used in the assembly of "SMPS" will be minimum 20% (of the total value of parts and components used in the manufacture of "SMPS").
Keyboard/Mouse	Domestic PCB Assembly* and the final assembly and testing from imported / domestically manufactured parts and components.
Bare PCB	Domestically manufactured from imported / domestically manufactured inputs.
(i) Final Assembly / Testing and (ii) Design / Development	(i) Domestically assembled/tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the CPU(s)/ GPU/ processor(s) parts/ components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

4.2 Thin Clients

(A) Definition:

For the purpose of this Notification, a Thin Client (TC) shall necessarily consist of a CPU, Memory, DOM, Keyboard, Mouse and a separate or integrated display unit and should be able to operate independently.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of a Thin Client would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM / stages for manufacture of Thin Client	Value addition / local content required for the input to be classified as domestic BOM
1	2
Main Board / Motherboard and CPU	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components, including the value of Semiconductors** and CPU** and excluding the value of bare PCB. However, the weightage of total value of CPU shall not exceed 30% of the total BOM of the Thin Client.
Memory Module/ DOM	Domestic PCB Assembly* and testing from imported/ domestically manufactured memory chips** and parts/ components on imported/ domestically manufactured bare PCB, excluding the value of bare PCB.
SMPS/ Power Adapter	Domestically PCB Assembly* and the final assembly and testing of SMPS/ Power Adapter from imported/ domestically manufactured parts and components, subject to the condition that the domestically manufactured parts and components used in the assembly of "SMPS" will be minimum 20% (of the total value of parts and components used in the manufacture of "SMPS").
LCD/ LED Monitor	Domestic assembly and testing of LCD/ LED Monitor from parts consisting of imported/ domestically assembled LCD/ LED Panel and Domestic PCB Assembly*, Domestic plastic moulding and Domestic stamping of metal parts, excluding the value of bare PCB.
Cabinet	Domestically manufactured.
Keyboard/Mouse	Domestic PCB Assembly* and the final assembly and testing from imported / domestically manufactured parts and components.
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
(i) Final Assembly/ Testing and (ii) Design/ Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the CPU(s)/ processor(s)/ parts/ components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

4.3 <u>Laptop Personal Computers (PCs)</u>

(A) Definition:

For the purpose of this Notification, a Laptop PC (commonly known in the market as Laptop/ Notebook/ Netbook/ Ultrabook, etc.) shall necessarily consist of a CPU, Memory, Hard disk drive, Keyboard, Touchpad and/ or Trackpoint, an integrated display unit, integrated battery and should be able to operate independently.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of a Laptop PC would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/ stages for manufacture of Laptop PC	**Value addition/ local content required for the input to be classified as domestic BOM**
1	**2**
Main Board/ Motherboard and CPU/ GPU	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components, including the value of Semiconductors** and CPU/ GPU** and excluding the value of bare PCB. However, the weightage of total value of CPU shall not exceed 35% of the total BOM of the Laptop PC.
Power Module	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components.
Memory Module	Domestic PCB Assembly* and testing from imported/ domestically manufactured memory chips** and parts/ components on imported/ domestically manufactured bare PCB, excluding the value of bare PCB.
Hard Disk Drive/ Solid State Drive (SSD)	Domestic PCB Assembly* and final product assembly and testing from imported/ domestically manufactured parts and components, excluding the value of bare PCB.
Display Panel (LCD, LED, etc.)	Domestic assembly into the Back Cover and Bezel and final testing from imported/ domestically assembled Display Panel, LED Back light, Antennae, excluding the value of the Back Cover and Bezel.
DVD Drive	Domestic assembly and testing from imported / domestically manufactured parts and components.
Power Adapter	Domestic assembly with domestic PCB Assembly* and final testing from imported/ domestically manufactured parts and components, subject to the condition that the value of domestically manufactured parts and components used in the assembly of "Power Adapter" will be minimum 40% (of the total value of parts and components used in the manufacture of "Power Adapter").
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
Keyboard/ Touchpad and/ or Trackpoint	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Battery	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Cabinet/ Chassis of Laptop including bottom cover, back cover, Keyboard face bezel and Display face bezel and Hinges	Domestically manufactured through injection moulding/ stamping using imported/ domestic inputs.
(i) Final Assembly/ Testing and (ii) Design/ Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the CPU(s)/ GPU/ processor(s)/ parts/ components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

4.4 Computer Monitors

(A) Definition:

For the purpose of this Notification, a Computer Monitor (commonly known in the market as Monitor) shall necessarily consist of a LCD/ LED Display assembly, Logic card, Power Supply Adaptor and Cabinet with Display Ports for connecting it to a Desktop PC or Laptop PC or Tablet PC.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of a Computer Monitor would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/ stages for manufacture of Computer Monitor	**Value addition/ local content required for the input to be classified as domestic BOM**
1	**2**
Logic Cards/ Micro controller	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components including value of Semiconductors** and excluding the value of bare PCB.
LCD/ LED Panel	Domestic integration and testing of imported/ domestically assembled LCD/ LED Panel*** into the Monitor Cabinet/ Chassis, excluding the value of the Cabinet/ Chassis. However, the weightage of total value of LCD/ LED Panel shall not exceed 35% of the total BOM of the Computer Monitor.
Cables	Domestically manufactured and tested from imported/ domestically manufactured inputs.
Power Adapter	Domestic assembly with domestic PCB Assembly* and final testing from imported/ domestically manufactured parts and components, subject to the condition that the value of domestically manufactured parts and components used in the assembly of "Power Adapter" will be minimum 40% (of the total value of parts and components used in the manufacture of "Power Adapter").
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
Cabinet/ Chassis	The Monitor Bezel, Back cover, Stand, Base and Chassis wherein the plastic mouldings and the stamping of metal parts is done domestically using imported/ domestic inputs.
(i) Final Assembly/ Testing and (ii) Design/ Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the CPU(s)/ GPU processor(s)/ parts/ components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

**** This shall be reviewed when LCD/LED Panel manufacturing of Computer Monitors (of each size and type) in India is operational.*

4.5 Tablet Personal Computers (PCs)

A. **Definition**:

For the purpose of this Notification, a Tablet PC shall necessarily consist of an Integrated Motherboard with on board CPU/Processor, Memory and Power Module; Display Panel (Touch Panel + LCD/LED Module) and integrated battery and should be able to operate independently.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of Tablet PC would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/ stages for manufacture of Tablet PC	**Value addition/ local content required for the input to be classified as domestic BOM**
1	**2**
Integrated Motherboard with on board CPU/ GPU/ Processor	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components, including value of Semiconductors** and Processor/CPU/ GPU** and excluding the value of bare PCB. However, the weightage of total value of CPU shall not exceed 35% of the total BOM of the Tablet PC.
Memory and Power Module	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components, excluding the value of bare PCB.
Display Panel (Touch Panel + LCD/ LED Module)	Domestic assembly and testing from imported/ domestically manufactured Touch Panel, LCD/ LED Module or combination, subject to the condition that backlight assembly and testing of Display Panel shall be done domestically.
Power Adapter	Domestic assembly and testing from imported/ domestically manufactured parts and components, subject to the condition that the value of domestically manufactured parts and components used in the assembly of "Power Adapter" will be minimum 40% (of the total value of parts and components used in the manufacture of "Power Adapter").
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
Casing	Domestically manufactured Casing.
Battery	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Accessories (Camera, Speaker, WiFi Antenna, etc.)	Domestic assembly and testing from imported/ domestically manufactured parts and components.
(i) Final Assembly/ Testing and (ii) Design/ Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the CPU/ GPU/ processor/ parts/ components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational*

4.6 Dot Matrix Printers

(A) Definition:

A Dot Matrix Printer is a type of impact printer that forms dot on paper by a metal pin of diameter 0.2 mm to 0.3 mm which is driven by electromagnet based on solenoid principle and required character matrix is produced by horizontal and vertical resolution of dot matrix printhead. Dot Matrix Printer can create carbon copies and carbonless copies based on mechanical pressure of pin.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of a Dot Matrix Printer would be the sum of the cost of main inputs specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/ stages for manufacture of Dot Matrix Printer	**Value addition/ local content required for the input to be classified as domestic BOM**
1	**2**
Main PCB	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components, subject to the condition that value of domestically manufactured parts and components used in the assembly of "Main PCB" will be minimum 15% (of the total value of parts and components used in the manufacture of "Main PCB"), excluding the value of bare PCB.
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
SMPS	Domestic assembly and testing from imported/ domestically manufactured parts and components, subject to the condition that value of domestically manufactured parts and components used in the assembly of "SMPS" will be minimum 20% (of the total value of parts and components used in the manufacture of "SMPS").
Carriage Motors and Paper Feed Motors	Imported as sub-assembly and tested domestically along with main Printer Mechanism.
Front Control Panel	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Home Position/Paper End Sensors	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Main Printer Cabinet and other small plastic components	Domestic moulding of Printer Cabinet and other parts.
Printer Mechanism Assembly	Domestic assembly using domestically manufactured Rubber Platens, small rubber parts, sheet metal components, plastic gears and other plastic parts with turned steel shafts and above mentioned sensors and Motors.
Print Heads and Interconnecting Cables	Imported as sub-assembly and tested domestically along with main Printer Mechanism.
(i) Final Assembly/ Testing and (ii) Design/ Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the parts/ components on the bare PCB using the SMT process should mandatorily be done in India.*

4.7 Smart Cards

(A) Definition:

For the purpose of this Notification, Smart Card is usually a Credit Card sized plastic Card with an Integrated Circuit (IC) contained inside. The IC contains a microprocessor and memory. Smart Cards can be contact, contactless or dual interface (both contact and contactless). Some of the applications of Smart Card are Identity Card, Banking Card, Health Card, Vehicle Registration Card etc.

I(B) Mechanism for calculation of local content for Contact Smart Cards

The domestic Bill of Material (BOM) of a Contact Smart Card would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/ stages for manufacture of Contact Smart Card	**Value addition/ local content required for the input to be classified as domestic BOM**
1	**2**
Plastic Card Body	Domestic manufacturing including sheet cutting & punching, printing, lamination and testing using imported/ domestically manufactured raw material, parts and components.
IC Chip Module	Domestic assembly, packaging and testing of IC Chip Module using imported/ domestically manufactured raw material, parts and components*.
Milling and Embedding of IC Chip Module on Plastic Card	Milling and Embedding of IC Chip Module on Plastic Card done domestically.
(i) Final Assembly and Testing (ii) Design/ Development	(i) Domestically assembled/tested and (ii) Intellectual Property (IP) resident in India for any of the above items, including fusion of domestically developed Operating System. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

** This shall be reviewed when the Semiconductor FAB in India is operational.*

II(B) Mechanism for calculation of local content for Contactless Smart Cards

The domestic Bill of Material (BOM) of Contactless Smart Card would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/stages for manufacture of Contactless Smart Card	**Value addition/local content required for the input to be classified as domestic BOM**
1	**2**
Plastic Card Body	Domestic manufacturing including sheet cutting & punching, printing, lamination and testing using imported/ domestically manufactured raw material, parts and components.
Card inlay (Antenna)	Domestic assembly and testing from imported/ domestically manufactured raw material, parts and components.
IC Chip Module	Domestic assembly, packaging and testing of IC Chip Module using imported/ domestically manufactured raw material, parts and components*.
Milling and Embedding of IC Chip Module on Plastic Card	Milling and Embedding of IC Chip Module on Plastic Card done domestically.

(i) Final Assembly and Testing (ii) Design/ Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items, including fusion of domestically developed Operating System. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

** This shall be reviewed when the Semiconductor FAB in India is operational.*

4.8 LED Products

(A) Definition:

For the purpose of this Notification, LED products are those whose function is to utilize light produced by LEDs and spanning applications in the areas of: (i) Illumination, (ii) Optical Displays including True LED TVs, (iii) Backlighting, (iv) Signalling & Indication and (v) Transportation.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of LED Products would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table. However, the weightage of total value of (d) Heat Sink or Thermal Management Solutions, (e) Secondary Optics and (f) System Fixture and Fitting shall not exceed 20% of the total BOM of the LED Product:

Main inputs in BOM/ stages for manufacture of LED Products	**Value addition/ local content required for the input to be classified as domestic BOM**
1	**2**
LED Emitter	Packaging from imported/ domestically fabricated Bare LED Die, subject to the condition that the Bare LED Die shall be domestically fabricated using imported/ domestically manufactured inputs.
Driving Electronics	Domestic assembly from imported/ domestically manufactured parts and components, subject to the condition that the value of domestically manufactured parts and components (excluding the value of bare PCB) used in the assembly of "Driving Electronics" will be minimum 30% of the total value of parts and components used in the manufacture of "Driving Electronics".
Bare PCB including MCPCB	Domestically manufactured using imported/ domestically manufactured inputs.
Heat Sink or Thermal Management Solutions	Domestically manufactured using imported/ domestically manufactured inputs.
Secondary Optics	Domestically manufactured using imported/ domestically manufactured inputs.
System Fixture and Fitting	Domestically manufactured.
Final Assembly/ Testing	Domestically assembled/ tested meeting Indian Standards as notified from time to time.

4.9 Biometric Access Control/Authentication Devices

(A) Definition:

For the purpose of this Notification, Biometric Access Control/ Authentication Device shall include *inter-alia* a Finger Print Sensor/ Iris Sensor, Controller Module and Power supply. It may or may not contain a display unit. Some of the applications of Biometric Access Control/ Authentication Device are Physical access control, Time and Attendance control etc.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of Biometric Access Control/ Authentication Device would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/ stages for manufacture of Biometric Access Control/ Authentication Device	**Value addition/ local content required for the input to be classified as domestic BOM**
1	**2**
Finger Print Sensor/ Iris Sensor	Domestically manufactured as notified (refer Paragraphs 4.10 and 4.11).
Main PCB (Controller Module)	Domestic assembly and testing from imported / domestically manufactured parts and components, excluding the value of bare PCB.
Bare PCB	Domestically manufactured from imported / domestically manufactured inputs.
Power Supply/ Battery(if separate)	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Display Unit	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Optional features such as Camera, Keyboard, RFID, Smart Card Reader, GPRS Module, Wi-Fi, Blue Tooth etc.	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Plastic Housing	Domestically manufactured from imported/ domestically manufactured inputs.
USB Cables	Domestically manufactured from imported/ domestically manufactured inputs.
(i) Final Assembly and Testing (ii) Design/ Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

4.10 <u>Biometric Finger Print Sensors</u>

(A) Definition:

For the purpose of this Notification, Biometric Finger Print Sensor consists of a Controller Module, CMOS Sensor and Optics. The applications of the Biometric Finger Print Sensor are personal identification and verification, etc.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of Biometric Finger Print Sensor would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/stages for manufacture of Biometric Finger Print Sensor	**Value addition/local required for the input to be classified as domestic BOM**
1	**2**
Main PCB (Controller Module)	Domestic assembly and testing using imported / domestically manufactured parts and components, excluding the value of bare PCB.

Optics	Domestically manufactured from imported / domestically manufactured inputs.
CMOS Sensor	Domestic assembly, packaging and testing of CMOS Sensor using imported/ domestically manufactured inputs*.
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
Plastic Housing	Domestically manufactured from imported / domestically manufactured inputs.
USB Cables	Domestically manufactured from imported / domestically manufactured inputs.
(i) Final Assembly and Testing (ii) Design/ Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

** This shall be reviewed when the Semiconductor FAB in India is operational.*

4.11 Biometric Iris Sensors

(A) Definition:

For the purpose of this Notification, Biometric Iris Sensor consists of a Controller Module, CMOS Sensor and Optics. The applications of the Biometric Iris Sensor are personal identification and verification, etc.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of Biometric Iris Sensor would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/stages for manufacture of Biometric Iris Sensor	**Value addition/local content required for the input to be classified as domestic BOM**
1	**2**
Main PCB (Controller Module)	Domestic assembly and testing using imported / domestically manufactured parts and components except value of bare PCB.
Optics	Domestically manufactured from imported/ domestically manufactured inputs.
CMOS Sensor	Domestic assembly, packaging and testing of CMOS Sensor using imported/ domestically manufactured inputs*.
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
Plastic Housing	Domestically manufactured from imported/ domestically manufactured inputs.
USB Cables	Domestically manufactured from imported/ domestically manufactured inputs.
(i) Final Assembly and Testing (ii) Design/ Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in domestic BOM.

**This shall be reviewed when the Semiconductor Fab in India is operational.*

4.12 Servers

(A) Definition:

For the purpose of this Notification, a Server shall necessarily consist of a Mother Board, CPU, Memory (RAM), Hard Disk Drive (HDD)/ Solid State Storage Drive (SSD), Power Supply Unit (SMPS), Chassis, Connecting Cables and Firmware & OS.

(B) Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of a Server would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/ stages for manufacture of Server	**Value addition /local content required for the input to be classified as domestic BOM**
1	**2**
Server Board/ Mother Board and CPU(s)/ GPU(s)	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components, including the value of Semiconductors** and CPU(s)/ GPU(s)** and excluding the value of bare PCB.
Memory Module	Domestic PCB Assembly* and testing from imported/ domestically manufactured memory chips** and parts/ components on imported/ domestically manufactured bare PCB, excluding the value of bare PCB.
Hard Disk Drive/ Solid State Storage Drive (SSD)	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Cabinet + SMPS	Domestically manufactured Cabinet and domestic assembly and testing of SMPS from imported / domestically manufactured parts and components, subject to the condition that value of domestically manufactured parts and components used in the assembly of "SMPS" will be minimum 25% (of the total value of parts and components used in the manufacture of "SMPS").
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
Accessories (Power Cables, Connectors, etc.)	Domestic assembly and testing from imported/ domestically manufactured parts and components.
(i) Final Assembly and Testing (ii) Design and Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in the domestic BOM.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the CPU(s)/ GPU(s)/ processor(s), parts/ components on the bare PCB using the SMT process should mandatorily be done in India*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

4.13 Cellular Mobile Phones

(A) Definition:

For the purpose of this Notification, a Cellular Mobile Phone (Feature Phone or Smart Phone) shall necessarily consist of a Main Printed Circuit Board (PCB), Battery Pack, Display Unit, Key Pad/ Touch Panel, Charger/ Adapter, Microphone & Receiver, Vibrator Motor/ Ringer and Mechanics.

(B) Mechanism for calculation of local content:

The domestic BOM of Cellular Mobile Phones would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Main inputs in BOM/stages for manufacture of Cellular Mobile Phone	**Value addition/ local content required for the input to be classified as domestic BOM**
1	**2**
Main PCB*	Domestic assembly and testing from imported/ domestically manufactured parts and components including Processor and Semiconductor** BOM (i.e the Semiconductor Chips and Modules on Main PCB), and excluding value of bare PCB.
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
Battery Pack	Domestic assembly and testing of imported/ domestically manufactured inputs.
Charger/ Adapter	Domestic assembly and testing from imported/ domestically manufactured inputs.
Wired Headset	Domestic assembly and testing from imported/ domestically manufactured inputs.
Mechanics***	Domestic assembly and testing from imported/ domestically manufactured inputs.
Die Cut Parts***	Domestic assembly and testing from imported/ domestically manufactured inputs.
Microphone and Receiver	Domestic assembly and testing from imported/ domestically manufactured inputs.
Key Pad	Domestic assembly and testing from imported/ domestically manufactured inputs
USB Cable	Domestic assembly and testing from imported/ domestically manufactured inputs.
Camera Module	Domestic assembly and testing from imported/ domestically manufactured inputs.
Connectors	Domestic assembly and testing from imported/ domestically manufactured inputs.
Display Unit	Domestic assembly and testing from imported/ domestically manufactured inputs.
Touch Panel/ Cover Glass Assembly	Domestic assembly and testing from imported/ domestically manufactured inputs.
Vibrator Motor/ Ringer	Domestic assembly and testing from imported/ domestically manufactured inputs.
(i) Final Assembly and Testing (ii) Design and Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in the domestic BOM.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the processor/ components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational*

****Refer Annexure*

5. The Notification comes into effect immediately. This Notification shall remain valid till the revised Notification is issued.

6. No Electronic Product Notification under the Public Procurement (Preference to Make in India) Order 2017 shall have retrospective effect.

7. Purchase Preference shall be provided as per the provisions cited in the Public Procurement (Preference to Make in India) Order 2017 dated 04.06.2020 for the procurement of aforesaid electronics products.

8. The notification would also be applicable to all Central Schemes (CS)/ Central Sector Schemes (CSS) for the procurement of electronic products made by States and local bodies, if project or scheme is fully or partially funded by Government of India.

9. Procedure for calculating local content/ domestic value addition

9.1 Bill of Material sourced from domestic manufacturers (Dom-BOM) may be calculated based on one of the followings depending on data available. Each of these calculations should provide consistent result.

a. Sum of the costs of all inputs which go into the product (including duties and taxes levied on procurement of inputs except those for which credit/ set-off can be taken) and which have not been imported directly or through a domestic trader or an intermediary.

b. Ex-Factory Price of product minus profit after tax minus sum of imported Bill of Material used (directly or indirectly) as inputs in producing the product (including duties and taxes levied on procurement of inputs except those for which credit/ set-off can be taken) minus warranty costs.

c. Market price minus post-production freight, insurance and other handling costs minus profit after tax minus warranty costs minus sum of Imported Bill of Material used as inputs in producing the product (including duties and taxes levied on procurement of inputs except those for which credit / set-off can be taken) minus sales and marketing expenses.

9.2 Total Bill of Material (Total-BOM) may be calculated based on one of the following depending on data available. Each of these calculations should provide consistent result.

a. Sum of the costs of all inputs which go into the product (including duties and taxes levied on procurement of inputs except those for which credit / set-off can be taken).

b. Ex-Factory Price of product minus profit after tax, minus warranty costs.

c. Market price minus post-production freight, insurance and other handling costs minus profit after tax, minus warranty costs minus sales and marketing expenses.

9.3 The percentage of domestic value-addition may be calculated based on information furnished as per the following formula:

$$\text{Percentage of local content/ domestic value-addition} = \frac{\text{Dom-BOM}}{\text{Total-BOM}} \times 100$$

It is recommended that each agency assessing should calculate the domestic local content/ value-addition using at least two of the above formulae so as to validate the assessments in this regard and ensure that the domestic value addition that is claimed is consistent.

10. Verification of local content/ Domestic Value Addition

a. The local supplier at the time of tender, bidding or solicitation shall provide self-certification that the item offered meets the minimum local content and shall give details of the location(s) at which the local value addition is made.

b. In cases of procurement for a value in excess of Rs. 10 crore, the local supplier shall provide a certificate from the statutory auditor or cost auditor of the company (in the case of companies) or from a practicing cost accountant or practicing chartered accountant (in respect of suppliers other than companies) giving the percentage of local content.

c. In case a complaint is received by the procuring agency or the concerned Ministry/Department against the claim of a bidder regarding local content/ domestic value addition in an electronic product, the same shall be referred to STQC.

d. Any complaint referred to STQC shall be disposed of within 4 weeks. The bidder shall be required to furnish the necessary documentation in support of the domestic value addition claimed in an electronic product to STQC. If no information is furnished by the bidder, such laboratories may take further necessary action, to establish the bonafides of the claim.

e. A complaint fee of Rs.2 Lakh or 1% of the value of the domestically manufactured electronic products being procured (subject to a maximum of Rs. 5 Lakh), whichever is higher, to be paid by Demand Draft to be deposited with STQC. In case, the complaint is found to be incorrect, the complaint fee shall be forfeited. In case, the complaint is upheld and found to be substantially correct, deposited fee of the complainant would be refunded without any interest.

f. False declarations will be in breach of the Code of Integrity under Rule 175(1)(i)(h) of the General Financial Rules for which a bidder or its successors can be debarred for up to two years as per Rule 151 (iii) of the General Financial Rules along with such other actions as may be permissible under law.

11. MeitY shall be the Nodal Ministry to monitor the implementation of the Electronic Products Notification.

12. In case of a question whether an item being procured is an electronic product to be covered under the Public Procurement (Preference to Make in India) Order 2017, the matter would be referred to the Ministry of Electronics and Information Technology for clarification.

SAURABH GAUR, Jt. Secy.

Annexure

Indicative List of Mechanics and Die Cut Parts

A. Mechanics

1. Battery Cover (HS 39209999)
2. Front Cover (HS 39209999)
3. Front Cover (With Zinc Casting) (HS 39209999)
4. Middle Cover (HS 39209999)
5. GSM Antenna/ Antenna of any technology (HS 39209999)
6. Side Key (HS 85389000)
7. Main Lens (HS 39209999)
8. Camera Lens (HS 39209999)
9. Screw (HS 73181500)
10. Mic Rubber Case (HS 40169990)
11. Sensor Rubber Case/ Sealing Gasket including sealing gaskets/ cases from Rubbers like SBR, EPDM, CR, CS, Silicone and all other individual rubbers or combination/ combinations of rubbers (HS 40 169990)
    - 11.1 PU Case/ Sealing Gasket (HS 39269091) - Other articles of Polyurethane foam like sealing gaskets/ cases.
    - 11.2 Sealing Gaskets/ Cases from PE, PP, EPS, PC and all other individual polymers or combination/ combinations of polymers (HS 39269099)
12. SIM Socket/ Other Mechanical items (Metal) (HS 73269099)
13. SIM Socket/ Other Mechanical items (Plastic) (HS 39269099)
14. Back Cover (HS 39209999)

B. Die Cut Parts

1. Conductive Cloth (HS 39269099)
2. Heat Dissipation Sticker Battery Cover (HS 39199090)
3. Sticker-Battery Slot (HS 39199090)
4. Protective Film for Main Lens (HS 39199090)
5. Mylar for LCD FPC (HS 39199090)
6. LCD Conductive Foam (HS 39269099)
7. Film-Front Flash (HS 39199090)
8. LCD Foam (HS 39269099)
9. BT Foam (HS 39269099)

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भारत का राजपत्र
The Gazette of India

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असाधारण
EXTRAORDINARY

भाग II—खण्ड 3—उप-खण्ड (ii)
PART II—Section 3—Sub-section (ii)

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PUBLISHED BY AUTHORITY

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No. 4417] NEW DELHI, THURSDAY, OCTOBER 19, 2023/ASVINA 27, 1945

इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय

(आईपीएचडब्ल्यू डिवीजन)

अधिसूचना

नई दिल्ली, 12 अक्तूबर, 2023

विषय: दिनांक 07.09.2020 के इलेक्ट्रॉनिक उत्पाद अधिसूचना फ़ा.सं.डब्ल्यू-43/4/2019-आईपीएचडब्ल्यू-एमईआईटीवाई के खंड संख्या 4.12 (सर्वर) में संशोधन संबंधी।

का.आ. 4591(अ).—दिनांक 15.06.2017 के उद्योग और आंतरिक व्यापार संवर्धन विभाग (डीपीआईआईटी) सार्वजनिक खरीद (मेक इन इंडिया को वरीयता) आदेश 2017 जिसे दिनांक 28.05.2018, 29.05.2019, 04.06.2020 और 16.09.2020 के आदेशों द्वारा यथा संशोधित किया गया, को आगे बढ़ाते हुए दिनांक 07.09.2020 की इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय (एमईआईटीवाई) की राजपत्र अधिसूचना फा.सं. डब्लयु-43/4/2049-आईपीएचडब्ल्यु-एमईआईटीवाई के द्वारा इलेक्ट्रॉनिक उत्पादों को अधिसूचित किया गया था।

2. सर्वर के मामले में, खंड 4.12 (ख): स्थानीय सामग्री की गणना के लिए तंत्र को निम्नानुसार संशोधित किया गया है और यह तत्काल प्रभाव से लागू होगा:

Page 107 of 684 : B649-095-VV-MR-6009-04-RFO : Rev. A

स्थानीय सामग्री की गणना के लिए संशोधित तंत्र:

सर्वर की घरेलू सामग्री बिल (बीओएम) निम्नलिखित तालिका के कॉलम 1 में विनिर्दिष्ट मुख्य इनपुट की लागत के योग के बराबर होगा, बशर्ते कि इनपुट अलग-अलग रूप से तालिका के कॉलम 2 में विनिर्दिष्ट मूल्य वर्धन आवश्यकता को पूरा करें। हालांकि, 'एड ऑन कंट्रोलर' के कुल मूल्य का वेटेज **सर्वर** के कुल बीओएम के **3%** से अधिक नहीं होगा।

बीओएम के प्रमुख इनपुट/ सर्वर के विनिर्माण के लिए चरण	**मूल्य वर्धन/इनपुट को घरेलू बीओएम के रूप में वर्गीकृत करने के लिए अपेक्षित स्थानीय सामग्री**
1	2
सर्वर बोर्ड/मदर बोर्ड और सीपीयू(एस)/जीपीयू(एस)	सेमीकंडक्टर्स** और सीपीयू (एस) / जीपीयू (एस)** को शामिल करके" और बेअर पीसीबी के मूल्य को छोड़कर आयातित/घरेलू स्तर पर विनिर्मित भागों और संघटकों से घरेलू पीसीबी असेंबली* और परीक्षण
मेमोरी मॉड्यूल	पीसीबी असेंबली* और बेअर पीसीबी के मूल्य को छोड़कर, आयातित/घरेलू स्तर पर विनिर्मित बेअर पीसीबी पर आयातित/घरेलू स्तर पर विनिर्मित मेमोरी चिप्स** और कलपुर्जों/संघटको से घरेलू परीक्षण।
हार्ड डिस्क ड्राइव/ सॉलिड स्टेट स्टोरेज ड्राइव (एसएसडी)	आयातित/घरेलू स्तर पर विनिर्मित भागों और संघटकों से घरेलू असेंबली और परीक्षण।
कैबिनेट + एसएमपीएस	आयातित/घरेलू स्तर पर विनिर्मित भागों और संघटकों से घरेलू रूप से घरेलू स्तर पर विनिर्मित कैबिनेट और एसएमपीएस का घरेलू असेंबली और परीक्षण, इस शर्त के अध्यधीन कि "एसएमपीएस" की असेंबली में उपयोग किए गए घरेलू स्तर पर विनिर्मित भागों और संघटकों का मूल्य न्यूनतम 25% होगा ("एसएमपीएस " के विनिर्माण में उपयोग किए जाने वाले भागों और संघटकों का कुल मूल्य)
बेयर पीसीबी	आयातित/घरेलू विनिर्मित इनपुट से घरेलू स्तर पर विनिर्मित
सहायक उपकरण (पावर केबल, कनेक्टर, आदि)	आयातित/घरेलू स्तर पर विनिर्मित भागों और संघटकों से घरेलू असेंबली और परीक्षण।
ऐड-ऑन कंट्रोलर I. आरएआईडी कंट्रोलर/एडाप्टर II. नेटवर्क इंटरफ़ेस कंट्रोलर	पीसीबी असेंबली* और बेअर पीसीबी के मूल्य को छोड़कर, आयातित/घरेलू स्तर पर विनिर्मित बेअर पीसीबी पर आयातित/घरेलू स्तर पर विनिर्मित चिप्स** और कलपुर्जों/संघटको से घरेलू परीक्षण।
(i) फाइनल असेंबली और टेस्टिंग (ii) डिजाइन और विकास	(i) घरेलू स्तर पर असेम्बल/परीक्षित और (ii) उपरोक्त आइटमों में से किसी आइटम के लिए बौद्धिक सम्पदा (आईपी) अधिकार रखने वाला व्यक्ति भारत का निवासी हो । उपरोक्त आइटमों में से किसी आइटम के लिए भारत में (आईपी) निवासी के मूल्य को घरेलू बीओएम में उसके मूल्य से घटाया जाएगा।

** यह आवश्यक है कि सीपीयू (एस)/जीपीयू (एस)/प्रोसेसर (एस) के प्रिंटेड सर्किट बोर्ड असेंबली (पीसीबीए), एसएमटी प्रक्रिया का उपयोग कर बेयर पीसीबी पर भागों/संघटकों को अनिवार्य **रूप से** भारत में किया जाना चाहिए।*

*** भारत में सेमीकंडक्टर फैब चालू होने पर इसकी समीक्षा की जाएगी।*

3. दिनांक 07.09.2020 की एमईआईटीवाई की राजपत्र अधिसूचना फ़ा.सं.डब्ल्यू-43/4/2019-आईपीएचडब्ल्यू-एमईआईटीवाई के अन्य सभी नियम और शर्तें अपरिवर्तित रहेंगी।

[फा. सं. डब्ल्यू-43/4/2019-आईपीएचडब्ल्यू-एमईआईटीवाई]

आशा नांगिया, समूह समन्वयक और वैज्ञानिक 'जी'

MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

(IPHW Division)

NOTIFICATION

New Delhi, the 12th October, 2023

Subject: Amendment to Electronic Products Notification F.No.W-43/4/2019-IPHW-MeitY dated 07.09.2020 w.r.t. Clause No. 4.12 (Servers) – reg.

S.O. 4591(E).— Electronic Products were notified vide Ministry of Electronics and Information Technology (MeitY) Gazette Notification F.No.W-43/4/2019-IPHW-MeitY dated 07.09.2020, in furtherance of the Department for Promotion of Industry and Internal Trade (DPIIT) Public Procurement (Preference to Make in India) Order 2017 dated 15.06.2017, as amended by Orders dated 28.05.2018, 29.05.2019, 04.06.2020 and 16.09.2020.

2. In case of Servers, Clause 4.12 (B): Mechanism for calculation of local content is amended as under and will be applicable with immediate effect:

Revised Mechanism for calculation of local content:

The domestic Bill of Material (BOM) of **Server** would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table. However, the weightage of total value of 'Add on Controller' shall not exceed **3%** of the total BOM of the **Server**.

Main inputs in BOM/ stages for manufacture of Server	**Value addition /local content required for the input to be classified as domestic BOM**
1	2
Server Board/ Mother Board and CPU(s)/ GPU(s)	Domestic PCB Assembly* and testing from imported/ domestically manufactured parts and components, including the value of Semiconductors** and CPU(s)/ GPU(s)** and excluding the value of bare PCB.
Memory Module	Domestic PCB Assembly* and testing from imported/ domestically manufactured memory chips** and parts/ components on imported/ domestically manufactured bare PCB, excluding the value of bare PCB.
Hard Disk Drive/ Solid State Storage Drive (SSD)	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Cabinet + SMPS	Domestically manufactured Cabinet and domestic assembly and testing of SMPS from imported / domestically manufactured parts and components, subject to the condition that value of domestically manufactured parts and components used in the assembly of "SMPS" will be minimum 25% (of the total value of parts and components used in the manufacture of "SMPS").
Bare PCB	Domestically manufactured from imported/ domestically manufactured inputs.
Accessories (Power Cables, Connectors, etc.)	Domestic assembly and testing from imported/ domestically manufactured parts and components.
Add-on Controllers i. RAID Controller/Adapter	Domestic PCB Assembly* and testing from imported/ domestically manufactured chips** and parts/ components on imported/ domestically

ii. Network Interface Controllers	manufactured bare PCB, excluding the value of bare PCB.
(i) Final Assembly and Testing (ii) Design and Development	(i) Domestically assembled/ tested and (ii) Intellectual Property (IP) resident in India for any of the above items. The value of IP resident in India for any of the above items shall be reduced from its value in the domestic BOM.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the CPU(s)/ GPU(s)/ processor(s), parts/ components on the bare PCB using the SMT process should **mandatorily** be done in India*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

3. All other terms and conditions of MeitY Gazette Notification F.No.W-43/4/2019-IPHW-MeitY dated 07.09.2020 remain unchanged.

[F. No. W-43/4/2019-IPHW-MeitY]

ASHA NANGIA, Group Coordinator & Scientist 'G'

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and Published by the Controller of Publications, Delhi-110054.

No. 13(7)/2022-EG-II
Government of India
Ministry of Electronics and Information Technology
Electronics Niketan, 6, CGO Complex, New Delhi-110003.

Dated: 17.08.2022

OFFICE MEMORANDUM

Subject: Exemption from Public Procurement (Preference to Make in India) Clause for procurement of various items – regarding

This has reference to the request made by the Unique Identification Authority of India (UIDAI), vide letter no. T-11014/07/2017-Tech/87 dated 07.06.2022, for seeking exemption from Public Procurement (Preference to Make in India) Clause for procurement of various information security tools/items.

2. The items requested by UIDAI for exemption along with their justification for exemption are listed in the table at **Annexure-1** (08 Items).

3. In this regard, the Government has issued Public Procurement (Preference to Make in India) Order 2017 vide the Department for Promotion of Industry and Internal Trade (DPIIT) Order No.P-45021/2/2017-B.E-II dated 15.06.2017, as amended by Orders dated 28.05.2018, 29.05.2019, 04.06.2020 and 16.09.2020 to encourage 'Make in India' and promote manufacturing and production of goods and services in India with a view to enhancing income and employment.

4. In furtherance of the Public Procurement (Preference to Make in India) Order 2017 [PPP-MII Order 2017], Ministry of Electronics and Information Technology (MeitY) has notified the mechanism for calculation of local content for 13 Electronic Products vide Electronic Products Notification vide F.No.43/4/2019-IPHW-MeitY dated 07.09.2020; and Public Procurement (Preference to Make in India) Order 2019 for Cyber Security Products vide Notification No.1(10)/2017-CLES dated 06.12.2019.

5. In this regard, the undersigned is directed to convey the approval of Hon'ble Minister of Electronics and Information Technology for exemption from Public Procurement (Preference to Make in India) for procurement of the items requested by UIDAI (refer **Annexure-1**) as per Clause 14 of the Public Procurement (Preference to Make in India) Order 2017 notified vide DPIIT File No. P-45021/2/2017-B.E.-II dated 16.09.2020 with following conditions:

i. UIDAI will do a thorough check on Government e-Marketplace (GeM) for availability of the aforesaid items / services as per PPP-MII Order 2017 conditions before going for the exempted category of procurement.

Dir (IS) 20 Aug 22
DDG (IS) 23/8
Dhawan 24/8/22
846/CEO/UIDAI 17/8/22
149/DDG (ES & Eng.)/UIDAI 22/08/2022
DD/S
SO/S

ii. Exemption shall be given for a maximum period of one year which may be reviewed any time after completion of six months.

iii. UIDAI shall adhere to the PPP-MII Order 2017 issued by DPIIT and the Electronic Products & Cyber Security Products Notifications issued by MeitY, as amended from time to time.

iv. Adherence to all codal formalities as per the provisions of General Financial Rules-2017, and Department of Expenditure and central Vigilance Commission's guidelines.

(D. K. Sagar)
Joint Director

Encl: As above.

To:

1. The Chief Executive Officer, UIDAI, Bangla Sahib Road, New Delhi.

2. The Secretary, DPIIT & Chairman, Standing Committee on PPP-MII Order 2017, New Delhi.

3. Group Coordinator (UIDAI), Ministry of Electronics and Information Technology, New Delhi.

Annexure-1

List of various hardware and software items requested by UIDAI for Exemption of Make in India (MII) clause for procurement through GeM			
SL. No.	**ICT Item Categories**	**Item Details**	**Justification for MII Exemption**
1	Network Intrusion Prevention system (NIPS)	Network Intrusion Prevention system works by examining incoming/outgoing traffic for intrusions, vulnerabilities, exploits and provides remediation to protect network servers and applications. IPS provides holistic view of intrusions across the network through a single dashboard (existing IPS manager) where it stores vulnerability patches and signatures in single storage which can be pushed to all the IPS sensors across the network. The IPS managers can be utilized as to troubleshoot customer/ network issues related to IPS. In the present setup the Data Centre has IPS Solutions of McAfee.	1. At present there is no OEM manufacturing this class of equipment in India. 2. Network Intrusion preventions works on signature based technology for basic solution coverage and hence it needs global and deep experience to have signature coverage for all latest attack patterns to provide efficient solution which is currently not available in India.

No. IV-24011/22/2020-Prov-I/270.
Government of India
Ministry of Home Affairs
Police Modernization Division
Prov-III Desk

26, Man Singh Road,
Jaisalmer House, New Delhi
Dated the 26 April, 2024.

Office Memorandum

Subject : Forwarding of MeitY's communications regarding CCTVs.

(i) Notification dated 06 March, 2024 on the amendments to Public Procurement Order (PPO)-2017 for CCTV/Video Surveillance Systems,
(ii) Amendment notification dated 09 April, 2024 to the "Electronics and information Technology Goods (Requirement of Compulsory Registration) Order, 2021
(iii) Advisory dated 11 March, 2024 on the Threat of Information through CCTV/Video Surveillance system."

The undersigned is directed to say that Ministry of Electronics and Information Technology has notified amendment to Public Procurement (Preference to Make in India) Order 2017 for CCTV/Video Surveillance System for Security and furtherance of the order vide Gazette Notification dated 6th March, 2024 (copy enclosed). MeitY has also issued amendment notification dated 09 April, 2024 to the "Electronics and Information Technology Goods (Requirement of Compulsory Registration) Order, 2021" for the security of a CCTV system (copy enclosed).

2. All Government agencies are advised that henceforth they may procure their CCTVs and allied solutions and hardware on the basis of the Public Procurement Order-2017 and ensure the security of CCTV systems as mentioned in the above said notifications.

3. Further, MeitY has issued Advisory vide O.M. dated W-43/11/2021-IPHW dated 11 Mar, 2024 (copy enclosed) on the Threat of Information through CCTV/Video Surveillance System (VSS)/Digital Video Recorders/Network Video Recorders. All Government agencies are advised to adhere to the guidelines outlined within the ambit of the Public Procurement Orders to safeguard the overall security and integrity of CCTV Cameras and IoT Devices.

Encl:A/a

26/04/24

(Anuj Kumar)
2IC (Prov.)
Tel: 23386034
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To:

1. Chief Secretaries of all States.
2. Administrators of all UTs.
3. Commissioner of Police, Delhi.
4. Directors : CBI, IB, SVP NPA-Hyderabad, NEPA-Shillong, NCRB, NICFS.
5. DsG : AR, BSF, BPR&D, CISF, CRPF, ITBP, NDRF, NSG, SSB & NIA

Internal:

1. All Divisional Heads, MHA.
2. SO(IT) with the request to upload the OM on MHA's website.
3. Director (Procurement)/Director (PM-I & II).
4. US (Prov-I)/US(Prov-II).

रजिस्ट्री सं. डी.एल.- 33004/99 REGD. No. D. L.-33004/99

भारत का राजपत्र
The Gazette of India

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EXTRAORDINARY

भाग II—खण्ड 3—उप-खण्ड (ii)
PART II—Section 3—Sub-section (ii)

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इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय

(आईपीएचडब्ल्यू प्रभाग)

अधिसूचना

नई दिल्ली, 6 मार्च, 2024

विषय: सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश 2017 के अनुक्रम में सुरक्षा के लिए क्लोज्ड सर्किट टेलीविजन (सीसीटीवी)/वीडियो निगरानी प्रणाली को अधिसूचित करना

सन्दर्भ:

(i) उद्योग और आंतरिक व्यापार संवर्धन विभाग (डीपीआईआईटी) के दिनांक 28.05.2018, 29.05.2019 और 04.06.2020 के आदेशों के जरिए यथासंशोधित आदेश संख्या पी-45021/2/2017-बीई-II दिनांक 15.06.2017

(ii) इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय (एमईआईटीवाई) की दिनांक 07.09.2020 की इलेक्ट्रॉनिक उत्पाद अधिसूचना सं. डब्ल्यू-43/4/2019-आईपीएचडब्ल्यू-एमईआईटीवाई

का.आ. 1119(अ).—सरकार ने 'मेक इन इंडिया' को प्रोत्साहित करने और आय और रोजगार बढ़ाने के उद्देश्य से भारत में वस्तुओं और सेवाओं के विनिर्माण और उत्पादन को बढ़ावा देने के लिए उद्योग और आंतरिक व्यापार संवर्धन विभाग (डीपीआईआईटी) के दिनांक **28.05.2018, 29.05.2019 और 04.06.2020 के आदेशों के जरिए यथासंशोधित आदेश संख्या पी-45021/2/2017-बीई-II दिनांक 15.06.2017 के तहत सार्वजनिक क्रय (मेक इन इंडिया वरीयता) आदेश, 2017 जारी किया है।**

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2. उपर्युक्त संदर्भ के जरिये अधिसूचित सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश 2017 [पीपीपी-एमआईआई आदेश 2017] के अनुक्रम में इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय (एमईआईटीवाई) एतद्वारा अधिसूचित करता है कि समय-समय पर संशोधित पूर्वोक्त आदेश के अनुसार स्थानीय रूप से विनिर्मित इलेक्ट्रॉनिक उत्पादों को सभी क्रय संस्थाओं द्वारा वरीयता प्रदान की जाएगी।

3. इस अधिसूचना के उद्देश्य से :

3.1 श्रेणी- I स्थानीय आपूर्तिकर्ता, श्रेणी-II स्थानीय आपूर्तिकर्ता और गैर-स्थानीय आपूर्तिकर्ता की परिभाषा दिनांक 16.09.2020 के डीपीआईआईटी पीपीपी-एमआईआई आदेश 2017 संख्या समय-समय पर यथा संशोधित पी-45021/2/2017-पीपी(बीई-II) के पैराग्राफ 2 के अनुसार होगी।, । इस अधिसूचना में प्रत्येक अधिसूचित इलेक्ट्रॉनिक उत्पाद के लिए स्थानीय सामग्री की गणना के लिए तंत्र निर्धारित किया गया है।

3.2 समय-समय पर यथासंशोधित दिनांक 16.09.2020 के डीपीआईआईटी पीपीपी-एमआईआई आदेश 2017 सं. पी-45021/2/2017-पीपी(बीई-II) के पैराग्राफ 3 क को खरीद के प्रतिशत के लिए संदर्भित किया जाएगा, जिसके लिए घरेलू रूप से निर्मित इलेक्ट्रॉनिक उत्पादों को वरीयता प्रदान की जानी है (मूल्य के संदर्भ में)।

4. वीडियो निगरानी प्रणाली के निम्नलिखित इलेक्ट्रॉनिक उत्पादों को सार्वजनिक खरीद (मेक इन इंडिया को वरीयता) आदेश 2017 के तहत अधिसूचित किया गया है:

4.1 वीडियो निगरानी प्रणाली

परिभाषा:

इस अधिसूचना के प्रयोजन के लिए, वीडियो निगरानी प्रणाली आमतौर पर एक प्रणाली है जिसमें *अन्य बातों के साथ-साथ* निम्नलिखित शामिल हैं: (i) क्लोज्ड सर्किट टेलीविजन (सीसीटीवी) कैमरा (एनालॉग/आईपी/एनालॉग स्पीड डोम/आईपी स्पीड डोम), (ii) डिजिटल वीडियो रिकॉर्डर (डीवीआर)/नेटवर्क वीडियो रिकॉर्डर (एनवीआर)। सुरक्षा और वीडियो निगरानी प्रणाली के कुछ अनुप्रयोग शहरों, स्कूलों, बैंकों, सरकारी कार्यालयों, सार्वजनिक स्थानों, यातायात निगरानी और गृह सुरक्षा आदि की निगरानी हैं।

4.2 सीसीटीवी कैमरा (एनालॉग) की स्थानीय सामग्री की गणना के लिए तंत्र:

सीसीटीवी कैमरा (एनालॉग) का घरेलू बीओएम निम्नलिखित तालिका के कॉलम 1 में यथा निर्दिष्ट मुख्य इनपुट की लागत का योग होगा, बशर्ते इनपुट व्यक्तिगत रूप से तालिका के कॉलम 2 में निर्दिष्ट मूल्य संवर्धन आवश्यकता को पूरा करते हों:

क्रम सं.	बीओएम में मुख्य इनपुट / सीसीटीवी कैमरा (एनालॉग) के निर्माण के लिए चरण	इनपुट को घरेलू बीओएम के रूप में वर्गीकृत करने के लिए अपेक्षित मूल्य संवर्धन/स्थानीय सामग्री
	(1)	(2)
1.0	मुख्य पीसीबी*	सेमीकंडक्टर के मूल्य सहित ** और बेयर पीसीबी के मूल्य को छोड़कर घरेलू पीसीबी असेम्बली * और आयातित / घरेलू रूप से निर्मित कलपुर्जो और संघटकों से परीक्षण,. हालांकि, वैकल्पिक बोर्डों के साथ/इसके बिना मुख्य पीसीबी के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एनालॉग) के कुल बीओएम के 10% से अधिक नहीं होगा
(i)	आई /ओ बोर्ड (वैकल्पिक)	
(ii)	इन्फ्रारेड (आईआर) बोर्ड (वैकल्पिक)	
(iii)	नियंत्रण बोर्ड (वैकल्पिक)	
2.0	बेयर पीसीबी	घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। बेयर पीसीबी के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एनालॉग) के कुल बीओएम के 5% से अधिक नहीं होगा
3.0	ऑप्टिक्स	घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। बेयर पीसीबी के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एनालॉग) के कुल बीओएम के 10% से अधिक नहीं होगा
4.0	इन-हाउस आर एंड डी #	घरेलू/इन हाउस आर एंड डी में स्वयं अपने संसाधन/विशेषज्ञता का उपयोग कर नये उत्पादों, प्रक्रियाओं, प्रौद्योगिकीयों या समाधानों के विकास के लिए वैज्ञानिक अनुसंधान, प्रयोग, प्रोटोटाइपिंग और परीक्षण अथवा भारत में वैज्ञानिक और औद्योगिक अनुसंधान विभाग (डीएसआईआर) के साथ पंजीकृत आर एंड डी इकाई के साथ कानूनी अनुबंध शामिल है। इसके वेटेज की गणना संबंधित वित्तीय वर्ष में घरेलू बीओएम के प्रतिशत के आधार पर की जाएगी जैसा कि नीचे दी गई तालिका में उल्लिखित है।

		2023-24 & 2024-25	2025-26	2026-27 के बाद
		7%	10%	12%
5.0	इन-हाउस डिजाइन/आईपीआर #	घरेलू/इन हाउस डिज़ाइनिंग में संगठन के अपने संसाधन/विषेशज्ञता का प्रयोग कर अपनी फैसिलिटी में ही डिज़ाइन पद्धतियों सेवाओं या समाधानों की प्रक्रिया अथवा भारत में तृतीय पक्षकार के साथ कानूनी अनुबंध शामिल हैं। भारत में बौद्धिक संपदा अधिकार (आईपीआर) के तहत कोई भी पेटेंट, ट्रेडमार्क, कॉपीराइट और व्यापार रहस्य उनकी रचनाओं या आविष्कारों के डिजाइन के पक्ष में कानूनी स्वामित्व स्थापित करता है। इसके वेटेज की गणना संबंधित वित्तीय वर्ष में घरेलू बीओएम के प्रतिशत के आधार पर की जाएगी जैसा कि नीचे दी गई तालिका में उल्लिखित है।		
		2023-24 & 2024-25	2025-26	2026-27 के बाद
		7%	10%	12%
6.0	हाउसिंग (प्लास्टिक/एल्यूमीनियम/एमएस) और कैमरा माउंट	घरेलू रूप से निर्मित इनपुट से घरेलू रूप से विनिर्माण । हाउसिंग और कैमरा माउंट के लिए इनपुट के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एनालॉग) के कुल बीओएम के 15% से अधिक नहीं होगा।		
7.0	कनेक्टिंग केबल्स और कनेक्टर्स	घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। कनेक्टिंग केबल्स और कनेक्टर्स के लिए इनपुट के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एनालॉग) के कुल बीओएम के 5% से अधिक नहीं होगा।		
8.0	अंतिम असेम्बली और परीक्षण	समय-समय पर अधिसूचित भारतीय मानकों को पूरा करने वाले घरेलू रूप से असेम्बल / परीक्षित. अंतिम संयोजन और परीक्षण के कुल मूल्य का वेटेज सीसीटीवी कैमरे के कुल बीओएम के 10% से अधिक नहीं होगा		

** यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करके बेयर पीसीबी पर कलपुर्जो और संघटकों के मुद्रित सर्किट बोर्ड की असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

*** भारत में सेमीकंडक्टर एफएबी के चालू होने पर इसकी समीक्षा की जाएगी।*

 # आवेदक संबंधित वित्तीय वर्ष के लिए नीचे दी गई तालिका में उल्लिखित न्यूनतम मूल्यवर्धन/स्थानीय सामग्री को पूरा करके मूल्य संवर्धन/स्थानीय सामग्री की गणना के लिए संबंधित वित्तीय वर्ष में क्रम संख्या 4.0 (इन-हाउस आर एंड डी) और क्रम संख्या 5.0 (इन-हाउस डिजाइन/आईपीआर) का लाभ उठाने के लिए पात्र होगा.

वित्तीय वर्ष	2023-24 & 2024-25	2025-26	2026-27 के बाद
आवश्यक के रूप में न्यूनतम मूल्य संवर्धन/स्थानीय सामग्री (क्रम सं. 1.0) मुख्य पीसीबी और (क्रम सं.8.0) अंतिम असेंबली और परीक्षण और अन्य वैकल्पिक के रूप में	25%	35%	45%

4.3 सीसीटीवी कैमरा (आईपी) की स्थानीय सामग्री की गणना के लिए तंत्र:

सीसीटीवी कैमरा (आईपी) का घरेलू बीओएम निम्नलिखित तालिका के कॉलम 1 में यथा निर्दिष्ट मुख्य इनपुट की लागत का योग होगा, बशर्ते इनपुट व्यक्तिगत रूप से तालिका के कॉलम 2 में निर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करते हों:

क्रम सं.	बीओएम में मुख्य इनपुट / सीसीटीवी कैमरा (आईपी) के निर्माण के लिए चरण	इनपुट को घरेलू बीओएम के रूप में वर्गीकृत करने के लिए आवश्यक मूल्य संवर्धन/स्थानीय सामग्री
	(1)	(2)
1.0	मुख्य पीसीबी* कैप्चर और/या प्रोसेसर कार्ड के साथ	सेमीकंडक्टर के मूल्य सहित ** और बेयर पीसीबी के मूल्य को छोड़कर घरेलू पीसीबी असेम्बली * और आयातित / घरेलू रूप से निर्मित कलपुर्जो और संघटकों से परीक्षण.

<table>
<tr><td>(i)</td><td>नेटवर्क इंटरफेस कार्ड (वैकल्पिक)</td><td rowspan="4">हालांकि, वैकल्पिक बोर्डों के साथ/इनके बिना मुख्य पीसीबी के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपी) के कुल बीओएम के 10% से अधिक नहीं होगा</td></tr>
<tr><td>(ii)</td><td>इन्फ्रारेड (आईआर) बोर्ड (वैकल्पिक)</td></tr>
<tr><td>(iii)</td><td>आई/ओ बोर्ड (वैकल्पिक)</td></tr>
<tr><td>(iv)</td><td>पावर ओवर इथरनेट (पी ओ ई) कार्ड (वैकल्पिक)</td></tr>
<tr><td>2.0</td><td>बेयर पीसीबी</td><td>घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। बेयर पीसीबी के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपी) के कुल बीओएम के 5% से अधिक नहीं होगा</td></tr>
<tr><td>3.0</td><td>ऑप्टिक्स</td><td>घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। प्रकाशिकी के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपी) के कुल बीओएम के 10% से अधिक नहीं होगा</td></tr>
<tr><td>4.0</td><td>इन-हाउस आर एंड डी #</td><td>घरेलू/इन हाउस आर एंड डी में स्वयं अपने संसाधन/विशेषज्ञता का उपयोग कर नये उत्पादों, प्रक्रियाओं, प्रौद्योगिकीयों या समाधानों के विकास के लिए वैज्ञानिक अनुसंधान, प्रयोग, प्रोटोटाइपिंग और परीक्षण अथवा भारत में वैज्ञानिक और औद्योगिक अनुसंधान विभाग (डीएसआईआर) के साथ पंजीकृत आर एंड डी इकाई के साथ कानूनी अनुबंध शामिल है। इसके वेटेज की गणना संबंधित वित्तीय वर्ष में घरेलू बीओएम के प्रतिशत के आधार पर की जाएगी जैसा कि नीचे दी गई तालिका में उल्लिखित है।
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<tr><td>5.0</td><td>इन-हाउस डिजाइन/आईपीआर #</td><td>घरेलू/इन हाउस डिज़ाइनिंग में संगठन के अपने संसाधन/विषेशज्ञता का प्रयोग कर अपनी फैसिलिटी में ही डिज़ाइन पद्धतियों सेवाओं या समाधानों की प्रक्रिया अथवा भारत में तृतीय पक्षकार के साथ कानूनी अनुबंध शामिल हैं। भारत में बौद्धिक संपदा अधिकार (आईपीआर) के तहत कोई भी पेटेंट, ट्रेडमार्क, कॉपीराइट और व्यापार रहस्य उनकी रचनाओं या आविष्कारों के डिजाइन के पक्ष में कानूनी स्वामित्व स्थापित करता है। इसके वेटेज की गणना संबंधित वित्तीय वर्ष में घरेलू बीओएम के प्रतिशत के आधार पर की जाएगी जैसा कि नीचे दी गई तालिका में उल्लिखित है।
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<tr><td>6.0</td><td>हाउसिंग (प्लास्टिक / एल्यूमीनियम / एमएस) और</td><td>घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। हाउसिंग और कैमरा माउंट के लिए इनपुट के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपी) के कुल बीओएम के 15% से अधिक नहीं होगा।</td></tr>
<tr><td>7.0</td><td>कनेक्टिंग केबल्स और कनेक्टर्स</td><td>घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। कनेक्टिंग केबल्स और कनेक्टर्स के लिए इनपुट के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपी) के कुल बीओएम के 5% से अधिक नहीं होगा।</td></tr>
<tr><td>8.0</td><td>अंतिम असेम्बली और परीक्षण</td><td>समय-समय पर यथा अधिसूचित भारतीय मानकों को पूरा करने वाले घरेलू रूप से असेम्बल / परीक्षित .फाइनल असेंबलिंग और टेस्टिंग के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपी) के कुल बीओएम के 10% से अधिक नहीं होगा.</td></tr>
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** यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करके बेयर पीसीबी पर कलपुर्जो और संघटकों की मुद्रित सर्किट बोर्ड असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

***भारत में सेमीकंडक्टर एफएबी के चालू होने पर इसकी समीक्षा की जाएगी।# आवेदक संबंधित वित्तीय वर्ष के लिए नीचे दी गई तालिका में उल्लिखित न्यूनतम मूल्यवर्धन/स्थानीय सामग्री को पूरा करके मूल्य संवर्धन/स्थानीय सामग्री की गणना के लिए संबंधित वित्तीय वर्ष में क्रम संख्या 4.0 (इन-हाउस आर एंड डी) और क्रम संख्या 5.0 (इन-हाउस डिजाइन/आईपीआर) का लाभ उठाने के लिए पात्र होगा.*

वित्तीय वर्ष	2023-24 & 2024-25	2025-26	2026-27 के बाद
आवश्यक के रूप में न्यूनतम मूल्य संवर्धन/स्थानीय सामग्री (क्रम सं. 1.0) मुख्य पीसीबी और (क्रम सं.8.0) अंतिम असेंबली और परीक्षण और अन्य वैकल्पिक के रूप में	25%	35%	45%

4.4 सीसीटीवी कैमरा (एनालॉग स्पीड डोम) की स्थानीय सामग्री की गणना के लिए तंत्र:

सीसीटीवी कैमरा (एनालॉग स्पीड डोम) का घरेलू बीओएम निम्नलिखित तालिका के कॉलम 1 में यथा निर्दिष्ट मुख्य इनपुट की लागत का योग होगा, बशर्ते इनपुट व्यक्तिगत रूप से तालिका के कॉलम 2 में निर्दिष्ट मूल्य संवर्धन आवश्यकता को पूरा करते हों:

<table>
<tr><th>क्रम सं.</th><th>बीओएम में मुख्य इनपुट / सीसीटीवी कैमरा (एनालॉग स्पीड डोम) के निर्माण के लिए चरण</th><th>इनपुट को घरेलू बीओएम के रूप में वर्गीकृत करने के लिए आवश्यक मूल्य संवर्धन</th></tr>
<tr><td></td><td>(1)</td><td>(2)</td></tr>
<tr><td>1.0</td><td>मुख्य नियंत्रक बोर्ड*</td><td rowspan="3">सेमीकंडक्टर के मूल्य सहित ** और बेयर पीसीबी के मूल्य को छोड़कर घरेलू पीसीबी असेम्बली * और आयातित / घरेलू रूप से निर्मित कलपुर्जो और संघटकों से परीक्षण. हालांकि, वैकल्पिक बोर्डों के साथ/के बिना मुख्य नियंत्रक बोर्ड/मुख्य पीसीबी के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एएसडी) के कुल बीओएम के 10% से अधिक नहीं होगा</td></tr>
<tr><td>(i)</td><td>आई /ओ बोर्ड (वैकल्पिक)</td></tr>
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</table>

	कैमरा माउंट	5% से अधिक नहीं होगा।
6.0	कनेक्टिंग केबल्स और कनेक्टर्स	घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। कनेक्टिंग केबल्स और कनेक्टर्स के लिए इनपुट के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एएसडी) के कुल बीओएम के 5% से अधिक नहीं होगा।
7.0	मोटर, हीटर और ब्लोअर, ज़ूम मॉड्यूल, बेल्ट, प्लास्टिक तंत्र भागों, ग्लास आदि	घरेलू रूप से आयातित/घरेलू रूप से निर्मित इनपुट भागों और यांत्रिक घटकों से निर्मित। हालांकि, इन सभी भागों/घटकों के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एएसडी) के कुल बीओएम के 5% से अधिक नहीं होगा
8.0	पावर एडाप्टर	घरेलू पीसीबी असेम्बली * और आयातित / घरेलू रूप से निर्मित कलपुर्जो और संघटकों से अंतिम असेम्बली और परीक्षण, बशर्ते कि "पावर एडाप्टर" की असेम्बली में इस्तेमाल घरेलू रूप से निर्मित कलपुर्जो और संघटकों का मूल्य न्यूनतम 40%होगा ("पावर एडाप्टर" के निर्माण में इस्तेमाल कलपुर्जो और संघटकों के कुल मूल्य का). हालांकि, "पावर एडॉप्टर" के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एएसडी) के कुल बीओएम के 5% से अधिक नहीं होगा
9.0	अंतिम असेम्बली और परीक्षण	समय-समय पर अधिसूचित भारतीय मानकों को पूरा करने वाले घरेलू रूप से असेम्बल / परीक्षित. अंतिम संयोजन और परीक्षण के कुल मूल्य का वेटेज सीसीटीवी कैमरा (एएसडी) के कुल बीओएम के 10% से अधिक नहीं होगा

** यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करके बेयर पीसीबी पर कलपुर्जो और संघटकों के मुद्रित सर्किट बोर्ड की असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

*** भारत में सेमीकंडक्टर एफएबी के चालू होने पर इसकी समीक्षा की जाएगी।*

 # आवेदक संबंधित वित्तीय वर्ष के लिए नीचे दी गई तालिका में उल्लिखित न्यूनतम मूल्यवर्धन/स्थानीय सामग्री को पूरा करके मूल्य संवर्धन/स्थानीय सामग्री की गणना के लिए संबंधित वित्तीय वर्ष में क्रम संख्या 3.0 (इन-हाउस आर एंड डी) और क्रम संख्या 4.0 (इन-हाउस डिजाइन/आईपीआर) का लाभ उठाने के लिए पात्र होगा.

वित्तीय वर्ष	2023-24 & 2024-25	2025-26	2026-27 के बाद
आवश्यक के रूप में न्यूनतम मूल्य संवर्धन/स्थानीय सामग्री (क्रम सं. 1.0) मुख्य पीसीबी और (क्रम सं.8.0) अंतिम असेंबली और परीक्षण और वैकल्पिक के रूप में अन्य	25%	35%	45%

4.5 सीसीटीवी कैमरा (आईपी स्पीड डोम) की स्थानीय सामग्री की गणना के लिए तंत्र:

सीसीटीवी कैमरा (आईपी स्पीड डोम) का घरेलू बीओएम निम्नलिखित तालिका के कॉलम 1 में निर्दिष्ट मुख्य इनपुट की लागत का योग होगा, बशर्ते इनपुट व्यक्तिगत रूप से तालिका के कॉलम 2 में निर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करते हों:

क्रम सं.	बीओएम में मुख्य इनपुट / सीसीटीवी कैमरा (आईपी स्पीड डोम) के निर्माण के लिए चरण	इनपुट को घरेलू बीओएम के रूप में वर्गीकृत करने के लिए आवश्यक मूल्य संवर्धन/स्थानीय सामग्री
	(1)	(2)
1.0	मुख्य नियंत्रक बोर्ड*	सेमीकंडक्टर के मूल्य सहित ** और बेयर पीसीबी के मूल्य को छोड़कर घरेलू पीसीबी असेम्बली * और आयातित / घरेलू रूप से निर्मित कलपुर्जो और संघटकों से परीक्षण. हालांकि, वैकल्पिक बोर्डों के साथ/के बिना मुख्य नियंत्रक बोर्ड के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपीएसडी) के कुल बीओएम के 10% से अधिक नहीं होगा
(I)	आई /ओ बोर्ड (वैकल्पिक)	
(ii)	इन्फ्रारेड (आईआर) बोर्ड (वैकल्पिक)	

<table>
<tr><td>(iii)</td><td>प्रोसेसर कार्ड (वैकल्पिक)</td><td rowspan="3"></td></tr>
<tr><td>(iv)</td><td>नेटवर्क इंटरफेस कार्ड (वैकल्पिक)</td></tr>
<tr><td>(v)</td><td>पावर ओवर इथरनेट (पी ओ ई) कार्ड (वैकल्पिक)</td></tr>
<tr><td>2.0</td><td>बेयर पीसीबी</td><td>घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। बेयर पीसीबी के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपीएसडी) के कुल बीओएम के 5% से अधिक नहीं होगा</td></tr>
<tr><td>3.0</td><td>इन-हाउस आर एंड डी #</td><td>घरेलू/इन हाउस आर एंड डी में स्वयं अपने संसाधन/विशेषज्ञता का उपयोग कर नये उत्पादों, प्रक्रियाओं, प्रौद्योगिकीयों या समाधानों के विकास के लिए वैज्ञानिक अनुसंधान, प्रयोग, प्रोटोटाइपिंग और परीक्षण अथवा भारत में वैज्ञानिक और औद्योगिक अनुसंधान विभाग (डीएसआईआर) के साथ पंजीकृत आर एंड डी इकाई के साथ कानूनी अनुबंध शामिल है। इसके वेटेज की गणना संबंधित वित्तीय वर्ष में घरेलू बीओएम के प्रतिशत के आधार पर की जाएगी जैसा कि नीचे दी गई तालिका में उल्लिखित है।
<table>
<tr><th>2023-24 & 2024-25</th><th>2025-26</th><th>2026-27 के बाद</th></tr>
<tr><td>7%</td><td>10%</td><td>12%</td></tr>
</table>
</td></tr>
<tr><td>4.0</td><td>इन-हाउस डिजाइन/आईपीआर #</td><td>घरेलू/इन हाउस डिज़ाइनिंग में संगठन के अपने संसाधन/विषेशज्ञता का प्रयोग कर अपनी फैसिलिटी में ही डिज़ाइन पद्धतियों सेवाओं या समाधानों की प्रक्रिया अथवा भारत में तृतीय पक्षकार के साथ कानूनी अनुबंध शामिल हैं। भारत में बौद्धिक संपदा अधिकार (आईपीआर) के तहत कोई भी पेटेंट, ट्रेडमार्क, कॉपीराइट और व्यापार रहस्य उनकी रचनाओं या आविष्कारों के डिजाइन के पक्ष में कानूनी स्वामित्व स्थापित करता है। इसके वेटेज की गणना संबंधित वित्तीय वर्ष में घरेलू बीओएम के प्रतिशत के आधार पर की जाएगी जैसा कि नीचे दी गई तालिका में उल्लिखित है।
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<tr><td>7%</td><td>10%</td><td>12%</td></tr>
</table>
</td></tr>
<tr><td>5.0</td><td>हाउसिंग (प्लास्टिक / एल्यूमीनियम / एमएस) और कैमरा माउंट</td><td>घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। हाउसिंग और कैमरा माउंट के लिए इनपुट के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपीएसडी) के कुल बीओएम के 5% से अधिक नहीं होगा।</td></tr>
<tr><td>6.0</td><td>कनेक्टिंग केबल्स और कनेक्टर्स</td><td>घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। कनेक्टिंग केबल्स और कनेक्टर्स के लिए इनपुट के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपीएसडी) के कुल बीओएम के 5% से अधिक नहीं होगा।</td></tr>
<tr><td>7.0</td><td>मोटर, हीटर और ब्लोअर, ज़ूम मॉड्यूल, बेल्ट, प्लास्टिक तंत्र भागों, ग्लास आदि</td><td>घरेलू रूप से आयातित/घरेलू रूप से निर्मित इनपुट भागों और यांत्रिक घटकों से निर्मित। तथापि, इन सभी भागों/संघटकों के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपीएसडी) के कुल बीओएम के 5% से अधिक नहीं होगा।</td></tr>
<tr><td>8.0</td><td>पावर एडाप्टर</td><td>घरेलू पीसीबी असेम्बली * और आयातित / घरेलू रूप से निर्मित कलपुर्जो और संघटकों से अंतिम असेम्बली और परीक्षण, बशर्ते कि "पावर एडाप्टर" की असेम्बली में इस्तेमाल घरेलू रूप से निर्मित कलपुर्जो और संघटकों का मूल्य न्यूनतम 40%होगा ("पावर एडाप्टर" के निर्माण में इस्तेमाल कलपुर्जो और संघटकों के कुल मूल्य का). हालांकि, "पावर एडॉप्टर" के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपीएसडी) के कुल बीओएम के 5% से अधिक नहीं होगा</td></tr>
<tr><td>9.0</td><td>अंतिम असेम्बली और परीक्षण</td><td>समय-समय पर यथाअधिसूचित भारतीय मानकों को पूरा करने वाले घरेलू रूप से असेम्बल /परीक्षित . अंतिम संयोजन और परीक्षण के कुल मूल्य का वेटेज सीसीटीवी कैमरा (आईपीएसडी) के कुल बीओएम के 10% से अधिक नहीं होगा</td></tr>
</table>

** यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करके बेयर पीसीबी पर कलपुर्जो और संघटकों के मुद्रित सर्किट बोर्ड की असेंबली (पीसीबीए) अनिवार्य रूप से भारत में की जानी चाहिए।*

*** भारत में सेमीकंडक्टर एफएबी के चालू होने पर इसकी समीक्षा की जाएगी।*

 # आवेदक संबंधित वित्तीय वर्ष के लिए नीचे दी गई तालिका में उल्लिखित न्यूनतम मूल्यवर्धन/स्थानीय सामग्री को पूरा करके मूल्य संवर्धन/स्थानीय सामग्री की गणना के लिए संबंधित वित्तीय वर्ष में क्रम संख्या 3.0 (इन-हाउस आर एंड डी) और क्रम संख्या 4.0 (इन-हाउस डिजाइन/आईपीआर) का लाभ उठाने के लिए पात्र होगा.

वित्तीय वर्ष	2023-24 & 2024-25	2025-26	2026-27 के बाद
आवश्यक के रूप में न्यूनतम मूल्य संवर्धन/स्थानीय सामग्री (क्रम सं. 1.0) मुख्य पीसीबी और (क्रम सं.9.0) अंतिम असेंबली और परीक्षण और अन्य वैकल्पिक के रूप में	25%	35%	45%

4.6 डीवीआर/एनवीआर की स्थानीय सामग्री की गणना के लिए तंत्र:

डीवीआर/एनवीआर का घरेलू बीओएम निम्नलिखित तालिका के कॉलम 1 में यथा निर्दिष्ट मुख्य इनपुट की लागत का योग होगा, बशर्ते इनपुट व्यक्तिगत रूप से तालिका के कॉलम 2 में निर्दिष्ट मूल्यवर्धन आवश्यकता को पूरा करते हों:

<table>
<tr><th>क्रम सं.</th><th>बीओएम में मुख्य इनपुट / डीवीआर/एनवीआर के निर्माण के लिए चरण</th><th>इनपुट को घरेलू बीओएम के रूप में वर्गीकृत करने के लिए आवश्यक मूल्य संवर्धन/स्थानीय सामग्री</th></tr>
<tr><td></td><td>(1)</td><td>(2)</td></tr>
<tr><td>1.0</td><td>मुख्य पीसीबी*</td><td rowspan="3">सेमीकंडक्टर के मूल्य सहित ** और बेयर पीसीबी के मूल्य को छोड़कर घरेलू पीसीबी असेम्बली * और आयातित / घरेलू रूप से निर्मित कलपुर्जो और संघटकों से परीक्षण. तथापि, वैकल्पिक बोर्डों के साथ/के बिना मुख्य पीसीबी के कुल मूल्य का वेटेज डीवीआर/एनवीआर के कुल बीओएम के 10 % से अधिक नहीं होगा.</td></tr>
<tr><td>(i)</td><td>फ्रंट पैनल और एलईडी बोर्ड (वैकल्पिक)</td></tr>
<tr><td>(ii)</td><td>आई/ओ बोर्ड (वैकल्पिक)</td></tr>
<tr><td>2.0</td><td>बेयर पीसीबी</td><td>घरेलू रूप से विनिर्मित इनपुट से घरेलू रूप से निर्मित। बेयर पीसीबी के कुल मूल्य का वेटेज सीसीटीवी कैमरा (डीवीआर / एनवीआर) के कुल बीओएम के 5% से अधिक नहीं होगा.</td></tr>
<tr><td>3.0</td><td>इन-हाउस आर एंड डी #</td><td>घरेलू/इन हाउस आर एंड डी में स्वयं अपने संसाधन/विशेषज्ञता का उपयोग कर नये उत्पादों, प्रक्रियाओं, प्रौद्योगिकीयों या समाधानों के विकास के लिए वैज्ञानिक अनुसंधान, प्रयोग, प्रोटोटाइपिंग और परीक्षण अथवा भारत में वैज्ञानिक और औद्योगिक अनुसंधान विभाग (डीएसआईआर) के साथ पंजीकृत आर एंड डी इकाई के साथ कानूनी अनुबंध शामिल है। इसके वेटेज की गणना संबंधित वित्तीय वर्ष में घरेलू बीओएम के प्रतिशत के आधार पर की जाएगी जैसा कि नीचे दी गई तालिका में उल्लिखित है।
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<tr><td>4.0</td><td>इन-हाउस डिजाइन/आईपीआर #</td><td>घरेलू/इन हाउस डिज़ाइनिंग में संगठन के अपने संसाधन/विषेशज्ञता का प्रयोग कर अपनी फैसिलिटी में ही डिज़ाइन पद्धतियों सेवाओं या समाधानों की प्रक्रिया अथवा भारत में तृतीय पक्षकार के साथ कानूनी अनुबंध शामिल हैं। भारत में बौद्धिक संपदा अधिकार (आईपीआर) के तहत कोई भी पेटेंट, ट्रेडमार्क, कॉपीराइट और व्यापार रहस्य उनकी रचनाओं या आविष्कारों के डिजाइन के पक्ष में कानूनी स्वामित्व स्थापित करता है। इसके वेटेज की गणना संबंधित वित्तीय वर्ष में घरेलू बीओएम के प्रतिशत के आधार पर की जाएगी जैसा कि नीचे दी गई तालिका में उल्लिखित है।
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<tr><th>2023-24 & 2024-25</th><th>2025-26</th><th>2026-27 के बाद</th></tr>
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</td></tr>
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		7%	10%	12%
5.0	हाउसिंग (प्लास्टिक / एल्यूमीनियम / एमएस) और कैमरा माउंट	घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। आवास और कैमरा माउंट के लिए इनपुट के कुल मूल्य का वेटेज डीवीआर/एनवीआर के कुल बीओएम के 5% से अधिक नहीं होगा।		
6.0	कनेक्टिंग केबल, कनेक्टर्स और यूएसबी माउस	घरेलू रूप से निर्मित इनपुट से घरेलू रूप से निर्मित। कनेक्टिंग केबल, कनेक्टर्स और यूएसबी माउस के लिए इनपुट के कुल मूल्य का वेटेज डीवीआर/एनवीआर के कुल बीओएम के 5% से अधिक नहीं होगा।		
7.0	रिमोट कंट्रोल	घरेलू असेंबली * और आयातित/घरेलू रूप से निर्मित कलपुर्जो और संघटकों से परीक्षण इस शर्त के अधीन है कि "रिमोट कंट्रोल" की असेंबली में उपयोग किए जाने वाले घरेलू रूप से निर्मित कलपुर्जो और संघटकों का मूल्य वर्ष 2 में न्यूनतम 20% होगा ("रिमोट कंट्रोल" के निर्माण में उपयोग किए जाने वाले कलपुर्जो और संघटकों के कुल मूल्य का), जो वित्तीय वर्ष 2025-26, 2026-27 और 2027-28 से आगे क्रमशः न्यूनतम 30%, 40% और 50% तक बढ़ जाएगा।		
8.0	पावर एडाप्टर	घरेलू पीसीबी असेम्बली * और आयातित / घरेलू रूप से निर्मित कलपुर्जो और संघटकों से अंतिम असेम्बली और परीक्षण, बशर्ते कि "पावर एडाप्टर" की असेम्बली में इस्तेमाल घरेलू रूप से निर्मित कलपुर्जो और संघटकों का मूल्य न्यूनतम 40%होगा ("पावर एडाप्टर" के निर्माण में इस्तेमाल कलपुर्जो और संघटकों के कुल मूल्य का). हालांकि, "पावर एडाप्टर" के कुल मूल्य का वेटेज डीवीआर/एनवीआर के कुल बीओएम के 5% से अधिक नहीं होगा।		
9.0	अंतिम असेम्बली और परीक्षण	समय-समय पर अधिसूचित वेटेज भारतीय मानकों को पूरा करने वाले घरेलू रूप से असेम्बल / परीक्षित. अंतिम संयोजन और परीक्षण के कुल मूल्य का वेटेज डीवीआर/एनवीआर के कुल बीओएम के 10% से अधिक नहीं होगा।		

** यह आवश्यक है कि एसएमटी प्रक्रिया का उपयोग करके बेयर पीसीबी पर कलपुर्जो और संघटकों के मुद्रित सर्किट बोर्ड असेंबली (पीसीबीए) अनिवार्य रूप से भारत में किया जाना चाहिए।*

*** भारत में सेमीकंडक्टर एफएबी के चालू होने पर इसकी समीक्षा की जाएगी।*

 # आवेदक संबंधित वित्तीय वर्ष के लिए नीचे दी गई तालिका में उल्लिखित न्यूनतम मूल्यवर्धन/स्थानीय सामग्री को पूरा करके मूल्य संवर्धन/स्थानीय सामग्री की गणना के लिए संबंधित वित्तीय वर्ष में क्रम संख्या 3.0 (इन-हाउस आर एंड डी) और क्रम संख्या 4.0 (इन-हाउस डिजाइन/आईपीआर) का लाभ उठाने के लिए पात्र होगा.

वित्तीय वर्ष	2023-24 & 2024-25	2025-26	2026-27 के बाद
आवश्यक के रूप में न्यूनतम मूल्य संवर्धन/स्थानीय सामग्री (क्रम सं. 1.0) मुख्य पीसीबी और (क्रम सं.9.0) अंतिम असेंबली और परीक्षण और अन्य वैकल्पिक के रूप में	25%	35%	45%

5. निगरानी उपकरणों यानी सीसीटीवी कैमरों (एनालॉग/आईपी/एनालॉग स्पीड डोम/आईपी स्पीड डोम/आईपी स्पीड डोम) को समय-समय पर यथा संशोधित परिशिष्ट 'क' के अनुसार वीएसएस/सीसीटीवी सिस्टम की सुरक्षा सुनिश्चित करने के लिए एमईआईटीवाई द्वारा निर्धारित सुरक्षा के लिए आवश्यक आवश्यकताओं (ईआर) का अनु पालन करना चाहिए। सीसीटीवी/वीएसएस के लिए सुरक्षा परीक्षण प्रतिवेदन मानकीकरण परीक्षण और गुणवत्ता प्रमाणन (एसटीक्यूसी) प्रयोगशाला या एमईआईटीवाई द्वारा समय-समय पर अधिसूचित किसी अन्य एजेंसी द्वारा जारी किया जाएगा। एसटीक्यूसी लैब द्वारा जारी परीक्षण रिपोर्ट की वैधता रिपोर्ट के जारी होने की तारीख से तीन वर्ष होगी. सीसीटीवी कैमरों/वीएसएस प्रणाली की सुरक्षा के लिए इन अनिवार्य आवश्यकताओं (ईआर) को इस अधिसूचना के जारी होने की तारीख से 3 महीने के बाद लागू किया जाएगा। डीवीआर और एनवीआर के लिए सुरक्षा मानदंडों को बाद में अधिसूचित किया जाएगा।

6. अधिसूचना इस अधिसूचना के जारी होने की तारीख से 3 महीने के बाद प्रभावी होगी। यह अधिसूचना संशोधित अधिसूचना जारी होने तक वैध रहेगी। इस अधिसूचना के प्रयोजन के लिए 1 वर्ष 31.03.2025 तक होगा।

7. सार्वजनिक खरीद (मेक इन इंडिया को वरीयता) आदेश 2017 के तहत किसी भी इलेक्ट्रॉनिक उत्पाद अधिसूचना का पूर्वव्यापी प्रभाव नहीं होगा।

8. पूर्वोक्त इलेक्ट्रॉनिक्स उत्पादों की खरीद के लिए समय-समय पर यथा संशोधित दिनांक 16.09.2020 के सार्वजनिक खरीद (मेक इन इंडिया को वरीयता) आदेश 2017 में उद्धृत प्रावधानों के अनुसार खरीद वरीयता प्रदान की जाएगी ।

9. यदि भारत सरकार द्वारा परियोजना या योजना पूरी तरह से या आंशिक रूप से वित्त पोषित होती है, तो राज्यों और स्थानीय निकायों द्वारा बनाए गए इलेक्ट्रॉनिक उत्पादों की क्रय के लिए अधिसूचना सभी केंद्रीय योजनाओं (सीएस)/केंद्रीय क्षेत्र की योजनाओं (सीएसएस) पर भी लागू होगी।

10. स्थानीय सामग्री/घरेलू मूल्यवर्धन की गणना करने हेतु प्रक्रिया

10.1 घरेलू विनिर्माताओं (डीओएम-बीओएम) से प्राप्त सामग्री के बिल की गणना उपलब्ध आंकड़ों के आधार पर निम्नलिखित में से किसी एक के आधार पर की जा सकती है।

क. सभी इनपुटों की लागतों का योग जो उस उत्पाद को बनाने के लिए आवश्यक है (इनपुटों की क्रय पर लगाए गए शुल्क और करों सहित, सिवाय उनको छोड़कर जिनके लिए क्रेडिट/सेट-ऑफ का लाभ लिया जा सकता है) और जिनका आयात प्रत्यक्ष रूप से या घरेलू व्यापारी या माध्यस्थ के माध्यम से नहीं किया गया है।

ख. उत्पाद के फैक्ट्री बाह्य मूल्य से उत्पाद के उत्पादन में इनपुटों के रूप में उपयोग की गई सामग्री के आयातित बिल (प्रत्यक्ष या अप्रत्यक्ष रूप से) को घटाकर (इनपुटों की क्रय पर लागए गए शुल्क और करों सहित, सिवाय उनको छोड़कर जिनके लिए क्रेडिट/सेट-ऑफ का लाभ लिया जा सकता है) और उसमें से वारंटी लागत को घटाकर।

ग. बाजार मूल्य से उत्पादन पश्चात मालभाडा, बीमा और अन्य हथालन लागतों को घटाकर, उसमें से कर पश्चात लाभ, वारंटी लागत को घटाया जाता है, उसमें से उत्पादों के उत्पादन (इनपुटों की क्रय पर लगाए गए शुल्क और करों सहित, सिवाय उनको छोड़कर जिनके लिए क्रेडिट/सेट-ऑफ लिया जा सकता है) में इनपुटों के रूप में उपयोग की गई सामग्री के आयातित बिल के योग को घटाया जाता है और बिक्री और विपणन खर्च को घटाया जाता है।

10.2 सामग्री के कुल बिल (कुल-बीओएम) की गणना उपलब्ध आंकडों पर आधारित निम्नलिखित में से किसी एक के आधार पर की जा सकती है।

क. सभी इनपुटों की लागतों का योग जो उस उत्पाद को बनाने के लिए आवश्यक है (इनपुटों की क्रय पर लगाए गए शुल्क और करों सहित, सिवाय उनको छोड़कर जिनके लिए क्रेडिट/सेट-ऑफ का लाभ लिया जा सकता है)।

ख. उत्पाद के फैक्टरी बाह्य मूल्य से कर पश्चात लाभ और वारंटी लागतों को घटाया जाता है।

ग. बाजार मूल्य से उत्पादन पश्चात मालभाडा, बीमा और अन्य हथालन लागतों को घटाकर उसमें से कर पश्चात लाभ, वारंटी लागत, बिक्री और विपणन खर्च घटाया जाता है।

10 .3 निम्नलिखित फॉर्मूला के अनुसार प्रस्तुत जानकारी के आधार पर घरेलू मूल्य-वृद्धि के प्रतिशत की गणना की जा सकती है:

$$\text{स्थानीय सामग्री/घरेलू मूल्य-वृद्धि का प्रतिशत} = \frac{\text{डीओएम-बीओएम}}{\text{कुल-बीओएम}} \times 100$$

11. स्थानीय सामग्री/घरेलू मूल्यवृद्धि का सत्यापन

11.1. निविदा, बोली प्रक्रिया या सिफ़ारिश के समय स्थानीय आपूर्तिकर्ता स्व-प्रमाणन प्रदान करेगा कि प्रसतावित की गई वस्तुएं न्यूनतम स्थानीय सामग्री की आवश्यकताओं को पूरा करती हैं और उस स्थान (स्थानों) का विवरण देगा, जिस पर स्थानीय मूल्य वृद्धि की गई है।

11.2. 10 करोड़ रुपए से अधिक मूल्य की क्रय के मामलों में, स्थानीय आपूर्तिकर्ता स्थानीय सामग्री का प्रतिशत देते हुए कंपनी के सांविधिक लेखा परीक्षक)कंपनियों के मामले में) या लागत लेखापरीक्षक या किसी पेशेवर लागत लेखाकार ये पेशेवर चार्टेड लेखाकार (कंपनियों के अलावा अन्य आपूर्तिकर्ताओं के संबंध में) से एक प्रमाणपत्र उपलब्ध कराएगा।

11.3. यदि किसी इलेक्ट्रॉनिक उत्पाद के संबंध में स्थानीय सामग्री/घरेलू मूल्य वृद्धि के संबंध में बोलीकर्ता के दावे के खिलाफ क्रय एजेंसी या संबंधित मंत्रालय/विभाग द्वारा शिकायत प्राप्त की जाती है, तो उसे एसटीक्यूसी के पास भेजा जाएगा।

11.4. एसटीक्यूसी को संदर्भित कोई भी शिकायत 4 सप्ताह के भीतर निपटा दी जाएगी। बोलीकर्ता को एसटीक्यूसी हेतु किसी इलेक्ट्रॉनिक उत्पाद में दावा की गई घरेलू मूल्य वृद्धि के समर्थन में आवश्यक दस्तावेज प्रस्तुत करने की आवश्यकता होगी। यदि बोलीकर्ता के द्वारा कोई जानकारी प्रस्तुत नहीं की जाती है, तो ऐसी प्रयोगशालाएं दावे के वास्तविकता को सिद्ध करने के लिए अगली आवश्यक कार्रवाई कर सकती हैं। **स्थानीय सामग्री की गणना में विसंगतियों के मामले में, स्थानीय सामग्री के प्रतिशत की गणना के लिए कंपनी के सांविधिक लेखा परीक्षक या लागत लेखा परीक्षक (कंपनियों के मामले में) या प्रैक्टिसिंग कॉस्ट एकाउंटेंट या प्रैक्टिसिंग चार्टर्ड एकाउंटेंट (कंपनियों के अलावा अन्य आपूर्तिकर्ताओं के संबंध में) से एक प्रमाण पत्र स्व-घोषणा प्रमाण पत्र पर अधिक्रमित होगा।**

11.5. 2 लाख रुपए या खरीदे जा रहे घरेलू रूप से विनिर्मात इलेक्ट्रॉनिक उत्पादों की कीमत का 1% शिकायत शुल्क (5 लाख रुपए की अधिकतदम सीमा के अध्यधीन), जो भी अधिक हो, का डिमांड ड्राफ्ट द्वारा भुगतान किया जाएगा, जो एसयीक्यूसी में जमा जमा होगा। यदि शिकायत गलत पाई जाती है, तो शिकायत शुल्क जब्त कर लिया जाएगा। यदि, शिकायत को बरकरार रखा जाता है और काफी हद तक सही पाया जाता है तो शिकायतकर्ता को जमा शुल्क बिना किसी ब्याज के वापस कर दिया जाएगा।

11.6. सामान्य वित्तीय नियम (जीएफआर) के नियम 175(1)(i)(एच) के तहत झूठी घोषणाएं सत्यनिष्ठा संहिता का उल्लंघन होंगी, जिसके लिए बोली लगाने वाले या उसके उत्तराधिकारियों को नियम 151 (iii) के अनुसार प्रतिबंधित किया जा सकता है। सामान्य वित्तीय नियमों के साथ-साथ बोली लगाने के ऐसे अन्य प्रावधान, एकल मंत्रालय/विभाग द्वारा रोक लगाना और व्यय विभाग के कार्यालय ज्ञापन संख्या एफ.1/20/2018-पीपीडी दिनांक 2 नवंबर, 2021 के अनुसार आदेशों को रद्द करना समय-समय पर संशोधित कानून के तहत अनुमन्य हो सकता है।

12. इलेक्ट्रॉनिक्स उत्पाद अधिसूचनाओं के कार्यान्वयन की निगरानी के लिए एमईआईटीवाई नोडल मंत्रालय होगा। एसटीक्यूसी द्वारा दिए गए निर्णय के विरुद्ध किसी विवाद/स्पष्टीकरण के मामले में, शिकायतों के सत्यापन/पुनरीक्षण, यादृच्छिक आधार पर स्व-घोषणाओं और लेखा परीक्षक/लेखाकार के प्रमाणपत्रों के स्वतंत्र सत्यापन, इलेक्ट्रॉनिकी उत्पादों के घरेलू विनिर्माताओं के विरुद्ध प्रतिबंधात्मक और भेदभावपूर्ण निबंधन और शर्तों के लिए निम्नलिखित घटकों वाली एक समिति गठित की जाती है: -

i)	समूह समन्वयक/वैज्ञानिक- 'जी' (आईपीएचडब्ल्यू)	- अध्यक्ष
ii)	एसटीक्यूसी से वैज्ञानिक-'जी'/वैज्ञानिक-'एफ'	- सदस्य
iii)	वैज्ञानिक- 'ई'/सर्ट-इन/साइबर डिवीजन अथवा वरिष्ठ.	- सदस्य
vi)	अध्यक्ष द्वारा सहयोजित कोई अन्य सदस्य (सदस्यों)	-सदस्य
v)	आईपीएचडब्ल्यू प्रभाग से वैज्ञानिक-'ई'/प्रतिनिधि	-सदस्य सचिव

13. यदि प्रश्न उठता है कि क्या खरीदी जा रही कोई वस्तु एक इलेक्ट्रॉनिक उत्पाद है, जिसे सार्वजनिक क्रय (मेक इन इंडिया को वरीयता) आदेश 2017 के तहत शामिल किया जाना है, तो इस मामले को स्पष्टीकरण के लिए इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय को संदर्भित किया जाएगा।

[फ़ा. सं. डब्ल्यू-43/11/2021-आईपीएचडब्ल्यू]

आशा नांगिया, समूह समन्वयक और वैज्ञानिक 'जी'

परिशिष्ट 'क'

सीसीटीवी की सुरक्षा के लिए अनिवार्य आवश्यकता

संवेदनशील जानकारी की सुरक्षा और सिस्टम को प्रभावी ढंग से संचालित करने के लिए सीसीटीवी (क्लोज-सर्किट टेलीविजन) प्रणाली को सुरक्षित करना महत्वपूर्ण है। परीक्षण के प्रमुख क्षेत्रों में एक्स्पोस्ड नेटवर्क सेवाएं, डिवाइस संचार प्रोटोकॉल, डिवाइस के यूएआरटी,जेटीएजी,एसडब्ल्युडी आदि तक भौतिक पहुंच, मेमोरी और फर्मवेयर निकालने की क्षमता, फर्मवेयर अपडेट प्रक्रिया सुरक्षा और डेटा का भंडारण और एन्क्रिप्शन शामिल हैं। सीसीटीवी सिस्टम की सुरक्षा के लिए यहां संक्षिप्त आवश्यकताएं दी गई हैं:

		और विकास प्रक्रियाओं के हिस्से के रूप में एक या अधिक अद्यतन मैलवेयर पहचान उपकरण नियोजित किए जाएंगे। अंतिम पैकेजिंग और प्रदायगी से पहले मैलवेयर पहचान तकनीकों का उपयोग किया जाएगा (उदाहरण के लिए, एक या अधिक अद्यतन मैलवेयर पहचान उपकरणों का उपयोग करके मैलवेयर के लिए तैयार उत्पादों और घटकों को स्कैन करना)।	पहचान टैनिंग/जालसाजी, सीएम टूल के ट्रैकिंग लक्ष्यों की आवश्यकता के रूप में की गई है। गुणवत्ता आश्वासन प्रक्रिया को प्रस्तुत करने और उसे प्रदर्शित करने की आवश्यकता है।	
		4.4 आपूर्ति श्रृंखला जोखिम की पहचान, मूल्यांकन, प्राथमिकता और शमन आयोजित किया जाएगा।		आपूर्ति श्रृंखला जोखिम / व्यापार निरंतरता योजना नीति दस्तावेज, प्लेबुक जो दर्शाती है कि आपूर्ति श्रृंखला व्यवधान को कैसे संभालना है, घटना के बाद सारांश दस्तावेजों को प्रस्तुत करने और उसी को प्रदर्शित करने की आवश्यकता है।

MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

(IPHW DIVISION)

NOTIFICATION

New Delhi, the: 6th March, 2024

Subject: Public Procurement (Preference to Make in India) Order 2017-Notifying CCTV/ Video Surveillance System for Security in furtherance of the Order

Reference:

(i) Department for Promotion of Industry and Internal Trade (DPIIT) Order No. P-45021/2/2017-B.E.-II dated 15.06.2017, as amended by Orders dated 28.05.2018, 29.05.2019, 04.06.2020 and 16.09.2020

(ii) Ministry of Electronics and Information Technology (MeitY) Electronic Products Notification No. W-43/4/2019-IPHW-MeitY dated 07.09.2020

S.O. 1119(E).—The Government has issued Public Procurement (Preference to Make in India) Order 2017 vide the Department for Promotion of Industry and Internal Trade (DPIIT) Order No.P-45021/2/2017-B.E.-II dated 15.06.2017, as amended by Orders dated 28.05.2018, 29.05.2019, 04.06.2020 and 16.09.2020, to encourage 'Make in India' and promote manufacturing and production of goods and services in India with a view to enhancing income and employment.

2. In furtherance of the Public Procurement (Preference to Make in India) Order 2017 [PPP-MII Order 2017] notified vide reference cited above, the Ministry of Electronics and Information Technology (MeitY) hereby notifies

that preference shall be provided by all procuring entities to locally manufactured Video Surveillance System for Security as per the aforesaid Order, as amended from time to time.

3. For the purpose of this notification:

3.1 The definition of Class-I local supplier, Class-II local supplier and Non-local supplier shall be as per paragraph 2 of the DPIIT PPP-MII Order 2017 No. P-45021/2/2017-PP(BE-II) dated 16.09.2020, as amended from time to time. The mechanism for calculation of local content is prescribed for each notified Electronic Product in this notification.

3.2 Paragraph 3A of the DPIIT PPP-MII Order 2017 No. P-45021/2/2017-PP(BE-II) dated 16.09.2020, as amended from time to time, shall be referred for percentage of procurement for which preference to domestically manufactured Electronic Products is to be provided (in value terms).

4. Following electronic products of Video Surveillance System are notified under the Public Procurement (Preference to Make in India) Order 2017:

4.1 Video Surveillance System

Definition:

For the purpose of this Notification, Video Surveillance System is usually a system that *inter-alia* includes: (i) Closed Circuit Television (CCTV) Camera (Analog/ IP/ Analog Speed Dome/ IP Speed Dome), (ii) Digital Video Recorder (DVR) / Network Video Recorder (NVR). Some of the applications of Security and Video Surveillance System are surveillance of Cities, Schools, Banks, Government Offices, public places, traffic monitoring and home security, etc.

4.2 Mechanism for calculation of local content of CCTV Camera (Analog):

The domestic BOM of CCTV Camera (Analog) would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

<table>
<tr><th>Sr. No.</th><th>Main inputs in BOM / stages for manufacture of CCTV Camera (Analog)</th><th>Value addition / local content required for the input to be classified as domestic BOM</th></tr>
<tr><td></td><td>(1)</td><td>(2)</td></tr>
<tr><td>1.0</td><td>Main PCB*</td><td rowspan="4">Domestic PCB Assembly* and testing from imported / domestically manufactured parts and components, including the value of Semiconductors** and excluding the value of bare PCB. However, the weightage of the total value of Main PCB with/without optional boards shall not exceed 10 % of the total BOM of the CCTV Camera (Analog)</td></tr>
<tr><td>(i)</td><td>I/O Board (optional)</td></tr>
<tr><td>(ii)</td><td>Infrared (IR) Board (optional)</td></tr>
<tr><td>(iii)</td><td>Control Board (optional)</td></tr>
<tr><td>2.0</td><td>Bare PCB</td><td>Domestically manufactured from domestically manufactured inputs. The weightage of the total value of Bare PCB shall not exceed 5 % of the total BOM of the CCTV Camera (Analog)</td></tr>
<tr><td>3.0</td><td>Optics</td><td>Domestically manufactured from domestically manufactured inputs. The weightage of the total value of Optics shall not exceed 10 % of the total BOM of the CCTV Camera (Analog)</td></tr>
<tr><td>4.0</td><td>In-house R&D #</td><td>Domestically/In-house R&D includes scientific research, experimentation, prototyping and testing for development of new products, process, technologies, or solutions using its own resources/expertise or legal contract with R&D unit registered with Department of Scientific and Industrial Research (DSIR) in India. The weightage of this will be calculated based on the percentage of the domestic BOM in respective Financial Year as mentioned in below table.:
<table>
<tr><td>2023-24 & 2024-25</td><td>2025-26</td><td>2026-27 onwards</td></tr>
<tr><td>7%</td><td>10%</td><td>12%</td></tr>
</table></td></tr>
<tr><td>5.0</td><td>In-house Design/IPR #</td><td>Domestically/In-house Designing involves the process of design practices, services, or solutions within the organization's own facilities, using its own resources/expertise or legal contract with third party in India. Any patents, trademarks, copyrights, and trade secrets registered under Intellectual Property</td></tr>
</table>

		Rights (IPR) in India in favor of the design for their creations or inventions establishes legal ownership. The weightage of this will be calculated based on the percentage of the domestic BOM in respective Financial Year as mentioned in below table.:

2023-24 & 2024-25	2025-26	2026-27 onwards
7%	10%	12%

6.0	Housing (Plastic /Aluminium/MS) and Camera Mount	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Housing and Camera Mount shall not exceed to 15 % of the total BOM of the CCTV Camera (Analog).
7.0	Connecting Cables & Connectors	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Connecting Cables and Connectors shall not exceed 5% of the total BOM of the CCTV Camera (Analog).
8.0	Final Assembly and Testing	Domestically assembled / tested meeting Indian Standards as notified from time to time. The weightage of the total value of final assembling and testing shall not exceed 10% of the total BOM of the CCTV Camera.

** It is essential that the Printed Circuit Board Assembly (PCBA) of the parts and components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

 # The applicant will be eligible to avail the benefit of Sr. No. 4.0 (In-house R&D) & Sr. No. 5.0 (In-house Design/IPR) in respective financial year for calculation of Value Addition/Local Content by meeting the minimum Value Addition/Local Content mentioned in below table for respective financial year.

Financial Year	2023-24 & 2024-25	2025-26	2026-27 onwards
Minimum Value Addition/Local Content of (Sr. No. 1) Main PCB and (Sr. No.8) Final Assembly and Testing as -Essential and others as optional	25%	35%	45%

4.3 Mechanism for calculation of local content of CCTV Camera (IP):

The domestic BOM of CCTV Camera (IP) would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Sr. No.	Main inputs in BOM / stages for manufacture of CCTV Camera (IP)	Value addition / local content required for the input to be classified as domestic BOM
	1	2
1.0	Main PCB* with Capture and/or Processor Card	Domestic PCB Assembly* and testing from imported / domestically manufactured parts and components, including the value of Semiconductors** and excluding the value of bare PCB. However, the weightage of the total value of Main PCB with/without optional boards shall not exceed 10 % of the total BOM of the CCTV Camera (IP)
(i)	Network Interface Card (optional)	
(ii)	Infrared (IR) Board (optional)	
(iii)	I/O Board (optional)	
(iv)	Power over Ethernet (PoE) Card (optional)	
2.0	Bare PCB	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of Bare PCB shall not exceed 5 % of the total BOM of the CCTV Camera (IP)
3.0	Optics	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of Optics shall not exceed 10 % of the total BOM of the CCTV Camera (IP)
4.0	In-house R&D #	Domestically/In-house R&D includes scientific research, experimentation,

		prototyping and testing for development of new products, process, technologies, or solutions using its own resources/expertise or legal contract with R&D unit registered with *Department of Scientific and Industrial Research (DSIR)* in India. The weightage of this will be calculated based on the percentage of the domestic BOM in respective Financial Year as mentioned in below table.:

2023-24 & 2024-25	2025-26	2026-27 onwards
7%	10%	12%

5.0	In-house Design/IPR #	Domestically/In-house Designing involves the process of design practices, services, or solutions within the organization's own facilities, using its own resources/expertise or legal contract with third party in India. Any patents, trademarks, copyrights, and trade secrets registered under Intellectual Property Rights (IPR) in India in favor of the design for their creations or inventions establishes legal ownership. The weightage of this will be calculated based on the percentage of the domestic BOM in respective Financial Year as mentioned in below table.:

2023-24 & 2024-25	2025-26	2026-27 onwards
7%	10%	12%

6.0	Housing (Plastic / Aluminium / MS) and	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Housing and Camera Mount shall not exceed 15 % of the total BOM of the CCTV Camera (IP).
7.0	Connecting Cables and Connectors	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Connecting Cables and Connectors shall not exceed 5% of the total BOM of the CCTV Camera (IP).
8.0	Final Assembly and Testing	Domestically assembled / tested meeting Indian Standards as notified from time to time. The weightage of the total value of Final Assembling and Testing shall not exceed 10% of the total BOM of the CCTV Camera (IP)

** It is essential that the Printed Circuit Board Assembly (PCBA) of the parts and components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

 # The applicant will be eligible to avail the benefit of Sr. No. 4.0 (In-house R&D) & Sr. No. 5.0 (In-house Design/IPR) in respective financial year for calculation of Value Addition/Local Content by meeting the minimum Value Addition/Local Content mentioned in below table for respective financial year.

Financial Year	2023-24 & 2024-25	2025-26	2026-27 onwards
Minimum Value Addition/Local Content of (Sr. No. 1.0) Main PCB and (Sr. No. 8.0) Final Assembly and Testing as -Essential and others as optional	25%	35%	45%

4.4 Mechanism for calculation of local content of CCTV Camera (Analog Speed Dome):

The domestic BOM of CCTV Camera (Analog Speed Dome) would be the sum of the costs of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Sr. No.	Main inputs in BOM / stages for manufacture of CCTV Camera (Analog Speed Dome)	Value addition required for the input to be classified as domestic BOM
	1	2
1	Main Controller Board*	Domestic PCB Assembly* and testing from imported / domestically manufactured parts and components, including the value of Semiconductors** and excluding the value of bare PCB. However, the weightage of the total value of Main Controller Board/Main PCB with/without optional boards shall not exceed 10 % of the total BoM of the CCTV Camera (ASD)
(i)	I/O Board (optional)	
(ii)	Infrared (IR) Board (optional)	
2.0	Bare PCB	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of Bare PCB shall not exceed 5 % of the total

<table>
<tr><td></td><td></td><td>BoM of the CCTV Camera (ASD)</td></tr>
<tr><td>3.0</td><td>Inhouse R&D#</td><td>Domestically/In-house R&D includes scientific research, experimentation, prototyping and testing for development of new products, process, technologies, or solutions using its own resources/expertise or legal contract with R&D unit registered with Department of Scientific and Industrial Research (DSIR) in India. The weightage of this will be calculated based on the percentage of the domestic BoM in respective Financial Year as mentioned in below table.:
<table>
<tr><td>2023-24 & 2024-25</td><td>2025-26</td><td>2026-27 onwards</td></tr>
<tr><td>7%</td><td>10%</td><td>12%</td></tr>
</table></td></tr>
<tr><td>4.0</td><td>In-house Design/IPR#</td><td>Domestically/In-house Designing involves the process of design practices, services, or solutions within the organization's own facilities, using its own resources/expertise or legal contract with third party in India. Any patents, trademarks, copyrights, and trade secrets registered under Intellectual Property Rights (IPR) in India in favor of the design for their creations or inventions establishes legal ownership. The weightage of this will be calculated based on the percentage of the domestic BoM in respective Financial Year as mentioned in below table.:
<table>
<tr><td>2023-24 & 2024-25</td><td>2025-26</td><td>2026-27 onwards</td></tr>
<tr><td>7%</td><td>10%</td><td>12%</td></tr>
</table></td></tr>
<tr><td>5.0</td><td>Housing (Plastic/ Aluminium/ MS) and Camera Mount</td><td>Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Housing and Camera Mount shall not exceed 5 % of the total BoM of the CCTV Camera (ASD).</td></tr>
<tr><td>6.0</td><td>Connecting Cables and Connectors</td><td>Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Connecting Cables and Connectors shall not exceed 5% of the total BoM of the CCTV Camera (ASD).</td></tr>
<tr><td>7.0</td><td>Motor, Heater & Blower, Zoom Module, Belt, Plastic Mechanism Parts, Glass etc.</td><td>Domestically manufactured from imported / domestically manufactured input parts and mechanical components. However, the weightage of the total value of all these parts/components shall not exceed 5% of the total BoM of the CCTV Camera (ASD)</td></tr>
<tr><td>8.0</td><td>Power Adapter</td><td>Domestic PCB assembly* and the final assembly and testing from imported / domestically manufactured parts and components, subject to the condition that the value of domestically manufactured parts and components used in the assembly of "Power Adapter" will be minimum 40% (of the total value of parts and components used in the manufacture of "Power Adapter"). However, the weightage of the total value of "Power Adapter" shall not exceed 5% of the total BoM of the CCTV Camera (ASD)</td></tr>
<tr><td>9.0</td><td>Final Assembly and Testing</td><td>Domestically assembled / tested meeting Indian Standards as notified from time to time. The weightage of the total value of final assembling and testing shall not exceed 10 % of the total BoM of the CCTV Camera (ASD)</td></tr>
</table>

** It is essential that the Printed Circuit Board Assembly (PCBA) of the parts and components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

 # The applicant will be eligible to avail the benefit of Sr. No. 3.0 (In-house R&D) & Sr. No. 4.0 (In-house Design/IPR) in respective financial year for calculation of Value Addition/Local Content by meeting the minimum Value Addition/Local Content mentioned in below table for respective financial year.

Financial Year	**2023-24 & 2024-25**	**2025-26**	**2026-27 onwards**
Minimum Value Addition/Local Content of (Sr. No. 1.0) Main PCB and (Sr. No. 9.0) Final Assembly and Testing as -Essential and others as optional	25%	35%	45%

4.5 Mechanism for calculation of local content of CCTV Camera (IP Speed Dome):

The domestic BOM of CCTV Camera (IP Speed Dome) would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

<table>
<tr><th>Sr. No.</th><th>Main inputs in BOM / stages for manufacture of CCTV Camera (IP Speed Dome)</th><th>Value addition / local content required for the input to be classified as domestic BOM</th></tr>
<tr><td></td><td>1</td><td>2</td></tr>
<tr><td>1.0</td><td>Main Controller Board*</td><td rowspan="6">Domestic PCB Assembly* and testing from imported / domestically manufactured parts and components, including the value of Semiconductors** and excluding the value of bare PCB. However, the weightage of the total value of Main Controller Borad with/without optional boards Shall not exceed 10 % of the total BoM of the CCTV Camera (IPSD)</td></tr>
<tr><td>(i)</td><td>I/O Board (optional)</td></tr>
<tr><td>(ii)</td><td>Infrared (IR) Board (optional)</td></tr>
<tr><td>(iii)</td><td>Processor Card (optional)</td></tr>
<tr><td>(iv)</td><td>Network Interface Card (Optional)</td></tr>
<tr><td>(v)</td><td>Power over Ethernet (PoE) Card (optional)</td></tr>
<tr><td>2.0</td><td>Bare PCB</td><td>Domestically manufactured from domestically manufactured inputs. The weightage of the total value of Bare PCB shall not exceed 5 % of the total BoM of the CCTV Camera (IPSD)</td></tr>
<tr><td>3.0</td><td>In-house R&D #</td><td>Domestically/In-house R&D includes scientific research, experimentation, prototyping and testing for development of new products, process, technologies, or solutions using its own resources /expertise or legal contract with R&D unit registered with Department of Scientific and Industrial Research (DSIR) in India. The weightage of this will be calculated based on the percentage of the domestic BoM in respective Financial Year as mentioned in below table.:
<table>
<tr><td>2023-24 & 2024-25</td><td>2025-26</td><td>2026-27 onwards</td></tr>
<tr><td>7%</td><td>10%</td><td>12%</td></tr>
</table></td></tr>
<tr><td>4.0</td><td>In-house Design/IPR #</td><td>Domestically/In-house Designing involves the process of design practices, services, or solutions within the organization's own facilities, using its own resources/expertise or legal contract with third party in India. Any patents, trademarks, copyrights, and trade secrets registered under Intellectual Property Rights (IPR) in India in favor of the design for their creations or inventions establishes legal ownership. The weightage of this will be calculated based on the percentage of the domestic BoM in respective Financial Year as mentioned in below table.:
<table>
<tr><td>2023-24 & 2024-25</td><td>2025-26</td><td>2026-27 onwards</td></tr>
<tr><td>7%</td><td>10%</td><td>12%</td></tr>
</table></td></tr>
<tr><td>5.0</td><td>Housing (Plastic / Aluminium / MS) and Camera Mount</td><td>Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Housing and Camera Mount shall not exceed 5 % of the total BoM of the CCTV Camera (IPSD).</td></tr>
<tr><td>6.0</td><td>Connecting Cables and Connectors</td><td>Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Connecting Cables and Connectors shall not exceed 5% of the total BoM of the CCTV Camera (IPSD).</td></tr>
<tr><td>7.0</td><td>Motor, Heater & Blower, Zoom Module, Belt, Plastic Mechanism Parts, Glass etc.</td><td>Domestically manufactured from imported / domestically manufactured input parts and mechanical components. However, the weightage of the total value of all these parts/components shall not exceed 5% of the total BoM of the CCTV Camera (IPSD).</td></tr>
<tr><td>8.0</td><td>Power Adapter</td><td>Domestic PCB assembly* and the final assembly and testing from imported / domestically manufactured parts and components, subject to the condition that the value of domestically manufactured parts and components used in the</td></tr>
</table>

		assembly of "Power Adapter" will be minimum 40% (of the total value of parts and components used in the manufacture of "Power Adapter"). However, the weightage of the total value of "Power Adapter" shall not exceed 5% of the total BoM of the CCTV Camera (IPSD)
9.0	Final Assembly and Testing	Domestically assembled / tested meeting Indian Standards as notified from time to time. The weightage of the total value of final assembling and testing shall not exceed 10 % of the total BoM of the CCTV Camera (IPSD)

** It is essential that the Printed Circuit Board Assembly (PCBA) of the parts and components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

 # The applicant will be eligible to avail the benefit of Sr. No. 3.0 (In-house R&D) & Sr. No. 4.0 (In-house Design/IPR) in respective financial year for calculation of Value Addition/Local Content by meeting the minimum Value Addition/Local Content mentioned in below table for respective financial year.

Financial Year	2023-24 & 2024-25	2025-26	2026-27 onwards
Minimum Value Addition/Local Content of (Sr. No. 1.0) Main PCB and (Sr. No. 9.0) Final Assembly and Testing as -Essential and others as optional	25%	35%	45%

4.6 Mechanism for calculation of local content of DVR/ NVR:

The domestic BOM of DVR/ NVR would be the sum of the cost of main inputs as specified in Column 1 of the following table, provided the inputs individually satisfy the value addition requirement specified in Column 2 of the table:

Sr No.	Main inputs in BOM / stages for manufacture of DVR / NVR	Value addition / local content required for the input to be classified as domestic BOM
	1	2
1.0	Main PCB*	Domestic PCB Assembly* and testing from imported / domestically manufactured parts and components, including the value of Semiconductors** and excluding the value of bare PCB. However, the weightage of the total value of Main PCB with/without optional boards shall not exceed 10 % of the total BoM of the DVR/NVR.
(i)	Front Panel & LED Board (optional)	
(ii)	I/O Board (optional)	
2.0	Bare PCB	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of Bare PCB shall not exceed 5 % of the total BoM of the CCTV Camera (DVR/NVR)
3.0	In-house R&D #	Domestically/In-house R&D includes scientific research, experimentation, prototyping and testing for development of new products, process, technologies, or solutions using its own resources/expertise or legal contract with R&D unit registered with *Department of Scientific and Industrial Research (DSIR)* in India. The weightage of this will be calculated based on the percentage of the domestic BOM in respective Financial Year as mentioned in below table.:
		2023-24 & 2024-25: 7% / 2025-26: 10% / 2026-27 onwards: 12%
4.0	In-house Design/IPR #	Domestically/In-house Designing involves the process of design practices, services, or solutions within the organization's own facilities, using its own resources/expertise or legal contract with third party in India. Any patents, trademarks, copyrights, and trade secrets registered under Intellectual Property Rights (IPR) in India in favor of the design for their creations or inventions establishes legal ownership. The weightage of this will be calculated based on the percentage of the domestic BOM in respective Financial Year as mentioned in below table.:

		2023-24 & 2024-25	2025-26	2026-27 onwards
		7%	10%	12%
5.0	Housing (Plastic / Aluminium / MS) and Camera Mount	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Housing and Camera Mount shall not exceed 5 % of the total BoM of the DVR/NVR.		
6.0	Connecting Cables, Connectors and USB Mouse	Domestically manufactured from domestically manufactured inputs. The weightage of the total value of inputs for Connecting Cables, Connectors and USB Mouse shall not exceed 5% of the total BoM of the DVR/NVR.		
7.0	Remote Control	Domestic assembly* and testing from imported / domestically manufactured parts and components subject to the condition that the value of domestically manufactured parts and components used in the assembly of "Remote Control" will be minimum 20% (of the total value of the parts and components used in the manufacture of "Remote Control") in Year 2, which will increase to minimum 30%, 40% and 50% in Financial Years 2025-26, 2026-27 and 2027-28 onwards respectively.		
8.0	Power Adapter	Domestic PCB assembly* and the final assembly and testing from imported / domestically manufactured parts and components, subject to the condition that the value of domestically manufactured parts and components used in the assembly of "Power Adapter" will be minimum 40% (of the total value of parts and components used in the manufacture of "Power Adapter"). However, the weightage of the total value of "Power Adapter" shall not exceed 5% of the total BoM of the DVR/NVR.		
9.0	Final Assembly and Testing	Domestically assembled/tested meeting Indian Standards as notified from time to time. The weightage of the total value of final assembling and testing shall not exceed 10 % of the total BOM of the DVR/NVR.		

** It is essential that the Printed Circuit Board Assembly (PCBA) of the parts and components on the bare PCB using the SMT process should mandatorily be done in India.*

*** This shall be reviewed when the Semiconductor FAB in India is operational.*

 # The applicant will be eligible to avail the benefit of Sr. No. 4.0 (In-house R&D) & Sr. No. 5.0 (In-house Design/IPR) in respective financial year for calculation of Value Addition/Local Content by meeting the minimum Value Addition/Local Content mentioned in below table for respective financial year.

Financial Year	2023-24 & 2024-25	2025-26	2026-27 onwards
Minimum Value Addition/Local Content of (Sr. No. 1.0) Main PCB and (Sr. No. 9.0) Final Assembly and Testing as -Essential and others as optional	25%	35%	45%

5. The surveillance devices i.e., CCTV Cameras (Analog/ IP/ Analog Speed Dome/ IP Speed Dome) should comply with the Essential Requirements (ERs) for security prescribed by MeitY to ensure the security of the VSS / CCTV systems, as per the Appendix 'A', as amended from time to time. The security testing report for CCTV/VSS to be issued by Standardisation Testing and Quality Certification (STQC) Laboratory or any other agency notified by MeitY from time to time. The validity of the test report issued by STQC Lab will be three years from the date of issue of the report. These Essential Requirements (ERs) for security of CCTV Cameras/VSS system will be enforced after 3 months from the date of issuance of this Notification. The security norms for the DVR and NVR will be notified subsequently.

6. The notification will come into effect after 3 months from the date of issuance of this Notification. This Notification shall remain valid till the revised Notification is issued. The Year 1 for the purpose of this notification would be upto 31.03.2025.

7. No Electronic Product Notification under the Public Procurement (Preference to Make in India) Order 2017 shall have retrospective effect.

8. Purchase Preference shall be provided as per the provisions cited in the Public Procurement (Preference to Make in India) Order 2017 dated 16.09.2020 and, as amended from time to time, for the procurement of aforesaid electronics products.

9. The notification would also be applicable to all Central Schemes (CS)/ Central Sector Schemes (CSS) for the procurement of electronic products made by States and local bodies, if project or scheme is fully or partially funded by Government of India.

10. Procedure for calculating local content/ domestic value addition

10.1 Bill of Material sourced from domestic manufacturers (Dom-BOM) may be calculated based on one of the followings depending on data available:

a) Sum of the costs of all inputs which go into the product (including duties and taxes levied on procurement of inputs except those for which credit/ set-off can be taken) and which have not been imported directly or through a domestic trader or an intermediary.

b) Ex-Factory Price of product minus profit after tax minus sum of imported Bill of Material used (directly or indirectly) as inputs in producing the product (including duties and taxes levied on procurement of inputs except those for which credit/ set-off can be taken) minus warranty costs.

c) Market price minus post-production freight, insurance and other handling costs minus profit after tax minus warranty costs minus sum of Imported Bill of Material used as inputs in producing the product (including duties and taxes levied on procurement of inputs except those for which credit / set-off can be taken) minus sales and marketing expenses.

10.2 Total Bill of Material (Total BOM) may be calculated based on one of the following depending on data available:

a) Sum of the costs of all inputs which go into the product (including duties and taxes levied on procurement of inputs except those for which credit / set-off can be taken).

b) Ex-Factory Price of product minus profit after tax, minus warranty costs.

c) Market price minus post-production freight, insurance and other handling costs minus profit after tax, minus warranty costs minus sales and marketing expenses.

10.3 The percentage of domestic value-addition may be calculated based on information furnished as per the following formula:

Percentage of local content/ domestic value-addition = [(Dom-BOM) / (Total BOM)]*100

11. Verification of local content/ Domestic Value Addition

11.1. The local supplier at the time of tender, bidding or solicitation shall provide self-certification that the item offered meets the minimum local content and shall give details of the location(s) at which the local value addition is made.

11.2. In cases of procurement for a value in excess of Rs.10 crore, the local supplier shall provide a certificate from the statutory auditor or cost auditor of the company (in the case of companies) or from a practicing cost accountant or practicing chartered accountant (in respect of suppliers other than companies) giving the percentage of local content.

11.3. In case a complaint is received by the procuring agency or the concerned Ministry/Department against the claim of a bidder regarding local content/ domestic value addition in an Electronic Product(s), the same shall be referred to STQC.

11.4. Any complaint referred to STQC shall be disposed of within 4 weeks. The bidder shall be required to furnish the necessary documentation in support of the domestic value addition claimed in an Electronic Product(s) to STQC. If no information is furnished by the bidder, such laboratories may take further necessary action, to establish the bonafides of the claim. In case of the discrepancies in computation of local content, a certificate from the statutory auditor or cost auditor of the company (in the case of companies) or from a practicing cost accountant or practicing chartered accountant (in respect of suppliers other than companies) for calculating the percentage of local content will supersede over the self-declaration certificate.

11.5. A complaint fee of Rs.2 Lakh or 1% of the value of the domestically manufactured Electronic Products being procured (subject to a maximum of Rs. 5 Lakh), whichever is higher, to be paid by Demand Draft to be deposited with STQC. In case, the complaint is found to be incorrect, the complaint fee shall be forfeited. In case, the complaint is upheld and found to be substantially correct, deposited fee of the complainant would be refunded without any interest.

11.6. False declarations will be in breach of the Code of Integrity under Rule 175(1)(i)(h) of the General Financial Rules (GFR) for which a bidder or its successors can be debarred as per Rule 151 (iii) of the General Financial Rules along with such other provisions of bidding, debarment by a single Ministry/Department and revocation of orders as per the Department of Expenditure O.M-No. F.1/20/2018-PPD dated 2nd Nov, 2021 may be permissible under law amended from time to time.

12. MeitY shall be the Nodal Ministry to monitor the implementation of the Electronic Products notifications. In case of any dispute/clarifications against the decision given by STQC, a committee is constituted with the following

compositions for verification/vetting the complaints, independent verification of self-declarations and auditor's/accountant's certificates on random basis, restrictive and discriminating terms and conditions against the domestic manufactures of Electronics Products(s): -

i)	Group Coordinator/Scientist- 'G' (IPHW)	-	Chairperson
ii)	Scientist-'G'/Scientist-'F' from STQC	-	Member
iii)	Scientist- 'E' or above from Cert-In/Cyber Division	-	Member
vi)	Any other member(s) as co-opted by the Chairperson	-	Member
v)	Representative from IPHW Division	-	Member Secretary

13. In case of a question whether an item being procured is an Electronic Product(s) to be covered under the Public Procurement (Preference to Make in India) Order 2017, the matter would be referred to the Ministry of Electronics and Information Technology for clarification.

[F. No. W-43/11/2021-IPHW]

ASHA NANGIA, Group Coordinator & Scientist 'G'

Appendix 'A'

Essential Requirement(s) for Security of CCTV

Securing a CCTV (Closed-Circuit Television) system is crucial to protect sensitive information and ensure the system operates effectively. Key areas of testing include exposed network services, device communication protocols, physical access to the device's UART, JTAG, SWD, etc., the ability to extract memory and firmware, firmware update process security and storage and encryption of data. Here are brief requirements for the security of a CCTV system:

1) Physical Security - Use tamper-resistant camera enclosures and locking mechanisms to deter physical tampering.
2) Access Control by Authentication, Role-Based Access Control (RBAC) and regularly review and update access permissions to reflect personnel changes.
3) Network Security by employing encryption of data transmission
4) Software Security by Regular Updates, Disable Unused Features and Strong Password Policies
5) Penetration Testing: Employ penetration testing to assess the system's resistance to cyberattacks and address vulnerabilities.

Essential Security Requirements

Sr. No.	Category	Testing Parameter	What to be tested	Documents Required
1)	Hardware Level Security Parameter (supported by software)	1.1 Verify that application layer debugging interfaces such USB, UART, and other serial variants are disabled or protected by a complex password.	1. Identification of the availability of debugging interfaces such as USB, UART, and other serial variants through the Datasheet of the SoC being used in the device under test 2. Verification and validation of the ports/interfaces enabled in the production devices and the related access control mechanism for protection of the same as declared in the vendor documentation 3. Testing, in presence of	1. Datasheet of the SoC being used in the device. 2. Documentation related to ports/interfaces enabled in the production devices and the related access control mechanism for protection of the same. 3. Process flow of the Manufacturing/Provisioning of the device

			OEM team, to verify the enabling/disabling of all the ports and debugging interfaces such as USB, UART, and other serial variants using their relevant hardware based debuggers and access control mechanisms in case the interface is enabled. 4. Process audit of the manufacturing facility to validate the vendor's claim regarding the debugging interfaces which are closed/disabled during provisioning. [For instance, through Block connection diagram depicting pin connections between the host microcontroller and its interactions with various sub components/peripherals.]	
		1.2 Verify that cryptographic keys and certificates are unique to each individual device.	Identifying all the keys and certificates being used in the device eco-system and verification through: • Testing, in presence of OEM team • Code review • Process audit of the key-life cycle process	1. List of all keys and certificates being used in the device ecosystem 2. Key management life cycle (purpose, generation, storage, destruction/zeroization, validity, key changeover/rotation)
		1.3 Verify that on-chip debugging interfaces such as JTAG or SWD are disabled or that available protection mechanism is enabled and configured appropriately.	1. Identification of the availability of debugging interfaces such as USB, UART, and other serial variants through the Datasheet of the SoC being used in the device under test 2. Verification and validation of the ports/interfaces enabled in the production devices and the related access control mechanism for protection of the same as declared in the vendor documentation 3. Testing, in presence of OEM team, to verify the enabling/disabling of all the	1. Datasheet of the SoC being used in the device. 2. Documentation related to ports/interfaces enabled in the production devices and the related access control mechanism for protection of the same. 3. Process flow of the Manufacturing/Provisioning of the device

			ports and debugging interfaces such as USB, UART, and other serial variants using their relevant hardware based debuggers and access control mechanisms in case the interface is enabled. 4. Process audit of the manufacturing facility to validate the vendor's claim regarding the debugging interfaces which are closed/disabled during provisioning. [For instance, through Block connection diagram depicting pin connections between the host microcontroller and its interactions with various sub components/peripherals.]	
		1.4 Verify that trusted execution is implemented and enabled, if available on the device SoC or CPU.	Identifying whether TEE/SE/TPM is available or not in the device through the SoC datasheet and technical documentation submitted by the vendor. Further assessment is done on the basis of scenarios as applicable to device as defined below: **CASE 1: TEE/SE/TPM is not available:** No further assessment **CASE 2: TEE/SE/TPM is available and enabled:** Verification through code-review that crypto functions are called through TEE/SE/TPM APIs. **CASE 3: TEE/SE/TPM is available but not enabled by the vendor:** Termed as non-conformance to the requirement. OEM is required to enable and implement the TEE/SE/TPM.	1. Datasheet of the SoC being used in the device. 2. User manual/ Technical specifications of the device 3. Code snippets of the TEE API call, wherever applicable
		1.5 Verify that sensitive data, private keys and certificates are stored securely in a Secure Element, TPM,	Identifying all the keys and certificates being used in the device eco-system, sensitive data and their storage mechanism(s); and verification through:	1. List of all keys and certificates being used in the device ecosystem 2. List of all the sensitive data with their intended usage and secure storage

		TEE (Trusted Execution Environment), or protected using strong cryptography.	• Testing, in presence of OEM team • Code review • Process audit of the key-life cycle process	mechanism(s) as implemented along with secure configurations to be enabled in the device. 3. Key management life cycle (purpose, generation, storage, destruction/zeroization, validity, key changeover/rotation) private keys and certificates.
		1.6Verify the presence of tamper resistance and/or tamper detection features.	Testing, in presence of OEM team, to verify the measures implemented in the device to prevent software and hardware tampering.	1. Measures available in the device to prevent software tampering. 2. Measures available in the device to prevent hardware tampering.
		1.7 Verify that any available Intellectual Property protection technologies provided by the chip manufacturer are enabled.	Testing, in presence of OEM team, to verify the enabling of the Intellectual Property protection technologies provided by the chip manufacturer, if available.	1. Datasheet of the SoC 2. Documentation regarding the Intellectual Property protection technologies provided by the chip manufacturer which have been enabled. 3. In case, no Intellectual Property protection technologies are being provided by the chip manufacturer, then a declaration stating the same.
		1.8 Verify the device validates the boot image signature before loading.	Testing, in presence of OEM team, to verify the following: 1. Device boots up successfully with the documented secure boot process when a valid boot image is provided. 2. Device does not boot up when a tampered boot image (like with missing signature, invalid signature) is provided.	1. Datasheet of the SoC 2.Technical specifications of the device regarding secure boot (should consist of keys involved and their management life cycle*, signature validation process and any other secure mechanisms if implemented.)
		1.9 Verify usage of cryptographically secure pseudo-random number generator on embedded device (e.g., using chip-provided random number	Verification of the documentation provided by the vendor regarding the random number generators being used in the device. Verification through code-review that random number generators or related	Documentation regarding the random generators (either hardware based or software based or both) being used in the device with their intended usage. In case, hardware based random number generators are being used, vendors

		generators).	libraries as applicable are being used in the device.	shall submit the following: 1. Datasheet of the SoC 2. Technical specifications of the device regarding random generators In case, software based random number generators are being used, vendors shall provide the libraries being used for the same.
2)	Software/Firmware	2.1 Verify that memory protection controls such as ASLR and DEP are enabled by the embedded/IoT operating system, if applicable.	Testing, in presence of OEM team, to verify the declared memory protection controls available and enabled in the device using command line-based tools/commands or any other open-source tool like DEP, EMET tool.	Declaration of the memory protection controls available and enabled in the device.
		2.2 Verify that the firmware apps protect data-in-transit using transport layer security.	1. Verifying that strong encryption algorithms and secure TLS version is supported by the device to establish secure communication. 2. Verifying that device properly validates the server's TLS certificate to ensure that it is trusted and has not been tampered with. 3. Testing for vulnerabilities which can affect the security of TLS connection such as padding oracle attacks, or weak cipher suites. 4. Using tools such as Nmap to identify open ports through which device can be accessed leading to unintended data retrieval. 5. Verifying that theTLS session(s) are resistant to attemptsof interception and decryption of network traffic using man-in-the-middle attacks using tools like Burpsuite.	Specifications and documentation related to the configurations available in the applications and firmware related to transport layer security.
		2.3 Verify that the firmware apps validate the digital signature of server connections.	1. Identifying the scenarios when the device establishes the server connections with the external world and verifying the following: • Security features, related to secure server connections and digital	Document mentioning the use-cases when the device establishes server connections with the external world, with detailed information about the security measures in place while validating the digital signatures of the

			signature validation as implemented like strong cipher suites, secure TLS version, SSL pinning etc. supported by code walkthrough. • Proper certificate validation, certificate chain validation and certificate revocation checks are implemented in the device. 2. Testing for vulnerabilities which can affect the security of TLS connection such as padding oracle attacks, or weak cipher suites. 3. Using tools such as Nmap to identify open ports through which device can be accessed leading to unintended data retrieval. 4. Verifying that TLS session(s) are resistant to attemptsof interception and decryption of network traffic using man-in-the-middle attacks using tools like Burpsuite.	server connections.
		2.4 Verify that any use of banned C functions are replaced with the appropriate safe equivalent functions.	Secure code review [both automated and manual], in presence of OEM team, using a licensed static analysis tool through any of the following approaches: 1. Visit to the evaluation agency by the vendor with the firmware code and installing the licensed static analysis tool available with the evaluation agency in their systems. [Recommended] 2. Visit to the evaluation agency by the vendor with the firmware code and any licensed static analysis tool available with them and demonstrating the code review activity in the presence of representatives of evaluation agency.	1. Firmware binaries for code review. 2. Internal code review reports

			3. Giving a remote access of the systems at vendor site to the evaluation agency for installing their licensed static analysis tool available with them. 4. Giving a remote access of the systems at vendor site to the evaluation agency containing the firmware code along with the licensed static analysis tool available with the vendors.	
		2.5 Verify that each firmware maintains a software bill of materials cataloging third party components, versioning, and published vulnerabilities.	Verification of the submitted list of third-party components by running automated tools like FACT on the firmware. Identifying vulnerabilities in the third-party component(s) through publically available vulnerability databases Verification and validation of the process defined by the vendor for providing regular security updates and patches for the firmware to address any known vulnerabilities in third-party components.	1. Documentation for information on software bill of materials, including third-party components and versions. 2. Organization process and policies for the following: • Addressing and patching any identified vulnerabilities in third-party components. • Informing the customers about the security issues or vulnerabilities and providing security updates and patches for the same. 3. Configuration management system and related policies for maintaining firmware and third-party binaries, libraries and frameworks along with the patches/fixes issued to the devices.
		2.6 Verify all code including third-party binaries, libraries, frameworks are reviewed for hardcoded credentials (backdoors).	Independent secure code review [both automated and manual] using a licensed static analysis tool through any of the following approaches: 1. Visit to the evaluation agency by the vendor with the firmware code and installing the licensed static analysis tool available with the evaluation agency in their systems.	1. Firmware binaries for code review. 2. Internal code review reports

			[Recommended] 2. Visit to the evaluation agency by the vendor with the firmware code and any licensed static analysis tool available with them and demonstrating the code review activity in the presence of representatives of evaluation agency. 3. Giving a remote access of the systems at vendor site to the evaluation agency for installing their licensed static analysis tool available with them. 4. Giving a remote access of the systems at vendor site to the evaluation agency containing the firmware code along with the licensed static analysis tool available with the vendors.	
		2.7 Verify that the firmware apps pin the digital signature to a trusted server(s).	1. Identifying the scenarios when the device establishes the server connections with the external world and verifying the following: • Security features, related to secure server connections and digital signature validation as implemented like strong cipher suites, secure TLS version, SSL pinning etc. supported by code walkthrough. • Proper certificate validation, certificate chain validation and certificate revocation checks are implemented in the device.	Document mentioning the use-cases when the device establishes server connections with the external world, with detailed information about the security measures in place while validating the digital signatures of the server connections.
		2.8 Verify security controls are in place to hinder firmware reverse engineering (e.g.removal of verbose debugging symbols).	Testing, in presence of OEM team, to verify the security controls as provided by the vendor to hinder firmware reverse engineering.	Documentation regarding the security controls in place to hinder firmware reverse engineering.
		2.9 Verify that	Testing, in presence of	Measures implemented in

		the firmware update process is not vulnerable to time-of-check vs time-of-use attacks.	OEM team, to verify the measures implemented in the device to make it resistant to time-of-check vs.time-of-use attacks.	the device to make it resistant to time-of-check vs. time-of-use attacks.
		2.10 Verify the device uses code signing and validates firmware upgrade files before installing.	Testing, in presence of OEM team, to verify the following: i) Device gets successfully updated with the documented secure upgrade process when a valid update package is provided. ii) Device does not boot up when a tampered update package (like with missing signature, invalid signature) is provided.	Process of achieving secure firmware upgrade which should consist of keys involved and their management life cycle*, signature validation process and any other secure mechanisms if implemented.
		2.11 Verify that the device cannot be downgraded to old versions (anti-rollback) of valid firmware.	Testing, in presence of OEM team, to verify that the device cannot be downgraded to old versions (anti-rollback) of valid firmware.	Process of achieving secure firmware upgrade which should consist of keys involved and their management life cycle*, signature validation process and any other secure mechanisms if implemented.
		2.12 Verify that firmware can perform automatic firmware updates upon a predefined schedule.	Verification shall be done as per the applicable scenario: **Case 1: Automatic OTA updates are available:** A standard operating procedure for issuing automatic updates/upgrades to the in-field devices is required to be submitted by the vendor which can then be evaluated by the evaluation agency as per C20, C21 and C22 security requirement. **Case 2: Automatic OTA updates are not available and vendor provides manual updates:** A standard operating procedure for issuing manual updates/upgrades to the in-field devices is required to be submitted by the vendor which can then be evaluated by the	1. Modes of updates available i.e. automatic, manual or both. 2. Organizational process and policies regarding the issuing of updates to the devices.

			evaluation agency as per C20, C21 and C22 security requirement.	
3)	Secure Process Conformance	3.1 Verify that wireless communications are mutually authenticated.	Testing, in presence of OEM team, to verify the process of mutual authentication as laid down in the documentation by the vendor.	The documentation regarding the process of mutual authentication as implemented in the device when wireless communications are initiated. In case, the device does not support wireless communications, the vendor shall provide a declaration for the same.
		3.2 Verify that wireless communications are sent over an encrypted channel.	Identifying all the security mechanisms being used in the communication process verification through: • Testing, in presence of OEM team • Code review • Process audit of the key-life cycle process	Documentation regarding the security measures implemented in the device to prevent tampering of the data being sent through wireless mode of communication. In case, the device does not support wireless communications, the vendor shall provide a declaration for the same.
		3.3 Verify that whether trusted sources are being used for sourcing the components of the device i.e. trusted supply chain through a managed Bill of materials for critical hardware components (related to security functions like SoC) is in use.		Bill of materials for critical hardware components (related to security functions like SoC).
		3.4 Supply chain risk identification, assessment, prioritization, and mitigation shall be conducted. Supply chain risk/business continuity planning policy documents, playbooks reflecting how to handle supply chain disruption,		Supply chain risk identification, assessment, prioritization, and mitigation documents. Supply chain risk/business continuity planning policy documents, playbooks reflecting how to handle supply chain disruption, post-incident summary documents.

		post-incident summary documents need to be submitted and demonstrate the same.		
		3.5 Verify the no proprietary network protocols are being used in the device. If yes, then complete implementation details and the source code for the same shall be provided.		Document for Network protocols used in the device.
4)	Security Conformance at product development stage	4.1 Design and architecture details till the PCBA and SoC level to be provided to aid in counterfeit mitigation and malware detection.		Design and architecture documents till the PCBA and SoC level.
		4.2 Threat mitigation strategies for tainted and counterfeit products shall be implemented as part of product development.	Process and method artifacts need to be submitted and demonstrate the same.	
		4.3 One or more up-to-date malware detection tools shall be deployed as part of the code acceptance and development processes. Malware detection techniques shall be used before final packaging and delivery (e.g., scanning finished products and components for malware using one or more up-to-date	List of components that have been identified as requiring tracking targets of tainting/counterfeiting, CM tool. Quality assurance process need to be submitted and demonstrate the same.	

		malware detection tools).		
		4.4 Supply chain risk identification, assessment, prioritization, and mitigation shall be conducted.		Supply chain risk/business continuity planning policy documents, playbooks reflecting how to handle supply chain disruption, post-incident summary documents need to be submitted and demonstrate the same.

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EXTRAORDINARY

भाग II—खण्ड 3—उप-खण्ड (ii)
PART II—Section 3—Sub-section (ii)

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इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय

(आईपीएचडब्ल्यू प्रभाग)

आदेश

नई दिल्ली, 9 अप्रैल, 2024

विषय: "इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी माल में संशोधन" (अनिवार्य पंजीकरण की आवश्यकता) आदेश, 2021"

का.आ. 1652(अ).—भारतीय मानक ब्यूरो अधिनियम, 2016, (2016 का 11) की धारा 25 की उपधारा (3) के साथ पठित धारा 16 की उप-धारा (1) और (2) द्वारा प्रदत्त शक्तियों का प्रयोग करते हुए, केंद्र सरकार का यह मत है कि सार्वजनिक हित में ऐसा करना आवश्यक या समीचीन है, इसके द्वारा "इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी माल (अनिवार्य पंजीकरण के लिए आवश्यकताएं) आदेश, 2021" में निम्नलिखित संशोधन किए जाते हैं:

2. सी.सी.टी.बी. कैमरे हेतु, कॉलम (5) की निम्नलिखित प्रविष्टि को "इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी माल (अनिवार्य पंजीकरण के लिए आवश्यकताएं) आदेश, 2021 की अनुसूची में क्रम संख्या 41 पर जोड़ा जाएगा।

क्रमांक (1)	माल या सामान (2)	भारतीय मानक (3)	भारतीय मानक का शीर्षक (4)	अपेक्षित आवश्यकता (आवश्यकताएँ) (5)
41	सीसीटीवी कैमरा	आईएस 13252 : भाग 1 : 2010	सूचना तकनीकी उपकरण - सुरक्षा सामान्य आवश्यकताएं--	अनुलग्नक के अनुसार सीसीटीवी हेतु अनिवार्य आवश्यकता (आवश्यकताएँ)

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3. "इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी माल (अनिवार्य पंजीकरण के लिए आवश्यकताएं) आदेश, 2021" के प्रावधान इस अधिसूचना के आधार पर उक्त आदेश की अनुसूची में जोड़े गए कॉलम (2) में निर्दिष्ट माल या सामान पर आधिकारिक राजपत्र में इस अधिसूचना के प्रकाशन की तारीख से छह महीने की समाप्ति पर, कॉलम (5) में निर्दिष्ट किए गए संबंधित अपेक्षित आवश्यकताओं के अनुरूप लागू होंगे। बीआईएस अनुरूपता मूल्यांकन विनियम, 2018 की योजना II के अनुसार बीआईएस मान्यता प्राप्त प्रयोगशालाओं से परीक्षण रिपोर्ट जमा करना मानक चिन्ह का उपयोग करने के लिए लाइसेंस प्राप्त करने हेतु एक पूर्व-आवश्यकता होगी।

[फा.सं. डब्ल्यू-43/11/2021-आईपीएचडब्ल्यू]
आशा नांगिया, समूह समन्वयक और वैज्ञानिक 'जी'

अनुलग्नक

सीसीटीवी की सुरक्षा के लिए अनिवार्य आवश्यकता

संवेदनशील जानकारी की सुरक्षा और सिस्टम को प्रभावी ढंग से संचालित करने के लिए सीसीटीवी (क्लोज-सर्किट टेलीविजन) प्रणाली को सुरक्षित करना महत्वपूर्ण है। परीक्षण के प्रमुख क्षेत्रों में एक्स्पोस्ड नेटवर्क सेवाएं, डिवाइस संचार प्रोटोकॉल, डिवाइस के यूएआरटी,जेटीएजी,एसडब्ल्युडी आदि तक भौतिक पहुंच, मेमोरी और फर्मवेयर निकालने की क्षमता, फर्मवेयर अपडेट प्रक्रिया सुरक्षा और डेटा का भंडारण और एन्क्रिप्शन शामिल हैं। सीसीटीवी सिस्टम की सुरक्षा के लिए यहां संक्षिप्त आवश्यकताएं दी गई हैं:

1. भौतिक सुरक्षा - भौतिक छेड़छाड़ को रोकने के लिए छेड़छाड़-प्रतिरोधी कैमरा एन्क्लोज़र और लॉकिंग तंत्र का उपयोग करें।
2. प्रमाणीकरण द्वारा अभिगम नियंत्रण, भूमिका-आधारित अभिगम नियंत्रण (आरबीएसी) और कर्मियों के परिवर्तनों को प्रतिबिंबित करने के लिए अभिगम अनुमतियों की नियमित रूप से समीक्षा और अद्यतनीकरण ।
3. डेटा ट्रांसमिशन के एन्क्रिप्शन को नियोजित करके नेटवर्क सुरक्षा
4. नियमित अपडेट द्वारा सॉफ़्टवेयर सुरक्षा, अप्रयुक्त सुविधाओं को अक्षम करना और सुदृढ़ पासवर्ड नीतियाँ
5. पेनीट्रेशन परीक्षण: साइबर हमलों के लिए सिस्टम के प्रतिरोध का आकलन करने और कमजोरियों को दूर करने के लिए पेनीट्रेशन परीक्षण को नियोजित करें।

अनिवार्य सुरक्षा आवश्यकताएँ

क्रमांक	वर्ग	परीक्षण पैरामीटर	क्या परीक्षण किया जाए	अपेक्षित दस्तावेज़
	हार्डवेयर स्तर सुरक्षा पैरामीटर (सॉफ़्टवेयर द्वारा समर्थित)	1.1 एक जटिल पासवर्ड द्वारा यह सत्यापित करें कि एप्लिकेशन लेयर डिबगिंग इंटरफ़ेस जैसे यूएसबी, यूएआरटी और अन्य सीरियल बेरिएंट अक्षम या संरक्षित हैं।	1. परीक्षण के तहत डिवाइस में उपयोग किए जा रहे एसओसी की डेटाशीट के माध्यम से यूएसबी, यूएआरटी और अन्य सीरियल बेरिएंट जैसे डिबगिंग इंटरफेस की उपलब्धता की पहचान करना 2. विक्रेता दस्तावेज़ीकरण में घोषित की गई सुरक्षा के लिए उत्पादन उपकरणों और संबंधित पहुंच नियंत्रण तंत्र में सक्षम पोर्ट/इंटरफ़ेस का सत्यापन और वेधता हार्डवेयर आधारित डिबगर्स और एक्सेस कंट्रोल तंत्र का उपयोग करके सभी पोर्ट और यूएसबी, यूएआरटी और अन्य सीरियल बेरिएंट जैसे डिबगिंग इंटरफेस को सक्षम/अक्षम करने को सत्यापित करने के लिए ओईएम	विक्रेता द्वारा निम्नलिखित को उपलब्ध करना होगा: 1. डिवाइस में उपयोग किए जा रहे एसओसी की डेटाशीट। 2. उत्पादन उपकरणों में सक्षम पोर्ट/इंटरफ़ेस से संबंधित दस्तावेज़ीकरण और उसकी सुरक्षा के लिए संबंधित एक्सेस नियंत्रण तंत्र। 3. डिवाइस के विनिर्माण/प्रावधान की प्रक्रिया प्रवाह

			टीम की उपस्थिति में परीक्षण। 4. डिबगिंग इंटरफेस के बारे में विक्रेता के दावे को मान्य करने के लिए विनिर्माण सुविधा की प्रक्रिया लेखा परीक्षा जो प्रावधान के दौरान बंद/अक्षम हैं। [उदाहरण के लिए, ब्लॉक कनेक्शन आरेख के माध्यम से होस्ट माइक्रोकंट्रोलर के बीच पिन कनेक्शन और विभिन्न उप घटकों/परिधीय के साथ इसकी बातचीत को दर्शाया गया है।]	
		1.2 सत्यापित करें कि क्रिप्टोग्राफ़िक कुंजी और प्रमाणपत्र प्रत्येक व्यक्तिगत डिवाइस के लिए अद्वितीय हैं।	डिवाइस इको-सिस्टम में उपयोग की जा रही सभी कुंजियों और प्रमाणपत्रों की पहचान करना और इनके माध्यम से सत्यापन करना: ओईएम टीम की उपस्थिति में परीक्षण कोड समीक्षा कुंजी-जीवन चक्र प्रक्रिया संबंधी प्रक्रिया लेखा परीक्षा	विक्रेता द्वारा निम्नलिखित को प्रस्तुत करना होगा: 1. डिवाइस इकोसिस्टम में उपयोग की जा रही सभी कुंजियों और प्रमाणपत्रों की सूची 2. मुख्य प्रबंधन जीवन चक्र (उद्देश्य, उत्पादन, भंडारण, विनाश/शून्यीकरण, वैधता, कुंजी परिवर्तन/रोटेशन)
		1.3 सत्यापित करें कि जेटीएजी या एसडब्ल्यूडी जैसे ऑन-चिप डिबगिंग इंटरफ़ेस अक्षम हैं या उपलब्ध सुरक्षा तंत्र सक्षम और उचित रूप से कॉन्फ़िगर किया गया है।	1. परीक्षण के तहत डिवाइस में उपयोग किए जा रहे एसओसी की डेटाशीट के माध्यम से यूएसबी, यूएआरटी और अन्य सीरियल वेरिएंट जैसे डिबगिंग इंटरफेस की उपलब्धता की पहचान करना 2. विक्रेता दस्तावेज़ में घोषित की गई सुरक्षा के लिए उत्पादन उपकरणों और संबंधित पहुंच नियंत्रण तंत्र में सक्षम पोर्ट/इंटरफ़ेस का सत्यापन और वेधता 3. इंटरफ़ेस सक्षम होने की स्थिति में उनके प्रासंगिक हार्डवेयर आधारित डिबगर्स और एक्सेस कंट्रोल तंत्र का उपयोग करके सभी पोर्ट और यूएसबी, यूएआरटी और अन्य सीरियल वेरिएंट जैसे डिबगिंग इंटरफेस को सक्षम/अक्षम करने को सत्यापित करने के लिए ओईएम टीम की उपस्थिति में परीक्षण। 4. डिबगिंग इंटरफेस के बारे में विक्रेता के दावे को मान्य करने के लिए विनिर्माण सुविधा की प्रक्रिया लेखा परीक्षा जो प्रावधान	विक्रेता द्वारा निम्नलिखित को उपलब्ध करना होगा: 1. डिवाइस में उपयोग किए जा रहे एसओसी की डेटाशीट। 2. उत्पादन उपकरणों में सक्षम पोर्ट/इंटरफ़ेस से संबंधित दस्तावेज़ीकरण और उसकी सुरक्षा के लिए संबंधित एक्सेस नियंत्रण तंत्र। 3. डिवाइस के विनिर्माण/प्रावधान की प्रक्रिया प्रवाह

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			के दौरान बंद/अक्षम हैं। [उदाहरण के लिए, ब्लॉक कनेक्शन आरेख के माध्यम से होस्ट माइक्रोकंट्रोलर के बीच पिन कनेक्शन और विभिन्न उप घटकों/परिधीय के साथ इसकी बातचीत को दर्शाया गया है।]	
		1.4 सत्यापित करें कि विश्वसनीय निष्पादन लागू और सक्षम है, यदि डिवाइस एसओसी या सीपीयू पर उपलब्ध है।	विक्रेता द्वारा प्रस्तुत एसओसी डेटाशीट और तकनीकी दस्तावेज के माध्यम से डिवाइस में टीईई/एसई/टीपीएम उपलब्ध है या नहीं, इसकी पहचान करना। आगे का मूल्यांकन डिवाइस पर लागू परिदृश्यों के आधार पर किया जाता है जैसा कि नीचे परिभाषित किया गया है: **स्थिति 1: टीईई/एसई/टीपीएम उपलब्ध नहीं है:** कोई और अग्रिम मूल्यांकन नहीं **स्थिति 2: टीईई/एसई/टीपीएम उपलब्ध और सक्षम है:** कोड-समीक्षा के माध्यम से सत्यापन कि क्रिप्टो फ़ंक्शन को टीईई/एसई/टीपीएम एपीआई के माध्यम से बुलाया जाता है। **स्थिति 3: टीईई/एसई/टीपीएम उपलब्ध है लेकिन विक्रेता द्वारा सक्षम नहीं किया गया है:** आवश्यकता के अनुरूप न होने के रूप में करार दिया गया। टीईई/एसई/टीपीएम को सक्षम और कार्यान्वित करने के लिए ओईएम की आवश्यकता होती है।	विक्रेता द्वारा निम्नलिखित को उपलब्ध कराना होगा: 1. डिवाइस में उपयोग किए जा रहे एसओसी की डेटाशीट। 2. डिवाइस का उपयोगकर्ता मैनुअल/तकनीकी विनिर्देश 3. टीईई एपीआई कॉल के कोड स्निपेट, जहां भी लागू हो
		1.5 सत्यापित करें कि संवेदनशील डेटा, निजी कुंजियाँ और प्रमाणपत्र एक सुरक्षित तत्व, टीपीएम, टीईई (विश्वसनीय निष्पादन पर्यावरण) में सुरक्षित रूप से संग्रहीत हैं, या सुदृढ़ क्रिप्टोग्राफी का उपयोग करके संरक्षित हैं।	डिवाइस इको-सिस्टम, संवेदनशील डेटा और उनके भंडारण तंत्र में उपयोग की जा रही सभी कुंजियों और प्रमाणपत्रों की पहचान करना; और इसके माध्यम से सत्यापन करना : • ओईएम टीम की उपस्थिति में परीक्षण • कोड समीक्षा • कुंजी-जीवन चक्र प्रक्रिया संबंधी प्रक्रिया लेखा परीक्षा	विक्रेता द्वारा निम्नलिखित को प्रस्तुत करना होगा: 1. डिवाइस इकोसिस्टम में उपयोग की जा रही सभी कुंजियों और प्रमाणपत्रों की सूची 2. डिवाइस में सक्षम किए जाने वाले सुरक्षित कॉन्फ़िगरेशन के साथ कार्यान्वित सभी संवेदनशील डेटा की उनके इच्छित उपयोग और सुरक्षित भंडारण तंत्र (ओं) के साथ सूची। 3. मुख्य प्रबंधन जीवन चक्र (उद्देश्य, उत्पादन, भंडारण, विनाश/शून्यीकरण, वैधता, कुंजी परिवर्तन/रोटेशन) निजी कुंजी

				और प्रमाणपत्र।
		1.6 टैम्पर रेसिस्टेंस और/या टैम्पर का पता लगाने वाली सुविधाओं की उपस्थिति सत्यापित करें।	सॉफ़्टवेयर और हार्डवेयर से टैंपरिंग को रोकने के लिए डिवाइस में लागू किए गए उपायों को सत्यापित करने के लिए ओईएम टीम की उपस्थिति में परीक्षण।	विक्रेता द्वारा निम्नलिखित को प्रस्तुत करना होगा: 1. सॉफ्टवेयर से टैंपरिंग रोकने के लिए डिवाइस संबंधी उपलब्ध उपाय। 2. हार्डवेयर से टैंपरिंग रोकने के लिए डिवाइस संबंधी उपलब्ध उपाय।
		1.7 चिप निर्माता द्वारा प्रदान की गई कोई भी उपलब्ध बौद्धिक संपदा सुरक्षा तकनीक सक्षम है।	यदि उपलब्ध हो तो चिप निर्माता द्वारा प्रदान की गई बौद्धिक संपदा सुरक्षा तकनीकों को सक्षम करने के लिए ओईएम टीम की उपस्थिति में परीक्षण।	विक्रेता द्वारा निम्नलिखित को प्रस्तुत करना होगा: 1. एसओसी की डेटाशीट 2. चिप निर्माता द्वारा प्रदान की गई बौद्धिक संपदा संरक्षण प्रौद्योगिकियों के संबंध में दस्तावेज़ीकरण, जिन्हें सक्षम किया गया है। 3. यदि चिप निर्माता द्वारा कोई बौद्धिक संपदा संरक्षण तकनीक प्रदान नहीं की जा रही है, तो एक घोषणा जिसमें समरूप बात हो।
		1.8 सत्यापित करें कि डिवाइस लोड करने से पहले बूट छवि हस्ताक्षर को सत्यापित करता है।	निम्नलिखित को सत्यापित करने के लिए ओईएम टीम की उपस्थिति में परीक्षण करना : 1. वैध बूट छवि प्रदान किए जाने पर डिवाइस दस्तावेज़ीकृत सुरक्षित बूट प्रक्रिया के साथ सफलतापूर्वक बूट हो जाता है। 2. छेड़छाड़ की गई बूट छवि (जैसे मिसिंग हस्ताक्षर, अमान्य हस्ताक्षर) प्रदान किए जाने पर डिवाइस बूट नहीं होता है।	विक्रेता द्वारा निम्नलिखित को प्रस्तुत करना होगा: 1. एसओसी की डेटाशीट 2. सुरक्षित बूट के संबंध में डिवाइस के तकनीकी विनिर्देश (इसमें शामिल कुंजियाँ और उनका प्रबंधन जीवन चक्र ', हस्ताक्षर सत्यापन प्रक्रिया और लागू होने पर कोई अन्य सुरक्षित तंत्र शामिल होना चाहिए।)
		1.9 एम्बेडेड डिवाइस पर क्रिप्टोग्राफ़िक रूप से सुरक्षित छद्म-यादृच्छिक संख्या जनरेटर के उपयोग को सत्यापित करें (उदाहरण के लिए, चिप-प्रदत्त यादृच्छिक संख्या जनरेटर का उपयोग करके)।	डिवाइस में उपयोग किए जा रहे यादृच्छिक संख्या जनरेटर के संबंध में विक्रेता द्वारा प्रदान किए गए दस्तावेज़ का सत्यापन करना। कोड-समीक्षा के माध्यम से सत्यापन कि डिवाइस में यादृच्छिक संख्या जनरेटर या संबंधित लाइब्रेरी का उपयोग किया जा रहा है।	विक्रेता को अपने इच्छित उपयोग के साथ डिवाइस में उपयोग किए जा रहे यादृच्छिक जेनरेटर (या तो हार्डवेयर आधारित या सॉफ़्टवेयर आधारित या दोनों) के संबंध में दस्तावेज़ प्रस्तुत करना होगा। यदि हार्डवेयर आधारित यादृच्छिक संख्या जनरेटर का उपयोग किया जा रहा है, तो विक्रेताओं को निम्नलिखित प्रस्तुत करना होगा: 1. एसओसी की डेटाशीट 2. यादृच्छिक जनरेटर के संबंध में डिवाइस की तकनीकी विशिष्टताएँ यदि सॉफ्टवेयर आधारित

				यादृच्छिक संख्या जनरेटर का उपयोग किया जा रहा है, तो विक्रेताओं को इसके लिए उपयोग की जाने वाली लाइब्रेरी प्रदान करनी होगी।
2.	सॉफ्टवेयर/फर्मवेयर	2.1 सत्यापित करें कि एएसएलआर और डीईपी जैसे मेमोरी सुरक्षा नियंत्रण एम्बेडेड/आईओटी ऑपरेटिंग सिस्टम द्वारा सक्षम हैं, यदि लागू हो।	कमांड लाइन-आधारित टूल/कमांड या डीईपी, ईएमईटी टूल जैसे किसी अन्य ओपन-सोर्स टूल का उपयोग करके डिवाइस में उपलब्ध और सक्षम घोषित मेमोरी सुरक्षा नियंत्रणों को सत्यापित करने के लिए ओईएम टीम की उपस्थिति में परीक्षण करना।	विक्रेता को डिवाइस में उपलब्ध और सक्षम मेमोरी सुरक्षा नियंत्रणों की घोषणा प्रस्तुत करनी होगी।
		2.2 सत्यापित करें कि फ़र्मवेयर ऐप्स ट्रांसपोर्ट लेयर सुरक्षा का उपयोग करके ट्रांज़िट में डेटा की सुरक्षा करते हैं।	1. यह सत्यापित करना कि सुरक्षित संचार स्थापित करने के लिए सुदृढ़ एन्क्रिप्शन एल्गोरिदम और सुरक्षित टीएलएस संस्करण डिवाइस द्वारा समर्थित है। 2. यह सत्यापित करना कि डिवाइस सर्वर के टीएलएस प्रमाणपत्र को ठीक से मान्य करता है ताकि यह सुनिश्चित हो सके कि यह विश्वसनीय है और इसके साथ छेड़छाड़ नहीं की गई है। 3. सुभेद्यताओं का परीक्षण जो टीएलएस कनेक्शन की सुरक्षा को प्रभावित कर सकता है जैसे पैडिंग ऑरिकल हमले, या सुभेद्य सिफर सुइट्स। 4. खुले पॉर्ट्स की पहचान करने के लिए एनएमएपी जैसे टूल का उपयोग करना जिसके माध्यम से डिवाइस तक पहुंचा जा सकता है जिससे अनपेक्षित डेटा पुनर्प्राप्ति हो सकती है। 5. यह सत्यापित करना कि टीएलएस सत्र बर्पसुइट जैसे टूल का उपयोग करके मैन-इन-द-मिडिल हमलों का उपयोग करके नेटवर्क ट्रैफ़िक के अवरोधन और डिक्रिप्शन के प्रयासों के लिए प्रतिरोधी हैं।	विक्रेता ट्रांसपोर्ट लेयर सुरक्षा से संबंधित एप्लिकेशन और फर्मवेयर में उपलब्ध कॉन्फ़िगरेशन से संबंधित विनिर्देश और दस्तावेज प्रस्तुत करेगा।
		2.3 सत्यापित करें कि फ़र्मवेयर ऐप्स सर्वर कनेक्शन के डिजिटल हस्ताक्षर को मान्य करते हैं।	1. उन परिदृश्यों की पहचान करना जब डिवाइस बाह्य दुनिया के साथ सर्वर कनेक्शन स्थापित करता है और निम्नलिखित की पुष्टि करता है:	विक्रेता को उपयोग के मामलों का उल्लेख करते हुए एक दस्तावेज़ प्रस्तुत करना होगा जब डिवाइस बाहरी दुनिया के साथ सर्वर कनेक्शन स्थापित करता है, जिसमें सर्वर कनेक्शन के डिजिटल

			• सुरक्षित सर्वर कनेक्शन और डिजिटल हस्ताक्षर सत्यापन से संबंधित सुरक्षा सुविधाएँ, जैसे सुदृढ़ साईफर सुइट्स, सुरक्षित टीएलएस संस्करण, एसएसएल पिनिंग आदि कोड वॉकथ्रू द्वारा समर्थित हैं। • डिवाइस में उचित प्रमाणपत्र सत्यापन, प्रमाणपत्र श्रृंखला सत्यापन और प्रमाणपत्र निरस्तीकरण जांच लागू की जाती हैं। 2. सुभेद्धताओं का परीक्षण जो टीएलएस कनेक्शन की सुरक्षा को प्रभावित कर सकता है जैसे पैडिंग ऑरेकल हमले, या सुभेद्ध सिफर सुइट्स। 3. खुले पोर्ट की पहचान करने के लिए एनएमएपी जैसे टूल का उपयोग करना जिसके माध्यम से डिवाइस तक पहुंचा जा सकता है जिससे अनपेक्षित डेटा पुनर्प्राप्ति हो सकती है। 4. यह सत्यापित करना कि टीएलएस सत्र बर्पसुइट जैसे उपकरणों का उपयोग करके मैन-इन-द-मिडिल हमलों का उपयोग करके नेटवर्क ट्रैफ़िक के अवरोधन और डिक्रिप्शन के प्रयासों के लिए प्रतिरोधी हैं।	हस्ताक्षरों को मान्य करते समय सुरक्षा उपायों के बारे में विस्तृत जानकारी होगी।
		2.4 सत्यापित करें कि प्रतिबंधित सी फ़ंक्शंस के किसी भी उपयोग को उचित सुरक्षित समकक्ष फ़ंक्शंस के साथ बदल दिया गया है।	निम्नलिखित में से किसी भी दृष्टिकोण के माध्यम से लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण का उपयोग करके ओईएम टीम की उपस्थिति में सुरक्षित कोड समीक्षा [स्वचालित और मैन्युअल दोनों]: 1. फ़र्मवेयर कोड के साथ विक्रेता द्वारा मूल्यांकन एजेंसी का दौरा करना और मूल्यांकन एजेंसी के पास उपलब्ध लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण को	विक्रेता द्वारा उपलब्ध करना होगा: 1. कोड समीक्षा के लिए फ़र्मवेयर बायनेरिज़। 2. आंतरिक कोड समीक्षा रिपोर्ट

			अपने सिस्टम में स्थापित करना। [अनुशंसित] 2. विक्रेता द्वारा फर्मवेयर कोड और उनके पास उपलब्ध किसी भी लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण के साथ मूल्यांकन एजेंसी का दौरा करना और मूल्यांकन एजेंसी के प्रतिनिधियों की उपस्थिति में कोड समीक्षा गतिविधि का प्रदर्शन करना। 3. मूल्यांकन एजेंसी को उनके पास उपलब्ध लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण स्थापित करने के लिए विक्रेता साइट पर सिस्टम की रिमोट एक्सेस प्रदान करना। 4. विक्रेताओं के पास उपलब्ध लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण के साथ फर्मवेयर कोड वाले मूल्यांकन एजेंसी को विक्रेता साइट पर सिस्टम की दूरस्थ अभिगम प्रदान करना।	
		2.5 सत्यापित करें कि प्रत्येक फर्मवेयर तृतीय पक्ष के घटकों, संस्करण और प्रकाशित सुभेद्धताओं को सूचीबद्ध करने वाली सामग्री का एक सॉफ्टबेयर बिल रखता है।	फ़र्मवेयर पर एफ़एसीटी जैसे स्वचालित उपकरण चलाकर तृतीय-पक्ष घटकों की प्रस्तुत सूची का सत्यापन करना। सार्वजनिक रूप से उपलब्ध सुभेद्यता डेटाबेस के माध्यम से तीसरे पक्ष के घटकों में सुभेद्धताओं की पहचान करना तृतीय पक्ष के घटकों में किसी भी ज्ञात सुभेद्धता को दूर करने के लिए फर्मवेयर के लिए नियमित सुरक्षा अपडेट और पैच प्रदान करने के लिए विक्रेता द्वारा परिभाषित प्रक्रिया का सत्यापन और वैधता।	विक्रेता द्वारा निम्नलिखित को प्रस्तुत करना होगा: 1. तृतीय पक्ष के घटकों और संस्करणों सहित सामग्री के सॉफ़्टबेयर बिल की जानकारी के लिए दस्तावेज़ीकरण करना। 2. निम्नलिखित के लिए संगठन प्रक्रिया और नीतियां: • तृतीय पक्ष के घटकों में पहचानी गई किसी भी सुभेद्धता को संबोधित करना और ठीक करना। • ग्राहकों को सुरक्षा मुद्दों या सुभेद्धताओं के बारे में सूचित करना और उसके लिए सुरक्षा अद्यतन और पैच प्रदान करना। 3. उपकरणों के लिए जारी किए गए पैच/फिक्स के साथ फर्मवेयर और तृतीय-पक्ष बाइनरी, लाइब्रेरी और फ्रेमवर्क को बनाए रखने के लिए कॉन्फ़िगरेशन प्रबंधन प्रणाली और संबंधित नीतियां।
		2.6 हार्डकोडेड क्रेडेंशियल्स (बैकडोर) के लिए तृतीय-पक्ष बायनेरिज़, लाइब्रेरीज़,	निम्नलिखित में से किसी भी दृष्टिकोण के माध्यम से लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण का उपयोग करके स्वतंत्र सुरक्षित	विक्रेता द्वारा उपलब्ध करना होगा: 1. कोड समीक्षा के लिए फ़र्मवेयर बायनेरिज़।

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		फ्रेमवर्क सहित सभी कोड की समीक्षा की जाती है।	कोड समीक्षा [स्वचालित और मैन्युअल दोनों]: 1. फ़र्मवेयर कोड के साथ विक्रेता द्वारा मूल्यांकन एजेंसी का दौरा करना और मूल्यांकन एजेंसी के पास उपलब्ध लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण को अपने सिस्टम में स्थापित करना। [अनुशंसित] 2. विक्रेता द्वारा फर्मवेयर कोड और उनके पास उपलब्ध किसी भी लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण के साथ मूल्यांकन एजेंसी का दौरा करना और मूल्यांकन एजेंसी के प्रतिनिधियों की उपस्थिति में कोड समीक्षा गतिविधि का प्रदर्शन करना। 3. मूल्यांकन एजेंसी को उनके पास उपलब्ध लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण स्थापित करने के लिए विक्रेता साइट पर सिस्टम की दूरस्थ पहुंच प्रदान करना। 4. विक्रेताओं के पास उपलब्ध लाइसेंस प्राप्त स्थैतिक विश्लेषण उपकरण के साथ फर्मवेयर कोड वाले मूल्यांकन एजेंसी को विक्रेता साइट पर सिस्टम की दूरस्थ पहुंच प्रदान करना ।	2. आंतरिक कोड समीक्षा रिपोर्ट
		2.7 सत्यापित करें कि फ़र्मवेयर ऐप्स डिजिटल हस्ताक्षर को किसी विश्वसनीय सर्वर पर पिन करते हैं।	1. उन परिदृश्यों की पहचान करना जब डिवाइस बाहरी दुनिया के साथ सर्वर कनेक्शन स्थापित करता है और निम्नलिखित की पुष्टि करता है: • सुरक्षित सर्वर कनेक्शन और डिजिटल हस्ताक्षर सत्यापन से संबंधित सुरक्षा सुविधाएँ, जैसे सुदृढ़ सिफर सुइट्स, सुरक्षित टीएलएस संस्करण, एसएसएल पिनिंग आदि कोड वॉकथ्रू द्वारा समर्थित हैं। • डिवाइस में उचित प्रमाणपत्र सत्यापन, प्रमाणपत्र श्रृंखला सत्यापन और प्रमाणपत्र निरस्तीकरण जांच लागू की जाती हैं	विक्रेता को उपयोग के मामलों का उल्लेख करते हुए एक दस्तावेज़ प्रस्तुत करना होगा जब डिवाइस बाहरी दुनिया के साथ सर्वर कनेक्शन स्थापित करता है, जिसमें सर्वर कनेक्शन के डिजिटल हस्ताक्षरों को मान्य करते समय सुरक्षा उपायों के बारे में विस्तृत जानकारी होगी।

		2.8 (वर्चोज़ डिबगिंग प्रतीकों को हटाने) में बाधा डालने के लिए सुरक्षा नियंत्रण मौजूद हैं।	फ़र्मवेयर रिवर्स इंजीनियरिंग में बाधा डालने के लिए विक्रेता द्वारा प्रदान किए गए सुरक्षा नियंत्रणों को सत्यापित करने के लिए, ओईएम टीम की उपस्थिति में परीक्षण करना।	फ़र्मवेयर रिवर्स इंजीनियरिंग में बाधा डालने के लिए विक्रेता को सुरक्षा नियंत्रण के संबंध में दस्तावेज़ प्रस्तुत करना होगा।
		2.9 सत्यापित करें कि फ़र्मवेयर अपडेट प्रक्रिया समय जांच बनाम उपयोग के समय के हमलों के प्रति संवेदनशील नहीं है।	डिवाइस में लागू किए गए उपायों को सत्यापित करने के लिए ओईएम टीम की उपस्थिति में परीक्षण किया गया, ताकि इसे समय- समय पर उपयोग किए जाने वाले हमलों के प्रति प्रतिरोधी बनाया जा सके।	विक्रेता को डिवाइस में लागू किए गए उपायों को प्रस्तुत करना होगा ताकि इसे समय-जांच बनाम उपयोग के समय के हमलों के प्रति प्रतिरोधी बनाया जा सके।
		2.10 सत्यापित करें कि डिवाइस इंस्टॉल करने से पहले कोड साइनिंग का उपयोग करता है और फ़र्मवेयर अपग्रेड फ़ाइलों को मान्य करता है।	निम्नलिखित को सत्यापित करने के लिए ओईएम टीम की उपस्थिति में परीक्षण करना : क. वैध अपडेट पैकेज उपलब्ध कराए जाने पर डिवाइस दस्तावेज़ीकृत सुरक्षित अपग्रेड प्रक्रिया के साथ सफलतापूर्वक अपडेट हो जाता है। ख. छेड़छाड़ किए गए अपडेट पैकेज (जैसे मिसिंग हस्ताक्षर, अमान्य हस्ताक्षर) प्रदान किए जाने पर डिवाइस बूट नहीं होता है।	विक्रेता को सुरक्षित फ़र्मवेयर अपग्रेड प्राप्त करने की प्रक्रिया प्रस्तुत करनी होगी जिसमें शामिल कुंजियाँ और उनका प्रबंधन जीवन चक्र * , हस्ताक्षर सत्यापन प्रक्रिया और लागू होने पर कोई अन्य सुरक्षित तंत्र शामिल होना चाहिए।
		2.11 सत्यापित करें कि डिवाइस को वैध फर्मवेयर के पुराने संस्करण (एंटी-रोलबैक) में डाउनग्रेड नहीं किया जा सकता है।	यह सत्यापित करने के लिए कि डिवाइस को वैध फर्मवेयर के पुराने संस्करणों (एंटी-रोलबैक) में डाउनग्रेड नहीं किया जा सकता है, ओईएम टीम की उपस्थिति में परीक्षण किया जा रहा है।	विक्रेता को सुरक्षित फ़र्मवेयर अपग्रेड प्राप्त करने की प्रक्रिया प्रस्तुत करनी होगी जिसमें शामिल कुंजियाँ और उनका प्रबंधन जीवन चक्र * , हस्ताक्षर सत्यापन प्रक्रिया और लागू होने पर कोई अन्य सुरक्षित तंत्र शामिल होना चाहिए।
		2.12 सत्यापित करें कि फ़र्मवेयर पूर्वनिर्धारित शेड्यूल पर स्वचालित फ़र्मवेयर अपडेट कर सकता है।	सत्यापन लागू परिदृश्य के अनुसार किया जाएगा: **स्थिति 1: स्वचालित ओटीए अपडेट उपलब्ध हैं:** इन-फील्ड उपकरणों को स्वचालित अपडेट/अपग्रेड जारी करने के लिए एक मानक संचालन प्रक्रिया विक्रेता द्वारा प्रस्तुत की जानी आवश्यक है जिसका मूल्यांकन, मूल्यांकन एजेंसी द्वारा सी20, सी21 और सी22 सुरक्षा आवश्यकता के अनुसार किया जा सकता है।	विक्रेता निम्नलिखित प्रदान करेगा: विक्रेता द्वारा निम्नलिखित को प्रस्तुत कराना होगा: 1. उपलब्ध अपडेट के तरीके यानी स्वचालित, मैन्युअल या दोनों। 2. उपकरणों को अपडेट जारी करने के संबंध में संगठनात्मक प्रक्रिया और नीतियां।

			स्थिति 2: स्वचालित ओटीए अपडेट उपलब्ध नहीं हैं और विक्रेता मैन्युअल अपडेट प्रदान करता है: इन-फील्ड डिवाइसों में मैन्युअल अपडेट/अपग्रेड जारी करने के लिए विक्रेता द्वारा एक मानक संचालन प्रक्रिया प्रस्तुत की जानी आवश्यक है जिसका मूल्यांकन, मूल्यांकन एजेंसी द्वारा सी20, सी21 और सी22 सुरक्षा आवश्यकता के अनुसार किया जा सकता है।	
3.	सुरक्षित प्रक्रिया अनुरूपता	3.1 सत्यापित करें कि वायरलेस संचार परस्पर प्रमाणित हैं।	विक्रेता द्वारा दस्तावेज़ में निर्धारित आपसी प्रमाणीकरण की प्रक्रिया को सत्यापित करने के लिए, ओईएम टीम की उपस्थिति में परीक्षण करना।	विक्रेताओं को वायरलेस संचार शुरू होने पर डिवाइस में लागू पारस्परिक प्रमाणीकरण की प्रक्रिया के संबंध में दस्तावेज़ प्रदान करना होगा। यदि डिवाइस वायरलेस संचार का समर्थन नहीं करता है, तो विक्रेता को इसके लिए एक घोषणा पत्र प्रदान करना होगा।
		3.2 सत्यापित करें कि वायरलेस संचार एक एन्क्रिप्टेड चैनल पर भेजा जाता है।	संचार प्रक्रिया सत्यापन में उपयोग किए जा रहे सभी सुरक्षा तंत्रों की पहचान करना: • ओईएम टीम की उपस्थिति में परीक्षण करना • कोड समीक्षा • कुंजी-जीवन चक्र प्रक्रिया संबंधी प्रक्रिया लेखा परीक्षा लेखा परीक्षा	संचार के वायरलेस मोड के माध्यम से भेजे जाने वाले डेटा से छेड़छाड़ को रोकने के लिए विक्रेताओं को डिवाइस में लागू सुरक्षा उपायों के संबंध में दस्तावेज उपलब्ध कराने होंगे। यदि डिवाइस वायरलेस संचार का समर्थन नहीं करता है, तो विक्रेता को इसके लिए एक घोषणा पत्र प्रदान करना होगा।
		3.3 सत्यापित करें कि क्या डिवाइस के घटकों की सोर्सिंग के लिए विश्वसनीय स्रोतों का उपयोग किया जा रहा है यानी महत्वपूर्ण हार्डवेयर घटकों (एसओसी जैसे सुरक्षा कार्यों से संबंधित) के लिए सामग्रियों के प्रबंधित बिल के माध्यम से विश्वसनीय आपूर्ति श्रृंखला का उपयोग किया जा रहा है।		विक्रेता को महत्वपूर्ण हार्डवेयर घटकों (एसओसी जैसे सुरक्षा कार्यों से संबंधित) के लिए सामग्री का बिल प्रस्तुत करना होगा।
		3.4 आपूर्ति श्रृंखला जोखिम की पहचान,		विक्रेता निम्नलिखित प्रस्तुत करेगा: आपूर्ति श्रृंखला जोखिम की

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		मूल्यांकन, प्राथमिकता और शमन आयोजित किया जाएगा। आपूर्ति शृंखला जोखिम/व्यवसाय निरंतरता योजना नीति दस्तावेज़, आपूर्ति शृंखला व्यवधान को संभालने के तरीके को दर्शाने वाली प्लेबुक, घटना के बाद के सारांश दस्तावेज़ प्रस्तुत करने और उन्हें प्रदर्शित करने की आवश्यकता है।		पहचान, मूल्यांकन, प्राथमिकता और शमन दस्तावेज़। आपूर्ति शृंखला जोखिम / व्यापार निरंतरता योजना नीति दस्तावेज, प्लेबुक जो दर्शाती है कि आपूर्ति शृंखला व्यवधान को कैसे संभालना है, घटना के बाद सारांश दस्तावेजों को प्रस्तुत करना।
		3.5 सत्यापित करें कि डिवाइस में कोई प्रप्राइइटेरी नेटवर्क प्रोटोकॉल का उपयोग नहीं किया जा रहा है। यदि हाँ, तो संपूर्ण कार्यान्वयन विवरण और उसके लिए स्रोत कोड प्रदान किया जाएगा।		डिवाइस में प्रयुक्त नेटवर्क प्रोटोकॉल के लिए दस्तावेज़।
4.	उत्पाद विकास चरण में सुरक्षा अनुरूपता	4.1 नकली शमन और मैलवेयर का पता लगाने में सहायता के लिए पीसीबीए और एसओसी स्तर तक डिजाइन और आर्किटेक्चर विवरण प्रदान किया जाएगा।		पीसीबीए और एसओसी स्तर तक डिज़ाइन और आर्किटेक्चर दस्तावेज़।
		4.2 उत्पाद विकास के हिस्से के रूप में खराब और नकली उत्पादों के लिए खतरा कम करने की रणनीतियों को लागू किया जाएगा।	प्रक्रिया और विधि के विरूपण साक्ष्य प्रस्तुत करने और उन्हें प्रदर्शित करने की आवश्यकता है।	
		4.3 कोड स्वीकृति और विकास प्रक्रियाओं के हिस्से के रूप में एक या अधिक अद्यतन मैलवेयर पहचान उपकरण नियोजित किए जाएंगे। अंतिम पैकेजिंग और प्रदायगी से पहले मैलवेयर पहचान	उन घटकों की सूची जिनकी पहचान टैनिंग/जालसाजी, सीएम टूल के ट्रैकिंग लक्ष्यों की आवश्यकता के रूप में की गई है। गुणवत्ता आश्वासन प्रक्रिया को प्रस्तुत करने और उसे प्रदर्शित करने की आवश्यकता है।	

		तकनीकों का उपयोग किया जाएगा (उदाहरण के लिए, एक या अधिक अद्यतन मैलवेयर पहचान उपकरणों का उपयोग करके मैलवेयर के लिए तैयार उत्पादों और घटकों को स्कैन करना)।		
		4.4 आपूर्ति श्रृंखला जोखिम की पहचान, मूल्यांकन, प्राथमिकता और शमन आयोजित किया जाएगा।		आपूर्ति श्रृंखला जोखिम / व्यापार निरंतरता योजना नीति दस्तावेज, प्लेबुक जो दर्शाती है कि आपूर्ति श्रृंखला व्यवधान को कैसे संभालना है, घटना के बाद सारांश दस्तावेजों को प्रस्तुत करने और उसी को प्रदर्शित करने की आवश्यकता है।

MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

(IPHW Division)

ORDER

New Delhi, the 9th April, 2024

Subject: Amendment to the "Electronics and Information Technology Goods (Requirement of Compulsory Registration) Order, 2021"

S.O. 1652(E).—In exercise of the powers conferred by sub-section (1) and (2) of section 16 read with sub section (3) of section 25 of the Bureau of Indian Standards Act, 2016, (11 of 2016), the Central Government is of the opinion that it is necessary or expedient so to do in the public interest, hereby makes the following amendments to the "Electronics and Information Technology Goods (Requirements for Compulsory Registration) Order, 2021":

2. For CCTV Camera, the following entry of Column (5) be added at S. No. 41 in the Schedule of the "Electronics and Information Technology Goods (Requirements for Compulsory Registration) Order, 2021.

Sr. No. (1)	Goods or Articles (2)	Indian Standard (3)	Title of Indian Standard (4)	Essential Requirement(s) (5)
41	CCTV Camera	IS 13252: Part 1: 2010	Information Technology Equipment - Safety General Requirements--	Essential Requirement(s) for CCTV as per Annexure

3. The provisions of "Electronics and Information Technology Goods (Requirements for Compulsory Registration) Order, 2021" shall apply on the Goods or articles as specified in the column (2) added to the schedule of the said Order by virtue of this notification, for conforming to the corresponding Essential Requirement(s) as specified in the column (5), on the expiry of six months from the date of publication of this notification in the Official Gazette. As per Scheme II of BIS Conformity Assessment Regulations, 2018, submission of test reports from BIS recognized labs, shall form a pre -requisite for obtaining license to use Standard Mark.

[F.No. W-43/11/2021-IPHW]

ASHA NANGIA, Group Coordinator & Scientist 'G'

Annexure

Essential Requirement(s) for Security of CCTV

Securing a CCTV (Closed-Circuit Television) system is crucial to protect sensitive information and ensure the system operates effectively. Key areas of testing include exposed network services, device communication protocols, physical access to the device's UART, JTAG, SWD, etc., the ability to extract memory and firmware, firmware update process security and storage and encryption of data. Here are brief requirements for the security of a CCTV system:

1. Physical Security - Use tamper-resistant camera enclosures and locking mechanisms to deter physical tampering.
2. Access Control by Authentication, Role-Based Access Control (RBAC) and regularly review and update access permissions to reflect personnel changes.
3. Network Security by employing encryption of data transmission
4. Software Security by Regular Updates, Disable Unused Features and Strong Password Policies
5. Penetration Testing: Employ penetration testing to assess the system's resistance to cyberattacks and address vulnerabilities.

Essential Security Requirements

Sr. No.	Category	Testing Parameter	What to be tested	Documents Required
1.	Hardware Level Security Parameter (supported by software)	1.1 Verify that application layer debugging interfaces such USB, UART, and other serial variants are disabled or protected by a complex password.	1. Identification of the availability of debugging interfaces such as USB, UART, and other serial variants through the Datasheet of the SoC being used in the device under test 2. Verification and validation of the ports/interfaces enabled in the production devices and the related access control mechanism for protection of the same as declared in the vendor documentation 3. Testing, in presence of OEM team, to verify the enabling/disabling of all the ports and debugging interfaces such as USB, UART, and other serial variants using their relevant hardware-based debuggers and access control mechanisms in case the interface is enabled. 4. Process verification of the manufacturing facility to validate the vendor's claim regarding the debugging interfaces which are closed/disabled during provisioning. [For instance, through Block connection diagram depicting pin connections between the host	The vendor shall provide the following: a. Datasheet of the SoC being used in the device. b. Documentation related to ports/interfaces enabled in the production devices and the related access control mechanism for protection of the same. c. Process flow of the Manufacturing/Provisioning of the device

			microcontroller and its interactions with various sub components/peripherals.]	
		1.2 Verify that cryptographic keys and certificates are unique to each individual device.	Identifying all the keys and certificates being used in the device eco-system and verification through: • Testing, in presence of OEM team • Code review • Process audit of the key-life cycle process	Vendor shall submit the following: 1. List of all keys and certificates being used in the device ecosystem 2. Key management life cycle (purpose, generation, storage, destruction/zeroization, validity, key changeover/rotation)
		1.3 Verify that on-chip debugging interfaces such as JTAG or SWD are disabled or that available protection mechanism is enabled and configured appropriately.	1. Identification of the availability of debugging interfaces such as USB, UART, and other serial variants through the Datasheet of the SoC being used in the device under test 2. Verification and validation of the ports/interfaces enabled in the production devices and the related access control mechanism for protection of the same as declared in the vendor documentation 3. Testing, in presence of OEM team, to verify the enabling/disabling of all the ports and debugging interfaces such as USB, UART, and other serial variants using their relevant hardware based debuggers and access control mechanisms in case the interface is enabled. 4. Process audit of the manufacturing facility to validate the vendor's claim regarding the debugging interfaces which are closed/disabled during provisioning. [For instance, through Block connection diagram depicting pin connections between the host microcontroller and its interactions with various sub components/peripherals.]	The vendor shall provide the following: a. Datasheet of the SoC being used in the device. b. Documentation related to ports/interfaces enabled in the production devices and the related access control mechanism for protection of the same. c. Process flow of the Manufacturing/Provisioning of the device

		1.4 Verify that trusted execution is implemented and enabled, if available on the device SoC or CPU.	Identifying whether TEE/SE/TPM is available or not in the device through the SoC datasheet and technical documentation submitted by the vendor. Further assessment is done on the basis of scenarios as applicable to device as defined below: **CASE 1: TEE/SE/TPM is not available:** No further assessment **CASE 2: TEE/SE/TPM is available and enabled:** Verification through code-review that crypto functions are called through TEE/SE/TPM APIs. **CASE 3: TEE/SE/TPM is available but not enabled by the vendor:** Termed as non-conformance to the requirement. OEM is required to enable and implement the TEE/SE/TPM.	The vendor shall provide the following: 1. Datasheet of the SoC being used in the device. 2. User manual/ Technical specifications of the device 3. Code snippets of the TEE API call, wherever applicable
		1.5 Verify that sensitive data, private keys and certificates are stored securely in a Secure Element, TPM, TEE (Trusted Execution Environment), or protected using strong cryptography.	Identifying all the keys and certificates being used in the device eco-system, sensitive data and their storage mechanism(s); and verification through: • Testing, in presence of OEM team • Code review • Process audit of the key-life cycle process	Vendor shall submit the following: 1. List of all keys and certificates being used in the device ecosystem 2. List of all the sensitive data with their intended usage and secure storage mechanism(s) as implemented along with secure configurations to be enabled in the device. 3. Key management life cycle (purpose, generation, storage, destruction/zeroization, validity, key changeover/rotation) private keys and certificates.
		1.6Verify the presence of tamper resistance and/or tamper detection features.	Testing, in presence of OEM team, to verify the measures implemented in the device to prevent software and hardware tampering.	Vendor shall submit the following: 1. Measures available in the device to prevent software tampering. 2. Measures available in the device to prevent hardware tampering.
		1.7 Verify that any available Intellectual	Testing, in presence of OEM team, to verify the enabling of the	Vendor shall submit the following: 1. Datasheet of the SoC

		Property protection technologies provided by the chip manufacturer are enabled.	Intellectual Property protection technologies provided by the chip manufacturer, if available.	2. Documentation regarding the Intellectual Property protection technologies provided by the chip manufacturer which have been enabled. 3. In case, no Intellectual Property protection technologies are being provided by the chip manufacturer, then a declaration stating the same.
		1.8 Verify the device validates the boot image signature before loading.	Testing, in presence of OEM team, to verify the following: 1. Device boots up successfully with the documented secure boot process when a valid boot image is provided. 2. Device does not boot up when a tampered boot image (like with missing signature, invalid signature) is provided.	Vendor shall submit the following: 1. Datasheet of the SoC 2. Technical specifications of the device regarding secure boot (should consist of keys involved and their management life cycle*, signature validation process and any other secure mechanisms if implemented.)
		1.9 Verify usage of cryptographically secure pseudo-random number generator on embedded device (e.g., using chip-provided random number generators).	Verification of the documentation provided by the vendor regarding the random number generators being used in the device. Verification through code-review that random number generators or related libraries as applicable are being used in the device.	Vendor shall submit the documentation regarding the random generators (either hardware based or software based or both) being used in the device with their intended usage. In case, hardware based random number generators are being used, vendors shall submit the following: 1. Datasheet of the SoC 2. Technical specifications of the device regarding random generators In case, software based random number generators are being used, vendors shall provide the libraries being used for the same.
2.	Software/Firmware	2.1 Verify that memory protection controls such as ASLR and DEP are enabled by the embedded/IoT operating system, if applicable.	Testing, in presence of OEM team, to verify the declared memory protection controls available and enabled in the device using command line-based tools/commands or any other open-source tool like DEP, EMET tool.	Vendor shall submit the declaration of the memory protection controls available and enabled in the device.
		2.2 Verify that the firmware apps protect data-in-transit using transport layer	1. Verifying that strong encryption algorithms and secure TLS version is supported by the device to establish secure	The vendor shall submit the specifications and documentation related to the configurations available in the applications and

		security.	communication. 2. Verifying that device properly validates the server's TLS certificate to ensure that it is trusted and has not been tampered with. 3. Testing for vulnerabilities which can affect the security of TLS connection such as padding oracle attacks, or weak cipher suites. 4. Using tools such as Nmap to identify open ports through which device can be accessed leading to unintended data retrieval. 5. Verifying that theTLS session(s) are resistant to attemptsof interception and decryption of network traffic using man-in-the-middle attacks using tools like Burpsuite.	firmware related to transport layer security.
		2.3 Verify that the firmware apps validate the digital signature of server connections.	1. Identifying the scenarios when the device establishes the server connections with the external world and verifying the following: • Security features, related to secure server connections and digital signature validation as implemented like strong cipher suites, secure TLS version, SSL pinning etc. supported by code walkthrough. • Proper certificate validation, certificate chain validation and certificate revocation checks are implemented in the device. 2. Testing for vulnerabilities which can affect the security of TLS connection such as padding oracle attacks, or weak cipher suites.	Vendor shall submit a document mentioning the use-cases when the device establishes server connections with the external world, with detailed information about the security measures in place while validating the digital signatures of the server connections.

			3. Using tools such as Nmap to identify open ports through which device can be accessed leading to unintended data retrieval. 4. Verifying that TLS session(s) are resistant to attemptsof interception and decryption of network traffic using man-in-the-middle attacks using tools like Burpsuite.	
		2.4 Verify that any use of banned C functions are replaced with the appropriate safe equivalent functions.	Secure code review [both automated and manual], in presence of OEM team, using a licensed static analysis tool through any of the following approaches: 1. Visit to the evaluation agency by the vendor with the firmware code and installing the licensed static analysis tool available with the evaluation agency in their systems. [Recommended] 2. Visit to the evaluation agency by the vendor with the firmware code and any licensed static analysis tool available with them and demonstrating the code review activity in the presence of representatives of evaluation agency. 3. Giving a remote access of the systems at vendor site to the evaluation agency for installing their licensed static analysis tool available with them. 4. Giving a remote access of the systems at vendor site to the evaluation agency containing the firmware code along with the licensed static analysis tool available with the vendors.	Vendor shall provide : 1. Firmware binaries for code review. 2. Internal code review reports
		2.5 Verify that each firmware maintains a software bill of materials cataloging third party	Verification of the submitted list of third-party components by running automated tools like FACT on the firmware. Identifying vulnerabilities	Vendor shall submit the following: 1. Documentation for information on software bill of materials, including third-party components and versions.

		components, versioning, and published vulnerabilities.	in the third-party component(s) through publically available vulnerability databases Verification and validation of the process defined by the vendor for providing regular security updates and patches for the firmware to address any known vulnerabilities in third-party components.	2. Organization process and policies for the following: • Addressing and patching any identified vulnerabilities in third-party components. • Informing the customers about the security issues or vulnerabilities and providing security updates and patches for the same. 3. Configuration management system and related policies for maintaining firmware and third-party binaries, libraries and frameworks along with the patches/fixes issued to the devices.
		2.6 Verify all code including third-party binaries, libraries, frameworks are reviewed for hardcoded credentials (backdoors).	Independent secure code review [both automated and manual] using a licensed static analysis tool through any of the following approaches: 1. Visit to the evaluation agency by the vendor with the firmware code and installing the licensed static analysis tool available with the evaluation agency in their systems. [Recommended] 2. Visit to the evaluation agency by the vendor with the firmware code and any licensed static analysis tool available with them and demonstrating the code review activity in the presence of representatives of evaluation agency. 3. Giving a remote access of the systems at vendor site to the evaluation agency for installing their licensed static analysis tool available with them. 4. Giving a remote access of the systems at vendor site to the evaluation agency containing the firmware code along with the licensed static analysis tool available with the vendors.	Vendor shall provide: 1. Firmware binaries for code review. 2. Internal code review reports

		2.7 Verify that the firmware apps pin the digital signature to a trusted server(s).	1. Identifying the scenarios when the device establishes the server connections with the external world and verifying the following: • Security features, related to secure server connections and digital signature validation as implemented like strong cipher suites, secure TLS version, SSL pinning etc. supported by code walkthrough. • Proper certificate validation, certificate chain validation and certificate revocation checks are implemented in the device.	Vendor shall submit a document mentioning the use-cases when the device establishes server connections with the external world, with detailed information about the security measures in place while validating the digital signatures of the server connections.
		2.7 Verify security controls are in place to hinder firmware reverse engineering (e.g.removal of verbose debugging symbols).	Testing, in presence of OEM team, to verify the security controls as provided by the vendor to hinder firmware reverse engineering.	Vendor shall submit the documentation regarding the security controls in place to hinder firmware reverse engineering.
		2.8 Verify that the firmware update process is not vulnerable to time-of-check vs time-of-use attacks.	Testing, in presence of OEM team, to verify the measures implemented in the device to make it resistant to time-of-check vs.time-of-use attacks.	Vendor shall submit the measures implemented in the device to make it resistant to time-of-check vs. time-of-use attacks.
		2.9 Verify the device uses code signing and validates firmware upgrade files before installing.	Testing, in presence of OEM team, to verify the following: 1. Device gets successfully updated with the documented secure upgrade process when a valid update package is provided. 2. Device does not boot up when a tampered update package (like with missing signature, invalid signature) is provided.	Vendor shall submit the process of achieving secure firmware upgrade which should consist of keys involved and their management life cycle*, signature validation process and any other secure mechanisms if implemented.

		2.10 Verify that the device cannot be downgraded to old versions (anti-rollback) of valid firmware.	Testing, in presence of OEM team, to verify that the device cannot be downgraded to old versions (anti-rollback) of valid firmware.	Vendor shall submit the process of achieving secure firmware upgrade which should consist of keys involved and their management life cycle*, signature validation process and any other secure mechanisms if implemented.
		2.11 Verify that firmware can perform automatic firmware updates upon a predefined schedule.	Verification shall be done as per the applicable scenario: **Case 1: Automatic OTA updates are available:** A standard operating procedure for issuing automatic updates/upgrades to the in-field devices is required to be submitted by the vendor which can then be evaluated by the evaluation agency as per C20, C21 and C22 security requirement of OWASP open standard. **Case 2: Automatic OTA updates are not available and vendor provides manual updates:** A standard operating procedure for issuing manual updates/upgrades to the in-field devices is required to be submitted by the vendor which can then be evaluated by the evaluation agency as per C20, C21 and C22 security requirement of OWASP open standard.	Vendor shall provide the following: 1. Modes of updates available i.e. automatic, manual or both. 2. Organizational process and policies regarding the issuing of updates to the devices.
3.	Secure Process Conformance	3.1 Verify that wireless communications are mutually authenticated.	Testing, in presence of OEM team, to verify the process of mutual authentication as laid down in the documentation by the vendor.	Vendors shall provide the documentation regarding the process of mutual authentication as implemented in the device when wireless communications are initiated. In case, the device does not support wireless communications, the vendor shall provide a declaration for the same.
		3.2 Verify that wireless communications are sent over an encrypted	Identifying all the security mechanisms being used in the communication process verification through:	Vendors shall provide the documentation regarding the security measures implemented in the device to prevent tampering of the data

		channel.	• Testing, in presence of OEM team • Code review • Process audit of the key-life cycle process	being sent through wireless mode of communication. In case, the device does not support wireless communications, the vendor shall provide a declaration for the same.
		3.3 Verify that whether trusted sources are being used for sourcing the components of the device i.e. trusted supply chain through a managed Bill of materials for critical hardware components (related to security functions like SoC) is in use.		Vendor shall submit Bill of materials for critical hardware components (related to security functions like SoC).
		3.4 Supply chain risk identification, assessment, prioritization, and mitigation shall be conducted. Supply chain risk/business continuity planning policy documents, playbooks reflecting how to handle supply chain disruption, post-incident summary documents need to be submitted and demonstrate the same.		Vendor shall submit the following: Supply chain risk identification, assessment, prioritization, and mitigation documents. Supply chain risk/business continuity planning policy documents, playbooks reflecting how to handle supply chain disruption, post-incident summary documents.
		3.5 Verify the no proprietary network protocols are being used in the device. If yes, then complete implementation details and the source code for the same shall be provided.		Document for Network protocols used in the device.
4.	Security Conformance at	4.1 Design and architecture details till the PCBA and SoC		Design and architecture documents till the PCBA and SoC level.

	product development stage	level to be provided to aid in counterfeit mitigation and malware detection.		
		4.2 Threat mitigation strategies for tainted and counterfeit products shall be implemented as part of product development.	Process and method artifacts need to be submitted and demonstrate the same.	
		4.3 One or more up-to-date malware detection tools shall be deployed as part of the code acceptance and development processes. Malware detection techniques shall be used before final packaging and delivery (e.g., scanning finished products and components for malware using one or more up-to-date malware detection tools).	List of components that have been identified as requiring tracking targets of tainting/counterfeiting, CM tool. Quality assurance process need to be submitted and demonstrate the same.	
		4.4 Supply chain risk identification, assessment, prioritization, and mitigation shall be conducted.		Supply chain risk/business continuity planning policy documents, playbooks reflecting how to handle supply chain disruption, post-incident summary documents need to be submitted and demonstrate the same.

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MANOJ KUMAR VERMA

W-43/11/2021-IPHW
Government of India
Ministry of Electronics and Information Technology
(IPHW Division)

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6, CGO Complex,
Lodhi Road, New Delhi-110003.

Dated: 11 Mar, 2024

OFFICE MEMORANDUM

Subject: -Advisory on the Threat of Information Leakage through CCTV/ Video Surveillance system (VSS)/ Digital Video Recorders /Network Video Recorders-reg

The reference is made to the concerns raised by various Ministries/Departments regarding the security implications associated with the deployment of Closed-Circuit Television (CCTV) Cameras and the conduct of cyber auditing and testing of hardware pertaining to CCTV cameras and other Internet of Things (IoT) devices. The Ministry of Electronics and Information Technology (MeitY) has formulated comprehensive security guidelines for CCTV cameras as included in **Annexure 'A'**.

2. In light of these concerns, Government Ministries/Departments are strongly advised to adhere to the guidelines outlined within the ambit of the Public Procurement Orders to safeguard the overall security and integrity of CCTV Cameras and IoT Devices.

3. This issues with the approval of the Competent Authority.

(Asha Nangia)
Group Coordinator & Scientist-'G'
Ph. 011-24301965

To,

i) Secretaries of All Ministries/Department of Government of India

ii) Chief Secretaries/Administrators of Union Territories/National Capital Territory of Delhi

Annexture- 'A'

A video surveillance system, also known as Closed-Circuit Television (CCTV) system, is a collection of cameras and other related equipment used to monitor and record activities in a specific area commonly used for security and surveillance purposes.

2. Key components of a video surveillance system typically include cameras, (Analog, Digital, IP Cameras), Video Management System (VMS) Software, Storage (Network Video Recorders (NVRs) or Digital Video Recorders (DVRs)), Power Supply etc.

3. While these surveillance technologies undoubtedly offer a range of benefits and are valuable tools for monitoring and security, they also raise certain concerns and risks. Some of the growing risks associated with CCTV systems include data security, privacy breach, hacking and cyber-attack etc. Various incidents have also been reported due to security flaw in the surveillance cameras.

4. The cybersecurity is an ongoing process, so staying vigilant and keeping system up to date with the latest security practices can significantly enhance the security of CCTV system and protect it from potential threats and unauthorized access. In this regard, the following measures are advised to minimize the risk associated with CCTV surveillance system:

i) The Rules and regulations as applicable, notified by the Government or procurement of goods and services must be followed e.g.
   a) Public procurement Order (Make in India), 2017
   b) Electronics and Information Technology Goods (Requirement of Compulsory Registration) Order, 2021

ii) BIS has formulated Blank Detail Specification (BDS) for IS 16910 for performance requirements of CCTVs. The procuring government agency can stipulate their own technical requirements for the parameters listed in the BDS and the testing can be done as per the test methods prescribed in the standard.

iii) The procurement of Video Surveillance System from the brand having history of security breaches and data leakages should be avoided.

iv) **Hardware Security**: For the Hardware Security testing of CCTV cameras, the government agencies should use the testing infrastructure available with Standardization Testing and Quality Certification (STQC) Laboratory or any other agency notified by MeitY from time to time for testing the CCTVs as per the Essential Requirement(s) notified under the PPO for CCTV.

v) **Network Security**: The general cyber security practices for installation and monitoring should be also be adopted. Maintain the network isolation (Air-Gap) from the public network to minimize the risk of unauthorized access and potential cyberattacks. Wherever, air gap is not possible, Network segmentation, secure tunnel/Virtual Private Network (VPN) /Dedicated Lease Line etc. should be used for restricting access to CCTV systems and isolate them from critical infrastructure and sensitive data. Use MAC address binding to prevent the unauthorized access by unidentified devices.

vi) **Secure Physical Access**: Restrict physical access to the CCTV control room and equipment. Only authorized personnel should have access to the system. Use locks, access control systems, and surveillance measures to protect the equipment.

vii) **Strong Passwords**: Change default passwords immediately upon installation and use strong, unique passwords for all cameras, recorders, and access points. Avoid using easily guessable information or common words.

viii) **Regular Firmware Updates**: Manufacturers often release updates that address security vulnerabilities. Regularly check for updates and apply them promptly keeps the firmware and software of your CCTV devices up to date.

ix) **Encryption of Data**: Ensure all communication between cameras, recorders, and viewing devices is encrypted. This prevents unauthorized individuals from intercepting and accessing sensitive information.

x) **Disable Unused Features**: Turn off or disable any features and services that are not necessary for the proper functioning of the CCTV system. Each enabled feature potentially introduces another security vulnerability.

xi) **Secure Remote Access**: If remote access is required for maintenance or monitoring, implement a secure VPN (Virtual Private Network) for remote connections. Avoid exposing the system directly to the internet whenever possible. IPBEUs (IP-Based Encryption Unit) to safeguard data transmission between cameras and recording devices, Lease Line for dedicated and secure

network connectivity and Implementation of data diodes to ensure unidirectional flow of information, enhancing security.

xii) **Regular Auditing and Monitoring:** Monitor the CCTV system logs for unusual activities and potential security breaches. Regularly audit the system to ensure that everything is functioning correctly and there are no unauthorized access attempts.

xiii) **Physical Camera Security:** Position cameras in a way that prevents tampering and vandalism. Use vandal-resistant camera housings and install them in high and secure locations where they are less likely to be tampered with.

xiv) **User Access Control:** Implement a strict access control policy to limit the number of individuals who can access the CCTV system and its data. Assign different levels of access based on roles and responsibilities.

xv) **Data Storage and Retention:** Ensure proper data storage and retention policies are in place. Securely store recordings and define how long data should be retained before it gets automatically deleted. Data Storage should be in terms of storage duration (number of Days) based on operational requirements rather than storage capacity. The data storage of all CCTVs installed at Government Establishment/Public Places should be mandated to be within the India even if it is stored in cloud platforms.

xvi) **Staff Training:** Provide comprehensive training to employees and system administrators on security best practices. Make sure they understand the potential risks and how to mitigate them effectively.

xvii) **Regular Security Assessments:** Conduct periodic security assessments and penetration tests to identify vulnerabilities and weaknesses in the CCTV system. Address any issues discovered promptly.

xviii) **Tender/RFP** should encompass both Hardware and Software parts of the Bill of Materials (BoM) combined presenting comprehensive specifications for these components to facilitate the interoperability of the HW/SW as a whole in the VSs system. Model Technical Specifications/Guidelines for CCTVs/VSS issued by MHA from time to time, should be adopted while formulating the technical specifications for procurement of CCTV/VSS.

xix) **CCTV Device testing and certification:** CCTV Cameras (Analog/ IP/ Analog Speed Dome/ IP Speed Dome) should comply with the Essential Requirements (ERs) notified as part of the PPO for CCTV in Gazatte of India (EXTRAORDINARY, PART II—Section 3—Sub-section (ii) dated 7th March,

2024, at Sr. No. No. 1062) to ensure the security of the VSS / CCTV systems, as amended from time to time. The security testing certificate for CCTV/VSS to be issued by Standardisation Testing and Quality Certification (STQC) Laboratory or any other agency notified by MeitY from time to time. The validity of the test report issued by STQC Lab will be three years from the date of issue of the report.

5. In this regard, the Government Ministries, Departments are advised to instruct the Chief Information Security Officers (CISOs) of their respective organizations and subordinate organizations for enforcing the above measures to address the security threats of the CCTV network vulnerability and to ensure the overall security and integrity of CCTV/Video Surveillance Systems.

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रजिस्ट्री सं. डी.एल.- 33004/99 REGD. No. D. L.-33004/99

भारत का राजपत्र
The Gazette of India

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असाधारण
EXTRAORDINARY

भाग I—खण्ड 1
PART I—Section 1

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संचार मंत्रालय

(दूरसंचार विभाग)

अधिसूचना

नई दिल्ली, 21 अक्तूबर, 2024

विषय: सरकारी अधिप्राप्ति (मेक इन इंडिया को प्राथमिकता) आदेश 2017- दूरसंचार उत्पादों, सेवाओं अथवा कार्यों संबंधी अधिसूचना - के संबंध में।

फा. सं. 18-10/2017-आईपी.—संदर्भ: i) औद्योगिक नीति एवं संवर्धन विभाग (डीआईपीपी) आदेश संख्या पी-45021/2/2017-बी.ई.-II दिनांक 15.06.2017

ii) औद्योगिक नीति और संवर्धन विभाग (डीआईपीपी) आदेश सं. पी-45021/2/2017-पीपी (बीई-II) संशोधन दिनांक 28.05.2018

iii) उद्योग संवर्धन और आंतरिक व्यापार विभाग (डीपीआईआईटी) आदेश सं. पी-45021/2/2017 (बी.ई.-II) संशोधन दिनांक 29.05.2019

iv) उद्योग संवर्धन और आंतरिक व्यापार विभाग (डीपीआईआईटी) आदेश सं. पी-45021/2/2017 (बी.ई.-II) संशोधन दिनांक 04.06.2020

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v) उद्योग संवर्धन और आंतरिक व्यापार विभाग (डीपीआईआईटी) आदेश सं. पी-45021/2/2017 (बी.ई.-II) संशोधन दिनांक 16.09.2020

vi) उद्योग संवर्धन और आंतरिक व्यापार विभाग (डीपीआईआईटी) आदेश सं. पी-45021/2/2017-पीपी (बी.ई.-II)-खंड(4) खंड.II संशोधन दिनांक 19.07.2024

सरकार ने पूर्ववर्ती औद्योगिक नीति और संवर्धन विभाग (डीआईपीपी) के दिनांक 15.06.2017 के आदेश संख्या पी-45021/2/2/2017-बीई-II के माध्यम से सरकारी अधिप्राप्ति (मेक इन इंडिया को प्राथमिकता), आदेश 2017 (इसके बाद "पीपीपी-एमआईआई आदेश" के नाम से जाना गया) जारी किया था जिसे 'मेक इन इंडिया' को प्रोत्साहन देने और आय तथा रोज़गार बढ़ाने के उद्देश्य से भारत में वस्तुओं और सेवाओं के विनिर्माण और उत्पादन को बढ़ावा देने के लिए दिनांक 28.05.2018, 29.05.2019, 04.06.2020, 16.09.2020 और 19.07.2024 के आदेशों द्वारा संशोधित किया गया था।

2. तदनुसार दूरसंचार विभाग ने एतद्वारा दिनांक 29.08.2018 और 31.08.2021 की अपनी पिछली अधिसूचना संख्या 18-10/2017-आईपी का अधिक्रमण करते हुए यह अधिसूचना जारी की है।

3. यह अधिसूचना कंपनी अधिनियम में यथा परिभाषित सरकारी कंपनियों सहित भारत सरकार के अधीन सभी संबद्ध अथवा अधीनस्थ कार्यालयों या स्वायत्त निकायों और/अथवा सभी केंद्रीय स्कीमों/केंद्रीय क्षेत्र की स्कीमों के अंतर्गत अधिप्राप्ति करने वाले राज्यों और स्थानीय निकायों द्वारा की गई अधिप्रापण के संबंध में लागू होगी जहां यह स्कीम पूर्ण रूप से या आंशिक रूप से भारत सरकार द्वारा वित्त पोषित की जाती है। यह अधिसूचना उन परियोजनाओं पर भी लागू होगी जिनमें डिजिटल भारत निधि द्वारा वस्तुओं, सेवाओं या कार्यों का वित्तपोषण किया जा रहा हो अथवा उन्हें बोलियों के माध्यम से आवंटित किया जा रहा हो।

4. डीपीआईआईटी के पीपीपी-एमआईआई आदेश दिन ंक 19.07.2024 के खंड 2, 3 और 11 के अनुसार अधिसूचित किया जाता है कि **तालिका-क** में सूचीबद्ध दूरसंचार उत्पादों, सेवाओं और कार्यों में पर्याप्त स्थानीय क्षमता और स्थानीय प्रतिस्पर्धा है। श्रेणी-I के स्थानीय आपूर्तिकर्ता को तालिका-क में प्रत्येक मद के लिए निर्दिष्ट न्यूनतम घरेलू सामग्री कंटेंट (एमएलसी) की आवश्यकताओं को पूरा करना चाहिए। तालिका-क में प्रत्येक मद के संबंध में कुल घरेलू सामग्री (एलसी) में से डिजाइन के लिए एलसी प्रतिशत की अधिकतम सीमा भी विनिर्दिष्ट की गई है। इनपुट को घरेलू सामग्री (एलसी) के रूप में योग्यता प्राप्त करने के लिए शर्तें तालिका-ख में विनिर्दिष्ट की गई हैं।

5. डीपीआईआईटी के पीपीपी-एमआईआई आदेश दिन ंक 19.07.2024 के खंड 2 क के अनुसार दूरसंचार और नेटवर्किंग उत्पादों के लिए पीएलआई स्कीम के अंतर्गत उत्पादों का निर्माण करने वाले और प्रोत्साहन प्राप्त करने वाले विनिर्माताओं की सूची समय-समय पर https://dot.gov.in/public-procurement-preference-make-india और https://pli-telecom.udyamimitra.in/ पर प्रकाशित और अपलोड की जाएगी। ऐसे विनिर्माताओं को विनिर्दिष्ट अवधि तक सूची में विनिर्दिष्ट उत्पादों के लिए श्रेणी-।। के स्थानीय आपूर्तिकर्ता माना जाएगा।

6. डीपीआईआईटी के पीपीपी-एमआईआई आदेश दिन ंक 19.07.2024 के खंड 9 (क) के संदर्भ में निविदा, बोली या अनुरोध करते समय आपूर्तिकर्ता फार्म-1 में स्व-प्रमाणन संबंधी सूचना प्रदान करेगा जिसमें यह विनिर्दिष्ट किया जाएगा कि प्रस्तावित मद (मदें) न्यूनतम घरेलू सामग्री (सामग्रियां) की आवश्यकताओं को पूरा करती है और उस स्थान (स्थानों) का विवरण देगा जहां घरेलू मूल्य संवर्धन किया गया है।

7. डीपीआईआईटी के पीपीपी-पीएमआई आदेश दिन ंक 19.07.2024 के खंड 9 (ङ) के संदर्भ में शिकायतों की जांच करने और यादृच्छिक आधार पर स्थानीय सामग्री के संबंध में स्व-घोषणाओं और लेखा परीक्षकों/लेखाकारों के प्रमाणपत्रों का स्वतंत्र रूप से सत्यापन करने के लिए एतद्वारा निम्नलिखित समिति का गठन किया गया है:

क) उप महानिदेशक (टीसी), टीईसी, नई दिल्ली	- अध्यक्ष
ख) निदेशक (तकनीकी), सी-डॉट	- सदस्य
ग) अध्यक्ष द्वारा चुना गया कोई अन्य सदस्य	- सदस्य

घ) डीजी, टीईपीसी - सदस्य

ङ) निदेशक (वित्त), दूरसंचार विभाग - सदस्य

च) निदेशक (आईपी) - सदस्य

छ) निदेशक, टीईसी - संयोजक

8. पीपीपी-एमआईआई के दिनांक 19.07.2024 के आदेश के खंड 9 (च) के संदर्भ में दो लाख रुपये का शिकायत शुल्क अथवा घरेलू रूप से आपूर्ति किए गए दूरसंचार उत्पादों, सेवाओं अथवा कार्यों के मूल्य का 1% (जो भी अधिक हो, अधिकतम 5 लाख रुपये तक) का भुगतान अनिवार्य रूप से डिमांड ड्राफ्ट अथवा ऑनलाइन द्वारा किया जाना चाहिए और शिकायतकर्ता की शिकायत के साथ दूरसंचार इंजीनियरिंग केंद्र (टीईसी) में जमा किया जाना चाहिए। यदि शिकायत गलत पाई जाती है तो शिकायत शुल्क जब्त कर लिया जाएगा। हालांकि यदि शिकायत आंशिक या पूर्ण रूप से सही पाई जाती है, तो शिकायतकर्ता का जमा शुल्क बिना किसी ब्याज के वापस कर दिया जाएगा।

9. समिति के निष्कर्षों को दिनांक 19.07.2024 के पीपीपी-एमआईआई के आदेश के खंड 9 के अनुसार उचित कार्रवाई के लिए प्रापण इकाई को भेजा जाएगा।

10. **तालिका-क** में सूचीबद्ध उत्पादों, सेवाओं अथवा कार्यों के लिए जिनके पास संबद्ध टीईसी जीआर / आईआर है, प्रापण इकाई को न्यूनतम आवश्यकता के रूप में संबंधित टीईसी जीआर / आईआर विनिर्दिष्ट करना होगा।

11. स्टार्ट-अप (उद्योग संवर्धन और आंतरिक व्यापार विभाग द्वारा पंजीकृत) के लिए पूर्व टर्नओवर और पूर्व अनुभव की शर्त में छूट दी जा सकती है बशर्ते कि स्टार्ट-अप आवश्यक गुणवत्ता और तकनीकी विशिष्टताओं को पूरा करता हो।

12. स्वदेशी प्रौद्योगिकियों का समर्थन और संवर्धन करने के लिए प्रूफ ऑफ कॉन्सेप्ट (पीओसी) को योग्यता अनुभव, जहां कहीं भी लागू हो, के विकल्प के रूप में निविदाओं और आरएफपी में शामिल किया जा सकता है।

13. उल्लेखनीय है कि देश में फिलहाल स्वदेशी प्रौद्योगिकी वाले 5जी उत्पादों का परीक्षण किया जा रहा है। तालिका-क की समय-समय पर समीक्षा की जाएगी और विनिर्माण क्षमता के आधार पर 5जी उत्पादों/वर्चुअल राउटर को शामिल करने पर विचार किया जाएगा। घरेलू विनिर्माताओं और उनके संघों (एसोसिएशन) को प्रोत्साहित किया जाता है ताकि वे दूरसंचार विभाग को समय-समय पर सक्रिय रूप से विनिर्माण क्षमता का साक्ष्य usip.hq-dot@gov.in पर ईमेल द्वारा उपलब्ध कराएं ताकि ऐसे उत्पादों को शामिल किया जा सके।

14. आयातित उत्पादों की पेशकश करने वाले बोलीदाता गैर-स्थानीय आपूर्तिकर्ताओं की श्रेणी में आएंगे। वे परिवहन, बीमा, स्थापना, कमीशनिंग, प्रशिक्षण और एएमसी/सीएमसी आदि जैसी बिक्री के बाद सेवा सहायता जैसी सेवाओं का स्थानीय मूल्यवर्धन के रूप में दावा करके स्वयं को श्रेणी-1 स्थानीय आपूर्तिकर्ता/श्रेणी-2 स्थानीय आपूर्तिकर्ता होने का दावा नहीं कर सकते।

15. यह अधिसूचना अधिसूचित होने की तारीख से लागू होगी। स्पष्ट किया जाता है कि इस अधिसूचना में निहित विशिष्ट शर्तों को छोड़कर समय-समय पर संशोधित सरकारी अधिप्राप्ति (मेक इन इंडिया को प्राथमिकता) आदेश 2017 की अन्य सभी शर्तें प्रभावी हैं और इनका सख्ती से पालन किया जाना है।

तालिका-क

क्र. सं.	**प्रोडक्ट केटेगरी**	**एचएसएन कोड**	**उत्पाद का विवरण**	**मिनिमम लोकल कंटेंट (एमएलसी) (% में)**	**कुल एलसी में से डिजाइन के लिए एलसी की अधिकतम सीमा (% में)**
1	एन्क्रिप्शन/ यूटीएम (यूनिफाइड थ्रेट मैनेजमेंट) प्लेटफ़ॉर्म (टीडीएम और आईपी)	85176290	टीडीएम (टाइम डिवीजन मल्टीप्लेक्स)	65	55
			आईपी (इंटरनेट प्रोटोकॉल)	65	55

2	आईपी/ एमपीएलएस कोर राउटर्स/ एज/एग्रीगेशन/ राउटर	85176290	आईपी एज, एग्रीगेशन और कोर राउटर (समग्र क्षमता <=1 टीबीपीएस)	60	40
			सेल साइट राउटर (सीएसआर); (कुल क्षमता ≤1.2 टीबीपीएस) जिसमें वर्चुअल राउटर शामिल हैं	60	40
			एसडी-डब्ल्यूएएन राउटर्स (एसडीएन केटेगरी) (इंटरफ़ेस =<10 जीबीपीएस)	60	40
3	मैनेज्ड लीज़्ड लाइन नेटवर्क इक्विपमेंट, आईपी बेस्ड एक्सेस सिस्टम-डीएसएल	85176290, 85176990	सेंट्रल ऑफिस इक्विपमेंट और डीएसएल/एचडीएसएल/एस-एचडीएसएल के सीपीई	60	40
4	ईथरनेट स्विचेज (एल2 और एल 3)	85176290	एल2/एल3 ईथरनेट स्विच 1-100 जीबीई स्टैकेबल / स्टैंडअलोन / मॉड्यूलर स्विच, कोर / डिस्ट्रीब्यूशन / एक्सेस लेयर्स (पीओई / पीओई + सहित) के लिए)	60	40
			इंडस्ट्रियल ग्रेड एलएएन 1, 10 जीबीई पीओई/पीओई+/ नॉन -पीओई स्विच (ईथरनेट पर पावर)	60	40
5	आईपी आधारित सॉफ्ट स्विच, आईएमएस, यूनिफाइड कम्युनिकेशन सिस्टम	85176290, 85176990	आईपी पीबीएक्स / यूनिफाइड कम्युनिकेशन सिस्टम (यूसीएस)	60	40
			यूसीएस (नॉन-आईएमएस बेस्ड):	60	40
6	वायरलेस पीएबीएक्स / आईपी पीबीएक्स	85176290	वायरलाइन पीएबीएक्स/सर्विस पीबीएक्स	65	45
7	मीडिया गेटवे	85176290, 85176990	मीडिया गेटवे	65	45
8	कस्टमर प्रीमाइसेस इक्विपमेंट (सीपीई) (वाई-फाई एक्सेस पॉइंटस, कम क्षमता वाले राउटर्स, मीडिया कन्वर्टर सहित),	85176290, 85176990	सीपीई (वाई - फाई अभिगम बिंदु और राउटर्स सहित),	50	30
			मीडिया कनवर्टर	50	30
			टीडीएम ओवर आईपी कनवर्टर	50	30

	सीपीई - एनएफवी / एसडीएन / लीज्ड-लाइन मोडेम		सीपीई - एनएफवी/एसडीएन (वर्चुअलाइज्ड सीपीई विथ क्लाउड मैनेजमेंट)	50	30
			लीज़्ड-लाइन मोडेम एन* 2एमबीपीएस/सर्विस	50	30
9	एसडीएच/ एनजी-एसडीएच	85176290	एसडीएच/एनजी-एसडीएच (एसटीएम 1/4/16/64)	65	45
10	ईथरनेट/एमपीएलएस- टीपी/ पैकेट ऑप्टिकल ट्रांसपोर्ट इक्विपमेंट /पीटीएन सिस्टम	85176290	एमपीएलएस- टीपी/ पैकेट ऑप्टिकल ट्रांसपोर्ट इक्विपमेंट/पीटीएन (विद और विदाउट ओटीएन इंटरफेसेस) सिस्टम्स	65	45
11	सीडब्ल्यूडीएम सिस्टम	85176290	सीडब्ल्यूडीएम (एनx100जी, एनx10जी)	60	40
12	डीडब्ल्यूडीएम सिस्टम	85176290, 85176270	डीडब्ल्यूडीएम (10जी तक) मक्स/डी मक्स ट्रांसपोंडर/मक्सपोंडर के साथ; आरओएडीएम (रेकॉन्फ़िगराबल ऑप्टिकल ऐड ड्राप मल्टीप्लेक्सर), पैसिव डीडब्ल्यूडीएम एमयूएक्स/डीएमयूएक्स	60	40
13	जीपीओएन/एक्सजीपीओएन इक्विपमेंट (ओएनटी और ओएलटी सहित)	85176290, 85172990	जीपीओएन/एक्सजीपीओएन इक्विपमेंट (ओएनटी और ओएलटी सहित)	60	40
14	एक्सजीएस-पीओएन इक्विपमेंट (ओएनटी और ओएलटी सहित)	85176290, 85172990	एक्सजीएस-पीओएन (10जी डीएस - 10जी यूएस) (ओएनटी और ओएलटी सहित)	60	40
15	एसडीएच/पीडीएच क्रॉस कनेक्टस	85176290	एसडीएच/पीडीएच क्रॉसकनेक्टेड (एसटीएम: 1/4/16/64; ई1/ई3/डीएस3; ईथरनेट इंटरफेसेस)	60	40
16	स्मॉल साइज रेडियो सब सिस्टम	85176290	माइक्रो 2जी बीटीएस	60	40
17		85176290, 85176990	मैक्रो ईनोड बी, टीईसी ईएनबी जीआर (टीडीडी/एफडीडी) के अनुसार	60	40
			माइक्रो इनोडबी, टीईसी ईएनबी जीआर (टीडीडी/एफडीडी) के अनुसार	60	40

	एलटीई/एलटीई-आर बेस्ड मोबाइल सिस्टम, इसके विभिन्न व्युत्पन्न के साथ		पिको इनोडबी, टीईसी ईएनबी जीआर (टीडीडी/एफडीडी) के अनुसार	60	40
			होम बेस स्टेशन/ फेमटो	60	40
			ईपीसी	60	40
			4जी एनआईबी स्मॉल सेल	60	40
18	वाई-फाई बेस्ड ब्रॉडबैंड वायरलेस एक्सेस सिस्टम - इनडोर	85176290, 85176990	इनडोर एपी (802.11 एक्स एमआईएमओ, 2.4 और 5 गीगाहट्र्ज बैंड)	55	35
			इनडोर एपी (802.11 एसी एमआईएमओ, 2.4 और 5 गीगाहट्र्ज बैंड)	55	35
19	वाई-फाई बेस्ड ब्रॉडबैंड वायरलेस एक्सेस सिस्टम - आउटडोर	85176290, 85176990	आउटडोर एपी (802.11एसी - एमआईएमओ 2.4 और 5 गीगाहट्र्ज बैंड) - आईपी67 रेटेड	55	35
			आउटडोर एपी (802.11एक्स एमआईएमओ, 2.4 और 5 गीगाहट्र्ज बैंड)	55	35
20	वाई-फाई एक्सेस कंट्रोलर (विद इंटरऑपरेबिलिटी और मल्टी-वेंडर एपी सपोर्ट)	85176290	टाइप 1: <= 2000 एपी सपोर्ट	55	35
			टाइप 2: 5000 एपी सपोर्ट तक	55	35
21	ऑथेंटिकेशन, ऑथॉरिज़ेशन एंड एकाउंटिंग (एएए) सर्वर	85176290	एएए ब्लॉक (रेडियस)	50	35
			एएए ब्लॉक (रेडियस/डायमीटर)	50	35
22	रेडियो सिस्टम (आईपी/हाइब्रिड)	85176290	रेडियो सिस्टम (आईपी/हाइब्रिड) (5 गीगाहट्र्ज बैंड - लाइसेंस छूट बैंड)	55	35
23	सॉफ्टवेयर डिफाइंड रेडियो कॉग्निटिव रेडियो सिस्टम	85176290	रगेडाइज्ड हैंडहेल्ड, रगेडाइज्ड या नॉन-रगेडाइज्ड व्हीकल माउंट, पी2पी या पी2एमपी या मोबाइल एडहॉक टोपोलॉजी, अनलाइसेंसड बैंड ऑपरेशन	55	35
24	रिपीटर्स (आरएफ/आरएफ-ओवर-ऑप्टिकल), आईबीएस	85176290, 85177100	रिपीटर्स (आरएफ/आरएफ-ओवर-ऑप्टिकल), आईबीएस और डिस्ट्रिब्युटेड एंटीना	60	40

	और डिस्ट्रिब्यूटेड एंटीना सिस्टम (इनडोर/आउटडोर)		सिस्टम (इनडोर/आउटडोर) जिसमें सक्रिय और निष्क्रिय सहायक उपकरण शामिल हैं		
25	वीसैट टर्मिनल सबसिस्टम - ओडीयू/	85176290, 85256091	एंटीना	50	25
26	वीसैट हब सबसिस्टम	85256091	एनएमएस एच/डब्ल्यू और एस/डब्ल्यू	50	25
			एंटीना (>=7 मीटर) और सहायक उपकरण आदि	50	25
27	सैटेलाइट फोन (एस बैंड)	85176990	वॉयस और डेटा दोनों (9.6 केबीपीएस तक)	50	25
28	डिजास्टर कम्युनिकेशन सिस्टम्स आदि, जिसमें बैकपैक सैटेलाइट उत्पाद (एस बैंड) शामिल हैं	85176990	पोर्टेबल टर्मिनल वॉयस और डेटा (2 एमबीपीएस यूएल)	50	25
29	सैटेलाइट बेस्ड सिस्टम, टर्मिनल जिसमें लोकेशन, रिसोर्सेज ट्रैकिंग (एल, एस और के यू , के ए बैंड) शामिल हैं	85176990	टू वे एमएसएस डेटा टर्मिनल (सैटेलाइट रिसीवर विद लोकेशन डेटा) एस - बैंड (विभिन्न एप्लीकेशनों के लिए सैटेलाइट आईओटी डिवाइस यानी पोजीशन ट्रैकिंग, संदेश भेजना आदि)	50	25
			वन वे रिसीवर एमएसएस डेटा टर्मिनल (सैटेलाइट ट्रांसमीटर और रिसीवर - वन वे) एस - बैंड (सिंप्लेक्स / हाफ-डुप्लेक्स, मोबाइल सैटेलाइट सर्विस (एमएसएस) टर्मिनल, सैटेलाइट आईओटी डिवाइस)	50	25
30	टेलीकॉम नेटवर्क मैनेजमेंट सिस्टम (एनएमएस) इसके विभिन्न व्युत्पन्न के साथ	85176290		65	50
31	ऑप्टिकल फाइबर	9001000	सिंगल मोड	50	0
32	ऑप्टिकल फाइबर केबल	85447090	ऑप्टिकल फाइबर केबल (यूजी और ओएच)	55	0
33	टेलीकॉम पावर सिस्टम (सौर ऊर्जा सहित)	85176990	यूपीएस, पावर प्लांट, इन्वर्टर (सौर ऊर्जा सहित) 0.5-1500 केवीए/ पावर प्लांट: 1-2000 केवीए (मॉड्यूलर)	55	30

34	टेलीकॉम बैटरी (लीड एसिड)	85176990	5000 एएच तक लीड एसिड (वीआरएलए सहित)	55	30
35	आईपी ऑडियो/वीडियो फोन/एडेप्टर	85176990	आईपी ऑडियो / वीडियो फोन / एडाप्टर	50	15
36	आईपी/इंटरनेट सेट-टॉप बॉक्स	85176990	आईपी / इंटरनेट सेट-टॉप बॉक्स	55	35

एमएलसी%- मेक इन इंडिया दूरसंचार उत्पाद, सेवा या कार्य के रूप में अर्हता प्राप्त करने के लिए कुल सामग्री बिल (उत्पादन लागत) के प्रतिशत के रूप में न्यूनतम स्थानीय सामग्री।

तालिका-ख

दूरसंचार उत्पादों के विनिर्माण के लिए मुख्य इनपुट/स्टेज और घरेलू सामग्री (कंटेंट) के रूप में पात्र बनने के लिए इनपुट संबंधी शर्त

दूरसंचार उत्पादों के विनिर्माण के लिए मुख्य इनपुट/स्टेज *	**इनपुट को स्थानीय सामग्री के रूप में योग्य बनाने के लिए पूरी की जाने वाली शर्तें**
1) डिजाइन हार्डवेयर डिजाइन सॉफ्टवेयर डिजाइन और विकास	डिजाइन के लिए अधिकतम घरेलू कंटेंट (सामग्री) (एलसी) का प्रतिशत **तालिका-क (अंतिम कॉलम)** के अनुसार अतिरिक्त शर्तों के अधीन है जिसका दावा स्थानीय विनिर्माता द्वारा भारत में आईपीआर सृजित करने के लिए वहन की गई की गई /परिशोधित इन-हाउस/देश में आर एंड डी लागतों के आधार पर दूरसंचार उत्पादों के लिए किया जा सकता है हार्डवेयर डिजाइन के लिए बौद्धिक संपदा अधिकार (आईपीआर) भारत के पास है; एवं सॉफ्टवेयर डिजाइन एवं विकास का कॉपीराइट भारत के पास है।
2) कंपोनेंट इंटीग्रेटेड चिपस (आईसी) - प्रोसेसर, मेमोरी आदि। एक्टिव कंपोनेंट - ट्रांजिस्टर, डायोड आदि। पेसिव कंपोनेंट – रेजिस्टर, केपेसिटर, इंडक्टर आदि।	भारत में विनिर्मित होना चाहिए
3) पीसीबी फेब्रिकेशन	भारत में विनिर्मित होना चाहिए
4) केबल/चेसिस आदि चेसिस केबल रैक हीट सिंक एंक्लोजर	भारत में विनिर्मित होना चाहिए
5) आरएफ कंपोनेंट/सबसिस्टम डुप्लेक्सर/फिल्टर एंटीना	भारत में विनिर्मित होना चाहिए
6) असेंबली/इन्टीग्रेशन/टेस्टिंग#	**तालिका-क** में सूचीबद्ध दूरसंचार उत्पादों के संबंध में असेंबली/इन्टीग्रेशन/टेस्टिंग के लिए घरेलू स्थानीय सामग्री (कंटेंट) (एलसी) की अधिकतम सीमा कुल उत्पाद सामग्री के बिल की 10% होगी।
* उत्पाद में उपर्युक्त उल्लिखित कुछ/सभी इनपुट/स्टेज शामिल हो सकते हैं। घरेलू सामग्री (कंटेंट) की गणना करते समय केवल उन इनपुट/स्टेज पर विचार किया जाएगा जो दूरसंचार उत्पाद के विनिर्माण में प्रासंगिक हैं।	

#यदि किसी सिस्टम या सबसिस्टम को केवल असेंबल/इन्टीग्रेट/टेस्ट किया गया है, तो वास्तविक घरेलू सामग्री (कंटेंट) को सिस्टम/सबसिस्टम की 10% लागत के रुप में माना जाएगा।

फॉर्म 1

दूरसंचार उत्पाद, सेवाओं या कार्यों के लिए घरेलू सामग्री (कंटेंट) (एलसी) के संबंध में स्व-प्रमाणन का प्रारूप 100/- रुपये के गैर-न्यायिक स्टाम्प पेपर पर प्रस्तुत किया जाना है।

दिनांक:

मैं _____**पुत्र/श्री, पुत्री/श्री, पत्नी/श्री**, निवासी _____ एतद्वारा सत्यनिष्ठा से पुष्टि करता/करती हूं और निम्नलिखित घोषणा करता/करती हूं:

कि मैं दूरसंचार विभाग, भारत सरकार के दिनांक_____ की अधिसूचना संख्या:_____ द्वारा जारी किए गए निबंधन और शर्तों का पालन करने के लिए सहमत हूं।

यहां दी गई जानकारी मेरे सर्वोत्तम ज्ञान और विश्वास के अनुसार सही है और मैं एल.सी. का मूल्यांकन करने के प्रयोजन से प्रायण यूनिट या दूरसंचार विभाग, भारत सरकार द्वारा नामित किसी अन्य प्राधिकारी के समक्ष प्रासंगिक रिकॉर्ड प्रस्तुत करने का वचन देता हूँ।

मेरे द्वारा उक्त दूरसंचार उत्पादों/सेवाओं/कार्यों सहित सभी इनपुट की एलसी का सत्यापन किया गया है और मैं इसमें किए गए दावों की सत्यता के लिए जिम्मेदार हूं।

कि यदि यहां उल्लिखित दूरसंचार उत्पाद/सेवाओं/कार्यों की एलसी गलत पाई जाती है और निर्धारित एलसी मानदंडों को पूरा नहीं करती है तो दूरसंचार विभाग, भारत सरकार द्वारा नामित किसी प्राधिकारी के आकलन के आधार पर **मैं और मेरे सांविधिक लेखा परीक्षक या लागत लेखा परीक्षक (यदि लागू हो) ऐसे सभी असत्य/गलत तथ्यों और आंकड़ों के लिए दिनांक 19.07.2024 के डीपीआईआईटी पीपीपी-एमआईआई के आदेश के खंड (9) में विनिर्दिष्ट कार्रवाई के लिए उत्तरदायी होंगे।**

मैं 2 वर्ष की अवधि के लिए कंपनी के रिकॉर्ड में एलसी के लिए अपने दावे की पुष्टि करने के लिए विस्तृत विवरण/सूचना (प्रत्येक उत्पाद के लिए अलग से) रखने के लिए सहमत हूं और इसे किसी भी प्राधिकारी को सत्यापन के लिए उपलब्ध कराऊंगा। मैं अन्य घरेलू विनिर्माताओं/व्यापारियों से खरीदी गई मदों से संबंधित स्थानीय सामग्री (कंटेट) का रिकॉर्ड भी रखूंगा।

[कृपया निम्नलिखित सूचना उपलब्ध करावें]

i स्थानीय आपूर्तिकर्ता का नाम और विवरण (पंजीकृत कार्यालय, विनिर्माण यूनिट का स्थान, कानूनी संस्था की प्रकृति)

ii. वह तिथि जिस दिन यह प्रमाणपत्र जारी किया गया है

iii. दूरसंचार उत्पाद/सेवाएं/कार्य जिसके लिए प्रमाणपत्र प्रस्तुत किया गया है

iv. वह प्रायण एजेंसी जिसे प्रमाण पत्र प्रस्तुत किया गया है

v. दावा किए गए एलसी का प्रतिशत

vi. विनिर्माता यूनिट का नाम और संपर्क विवरण

vii. उत्पाद का विक्रय मूल्य

viii. उत्पाद की फैक्ट्री से बाहर कीमत

ix. माल ढुलाई, बीमा और हैंडलिंग

x. सामग्री का कुल बिल

xi. दूरसंचार उत्पाद/सेवाओं/कार्यों के विनिर्माण के लिए प्रयुक्त इनपुट की सूची और कुल लागत मूल्य

xii. स्थानीय रूप से प्राप्त इनपुट की सूची और कुल लागत। यदि इनपुट का उत्पादन इन-हाउस नहीं किया गया है तो कृपया स्थानीय आपूर्तिकर्ताओं से एलसी प्रमाणपत्र संलग्न करें।

xiii. प्रत्यक्ष या अप्रत्यक्ष रूप से आयातित इनपुट की सूची और लागत

मैं एतद्वारा प्रमाणित करता हूं कि उपर्युक्त आदेश के सभी प्रावधानों और सिद्धांतों/गणना के आधार को पढ़ने के बाद स्थानीय सामग्री (कंटेट) की गणना में निम्नलिखित शामिल नहीं है:

(क) आयातित वस्तुएं स्थानीय स्तर पर पुनर्विक्रेताओं (रिसेलर)/वितरक से प्राप्त की जाती हैं।

(ख) भारत से बाहर भुगतान की गई लाइसेंस फीस / रॉयल्टी / तकनीकी प्रभार

(ग) पुनःपैक किए गए/नवीनीकृत/पुनःब्रांडेड आयातित उत्पादों की खरीद/आपूर्ति

मैं अपने सर्वोत्तम ज्ञान और विश्वास के अनुसार यह भी प्रमाणित करता हूँ कि ऊपर दिए गए सभी विवरण सही और पूर्ण हैं। मैं दिनांक 19.07.2024 के डीपीआईआईटी पीपीपी-एमआईआई के आदेश और दिनांक ..... के डीओटी की पीपीपी-एमआईआई अधिसूचना के निबंधन और शर्तों का पालन करने के लिए सहमत हूँ।

मैं सहमत हूँ कि घरेलू सामग्री (कंटेट) के संबंध में कोई भी गलत घोषणा देने या एलसी के दावे को प्रमाणित करने में विफल होने पर दिनांक 19.07.2024 के डीपीआईआईटी पीपीपी-एमआईआई के आदेश के खंड (9) में विनिर्दिष्ट दंड लगाया जाएगा।

मैं यह भी प्रमाणित करता हूं तथा व्यक्तिगत जिम्मेदारी लेता हूं कि मैंने इस आदेश में निर्दिष्ट एलसी की गणनाओं और सिद्धांतों पर ध्यान दिया है तथा एलसी की घोषणा करने के बाद मैं किसी भी आधार पर इसमें परिवर्तन करने का प्रयास नहीं करूंगा। बोली प्रस्तुत करने के बाद एलसी की बोली में किसी भी आधार पर मेरे द्वारा किए गए कोई भी परिवर्तन से मेरी बोली अप्रसांगिक हो जाएगी और मैं ऐसे किसी भी परिवर्तन से उत्पन्न होने वाली सिविल/आपराधिक कार्रवाई के लिए स्वयं को उत्तरदायी मानूंगा। मुझे ज्ञात है मैं सहमत हूं कि बोली के बाद एलसी सामग्री में किया गया कोई भी परिवर्तन भविष्य के संविदाओं/बोलियों से फर्म को ब्लेकलिस्ट करने का वैध आधार भी होगा।

हस्ताक्षर:
नाम:
पदनाम: सीईओ/एमडी
पता:
ईमेल एड्रेस:
मोबाइल नंबर:
स्थान:
दिनांक:

<u>अधिसूचना में प्रयुक्त संक्षिप्ताक्षर की सूची</u>

क्र. सं.	संक्षिप्ताक्षर	पूर्ण रूप
1.	एएए	प्रमाणीकरण प्राधिकरण एवं लेखांकन
2.	एएमसी	वार्षिक रखरखाव अनुबंध
3.	एपी	एक्सेस प्वाइंट
4.	बीबीयू	बेस बैंड यूनिट
5.	बीआईएस	भारतीय मानक ब्यूरो
6.	बी-आरएएस	ब्रॉडबैंड रिमोट एक्सेस सर्वर
7.	बीटीएस	बेस ट्रांसीवर स्टेशन

8.	सीडॉट	टेलीमेटिक्स विकास केंद्र
9.	सीएमसी	व्यापक रखरखाव अनुबंध
10.	सीपीई	उपभोक्ता परिसर उपकरण
11.	सीएस	केंद्रीय स्कीम
12.	सीएसआर	सेल साइट राउटर
13.	सीएसएस	केंद्रीय क्षेत्र के स्कीम
14.	सीडब्ल्यूडीएम	कोआर्स वेवलेंथ डिवीजन मल्टीप्लेक्सिंग
15.	डीआईपीपी	औद्योगिक नीति और संवर्धन विभाग
16.	डीओटी	दूरसंचार विभाग
17.	डीपीआईआईटी	उद्योग संवर्धन और आंतरिक व्यापार विभाग
18.	डीएसएल	डिजिटल सब्सक्राइबर लाइन
19.	डीएसएलएएम	डिजिटल सब्सक्राइबर लाइन एक्सेस मल्टीप्लेक्सर
20.	डीडब्ल्यूडीएम	डेन्स वेवलेंथ डिवीजन मल्टीप्लेक्सिंग
21.	ई-नोड बी (ईएनबी)	ईवाल्व नोड बी
22.	ईपीसी	इंजीनियरिंग प्रोक्योरमेंट कंस्ट्रक्शन
23.	एफडीडी	फ़्रीक्वेंसी डिवीज़न डुप्लेक्सिंग
24.	जीएनबी	नेक्स्ट जेनरेशन नोड बी
25.	जीएफआर	सामान्य वित्तीय नियम
26.	जीपीओएन	गीगाबिट पैसिव ऑप्टिकल नेटवर्क
27.	जीआर	सामान्य आवश्यकताएँ
28.	जीएसएम	ग्लोबल सिस्टम फॉर मोबाइल
29.	एचडीएसएल	हाई बिट-रेट डिजिटल सब्सक्राइबर लाइन
30.	आईबीएस	एकीकृत बिल्डिंग सिस्टम
31.	आईसी	इंटीग्रेटेड चिप
32.	आईएमएस	आईपी मल्टीमीडिया सबसिस्टम
33.	आईओटी	इंटरनेट ऑफ़ थिंग्स
34.	आईपी	इंटरनेट प्रोटोकॉल
35.	आईपीआर	बौद्धिक संपदा अधिकार
36.	आईआर	इंटरफ़ेस आवश्यकताएँ
37.	लैन	लोकल एरिया नेटवर्क
38.	एल-2	लैअर-2
39.	एल-3	लैअर-3
40.	एलसी	लोकल कंटेन्ट
41.	एलआई-आयन	लिथियम-आयन
42.	एलटीई	दीर्घकालिक विकास
43.	एलटीई-आर	दीर्घकालिक विकास-रेलवे

44.	एमआईएमओ	मल्टीपल इनपुट मल्टीपल आउटपुट
45.	एमएलसी	मिनिमम लोकल कंटेन्ट
46.	एमपीएलएस	मल्टी-प्रोटोकॉल लेबल स्विचिंग
47.	एमपीएलएस-टीपी	मल्टीप्रोटोकॉल लेबल स्विचिंग-ट्रांसपोर्ट प्रोफ़ाइल
48.	एमएसएस	मोबाइल सैटेलाइट सर्विस
49.	एमयूएक्स	मल्टीप्लेक्सर
50.	एनएफवी	नेटवर्क फ़ंक्शन वर्चुअलाइज़ेशन
51.	एनजी-पीओएन2	नेक्स्ट जेनरेशन गीगाबिट पैसिव ऑप्टिकल नेटवर्क 2
52.	एनआईबी	नेटवर्क इन बॉक्स
53.	एनएमएस	नेटवर्क प्रबंधन प्रणाली
54.	ओएडीएम	ऑप्टिकल ऐड-ड्रॉप मल्टीप्लेक्सर
55.	ओडीयू	आउटडोर यूनिट
56.	ओएलटी	ऑप्टिकल लाइन टर्मिनल
57.	ओएनटी	ऑप्टिकल नेटवर्क टर्मिनल
58.	ओटीएन	ऑप्टिकल ट्रांसपोर्ट नेटवर्क
59.	पी2पी	पॉइंट टू पॉइंट
60.	पी2एमपी	पॉइंट टू मल्टी पॉइंट
61.	पीएबीएक्स	प्राइवेट ऑटो ब्रांच एक्सचेंज
62.	पीबीएक्स	प्राइवेट ब्रांच एक्सचेंज
63.	पीसीबी	प्रिंटेड सर्किट बोर्ड
64.	पीडीएच	प्लेसियोक्रोनस डिजिटल हाइरार्की
65.	पीएमआई	मेक इन इंडिया को प्राथमिकता
66.	पीटीएन	पैकेट ट्रांसपोर्ट नेटवर्क
67.	पीओसी	प्रूफ ऑफ कॉन्सेप्ट
68.	पीओई	पावर ओवर ईथरनेट
69.	पीपीपी-एमआईआई	सार्वजनिक खरीद (मेक इन इंडिया को प्राथमिकता)
70.	पीटीएन	पैकेट ट्रांसपोर्ट नेटवर्क
71.	आरओएडीएम	रीकॉन्फिगरेबल ऑप्टिकल ऐड-ड्रॉप मल्टीप्लेक्सर
72.	आरएंडडी	अनुसंधान और विकास
73.	आरएफ	रेडियो फ्रीक्वेंसी
74.	एस-एचडीएसएल	सिंगल पेयर हाई स्पीड डिजिटल सब्सक्राइबर लाइन
75.	एसडीएच	सिंक्रोनस डिजिटल हाइरार्की
76.	एसडीएन	सॉफ्टवेयर डिफाइन्ड नेटवर्क
77.	एसडी-वैन	सॉफ्टवेयर डिफाइन्ड - वाइड एरिया नेटवर्क
78.	एसटीएम	सिंक्रोनस ट्रांसपोर्ट मॉड्यूल
79.	टीडीडी	टाइम डिवीजन डुप्लेक्सिंग

80.	टीडीएम	टाइम डिवीजन मल्टीप्लेक्सिंग
81.	टीईसी	दूरसंचार अभियांत्रिकी केंद्र
82.	टीईपीसी	दूरसंचार निर्यात संवर्धन परिषद
83.	यूसीएस	एकीकृत संचार प्रणाली
84.	यूएल	एकीकृत लाइसेंस
85.	यूएसओएफ	सार्वभौमिक सेवा दायित्व निधि
86.	यूटीएम	एकीकृत खतरा प्रबंधन
87.	वीआरएलए	वाल्व रेगुलेटेड लीड एसिड
88.	वीसैट	वेरी स्मॉल एपर्चर टर्मिनल
89.	वाई-फाई	वायरलेस फ़िडेलिटी
90.	एक्सजी-पीओएन	10 गीगाबिट पैसिव ऑप्टिकल नेटवर्क
91.	एक्सजीएस-पीओएन	10 गीगाबिट सममित पैसिव ऑप्टिकल नेटवर्क

अशोक कुमार जैन, उप महानिदेशक

MINISTRY OF COMMUNICATIONS

(Department of Telecommunications)

NOTIFICATION

New Delhi, the 21st October, 2024

Subject: Public Procurement (Preference to Make in India) Order 2017- Notification of Telecom Products, Services or Works - regarding.

F. No. 18-10/2017-IP.—Reference: i) Department of Industrial Policy & Promotion (DIPP) Order No. P-45021/2/2017-B.E.-II dated 15.06.2017

ii) Department of Industrial Policy & Promotion (DIPP) Order No. P-45021/2/2017-PP (BE-II) dated 28.05.2018 – Revision

iii) Department for Promotion of Industry & Internal Trade (DPIIT) Order No. P-45021/2/2017 (B.E.-II) dated 29.05.2019 – Revision

iv) Department for Promotion of Industry & Internal Trade (DPIIT) Order No. P-45021/2/2017 (B.E.-II) dated 04.06.2020 – Revision

v) Department for Promotion of Industry & Internal Trade (DPIIT) Order No. P-45021/2/2017 (B.E.-II) dated 16.09.2020 – Revision

vi) Department for Promotion of Industry & Internal Trade (DPIIT) Order No. P-45021/2/2017-PP (B.E.-II)-Part(4) Vol.II dated 19.07.2024 – Revision

The Government had issued the Public Procurement (Preference to Make in India), Order 2017 (hereinafter called as "PPP-MII Order") vide the erstwhile Department of Industrial Policy and Promotion (DIPP) Order No. P-45021/2/2017-B.E.-II dated 15.06.2017 which was further revised vide Orders dated 28.05.2018, 29.05.2019, 04.06.2020, 16.09.2020 and 19.07.2024 to encourage 'Make in India' and to promote manufacturing and production of goods and services in India with a view to enhancing income and employment.

2. Accordingly, the Department of Telecommunications, in supersession of its previous notifications No. 18-10/2017-IP dated 29.08.2018 and 31.08.2021, hereby issues this notification.

3. This Notification shall be applicable in respect of the procurement made by all attached or subordinate offices or autonomous bodies under the Government of India including Government Companies as defined in the Companies Act, and/or the States and Local Bodies making procurement under all Central Schemes/ Central Sector Schemes where the Scheme is fully or partially funded by the Government of India. This Notification shall also be applicable in respect of projects wherein funding of goods, services or works is by the Digital Bharat Nidhi to the extent of having being awarded through bids.

4. In terms of Clause 2, 3 and 11 of the DPIIT's PPP-MII Order dated 19.07.2024, it is hereby notified that telecom products, services and works listed in **Table-A** have sufficient local capacity and local competition. A Class-I

local supplier must meet the Minimum Local Content (MLC) requirement specified against each item in Table-A. Table-A also specifies the maximum ceiling of LC percentage for design out of total LC against each item. The conditions for the inputs to be qualified as Local Content (LC) are specified in Table-B.

5. In accordance with clause 2A of the DPIIT's PPP-MII Order dated 19.07.2024, the list of manufacturers manufacturing products under the PLI scheme for Telecom and Networking products and having received incentive will be published and uploaded from time to time on https://dot.gov.in/public-procurement-preference-make-india and https://pli-telecom.udyamimitra.in/. Such manufacturers will be treated as deemed Class-II local suppliers for the products specified in the list and upto the specified period.

6. In terms of clause 9 (a) of DPIIT's PPP-MII Order dated 19.07.2024, a supplier at the time of tender, bidding or solicitation shall provide self-certification in Form-1 specifying that the item(s) offered meet the minimum local content requirement and shall give details of the location(s) at which the local value addition is made.

7. In terms of clause 9 (e) of the DPIIT's PPP-PMI Order dated 19.07.2024, the following Committee is hereby constituted to examine complaints and conduct independent verification of self-declarations and auditor's/accountant's certificates regarding Local content on random basis:

a) DDG (TC), TEC, New Delhi -Chairperson
b) Director (Technical), C-DOT - Member
c) Any other member(s) as co-opted by the chairperson - Member
d) DG, TEPC - Member
e) Director (Finance), DoT - Member
f) Director (IP) - Member
g) Director, TEC - Convener

8. In terms of clause 9 (f) of DPIIT's PPP-MII Order dated 19.07.2024, a complaint fee of Rs. 2 Lakh or 1% of the value of the locally supplied telecom products, services or works being procured (whichever is higher, up to a maximum of Rs. 5 Lakh) must be paid by Demand Draft or online, and to be deposited with Telecommunications Engineering Centre (TEC), along with the complaint by the complainant. If the complaint is found to be incorrect, the complaint fee shall be forfeited. However, if the complaint is upheld in part or full, deposited fee of the complainant will be refunded without any interest.

9. The Committee's findings shall be forwarded to the procuring entity for appropriate action in accordance with Clause 9 of the DPIIT's PPP-MII Order dated 19.07.2024.

10. For the products, services or works listed in **Table-A** that have an associated TEC GR/IR, the procuring entity must specify the relevant TEC GR/IR as the minimum requirement.

11. The condition of prior turnover and prior experience may be relaxed for Start-ups (registered by Department for Promotion of Industry and Internal Trade), provided that the Start-up meet the necessary quality & technical specifications.

12. To support and promote indigenous technologies, Proof of Concept (PoC) may be included in tenders and RFPs as a substitute for qualifying experience, wherever applicable.

13. It is noted that 5G products with indigenous technology are presently being tested in the country. Table-A shall be reviewed from time to time and addition of 5G products/virtual routers shall be considered based on manufacturing capacity. Domestic manufacturers and their associations are encouraged to pro-actively provide evidence of manufacturing capacity from time to time to DoT to enable inclusion of such products by email to usip.hq-dot@gov.in.

14. The bidders offering imported products will fall under the category of non-local suppliers. They can not claim themselves as Class-I local suppliers/Class-II local suppliers by claiming the services such as transportation, insurance, installation, commissioning, training and after sales service support like AMC/CMC etc. as local value addition.

15. This Notification comes into effect from the date of notification. It is clarified that, except specific conditions contained in this notification, all the other conditions of the Public Procurement (Preference to Make in India) Order 2017 as amended from time to time shall prevail and are to be followed strictly.

TABLE-A

S. No.	Product Category	HSN code	Product Details	Minimum Local Content (MLC) (in %)	Ceiling of LC for Design out of total LC (in %)
1	Encryption/ UTM (Unified threat management) platforms (TDM and IP)	85176290	TDM (Time Division Multiplex)	65	55
			IP (Internet Protocol)	65	55

2	IP/ MPLS Core routers/ Edge/Aggregation/ Router	85176290	IP Edge, Aggregation & Core Routers (overall capacity <=1 Tbps)	60	40
			Cell Site Router (CSR); (Overall capacity ≤1.2 Tbps) including virtual routers	60	40
			SD-WAN Routers (SDN Category) (Interface Capacity =<10 Gbps)	60	40
3	Managed Leased Line Network Equipment, IP Based access systems-DSL	85176290, 85176990	Central Office Equipment and CPEs of DSL/HDSL/S-HDSL	60	40
4	Ethernet Switches (L2 and L3)	85176290	L2/L3 Ethernet Switches 1-100 GbE interface Switches (stackable / standalone / modular switches), for Core / Distribution / Access Layers (including PoE/PoE+)	60	40
			Industrial Grade LAN 1, 10 GbE PoE/PoE+/Non-PoE Switches (Power over Ethernet)	60	40
5	IP based Soft Switches, IMS, Unified Communication Systems	85176290, 85176990	IP PBX / Unified Communication Systems(UCS)	60	40
			UCS (non-IMS based):	60	40
6	Wireline PABXs / IP PBX	85176290	Wireline PABXs / IP PBX	65	45
7	Media Gateways	85176290, 85176990	Media Gateways	65	45
8	Customer Premises Equipment (CPEs) (including Wi-Fi Access points, low capacity Routers, Media Converters), CPE - NFV/ SDN/ Leased-line Modems	85176290, 85176990	CPE (including Wi-Fi Access points and Routers),	50	30
			Media convertor	50	30
			TDM over IP Converter	50	30
			CPE - NFV/ SDN (Virtualized CPE with cloud management)	50	30
			Leased-line Modems n* 2Mbps / IP	50	30
9	SDH/ NG-SDH	85176290	SDH/ NG-SDH (STM 1/4/16/64)	65	45
10	Ethernet/MPLS- TP/ Packet Optical Transport equipment/ PTN systems	85176290	MPLS- TP / Packet Optical Transport equipment/ PTN (with or without OTN interfaces) systems	65	45
11	CWDM systems	85176290	CWDM (Nx100G, Nx10G)	60	40
12	DWDM systems	85176290, 85176270	DWDM (upto 10G) with Mux/De MUX with Transponder / Muxponder; ROADM (Reconfigurable Optical Add Drop Multiplexer), Passive DWDM MUX/DeMUX	60	40
13	GPON/XGPON equipment (including ONT and OLT)	85176290, 85172990	GPON/ XGPON equipment (including ONT and OLT)	60	40
14	XGS-PON equipment (including ONT and OLT)	85176290, 85172990	XGS-PON (10G DS – 10G US) (including ONT and OLT)	60	40

15	SDH/ PDH Cross Connects	85176290	SDH/ PDH Cross Connects (STM : 1/4/16/64; E1/E3/DS3; Ethernet interfaces)	60	40
16	Small Size Radio Sub System	85176290	Micro 2G BTS	60	40
17	LTE/ LTE-R Based Mobile Systems, with its various derivatives	85176290, 85176990	Macro eNode B, as per TEC ENB GR (TDD/FDD)	60	40
			Micro eNodeB, as per TEC ENB GR (TDD/FDD)	60	40
			Pico eNodeB, as per TEC ENB GR (TDD/FDD)	60	40
			Home Base Station/ Femto	60	40
			EPC	60	40
			4G NIB Small cell	60	40
18	Wi-Fi based broadband wireless access systems -indoor	85176290, 85176990	Indoor AP (802.11 ax MIMO, 2.4 & 5 GHz bands)	55	35
			Indoor AP (802.11 ac MIMO, 2.4 & 5 GHz bands)	55	35
19	Wi-Fi based broadband wireless access systems -Outdoor	85176290, 85176990	Outdoor AP (802.11ac - MIMO 2.4 and 5 GHz bands) - IP67 Rated	55	35
			Outdoor AP (802.11ax MIMO, 2.4 & 5 GHz bands)	55	35
20	Wi-Fi Access Controller (with interoperability and multi-vendor AP support)	85176290	Type 1: <= 2000 AP support	55	35
			Type 2: upto 5000 AP support	55	35
21	Authentication, Authorization and Accounting(AAA) Server	85176290	AAA Block (Radius)	50	35
			AAA Block (Radius/ Diameter)	50	35
22	Radio systems (IP/ Hybrid)	85176290	Radio systems (IP/ Hybrid) (5 GHz band - License Exempt band)	55	35
23	Software Defined Radio, Cognitive Radio systems	85176290	Ruggedised handheld, Ruggedised or non-ruggedised vehicle mounts, P2P OR P2MP or Mobile adhoc topology, Unlicensed band operation	55	35
24	Repeaters (RF/RF-over-Optical), IBS and Distributed Antenna systems (indoor / outdoor)	85176290, 85177100	Repeaters (RF/RF-over-Optical), IBS and Distributed Antenna systems (indoor / outdoor) including Active & Passive Accessories	60	40
25	VSAT Terminal Subsystem – ODUs/	85176290, 85256091	Antenna	50	25
26	VSAT Hub Subsystems	85256091	NMS H/W & S/W	50	25
			Antenna (>=7m) and accessories etc.	50	25
27	Satellite Phones(S Bands)	85176990	Both Voice and Data (upto 9.6 Kbps)	50	25
28	Disaster communication Systems etc., including backpack satellite products (S Band)	85176990	Portable Terminals Voice and Data (2 Mbps UL)	50	25
29	Satellite based Systems, terminals including location, resources tracking (L, S and Ku, Ka bands)	85176990	Two way MSS Data Terminals (Satellite Receivers with location data) S - Band (Satellite IoT devices for various applications i.e.	50	25

			position tracking, messaging etc.)		
			One Way Receivers MSS Data Terminals (Satellite Transmitters & Receivers - one way) S – Band (Simplex / Half-duplex, Mobile Satellite Service (MSS) terminal, Satellite IoT devices)	50	25
30	Telecom Network Management systems (NMS) with its various derivatives	85176290		65	50
31	Optical Fibre	9001000	Single Mode	50	0
32	Optical Fiber Cable	85447090	Optical Fiber Cable (UG & OH)	55	0
33	Telecom Power System (Including Solar Power)	85176990	UPS, Power Plant, Invertor (Including Solar Power) 0.5-1500 KVA/ Power Plant: 1-2000 KVA (Modular)	55	30
34	Telecom Batteries (Lead Acid)	85176990	Lead Acid up to 5000 AH (Including VRLA)	55	30
35	IP Audio/ Video phones / Adaptor	85176990	IP Audio/ Video phones / Adaptor	50	15
36	IP/ Internet Set-top Box	85176990	IP/ Internet Set-top Box	55	35

MLC%- Minimum Local Content as a percentage of total Bill of Material (cost of production) to qualify as Make in India Telecom Products, Services or Works.

TABLE-B

Main Inputs /stages for manufacture of telecom products & conditions for the inputs to be qualified as Local Content

Main Inputs /stages for manufacture of telecom products *	**Conditions to be met for the inputs to be qualified as Local Content**
1) Design a) Hardware design b) Software Design & Development	The maximum Local Content (LC) percentage for Design which can be claimed by a Local manufacturer for the telecom products based on in-house/in country R&D costs incurred/amortized to create IPR in India are as per **Table-A (last column)** subject to additional conditions that: a) The Intellectual Property Right (IPR) reside in India for Hardware Design; and b) The Copyright is in India for the software Design & Development.
2) Components a) Integrated chips (ICs)- Processor, Memory etc. b) Active components – Transistors, Diodes etc. c) Passive Components – Resistors, Capacitors, Inductors etc.	Should be Manufactured in India
3) PCB Fabrication	Should be Manufactured in India
4) Cables/Chassis etc. a) Chassis b) Cables c) Racks d) Heat sinks e) Enclosures	Should be Manufactured in India
5) RF Components/Subsystem a) Duplexers/Filters	Should be Manufactured in India

b) Antenna	
6) Assembly/Integration/Testing#	The upper ceiling limit of Domestic Local Content (LC) for Assembly/ Integration/ Testing in respect of the telecom products listed in **Table-A** would be 10% of the total product Bill of Material.
* The product may include some/all of the input/stage as mentioned above. While calculating the local content only those inputs/stages will be considered which are relevant in the manufacturing of the telecom product. #In case a system or subsystem is merely assembled / integrated / tested, then actual Local Content shall be taken as up to 10% of the cost of system / subsystem.	

Form 1

Format for Self-Certification regarding Local Content (LC) for Telecom Product, Services or Works to be submitted on non-judicial stamp paper of the value Rs. 100/-

Date:

I__ _____**S/o, D/o, W/o,** Resident of _____ do hereby solemnly affirm and declare as under:

That I agree to abide by the terms and conditions of Department of Telecommunications, Government of India issued vide Notification No:_____ dated_____.

That the information furnished hereinafter is correct to the best of my knowledge and belief and I undertake to produce relevant records before the procuring entity or any other authority so nominated by the Department of Telecommunications, Government of India for the purpose of assessing the LC.

That the LC for all inputs which constitute the said Telecom Product/Services/Works has been verified by me and I am responsible for the correctness of the claims made therein.

That in the event of the LC of the Telecom Product/Services/Works mentioned herein is found to be incorrect and not meeting the prescribed LC norms, based on the assessment of an authority so nominated by the Department of Telecommunications, Government of India, **I and my Statutory auditor or cost auditor (if applicable) will be liable for actions as specified in Clause (9) of the DPIIT PPP-MII Order dated 19.07.2024 for all incorrect/false facts and figures.**

I agree to maintain detailed breakup / information (separately for each product) to substantiate my claim for LC in the Company's record for a period of 2 years and shall make this available for verification to any authority. I shall also maintain records of local content pertaining to items bought from other domestic manufacturers / traders.

[Please provide following information]

i Name and details of the local supplier (Registered Office, Manufacturing unit location, nature of legal entity)
ii. Date on which this certificate is issued
iii. Telecom Product/Services/Works for which the certificate is produced
iv. Procuring agency to whom the certificate is furnished
v. Percentage of LC claimed
vi. Name and contact details of the unit of the manufacturer
vii. Sale Price of the product
viii. Ex-Factory Price of the product
ix. Freight, insurance and handling
x. Total Bill of Material
xi. List and total cost value of inputs used for manufacture of the Telecom product/Services/Works
xii. List and total cost of inputs which are locally sourced. Please attach LC certificates from local suppliers, if the input is not produced in-house.
xiii. List and cost of inputs which are imported, directly or indirectly

I hereby certify that, having read all the provisions of the above order and principles / basis of calculations, the local content calculation does not include the following:

a. Imported items sourced locally from resellers/distributors.

b. The license fees / royalties paid/ technical charges paid out of India

c. Procurement / supply of repackaged / refurbished/rebranded imported products

I hereby also certify to the best of my knowledge and belief that all the particulars furnished above are correct and complete. I agree to comply with the terms and conditions of the DPIIT PPP-MII order dated 19.07.2024 and DoT PPP-MII Notification dated……..…...

I understand that any incorrect declaration regarding the local content or failure to substantiate the claim of LC will result in penalties as specified in Clause (9) of the DPIIT PPP-MII Order dated 19.07.2024.

I further certify and take personal responsibility that I have applied my mind to the calculations and principles of LC as specified in this order and I shall, having declared the LC shall not seek recourse to change it on any ground. Any changes made by me on any grounds in a bid in LC after bid submission shall make my bid non-responsive and I shall hold myself liable for civil/criminal action arising out of any such change. I understand and agree that any such post bid change in LC content shall also be a valid ground for blacklisting of the firm from future contracts/bids.

Signature:

Name:

Designation: CEO/MD

Address:

Email Address:

Mobile No.:

Place:

Date:

List of Acronyms used in the notification

Sl. No.	Acronyms	Expansion
1.	AAA	Authentication Authorization & Accounting
2.	AMC	Annual Maintenance contract
3.	AP	Access Point
4.	BBU	Base Band Unit
5.	BIS	Bureau of Indian Standard
6.	B-RAS	Broadband Remote Access Server
7.	BTS	Base Transceiver Station
8.	CDOT	Centre for Development of Telematics
9.	CMC	Comprehensive Maintenance Contract
10.	CPE	Customer Premise Equipment
11.	CS	Central Scheme
12.	CSR	Cell Site Router
13.	CSS	Central Sector Scheme
14.	CWDM	Coarse Wavelength Division Multiplexing
15.	DIPP	Department of Industrial Policy and Promotion
16.	DoT	Department of Telecommunications
17.	DPIIT	Department for Promotion of Industry and Internal Trade
18.	DSL	Digital Subscriber Line
19.	DSLAM	Digital Subscriber Line Access Multiplexer
20.	DWDM	Dense Wavelength Division Multiplexing
21.	e-node B (ENB)	Evolved node B
22.	EPC	Engineering Procurement Construction
23.	FDD	Frequency Division Duplexing
24.	gNB	Next Generation Node B
25.	GFR	General Financial Rules
26.	GPON	Gigabit Passive Optical Network
27.	GR	Generic Requirements
28.	GSM	Global System for Mobile
29.	HDSL	High bit-rate Digital Subscriber Line

30.	IBS	Integrated Building System
31.	IC	Integrated Chip
32.	IMS	IP Multimedia Subsystem
33.	IOT	Internet of Things
34.	IP	Internet Protocol
35.	IPR	Intellectual Property Right
36.	IR	Interface Requirement
37.	LAN	Local Area Network
38.	L-2	Layer Two
39.	L-3	Layer Three
40.	LC	Local Content
41.	Li-ion	Lithium-ion
42.	LTE	Long Term Evolution
43.	LTE-R	Long Term Evolution-Railway
44.	MIMO	Multiple Input Multiple Output
45.	MLC	Minimum Local Content
46.	MPLS	Multi-protocol Label Switching
47.	MPLS-TP	Multiprotocol Label Switching-Transport Profile
48.	MSS	Mobile Satellite Service
49.	MUX	Multiplexer
50.	NFV	Network Function Virtualization
51.	NG-PON2	Next Generation Gigabit Passive Optical Network 2
52.	NIB	Network in Box
53.	NMS	Network Management System
54.	OADM	Optical Add-Drop Multiplexer
55.	ODU	Outdoor Unit
56.	OLT	Optical Line Terminal
57.	ONT	Optical Network Terminal
58.	OTN	Optical Transport Network
59.	P2P	Point to Point
60.	P2MP	Point to Multi Point
61.	PABX	Private Auto Branch Exchange
62.	PBX	Private Branch Exchange
63.	PCB	Printed Circuit Board
64.	PDH	Plesiochronous Digital Hierarchy
65.	PMI	Preference to Make in India
66.	PTN	Packet Transport Network
67.	PoC	Proof of Concept
68.	PoE	Power over Ethernet
69.	PPP-MII	Public Procurement (Preference to Make in India)
70.	PTN	Packet Transport Network
71.	ROADM	Reconfigurable Optical Add-Drop Multiplexer
72.	R&D	Research and Development
73.	RF	Radio Frequency
74.	S-HDSL	Single pair High speed Digital Subscriber Line
75.	SDH	Synchronous Digital Hierarchy
76.	SDN	Software Defined Network
77.	SD-WAN	Software Defined - Wide Area Network

78.	STM	Synchronous Transport Module
79.	TDD	Time Division Duplexing
80.	TDM	Time Division Multiplexing
81.	TEC	Telecommunication Engineering Centre
82.	TEPC	Telecom Export Promotion Council
83.	UCS	Unified Communication System
84.	UL	Unified License
85.	USOF	Universal Service Obligation Fund
86.	UTM	Unified Threat Management
87.	VRLA	Valve Regulated Lead Acid
88.	VSAT	Very Small Aperture Terminal
89.	Wi-Fi	Wireless Fidelity
90.	XG-PON	10 Gigabit Passive Optical Network
91.	XGS-PON	10 Gigabit Symmetrical Passive Optical Network

ASHOK KUMAR JAIN, Dy. Director General

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APPENDIX-V TO ITB

POLICY TO PROVIDE PURCHASE PREFERENCE UNDER PUBLIC PROCUREMENT (PREFERENCE TO MAKE IN INDIA) (PPP-MII) ORDER, 2017

(EVALUATION AND AWARD PHILOSOPHY)

EVALUATION AND AWARD PHILOSOPHY

Ministry of Finance (MoF), Department of Public Enterprises, vide **DO ref. F.No. DPE/3(3)/10-Fin dated 29.05.2023** have forwarded the copy of Department of Expenditure (DoE) O.M. No. F.1/4/2021-PPD dated 18.05.2023, in which the guidelines on **"Concurrent application of Public Procurement Policy for Micro and Small Enterprises Order, 2012 and Public Procurement (Preference to Make in India) Order, 2017"** are issued for information and strict compliance.

Guidelines for Purchase preference:

1. The Class-I local suppliers, under PPP-MII Order, participating in any government tender, may or may not be MSEs, as defined under the MSME Act. Similarly, MSEs participating in any Government Tender, may or may not be Class-I local suppliers. Suppliers may be categorised in following four broad categories for consideration or applicability of purchase preference:

Sl. No.	Category	Terminology
1.	Supplier is both MSE & Class-I local supplier.	"MSE Class-I local supplier"
2.	Supplier is MSE but not Class-I local supplier.	"MSE but non-Class-I local supplier"
3.	Supplier is not MSE but is Class-I local supplier.	"Non-MSE but Class-I local supplier"
4.	Supplier is neither MSE nor Class-I local.	"Non-MSE non-Class-I local supplier"

2. The applicability of PPP-MSE Order and PPP-MII Order in various scenarios, involving simultaneous purchase preference to MSEs and Class-I local suppliers under PPP-MSE Order and PPP-MII Order respectively, shall be as under:

Sl. No.	Description	Evaluation Methodology
1)	**Items covered under Para 3(a) of PPP- MII Order, 2017 for which Nodal Ministry has notified sufficient local capacity and competition:**	For these items, only Class-I local suppliers are eligible to bid irrespective of purchase value. Hence, **Class-II local suppliers or Non-local suppliers, including MSEs which are Class-II local suppliers/ Non-local suppliers, are not eligible to bid.**
	Possible scenarios under Sl. No. 1)above can be as under:	
(i)	**Case-I:** L-1 is "MSE Class-I local supplier"	100% of the tendered quantity is to be awarded to L-1.
(ii)	**Case-II:** L-1 is "Non-MSE but Class-I local supplier"	Purchase preference is given to MSEs as per PPP-MSE Order. Balance quantity is to be awarded to the L-1 Bidder
2)	**Items reserved exclusively for procurement from MSEs as per PPP-MSE Order:**	These items are reserved exclusively for purchase from MSEs. Hence, **non-MSEs are not eligible to bid for these items.**
	Possible scenarios under Sl. No. 2)above can be as under:	
(i)	**Case-I:** L-1 is "MSE Class-I local supplier"	100% of the tendered quantity is to be awarded to L-1.
(ii)	**Case-II:** L-1 is "MSE non-Class-I local supplier"	Purchase preference is to be given to MSE Class-I local supplier as per PPP-MII Order. Balance quantity, is to be awarded to L-1 Bidder

Sl. No.	Description	Evaluation Methodology
3)	**If items are neither notified for sufficient local capacity nor reserved for MSEs, then the process will be as follows:**	
a)	Items covered under Para 3A(b) of PPP-MII Order are **divisible items and both MSEs as well as Class-I local suppliers are eligible for purchase preference**.	
	Possible scenarios under Sl. No. 3)a) above can be as under:	
(i)	**Case-I:** L-1 is "MSE Class-I local supplier"	100% of the tendered quantity is to be awarded to L-1.
(ii)	**Case-II:** L-1 is "Non-MSE but Class-I local supplier"	Purchase preference is to be given to MSEs, if eligible, as per PPP-MSE Order. Balance quantity is to be awarded to L-1 Bidder.
(iii)	**Case-III:** L-1 is "MSE but non-Class-I local supplier"	Purchase preference is to be given to Class-I local suppliers, if eligible, as per PPP-MII Order. Balance quantity is to be awarded to L-1 Bidder.
(iv)	**Case-IV:** L-1 is "Non-MSE non-Class-I local supplier"	Purchase preference is to be given to MSEs as per PPP-MSE Order. Thereafter, purchase preference is to be given to Class-I local suppliers for "50% of the tendered quantity minus quantity allotted to MSEs above" as per PPP- MII Order. For the balance quantity, Contract is to be awarded to L-1 Bidder. (refer to the illustrative example mentioned below).
b)	Items covered under Para 3A(c) of PPP-MII Order, 2017 are **non-divisible items and both MSEs as well as Class-I local suppliers are eligible for purchase preference**.	
	Possible scenarios under Sl. No. 3)b) above can be as under:	
(i)	**Case-I:** L-1 is "MSE Class-I local supplier"	Contract is awarded to L-1.
(ii)	**Case-II:** L-1 is not "MSE Class-I local supplier" but the "MSE Class-I local supplier" falls within 15% margin of purchase preference	Purchase preference is to be given to lowest quoting "MSE Class-I local supplier". If lowest quoting "MSE Class-I local supplier" does not accept the L-1 rates, the next higher "MSE Class-I local supplier" falling within 15% margin of purchase preference is to be given purchase preference and so on.
(iii)	**Case-III:** If conditions mentioned in sub paras **3)b)(i) and 3)b)(ii)** above are not met i.e. L-1 is neither "MSE Class-I local supplier" nor is the "MSE Class-I local supplier" eligible to take benefit of purchase preference:	Contract is to be awarded/ purchase preference to be given in different possible scenarios as under: A. L1 is "MSE but non-Class-I local supplier" or "Non-MSE but Class-I local supplier" - Contract is be awarded to L1. B. L1 is "Non-MSE non-Class-I local supplier" - First purchase preference to be given to MSE as per PPP-MSE Order. If MSE not eligible/ does not accept - purchase preference is to be given to Class- I Local supplier as per PPP-MII Order. If Class-I Local supplier is also not eligible/ does not accept - contract to be awarded to L-1
4)	Items reserved for both MSEs and Class-I local suppliers:	These items are reserved exclusively for purchase from MSEs as well as Class-I local suppliers. **Hence,**

Sl. No.	Description	Evaluation Methodology
		only "MSE Class-I local suppliers" are eligible to bid for these items. Non-MSEs/Class-II local suppliers/ Non-local suppliers cannot bid for these items. Hence the question of purchase preference does not arise
5)	**Non-local suppliers, including MSEs falling in the category of Non-local suppliers, shall be eligible to bid only against Global Tender Enquiry**	

3. Verification of Local Content:

3.1 The local supplier shall provide following documents

3.1.1 **At the bidding stage**:

i. The 'Class-I local supplier'/ 'Class-II local supplier' / 'Non-Local Suppliers' shall be required to indicate percentage of local content & details of the location(s) of the local value addition, and also provide self-certification (as per **FORM-2**) along with their bid. The percentage of local content to be declared by the bidder shall correspond to the basis of evaluation level specified for the MR/Tender.

ii. In addition to above-

a. In case the Bill of Material of MR/Tender includes Telecom Products as listed in Table-A of DOT order File No. 18-10/2017-IP dated 21.10.2024, the bidder(s) shall provide self-certification in **Form- 2B** (duly notarized from public notary) specifying that the item(s) offered, meet the minimum local content requirement and shall give details of the location(s) at which the local value addition is made.

b. In case of Bill of Material of MR/Tender includes cyber security Products-

i. The local supplier shall provide self-declaration that the item offered meets the definition of 'local supplier' of domestically developed/manufactured/ produced Cyber Security Products, as per para 4 of Notification File No.1(10)/2017-CLES dated 06.12.2019 issued by Meity. Format of self-certification is enclosed as **Form-2C**.

ii. In cases of procurement for a value in excess of Rs. 10 crore, the local supplier shall provide a certificate from the statutory auditor or cost auditor of the company (in the case of companies) [or from a practicing Chartered Accountants or practicing Cost Accountants in case of suppliers other than companies] **as per Form-2D** that, the item offered meets the definition of 'local supplier' of domestically manufactured/produced Cyber Security Products, as per para 4 of Notification File No.1(10)/2017-CLES dated 06.12.2019 issued by Meity.

3.1.2 **During execution stage:**

For all contracts up to INR 10 Crore Contractor/ Supplier shall submit **Form-2A**. However, for contracts above INR 10 Crore, the Bidder shall be required to submit local content certification (**as per FORM-3 & 3A**) duly certified by Statutory Auditor or Cost Auditor (in the case of companies) or from a Practicing Cost Accountant or Practicing Chartered accountant (in respect of contractors/suppliers other than companies). For cases where it is not possible to provide such certification, the Contractor / supplier is permitted to provide the certificate after completion of contract but not later than submission of Final Bill. In that case, an amount equivalent to 10% of invoice value shall be retained from each invoice till submission of **Form-3 & 3A**.

3.2 In case, the Contractor/Supplier submit False declarations And/ Or does not meet the stipulated local content requirement and the category of the Contractor/ Supplier changes from Class-I to Class-II/Non-local or from Class-II to Non-local, default and penalty actions shall be taken as per the provisions of Public Procurement (Preference to Make in India) (PPP-MII) Order 2017 and its subsequent amendments.

4. 'Purchase preference' means the opportunity given to 'MSE bidder' or 'Class-I local supplier' for matching the Price of L1 Bidder, despite its Price being not the lowest valid Price Bid (i.e. not L1) but because it is within the margin of purchase preference prescribed as follows:

a. As per PPP-MSE policy 2012 : 15% of L1 price

b. As per PPP-MII policy 2017 : 20% of L1 price

5. Eligibility of Class-I local Supplier, Class-II local Supplier and Non-local Suppliers to bid for different types of procurement of Goods, services or works -

Table-1

S. no	Estimated Cost	Notification as per clause no 3(a) of PPP-MII Policy	Eligibility
i.	Cost Estimate <=Rs.1 Cr.	items are notified as having sufficient local capacity and local competition under clause 3 (a) of the PPP-MII Order 2017	PPP-MII policy is not applicable.
		items are not notified as having sufficient local capacity and local competition under clause 3 (a) of the PPP-MII Order 2017	PPP-MII policy is not applicable.
ii.	Cost Estimate >Rs.1 Cr.	items are notified as having sufficient local capacity and local competition under clause 3 (a) of the PPP-MII Order 2017	Only Class-I local Suppliers can participate
		items are not notified as having sufficient local capacity and local competition under clause 3 (a) of the PPP-MII Order 2017	Both Class-I & Class-II Suppliers can participate. Class-I local Suppliers eligible for preference as per policy
iii.	-	Global Procurement	Class-I, Class-II & Non-Local Suppliers can participate. Class-I local Suppliers are eligible for preference as per policy.

6. Reverse Auction

6.1 Eligibility of MSE and Class-I Local Suppliers to participate in RA:

The Reverse Auction shall be conducted in accordance with the terms and conditions specified in the Tender Document. However, wherever the "H1 elimination rule" or the "50% elimination rule" is prescribed, participation of MSEs and Class-I Local Suppliers shall be determined as per the following methodology, irrespective of their ranking. Such MSE/Class-I Local bidders shall be included over and above the bidders shortlisted as per the RA criteria.

a) Where the Public Procurement Policy for Micro and Small Enterprises (MSEs), 2012 is applicable and the L1 bidder (prior to RA) is a non-MSE, all eligible MSE bidders whose evaluated prices (prior to RA) are within the L1 + 15% band shall be invited to participate

in the RA.

b) Where the Public Procurement (Preference to Make in India) Order, 2017 is applicable and the L1 bidder (prior to RA) is a non-Class-1 Local Supplier, all eligible Class-I Local Suppliers whose evaluated prices (prior to RA) are within the L1 + 20% band shall be invited to participate in the RA.

c) Where both the above policies are applicable, the provisions of each policy shall be applied independently within their respective domains.

6.2 Application of Purchase Preference after RA:

In cases where the L1 bidder is a non-MSE or non-Class-I Local Supplier, eligible MSEs or Class-I Local Suppliers, as per the terms and conditions of the Tender Document, shall be given an opportunity to match the L1 price discovered after the Reverse Auction, provided their final RA prices fall within the permissible percentage band stipulated in Clause 4 above.

Purchase preference shall be extended to eligible MSE bidder(s) and/or eligible Class-I Local Suppliers in accordance with the evaluation methodology specified in Clause 2 above.

If a bidder does not participate in the Reverse Auction, the price quoted in the financial bid prior to RA shall be treated as the bidder's valid price. After RA, bidder's ranking (L1, L2, etc.) shall be determined based on the lower of (i) the price submitted in the Reverse Auction, or (ii) the price quoted in the financial bid.

Annexure -1 to Appendix –V to ITB

Example explaining applicability in scenario explained in Para B Sl. No. 3(a) (iv)

(Scenario: Divisible items, both MSEs as well as Class-I local suppliers eligible for purchase preference and L-1 is “Non-MSE non-Class-I local supplier”)

Item – Desktop computer

Qty – 50 Nos.

Details of bids received

Sr. No.	Name of Bidder	Rates quoted	Price Ranking	Status of Bidder
1.	A	100	L1	“Non-MSE non-Class-I local supplier”
2.	B	110	L2	“Non-MSE but Class-I local supplier”
3.	C	112	L3	“MSE but non-Class-I local supplier”
4.	D	115	L4	“Non-MSE but Class-I local supplier”
5.	E	118	L5	“MSE but non-Class-I local supplier”
6.	F	120	L6	“MSE Class-I local supplier”

1. In this case, first purchase preference is to be given to MSEs as per PP-MSE Order for 25% of tendered quantity of 50 Nos. i.e. 12.5 Nos. (rounded off to the next whole number say 13 Nos.). Accordingly, invite L3 (bidder C), whose quoted rates falls within 15% margin of purchase preference to match L1 price i.e. Rs. 100/- for quantity of 13 Nos. Bidder “E” and “F”, although MSEs, will not get purchase preference since their quoted rates don’t fall within 15% margin of purchase preference. Bidder C will be considered for order of 13 Nos. on confirmation of reduction of price.
2. For 50% of balance quantity of 37 number (tendered quantity of 50 – 13 awarded to bidder C; assuming bidder C has confirmed to accept l1 rates), purchase preference will be given to lowest Class-I local supplier as per PPP-MII Order. Accordingly, bidder B will be invited to match L-1 price for 50% of 37 Nos. i.e. 18.5 (say 19 Nos. of computers). If bidder “B” does not accept the L1 price i.e. price of Rs. 100/- per unit, next higher Class-I local supplier falling within 20% margin of purchase preference, i.e. bidder “D”, may be invited to match L-1 price for 19 Nos. of computers and so on.
3. For remaining quantity i.e. 18 Nos. (50-13-19), the contract will be awarded to lowest quoting bidder i.e. Bidder “A”, who is L-1 in the example.

APPENDIX-VII TO ITB

PROCUREMENT FROM BIDDERS FROM A COUNTRY OR COUNTRIES SHARING LAND BORDER WITH INDIA

1.0 Department of Expenditure (DoE) vide **O.M. No. 7/10/2021-PPD(1) dated 23.02.2023** has issued the revised guidelines for Restrictions on Procurement from Bidders from Country(ies) sharing Border with India. These guidelines are available on the website of DoE (https://doe.gov.in).

2.0 ***Requirement of registration from Competent Authority:***

i). Any Bidder from a country which shares a land border with India will be eligible to bid in any procurement whether of goods, services (including consultancy services and non-consultancy services) or works (including turnkey projects) only if the Bidder is registered with the Competent Authority, specified in **Annexure-I of DoE O.M. No. 7/10/2021-PPD(1) dated 23.02.2023.**.

ii). Any Bidder (including an Indian bidder) who has a Specified Transfer of Technology (ToT) arrangement with an entity from a country which shares a land border with India will be eligible to bid in any procurement whether of goods, services (including consultancy services and non-consultancy services) or works (including turnkey projects) only if the Bidder is registered with the Competent Authority, specified in **Annexure-I of DoE O.M. No. 7/10/2021-PPD(1) dated 23.02.2023.**

3.0 ***This Order shall not apply to the following special cases:***

i). In projects which receive international funding with the approval of the Department of Economic Affairs (DEA), Ministry of Finance, the procurement guidelines applicable to the project shall normally be followed, notwithstanding anything contained in this order and without reference to the Competent Authority. Exceptions to this shall be decided in consultation with DEA.

ii). This order shall not apply to procurement by Indian missions and by offices of Government agencies/ undertakings located outside India.

iii). This order will not apply to Bidders (or Entities) from those countries (even if sharing a land border with India) to which the Government of India has extended lines of credit or in which the Government of India is engaged in development projects. Updated lists of countries to which lines of credit have been extended or in which development projects are undertaken are given on the website of the Ministry of External Affairs.

iv). Procurement of spare parts and other essential service support like Annual Maintenance Contract (AMC)/ Comprehensive Maintenance Contract (CMC), including consumables for closed systems, from Original Equipment Manufacturers (OEMs) or their authorized agents, shall be exempted from the requirement of registration.

4.0 ***Definitions****:*

i). **"Bidder"** (including the term 'Tenderer', 'Consultant' 'Vendor' or 'Service Provider' in certain contexts) means any person or firm or company, including any member of a Consortium or Joint Venture (that is an association of several Persons, or Firms or Companies), every artificial juridical person not falling in any of the descriptions of Bidders stated hereinbefore, including any Agency, Branch or Office controlled by such person, participating in a procurement process.

ii). **"Tender"** will include other forms of procurement, except where the context requires otherwise.

iii). **"Transfer of Technology"** means dissemination and transfer of all forms of commercially usable knowledge such as transfer of know-how, skills, technical expertise, designs, processes and procedures, trade secrets, which enables the acquirer of such technology to perform activities using the transferred technology independently. (Matters of interpretation of this term shall be referred to the Registration Committee constituted by the Department for Promotion of Industry and Internal Trade, and the interpretation of the Committee shall be final.)

iv). **"Specified Transfer of Technology"** means a transfer of technology in the sectors and/ or technologies, specified in **paragraph 5.0**, occurring on or after 23.07.2020.

v). ***"Bidder (or entity) from a country which shares a land border with India"*** *means*

(a) An entity incorporated, established or registered in such a country; or

(b) A subsidiary of an entity incorporated, established or registered in such a country; or

(c) An entity substantially controlled through entities incorporated, established or registered in such a country; or

(d) An entity whose beneficial owner is situated in such a country; or

(e) An Indian (or other) agent of such an entity; or

(f) A natural person who is a citizen of such a country; or

(g) A Consortium or Joint Venture where any member of the Consortium or Joint Venture falls under any of the above.

vi). **Beneficial Owner** for the purposes of **paragraph 4.0 v) (d)** will be as under:

a) **In case of a Company or Limited Liability Partnership**, the beneficial Owner is the natural person(s), who, whether acting alone or together, or through one or more juridical person(s), has a controlling ownership interest or who exercises control through other means. Explanation: -

1. "Controlling ownership interest" means ownership of, or entitlement to, more than twenty-five per cent of shares or capital or profits of the company;

2. "Control" shall include the right to appoint the majority of the Directors or to control the management or policy decisions, including by virtue of their shareholding or management rights or shareholders agreements or voting agreements;

b) **In case of a Partnership Firm**, the beneficial Owner is the natural person(s) who, whether acting alone or together, or through one or more juridical person, has ownership of entitlement to more than fifteen percent of capital or profits of the partnership;

c) **In case of an Unincorporated Association or body of Individuals**, the beneficial Owner is the natural person(s), who, whether acting alone or together, or through one or more juridical person, has ownership of or entitlement to more than fifteen percent of the property or capital or profits of such association or body of individuals;

d) Where no natural person is identified under a) or b) or c) above, the beneficial Owner is the relevant natural person who holds the position of senior managing official;

e) **In case of a Trust**, the identification of beneficial Owner(s) shall include identification of the author of the trust, the trustee, the beneficiaries with fifteen percent or more interest in the trust and any other natural person exercising ultimate effective control over the trust through a chain of control or ownership.

vii). **"Agent'** is a person employed to do any act for another, or to represent another in dealings with third persons.

[Note:

i). A person who procures and supplies finished goods from an entity from a country which shares a land border with India will, regardless of the nature of his legal or commercial relationship with the producer of the goods, be deemed to be an Agent for the purpose of this Order.

ii). However, a Bidder who only procures raw material, components etc. from an entity from a country which shares a land border with India and then manufactures or converts them into other goods will not be treated as an Agent.]

5.0 ***Sensitive Sectors/ Technologies (relevant only for the provisions on ToT arrangements):***

i). Certain sectors and technologies have been identified as sensitive from the national security point of view. **The sectors listed in Schedule I (copy attached) are considered Category-I sensitive sectors**. **The sectors listed in Schedule II (copy attached) are considered Category-II sensitive sectors**. The technologies listed in **Schedule III** (copy attached) are considered sensitive technologies.

ii). For **Category-I sensitive sectors**, Bidders with ToT arrangement in any technology with an entity from a country which shares a land border with India shall require registration.

iii). For **Category-II sensitive sectors**, Bidders with ToT arrangement in the sensitive technologies listed in **Schedule III**, with an entity from a country which shares a land border with India shall require registration.

iv). In **Category-II sensitive sectors**, the Secretary (or an officer not below the rank of Joint Secretary to Government of India, so authorized by the Secretary) of the Ministry/ Department of the Government of India is empowered, after due consideration, to waive the requirement of registration for a particular item/ application or a class of items/ applications from the requirement of registration, even if included in Schedule-III. The Ministry/ Department concerned shall intimate the Department for Promotion of Industry and Internal Trade (DPIIT) and National Security Council Secretariat (NSCS) of their decision to waive the requirement of registration.

Ministries/ Departments of the Government of India are not required to consult the DPIIT/ NSCS before deciding and are only required to intimate the decision to DPIIT/ NSCS. If any point is raised by DPIIT/ NSCS, it should be considered in future procurements; ongoing procurement for which the waiver was granted need not be interrupted or altered.

Based on security considerations, a Ministry/ Department in a **Category-II sensitive sector** or other Ministries/ Departments may recommend to DPIIT inclusion of any other technology in the list of sensitive technologies, either generally or for their Ministry/ Department.

6.0 ***Sub-contracting in works contracts***

In works contracts, including turnkey contracts, Contractors shall not be allowed to sub-contract works to any Contractor from a country which shares a land border with India unless such contractor is registered with the Competent Authority. The definition of **"Contractor from a country which shares a land border with India"** shall be as given in above **paragraph 4.0 v)**.

[Note: Procurement of raw material, components, etc. does not constitute sub- contracting.]

7.0 ***Certificate regarding compliance***

Bidder shall submit a certificate of Compliance in the enclosed **Form-I A/ Form-I B/ Form - II** (as applicable).

This certificate shall be on the Bidder's Letter head and shall be duly signed & stamped by the authorised signatory of the Bidder.

In case at any stage, it is found that the certification given by the Bidder is false, their Bid shall be rejected and shall be liable for other penal actions like placement on Suspension/ Banning or forfeiture of EMD (if applicable). However, if this is found after order placement, this would be ground for immediate termination and further legal actions in accordance with law/provisions of Bidding Document including Suspension/Banning and forfeiture of CPBG/ Security Deposit.

Form- I A

BIDDER'S UNDERTAKING

(On Company's Letter Head)

To,

M/s Engineers India Limited/ _____(Name of the client)

Tender Name : _____

Tender No. : _____

"I have read the clause regarding restrictions on procurement from a Bidder of a country which shares a land border with India; I hereby certify that, Bidder M/s_____ (Name of the Bidder) is:

i).	Not from such a country	[ ]
ii).	If from such a country, has been registered with the Competent Authority	[ ]

Notes:

a) Bidder to tick appropriate option (√) in either i) or ii) above)
b) In case of sl. no. ii) above, Bidder to attach evidence of valid registration by the Competent Authority.

I hereby certify that the Bidder fulfills all requirements in this regard and eligible to be considered.

Place :

Date :

Signature (#):

Name :

Designation :
Seal :

(#) Undertaking shall be signed by the authorized signatory of the Bidder.

Form- I B

BIDDER'S UNDERTAKING IN CASE OF SUB-CONTRACTING IN WORK CONTRACTS

(On Company's Letter Head)

To,

M/s Engineers India Limited/ _____(Name of the client)

Tender Name : _____

Tender No. : _____

"I have read the clause regarding restrictions on procurement from a Bidder of a country which shares a land border with India and on sub-Contracting to Contractors from such countries; I hereby certify that, Bidder M/s_____(Name of the Bidder) is:

i).	Not from such a country and will not sub-Contract work to a Contractor from such countries unless such Contractor is registered with the Competent authority	[ ]
ii).	If from such a country, has been registered with the Competent Authority and will not sub-Contract work to a Contractor from such countries unless such Contractor is registered with the Competent authority	[ ]
Notes: a) Bidder to tick appropriate option (√) in either i) or ii) above) b) In case of sl. no. ii) above, Bidder to attach evidence of valid registration by the Competent Authority.		

I hereby certify that the Bidder fulfills all requirements in this regard and eligible to be considered.

Place :

Date :

Signature (#):

Name :

Designation :

Seal :

(#) Undertaking shall be signed by the authorized signatory of the Bidder.

FORM-II

BIDDER'S UNDERTAKING IN CASE OF TRANSFER OF TECHNOLOGY (ToT) ARRANGEMENT

(On Company's Letter Head)

To,
M/s Engineers India Limited/ _____(Name of the client)

Tender Name : _____

Tender No. : _____

"I have read the clause regarding restrictions on procurement from a Bidder having Transfer of Technology (ToT) arrangement. Bidder M/s_____ (Name of the Bidder), hereby certify that,

i).	Bidder does not have any ToT arrangement requiring registration with the competent authority	[ ]
ii).	Bidder has valid registration to participate in this procurement	[ ]

Notes:

a) Bidder to tick appropriate option (√) in either i) or ii) above)
b) In case of sl. no. ii) above, Bidder to attach evidence of valid registration by the Competent Authority.

I hereby certify that the Bidder fulfills all requirements in this regard and eligible to be considered.

Place :

Date :

Signature (#):

Name :

Designation :
Seal :

(#) Undertaking shall be signed by the authorized signatory of the Bidder

POLICY FOR PROCUREMENT FROM BIDDERS FROM A COUNTRY OR COUNTRIES SHARING LAND BORDER WITH INDIA

No.F.7/10/2021-PPD (1)
Government of India
Ministry of Finance
Department of Expenditure
Procurement Policy Division
264-C, North Block, New Delhi.
23.02.2023.

Order (Public Procurement No. 4)

Subject: Restrictions under Rule 144(xi) of the General Financial Rules (GFRs), 2017.

Attention is invited to Order (Public Procurement No. 1) issued vide F.6/18/2019-PPD dated 23.07.2020, Order (Public Procurement No. 2) issued vide F.6/18/2019-PPD dated 23.07.2020, Order (Public Procurement No. 3) issued vide F.6/18/2019-PPD dated 24.07.2020, Office Memorandum (OM) No. F.18/37/2020-PPD dated 08.02.2021, OM No. F.12/1/2021-PPD(Pt.) dated 02.03.2021 and OM No. F.7/10/2021-PPD dated 08.06.2021. In this regard, the following is hereby ordered under Rule 144(xi) (as amended vide OM No. F.7/10/2021-PPD dated 23.02.2023) on the grounds stated therein, in supersession to all of the above mentioned Orders/ clarifications:

Requirement of registration:

2. Any bidder from a country which shares a land border with India will be eligible to bid in any procurement whether of goods, services (including consultancy services and non-consultancy services) or works (including turnkey projects) only if the bidder is registered with the Competent Authority, specified in **Annexure I**.

3. Any bidder (including an Indian bidder) who has a Specified Transfer of Technology (ToT) arrangement with an entity from a country which shares a land border with India will be eligible to bid in any procurement whether of goods, services (including consultancy services and non-consultancy services) or works (including turnkey projects) only if the bidder is registered with the Competent Authority, specified in **Annexure I**.

4. The requirement of registration for cases covered by paragraph 2 above has been applicable since 23.07.2020. The requirement of registration for bidders covered by paragraph 3 above will be applicable for all procurements where tenders are issued/ published after 01.04.2023.

5. In tenders issued after 23.07.2020 or 01.04.2023, as the case may be, the provisions of requirement of registration of bidders and of other relevant provisions of this Order shall be incorporated in the tender conditions.

Applicability:

6. Apart from Ministries/ Departments, attached and subordinate bodies, notwithstanding anything contained in Rule 1 of the GFRs 2017, the Order shall also be applicable

a) to all Autonomous Bodies;

b) to all public sector banks and public sector financial institutions;

c) to all Central Public Sector Enterprises;

d) to all procurement in Public Private Partnership projects receiving financial support from the Government or public sector enterprises/ undertakings; and

e) Union Territories, National Capital Territory of Delhi and all agencies/ undertakings thereof.

7. This order will not be applicable for cases falling under **Annexure II**.

Definitions:

8. "*Bidder*" for the purpose of the Order (including the term 'tenderer', 'consultant' 'vendor' or 'service provider' in certain contexts) means any person or firm or company, including any member of a consortium or joint venture (that is an association of several persons, or firms or companies), every artificial juridical person not falling in any of the descriptions of bidders stated hereinbefore, including any agency, branch or office controlled by such person, participating in a procurement process.

9. "*Tender*" for the purpose of the Order will include other forms of procurement, except where the context requires otherwise.

10. *"Transfer of Technology"* means dissemination and transfer of all forms of commercially usable knowledge such as transfer of know-how, skills, technical expertise, designs, processes and procedures, trade secrets, which enables the acquirer of such technology to perform activities using the transferred technology independently. (Matters of interpretation of this term shall be referred to the Registration Committee constituted by the Department for Promotion of Industry and Internal Trade, and the interpretation of the Committee shall be final.)

11. *"Specified Transfer of Technology"* means a transfer of technology in the sectors and/ or technologies, specified in paragraph 15, occurring on or after 23.07.2020.

12. *"Bidder (or entity) from a country which shares a land border with India"* for the purpose of the Order means

(a) An entity incorporated, established or registered in such a country; or

(b) A subsidiary of an entity incorporated, established or registered in such a country; or

(c) An entity substantially controlled through entities incorporated, established or registered in such a country; or

(d) An entity whose beneficial owner is situated in such a country; or

(e) An Indian (or other) agent of such an entity; or

(f) A natural person who is a citizen of such a country; or

(g) A consortium or joint venture where any member of the consortium or joint venture falls under any of the above

13. *Beneficial owner* for the purposes of Para 12 (d) will be as under:

(i) In case of a company or Limited Liability Partnership, the beneficial owner is the natural person(s), who, whether acting alone or together, or through one or more juridical person(s), has a controlling ownership interest or who exercises control through other means.

Explanation:-

a. "Controlling ownership interest" means ownership of, or entitlement to, more than twenty-five per cent of shares or capital or profits of the company;

b. "Control" shall include the right to appoint the majority of the directors or to control the management or policy decisions, including by virtue of their shareholding or management rights or shareholders agreements or voting agreements;

(ii) In case of a partnership firm, the beneficial owner is the natural person(s) who, whether acting alone or together, or through one or more juridical person, has ownership of entitlement to more than fifteen percent of capital or profits of the partnership;

(iii) In case of an unincorporated association or body of individuals, the beneficial owner is the natural person(s), who, whether acting alone or together, or through one or more juridical person, has ownership of or entitlement to more than fifteen percent of the property or capital or profits of such association or body of individuals;

(iv) Where no natural person is identified under (i) or (ii) or (iii) above, the beneficial owner is the relevant natural person who holds the position of senior managing official;

(v) In case of a trust, the identification of beneficial owner(s) shall include identification of the author of the trust, the trustee, the beneficiaries with fifteen percent or more interest in the trust and any other natural person exercising ultimate effective control over the trust through a chain of control or ownership.

14. "*Agent*" for the purpose of the Order is a person employed to do any act for another, or to represent another in dealings with third persons.

[Note:

i. A person who procures and supplies finished goods from an entity from a country which shares a land border with India will, regardless of the nature of his legal or commercial relationship with the producer of the goods, be deemed to be an Agent for the purpose of this Order.

ii. However, a bidder who only procures raw material, components etc. from an entity from a country which shares a land border with India and then manufactures or converts them into other goods will not be treated as an Agent.]

Sensitive Sectors/ Technologies (relevant only for the provisions on ToT arrangements):

15. (i) Certain sectors and technologies have been identified as sensitive from the national security point of view. The sectors listed in Schedule I to this Order are considered Category-I sensitive sectors. The sectors listed in Schedule II to this Order are considered Category-II sensitive sectors. The technologies listed in Schedule III are considered sensitive technologies.

(ii) For Category-I sensitive sectors, bidders with ToT arrangement in any technology with an entity from a country which shares a land border with India shall require registration.

(iii) For Category-II sensitive sectors, bidders with ToT arrangement in the sensitive technologies listed in Schedule III, with an entity from a country which shares a land border with India shall require registration.

(iv) In Category-II sensitive sectors, the Secretary (or an officer not below the rank of Joint Secretary to Government of India, so authorized by the Secretary) of the Ministry/ Department of the Government of India is empowered, after due consideration, to waive the requirement of registration for a particular item/ application or a class of items/ applications from the requirement of registration, even if included in Schedule III. The Ministry/ Department concerned shall intimate the Department for Promotion of Industry and Internal Trade (DPIIT) and National Security Council Secretariat (NSCS) of their decision to waive the requirement of registration. Ministries/ Departments of the Government of India are not required to consult the DPIIT/ NSCS before deciding and are only required to intimate the decision to DPIIT/ NSCS. If any point is raised by DPIIT/ NSCS, it should be considered in future procurements; ongoing procurement for which the waiver was granted need not be interrupted or altered.

16. Based on security considerations, a Ministry/ Department in a Category II sensitive sector or other Ministries/ Departments may recommend to DPIIT inclusion of any other technology in the list of sensitive technologies, either generally or for their Ministry/ Department.

Sub-contracting in works contracts

17. In works contracts, including turnkey contracts, contractors shall not be allowed to sub-contract works to any contractor from a country which shares a land border with India unless such contractor is registered with the Competent Authority. The definition of "contractor from a country which shares a land border with India" shall be as in paragraph 12 above. This shall not apply to sub-contracts already awarded on or before 23.07.2020.

[Note: Procurement of raw material, components, etc. does not constitute sub-contracting.]

Certificate regarding compliance

18. An undertaking shall be taken from bidders in the tender documents **(Annexure III)** that the extant guidelines for participation in the tenders (which should include conditions for implementation of this Order) have been complied with. If such certificate given by a bidder whose bid is accepted is found to be false, this would be a ground for debarment and further legal action in accordance with law.

Validity of registration

19. In respect of tenders, registration should be valid at the time of submission of bids and at the time of acceptance of bids. In respect of supply otherwise than by tender, registration should be valid at the time of placement of order. If the bidder was validly registered at the time of acceptance / placement of order, registration shall not be a relevant consideration during contract execution.

Government e-Marketplace (GeM)

20. GeM shall remove non-compliant entities from GeM unless/ until they are registered in accordance with this Order.

Model Clauses/ Certificates

21. Model Clauses and Model Certificates which may be inserted in tenders/ obtained from Bidders are given at Annexure-III. While adhering to the substance of the Order, procuring entities are free to appropriately modify the wording of these clauses based on their past experience, local needs etc.

23/02/2023

(Kanwalpreet)
Director(PPD)
Tel.No. 2309 3811; email: kanwal.irss@gov.in

To

1. Secretaries of all Ministries/ Departments of Government of India for information and necessary action. They are also requested to inform these provisions to all procuring entities.
2. Secretary, Department of Public Enterprises with a request to immediately reiterate these orders in respect to public enterprises.
3. Secretary, Department of Financial Services with a request to immediately reiterate these orders in respect to public sector and public sector financial institutions.
4. Secretary DPIIT with a request to take action as provided under Annexure I.
5. Chief Secretaries/ Administrators of Union Territories/ National Capital Territory of Delhi.
6. CEO/ GeM with a request to ensure implementation of this order on GeM.

Schedule I

List of Category-I Sensitive sectors:

Sr.No.	Sector
(i)	Atomic Energy
(ii)	Broadcasting/ Print and Digital Media
(iii)	Defence
(iv)	Space
(v)	Telecommunications

Schedule II

List of Category-II Sensitive sectors:

Sr.No.	Sector
(i)	Power and Energy (including exploration/ generation/ transmission/ distribution/ pipeline)
(ii)	Banking and Finance including Insurance
(iii)	Civil Aviation
(iv)	Construction of ports and dams & river valley projects
(v)	Electronics and Microelectronics
(vi)	Meteorology and Ocean Observation
(vii)	Mining and extraction (including deep sea projects)
(viii)	Railways
(ix)	Pharmaceuticals & Medical Devices
(x)	Agriculture
(xi)	Health
(xii)	Urban Transportation

Schedule-III

List of Sensitive Technologies:

Sr.No.	Sensitive Technologies
(i)	Additive Manufacturing (e.g. 3D Printing)
(ii)	Any equipment having electronic programmable components or autonomous systems (e.g. SCADA systems)
(iii)	Any technology used for uploading and streaming of data including broadcasting, satellite communication etc.
(iv)	Chemical Technologies
(v)	Biotechnologies including Genetic Engineering and Biological Technologies
(vi)	Information and Communication Technologies
(vii)	Software

ANNEXURE I

Competent Authority and Procedure for Registration

A. The Competent Authority for the purpose of registration under this order shall be/ continue to be the Registration Committee constituted by the Department for Promotion of Industry and Internal Trade (DPIIT)*. [This Committee was already constituted under Order (Public Procurement) No.1].

B. The Registration Committee shall have the following members*:

   i. An officer, not below the rank of Joint Secretary, designated for this purpose by DPIIT, who shall be the Chairman;

   ii. Officers (ordinarily not below the rank of Joint Secretary) representing the Ministry of Home Affairs, Ministry of External Affairs, and of those Departments whose sectors are covered by applications under consideration;

   iii. Any other officer whose presence is deemed necessary by the Chairman of the Committee.

   iv. With effect from 01.04.2023, an officer (ordinarily not below the rank of Joint Secretary) representing the National Security Council Secretariat.

C. DPIIT shall lay down the method of application, format etc. for such bidders as covered by the Order.

D. On receipt of an application seeking registration from a bidder covered by Para 2 and 3 of this order, the Competent Authority shall first seek political and security clearances from the Ministry of External Affairs and Ministry of Home Affairs, as per guidelines issued from time to time. Registration shall not be given unless political and security clearance have both been received.

E. The Ministry of External Affairs and Ministry of Home Affairs may issue guidelines for internal use regarding the procedure for scrutiny of such applications by them.

F. The decision of the Competent Authority, to register such bidder may be for all kinds of tenders or for a specified type(s) of goods or services, and may be for a specified or unspecified duration of time, as deemed fit. The decision of the Competent Authority shall be final.

G. Registration granted by the Competent Authority of the Government of India shall be valid not only for procurement by the Central Government and its bodies specified in para 6 of this order, but also for procurement by State Governments and their agencies/ public enterprises etc. No fresh registration at the State level shall be required.

H. The Competent Authority is empowered to cancel the registration already granted if it determines that there is sufficient cause. Such cancellation by itself, however, will

not affect the execution of contracts already awarded. Pending cancellation, it may also suspend the registration of a bidder, and the bidder shall not be eligible to bid in any further tenders during the period of suspension.

I. For national security reasons, the Competent Authority shall not be required to give reasons for rejection/cancellation of registration of a bidder.

[*Note:

(i) In respect of application of the Order to procurement by/ under State Governments, all functions assigned to DPIIT shall be carried out by the State Government concerned through a specific department or authority designated by it. The composition of the Registration Committee shall be as decided by the State Government. However, the requirement of political and security clearance as per para D shall remain and no registration shall be granted without such clearance.

(ii) Registration granted by State Governments shall be valid only for procurement by the State Government and its agencies/ public enterprises etc. and shall not be valid for procurement in other states or by the Government of India and their agencies/ public enterprises etc.]

Special Cases

A. In projects which receive international funding with the approval of the Department of Economic Affairs (DEA), Ministry of Finance, the procurement guidelines applicable to the project shall normally be followed, notwithstanding anything contained in this order and without reference to the Competent Authority. Exceptions to this shall be decided in consultation with DEA.

B. This order shall not apply to procurement by Indian missions and by offices of government agencies/ undertakings located outside India.

C. This order will not apply to bidders (or entities) from those countries (even if sharing a land border with India) to which the Government of India has extended lines of credit or in which the Government of India is engaged in development projects. Updated lists of countries to which lines of credit have been extended or in which development projects are undertaken are given on the website of the Ministry of External Affairs.

D. Procurement of spare parts and other essential service support like Annual Maintenance Contract (AMC)/ Comprehensive Maintenance Contract (CMC), including consumables for closed systems, from Original Equipment Manufacturers (OEMs) or their authorized agents, shall be exempted from the requirement of registration.

ANNEXURE III

Model Clause/ Certificate/ Undertaking to be inserted in tenders etc.

[While adhering to the substance of the Order, procuring entities and GeM are free to appropriately modify the wording of the clause/ certificate based on their past experience, local needs etc.

The conditions relating to specified ToT (as shown in italics) should be incorporated only in the tenders which attract the restrictions due to specified ToT.]

A. Model Clauses for Tenders (including tenders issued manually or any electronic portal including GeM):

I. Any bidder from a country which shares a land border with India will be eligible to bid in any procurement whether of goods, services (including consultancy services and non-consultancy services) or works (including turnkey projects) only if the bidder is registered with the Competent Authority. *Further, any bidder (including bidder from India) having specified Transfer of Technology (ToT) arrangement with an entity from a country which shares a land border with India, shall also require to be registered with the same competent authority.*

II. "Bidder" (including the term 'tenderer', 'consultant' or 'service provider' in certain contexts) means any person or firm or company, including any member of a consortium or joint venture (that is an association of several persons, or firms or companies), every artificial juridical person not falling in any of the descriptions of bidders stated hereinbefore, including any agency branch or office controlled by such person, participating in a procurement process.

III. "Bidder (or entity) from a country which shares a land border with India" for the purpose of this Order means: -

(a) An entity incorporated, established or registered in such a country; or
(b) A subsidiary of an entity incorporated, established or registered in such a country; or
(c) An entity substantially controlled through entities incorporated, established or registered in such a country; or
(d) An entity whose beneficial owner is situated in such a country; or
(e) An Indian (or other) agent of such an entity; or
(f) A natural person who is a citizen of such a country; or
(g) A consortium or joint venture where any member of the consortium or joint venture falls under any of the above

IV. The beneficial owner for the purpose of (iii) above will be as under:

1. In case of a company or Limited Liability Partnership, the beneficial owner is the natural person(s), who, whether acting alone or together, or through one or more juridical person, has a controlling ownership interest or who exercises control through other means.

Explanation—

a. "Controlling ownership interest" means ownership of or entitlement to more than twenty-five per cent. of shares or capital or profits of the company;

b. "Control" shall include the right to appoint majority of the directors or to control the management or policy decisions including by virtue of their shareholding or management rights or shareholders agreements or voting agreements;

2. In case of a partnership firm, the beneficial owner is the natural person(s) who, whether acting alone or together, or through one or more juridical person, has ownership of entitlement to more than fifteen percent of capital or profits of the partnership;

3. In case of an unincorporated association or body of individuals, the beneficial owner is the natural person(s), who, whether acting alone or together, or through one or more juridical person, has ownership of or entitlement to more than fifteen percent of the property or capital or profits of such association or body of individuals;

4. Where no natural person is identified under (1) or (2) or (3) above, the beneficial owner is the relevant natural person who holds the position of senior managing official;

5. In case of a trust, the identification of beneficial owner(s) shall include identification of the author of the trust, the trustee, the beneficiaries with fifteen percent or more interest in the trust and any other natural person exercising ultimate effective control over the trust through a chain of control or ownership.

V. An Agent is a person employed to do any act for another, or to represent another in dealings with third person.

VI. [To be inserted in tenders for Works contracts, including Turnkey contracts] The successful bidder shall not be allowed to sub-contract works to any contractor from a country which shares a land border with India unless such contractor is registered with the Competent Authority.

VII. The registration shall be valid at the time of submission of bid and at the time of acceptance of bid.

VIII. If the bidder was validly registered at the time of acceptance / placement of order, registration shall not be a relevant consideration during contract execution

Model Certificate for Tenders:

"I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India; I certify that this bidder is not from such a country or, if from such a country, has been registered with the Competent Authority. I hereby certify that this bidder fulfills all requirements in this regard and is eligible to be considered. [Where applicable, evidence of valid registration by the Competent Authority shall be attached.]"

Model Certificate for Tenders for Works involving possibility of sub-contracting:

"I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India and on sub-contracting to contractors from such countries; I certify that this bidder is not from such a country or, if from such a country, has been registered with the Competent Authority and will not sub-contract any

work to a contractor from such countries unless such contractor is registered with the Competent Authority. I hereby certify that this bidder fulfills all requirements in this regard and is eligible to be considered. [Where applicable, evidence of valid registration by the Competent Authority shall be attached.]"

Model additional certificate by Bidders in the cases of specified ToT:

"I have read the clause regarding restrictions on procurement from a bidder having Transfer of Technology (ToT) arrangement. I certify that this bidder does not have any ToT arrangement requiring registration with the competent authority."

OR

"I have read the clause regarding restrictions on procurement from a bidder having Transfer of Technology (ToT) arrangement. I certify that this bidder has valid registration to participate in this procurement."

B. Model Certificate for GeM (to be taken by the GeM from seller during registration on GeM. GeM should also obtain this certificate from all existing bidders as soon as possible):

"I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India; I certify that this vendor/ bidder is not from such a country and does not have any specified Transfer of Technology (ToT) from such a country or, if from such a country or if having specified ToT from such a country has been registered with the Competent Authority. I hereby certify that this vendor/ bidder fulfills all requirements in this regard and is eligible to be considered for procurement on GeM. [Where applicable, evidence of valid registration by the Competent Authority shall be attached.]"

***

APPENDIX-VIII TO ITB

STARTUP INDIA POLICY

[NOT APPLICABLE]

APPENDIX - IX TO ITB

FOREIGN TRADE POLICY ON IMPORTS & EXPORTS TO DEMOCRATIC PEOPLES REPUBLIC OF KOREA

(To be published in the Gazette of India Extra Ordinary Part-II, Section-3, Sub section (ii))

Government of India
Ministry of Commerce and Industry
Department of Commerce
Directorate General of Foreign Trade
Udyog Bhawan

NOTIFICATION No.41/2015-2020
New Delhi, Dated: 21 March, 2017

Subject:- Amendment in Para 2.17 of the Foreign Trade Policy 2015-20 on Imports and Exports to Democratic People's Republic of Korea

S.O.(E) In exercise of the powers conferred by Section 5 read with Section 3(2) of the Foreign Trade (Development & Regulation) Act, 1992, as amended, read with Para 1.02 and Para 2.01 of the Foreign Trade Policy, 2015-2020, as amended from time to time, the Central Government hereby makes the following amendment in the Foreign Trade Policy, 2015-2020 with immediate effect:

1. Paragraph 2.17 of the Foreign Trade Policy 2015-20 stands substituted as follows:-

"2.17 Prohibition on Direct or Indirect Import and Export from/to Democratic People's Republic of Korea

A. Direct or indirect export and import of following items, whether or not originating in Democratic People's Republic of Korea (DPRK), to/from, DPRK is 'Prohibited':

(i) All items, materials, equipment, goods and technology as set out in the following UNSC and IAEA documents:
   a) S/2006/853* and S/2006/853/Corr.1;
   b) S/2009/364;
   c) Annex- III of S/RES/2094(2013);
   d) Committee decision of July 16, 2009 (S/2009/364) and resolution 2094(2013);
   e) Annex-A to INFCIRC/254/Rev.12/Part1;
   f) Annex to INFCIRC/254/Rev.9/Part2;
   g) S/2014/253;
   h) S/2016/308;
   i) S/RES/2321(2016) and
   j) Any other item as determined by Central Government, which could contribute to DPRK's nuclear-related, ballistic missile-related or other weapons of mass destruction related programmes;

(ii) Any battle tanks, armoured combat vehicles, large calibre artillery systems, combat aircraft, attack helicopters, warships, missiles or missile systems as defined for the purpose of the United Nations Register on Conventional Arms or related materiel including spare parts, as well as all arms and related materiel, including small arms and light weapons and their related materiel.

B. Direct or indirect export of following items to DPRK is 'Prohibited':

(i) Luxury goods including, but not limited to, the items specified in Annex-IV of S/RES/2094(2013), Annex-IV of S/RES/2270(2016); and Annex-IV of S/RES/2321(2016);

(ii) Aviation fuel, including aviation gasoline, naphtha-type jet fuel, kerosene-type jet fuel, and kerosene-type rocket fuel subject to the provisions of Paragraph 31 of UNSC Resolution 2270 (2016) and paragraph 20 of UNSC Resolution 2321 (2016); and

(iii) New helicopters and vessels, except as approved in advance by the Committee on a case-by-case basis.

C. Direct or indirect import of following items, whether or not originating in Democratic People's Republic of Korea (DPRK), from DPRK is 'Prohibited':

(i) Coal, iron, and iron ore, subject to the provisions of Paragraph 26 of UNSC Resolution 2321(2016);

(ii) Gold, titanium ore, vanadium ore, and rare earth minerals;

(iii) Statues, except as approved in advance by the Committee on a case-by-case basis and

(iv) Copper, nickel, silver and zinc.

Explanation:

a) IAEA refers to the International Atomic Energy Agency;

b) UNSC refers to the United Nations Security Council;

c) Committee refers to "Committee" of the United Nations Security Council set up in terms of Paragraph 12 of the United Nations Security Council Resolution 1718 (2006)and Paragraph 39 of 2321(2016).

2. **Effect of this notification:**

This notification seeks to update the Foreign Trade Policy, 2015-2020 to account for current UN Security Council Resolutions concerning Democratic People's Republic of Korea (DPRK) up to Resolution 2321(2016) of 30th November 2016.

Ajay Kumar Bhalla
Director General of Foreign Trade
E-mail: dgft@nic.in

[Issued from File No. 01/91/180/991/AM07/PC-III/EC]

Project / Client : **GROUP-II/III LOBS PROJECT AT MANALI REFINERY, CHENNAI**

RFQ No.:

Item:

Notes:

a) **Use capital letters and English language only**.

b) Bidder to submit it duly signed and stamped on each page, as a token of their acceptance and confirmation(s) in respect of RFQ.

c) Bidder to note that fill in the blanks - data / details is to be entered by the bidder and correct option ticked in respect of Sl. Nos. 1 to 5, 13, 17 to 19 (the same are also highlighted in shaded grey in the table below).

d) Bidders are required to carefully fill their responses to below points along with submission of relevant documentary evidence as per the provisions of RFQ.

e) In case the responses are incomplete/ inconsistent and / or not substantiated by relevant document(s) the bidders are at risk of evaluation of their bid being concluded without affording them another opportunity to complete / correct / comply their submission with respect to RFQ requirements. In such a case EIL/ Owner reserves the right to complete the evaluation based on the available information and EIL/ Owners decision shall be final and binding.

SL. NO.	DESCRIPTION	BIDDER'S / SUPPLIER'S CONFIRMATION (Strike out response which is not applicable)
1.	a) Bidder/Supplier Name	
	b) Status of Bidder's Firm: Proprietorship/ Partnership/ Limited Liability Partnership / Public Ltd./ Pvt Ltd./ Govt Deptt./ PSU/ Others (Specify)	
	c) Supplier's Offer Ref No.& Date	
2.	Name and contact details of Bidder's/ Supplier's authorized signatory: a) Name: b) Tel. No.: c) Mob. No.: d) E-mail:	
3.	a) Specify Despatch Point b) GSTIN no.	a) …….. b) …… In case different despatch points are applicable, specify the same as under:- …… …… ……
4.	Details of GST on Supply & Transportation Whether IGST or CGST and SGST/UTGST is applicable (Pls. tick ✓whichever is applicable)	☐ IGST ☐ CGST & SGST/UTGST

BIDDER'S SIGNATURE WITH STAMP/SEAL

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<table>
<tr><th>SL. NO.</th><th>DESCRIPTION</th><th>BIDDER'S / SUPPLIER'S CONFIRMATION
(Strike out response which is not applicable)</th></tr>
<tr><td>5.</td><td>Rate of GST</td><td>If a single rate of GST is applicable:
GST Rate extra @_____%

If different rates of GST are applicable, specify the GST rates in the table below:
<table><tr><th>Description</th><th>GST rate in %</th></tr><tr><td>Supply</td><td></td></tr><tr><td>Freight</td><td></td></tr><tr><td>Any other</td><td></td></tr></table></td></tr>
<tr><td>6.</td><td>Bidder shall ensure correct rate of GST as per CGST Act 2017 considering the scope of supply and services as per bidding document.</td><td>Confirmed</td></tr>
<tr><td>7.</td><td>Confirm that bidder has quoted the prices in the BoQ Format in e-tendering portal as per MR (Material Requisition) and following has been quoted in the finance folder as per applicability in RFQ/MR/Price Schedule:
a) 2 years O&M Spares
b) Unit Rates for Addition/ Deletion (other than pre-filled)
c) Unit rates for specific test
d) Overtime Rates for Supervision</td><td>Confirmed</td></tr>
<tr><td>8.</td><td>Confirm that firm transportation charges upto Project Site have been quoted by the bidder in the BoQ/Price Schedule format</td><td>Confirmed</td></tr>
<tr><td>9.</td><td>Confirm that quoted prices will remain firm and fixed (except where price escalation/variation is allowed in the RFQ/BDS) till complete execution of the order.</td><td>Confirmed</td></tr>
<tr><td>11.</td><td>Confirm that the offer is valid for the period specified in RFQ/BDS, from the final 'Bid Due Date'.</td><td>Confirmed</td></tr>
<tr><td>12.</td><td>Confirm submission EMD/Bid Security (as applicable) as per provision of RFQ.</td><td>Confirmed</td></tr>
</table>

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SL. NO.	DESCRIPTION	BIDDER'S / SUPPLIER'S CONFIRMATION (Strike out response which is not applicable)
13.	MSE Status: a) Confirm whether the Bidder is a MSE Bidder	Yes / No If yes, mention 19 digit UDYAM No. UDYAM-……
	b) If the owner of Bidder's organization belongs to MSE social category: SC/ST as per Udyam registration, submit SC/ST Certificate issued by District Authority, duly certified by Statutory Auditor or Notarized or any other specified document.	(Tick whichever is applicable) ☐ Yes, document attached ☐ Not Applicable
	c) If the owner of Bidder's organization belongs to the MSE Gender category – Woman, submit requisite documents as required in the RFQ.	(Tick whichever is applicable) ☐ Yes, document attached ☐ Not Applicable
	Notes: **In case of bidding on other than GeM portal, if no specific declaration towards MSE / SC / ST / Female is made above in the ATC (i.e. no response is marked made against (a), (b), and (c) above) and no relevant documentary evidence is submitted in their Un-priced Bid, it will be considered that the bidder does not belong to above preferential categories & no benefits shall be extended to such bidder.**	
14.	Public Procurement (Preference to Make in India) (PPP-MII) Order 2017 and its amendments thereof: Bidder has submitted FORM-2 and FORM-2B, 2C, 2D (wherever applicable) (appended below as **Annexure-I** to ATC) duly filled and completed.	Submitted
15.	Compliance to DMI&SP Policy: Bidder to confirm compliance to Domestically Manufactured Iron & Steel Policy (If applicable as per RFQ). Format(s) for compliance to DMI&SP Policy is appended as **Annexure-II** to this ATC. Formats to be submitted during bidding Stage: Annexure-2A, Annexure-3 and Annexure-4, as applicable, of **Annexure-II** to this ATC.	Confirmed Note: It shall be automatically construed that the signatory is the authorized representative of the bidder duly empowered to sign the affidavit.
16.	Bidder's Undertaking Regarding Sharing Land Border with India: Format(s) for Undertaking is appended as Form-IA and Form-II under **Annexure-III** to this ATC duly completed to be submitted on company letterhead.	Submitted

BIDDER'S SIGNATURE WITH STAMP/SEAL

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SL. NO.	DESCRIPTION	BIDDER'S / SUPPLIER'S CONFIRMATION (Strike out response which is not applicable)
17.	a) Confirm that Bidder is not on Holiday/ Blacklist/ Negative/ Suspension/ Banning list of EIL/ CPCL as on date of submission of our bid.	Confirm / Not Confirm Note: In case response is not marked, "Confirm" shall be considered as Bidder's response Bidder to give correct information, otherwise the bid shall fall under 'misrepresentation of facts' provision of the RFQ.
	b) Confirm that Bidder is not banned or delisted or black listed by any the Administrative Ministry of EIL/CPCL.	Confirm / Not Confirm Note: In case response is not marked, "Confirm" shall be considered as Bidder's response. Bidder to give correct information, otherwise the bid shall fall under 'misrepresentation of facts' provision of the RFQ.
	c) Confirm that Bidder is not under liquidation, court receivership or similar proceedings.	Confirm / Not Confirm Note: In case response is not marked, "Confirm" shall be considered as Bidder's response Bidder to give correct information, otherwise the bid shall fall under 'misrepresentation of facts' provision of the RFQ.
18.	a) Is Bidder involved in any Litigation / Arbitration? b) In case response to a) above is 'Yes', Bidder hereby confirm that the current Litigation / Arbitration, in which they are involved will not have any impact in work being tendered or in entering into contract during the validity of offer and performing the contract till all the contractual obligations under contract are performed.	a) Yes / No *Note: In case response is not marked, "No" shall be considered as Bidder's response* b) Yes / No *Note: In case response is not marked, "Yes" shall be considered as Bidder's response* Bidder to give correct information, otherwise the bid shall fall under 'misrepresentation of facts' provision of the RFQ.
19.	Bidder confirms that: a) None of Directors (in Board of Director) of Bidder is a relative of any Director (in Board of Director) of CPCL/ EIL OR b) The bidder is not a firm in which any Director (in Board of Director) of CPCL/EIL or their relative is a partner.	a) Confirmed / Not Confirmed b) Confirmed / Not Confirmed If response is marked as "Not Confirmed", then provide relevant details…… c) *Note: In case response is not marked, "Confirmed" shall be considered as Bidder's response.*

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SL. NO.	DESCRIPTION	BIDDER'S / SUPPLIER'S CONFIRMATION (Strike out response which is not applicable)
20.	Bidder confirms compliance to the provisions for no conflict of interest / multiple bids / alternative bids, as per provisions of RFQ Document. Accordingly bidder to submit declaration(s) as per **Annexure-IV to ATC**.	Confirmed
21.	Integrity Pact (if applicable as per BDS): Bidder confirms that they comply with the provisions of Integrity Pact (IP) and have submitted IP duly signed and stamped. Bidder has ensured that: a) They have submitted the IP duly signed by authorised POA holder of the Bidder. b) IP is signed on all the pages. c) IP is signed by two witnesses. d) In case of 'Proprietorship firm', IP is signed by the Proprietor. e) In case of Partnership firm, IP is signed by all the partners of the firm. f) The correct name and number of RFQ is mentioned in the IP.	Confirmed
22.	Bidder hereby confirms that they will comply with the policy requirements under "Amendment in para 2.17 of the Foreign Trade Policy 2015-20 on Imports & Exports to People's Republic of Korea" as per Appendix-IX to ITB.	Confirmed
23.	Bank Mandate We hereby undertake to submit the Bank mandate Form (refer elsewhere for format) along with first invoice/ bill submission, duly filled and certified by bank along with cancelled cheque. In case the same is not submitted along with first invoice, the subsequent date of submission of mandate form shall be considered as the date of invoice submission.	Confirmed
24.	Compliance to RFQ Documents We hereby agree to fully comply with, abide by and accept the RFQ/Enquiry documents including but not limited to technical, commercial and other conditions, technical specifications including general/ technical notes and scope of supply/work including documentation as per material requisition (MR) and subsequent technical/ commercial amendment, if any, and technical/commercial corrigendum, if any, without any variation, deviation or reservation.	Confirmed
25.	Bidder understand that being a "Zero Deviation" bidding & in the case of submission of deviations, the evaluation of bid shall be done as per provisions of RFQ and the bid is liable for rejection.	Confirmed

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26.	All commercial terms / compliance / salient details are given /confirmed in this ATC and not elsewhere in the Offer. Bidder to note that the terms and conditions mentioned elsewhere in their offer or Printed terms and conditions of the bidder, if any, appearing in quotation/offer or Bidder's own terms and conditions, shall not be given any cognizance.	Confirmed

DECLARATION BY THE BIDDER:

We further confirm that the above declaration/confirmation (including the pre-filled / default response to be considered in case relevant option is not ticked/ not filled) are true and correct in all respects and not in conflict with the declaration mentioned in any part of our bid /any other document. In case of contradiction, the confirmation given herein by the bidder shall prevail over the declaration/ statement/ confirmation given in other places by the Bidder.

We also declare that in case, at a later date, if the above declaration(s) / confirmation(s) submitted in our bid are found to be false, bidder shall be held responsible for the same and EIL/Owner has every right to take action against the bidder / company, as deemed fit as per law of the land and provisions of RFQ Document including but not limited to forfeiture of EMD/ Bid Security and EIL/Owner's right to put our Company on Suspension/Banning list for future business with EIL/Owner.

BIDDER'S SIGNATURE WITH STAMP/SEAL

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ANNEXURE-I to ATC

FORM – 2

SELF CERTIFICATION BY BIDDER UNDER PPP-MII ORDER, 2017

To
M/s _____ <Insert name of Client>

NAME OF WORK / ITEM:

TENDER /RFQ NO. :

Dear Sir,

We, M/s _____ *<Name of Bidder>* hereby confirm that we meet the mandatory minimum requirement of Local Content (LC) for **Class-I local supplier / Class-II local supplier** *(strike out which is not applicable)* under Public Procurement (Preference to Make in India), Order 2017 (PPP-MII, 2017) and the work / Item shall have **LC of** _____**%.**

The details of the location(s) at which the local value addition is/shall be made, is/ areas under:

……..
……..
……..

We confirm compliance of Public Procurement (Preference to Make in India) (PPP-MII) Order 2017, vide Order No P-45021/2/2017-PP(BE-II)-Part-4)Vol.II, Notifications with respect to Electronics Products [File No. W-43/4/2019-IPHW-Meity dated 07.09.2020 and S.O. No 4591(E) dated 12.10.2023 issued by Meity], Cyber Security Products [File No.1(10)/2017-CLES dated 06.12.2019 issued by Meity] and Telecom Products [File No FP-18-10/2017-IP dated 21.10.2024 issued by Ministry of Telecommunication]. {Strike out which is not applicable}.

We also confirm that, the imported items sourced locally from resellers/distributors are excluded from calculation of local content. The cost of such locally sourced imported items (inclusive of taxes) along with break up on license/royalties paid/technical expertise etc. sourced from outside India shall be submitted at the time of execution as per Form-2A.

We further confirm that in case we fail to meet the minimum local content/domestic value addition, the same shall be treated as false information and Owner reserves the right to take appropriate action for debarment, Penalty after award etc. as per provision of the PPP-MII Order, 2017 and its subsequent Amendments and as more specifically mentioned in the Bidding/ RFQ document.

We also confirm that we are not debarred by any other procuring entity (GOI/PSU/PSE etc.)
or

Our debarment period is completed as on _____ <insert date>, and as such we are eligible to avail purchase preference under this Order.
(## Strike out which is not applicable)

_____
Signature of Bidder

Place:

Name :

Date:

Designation

Seal :

BIDDER'S SIGNATURE WITH STAMP/SEAL

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ANNEXURE-I to ATC

Form-2B

Format for Self-Certification regarding Local Content (LC) for Telecom Product, Services or Works to be submitted on non-judicial stamp paper of the value Rs. 100/-

Date:

I__ _____S/o, D/o, W/o, Resident of _____ do hereby solemnly affirm and declare as under:

That I agree to abide by the terms and conditions of Department of Telecommunications, Government of India issued vide Notification No:_____ dated_____.

That the information furnished hereinafter is correct to the best of my knowledge and belief and I undertake to produce relevant records before the procuring entity or any other authority so nominated by the Department of Telecommunications, Government of India for the purpose of assessing the LC.

That the LC for all inputs which constitute the said Telecom Product/Services/Works has been verified by me and I am responsible for the correctness of the claims made therein.

That in the event of the LC of the Telecom Product/Services/Works mentioned herein is found to be incorrect and not meeting the prescribed LC norms, based on the assessment of an authority so nominated by the Department of Telecommunications, Government of India, I and my Statutory auditor or cost auditor (if applicable) will be liable for actions as specified in Clause (9) of the DPIIT PPP-MII Order dated 19.07.2024 for all incorrect/false facts and figures.

I agree to maintain detailed breakup / information (separately for each product) to substantiate my claim for LC in the Company's record for a period of 2 years and shall make this available for verification to any authority. I shall also maintain records of local content pertaining to items bought from other domestic manufacturers / traders.

[Please provide following information]

i	Name and details of the local supplier - Registered Office, Manufacturing unit location, nature of legal entity
ii	Date of certificate issuance - Date on which this certificate is issued
iii	Telecom Product/Services/Works - Telecom Product/ Services/ Works for which the certificate is produced
iv	Procuring agency - Procuring agency to whom the certificate is furnished
v	Percentage of Local Content claimed - Percentage of LC claimed
vi	Manufacturer unit details - Name and contact details of the unit of the manufacturer
vii	Sale Price of the product - Sale Price of the product
viii	Ex-Factory Price of the product - Ex-Factory Price of the product
ix	Freight, insurance and handling - Freight, insurance and handling charges
x	Total Bill of Material - Total Bill of Material
xi	Local inputs cost - List and total cost value of inputs used for manufacture of the Telecom product/Services/Works

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xii	Locally sourced inputs - List and total cost of inputs which are locally sourced. Please attach LC certificates from local suppliers, if the input is not produced in-house
xiii	Imported inputs cost - List and cost of inputs which are imported, directly or indirectly

I hereby certify that, having read all the provisions of the above order and principles / basis of calculations, the local content calculation does not include the following:

a. Imported items sourced locally from resellers/distributors.
b. The license fees / royalties paid/ technical charges paid out of India
c. Procurement / supply of repackaged / refurbished/rebranded imported products

I hereby also certify to the best of my knowledge and belief that all the particulars furnished above are correct and complete. I agree to comply with the terms and conditions of the DPIIT PPP-MII order dated 19.07.2024 and DoT PPPMII Notification dated……..…...

I understand that any incorrect declaration regarding the local content or failure to substantiate the claim of LC will result in penalties as specified in Clause (9) of the DPIIT PPP-MII Order dated 19.07.2024.

I further certify and take personal responsibility that I have applied my mind to the calculations and principles of LC as specified in this order and I shall, having declared the LC shall not seek recourse to change it on any ground. Any changes made by me on any grounds in a bid in LC after bid submission shall make my bid non-responsive and I shall hold myself liable for civil/criminal action arising out of any such change. I understand and agree that any such post bid change in LC content shall also be a valid ground for blacklisting of the firm from future contracts/bids.

Signature:
Name:
Designation: CEO/MD
Address:
Email Address:
Mobile No.:
Place:
Date:

BIDDER'S SIGNATURE WITH STAMP/SEAL

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ANNEXURE-I to ATC

Form-2C

Format for Self-Declaration regarding local supplier' for Cyber Security Products

This is to Certify that the organization_____ registered as_____with registration number_____ do hereby solemnly affirm and declare as under:

That we agree to abide by the terms and conditions of the Notification issued by Ministry of Electronics and Information Technology (MeitY), Government of India no: _____dated_____.

That the information furnished hereinafter is correct and we undertake to produce relevant records before the procuring entity or any other authority so nominated by the Ministry of Electronics and Information Technology (MeitY), Government of India for the purpose of establishing ownership of the Intellectual Property Rights(IPR), legal existence and revenue accrual, local content for the cyber security products nominated by the aforesaid organization.

That all IPR which constitute the said cyber security product has been verified by us and we are responsible for the correctness of the claims made therein and we take complete responsibility of the same.
We agree to maintain all information regarding our claim(s) for IPR ownership, legal existence and revenue accrual, local content in the organization's record throughout the existence of the product and shall make this available for verification to any authorities specified by Government of India.

In case of any change in the IPR ownership, legal existence and revenue accrual, local content we will be accountable to inform Ministry of Electronics and Information Technology, Government of India within one week or before applying for any public procurement or before referring this order for taking any advantage which so ever occurs first.

That in the event of the cyber security product mentioned herein is found to be incorrect and not meeting the prescribed norms, based on the assessment of an authority so nominated by the MeitY, Government of India and we will be liable as under clause 7 (f) of Public Procurement (Preference to Make in India) Order 2019 for cyber security product.

1. Name and details of the organization nominating product under this order (Registered Office, Manufacturing unit location, nature of legal entity)

2. Entity Registration Certificate number
   a. Type of registration

3. Date on which this certificate is issued

4. Percentage of Royalty/License fee to be paid to other entity with respect to estimated cost of the product

5. Name and contact details of the unit of the manufacturer

For and on behalf of_____ (Name of firm/entity)

Authorize signatory (To be duly authorized by the Board of Directors)

BIDDER'S SIGNATURE WITH STAMP/SEAL

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ANNEXURE-I to ATC

FORM – 2D

CERTIFICATE UNDER DOMESTICALLY MANUFACTURED/PRODUCED CYBER SECUIRTY PRODUCTS (HAVING VALUE OF PROCUREMENT MORE THAN INR 10 CRORE)

To
M/s _____ <Insert name of Client/Owner>

NAME OF WORK / ITEM:

TENDER /RFQ NO.:

Dear Sir,

We, _____, the Statutory Auditor or Cost Auditor of the Company (in case of Companies) or are a practicing Cost Accountant or practicing Chartered Accountant (in respect of supplier other than companies) of M/s. _____ *<Name of Bidder>*, hereby certify offered product(s) by M/s _____ *<Name of Bidder>* meets the definition of 'local supplier' of domestically manufactured/produced Cyber Security Products as per para 4 of Public Procurement (Preference to Make in India) Order 2019 for Cyber Security Products.

_____
Signature of certificate issuing authority

Place:

Name :

Date:

Name of Audit Firm:

Designation

Seal :

BIDDER'S SIGNATURE WITH STAMP/SEAL

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ANNEXURE-II to ATC

Annexure-2A

(to be submitted in Bidder's letter head)

[for Bidders who are selling agents/authorized distributors/authorized dealers/authorized supply houses]

Item:

Bidding Document no:

This has reference to "Policy for providing Preference to Domestically Manufactured Iron & Steel Products in Government Procurement" issued by Ministry of Steel, Govt. of India, vide their notification "The Gazette of India, Extraordinary Part -…… dated…… including its amendments thereafter (if any)".

We confirm:

- that we will comply the Policy for providing Preference to Domestically Manufactured Iron & Steel Products in Government Procurement" issued by Ministry of Steel, Govt. of India.

- that we will submit authorization certificate issued by the domestic manufacturer at the time of delivery of related iron and steel products required under the enquiry document.

Sign & Stamp of bidder

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ANNEXURE-II to ATC

Annexure-3

DECLARATION OF NON-APPLICABILITY OF DMI&SP POLICY

(to be submitted in Bidder's letter head)
UNDERTAKING ON LETTERHEAD

Item / Works:
Bidding Document no:

This has reference to "Policy for providing Preference to Domestically Manufactured Iron & Steel Products DMI&SP) in Government Procurement" issued by Ministry of Steel, Govt. of India, vide their notification "The Gazette of India, Extraordinary Part …….. dated …….. including its amendments thereafter (if any)".

We confirm that, we have understood the scope and terms & conditions of above mentioned DMI&SP Policy and also confirm that, based on the Prices quoted by us for each individual item, DMI&SP Policy is not applicable for the following items –

Where evaluation is on Item wise basis	
S. No	MR Item No
1	
2	
3	

Where evaluation is on Group wise basis		
S. No	Group	Item no of the respective group
1		
2		
3		

Further, in case at any stage before or after award, if it is observed by the EIL/OWNER the value of the above items is more than INR 5 Lakhs (Excluding Freight and GST) it shall be construed as the false/wrong declaration and procuring entity is entitled to take penal action as per provision of Bidding Document.

Sign & Stamp of Bidder

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ANNEXURE-II to ATC

Annexure-4

Item / Works:

RFQ/Bidding Document no:

Form-1

Affidavit (Notarized) of Self-Certification for Compliance with Melt & Pour Requirements (Appendix-A) and/or Domestic Value Addition Requirements (Appendix-A & Appendix-B)*
**For Appendix-B capital goods, this affidavit must accompany the statutory auditor's / cost auditor's / CA certificate, as mandated under the policy.*

I,_____, Resident of _____, being the authorized signatory of M/s _____, do hereby solemnly affirm and declare as under:

1. I have read and understood the Domestically Manufactured Iron & Steel Products (DMI&SP) Policy, 2025, issued vide Notification No. _____ dated _____. I agree to abide by all terms and conditions of the said policy and the tender in which this affidavit is submitted.

2. This affidavit pertains to the following iron & steel product(s)/capital good(s):-

   Name of product (s):……
   HS Code (s):……..
   Appendix classification: Appendix-A / Appendix-B (select as applicable)
   Procuring agency: ……..

3. Melt & Pour - Applicable only for Appendix-A items with Melt & Pour condition
   I hereby declare that the above product(s) satisfy the Melt & Pour in India requirement as notified in Appendix-A. Specifically: The steel was melted and poured in India.

   The first solid shape (slab/billet/ingot/other) was produced at the following location(s): as specified below / as per Form-1A *(strike through which is "Not Applicable")*.
   Steelmaking facility: ……
   Location: ……..
   Heat numbers:……..
   Batch details: ……

   I undertake to produce mill test certificates, heat records, and any other documentary proof to substantiate this declaration whenever required by the procuring agency. *(If not applicable, write "Not Applicable".)*

4. Domestic Value Addition - Applicable for Appendix-A DVA items and Appendix-B capital goods:
   I hereby declare that the Domestic Value Addition (DVA) for the product(s) is: _____% / as per Form-1A *(strike through which is "Not Applicable")*. The DVA has been computed as per DPIIT guidelines and the formula specified in the DMI&SP Policy, 2025.

   4A. For Appendix-B Capital Goods (Mandatory): A certificate issued by the statutory auditor / cost auditor / chartered accountant is enclosed herewith, confirming the DVA

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of the capital good package in accordance with the policy. (If Appendix-A item only, strike this clause out.)

5. I confirm that: All information furnished is true and correct. All input steel costs, imported content, and domestic content calculations have been verified. All manufacturing, purchase, and pricing records are available for examination.
   If any information is found incorrect or misleading: The procuring agency may disqualify me / my firm from Government tenders as per the procuring agency norms. I agree to pay all assessment costs and penalties, including forfeiture of EMD, as per policy and tender conditions.

6. I undertake to maintain the relevant records, and make them available to any statutory authority.

7. Declaration: I do hereby declare that the statements made above are true to the best of my knowledge and belief. I understand that any misrepresentation shall attract penalties under the DMI&SP Policy, applicable laws, and tender conditions.

   For and on behalf of M/s _____

   (Authorised Signatory)
   (Place & Date)

Note: Bidder may submit information sought against clause no 2, 3 and 4 in Form-1A if required. In that case, Form-1A also shall be part of Affidavit.

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ANNEXURE-II to ATC

Annexure-4

Item / Works:

RFQ/Bidding Document no:

FORM-1A

(Applicable for the iron and Steel products specified in Appendix-A only)

BIDDER:	
TENDER / RFQ NO:	
PROCURING AGENCY:	
APPENDIX CLASSIFICATION:	Appendix-A

MR Item No	Name of product (s):	HS Code (s)	Steelmaking facility:	Location:	Heat numbers	Batch details:	Domestic Value Addition (DVA) for the product(s) in %.
shall be referred from MR		These columns shall be filled for items with required condition as "Melt & Pour"					This column is applicable for items "condition" with specific DVA.
							……..%
							……..%
							……..%
							……..%
							……..%

Note –

a. Bidder to refer "Condition" and "Domestic Value Addition" specified in Appendix-A while certification of the above Form-1 and Form-1A.
b. Form-1A shall form part of Affidavit (Form-1)

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ANNEXURE-III to ATC

Form-I A

BIDDER'S UNDERTAKING

(On Company's Letter Head)

To,

M/s Chennai Petroleum Corporation Limited / Engineers India Limited

Item : _____

Bidding Document No.: _____

"I have read the clause regarding restrictions on procurement from a Bidder of a country which shares a land border with India; I hereby certify that, Bidder M/s_____(Name of the Bidder) is:

i)	Not from such a country	[ ]
ii)	If from such a country, has been registered with the Competent Authority	[ ]

Notes:

a) Bidder to tick appropriate option (√) in either i) or ii) above)

b) In case of sl. no. ii) above, Bidder to attach evidence of valid registration by the Competent Authority.

I hereby certify that the Bidder fulfills all requirements in this regard and eligible to be considered.

Place :
Date :

Signature (#):
Name :
Designation :
Seal :

(#) Undertaking shall be signed by the authorized signatory of the Bidder.

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ANNEXURE-III to ATC

Form-II

(On Company's Letter Head)

BIDDER'S UNDERTAKING IN CASE OF TRANSFER OF TECHNOLOGY (ToT) ARRANGEMENT

To,

M/s Chennai Petroleum Corporation Limited / Engineers India Limited

Item : _____

Bidding Document No.: _____

"I have read the clause regarding restrictions on procurement from a Bidder of a country having Transfer of Technology (ToT) arrangement.; Bidder M/s_____ (Name of the Bidder), hereby certify that,

i)	Bidder does not have any ToT arrangement requiring registration with the competent authority	[ ]
ii)	Bidder has valid registration to participate in this procurement	[ ]

Notes:

a) Bidder to tick appropriate option (√) in either i) or ii) above)

b) In case of sl. no. ii) above, Bidder to attach evidence of valid registration by the Competent Authority.

I hereby certify that the Bidder fulfills all requirements in this regard and eligible to be considered.

Place :
Date :

Signature (#):
Name :
Designation :
Seal :

(#) Undertaking shall be signed by the authorized signatory of the Bidder.

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ANNEXURE-IV to ATC

ANNEXURE-A

[to be submitted in Bidder's Letter Head]

SELF DECLARATION OF CONFLICT OF INTEREST/ ALTERNATE OFFERS/ MULTIPLE BIDS

(to be submitted in Bidder's Letter Head)

Item Description:

RFQ/Bidding Document No.:

Bidder: ……..

I am / We are participating in this Enquiry/RFQ…… (whole or part of the Enquiry/RFQ against which separate contracts will be awarded to individual bidders). We understand the terms & conditions of Conflict of Interest/ Alternate Offers/Multiple Bids as stipulated in Enquiry/RFQ Document.

I / We hereby declare that there is no other participating bidder in the same (whole or part of the Enquiry/RFQ), which may create conflict of interest and /or alternate offers/ Multiple bids as described above and in Enquiry/RFQ Document to the best of our knowledge and understanding.

I/We understand that in case any conflict of interest and/or alternate offers/ Multiple bids is found to have occurred, EIL will disqualify/reject the bids from the bidders deemed to have conflict of interest as per provision of Enquiry/RFQ Document.

*I/We hereby declare that we are selling agents/ authorized distributors/ authorized dealers/ authorized supply houses/Stockist/ Traders/ Resellers of offered product(s) as specified in Annexure-B and EIL shall have the right to reject our offer if declared manufacturer has participated for the same item(s)/Group(s)/Part(s) in the subject RFQ**(#).***

I / We further declare that in case I / We observe that any conflict of interest and/or alternate offers/ Multiple bids is found to have occurred after the unpriced bid opening, I / We shall immediately inform EIL in writing regarding such conflict of interest within 3 days from unpriced bid opening; and EIL shall have the right to disqualify the bids from the bidders deemed to have conflict of interest.

It is understood that any wrong declaration / non-declaration in this context shall make me / my agency / company liable for further action as per provision of Enquiry/RFQ document.

Place: _____

Date: (SIGNATURE OF BIDDER)

 # [strike through the paragraph if not applicable]

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ANNEXURE-IV to ATC

ANNEXURE-B

[to be submitted by selling agents/ authorized distributors/ authorized dealers/ authorized supply houses/Stockist/ Traders/ Resellers]

(Applicable for the procurement of items under Domestically manufactured iron & steel products Policy)

BIDDER:			
RFQ NO:			
MR Item No	Name of product (s):	HS Code (s)	Name of the Manufacturer**(##)**
shall be referred from MR			

(##) selling agents/ authorized distributors/ authorized dealers/ authorized supply houses/ Stockist/ Traders/ Resellers can offer more than one Manufacturer.

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INTEGRITY PACT

(ANNEXURE-V to ATC)

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SPECIAL PURCHASE CONDITIONS

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1.0 INTRODUCTION

1.1 These Special Purchase conditions (SPC) shall be read in conjunction with the General Purchase Conditions (GPC), Material Requisition and any other document forming part of this RFQ wherever the context so requires.

1.2 Where any portion of the GPC is repugnant to or at variance with any provisions of the SPC & Bid Data Sheet, then unless a different intention appears, the provision(s) of the SPC & Bid Data Sheet shall be deemed to override the provision(s) of GPC only to the extent that such repugnancy of variations in the SPC & Bid Data Sheet are not possible of being reconciled with the provisions of GPC.

1.3 The terms fully capitalized and/or initial capitalized shall be interchangeable and shall have the meaning as assigned to fully capitalized term or initial capitalised term.

2.0 PAYMENT TERMS

2.1 In partial modification to the payment terms as mentioned in the General Purchase Conditions, the payment terms for the scope of supply/ services shall be as per Bid Data Sheet.

2.2 **NOTES TO PAYMENT TERMS:**

2.2.1 1st Milestone payment shall be released only after receipt of CPBG (wherever applicable) and manufacturing schedule (in template available on portal https://www6.eil.co.in/epsinspection/Supp/).

2.2.2 Wherever ABG is required to be submitted by the supplier against milestone progress payment, validity of ABG shall be valid till CDD plus a 3 (three) months claim period which shall be further extended, as required till last date of receipt of all material at site. However, ABG will be released on dispatch/ receipt of material at site (as applicable).

2.2.3 Prorate payments shall be applicable on dispatch of material. In case of equipment, prorate shall not be on part of equipment.

2.2.4 In case of indigenous SUPPLIERs, wherever taxes/duties are separately indicated, the order value shall be exclusive of SGST & CGST/ IGST. 100% payment of taxes/duties shall be paid along with the payment released against dispatch documents on receipt of GST Invoice as detailed in clause 16.

2.2.5 Certification for placement of sub-order and/or identification of major raw material, in the above payment terms, shall be carried out by TPI agency, as applicable.

2.2.6 All final technical documents as per technical specifications provided in PR, in the required number of copies, MUST be sent to Engineers India Limited before shipment of the material/equipment. LC shall specify that payment shall be released with EIL certificate of receipt of technical documents as one of the negotiable documents. In case the SUPPLIER does not submit these documents to EIL prior to despatch clearance, SUPPLIER will claim payment after deducting 5% of their payment due against shipment. This 5% payment shall be made through wire transfer against certification by EIL.

2.2.7 All dispatches shall be made only after receipt of Despatch Clearance from CPCL/EIL.

2.2.8 No initial advance payment along with order shall be made by Owner against supplies as well as services (i.e. supply, transportation, erection, site work etc.).

2.2.9 Total progressive payments shall be as per the payment terms indicated elsewhere in the

tender document. No progressive payments at any stage other than those milestones specifically mentioned in payment term shall be payable by CPCL.

2.2.10 All Bank guarantee (or e-BG) will be issued directly to CPCL by the Bank and SUPPLIER shall enclose copy of the same along with invoice. BG shall be accompanied by a covering letter from the issuing bank if BG is submitted by the bidder directly. As per RBI guidelines, Bank guarantees issued for Rs.50,000/- and above should be signed by two officials jointly. The Bank Guarantee would be accepted by CPCL only on confirmation of receipt of SFMS by our bankers, SBI, CAG Branch, Chennai.

2.2.11 All payments (other than through L/C) shall be released within 30 days of receipt of invoice and all requisite documents, complete in all respects.

All bank charges of respective bankers shall be to respective account.

2.2.12 CPCL shall establish the Letter of Credit only on receipt of SUPPLIER's acknowledgement to Fax of Acceptance / Purchase Order (whichever issued earlier) and on submission of 10% Contract Performance Bank Guarantee (CPBG) by SUPPLIER to CPCL.

2.2.13 Format for Advance Bank Guarantee (ABG) and Format for Contract Performance Bank Guarantee (CPBG) is enclosed elsewhere in the Bidding Document.

2.2.14 Indigenous Order ABGs and CPBGs shall be addressed to Chennai Petroleum Corporation Limited, Project Site Address (which will be indicated in the Purchase Order) and shall be sent to Project Site only.

2.2.15 For indigenous SUPPLIERs, in case of direct negotiation of despatch documents payment shall be released only after receipt of materials at site.

2.2.16 Supplier shall submit Billing Schedule within three weeks after placement of LOA/PO (whichever is issued earlier) for EIL/OWNER's approval (wherever applicable). Suppliers requiring multiple dispatches will restrict the number of dispatches to maximum three, unless agreed otherwise by EIL/ CPCL.

2.2.17 In case, delivery basis is on FOT Project Site, deduction on account of delay in delivery (if any) shall be made from milestone payment due immediately after the receipt of material at site.

2.2.18 Letter of Credit for foreign supplier shall be opened through a Govt. of India Nationalized / Scheduled Bank and hence need not be confirmed. However, if supplier still insists for confirmed L/C, confirmation charges shall be borne by supplier.

2.2.19 Payment to indigenous SUPPLIERs shall be released through Electronic Clearing System (ECS) and to foreign SUPPLIERs, payments other than against shipping documents, i.e., for Site Work and Supervision/Training etc. shall be through wire transfer.

2.2.20 For payment to Foreign suppliers: L/C shall be established in favour of the supplier in a bank in his country, for an appropriate amount (in line with Payment Terms). L/C shall be opened within 45 days of receipt of acknowledgement of LOA/PO and receipt of CPBG (wherever applicable).

2.2.21 In case site front is not available up to 6 (Six) months (wherever site work is applicable), after receipt of all supplies at site, the payment against the site work shall be released on submission of additional Bank Guarantee of equivalent amount valid for 1 (One) year or such extended period as may be required.

2.2.22 Wherever the payment provision is there against ABG, and Vendor is not seeking payment against these milestones, those milestone payments shall be released along with milestone of despatch/ receipt of material at site without seeking any ABG for earlier milestone payment.

2.2.23 Payment against Billing Breakup shall be made upto 90% of the payment due against dispatch. Beyond that, payment shall be made only after completion of the entire supply. Maximum number of invoice, for payment purpose, against a billing breakup shall be fixed.

2.2.24 Uploading of and Tracking of invoice in CPCL invoice portal:

Suppliers and Service Providers shall upload the scan copy of invoices and other documents as required in the Purchase Order/Work Order on the CPCL Invoice Portal (https://pms.cpcl.co.in/cpclbts/) to enable CPCL to speed up and cut down on the processing and payment cycle. Post uploading, all correspondence between CPCL and Suppliers/Service Providers will be through the portal. Suppliers/Service providers can track the status of the uploaded invoices through the portal.

3.0 SPARES

3.1 Quoted prices shall be inclusive of all Mandatory Spares and Commissioning spares and special tools and tackles, as per the technical specifications and scope provided in MR.

3.2 All spares quoted by suppliers against the specified MR requirement will be evaluated and recommended for ordering with the main equipment except recommended two years operation and maintenance spares. In case a supplier does not quote for all spares, they shall absorb the price of spares and confirm the complete scope of MR in quoted price.

3.3 Commissioning spares and special tools and tackles, if required and mentioned in MR, will be considered for price evaluation and recommended for ordering as quoted by Suppliers. However, Supplier shall be fully responsible and no additional cost shall be paid by CPCL in case commissioning spares/ special tools & tackles over & above those quoted by supplier are required.

3.4 Bidders are requested to quote for their recommended two years operation and maintenance spares but the same will not be considered for price evaluation. The prices quoted for recommended two years operation and maintenance spares shall remain valid upto 2 years validity after date of commissioning of the equipment. Order of recommended two years operation and maintenance spares may be placed by CPCL separately, if required. Spares and its quantity shall be decided by CPCL at the time of Order for two years operation and maintenance spares.

4.0 TESTING REQUIREMENT

4.1 The quoted Prices shall be inclusive of all testing requirement as per Material Requisition.

5.0 INSPECTION

5.1 All Material (Both Indigenous and Imported), including items sourced from foreign country but excluding Chemicals & Catalyst, shall be inspected by any Third Party Inspection Agency (TPIA) out of the following:

i) TÜV RHEINLAND (INDIA) PVT.LTD.
ii) LRQA INSPECTION SERVICES INDIA LLP
iii) INDUSTRIAL INSPECTION & VERIFICATION SERVICES (INDIA) PVT. LTD.
iv) PROJECTS AND DEVELOPMENT INDIA LIMITED
v) EDLIPSE ENGINEERING GLOBAL PRIVATE LIMITED
vi) APAVE TIV INDIA PVT. LTD.
vii) INTERNATIONAL CERTIFICATION SERVICES PVT. LTD
viii) SGS INDIA PRIVATE LIMITED

ix) MEENAAR GLOBAL CONSULTANTS LLP
x) BUREAU VERITAS (INDIA) PVT. LTD.
xi) TQ CERT SERVICES PVT. LTD.
xii) TUV SUD SOUTH ASIA PVT. LTD.
xiii) TUV INDIA PRIVATE LIMITED
xiv) CERTIFICATION ENGINEERS INTERNATIONAL LTD.
xv) IRCLASS SYSTEMS AND SOLUTIONS PRIVATE LIMITED
xvi) INTERTEK INDIA PVT. LTD.

The validity of the above TPI agencies are up to 30.04.2029.

5.2 Bidder's quoted price shall be inclusive of TPI charges.

5.3 The price quoted by the bidder shall be inclusive of all IBR/IGC/NACE charges (if applicable) and all statutory certification charges PESO/CCOE etc. (if applicable) as required in the Material Requisition.

5.4 Inspection shall be as per following:

5.4.1 QAP will be approved by TPIA incorporating requirements of Inspection and Test Plan, Technical specifications provided with Material / Purchase Requisition.

5.4.2 The Inspection release note issued by the Third Party Inspection agency shall clearly stipulate that materials/equipment have been inspected as per Code-3 approved drawings and approved QAP.

5.4.3 Certification of final documents, wherever applicable, for release of final payment, shall be carried out by Inspection Authority issuing the final inspection certificate.

5.4.4 TPIA shall Review/ Approve SUPPLIER"s various procedures like NDT Procedure, Performance Test procedure, Pressure test procedure, PMI procedure and PWHT procedure etc. prior to start of inspection activities.

5.4.5 TPIA shall carry out inspection of all Project items as per the following documents:

a) Purchase order placed on SUPPLIER
b) Purchase Requisition issued to the SUPPLIER
c) Approved Drawings by EIL / CPCL
d) Approved QAP/ ITP
e) Approved Supplier Documents
f) Deviations if any duly approved by EIL/CPCL
g) Applicable Codes & Standards
h) Procedures approved by TPIA and accepted by EIL / CPCL

5.4.6 TPIA shall carry out physical verification of spares for quantity, quality and compliance to requirements.

5.4.7 EIL/CPCL reserves the right to visit supplier/ sub vendor works at any stage of execution.

5.4.8 All packing list with the equipment, i.e. manuals, drawings, equipment catalogues, etc. shall be certified by TPIA Inspection engineer as packed. TPIA shall verify Packing as per PMS (Packing & Marking Specifications) attached with PR. Special attention shall be paid to sea worthy packing for all goods accordingly. Packing inspection report shall also be furnished by TPIA.

5.4.9 TPIA shall verify availability of statutory approvals for use of Electrical / Instrumentation items for related hazardous area classification defined in the documents. TPIA shall issue IRN after verification of all such statutory approved documents.

5.4.10 TPIA shall sign "As-Built" drawings and Certify completeness of Final Documents by ensuring required numbers of data folders including soft copies submitted by the SUPPLIER to CPCL /

EIL as per approved SUPPLIER data/Document control Index (DCI).

5.4.11 During course of order execution, if any TPIA is blacklisted / holiday listed from the list provided in the tender. The supplier shall not engage the TPIA upon receipt of written instruction from CPCL/EIL.

6.0 TRANSPORTATION

6.1 Indigenous Bidders

6.1.1 Indigenous suppliers are required to dispatch the goods on FOT/ FOR dispatch point basis / FOT Site basis, as mentioned in Bid Data Sheet (BDS), through any of the registered Transporters. Any deviation shall be subject to prior approval of Owner/EIL.

6.1.2 Transportation charges for the items may be sought separately as per requirement / scope of MR / RFQ.

6.1.3 It shall be responsibility of the seller to send intimation immediately on dispatch of the material so that necessary arrangements can be made at site. Delays on account of the same shall solely be attributable to the Supplier

6.2 Foreign Bidder

6.2.1 Ocean transportation from FOB Major International Gateway Seaport of Exit to Port of Entry in India shall be arranged by CPCL (for FOB Orders) through their nominated freight forwarder. Similarly, for air freighted items on FCA International Airport of Exit basis, arrangement shall be done by CPCL through their nominated consolidation agent.

6.2.2 Supplier shall arrange handing over the material to CPCL designated freight forwarder/Consolidation agent at the designated port of exit.

6.2.3 Following shall be the seaport / Airport of Entry in India:

Chennai / Kamarajar / Kattupalli Port(s) in case of Ocean Transport
AND
Chennai Airport in case of Air Transport

However, for ODC Consignment, Chennai seaport shall be considered.

6.2.4 Bidder shall also be required to quote the ocean/ Air freight charges, as applicable from nearest Seaport/Airport of Exit to above Seaport/Airport of Entry in India. The quote for ocean freight up to above mentioned Seaport of entry is for comparison of bids and will not in any way, limit the Owner's right to contract on different terms.

6.2.5 List of major international seaports / airports for containerized cargo is enclosed as Annexure to SPC. Bidders shall have to choose from the nearest / most convenient seaport to them and indicate in their offer. No other port shall be acceptable. However, for despatch from North America / Africa / Australia / South America, vendors shall be allowed to choose any major international gateway seaport of exit in the respective area.

6.2.6 Owner reserves the right to place the order on any of the specified price basis (FOB or CFR). In case of award, initially the Fax of Acceptance (FOA) shall be placed on FOB basis and Owner / EIL reserves the right to convert the same to CFR basis at a later date. Therefore, bidder's quoted ocean freight charges should be valid for the entire duration of contract.

6.2.7 In case of CFR contract, all shipments of break bulk material and containerized cargo shall be made by suitable vessels certified for sea worthiness and complying with National/ International safety regulations and age of the vessel being less than 25 years. Supplier to coordinate with CPCL nominated Customs House Agent for smooth discharge / delivery of the cargo at the Port of arrival. Supplier shall inform 07 days prior to the expected date of arrival of vessel in port of entry in India. Detention & Demurrages, if any, due to early arrival of vessel, shall be to supplier's account.

6.2.8 Ocean freight for Break Bulk cargo delivery shall be on FLT (Full Liner Terms), i.e., all

charges at destination port shall be borne by the supplier till the cargo is unloaded on to the wharf /vehicle as the case may be. Trans-shipment shall be permitted only for containerized Cargo and shall not be permitted on any other type of cargo including break bulk.

6.2.9 The BIDDER will be required to furnish firm cargo details containing weight, dimensions, no. of packages, no. & type of containers required & Port of shipment. In case the actual cargo dispatched exceeds the cargo details furnished, the extra freight incurred by OWNER will be to SUPPLIER"s account. However, SUPPLIER shall furnish the shipping details viz. Net/ Gross weight, Volume of consignment, number of packages with dimensions at least two month before the expiry of validity of Ocean Freight. In the event of any delay by SUPPLIER, the quoted Ocean Freight charges shall remain valid till the shipment. CPCL reserves the option to ask the bidder to arrange shipment at the quoted freight rates.

6.2.10 In case of break-bulk consignment, bidders shall be allowed to quote from the nearest major International Gateway Seaport of Exit.

6.2.11 The freight charges shall be quoted by bidder considering inclusion of the following:

- 21 Days Free Period for Containers
- Destination Charges like Terminal Handling Charges, Delivery Order Charges.

7.0 COMPREHENSIVE MARINE-CUM-ERECTION INSURANCE

7.1 Comprehensive Insurance (Transit/ Marine cum storage, erection, till handing over of equipment) shall be arranged by CPCL. Therefore, the same shall not be included in the quoted prices. However, all transit insurance up to FOB / FCA International Sea / Air Port of Exit must be included in the quoted Price.

8.0 CURRENCY OF BID

8.1 FOREIGN SUPPLIERS

8.1.1 Foreign SUPPLIERs are required to submit prices in single currency out of **US Dollar / Euro / JPY / SGD /GBP.** In case a foreign SUPPLIER is sourcing part of equipment/ services from India, prices of such equipment/ services shall be quoted only in **Indian Rupees** only.

8.1.2 Change in currency once quoted will not be allowed.

8.1.3 Owner / EIL shall not compensate due to any exchange rate fluctuation.

8.2 INDIGENOUS SUPPLIERS

8.2.1 Indigenous SUPPLIERs are required to submit price in **Indian Rupees** only.

9.0 SINGLE POINT RESPONSIBILITY OF THE SUPPLIER

9.1 FOREIGN SUPPLIERS

9.1.1 In case of any Indian supply or services quoted by foreign supplier in their bid (Applicable in case of global enquiry), the same shall be quoted only in INR clearly indicating in the offer the name and contact details of the Indian party on whom the order has to be placed.

For this purpose, the Indian sub-supplier's name, material/services and corresponding price in Indian Rupees, shall be clearly spelt out in the bid of foreign SUPPLIER. In case a Foreign supplier sources supplies/services within India, the applicable terms & conditions shall be as per Annexure to SPC. However, Separate Placement of Order for Indian supply or services shall be subject to EIL/CPCL's acceptance.

In case, a foreign SUPPLIER order involving part supply/site work/site services in INR separate orders shall be issued with following indicative interlinking clause:

"In addition to this purchase order No. ……, dt…… on M/s ……., M/s……, is aware that CPCL has awarded another purchase order No…….. dt….. for …… , on M/s ……, these two orders are intended for the Completion of complete scope of work as per Purchase Requisition Document, as an integrated package. M/s …….. (Principal bidder)

assumes complete responsibility for smooth execution, delivery, performance guarantee, back-up guarantees, project management, and Price reduction for delay, liabilities and responsibilities arising out of both the orders in such a manner appropriate to the Completion of complete order as per Purchase Requisition Document. Both these orders are linked to each other and that the splitting of order into two orders, will not lead to any kind of understanding that these two orders, are separate from each other. M/s …. (Principal Bidder) will be responsible for the technical back-up guarantee for both the orders as well as performance bank guarantee. In case of delay in execution of either of the two orders the price reduction for delay in delivery shall be computed on the total order value of both the orders.

9.2 INDIGENOUS SUPPLIERS

9.2.1 In case of MRs containing supplies as well as site work/ services, a single order, covering the supply part as well as site work/ services, shall be placed.

10.0 GUARANTEE/WARRANTY PERIOD

10.1 Guarantee period shall be as per Defect Liability Period of General Purchase Condition (GPC) and shall be as per clause no. “14. Warranty of Title” & “16. Acceptance of Materials & Guarantees” of GPC.

11.0 CONTRACT CUM PERFORMANCE BANK GUARANTEE (CPBG)

11.1 **For cases of procurement of Bulk Materials**, vendor shall furnish PBG equivalent to 5% of the order value at the time of dispatch and the same shall be valid till delivery period plus six months. In case a vendor does not furnish PBG, prorata payment of 5% shall be deducted from vendor’s invoice and retained for a period of final delivery date plus 6 months.

11.2 **For cases of procurement of Other than Bulk Materials, Equipments and Packages**, Performance Bank Guarantee shall be furnished for 5% of basic order value (for Supply plus Site Work, if applicable) unless otherwise specified in the tender documents within one month from the date of placement of order and the same shall be valid till the end of warranty/guaranty period plus three months claim period.

11.3 MSEs owned by SC/ST and Women Entrepreneurs shall be required to submit 50% of PBG amount as stipulated under 11.1 and 11.2 above.

11.4 Total Order Value shall mean supply price (FOT dispatch point price for Indian bidder and FOB/FCA price for foreign bidders) plus site work (wherever applicable) but excluding taxes & duties, transportation, supervision services and any other charges.

11.5 Indemnity Bond/Corporate Guarantee in lieu of CPBG shall not be accepted.

11.6 The proceeds of the CPBG shall be appropriated by the Owner as compensation for any loss resulting from the Seller’s failure to complete his obligations under the Contract without prejudice to any of the rights or remedies the OWNER may be entitled to as per terms and conditions of CONTRACT.

11.7 This CPBG shall also govern the successful performance of Goods and Services during the entire period of Contractual Warrantee/Guarantee/Defect Liability.

11.8 The CPBG shall be denominated in the currency of order only. Bill selling foreign exchange rate of State Bank of India, New Delhi on the day of price bid opening shall be considered for currency conversion. In case the rate is not available as on the day of price bid opening, the latest available rate prior to the day of price bid opening shall be considered.

11.9 The Contract cum Performance Bank Guarantee shall be valid for the duration of 3 months beyond the expiry of Warranty/Guarantee/Defect Liability period. The Bank Guarantee will be discharged by OWNER not later than 2 months from the date of expiration of the Seller’s entire obligations, including any warrantee obligations, under the CONTRACT.

11.10 It will be the responsibility of the Seller to keep the Bank Guarantee fully subscribed. Any

shortfall in the value of the Bank Guarantee as a result of encashment by the Owner either in full or in part in terms of Seller's Performance shall be replenished by the Seller within two weeks thereof.

11.11 Failure of the successful bidder to submit the above-mentioned **Contract cum Performance Bank Guarantee** shall constitute sufficient ground for the annulment of the award.

11.12 Supplier shall submit CPBG in the form of SFMS based **Contract Performance Bank Guarantee / Electronic Bank Guarantee (e-BG)/ Demand Draft**.

12.0 TAXES/DUTIES

12.1 Taxes and Duties shall be as per Annexure to this SPC (Special Conditions of Purchase).

13.0 FIRM PRICES

13.1 The prices shall be firm and fixed and not subject to any variation except as indicate in Bid Data Sheet and Annexure to SPC.

14.0 PRICE REDUCTION SCHEDULE

14.1 This clause shall be as per Cl. 12 of CPCL's General Purchase Condition.

14.2 Wherever the supply period and site work are indicated as two separate periods and notice of site readiness is required for site work, intervening period, if any, shall be excluded, while calculating the price reduction for delay.

14.3 For the purpose of application of Price Reduction Schedule, part thereof shall be considered as one full week. Further, Total Order Value for the purpose of PRS shall mean supply price (FOT dispatch point price for Indian bidder and FOB/FCA price for foreign bidders) plus site work (wherever applicable) considering any amendments/ change order, except cases of Short Closure/ Order cancellation. However, in case of Short Closure/ Order cancellation, the order Value shall be executed/ supplied value plus the actual cost of work done at the risk & cost of the supplier plus Owner/EIL overhead charges.

14.4 In case of delay in delivery, bidder shall submit credit note/adjustment in GST invoice for PRS, if applicable, on account of delay in delivery.

15.0 LOADING/REJECTION CRITERIA

15.1 Bids are invited on Zero Deviation bidding basis. Deviation to terms & conditions of RFQ shall lead to rejection of offer.

16.0 INVOICING INSTRUCTIONS

16.1 For imported goods, the Invoices shall be billed in the name of CPCL. Consignee on the invoices and bill of lading/ Airway Bill shall be in the name of CPCL.

16.2 For Indigenous Goods, the invoices shall be billed to CPCL. Consignee in the tax invoices and L/R shall be in the name of CPCL.

16.3 CGST & SGST or IGST shall be released only on receipt of GST Invoice containing the following details:-

(a) Name, address and GSTIN of the supplier;
(b) A serial number of the invoice;
(c) Date of issue;
(d) Name, address and GSTIN or UIN, if registered of the recipient;
(e) Name and address of the recipient and the address of the delivery, along with the State and its code,
(f) HSN Codes or Accounting Code of services;
(g) Description of goods or services;

(h) Total value of supply of goods or services;

(i) Taxable value of supply of goods or services taking into discount or abatement if any;

(j) Rate of tax (Central Tax, State Tax, Integrated Tax (for inter-state supply), Union Territory Tax or cess);

(k) Amount of tax charged in respect of taxable services (Central Tax, State Tax, Integrated Tax (for inter-state supply), Union Territory Tax or cess);

(l) Place of supply along with the name of State, in case of supply in the course of inter-state trade or commerce;

(m) Address of the dispatch point where the same is different from the place of supplier;

(n) Signature or digital signature of the supplier or his authorized representative;

(o) It's the obligation on the part of Bidder / Vendor / Contractor / Consultant to discharge their liability by payment of GST to Government of India in cash OR utilization of Input Tax credit in respect of such supply of services through GST Invoice under this Contract, so that Owner will avail Input Tax credit on such supply. Also it shall be the obligation on the part of the bidder / vendor to include the tax invoices in the return ensuring matching of the tax amount and to ensure timely filing of the return in the GSTN portal. In the event that the input tax credit of the GST charged by the Bidder / Vendor/ Contractor / Consultant is denied by the tax authorities to Owner due to reasons attributable to Bidder / Vendor, Owner shall be entitled to recover such amount from the Bidder / Vendor/ Contractor / Consultant by way of adjustment from the next invoice or from Bank Guarantee. In addition to the amount of GST, Owner shall also be entitled to recover interest and penalty, in case same is imposed by the tax authorities on Owner.

(p) Any deductions in lieu of deviations to the terms and conditions of the purchase order shall be suitably incorporated by the vendor in the invoice presented, else the vendor shall have to necessarily provide a credit note upon intimation by CPCL within a stipulated time period and any delay in payment due to the delay in providing credit note shall be to the vendor's account.

(q) Above details are only for illustration purposes and it is obligatory for the bidder to issue GST Invoice as per provisions of GST Act and Rules made there under

16.4 DISPATCH NUMBER

In addition to the details furnished in the invoice, supplier shall generate DISPATCH NUMBER (DN) against each invoice. The DN shall be 3 digit continuous number starting from 001 suffixed to purchase Order Number and shall be unique to each invoice. Same shall also be printed on each consignment / box for easy co-relation. DN shall be quoted in all further communications viz dispatch detail to inspection, correspondence with site as well as for payment purpose. Mentioning the DN on each invoice shall be mandatory.

17.0 REPEAT ORDER (For Applicability of this clause, Refer BDS)

17.1 The purchase reserves the right to place Repeat Order (RO) for not exceeding 100% of the ordered quantities within final delivery date (or extended delivery date for reasons not attributable to supplier) or 6 months from the date of placement of order, whichever is later.

17.2 Non-Acceptance of Repeat Order conditions by bidder shall not be the reason for rejection of bid.

18.0 FRAUDULENT PRACTICES

18.1 The Owner requires that Bidders/ Vendors/ Contractors observe the highest standard of ethics during the award/ execution of Contract. "Fraudulent Practice" means a misrepresentation of facts in order to influence the award of a Contract to the detriment of the Owner, and includes collusive practice among bidders (prior to or after bid submission) designed to establish bid prices at artificial non-competitive levels and to deprive the owner of the benefits of free and open competition.

18.2 The owner will reject a proposal for award if it determines that the bidder recommended for award has engaged in fraudulent practices in competing for the Contract in question.

18.3 Bidder/Seller is required to furnish the complete and correct information/ documents required for evaluation of their bids. If the information/ documents forming basis of evaluation is found to be false/ forged, the same shall be considered adequate ground for rejection of bids and forfeiture of Earnest Money Deposit.

18.4 In case, the information/ document furnished by the Seller/ Vendor/ Contractor forming basis of evaluation of his bid is found to be false/ forged after the award of the contract, Owner shall have full right to terminate the contract and get the remaining job executed at the risk & cost of such Seller/ Vendor/ Contractor without any prejudice to other rights available to Owner under the contract such as forfeiture of CPBG/ Security Deposit, withholding of payment etc.

18.5 In case, this issue of submission of false documents comes to the notice after execution of work, Owner shall have full right to forfeit any amount due to the Seller/ Vendor/ Contractor along with forfeiture of CPBG/ Security Deposit furnished by the Seller/vendor/ contractor.

18.6 Further, such Seller/ vendor/ contractor shall be put on Blacklist/ Holiday/ Suspension / Banning list / Negative List of owner debarring them from future business with Owner & EIL for a time period, as per the prevailing policy in EIL & Owner.

19.0 ACCEPTANCE OF BANK GUARANTEE (BG)

The performance bank guarantee(s) shall be as per the Proforma appended with GPC. All bank guarantees should be submitted by Seller's bankers directly to the Owner. Seller shall enclose copy of bank guarantee(s) along with the invoice. Bank Guarantee(s) shall be submitted as per the following details:-

19.1 Selection of Bank:-

19.1.1 BG upto INR 20 Million can be accepted if it is issued by an Indian branch of any scheduled bank appearing in the Second Schedule to the RBI Act, 1934.

19.1.2 BG of above INR 20 Million can be accepted if it is issued by an Indian branch of:

(i) Any Nationalized / PSU bank appearing in the Second Schedule to the RBI Act, 1934

Or

(ii) Any scheduled bank (other than a Nationalized Bank / PSU bank) having at least Desired Credit Rating at the time of acceptance of BG

Desired credit rating is defined as under:

Bank	If the tenor of BG is more than1 year, credit rating of	If the tenor of BG is upto 1 year, credit rating of
In case of foreign banks	A' of Moody's or equivalent	'**P-1**' of Moody's or equivalent i.e. highest short term rating
In case of Indian banks	'**AA**' of CRISIL or equivalent	'**A 1 +**' of CRISIL or equivalent i.e. highest short term rating

19.1.3 Apart from above, BG, irrespective of its amount, issued by any other bank including but not limited to non-scheduled banks, foreign branches of scheduled banks and foreign branches of foreign banks, can be accepted provided such BG is confirmed / counter guaranteed by any bank mentioned above at 19.1.2.

19.2 Credit rating

The vendor shall note that, in case of acceptance of BG issued or confirmed / counter guaranteed by a bank mentioned at para 19.1.2 (ii), if the credit rating of such bank falls below the Credit Rating mentioned under clause 19.1.2 (ii) during the validity period of BG, the Contractor shall either submit a fresh BG or get the existing BG confirmed/counter guaranteed, at its own cost, through a bank mentioned above at 19.1.2 (having at least desired Credit Rating as mentioned above, if applicable). In case of non-submission of bank guarantee(s), without prejudice to any other right or remedy available to the owner, the owner shall be entitled to en-cash the bank guarantee(s).

19.3 The vendor at the request of the owner shall extend the validity of the Bank Guarantee(s) for such further period(s) as may be required, failing which, without prejudice to any other right or remedy available to the owner, the owner shall be entitled to en-cash the bank guarantee(s) .

19.4 The vendor to ensure the validity of all bank guarantee(s) as stipulated else-where in the bidding documents/contract and no payments shall be released to the contractor, if the validity of the bank guarantee(s) is less than 30 days unless otherwise specifically intimated to the contractor. “

Beneficiary Bank details are as follow, vendor can use these details for issuance of EMD /ABG / PBG (if any).

S.NO.	PARTICULARS	DETAILS
1	Name of Beneficiary Bank	STATE BANK OF INDIA Address: III FLOOR SIGAPPI ACHI BUILDING, 18/3 RUKMINI LAKSHMIPATHY SALAI, EGMORE, CHENNAI 600008
2	Branch Name / CODE	CAG CHENNAI
3	Name of Beneficiary	CHENNAIN PETROLEUM CORPORATION LIMITED
4	SWIFT / RTGS / ISFC / MICR CODE	RTGS / ISFC CODE: SBIN0009999 SWIFT CODE: SBININBB174 MICR Code: 600002098

20.0 VERIFICATION OF AUTHENTICITY OF DOCUMENTS:-

20.1 Verification of authenticity of BQC documents shall be done for the successful bidders only in case of all open Tenders/ Enquiries.

Submission of authentic documents is the prime responsibility of the bidder. Wherever CPCL has concern or apprehension regarding the authenticity/ correctness of any document, CPCL reserves the right to get the documents verified from issuing authority/any relevant source. If documents (part or full) are found forged, such offers will be summarily rejected , EMD will be forfeited, orders if any placed against subject tender will be terminated and may be debarred from future tenders. For the purpose of verification bidders shall submit complete client details with names, address, phone number, e-mail ID etc.

21.0 ORDER OF PRECEDENCE

21.1 The documents forming the Purchase Order are to be taken as mutually explanatory of one another. If there is an ambiguity or discrepancy in the Term & Conditions defined in the various documents, the order of Precedence shall be as follows:

i Purchase Order and its enclosures along with Purchase Requisition
ii Amendment to Bidding Document, if any
iii Invitation for Bids (IFB)/ Request for Quotation (RFQ)
iv Bid Data Sheet

v Job Specifications
vi Drawings
vii Technical specifications
viii Special Purchase Conditions (SPC)
ix Instructions to Bidders (ITB)
x General Purchase Conditions (GPC) & Packing, Marking, Shipping and Documentation Specification
xi Other Documents

A variation or amendment issued after the execution of the formal contract shall take precedence over respective clauses of the formal contract and its Annexure.

22.0 CONCILIATION AND ARBITRATION

22.1 Clause No. 34.3 of GPC pertaining to Conciliation stands modified to the following extent:

22.1.1 Conciliation Rules shall be as per Chennai Petroleum Corporation Limited Conciliation Rules available in the CPCL's website www.cpcl.co.in.

22.2 Clause No. 34.1 and Clause No. 34.2 of GPC pertaining to Arbitration stands modified to the following extent:

22.2.1 Any dispute or difference or difference whatsoever arising between the parties and of or relating to the construction, interpretation, application, meaning, scope, operation or effect of this contract or the validity or the breach thereof, shall be settled by arbitration in accordance with the “SCOPE Forum of Arbitration Rules” framed by the Standing Conference of Public Enterprises (SCOPE), and the award made in pursuance there of shall be final and binding on the parties.

22.2.2 **Settlement of disputes between two PSUs or between a PSU and Govt. Department:**

a) In the event of any dispute or difference relating to the interpretation and application of the provisions of commercial contract(s) between Central Public Sector Enterprises (CPSEs) / Port Trusts inter se and also between CPSEs and Government Departments/Organizations (excluding disputes relating to Railways, Income Tax, Customs & Excise Departments), such dispute or difference shall be taken up by either party for its resolution through AMRCD as mentioned in DPE OM No. 05/0003/2019-FTS-10937 dated 14th December, 2022 and the decision of AMRCD on the said dispute will be binding on both the parties.

b) Further, vide DPE OM no. DPE-05/0002/2023-AMRCD, dated 25.07.2024, it is clarified that the interpretation and application of the provisions of commercial contract(s) in case of dispute/s between the following Organizations/ Departments shall be taken up under AMRCD only:

i) CPSE and CPSE;
ii) CPSE and Ministry / Department of Central Government;
iii) CPSE and Public Authority/ Organization or University under the Central Government;
iv) CPSE and State Governments/State PS Us/· Public authority/ Organization or University under any State Government

22.2.3 **Settlement of disputes with other than PSUs and Govt. Departments:**

Any dispute or differences whatsoever arising between the Parties and of or relating to the construction, interpretation, application meaning, scope, operation or effect of this Agreement or the validity or the breach thereof, shall be settled through mutual discussion. In case Parties are unable to resolve the dispute through mutual discussion, either Party may issue a notice for referring the same for arbitration in accordance with the SCOPE Forum of Arbitration Rules framed by the Standing Conference of Public Enterprises (SCOPE) and the

award made in pursuance thereof shall be final and binding on the parties. Notwithstanding the amount of claim, all the disputes or differences shall be settled by a Sole Arbitrator in accordance with SCOPE Forum of Arbitration Rules. The venue of Arbitration shall be Chennai, Tamilnadu, India and the Arbitration proceedings shall be conducted and the award shall be rendered in English. The Arbitrator shall give a speaking award. The Arbitrator shall render the award within six months of the commencement of the Arbitral Proceedings which period may be enlarged by consent of both the Parties. The Cost of Arbitration Proceedings shall be borne by both the Parties equally. Interest, if awarded by Arbitrators, shall be at a rate not exceeding the cash credit rate prevailing on the date of the award. This Agreement, and the rights and obligations of the Parties, shall remain in full force and effect pending the award in any arbitration proceedings. For the purposes of this clause, the term 'dispute' shall include a demand or difference of any kind whatsoever, arising out of this Agreement and respecting the performance thereof, whether during the Term of this Agreement including extensions, if any, or after completion, and whether before or after termination, abandonment or breach of the Agreement.

22.3 Notwithstanding the existence of any dispute or arbitration in terms hereof or otherwise, the Supplier shall continue and be bound to continue and perform the Works to completion in all respects according to the Contract (unless the Contract or Works be determined by CPCL) and the Supplier shall remain liable and bound in all respects under the Contract.

23.0 SUB-CONTRACTS / ASSIGNMENT

23.1 The provisions of Cl. No. GPC-10 is being modified to the extent below:

"The supplier/vendor shall not assign, novate or otherwise transfer any of the rights and obligations under the Contract finalised out of this tender in any manner whatsoever. However, OWNER shall have the right to assign, novate or transfer any of its rights or obligations under the Contract finalised out of this Tender to an affiliate or any other Party decided by OWNER. Any such assignment shall be informed in writing to the CONTRACTOR. In such case, either Party shall perform their respective obligations agreed in the Contract finalised out of this Tender.

All assignments shall be in terms so as not to prejudice the confidentiality obligations of the parties."

24.0 TERMINATION

24.1 Following provisions are added as cl. No. 26.6 of GPC:

"The PURCHASER may, by written notice of 10 days sent to the SELLER, Terminate the CONTRACT, in whole or part, at any time for his convenience. The notice of termination shall specify that termination is for the PURCHASER's convenience, the extent to which performance of work under the CONTRACT is terminated and the date upon which such termination becomes effective.

The GOODS that are complete and ready for shipment within 30 days after the SELLER's receipt of notice of termination shall be purchased by the PURCHASER at the CONTRACT terms and prices.

For the remaining GOODS, the PURCHASER may opt:

a) to have any portion completed and delivered at the CONTRACT terms and prices, and /or

b) to terminate the remainder and pay to the SELLER an agreed amount for partially completed GOODS and for materials and parts previously procured by the SELLER.

25.0 TRADE RECEIVABLE e-DISCOUNTING SYSTEM (TReDS):

25.1 TReDS is an online electronic institutional mechanism for facilitating the financing of trade receivables of MSEs through multiple financiers. The TReDS Platform will enable discounting

of invoices/bills of exchange of MSMEs Sellers/ Contractors against large Corporates including Govt. Departments and PSUs, through an auction mechanism, to ensure prompt realization of trade receivables at competitive market rates.

25.2 CPCL has registered itself on TReDS Platform. You are requested to kindly on-board/ register on TReDS Platform at the earliest and confirm to us. Details of service providers and brief about TReDS is available in our website www.cpcl.co.in under "Quick links" of "Tenders" tab.

26.0 PROMOTION AND DEVELOPMENT OF MSE SUPPLIER

26.1 Owner/EIL is committed to promotion and development of micro and small enterprises. Accordingly, Suppliers are also encouraged to promote the same by considering MSE sub-suppliers/ sub-contractors to the extent possible under the Orders/Contracts awarded on them by Owner/EIL. All efforts shall be made by Suppliers to encourage procurement thru MSEs (holding valid certificate) through their Sub orders/ Sub contract procurement. The successful bidder shall provide the following only for the purpose of reporting.

i) Monthly Reports for purchase of material/services/ sub-contracts (from MSE vendors) shall be submitted by bidder to .....(Name of CPSU).

i) The report shall be signed by authorised person of contractor (Successful Bidder).

ii) Report shall be submitted in the following format.

Sl. No.	Sub-Supplier/ Subcontractor	Work Order no.& date	Item Description	Amount (In Rs)	Category of MSE Gen/ SC/ ST	MSME Status Micro/ Small with code	Remarks

27.0 CONFIDENTIALITY CLAUSE W.R.T. COMPLIANCE WITH CPCL's INSIDER TRADER CODE

27.1 During the contract / engagement with CPCL, the organization, its employees, agents, other contractors, sub-contractors and their employees, etc., may be exposed to certain Unpublished Price Sensitive Information (UPSI), that is not generally available to the public or in public domain and which upon becoming generally available is likely to materially affect the price of the securities of CPCL traded in Stock Exchanges. Such UPSI shall be held by the organization, its employees, agents, other contractors, sub-contractors and their employees etc., in strictest confidence and shall not be disclosed to any other party except on need to know basis and with the prior consent of CPCL.

27.2 It is hereby further cautioned that, the organization, its employees, agents, other contractors, sub-contractors, their employees and immediate relatives thereof, should not trade in the securities of CPCL while in possession of such UPSI or communicate such UPSI to any person except in furtherance of legitimate purposes, performance of duty or discharge of legal obligations. Any trades in securities of CPCL, while in possession of any UPSI would be presumed to have been motivated by the knowledge and awareness of the UPSI and liable for penal action under this contract / engagement as well as under SEBI (Prohibition and Insider Trading) Regulations, 2015. It is therefore advised to familiarize with the "Code for Prevention of Insider Trading in the Securities of CPCL", copy of which is hosted on the website https://cpcl.co.in/company/overview/our-policies/.

Accordingly, it must be ensured that any UPSI, whenever received from the CPCL,

(a) shall be used solely for the purpose for which it is being disclosed;

(b) shall be preserved and the secrecy of such information shall be maintained;

(c) shall not be disclosed to any third party;

(d) shall be kept securely and properly protected against theft, damage, loss and unauthorized access (including access by electronic means) by deploying means similar to those being used to secure their own confidential information;

(e) to notify CPCL immediately upon becoming aware that any of the confidential information has been disclosed to or obtained by a third party.

27.3 The organization shall undertake that its employees & their immediate relatives, agents, sub-contractors and any other person associated with said contract / engagement does not violate any of the provision of the SEBI (Prohibition of Insider Trading) Regulation, 2015. Any violation shall be considered as breach of terms of contract / engagement and CPCL shall take necessary action for such breach in addition to reporting it to SEBI for taking necessary action under SEBI (Prohibition of Insider Trading) Regulations, 2015.

28.0 TERMS & CONDITIONS FOR SUPERVISION OF ERECTION, TESTING & COMMISSIONING / TRAINING AT SITE

28.1 Terms & conditions for supervision of erection, testing & commissioning / training at site shall be as per attached **Annexure to SPC**.

29.0 TERMS & CONDITIONS FOR SITE WORK

29.1 Terms & conditions for installation/ site work shall be as per attached **Annexure to SPC**.

30.0 TERMS & CONDITIONS FOR ANNUAL MAINTENANCE CONTRACT (AMC)

30.1 Terms & conditions for Annual Maintenance Contract (AMC) shall be as per attached **Annexure to SPC**.

31.0 TERMS & CONDITIONS FOR TRAINING OF OWNER PERSONNEL AT VENDOR'S/OEM'S WORKS / TRAINING FACILITY

31.1 Terms & conditions for training of owner personnel as per MR at vendor's works/ training facility shall be as per attached **Annexure to SPC**.

Foreign bidder may source components / sub-supplies and services from India, if allowed as per Bidding Document / Material Requisition. For such sub-supplies / services, Commercial Terms & Conditions of the RFQ shall stand modified to the following extent:

1. PRICING:

1.1 Unit and total price on FOT Despatch Point basis including packing and forwarding charges.

1.2 Details of Taxes and duties (rates) payable extra on the finished goods, applicable on Indian sub-suppliers.

1.3 Separate and clear break-up of charges for inland transportation is to be quoted by the bidder.

1.4 Materials if ordered against the RFQ are required to be dispatched on door delivery basis.

1.5 Transit insurance of Indian sub supplies shall be borne and arranged by the PLL.

2. CURRENCY OF QUOTE:

The quoted price of sub-supplies/services shall be in Indian Rupees only.

3. AWARD OF ORDER:

Foreign Principal shall be the single point responsible SUPPLIER for entire scope of MR.

However separate Purchase Order with interlinking clause as per SPC may be agreed subject to EIL/ CPCL's acceptance.

4. DELIVERY:

The delivery period for sub-supplies shall be on FOT site basis within specified delivery period for foreign bidder as per Terms of RFQ/Bidding document.

5. PRICE REDUCTION SCHEDULE:

PRS shall be applicable in totality on foreign bidder scope + Indian sourced component / site work, for delay in supply of goods/ services either by foreign bidder or Indian counterpart.

6. PAYMENT TERMS:

Payment of Indian sub supplies:

- 85% through E-Payment with taxes and duties will be paid against receipt of Indian sourced goods/material at Project Site.
- 10% against receipt of Indian supplies at site and delivery of main equipment by foreign bidder, as per Terms of bidding document and completion of site work, wherever applicable.
- 5% on submission of Final/As Built Drawings/Documents/Data/Manual in requisite number of copies/sets/CDs as per VDR specified in Purchase Requisition, by foreign bidder (principal).

Payment of Indian services:

- Shall be as per BDS.

Note:

- Payment of Indian sub supplies/Services shall be released directly to Indian counterpart against Invoices raised by Indian counterpart duly certified by foreign bidder (principal).
- In case of Foreign bids with Indian sub supplies, the payment terms of Foreign bidder against shipping documents stands reduced by 5% which shall be released as follows:

  5% payment shall be released thru wire transfer, on delivery of main equipment by foreign principal (as per terms of enquiry document) as well as receipt of Indian supplies at site.

- The payments shall be made after adjustment of Price Reduction Schedule.
- If foreign bidder (principal) delivers all their foreign sourced goods while Indian sourced delivery are yet to be completed and further requests for release of last 10% payment, then this last 10% shall be released to foreign bidder (principal) by Owner against the submission of bank guarantee for this 10% of Total order value valid for 03 months from CDD plus 1 month claim period or till completion of Indian sourced items, whichever is later.

7. CONTRACT PERFORMANCE BANKGUARANTEE:

Foreign bidder (principal) shall submit the Contract performance bank guarantee for 10% of Total Order Value including value of foreign portion and Indian sourced components / site work.

FOR INDIAN BIDDERS

(A) Terms & Conditions for Indigenous Bidders.

Bidder shall quote per diem charges considering the following:

- The per diem rates shall **include** lodging, boarding, transport, medical, all insurance including third party, income tax, all taxes & all other incidental charges.
- The per diem rates shall exclude GST which shall be payable extra
- Per diem rates shall apply from the date the supervisory personnel reach site and upto the day they leave site. No payment shall be made for the transit period.
- Working hours as those prevailing at site – upto eight hours a day, Monday through Saturday and holidays as applicable to site personnel shall apply.
- In case bidder's representative is not called for during Intervening weekly off/Holidays, the same shall be payable at the normal rate of site services. However, in case Seller's representative is called at site during weekly off/Holiday, the same shall be paid @ overtime rate quoted extra by bidder.
- In case bidder's representative takes leave, per diem charges for that day shall not be payable.
- Vendor's supervisory personnel shall observe/abide by site working conditions prevailing in the existing refinery, safety codes and Indian laws.
- A notice period of 15 days will be given for site supervision services.
- No. of mandays as defined elsewhere in the Enquiry Document shall be considered for Evaluation for Supervision Services & Training (at Site) Services.

  However, payment shall be made as per actual number of man-days, unless otherwise specified elsewhere in the bidding document.
- Bidder to ensure compliance to guidelines on PPE (Personal Protective Equipment) & Other Safety Norms.
- Validity period of per diem rate shall be upto the end of guarantee period. Further, in case services for supervision are required after the guarantee period, 8% annual escalation shall be applicable.

(B) Charges for Site Supervision of erection, testing & commissioning:

Sr. No.	Description	Rates in INR
1	Per diem charges for upto 8 hours work on normal working days.	To be filled in price Form (BOQ)
2	Overtime rate/ hour beyond normal 8 hours work on normal working days.	
3	Overtime rate/ hour for work on weekly off days / holidays (as applicable at Project site).	

(C) Training charges

Sr. No	Description	Rates in INR
1	Per diem charges of Training at CPCL site.	To be filled in price Form (BOQ)

FOR FOREIGN BIDDERS

(A) Terms & conditions for Foreign Bidders.

Bidder shall quote per diem charges considering the following:

- The per diem rates shall **include** lodging, boarding, transport, medical, all insurance including third party, withholding tax, Corporate Tax (as may be applicable) etc. & all other incidental charges.
- Taxes at source shall be deducted by CPCL as per prevalent Government policy while making payments against each invoice and tax deduction certificate shall be furnished by CPCL.
- The per diem rates shall exclude GST which shall be payable extra by CPCL.
- Per diem rates shall apply from the date the supervisory personnel reach site and upto the day they leave site. No payment shall be made for the transit period.
- Working hours as those prevailing at site–upto eight hours a day, Monday through Saturday and holidays as applicable to site personnel shall apply.
- In case bidder's representative is not called for during Intervening weekly off/Holidays, the same shall be payable at the normal rate of site services. However, in case Seller's representative is called at site during weekly off/Holiday, the same shall be paid @ overtime rate quoted extra by bidder.
- In case bidder's representative takes leave, per diem charges for that day shall not be payable.
- Vendor's supervisory personnel shall observe/ abide by site working conditions prevailing in the existing refinery, safety codes and Indian laws.
- **Requirement of Employment Visa for Foreign Nationals**
  All foreign nationals coming to India for execution of Projects / Orders will have to apply for Employment Visa only and that grant of Employment Visa would be subject to strict adherence of following norms:
  i) Employment Visa is granted only for the skilled and qualified professional or to a person who is being engaged or appointed by a company, organization, Industry or undertaking etc. in India on contract or employment basis at a senior level, skilled position such as technical expert, senior executive or in managerial position etc.
  ii) Request for employment Visa for jobs for which large number of qualified Indian are available, is not considered.
  iii) Under no circumstances an Employment Visa is granted for routine, ordinary secretarial / clerical jobs.
  iv) Visa should not contain the clause "Not valid for Prohibited, Restricted and Cantonment Areas"
- Tentative schedule shall be given **15 days** in advance so that vendor's personnel reach site within **7 days** from final intimation for site supervision services.
- No. of mandays as defined elsewhere in the Enquiry Document shall be considered for Evaluation for Supervision Services & Training (at Site) Services.
- However, payment shall be made as per actual number of man-days, unless otherwise specified elsewhere in the bidding document.
- Bidder to ensure compliance to guidelines on PPE (Personal Protective Equipment) & Other Safety Norms.
- Validity period of per diem rate shall be upto the end of guarantee period. Further, in case services for supervision are required after the guarantee period, 8% annual escalation shall be applicable.

(B) Charges for Site Supervision of erection, testing & commissioning:

Sl. No.	Description	Charges in currency of quote
1	Per Diem charges upto 8 hours on normal working days.	To be filled in price Form (BOQ)
2	Overtime rate per hour (beyond normal 8 hours on normal working days).	
3	Overtime rate per hour for work on weekly off days/ holidays (as applicable to Project Site).	

(C) Training charges

Sl. No.	Description	Rates in INR
1	Per diem charges of training at CPCL site.	To be filled in price Form (BOQ)

1. WATER, POWER & OTHER UTILITIES:

1.1 CONSTRUCTION WATER:

Subject to availability, Construction water shall be provided by OWNER free of Cost.

However, in case OWNER is not able to provide construction water due to any reason whatsoever, supplier shall be responsible for making all arrangements for Construction water at his cost. Any statutory requirements/ documentation etc. to this effect shall be met by the supplier.

Non-availability of water due to any reason shall not entitle the supplier for any claim against OWNER on account of cost and time implications.

1.2 CONSTRUCTION POWER:

i. The Owner shall provide construction power free of cost at designated source point(s) within CPCL refinery premises, subject to availability.

ii. The Contractor shall make necessary arrangement for further distribution from the main source point. Contractor is responsible for providing, installing, operating, maintaining all temporary electrical distribution systems, including cables, switchboards, distribution panels, protective devices, earthing systems, lighting arrangements, and associated accessories beyond the Owner's supply point.

iii. The Contractor shall, at his own cost and risk, arrange suitable Diesel Generator (DG) sets, fuel, operators, cables, distribution equipment, and all associated facilities whenever:

a) Construction power is required outside the refinery premises, or nearest power source is not available at the required location inside the refinery premises.

b) Owner's power supply is interrupted for any reason.

iv. The Owner will not provide construction power supply outside the battery limit of the Refinery premises

v. The Owner does not guarantee uninterrupted, continuous, construction power. Temporary outages, shutdowns, maintenance activities, operational constraints, load restrictions, or emergencies may result in interruption of power supply.

vi. The Contractor shall arrange, at his own cost, suitably rated DG set(s) and all associated facilities whenever Owner-supplied construction power is unavailable, inadequate, interrupted, or insufficient for the Contractor's requirements, to ensure uninterrupted execution of the Works.

vii. No claim for additional payment, idle charges, loss of productivity, compensation, escalation, extension of time, or damages shall be entertained on account of non-availability, inadequacy, fluctuation, or interruption of Owner-supplied construction power.

viii. The Contractor shall ensure that all temporary electrical installations and DG set operations comply with applicable statutory requirements, standards, safety regulations, environmental norms, pollution control requirements, and site safety procedures.

ix. The Contractor shall obtain all necessary approvals, permits, and clearances required for installation and operation of DG sets and temporary electrical systems.

x. The contractor shall include adequately rated sub distribution boards, power supply cables, and other associated materials for feeding loads to carry out construction and fabrication activities at his own cost. Meters shall be provided in the switch board installed by the contractor for measuring current and power.

xi. The Contractor shall provide adequate area lighting at site of construction, fabrication yards and office etc. by means of suitable lighting fixture, lighting masts, flood lighting poles etc. which are to be supplied and maintained by the contractor.

xii. All switch boards/welding machines shall be kept in well-ventilated & covered shed. The shed shall be elevated to avoid water logging.

xiii. All necessary safety precautions as per the normal safety practices and as directed by the Engineer-in-charge shall be followed by the supplier.

1.3 **LAND FOR VENDOR's OFFICE & FABRICATION YARD:**

Space for Site Office, fabrication yard & Warehouse etc. shall be provided free of cost by OWNER.

2. ACCOMODATION FOR VENDOR'S SUPERVISORY STAFF AND LABOUR ETC:

The Vendor shall make his own arrangements for the accommodation of labour, Engineers/Supervisors deputed for carrying out Site Works.

The OWNER shall not permit any land at site to be used for residential purposes nor shall the OWNER otherwise provide or arrange any land for temporary accommodation of the Vendor personnel or labour. The Vendor shall at its own cost and initiative arrange land for residential accommodation for its staff and workers and the price of services shall be deemed to include the same.

Vendor shall arrange for the necessary transport, accommodation, medical, canteen and other facilities for their employees / staff at their own cost and abide by all labour laws, safety codes and statutory regulations and keep Owner indemnified in respect thereof.

3. EQUIPMENTS, TOOLS AND PLANT:

All labour (both skilled and unskilled), tools, tackles and consumables shall be arranged by Vendor at his own cost except anything defined in MR / PR.

All tools & tackles, cranes, handling equipment etc., as required, shall be arranged and provided by vendor at his cost. All equipment, vehicles, tools and plants and materials brought to site by the Vendor from his resources shall have distinctive identification marks and the description and quantity of such inflow shall be intimated to OWNER in writing by the Vendor. All construction materials brought by the Vendor shall have prior approval regarding quality and quantity by OWNER. The Vendor shall also provide without extra cost necessary enclosures, containers and protective materials for proper storage of materials inside, whenever so instructed by OWNER without any extra cost.

No material or equipment or tools etc. shall be taken out of the work-site without the written consent of OWNER.

OWNER shall not be responsible for the safety to protection of the materials of the Vendor and the Vendor shall make his own arrangements for proper watch and ward for his materials.

4. RULES AND REGULATIONS AT SITE:

The Vendor shall ensure that all rules and regulations as effected from time to time by the Owner and/or EIL are strictly observed by the Vendor 's personnel.

The Vendor shall not permit the entry to the site of any person not directly connected with the execution of the Contract without prior permission of the Engineer-in charge.

5. ENTRY/ GATE PASSES

The Vendor shall follow regulations of Owner for arranging entry / Gate passes for his workers and staff in line with the existing rules of OWNER.

6. WORKMANSHIP:

The entire job shall be done in accordance with the approved drawings and specifications. The drawings and specifications are intended to be self-explanatory but should any discrepancy arise on anything contained therein, clarification shall be sought from OWNER.

All the work covered under this requisition shall be done to the entire satisfaction of OWNER who shall determine the quality, acceptability and fitness of the work. The Vendor shall provide necessary and sufficient tools, plant, equipment etc. for proper execution of work to the satisfaction of OWNER.

7. WORK ON HOLIDAYS AND OUTSIDE NORMAL WORKING HOURS:

The Vendor shall generally restrict his working hour to the normal working hours but may be permitted to work extra hours, on working days, on Sunday and holidays and round the clock depending upon the urgency of the work, with prior request at least 24 hours in advance.

8. OTHER AGENCIES AT WORKS:

The Vendor may have to execute work in such a place and condition where other agencies also will be engaged for other work. No claim shall be entertained for execution of work under such circumstances.

9. INSPECTION OF WORKS:

OWNER and or EIL will have full power and authority to inspect the works at any time, either on the site or at the Vendor's premises. The Vendor shall afford or procure for the purchase every facility and assistance to carry out such inspection. On no account will the Vendor be allowed to proceed with work of any type unless such work has been inspected and entries are made in the site inspection register by OWNER / EIL.

10. ASSISTANCE TO OWNER:

The Vendor shall render such assistance and supply such labour as may be required by OWNER in connection with the works, cost of all such assistance and requirements shall be deemed to be included and covered by the prices quoted in the offer. The Vendor shall make available to OWNER all necessary instruments and assistance in checking the setting out of works of and

taking measurements of works without extra cost to OWNER.

11. ACTION IN CASE OF BAD OR DELAYED WORK:

Where it is found by OWNER, the performance of work by the Vendor is not satisfactory in respect of workmanship, deployment of sufficient labour or equipment delay, in execution of work or any other matter, OWNER shall have the right to engage labour at the risk and cost of the Vendor and get the work executed through his own or through any other agency and debit the cost in excess of the value as per order to the Vendor and the Vendor shall have no right to claim compensation thereof. In such a case OWNER shall have the right to utilise the materials and tools brought by the Vendor for the same work.

12. INSURANCE:

Owner shall arrange Comprehensive Marine Cum Erection Insurance (MCE) policy, therefore insurance shall be excluded from the Vendor's scope. Insurance during transit (for the goods being dispatched to Client's Project site), storage at site and erection of all the material and equipment's to be incorporated in permanent works shall be covered under above MCE insurance policy by Owner. However, Building construction materials shall not be considered as a part of permanent works and hence shall not be covered under above MCE insurance policy by Owner. Vendor is, therefore, requested to arrange insurance towards the full value of Building Construction Materials at his own cost & expense in the joint names of Owner & Vendor (Owner shall be the first beneficiary).

All other insurance required under prevailing Indian laws for Vendor's personnel, equipment as well as Third Party Insurance shall be arranged by Vendor.

13. PERMITS, LICENCES ETC.:

The Vendor shall obtain all permits, licenses or other authorisations as are required for his work in terms of the order, from local / State or Central Authorities. He shall comply with all rules, regulations and other statutory obligations under the State of Union Legislative enactments. The Vendor shall pay all fees and other legally demandable compensation to any person, body or firm. OWNER shall not be responsible, either directly or indirectly, in any manner whatsoever for any omissions of the Vendor under this clause.

14. LOCAL AND OTHER AUTHORITIES NOTICES AND FEES:

The Vendor shall comply with and give notices required by any Act or by any Regulation or bye-law of any local or other public authority who have any jurisdiction with regard to the Vendor's work and shall pay and indemnify OWNER against any fees/or charges legally demandable under such Act, regulation or bye law in respect of the works.

15. PROTECTION TO WORK NOT TAKEN OVER BY OWNER/EIL:

Until such time the work is taken over by Owner/EIL, the Vendor shall be responsible for proper protection including proper fencing, guarding, lighting, flagging, watching and for the provision of temporary foot-ways, guards, and fences as far as may be rendered necessary by reason of the work, for the protection of foot passengers, vehicles, animals and to the properties of the Owner/EIL, occupiers of adjoining property and the public. The Vendor shall during the progress of work properly cover up and protect any part of the work liable to damage by exposure to the weather and shall take every reasonable precaution against accident or damage to the work from any cause.

16. OTHER FACILITIES:

Unless specifically agreed elsewhere in the Contract, the Owner/EIL shall not provide any other facilities to the Vendor and the Vendor shall arrange at his own cost and expense for plumbing, lighting, construction air or steam, sanitary facilities, petrol and other fuels and lubricants for all Vendor's equipments, temporary construction of buildings, construction and fabrication equipment, storage materials, tools, machinery (including consumables, costly and sophisticated fittings, instruments etc.), scaffolding, temporary erection bolts, all hand tools as required, welding equipment, packers and shims, and all other items required to complete the work.

17. DEWATERING:

The Vendor will keep his assigned areas free of water and must assume the costs for all pumping conditions including shoring and shuttering as required. The Vendor will tie in all his dumping operations to the existing sewer or drain system all as directed by the Engineer-in-charge. The Vendor will be held responsible for all damage which may be caused due to back water backing up or flowing over through floor or any part of the work or which any of his operation may cause to flow elsewhere.

18. HSE

The Vendor shall follow Health, Safety, Environment norms as per HSE document attached in the Enquiry Document.

19. COMPLIANCE WITH STATUTORY REGULATIONS:

Vendor shall comply with the following:

a) The Vendor shall be assumed to be familiar with all central, state and local regulations in force as applicable to the operations he is performing.

b) It shall be sole responsibility of the Vendor to execute the Contract in strict accordance with such statutory regulations in force and or to obtain necessary exemption certificates.

c) Any modifications/alterations deletion and/or addition that has to be carried out to meet statutory requirements shall be done at the cost and expense of the Vendor. The Vendor shall not be entitled to any extension in completion date to carry out changes, in order to meet statutory requirements.

d) The Vendor shall at his cost and expense, obtain and furnish Owner/EIL with all statutory certificates/permits, and/or exemption certificates by that are required, to put into operation, the work completed by the Vendor.

1. The Bidder shall quote price for the Post Warranty Annual Maintenance Contract (PWAMC) in his bid for the listed Item as detailed in technical specifications & bidding document.

2. Scope of Post Warranty Annual Maintenance Contract (PWAMC) shall be as per Material Requisition.

3. The PWAMC Charges quoted in Price Part shall be inclusive of all the charges for Transportation, Lodging, Boarding, all insurances including third party insurance, all Taxes/Duties/Cess/Levies/Fees and all other incidental charges etc. but excluding GST which shall be payable extra on Submission of Invoice as per GST Invoicing rules. Owner will not have any liability, whatsoever, over and above the quoted prices.

4. The payment terms for the PWAMC shall be on quarterly basis after completion of each quarter against submission of Bills duly certified by the Engineer-in-charge.

5. The order for PWAMC will be placed in line with the relevant provisions mentioned in Material Requisition/Bidding Document. The price of PWAMC services will remain firm & fixed for the complete execution of PWAMC contract.

6. LOA for PWAMC indicating rates & terms & conditions of PWAMC will be released along with the original supply order. Final SAP order for PWAMC will be released by site.

7. Bidder shall submit PBG for 5% of complete PWAMC order value at the time of main supply along with dispatch documents valid for complete PWAMC period + 3 months claim period. In case successful bidder finds it difficult to get BG with complete PWAMC period, they can submit BG with interim validity of min 2 years & subsequently the same to be renewed before expiry. In absence of a valid PBG, payment for PWAMC shall not be released.

8. Periodic and Preventive maintenance: Supplier shall depute their engineers to installation site as & when required for regular check-ups, as part of periodic maintenance. Supplier shall also depute their service engineers periodically for preventive maintenance, major repairs / replacement and reporting. Price shall also include emergency visits as necessary.

1. Bidder shall quote Training charges of Owner's Operation and Maintenance personnel at Bidder's/OEM's works/training facility on lumpsum basis (as per scope defined in the Material Requisition).
2. All expenses including but not limited to Air-fare, accommodation, local conveyance, living expenses etc. of Owner's personnel shall be borne by Purchaser.
3. Quoted prices shall be inclusive of all taxes & duties except GST which shall be payable extra at actuals against submission of documents as per GST rules.
4. A notice period of 15 days will be given by EIL/Owner for prior arrangement.
5. PAYMENT TERMS: Shall be as per Bid Data Sheet (BDS).

A. LUMPSUM CHARGES

Sl. No.	Description	Rate in Currency
1	Lumpsum Charges	Quoted in Price Schedule (BoQ)

A. Price basis w.r.t inclusion/exclusion of taxes and duties

1. The quoted price(s) shall be inclusive of all the taxes and duties except GST (i.e. IGST or CGST and SGST/UTGST applicable in case of interstate supply or intra state supply respectively) and GST compensation cess, (if applicable) leviable on sale of finished goods/services.
2. In case of foreign bidders, bidders shall submit their prices on FOB international sea port of exit basis/CFR on destination port basis *Note-1 (INCOTERM XXXX *Note-2), unless otherwise stated in the bid document. FOB/CFR *Note-1 prices quoted shall be inclusive of all applicable taxes up to FOB international sea port of exit/CFR on destination port basis *Note-1.
3. Transportation charges (if applicable) up to respective project site shall be exclusive of GST.
4. Site work prices, if applicable shall be inclusive of all taxes and duties except GST. All necessary registrations, if required, for carrying out the site activities shall be done by the bidder and costs towards the same shall be included in quoted site work prices.
5. Prices of per diem supervision charges, training/AMC, HAZOP Study etc. (as applicable) shall be exclusive of GST. However, GST shall not be applicable on training provided at foreign bidder's work/training facilities in any territory other than India.

B. TAX CLAUSES

1. Goods and Services Tax

    a. The supplier/contractor will be liable to register with the respective tax authorities and to submit self-attested copy of such registration certificates and the supplier/contractor will be responsible for procurement of goods/services in its own registration (GSTN).

    b. Supplier/Contractor shall be required to issue tax invoice in the form and manner prescribed under GST Act read with Rules thereunder including E-Invoicing provisions so that input tax credit under GST can be availed by Owner. In the event that the supplier/contractor fails to provide the Tax Invoice/E-Invoice in the form and manner prescribed under the GST act read with GST invoicing rules thereunder, Owner shall not be liable to make any payment on account of GST against such invoice.

c. In case supplier/contractor is not subject to E-Invoicing provisions under the GST Laws as amended from time to time, then a declaration to this effect shall be furnished by the supplier/contractor along with the Invoice.

d. GST shall be paid to supplier/contractor against receipt of Tax Invoice/E-Invoice and on auto-population of input tax credit on GSTN portal. In case of non-receipt of Tax Invoice/E-Invoice and/or non-auto-population of input tax credit on GSTN portal, Owner shall with hold the payment of GST.

e. The supplier/contractor shall always comply with the requirements of applicable laws and provide necessary documents as prescribed under the rules and regulations, as applicable from time to time. In particular, if any tax credit, refund or any other benefit Is denied/delayed to Owner or any interest/penalty is charged to Owner due to any non-compliance/ delayed compliance by the supplier/contractor under the GST Law (including but not limited to failure to upload the details of the sale on GSTN portal, failure to pay GST to the government or due to non-furnishing or furnishing of incorrect or incomplete documents, non-filing of GST return by the supplier/contractor), the supplier/contractor shall be liable to reimburse Owner for all such losses and other consequences including but not limited to the tax loss, interest and penalty. Owner shall be entitled to recover such amount from the supplier/contractor by way of adjustment from the next invoice, encashment of BG or by way of any other means either of same project or any other project of the Owner.

f. In case of any Advance including Mobilization/Secured/Other Advance is granted to supplier/contractor as per the provisions of Contract, then the supplier/contractor shall issue a Receipt Voucher / GST Invoice in the form and manner prescribed under GST Act read with Rules thereunder including e-invoicing provisions.

g. In case of any Price Variation as per the provisions of Contract, the supplier/contractor shall issue an Invoice in the form and manner prescribed under the GST Act read with Rules thereunder including e-invoicing provisions.

h. GST payable under reverse charge for specified services/goods under GST Act read with Rules thereunder, if any, shall not be paid to the supplier/contractor but will be directly deposited by Owner to the appropriate Government Tax Authority. If the same has already been reimbursed / paid to the supplier/contractor for any reason whatsoever, then Owner shall be

entitled to deduct/set off /recover such amount against any amounts paid/payable by Owner to supplier/contractor.

i. Where Owner has the obligation to discharge GST liability under reverse charge mechanism and Owner has paid or is liable to pay GST to the government on which interest/penalty becomes payable as per GST law for any reason which is not attributable to Owner or Input Tax Credit w.r.t. such payment is not available to Owner for any reason which is not attributable to Owner, then Owner shall be entitled to deduct/set off /recover such amount against any amounts paid/payable by Owner to contractor/supplier.

j. TDS under GST, if applicable shall be deducted from supplier/contractor bills at applicable rates and a certificate as per rules for tax so deducted shall be provided to supplier/contractor.

k. The supplier/contractor will be under obligation for charging correct rate of tax as prescribed under the respective tax laws. Further the supplier/contractor shall avail and pass on benefits of all exemptions/concessions available under applicable tax laws.

l. In case bidder is covered under composition scheme under GST law, then bidder shall quote the price inclusive of GST. Further, such bidder should mention “Covered under GST Composition Scheme” in column for GST of relevant price schedule(s). In case subsequently such bidder gets covered under regular GST regime, the prices including GST under regular GST regime shall not exceed the prices quoted by the bidder under composition scheme.

m. In case of foreign bidders, for supervision/site-work/training, services provided by foreign bidder at project site, GST shall be paid by Owner directly to appropriate Government Tax Authorities, under Reverse Charge Mechanism, if applicable and prices of such services will be loaded with applicable GST while evaluation of bid as per clause E (Evaluation methodology).

n. Bidder will be required to quote applicable tax rate (along with applicable HSN/SAC Code) as per provisions of GST Laws for all the line items forming part of the enquiry. Any higher rate of tax actually invoiced in excess of quoted rate of tax (except in compliance with provisions of Statutory Variation clauses) shall be adjusted in basic price.

o. E-way bills shall be issued directly by supplier/contractor in all cases except in cases of direct imports by Owner where E-way bill shall be issued by Owner.

p. In case of applicability of any recovery as per provisions of the contract, Owner shall raise Invoice on supplier/contractor after charging GST at applicable rates as per prevailing provisions of GST Laws.

2. IMPORT DUTIES

a. Direct supplies by foreign suppliers where, client becomes consignee:

i. Consignee shall be Owner, bill of entry shall be filed by Owner, all import duties, Social Welfare Surcharge, IGST and GST Compensation Cess etc. as applicable in India shall be paid by Owner.

b. Imported supplies by Indian/ Foreign suppliers/Contractors as built in import content where goods are cleared by filing into Bill of Entry in the name of Owner

i. High seas sale shall be effected between, supplier/contractor and Owner.

ii. Bill of entry shall be filed in the name of Owner by supplier/contractor, all import duties, Social Welfare Surcharge, IGST and GST Compensation Cess etc. as applicable in India shall be discharged by Owner.

iii. Custom duties, Social Welfare Surcharge, IGST, GST Compensation Cess, etc. as applicable in India on the imported materials shall not be included in the quoted prices by the bidder. All harbour dues/ pilotage fees, port fees, wharfage, unloading costs, demurrage charges etc. incurred in India in respect of any imported goods shall be to supplier's/contractor's account.

iv. Bidder to indicate CIF value of built-in import content and applicable merit rate, Social Welfare Surcharge on custom in the relevant price schedule.

v. The supplier/contractor shall be responsible for and shall exercise due diligence in properly classifying the goods and materials, undertaking the payment of custom duties, and/or otherwise complying with all applicable laws w.r.t. import of the goods and materials.

vi. If the total custom duties (BCD, SWS & IGST) paid or payable on the import of goods into India exceeds the total value thereof calculated as per the CIF value and rates indicated by supplier/contractor as above as increased or decreased for any increase or reduction in applicable

rates or CIF valuation, the Supplier/contractor shall pay and bear and/or reimburse to the Owner such excess custom duties.

c. General points:

i. In case Owner is exposed to any penal action, interest /penalties by the custom authorities for incorrect declaration and / or valuation of the goods or material by the supplier/contractor, or otherwise on account of any breach of applicable laws in the course of the import of the goods and material by supplier/contractor, the supplier/contractor shall indemnify and hold harmless the Owner for any and all costs, expenses or losses suffered or incurred by Owner in this regard.

ii. If bidder intends to source import of goods from a country with which India has Free Trade Agreement (FTA) or Comprehensive Economic Partnership Agreement (CEPA) or any such multi-lateral / bi-lateral Agreement or Treaty with India or under Generalized System of Preference (GSP); or under any other notification (allowing lower rate of custom duty), then the bidder is advised to ascertain and confirm its applicability along with supporting documents in the bid and ensure that conditions of such agreement/treaty etc. are strictly complied with.

iii. The bidder shall be liable to provide all documentation to ensure availment of the exemption/waiver. In case the bidder defaults on this due to any reason, whatsoever, he shall be liable to bear the incremental custom duty applicable if any.

iv. Any additional custom duty applicability on account of any change in the notification (allowing lower rate of custom duty)/ CEPA/FTA/multi-lateral/bi-lateral trade agreement shall be to bidder's account.

v. Documentation to be furnished for availing the exemption/waiver of custom duty shall be specifically listed in the letter of credit also as a prerequisite for release of payment against shipping documents and this documentation shall necessarily form a part of shipping documents.

vi. Owner will not bear any liability towards payment of Safeguard Duty, Anti-Dumping Duty, Protective Duty and applicable IGST including GST Cess on same or Countervailing Duty on subsidized articles or any other such duties of Customs imposed by Government under Customs Tariff Act, 1975.

vii. In case custom duty rate as quoted by foreign bidder is less than Actual rate applicable on due date of submission of last price bid and payment of custom duty is on Owner's Account, differential amount on account of above-mentioned rate variation will be recovered from foreign bidder.

3. INCOME TAX

a. Foreign supervisors/suppliers/contractors

Prices of site work, contracts and other services of foreign supplier/contractor shall be gross of income tax i.e., inclusive of Indian income tax at the applicable prevailing rate as per Indian Income Tax Act and Rules there under.

Owner shall deduct withholding tax at source (TDS) as applicable while making payments against each invoice. Certificate for TDS shall be provided to the supplier/contractor.

In case foreign bidder quotes Prices of site work, contracts and other services "net of Income Tax" i.e., exclusive of Indian income tax, Owner shall deduct and deposit TDS as applicable after grossing up the sums due while making payments against each invoice at the applicable prevailing rate as per Indian Income Tax Act and Rules thereunder.

Foreign bidder shall furnish the information required in line with the prevalent provisions of Income tax act read with rules thereunder including but not limited to Tax residency certificate, Form 10F, No permanent establishment declaration, No business connection certificate, PAN in India if available.

Failing to provide above documents/information, tax will be withheld at rates applicable as per the provisions of Income Tax Act, 1961 read with rules thereunder and the benefit of reduced rates will not be given.

If the non-resident is unable to obtain & submit Tax Residency Certificate to the Owner within a reasonable time, the bidder should furnish Form 10F along with an undertaking to the effect that the bidder is a tax resident of (the specified country) and that they shall obtain and provide the tax residency certificate (TRC) to the Owner before 30 days of submission of first Invoice

by them or within 3 months from the date of entering into contract whichever is earlier. However, payment will only be made after receipt of TRC as same is to be updated in form 145(earlier form 15CA).

In case of contracts with term exceeding one financial year, the relevant documents like TRC, Form 10 F, No PE / Business Connection Certificate etc. as applicable, as per Income Tax Act, 1961 must be submitted at the beginning of each financial year and the Owner will not bear additional tax liability (if any) arising from non-submission of documents.

The bidder shall be liable to intimate the subsequent changes (if any) to the information submitted against any of the above particulars, along with full details.

In addition to the above particulars, the bidder should also provide any other information as may be required at a later stage for determining the taxability of the amount to be remitted to the non-residents.

In case, any additional tax liability arises on the Owner due to any action of bidder like delay in submission / non-submission of information / documents required as above or change in residential status post submission of documents or change in any document (like TRC, Form 10F etc.) after its submission shall be recovered from the supplier/contractor.

b. Indian resident supervisors/ suppliers/contractors

Prices for supply and services of Indian suppliers/ contractors shall be inclusive of income tax.

Tax shall be deducted at source (TDS) by Owner on all sums due to supplier/contractor in accordance with the provisions of Indian Income Tax act read with rules thereunder as in force at the relevant point of time.

Owner shall issue a Tax Deduction Certificate to the supplier/contractor evidencing the tax deducted or withheld and deposited by Owner on payments made to the supplier/contractor to enable the supplier/contractor to claim the credit of Tax deducted by Owner.

c. INCOME TAX (General Points):-

i. The supplier/contractor shall also be responsible for ensuring compliance with all applicable provisions of the Direct Tax Laws of India including, but not limited to, the filing of relevant Tax Returns and shall promptly provide all information required by the Owner for discharging any of its responsibilities/liabilities under such laws in relation to or arising out of the Contract. For the lapses/failure, if any, on the part of the supplier/contractor and consequential penal action taken by the Income Tax department, Owner shall not take any responsibility whether financial or otherwise and shall be indemnified by the supplier/contractor.

ii. The Indian Income Tax Act and Rules made thereunder contains provisions permitting deduction of tax at source (TDS) at lesser rate if the supplier/contractor is able to justify to the Income Tax Authorities such lesser rate of deduction. However, a deduction once made has to be deposited by the Owner with the Income Tax Authorities in India and will not be adjustable by the Owner. It is therefore in the interest of the supplier/contractor that prior to release of any payment due to the supplier/contractor under the Contract that the supplier/contractor obtains from the relevant Income Tax Authorities in India, a certificate specifying the rate of deduction/withholding of Income tax at source, failing which, payment to the supplier/contractor shall be made by the Owner after withholding/deduction at the rate as may be applicable to the supplier/contractor as per provisions of Income Tax Act, 1961.

iii. TDS on interest recovered on Mobilization/Other Advance will be deposited by the supplier/contractor and same would be reimbursed by the Owner against submission of Form-16A (TDS certificate).

C. INVOICING METHODOLOGY W.R.T. TAXATION

Following invoicing methodology shall be followed :

1. In case of domestic / foreign PO's (Other than 2 below), Invoice/E-Invoice shall be raised by the supplier/contractor in the name of Owner and consignee shall be Owner.

2. In case of package MR's (where title transfer or take over is envisaged to take place at site after completion of site work) and tenders, supplier/contractor shall bring material at project site in their own name and they themselves shall be

consignee. As per provisions of contract between Owner and supplier/contractor, Invoice/E-Invoice shall be raised by supplier/contractor to Owner after charging GST.

D. APPLICABILITY OF TAX CREDIT

100% GST input credit is available to owner, except for Sec 17(5) disallowances.

E. EVALUATION METHODOLOGY W.R.T. TAXES

Following shall be loaded for evaluation:

1. Applicable Custom Duty, Social Welfare Surcharge etc. (Non-MOOWR cases)/Present value of deferred import duties (MOOWR cases), (excluding IGST & GST Compensation Cess (if applicable) on imported material which shall be dealt as per bid evaluation methodology given in E2 below) shall be loaded for price bid evaluation in respect of supplies for which bill of entry has to be filed by Owner/in the name of Owner (MOOWR cases).

2. a) GST including GST Compensation Cess (if applicable) on which input credit is not available to Owner [Excluding (b) below]
   b) Present Value of GST including GST Compensation Cess (if applicable) deferred under MOOWR scheme

3. Applicable Income tax, in case foreign bidder quotes prices net of taxes.

F. TAXATION TERMS FOR INDIAN SOURCED SUPPLIES (Applicable in case of foreign supplier)

Shall be the same as applicable for Indian bidders.

G. TAX IMPLICATION WHERE FABRICATION YARD IS OUTSIDE THE FACTORY PREMISES, WHEREVER APPLICABLE

In case of package MRs and tenders, since contractors/suppliers shall bring material at project site in their own name and raise invoice to Owner (as per payment milestone achieved) after charging GST, location of fabrication yard whether inside or outside factory premises will not attract any additional liability.

H. TAX IMPLICATION WHERE OWNER WILL ISSUE FREE ISSUE MATERIAL (FIM) TO CONTRACTORS/SUPPLIERS

Where Owner issues FIM's to suppliers/contractors, material will be released against delivery challan and such FIM shall be received back in factory premises as per provision of GST.

I. STATUTORY VARIATION (IN CASE OF MRS & TENDERS)

No variation on account of taxes and duties, statutory or otherwise, shall be payable to Supplier/Contractor except for the following:

1. GST: If after the due date of submission of last price bid and up to the contractual delivery/completion period (*Note 4), any increase/decrease occurs in the applicable rate of GST, the variation in such GST shall be to Owner's account and shall be adjusted (increase / decrease) to / from the Supplier's/Contractor's invoices based on the documentary evidence.

   Any increase in GST after the contractual delivery/completion period (*Note 4) shall be to Supplier's/Contractor's account. However, any decrease in the rate of GST shall be passed on to Owner.

   For calculating Statutory Variations ceiling amount as declared by the Bidder in price schedule shall only be considered.

2. Any new output taxes, duties, cess, levies notified or imposed after the due date of submission of last price bid but upto the contractual date of delivery/completion (*Note 4) shall be to Owner's account. These shall be reimbursed against documentary evidence. However, in case of delay attributable to supplier/contractor, any new output taxes, duties, cess, levies notified or imposed after Time for Completion, defined as above, shall be to supplier's/Contractor's account.

J. TAX INDEMNITY

Any omission/ errors of interpretation of applicability of taxes, duties, cesses and levies, whatsoever named, howsoever named, as are payable to any government, local or statutory authority in India or in any country other than India by the supplier/contractor shall be to supplier's/contractor's account.

If any fine/penalty/any other levy is required to be met by the Owner

arising out of any non-compliance/fault/fraud/willful suppression/misstatement of facts/information/documents, whatsoever, of any applicable laws of India or any country other than India by the supplier/contractor/their personnel/sub-contractors/agencies, the same shall be recovered from any amount payable to the supplier/contractor under this contract or under any other contract of the Owner.

If any tax is paid / required to be paid by supplier/contractor in pursuance of any demand of any law enforcement agency/Tax Authority on account of non-compliance/fault/fraud/willful suppression/misstatement of facts/information/documents, whatsoever, of any applicable laws of India or any country other than India by the supplier/contractor/their personnel/sub-contractors/agencies, the same shall be to supplier's/contractor's account only.

GENERAL:

1. *Note-1: Mode of Transport (i.e. Sea / Air or otherwise) and Price Basis (i.e. FOB/CFR/CIF or otherwise) shall be as defined in Request for Quotation (RFQ) / Notice Inviting Tender (NIT) / Letter Inviting Bid (LIB).
2. *Note-2: Applicable INCOTERM shall be as defined in Instruction to Bidders (ITB).
3. *Note-3: Deleted
4. *Note-4: Contractual delivery/completion period shall include extended contractual delivery/completion period for the reasons attributable to Owner or due to Force Majeure condition.

LIST OF SEAPORTS

A. EUROPE: FELIXSTOWE / ANTWERP / ROTTERDAM / HAMBURG / BARCELONA / GENOA / LA SPEZIA / LE HAVRE / FOS SUR MER / GOTHENBURG / COPENHAGEN / GIO TAURO

B. FAR EAST

JAPAN: YOKOHAMA / KOBE / MOJI / NAGOYA / OSAKA
CHINA: DALIAN / XINGANG / QINGDAO / HONG KONG / SHENZHEN / SHANGHAI
SINGAPORE
MALAYSIA PORT KELANG
KOREAN SEAPORT BUSAN

C. MIDDLE EAST
JEBEL ALI

NOTE: From other continents (Africa, North and South America, Australia), vendors will quote considering FOB major International Gateway Seaport of Exit.

LIST OF AIRPORTS

Sl. No.	COUNTRY	LOAD PORT
1	US	ATLANTA
		BOSTON
		CHICAGO
		NEWYORK
		DALLAS
		HOUSTON
		LOS ANGLES
2	UK	LONDON & Other Airports
3	UAE	DUBAI
4	NETHERLANDS	AMSTERDAM
5	SWEDEN	STOCKHOLM
6	SINGAPORE	SINGAPORE
7	LUXEMBORG	LUXEMBORG
8	JAPAN	TOKYO
		OSAKA
9	ITALY	MILAN & Other Airports
10	GERMANY	FRANKFURT & Other Airports
11	FRANCE	PARIS & Other Airports
12	CANADA	MONTREAL
13	BELGIUM	BRUSELLS

NOTE: From other countries vendors will quote considering FCA major International Gateway Airport of Exit.

GENERAL PURCHASE CONDITIONS

GENERAL PURCHASE CONDITIONS

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1.0 DEFINITIONS

1.1 Unless repugnant to the subject or context thereof, the following expressions herein used shall carry the meaning hereunder respectively assigned to each, namely:

(a) **"Bulk Consumables"** mean items specifically defined in the Contract Documents to constitute bulk consumables.

(b) **"Contract"** shall mean the contract as derived from:

i. The Tender Documents;
ii. Agreed Variations to the Tender Documents;
iii. Vendor Priced bid; and
iv. The Purchase Order.

(c) **"Contract Document(s)"** shall mean individually and collectively the documents constituting the contract.

(d) **"Defect Liability Period"** in respect of:

i. Bulk Consumables shall be the date of delivery plus 6 (six) months;

ii. In the case of other Material(s) shall be 18 (eighteen) months from the date of delivery or 12 (twelve) months after the same have been put in service or commissioned, whichever is earlier;

iii. In the case of altered or replaced Material(s):

i. For bulk consumables:

In case of repair / replacement, Vendor shall extend the warranty of such part further for a period of 6 (six) months from the date of supply of repaired/replaced material. However, in no case the warranty of repaired/ replaced part shall exceed 12 (twelve) months from last supply.

ii. In case of other materials:

In case of repair/ replacement, Vendor shall extend the warranty of such part further for a period of 12 (twelve) months for other materials from the date of repair/ replacement. However, in no case the warranty of repaired / replaced part shall exceed 24 (twenty- four) months from the date of commissioning or 30 (thirty) months from last supply, whichever is earlier.

In case of extended Warranty period, extension of Performance Bank Guarantee shall not be required.

(e) **"Delivery"**

i. with respect to Imported Material(s) means the date of completing shipment of the Material(s) on board the designated vessel or aircraft at the designated port or place of shipment, securely packed and unless otherwise determined, shall be deemed to be the date of the relative Bill of Lading or Airway Bill; and

ii. with respect to Indigenous Material(s) means the date of completing shipment of Material(s) F.O.R./F.O.T. securely packed and loaded and unless otherwise determined, shall be deemed to be the date of the relative Truck/lorry Receipt or Railway Receipt.

(f) **"Earnest Money Deposit (EMD)"** means the bank guarantee furnished in lieu of EMD {where the amount of EMD is more than Rs. 100,000/- (Rupees One hundred thousand only)} or online EMD submission (wherever allowed) on e-tendering portal by the Vendor in support of his/ its bid as required by the Bid Document.

(g) **"Equipment"** means plant, machinery, equipment, instruments, computer, control and other electronic and electrical systems, and shall include parts, components, assemblies and sub-assemblies thereof.

(h) **"Free Issue Material(s)"** means any equipment, parts or components or spares to be supplied by CPCL to the Vendor which are to be incorporated in any supply of Indigenous Material(s).

(i) **"CPCL"** means Chennai Petroleum Corporation Limited, a company incorporated in India and having its registered office at "New No. 536, Anna Salai, Teynampet, Chennai – 600018" and includes its successors and assigns and all persons through whom it acts in any matter for the purpose of the Tender or the Contract.

(j) **"Imported Material(s)"** mean(s) the materials to be fabricated, manufactured or procured by the Vendor outside India for shipment to India under the Contract.

(k) **"Indigenous Material(s)"** mean(s) materials to be fabricated, manufactured or procured by the Vendor within India for supply under the Contract.

(l) **"Inspectors"** means Inspectors nominated, appointed, approved or deputed by CPCL for inspection of the Material(s) prior to Delivery.

(m) **"Material(s)"** means any and all raw materials, manufactured articles, equipment, spares and other goods and supplies whatsoever and includes wherever applicable drawings, data, specifications and intellectual property rights and all services (including but not limited to design, fabrication, inspection, delivery and testing) required to be supplied, done, performed, prepared or undertaken to meet the requirements of the Contract

(n) **"Procurement Coordinator"** means the representative or agency appointed by CPCL for managing, expediting and/or coordinating the supply of Material(s).

(o) **"Project"** means the Project or Refinery for which the Material(s) is/are required.

(p) **"Project Site"** means the site of the Refinery unit or site of the Project for which the Material(s) is/are required.

(q) **"Purchase Order"** means CPCL's acceptance of the Vendors' offer/bid and includes any formal or detailed Purchase Order issued by CPCL pursuant to the acceptance of the bid.

(r) **"Stipulated Delivery Period"** means the date(s) for delivery of the Material(s) as stipulated in the Contract and failing such stipulation, shall mean the date(s) for such delivery(ies) as agreed between the Vendor and CPCL.

(s) **"Tender Documents"** with reference to the Purchase Order mean:

i. Material Requisition/Request for Quote;
ii. General Terms and Conditions of Purchase;
iii. Technical Specifications;
iv. Special Conditions of Purchase (if any);
v. Addendum (a) (if any) to the Tender Documents.

(t) **"Total Contract Value"** means total value of the Material(s) and services to be supplied as specified in the Purchase Order, exclusive of reimbursable taxes and duties.

(u) **"Vendor"** means the successful bidder on whom the Purchase Order is placed.

1.2 Interpretation of Contract Documents

1.2.1 The several Contract Documents forming the Contract are to be read together as a whole and are to be taken as mutually explanatory.

1.2.2 Should there be any doubt or ambiguity in the interpretation of the Contract Documents or in any of them, the Vendor shall prior to commencing the relative supply or work for supply under the Contract apply in writing to CPCL for resolution of the doubt or ambiguity. Should the Vendor fail to apply to CPCL within 7 days from the date of receipt of the Order for its clarification as aforesaid, the Vendor shall perform the relative work and/or make the relative supply at his own risk.

1.2.3 Any item of supply or service relative thereto shown, indicated or included by expression or implication in any document forming part of the Contract shall be deemed to form part of the Scope of Supply with the intent that the indication or inclusion of the supply or service within any of the said documents shall be a sufficient indication of the Scope of Supply or service covered by the Contract.

1.2.4 No verbal agreement or assurance, representation or understanding given by any employee or officer of CPCL or so understood by the Vendor shall anywise bind CPCL or alter the Contract Documents unless specifically given in writing and signed by or on behalf of CPCL as an Agreed Variation to the relative term(s) in the Contract Document(s).

1.2.5 Clause headings given in this or any other Contract Documents are intended only as a general guide for convenience in reading and segregating the general subject of the various clauses, but shall not govern the meaning or import of the clauses there under appearing or confine or otherwise affect the interpretation thereof.

1.3 **Irreconcilable Conflicts**

Subject to the provisions of Clause 1.2 hereof, in the event of an irreconcilable conflict between the provision of these General Terms and Conditions of Purchase and/or the Special Conditions of Purchase and/or Addendum (a) and/or the Agreed Variations to the Tender Documents and/or the Purchase Order and/or between any of the other said documents so that the conflicting provision(s) cannot co-exist, to the extent of such irreconcilable conflict, the following order of precedence shall apply so that the conflicting provision(s) in the document lower in the order of precedence set out below shall give way to the conflicting provision(s) in the document higher in the order of precedence, namely:

i. Purchase Order;
ii. Agreed Variations to the Tender Documents;
iii. Addendum/Addenda (a) to the Tender Document;
iv. Special Conditions of Purchase;
v. General Terms and Conditions of Purchase;
vi. Other Contract Documents.

2.0 CONFIRMATION OF ORDER

2.1 **Wherever applicable, signed Letter of Acceptance (also named as Fax of Acceptance in some cases) shall, normally, be issued through e-mail (scan of the signed document). SAP Purchase Order shall, normally, be issued as an unsigned, system generated document through e-mail.**

Without prejudice to the formation of contract by acceptance of bid, the Vendor shall acknowledge the receipt of the Letter of Acceptance (wherever applicable) and SAP Purchase Order within 7 (seven) days following receipt of the Letter of Acceptance or SAP Purchase Order.

3.0 PRICE

3.1 Unless otherwise specifically stipulated, the price shall be firm and shall not be subject to escalation for any reason.

3.2 Unless otherwise specifically stipulated, the price for Indigenous Material(s):

i. shall be inclusive of road/rail worthy water-proof packing and forwarding charges upto effecting delivery at F.O.T./F.O.R despatch point and shall also be inclusive of inland freight and local taxes (if any) as leviable on the transportation or entry of goods into any local area or limits pursuant to the Contract; and

ii. shall be exclusive of transit insurance, CGST or SGST/IGST and/or such other imposts which are leviable by law on the supply of goods or their sale to CPCL pursuant to the Contract.

3.3 (a) If it is stipulated that local taxes (if any) are to be borne by CPCL, the Vendor shall arrange for the transporter to pay the same, if any leviable and claim reimbursement thereof from CPCL against proof of payment.

(b) If it is stipulated that dispatch shall be on “freight to pay basis”, the Vendor shall advise the transporter to collect the freight from CPCL after the full quantity of the Material(s) has/have been delivered in good condition to the Project Site.

3.4 Unless otherwise stipulated, the price of Imported Material(s) shall be the FOB/ FCA (as per prevailing ICC INCOTERMS at the time of ordering) price of Material(s) and shall be inclusive of sea/ air worthy water-proof packing and forwarding charges (as the case may be).

4.0 EARNEST MONEY

4.1 A bid is liable to be rejected unless it is supported by earnest money of a value as provided in the Purchase Requisition/ Request for Quote.

4.2 Earnest Money by the tenderers shall be accepted only in the form of online EMD payment on CPCL’s e-tender portal or a Bank Guarantee. Bank Guarantee in the format set forth in Annexure "A" hereto may be furnished in lieu thereof if the amount of Earnest Money Deposit exceeds Rs. 100,000/- (Rupees One hundred thousand only).

4.3 If the Earnest Money is in the form of a Bank Guarantee, the validity of the Bank Guarantee shall be extended by the Vendor at the Vendor’s cost and initiative for a period 3 (three) months beyond the date of the acceptance of bid by CPCL, failing which the Bank Guarantee may be encashed by CPCL and the proceeds held as security for the performance of Vendor obligation and the due discharge of Vendor’s liability under the resultant Contract until the Vendor acknowledges the acceptance of the Purchase Order and furnishes the Performance Guarantee. Should the Vendor fail to accept the Purchase Order and/or furnish the Performance Guarantee within the time specified in this behalf, or specifically permitted by CPCL for the purpose, CPCL may encash the Bank Guarantee furnished by the Vendor by way of Earnest Money Deposit and/or forfeit such proceeds or other encashable Earnest Money Deposit held by it in cash without prejudice to any other right or remedy available to it.

4.4 EMD of bidders whose price bids are not opened/ disqualified during techno-commercial bid evaluation (i.e. unsuccessful Bidder) shall be released after price bid opening.

EMD of bidders qualified in the techno-commercial bid but unsuccessful for placement of Purchase Order shall be released after placement of Purchase Order on successful Bidder.

EMD of the successful bidder shall be released on receipt of acceptable Performance Bank Guarantee (PBG), wherever applicable. Wherever PBG is not applicable, EMD of successful bidder shall be returned after order acceptance.

4.5 Earnest Money furnished by a tenderer may also be forfeited in the following circumstances:

a) If the tenderer alters or modifies or withdraws their bid prior to opening of the price bid and within the specified validity period of the Tender; or

b) If the tenderer withdraws their bid after the Tender is opened. c) In case of submission of false/ fraudulent / forged documents.

5.0 TERMS OF PAYMENT

5.1 **For Imported Material(s):**

a) The price of Imported Material(s) shall be paid in the currency specified in the Contract.

b) 100% (one hundred percent) of the price of the relative Imported Material(s) (after considering price reduction for delay, if any, as per clause 12.0) will be paid under an irrevocable Letter of Credit against submission of documents specified in the Letter of Credit. The Letter of Credit shall be substantially in the format set forth in the Annexure-"B" hereto and shall be established either by the State Bank of India or other bankers of CPCL in India.

c) Unless otherwise specified, the Vendor may draw against the Letter of Credit on presentation of all the following documents:

i) Invoice
ii) Clear Bill of Lading/Airway Bill covering the Material(s) invoiced
iii) Packing list for the consignment
iv) Third Party Inspector's Certificate covering the invoiced Material(s) wherever applicable.
v) Test/Composition Certificate
vi) Certificate of origin
vii) Drawing(s)/Catalogue(s) covering the Material(s), wherever applicable.
viii) Export Certificate, wherever applicable.
ix) Invoice of Inspector's charges, wherever applicable.
x) CPCL's acknowledgement of Performance Bank Guarantee wherever applicable.

d) Bank charges payable to CPCL's banker for opening of the Letter of Credit shall be borne by CPCL and bank charges payable to the Vendors' banker shall be borne by the Vendor.

e) Should the Vendor desire to get the Letter of Credit confirmed by any other banker, confirmation charges will be borne by the Vendor.

f) Unless otherwise agreed, the Letter of Credit shall not permit drawing in more than 3 (three) tranches.

g) Unless otherwise specified, the Vendor shall furnish a Bank Guarantee towards performance favoring CPCL within 15-days of the Purchase Order for an amount equivalent to 10% (ten) of the Price of Material(s) from a Scheduled bank in India (including an Indian branch of a foreign bank) acceptable to CPCL in the format set forth in Annexure "C" hereto valid (in the first instance) for the period specified in Clause 16.12.

h) In the event that CPCL requests the Vendor to hold or to warehouse the Material(s) for any period after the Material(s) are ready for shipment, the storage charges as agreed, shall be borne by CPCL in addition to the Price.

5.2 **For Indigenous Supply:**

a) Unless otherwise specified in the Contract, where the total contract value does not exceed Rs. 50,000/- (Rupees fifty thousand only), CPCL shall release 100% of the relative Price of Material(s) within 30 days of receipt of the Material(s) at Project Site and their acceptance.

b) Unless otherwise specified in the Contract, where the total contract value exceeds Rs. 50,000/- (Rupees fifty thousand only) but is less than Rs. 1,00,000/- (Rupees one hundred thousand only), CPCL shall release 90% of the relative Price of Material(s) on delivery of the documents specified in (c) hereof below relative hereto, and will release the balance 10% of the Vendor's invoice within 30 (thirty) days of receipt of Material(s) at Project Site and their acceptance.

c) Unless otherwise specified, where the total contract value of the Material(s) is Rs. 1,00,000/- (Rupees one hundred thousand only) and above, CPCL shall release 90% of the relative price against the documents specified here below relative hereto through CPCL's bankers and will release the balance 10% of the Vendor's invoice within 30 (thirty) days of receipt of Material(s) at the Project Site and its/their acceptance. Unless otherwise mentioned, the specified documents are:

i) Invoice
ii) Clear Railway Receipt/Truck Receipt/Goods Receipt covering the Material(s) invoiced
iii) Packing list for the consignment
iv) Third Party Inspector's Certificate covering the invoiced Material(s)/ Release Note, wherever applicable

v) Test/Composition Certificate, wherever applicable
vi) IBR Certificate/CMRS Certificate, wherever applicable
vii) Drawing(s)/Catalogue(s) covering the Material(s), wherever applicable
viii) Guarantee/Warranty Certificate(s), wherever applicable
ix) Invoice of Inspector's charges, wherever applicable.
x) Freight Memo(s) if freight is not included in the Price and the RR/TR/GR does not give the freight particulars.
xi) Acknowledgement by CPCL of receipt of Performance Bank Guarantee (wherever applicable)

d) The financial settlement of the Vendor's invoice is liable to be withheld in the event the Vendor fails to submit the drawings, data and all other documents as called for in the Purchase Order.

e) Unless otherwise specified, the Vendor shall furnish a Bank Guarantee towards performance favoring CPCL within 15-days of the Purchase Order for an amount equivalent to 10% (ten) of the Price of Material(s) from a Scheduled bank in India (including an Indian branch of a foreign bank) acceptable to CPCL in the format set forth in Annexure "C" hereto valid (in the first instance) for the period specified in Clause 16.12.

6.0 VENDORS' DRAWINGS AND DATA REQUIREMENT

The Vendor shall submit drawings, data and documentation in accordance with (but not limited to) what is specified in the Purchase Requisition/Tender documents and/or Vendor's drawing and data form attached to the Purchase Order, within 30 (thirty) days of the Purchase Order. The types, quantities and time limits for submitting these must be respected by the Vendor and the Material(s) shall be deemed not to have been delivered for all purposes (including payment) until completion of the said submissions to the satisfaction of CPCL.

7.0 FREE ISSUE MATERIALS (for incorporation in the Indigenous supply)

If the Purchase Order involves the incorporation of any Free Issue Material(s):

a) The Vendor shall prior to taking delivery of the Free Issue Material(s) arrange for a Bank Guarantee for the full value of the Free Issue Material in the format set forth in Annexure "D" hereto valid from the date of the receipt of the Issue Material(s) until delivery of the Material(s) in which the Free Issue Material(s) has/have been incorporated.

b) The Vendor shall inspect the Free Issue Material(s) at the time of taking delivery thereof and satisfy itself of the quality, quantity and condition of the Free Issue Material(s). CPCL shall not be liable for any claims or complaints whatsoever in respect of the quality, quantity or condition of the Free Issue Material(s) once the Vendor has taken delivery thereof.

c) All Free Issue Material(s) shall be taken delivery of, transported, held, stored and utilized by the Vendor as trustee of CPCL, and delivery of the Free Issue

Material to the Vendor shall constitute an entrustment thereof by CPCL to the Vendor with the intent that any transportation, utilization, application or disposal thereof by the Vendor otherwise than for incorporation in the Indigenous Material(s) shall constitute a breach by the Vendor.

d) The Vendor shall transport the Free Issue Material(s) only by such transportation as is suitable and shall hold and store the Free Issue Material(s) only at such place and/or premises that are air and water tight and otherwise suitable for the storage of the Free Issue Material(s) so as to prevent damage or deterioration or theft or other loss, and shall arrange such watch and ward as shall be necessary to ensure the safety thereof.

e) Notwithstanding the Bank Guarantee mentioned in sub-paragraph (a) above, the Vendor shall replace any Free Issue Material(s) which is/are lost, damaged, misused, stolen or deteriorated with other Material(s) of equivalent quantity and quality and condition, and the same shall be deemed to constitute Free Issue Material(s) and the provisions of sub paragraphs (a) to (f) hereof shall apply thereto in the same manner as to the originally supplied Free Issue Material(s).

f) Unused Material(s) from the Free Issue Material(s) shall be returned by the Vendor to CPCL and if CPCL so directs, the Vendor shall dispose of the same by sale or otherwise on such terms and conditions as CPCL may stipulate or approve and the Vendor shall pay to CPCL the sale proceeds of the Material(s) so disposed of by sale.

8.0 THE BILL OF MATERIAL(S)

8.1 Where the price of Material(s) is a lumpsum price and pro-rata payment is envisaged in the Purchase Order, the Vendor shall within 60 (sixty) days of the issue of the Purchase Order furnish to CPCL for approval, a priced and detailed Bill of Material(s)/ Billing Schedule as required covering all Material(s), which shall conform to the price break-up and Total Contract Value given in the Purchase Order. The Bill of Material(s) shall operate as the Billing Schedule for payment of the price of the Material(s). In preparing the Bill of Material(s), the Vendor shall ensure that all contracted Material(s) are included in the Bill of Material(s) so as to ensure that CPCL is not required, due to any oversight or omission, to pay any taxes and duties on a value in excess of the total Value indicated in the Contract. Should CPCL be required to pay taxes or customs duties on account of such oversight or omission, the Vendor shall reimburse such excess payments to CPCL.

8.2 The Material Safety Data Sheets in the case of catalysts and chemicals and other items where ever applicable shall also be submitted within 30 days after receipt of the Purchase Order.

9.0 MODIFICATION

9.1 CPCL shall have the right to request changes or modifications in the technical documents and/or specifications comprised in the Contract, subject to the Vendor's approval thereto. CPCL shall bear any additional cost and shall be

entitled to the benefit of any reduced cost resultant upon any such change or modification.

9.2 As soon as possible after receipt of a written request from CPCL for change(s), the Vendor shall furnish in writing to CPCL an estimate of the additional cost or benefit for the change(s) and/or modification(s) requested and its effect on the delivery date. On agreement with respect to the enhanced/reduced cost and modified delivery time, which shall be finalized within 10 (ten) days of the request for the modification, CPCL shall issue an amendment to the Purchase Order, and the Vendor shall promptly proceed with the change(s)/modification(s) contemplated by the amended Purchase Order.

10.0 SUB-CONTRACTS / ASSIGNMENT

10.1 The Vendor shall not assign the Contract in whole or part without obtaining the prior written consent of CPCL.

10.2 The Vendor shall not sub-contract the Contract in whole or part to any entity without obtaining the prior written consent of CPCL.

10.3 The Vendor shall, notwithstanding the consent and assignment/sub-contract, remain jointly and severally liable and responsible to CPCL together with the assignee/ sub-contractor, for and in respect of the due performance of the Contract and the Vendor's obligations there under.

11.0 EXPEDITING

11.1 CPCL may appoint a Procurement Coordinator to manage, expedite and coordinate the manufacture, shipment and/or despatch of Material(s) covered by the Contract.

11.2 The Vendor shall furnish to the Procurement Coordinator within 30 (thirty) days of receiving the Purchase Order, the required number of copies of documents including but not limited to Schedule of manufacture/PERT chart, unpriced copies of sub-orders, phased programme of item-wise manufacture, testing and delivery and any other information and/or documents as may be called for by the Procurement Coordinator.

11.3 The Procurement Coordinator shall have free access to the Vendor's shop and sub-supplier's shop during normal working hours and shall be provided all the necessary assistance and information to help him perform his job.

12.0 RESPECT FOR DELIVERY DATES AND PRICE DISCOUNT

12.1 The time and date of Delivery of Material(s) as stipulated in the Contract shall be adhered to on the clear understanding that the Price(s) of the Material(s) has/have been fixed with reference to the said Delivery date(s).

12.2 If any delay is anticipated by the Vendor in the delivery of the Material(s) or any of them beyond the stipulated date(s) of Delivery, the Vendor shall forthwith inform CPCL in writing of such anticipated delay and of the steps being taken by the Vendor to remove or reduce the anticipated delay, and shall promptly keep CPCL informed of all subsequent developments.

12.3 **(A) In case scope includes only supply**

If any Material(s) is/are not delivered within the Delivery date(s) stipulated in respect thereof, CPCL shall be entitled to a discount by way of price adjustment in a sum equivalent to 0.5% (one half percent) of the price of such Material(s) per week or part thereof that the Material(s) remain(s) undelivered beyond the stipulated Delivery period in respect thereof, subject to a maximum discount of 5% (five percent) of the Total Contract Value. Such discount shall be given by the Vendor by equivalent reduction in the invoice value before presentation of documents to the Bank/CPCL for payment. Should the Vendor fail to deliver the Material(s) or to make such adjustment, the discount may be recovered by any other means.

(B) In case scope of work includes supply and site work

For delay in supply:

The price adjustment shall be applicable @ 0.5% of Total Supply Order Value (excluding site work) per week of delay or part thereof subject to maximum of 5% of Total Supply Order Value (excluding site work).

For Delay in site work:

The price adjustment shall be applicable @ 0.5% of Total Order Value (supply + site work value) per week of delay or part thereof subject to maximum 5% of Total Order Value (supply + site work value).

In no case total price adjustment shall exceed 5% of Total Order Value (Supply + site work).

Note:
In case of package items (ordered as a complete system), the price adjustment shall be applicable on the entire order value of that package and not on the value of the undelivered portions (even though a billing breakup has been approved).

In case of purchase of bulk items where tolerance limit is specified (e.g. pipes, cables etc.), price adjustment clause shall be applicable on the actual quantity supplied, within the tolerance limit, instead of Purchase Order quantity.

12.4 Without prejudice to its rights under Clause 12.3 hereof and to entitlement to discount(s) accrued in terms thereof and in addition thereto, CPCL may at any time after the expiry of the stipulated date(s) of Delivery in respect of any the undelivered Material(s) or any of them and either purchase such Material(s) from any other available source at the risks and costs of the Vendor and recover from the Vendor any additional cost incurred by it on such

purchase or recover from the Vendor without such purchase the difference between the market and contract price of such Material(s) on the date of termination of Contract relative thereto.

The maximum liability against risk & cost sourcing shall not be beyond the total Contract Value for the undelivered material. Provided, this shall not restrict CPCL's claim for damages or compensation, as the case may be, for acts of fraud, deliberate default, negligence or misconduct by the vendor.

13.0 DELAYS DUE TO FORCE MAJEURE

13.1 If a force majeure event as defined below, affecting the Vendor, arises prior to the expiry of the stipulated Delivery period in respect of any Material(s) and the Vendor intends to claim extension of the stipulated date of delivery in respect of such Material(s) or any of them, the Vendor must advise CPCL by notice in writing of such event by means of communication which secures undisputed service of the notice not later than 10 (ten) days of the occurrence of the event. Such occurrence shall be duly certified by a local Chamber of Commerce or statutory authority. The Vendor shall within 10 (ten) days of the end of the Force Majeure event similarly notify CPCL of such cessation, and of the period and Material(s) for which an extension of Delivery date(s) is consequently claimed. Such notification shall be a mandatory pre-condition to a claim for such extension.

13.2 No failure, delay or omission by Vendor to fulfill any of its obligations under Contract (other than the obligation to make payments when due) shall give rise to any claim against Vendor or CPCL or be deemed to be a breach of a Contract if and to the extent such failure, delay or omission arises from any of the following events not within the reasonable control of Vendor and not attributable to Vendor's fault, negligence or misconduct (each an event of "Force Majeure"):

a) Act of terrorism
b) Riot, war, invasion, act of foreign enemies, hostilities (whether war declared or not), civil war, rebellion, revolution, insurrection of military or usurped power;
c) Ionizing radiation or contamination, radioactivity from any nuclear fuel/ nuclear waste from reaction of nuclear fuel or any other hazardous radioactivity.
d) Epidemics, tsunami, earthquakes, flood, fire, hurricanes/typhoons or other natural disaster.
e) Freight embargoes, strikes at national/ state wide level (for more than 7 consecutive days) where the supplier's works is located.

13.3 For the avoidance of doubt, inclement weather, third party breach, delay in supply hardship, strike, shutdown or lockout other than as specified above shall not constitute a Force Majeure event.

13.4 In the event of Force Majeure, Vendor shall bear any costs incurred by it resulting there from. The Vendor affected by Force Majeure shall use all reasonable efforts to prevent and reduce to a minimum and mitigate the effect of delays occasioned by such Force Majeure.

13.5 If the Vendor is prevented from fulfilling its contractual obligations for a continuous

period of three (3) months because of Force Majeure, then the Vendor and CPCL shall consult with each other with a view to agreeing on the action to be taken under the circumstances, and failing such agreement, CPCL shall be entitled to terminate the contract in whole or to the extent that its performance is prevented by Force Majeure.

14.0 WARRANTY OF TITLE

14.1 The Vendor warrants that the Material(s) sold and supplied by it to CPCL pursuant to the Contract shall be free from any and all defects in title including but not limited to any charge, third party claim, mortgage, hypothecation, foreclosure, lien, restriction, injunction, attachment or encumbrance whatsoever and shall hold and keep CPCL indemnified from and against any and all contrary claims, demands, actions and proceedings and all costs (including legal costs), charges, expenses and losses suffered or incurred by CPCL as a consequence thereof and/or to defend any such claim, demand, action or proceeding.

14.2 The Vendor shall be understood to have represented to CPCL that the use by CPCL of the Material(s) supplied by the Vendor will not infringe any third party patent rights or pending patent applications or other intellectual property rights. Accordingly, the Vendor will hold harmless and indemnify CPCL against all costs (including legal costs), charges and expenses incurred or any damages or other sums that may be assessed or become payable under any decree or judgment of any court or under any settlement resulting from any suit, claim or action for infringement of third party patents or other third party intellectual property.

15.0 INSPECTION AND TESTING

15.1 In addition to any tests to be conducted by the Vendor under the Contract or any applicable codes or standards, the Material(s) shall be subject to inspection and/or testing by Inspector(s) (including Third Party Inspector(s)) at any time prior to shipment and/or despatch and to final inspection within a reasonable time after arrival at the Project Site. The Inspector(s) shall have the right to carry out the inspection or testing, which will include inspection and testing of the raw materials at manufacturers shop, at fabricators shop and at the time of actual despatch before and/or after completion of packing.

15.2 In addition to testing and inspection by Inspectors, CPCL may nominate an institutional agency like Boiler-Inspectorate for official testing of coded equipment. The Vendor shall ensure that all procedures for preparation and performance of tests prescribed by such institution shall be scrupulously complied and observed.

15.3 Unless otherwise specified in the Contract, the inspection shall be carried out as per the relevant standards/scope of inspection provided along with the Tender Enquiry/Purchase Order.

15.4 All charges for Third Party Inspectors shall be paid by the Vendor. CPCL shall pay these charges against vendor's invoice, unless agreed otherwise. However, no charges will be payable by CPCL in the event the inspection has become

infructuous for any cause.

15.5 All manufacturers' mill test certificates and analytical reports from material laboratories in respect of raw materials employed and components incorporated shall have to be presented by the Vendor.

15.6 Before shipping or despatch, the Material(s) will have to be checked and stamped by the Inspector(s) who may forbid the use and dispatch of any equipment and/or Material(s) which during tests and inspection fail(s) to comply with the specifications, codes and testing or other contractual requirements applicable thereto, and the Vendor shall not tender such rejected Material(s) for supply to CPCL nor shall incorporate the same in any Material(s) to be tendered for supply to CPCL.

15.7 The Vendor will inform CPCL at least eight (8) days in advance of the exact place, date and time of tendering the Material(s) for required inspection and provide free access to the Inspector(s) during normal working hours at Vendor's or his/its sub-Suppliers' works, and place at the disposal of the Inspector(s) all useful means for undertaking the Inspection, checking the results of tests performed, marking the Material(s), getting additional tests conducted and final stamping of the Material(s).

15.8 All tests will be performed at the Vendors' expense and if required by the Inspector(s), shall be conducted in accordance with the Inspector's Instructions. The Vendor shall also bear the expense for the preparation and rendering of tests required by the Boiler Inspectorate or other statutory testing or certifying agencies/institutions.

15.9 Unless otherwise specified, all charges for the Inspection shall be borne by the Vendor.

15.10 CPCL may, at its own expense, have its representative(s) witness any test or inspection. In order to enable CPCL's representative(s) to witness the tests/inspections, the Vendor shall notify CPCL at least 30 (thirty) days in advance, of the schedule of all inspection hold points prior to the initiation of equipment fabrication. CPCL shall be notified eight (8) calendar days in advance of any changes in the schedule of inspection. CPCL will advise the Vendor in advance whether it intends to have its representative(s) be present at any of the inspections.

15.11 Even if the inspection and tests are fully carried out, the Vendor shall not be absolved from its responsibilities to ensure that the Material(s), raw materials, components and other inputs are supplied strictly to conform and comply with all the requirements of the Contract at all stages, whether during manufacture and fabrication, or at the time of Delivery as on arrival at site and after its erection or start up or consumption, and during the defect liability period. The inspections and tests are merely intended to prima facie satisfy CPCL that the Material(s) and the parts and components comply with the requirements of the Contract.

15.12 The Vendor's responsibility shall also not be anywise reduced or discharged because CPCL or CPCL's representative(s) or Inspector(s) shall have examined or commented on the Vendor drawings or specifications or shall have witnessed the

tests or required any chemical or physical or other tests or shall have stamped or approved or certified any Material(s).

15.13 Unless otherwise specifically permitted by the Contract, no Material(s) shall be dispatched for delivery or delivered under the Contract without being stamped or otherwise approved for delivery by the Inspector(s).

15.14 Notwithstanding approval by the Inspector(s), if on testing and/or inspection after receipt of the Material(s) at Project Site, any Material(s) is/are found not to be in strict conformity with the contractual requirements or specifications, CPCL shall have the right to reject the same and hold the Vendor liable for non- performance of the Contract. The provision of Clause 16.5 to 16.11 shall mutatis mutandis apply to such rejected Materials.

16.0 ACCEPTANCE OF MATERIALS & GUARANTEES

16.1 The Vendor acknowledges that notwithstanding the provision or approval of any drawings, designs, specifications, source of supply or other data relative thereto by CPCL and/or the testing of Material(s) in accordance with the requirements of the Contract or any applicable code or specification and/or any inspection of the input or Material(s) by the Inspector(s) or issue of an Inspection Certificate relative thereto and/or any other act, matter or thing done or required by CPCL to satisfy itself of the quality, quantity, sufficiency or efficiency of the Material(s) prior to delivery thereof and/or the transfer of title and/or risks in relation to the Material(s), shall not be deemed or understood to constitute acceptance of the Material(s) by CPCL nor shall CPCL be understood to have accepted any Material(s) other than plant, machinery, equipment and parts and components unless such Material(s) have been received at the Project Site of CPCL and found to be acceptable as evidenced by a Certificate of Acceptance issued by CPCL, and in case of plant, machinery, equipment and parts and components, unless they have been incorporated into the relative Project Unit and the said Unit has been tested and the relative plant, machine, equipment, part or component has successfully functioned without patent defect.

16.2 To this end, the Vendor guarantees that:

i) All materials used in the execution of the Contract and all Material(s) used in performance thereof shall be in strict compliance and conformity to the characteristics, requirements and specifications of the Contract and suitable for the purpose for which such Material(s) are intended to be used if such purpose has been disclosed or is/are suitable for use to which such Material(s) are ordinarily put to use, if such purpose has not been disclosed.

ii) In the case of machinery, plant or equipment with rated capacities, outputs or other characteristics, that the machinery, plant or equipment as the case may be, shall function to such capacities and/or outputs and shall meet the other characteristics required in respect thereof.

16.3 The Vendor further undertakes to replace any Material(s) if found not to conform to the guarantees aforesaid at any time during the defect liability period applicable

thereto. CPCL shall give written notice of the defect to the Vendor and of the rejection of the defective Material(s).

16.4 If the defect can be rectified or repaired without diminishing the quality, utility, efficiency or life of the Material(s), instead of outright rejection of the Material(s), CPCL may at its discretion permit the Vendor to rectify the defect(s) within a period to be specified by CPCL in this behalf in the notice. In case Vendor fails to take action to rectify the defect(s) within the period specified to the satisfaction of CPCL, CPCL may at its discretion, at the risk and cost of the Vendor in all respects, rectify or repair or cause to be rectified or repaired the defect(s) either by itself or through any other source or agency, or reject the defective Material(s).

16.5 Should CPCL, notwithstanding the endeavour to do so, be unable to rectify or repair or get rectified or repaired the defect(s) within a reasonable time, CPCL may, notwithstanding such endeavour reject the defective Material(s).

16.6 The Vendor shall repair, rectify and/or replace, as the case may be, the defective and rejected Material(s) without entitlement to any extra payment. Prevailing DDP INCOTERMS (as per ICC) or **any other Incoterm which may replace DDP** shall apply for such replacement parts or components or Material(s) at Project Site with respect to imported parts or components or materials.

16.7 The Vendor shall at its own risk and cost remove any rejected Material(s) from the Project Site, and in case of plant, machinery, equipment, parts or components which have been installed, cause the same to be dismantled and removed from the Project Site subject to the Vendor in all cases prior to the removal of the rejected Material(s) from the Project site:

i) furnishing a bank guarantee to CPCL from a Scheduled bank in India (including an Indian branch of a foreign bank acceptable to CPCL) and in a format set forth in Annexure “E” hereto for the value paid by CPCL on the Material(s) rejected; and

ii) undertaking to replace the rejected Material(s) with other Material(s) conforming to the Vendor’s guarantees aforesaid applicable thereto.

CPCL shall be responsible for providing any on-site access to the vendor to perform warranty related work.

16.8 The Vendor shall not without the prior written consent of CPCL utilize any rejected Material(s) in the re-supply.

16.9 The Defect Liability Period with respect to any Material(s) replaced, repaired, altered and/or rectified shall be reckoned in accordance with Clause No. 1.1(d) (iii).

16.10 Should the Vendor fail to dismantle and/or remove any rejected Material(s) from the Project Site within the time specified in the notice of rejection, CPCL may without prejudice to any other right or remedy, at the risk and cost of the Vendor cause the rejected Material(s) to be dismantled and sold by public auction or private treaty as it deems fit and hold or adjust the sale proceeds for the recovery of the cost of

dismantling, sale and removal of the rejected Material(s) and any amount paid by CPCL towards the price of the rejected Material(s). In so doing, CPCL shall not act as a trustee or constructive trustee of the Vendor and shall be entitled to act solely on the basis of its best judgment without being accountable or liable to the Vendor in any manner except for the proceeds of the sale.

16.11 The time taken for the repair, rectification or replacement of Material(s) will not be added to the stipulated Delivery date for the purpose of calculating price discount, and delivery of such Material(s) shall be the date of Delivery of the repaired, rectified or replaced Material(s).

16.12 As security for the due performance of its obligations and the due discharge of its liabilities under the Contract, the Vendor shall within 15 (fifteen) days of the issue of the Purchase Order furnish to CPCL a Bank Guarantee issued by a Scheduled Bank in India acceptable to CPCL, in the format set forth hereto and marked Annexure “C” hereinbefore. The Bank guarantee shall remain in force for the entire period required for the performance of the contract and the defect liability period plus 3 (three) months claim period thereafter. Any shortfall in the value of the Bank Guarantee, as a result of encashment by CPCL either in full or in part, shall be made good by the Vendor within 7 (seven) days of notice by CPCL to the Vendor in this behalf. Any failure by the Vendor to furnish the Bank Guarantee or to enhance the Value of the Bank guarantee as stated above shall constitute a default by the Vendor for which CPCL shall, without prejudice to any other right or remedy available to it, be entitled to terminate the Contract with consequences as indicated in clause 12.4, the provisions whereof shall mutatis mutandis apply.

17.0 FREIGHT, TAXES AND DUTIES

17.1 Subject to the provision of Clause 17.2 hereunder, CGST & SGST / IGST payable on the supply and delivery of Material(s) pursuant to the contract will be paid at actual within the contractual delivery date. Any increase in the rates of the CGST & SGST/ IGST within the contractual completion date or approved extended contractual completion date will be borne by CPCL but not beyond the contractual delivery date or extended contractual delivery date. However, the benefit of any reduction must be passed on to CPCL.

17.2 Central GST (CGST) & State GST (SGST) / Integrated GST (IGST) payable or reimbursable by CPCL to the vendor on supply of indigenous Materials shall be included in and shown separately in the vendor's Tax invoice for the Material(s). The vendor shall prior to despatch of the Material(s) obtain from CPCL a list of the documents required by CPCL to enable it to avail of the relative benefits. Payment or reimbursement of CGST & SGST/IGST to CPCL shall be made upon the vendor furnishing the relevant documents.

17.3 Freight, if any, along with CGST & SGST/IGST thereon payable or reimbursable by CPCL shall be invoiced and shall be paid/reimbursed by CPCL after receipt of the material(s) at the project site.

17.4 Taxes and duties are not intended to operate as a profit centre but are intended only to meet the relevant costs incurred on this account. If any reimbursement or

collection of the taxes or duties by the vendor from the CPCL is in excess of the taxes and/or duties actually paid by the vendor, the vendor shall forthwith refund such excess to CPCL together with interest thereon at 1% (one percent) per annum above the MCLR rate (or rate that replaces it) of SBI from the date of collection until the date of refund.

18.0 WEIGHTS AND MEASUREMENTS

18.1 The shipping documents, invoices, packing lists and all other relevant documents shall contain the same units of weights and measurements as given in the Contract Documents, in respect to the following data:

a. Unit net weight
b. Unit gross weight (including packing)
c. Dimensions of packing

18.2 All weights and measurements recorded by the Procurement Coordinator or Inspector(s) on receipt of the Material(s) at the Project site will be treated as final.

19.0 PACKING & MARKING

19.1 All Material(s) shall be suitably packed in weatherproof seaworthy/airworthy packing for ocean/air transport under tropical conditions and/or for rail and road or other appropriate transport within India. The Vendor shall ensure that the packing is strong enough to ensure safety and preservation of the Material(s) upto the Project Site or other point of final destination.

19.2 Material(s) shall be protected by a suitable coat of paint and all bright parts shall be protected from rust by application of rust preventives as may be necessary. All machinery surfaces shall be suitably protected.

19.3 For uniform Material(s) when packed in several cases/crates, progressive serial numbers shall be indicated on each end. In case of bundles, the shipping marks shall be embossed on metal or tag and wired securely on each end.

19.4 A distinct colour splash in say red-black around each package/crate/bundle shall be given for identification.

19.5 All nozzle holes and openings as also all delicate surfaces shall be carefully protected against damage and bad weather. Flange faces of all nozzles shall be protected by blanks. All manufactured surfaces shall be painted with rust proof paint or as specified in the specification.

19.6 All threaded fittings shall be greased and provided with a plastic cap. All pipes and sheets shall be marked with strips bearing progressive numbers.

19.7 All small pieces shall be packed in cases. All fragile and exposed parts will be packed with care and packages will bear the words “HANDLE WITH CARE” in

English and in the case of Indigenous Supply, in Hindi also.

19.8 The Vendor shall be held liable for all damages or breakages to the Material(s) due to defective or insufficient packing as well as for corrosion due to insufficient greasing/protection.

19.9 On three sides of the packages, the Vendor shall affix or cause to be affixed the following marks clearly visible in indelible paint

FROM: VENDOR
TO: CHENNAI PETROLEUM CORPORATION LTD.
[Address]INDIA
PURCHASE ORDER NO.: [ ] Rev. No.: [ ]
10 DIGIT ITEM CODE : [ ]
EQUIPMENT NOMENCLATURE: [ ]
NET WEIGHT: [ ]kg/lb
GROSS WEIGHT: [ ]kg/lb
CASE NO.: [ ] OF [ ] TOTAL CASES
DIMENSIONS: [ ]
IMPORT LICENCE NO. [ ]

NOTE: Marking shall be bold with a minimum letter height of 5 cm.

19.10 a) For every shipment, packages must be marked with serial progressive numbering. The numbering will be progressively continued for each subsequent shipment covering the Contract.

b) All packages will bear warning signs on the outside denoting the center of gravity and sling marks. Packages that require special handling and transport shall have their centers of gravity and points at which they may be gripped clearly indicated and marked “Attention Special Load – Handle With Care” in English Language. Any other direction for handling shall also be clearly indicated on the package.

c) Top heavy containers will be marked either “TOP HEAVY” or “HEAVY ENDS’.

d) When packing is clean and light colored, a dark black stencil paint shall be acceptable. However, where packing is soiled or dark, a coat of flat Zinc white paint shall be applied and allowed to dry before applying the specific marking(s).

e) Colour codification shall be used to identify different items e.g., IBR, NACE, Fire Safety Items etc.

19.11 In case of large equipments like vessels, heat exchangers etc., documents contained in a waterproof envelope shall be fastened inside a shell connection with an identifying arrow sign “DOCUMENTS” applied with indelible paint.

20.0 DESPATCH INSTRUCTIONS

For Indigenous Material(s)

20.1 Unless otherwise advised by CPCL or the Procurement Coordinator in writing, Material(s) shall not be despatched without prior inspection and/or testing and Release Order/Material(s) Acceptance Certificate issued by the Inspector(s).

20.2 The Vendor shall exercise due care to ensure that the consignment is booked under appropriate railway classification, failing which any additional freight incurred by CPCL due to the Vendor booking the Material(s) under a wrong railway classification shall be borne by the Vendor.

20.3 The Material(s) shall be consigned in the name of the consignee viz.

Chennai Petroleum Corporation Limited.,

[Site address]

20.4 The Material(s) shall be transported only through bank approved transporters by the most economical and expeditious mode of transport to the destination as applicable for respective mode of despatches as follows:

a. By Rail in wagon load consignment to: *[Name and address of Refinery / Project to be specified]*

b. By road transport to: *[Name and address of Refinery / Project to be specified]*

CPCL and the Procurement Coordinator shall have the right to advise any change in despatch point or destination and/or mode of transport in respect of any Material(s). Any extra expenditure incurred by the Vendor on this account supported by satisfactory documentary evidence, will be reimbursed to the Vendor by CPCL.

21.0 SHIPMENT AND SHIPMENT NOTICES

21.1 Imported Material

The Vendor shall make shipment only after prior approval of the Inspector(s) unless otherwise specifically authorized in writing by CPCL or the Procurement Coordinator. As soon as any shipment is made, the Vendor shall send advance information by way of FAX message [Fax No : _____] to the Dy. General Manager, Chennai Petroleum Corporation Limited, for Chennai consignments (or other specified authority in respect of Material(s) consigned to other Ports), Giving particulars of the shipments, vessel's name, port of shipment, Bill of Lading number and date for ocean shipment and Airway Bill number & Date & Flight details for air consignment, total FOB and freight value with confirmation copy to CPCL, addressed to [designation and address]

21.2 Indigenous Material

Immediately after shipment, the Vendor shall inform despatch particulars to CPCL/ Procurement Coordinator hereunder:

1 copy to Dy. General Manager (Materials), [Address & Fax no]

1 copy to CPCL (Stores) addressed to [Designation, Address & Fax no.]

In the case of project orders, in addition to the above, a copy to the Dy. General Manager (Project) addressed to: [Address & Fax no.]

(PMC as per Purchase Order) addressed to: [Designation, Address & Fax no.]

22.0 MARINE AND TRANSIT RISK INSURANCE

22.1 Marine/Air and Transit Risk Insurance shall be covered by CPCL against its Open General Policy.

22.2 The Vendor shall send CPCL information of the proposed shipment/consignment well in advance by telegram/fax/e-mail/courier to enable CPCL to take necessary action for the marine/air/transit insurance of the shipment/consignment.

22.3 The Vendor shall advise the despatch particulars of the shipment/consignment to the Insurance Company by fax / email under advice to the Procurement Coordinator promptly after shipment to ensure that the consignment is fully covered by insurance. Any failure by the Vendor to do so shall place the consignment at the Vendor's risk.

23.0 SHIPPING AND SHIPPING DOCUMENTS

23.1 **For Imported Materials(s)**

a. The Vendor shall arrange with Vessel owner(s) or Forwarding Agent(s) specified in the Contract documents for proper storage of the Material(s) in a manner so as to facilitate the handling and off-loading at the port of destination and to avoid any over carriage or discharge.

b. The Bill(s) of Lading/Airway Bill shall be made out in favor of "Chennai Petroleum Corporation Limited" [Refinery/Unit] or to the order of the L/C opening bank, and the notify column should indicate [Designation] Chennai Petroleum Corporation Limited. [Insert Address] for port of discharge Mumbai and to the Dy. General Manager, Chennai Petroleum Corporation Limited [Insert Address] for port of discharge Chennai or other specified authority in respect of Material(s) consigned to other ports.

c. All columns in the body of the Bill of Lading/Airway Bill namely marks and Numbers, material description, weight particulars, etc. should be completed accurately and such statement should be uniform in all the shipping documents. The freight particulars should mention the basis of freight tonnage, heavy lift charges, if any, surcharge, discount, etc. clearly and separately and the net total freight payable, shown at the bottom.

d. The Bill of Lading/Airway Bill shall be free of any liability of CPCL to the carrier for demurrage.

e. The Bill of Lading/Airway Bill shall indicate the following:

Consignee : Chennai Petroleum Corporation Limited.

f. All documents viz. Bill of Lading/Airway Bill, invoices, packing list, freight memos, country of origin certificate, Third Party Inspection Release Certificate, drawings and catalogues should be in the English language.

g. i In addition to the Bill of Lading/Airway Bill, which should be obtained in 3 (three) stamped originals plus as many copies as required, invoices, packing lists, freight memos (if the freight particulars are not shown in the Bill of Lading), country of origin certificate(s), Third Party Inspection Release test/composition certificate, shall be made out against each shipment in as many number of copies as are shown in Clause 23.3.

ii. The Bill of Lading/Airway Bill, invoice and packing list must specifically show uniformly, the marks and numbers, material description, contents case-wise, country of origin, consignee's name, port of destination and all other indicated particulars. The invoice must show the unit rates and net total FOB price. The invoice must cover also items packed separately and the value shown accordingly.

iii. The packing list must show, apart from other particulars, the actual contents in each case, net and gross weights and dimensions and the total number of packages.

iv. All documents must be duly signed by an authorized representative of the Vendor.

23.2 In case of FOB contracts:

i) Shipping Arrangements shall be made through nominated freight forwarders (in the country of exit) as detailed in the Purchase Order and freight will be accordingly paid by CPCL in Indian Rupees.

ii) The Vendor shall furnish to the respective nominated freight forwarder, the full details of consignment such as outside dimensions, weights (both gross and net), No. of packages, technical description and drawings, name of the supplier, ports of loading etc. two weeks prior to the proposed date of shipment to enable the concerned agency to arrange the shipping space.

23.3 The Vendor shall obtain the shipping documents in required number of sets including three original stamped copies of the Bill of Lading/Airway Bill immediately after the shipment is made and airmail the shipping documents in the manner stipulated hereunder to ensure that the documents so forwarded are received at least one week before the vessel's arrival. The Vendor shall be fully responsible for any delay and/or demurrage in clearance of the consignment at the port due to delay in forwarding the shipping documents. If in terms of the Letter of Credit, the complete original set of documents are required to be sent to CPCL through the bank, the distribution indicated below will be confined to obtaining copies of documents only.

Documents	CPCL (Office that issued the PO)	CPCL (Port Office)	CPCL (Project Site)
Bill of lading/Airway Bill	3 (including 1 original)	1	1
Invoice	2	2	1
Packing List	2	2	1
Certificate of Origin	2	2	1
Test/Composition Certificate	2	2	1
Third Party Inspection Release Certificate	1	1	1 (original)
Drawing/Catalogue	1	1	1
Invoice of Third Party/Lloyds for Inspection Charges, wherever applicable.	2	1	1
Export Certificate (wherever applicable)	1	1	1

24.0 INVOICING & NEGOTIATION OF DOCUMENTS

24.1 Indigenous Material(s)

a. Unless otherwise directed by CPCL, Invoices and other documents shall be forwarded by the Vendor to CPCL as stipulated hereunder:

i. Original Invoice+1 copy of document as per Purchase Order to the Dy. General Manager (Finance), [Address, Fax No., e-mail id]

ii. 1 copy of Invoice with original documents as per Purchase Order to Dy. General Manager (Materials), [Address, Fax No., e-mail id] along with technical catalogues, Inspection certificate and Inspectors Release Note.

b. Where payment is to be released to the Vendor against despatch documents through the bank, the Vendor shall forward two complete original sets of the documents specified in clause 5.2(c) to the specified banker of CPCL with a stipulation that the banker shall forward one set of the documents to CPCL with LSC intimation and acceptance before retirement of the documents from the Bank. One complete set of all of the abovementioned documents shall be sent to the Dy. General Manager (Materials) [Address, Fax No., e-mail id]. Documents will not be retired by the bank failing strict compliance by the Vendor of the above mentioned despatch requirements.

c. Where payment is to be released to the Vendor directly by CPCL, two sets of all the documents mentioned in (b) above shall be submitted/forwarded directly to the Dy. General Manager (Materials) [Address, Fax No., e-mail id] and upon processing, the payment shall be released through the Finance Department, by cheque/RTGS/NEFT.

25.0 TRANSFER OF TITLE AND RISK OF LOSS

25.1 For Indigenous Material(s):

The transfer of property and risk in Indigenous Material(s) shall be deemed to take place as follows:

a) For delivery F.O.R. or F.O.T. despatch point: On handing over the Material(s) to the carrier against receipt of clean Railway Receipt/Truck or Lorry Receipt and such receipt having been handed over to CPCL.

b) For despatch F.O.R destination station: On removal of the Material(s) by CPCL from the railway authorities at the destination station.

c) Equipment sent freight/carriage paid to the Project Site: On receipt of the Material(s) by CPCL at the Project Site.

25.2 For Imported Material(s):

The transfer of property and risks in Imported Material(s) shall be deemed to take place as follows:

a) For FOB / FCA deliveries: On handing over the Material(s) to the carrier and issue of clean Bill of Lading/Airway Bill (except for freight to pay) and its transmission to CPCL.

b) For CFR deliveries: Transfer of title and risk of loss passes to CPCL when delivered on board the ship by vendor who pays the transportation cost upto the destination port or to the designated destination airport (as per INCOTERMS).

26.0 TERMINATION

26.1 Without prejudice to CPCL's right to price adjustment by way of discount or any other right or remedy available to CPCL, CPCL may terminate the Contract or any part thereof by a written notice to the Vendor, if:

i) The Vendor fails to comply with any material term of the Contract.

ii) The Vendor informs CPCL of its inability to deliver the Material(s) or any part thereof within the stipulated Delivery Period or such inability otherwise becomes apparent.

iii) The Vendor fails to deliver the Material(s) or any part thereof within the stipulated Delivery Period and/or to replace/rectify any rejected or defective Material(s) promptly.

iv) The Vendor becomes bankrupt or goes into liquidation.

v) The Vendor makes a general assignment for the benefit of creditors.

vi) A receiver is appointed for any substantial property owned by the Vendor.

vii) The Vendor has misrepresented to CPCL, acting on which misrepresentation CPCL has placed the Purchase Order on the Vendor.

viii) Violation of Integrity Pact, if applicable.

26.2 Upon receipt of said termination notice, the Vendor shall discontinue the work on the Contract so far as terminated, and matters connected therewith.

26.3 On termination of the Contract, without prejudice to any other right or remedy available to CPCL under the contract, in the event of CPCL suffering any loss on account of delayed delivery or non-delivery, CPCL reserves the right to claim and recover damages from the Vendor in respect thereof.

26.4 Notwithstanding anything to the contrary herein contained, CPCL will be at liberty to take independent administrative action to place the Vendor under 'holiday list' for delay or non-performance of its contractual obligations or any of them.

26.5 Action in case of submission of fraudulent documents:

If any information given by a bidder/ tenderer is found to be incorrect in any particular considered by CPCL to be relevant for the evaluation of the bid/ tender, or is found by CPCL to misrepresent or conceal facts, or if any of the documents submitted by the bidder in support of or relevant to the bid/ tender is found by CPCL to be forged, false or fabricated, CPCL may reject the bid, and without prejudice to any other right(s) of action or remedy available to CPCL, CPCL may forfeit the Earnest Money given by the bidder in order to compensate CPCL for the expenses incurred by it in considering the bid (and not by way of penalty) and take action for putting the bidder on holiday list for such period as CPCL in this behalf considers warranted and / or remove the bidder from any approved list of vendors / contractors. If prior to the discovery of the incorrect information, misrepresentation or forged, false or fabricated document(s), the bid has resulted in a contract, the Contract shall be liable to be terminated by CPCL with consequences of termination as provided in Cl 26 of the General Purchase Conditions.

27.0 RECOVERY OF SUMS DUE

27.1 Whenever there is any claim against the Vendor for payment of sums of money arising out of or under the execution of the Purchase Order, CPCL may, without prejudice to any other mode or source of recovery available, recover the same from any sum(s) then due or which at any time thereafter may become due to the Vendor under this or any other contract with CPCL and/or by recourse to any bank guarantee available to CPCL for this purpose, and should these sum(s) be not sufficient to cover the recoverable amount, the Vendor shall pay CPCL on demand, the balance.

28.0 NON-WAIVER

28.1 Failure of CPCL/CPCL's representative(s) to insist upon adherence to any of the terms or conditions incorporated in the Contract or failure or delay to exercise any rights or remedies herein or by law accruing, or failure to promptly notify the

Vendor in the event of breach or the acceptance of or the payment for any Material(s) hereunder or approval of any design or Material(s) shall not release the Vendor and shall not be deemed a waiver of any right of CPCL to insist upon the strict performance thereof or of any of its rights or remedies as to any such Material(s) regardless of when the Material(s) are shipped, received or accepted nor shall any purported oral modifications or revisions of the Contract by CPCL or CPCL's representative(s) act as a waiver of the terms hereof.

29.0 COMPLETE AGREEMENT

29.1 The Contract Document(s) including, technical documents, drawings and specifications and other Annexures to the Contract documents constitute the entire agreement between the Vendor and CPCL in relation to the Contract arising out of the Purchase Order. Changes or amendments will be binding only if the amendments/modification are agreed to in writing as an amendment/variation to the relative Contract Document and signed by an authorised representative of CPCL.

30.0 EXCLUSION OF THE GOVT. OF INDIA'S LIABILITY

30.1 It is expressly understood and agreed that CPCL has entered into the Contract solely on its own behalf and not on behalf of any other person or entity. In particular, it is expressly understood and agreed that the Government of India is not a party to the Contract and will have no liability, obligation or right whatsoever under the Contract. It is expressly understood and agreed that CPCL is an independent legal entity with power and authority to enter into the Contract solely on its own behalf under the applicable laws of India. The Vendor expressly agrees, acknowledges and undertakes and accepts that CPCL is not an agent, representative or delegate of the Government of India and that the Government of India is not and shall not be liable for any act, omission, commission, breach or other wrong or liability arising out of the Contract. The Vendor hereby expressly waives, releases and forgoes any and all actions or claims, including cross claims, impleader claims or counter claims, against the Government of India arising out of the Contract and covenants not to sue the Government of India for any manner of claim, cause of action or thing whatsoever arising out of or under the Contract.

31.0 TECHNICAL INFORMATION/CONFIDENTIALITY

31.1 Drawings, specifications and details shall at all times be the property of CPCL and shall be returned by the Vendor to CPCL on demand. The Vendor shall not make use of any document, drawing, specification, data or any other information connected with the Contract, for any purpose at any time, save and except in the implementation of the Contract.

31.2 The Vendor shall not disclose the technical information furnished or gained by the Vendor under or by virtue of or as a result of the implementation of the Contract and shall make all efforts to ensure that the technical information is kept confidential.

31.3 Notwithstanding anything contained in this contract to the contrary, nothing contained herein shall be construed as transferring between the parties any patent, trademark, copyright or other intellectual property rights relating to the supply of equipment and / or materials, or provisions of services covered by this contract, and all such rights are hereby expressly reserved to the true and lawful owners thereof.

32.0 MODE OF COMMUNICATION

32.1 It is understood that CPCL is a multifaceted and multi-locational organization and that for any communication to receive attention it is a prerequisite that the communication is properly addressed and that the subject of the communication is properly identified. To this end, except where otherwise provided, all communications, drawings, invoices, shipping notes, packages, packing lists and other document connected with reference to the Contract shall in the heading boldly set out the Purchase Order Number and date, and shall be addressed to the following:

Dy. General Manager (Materials) [Office issuing the Purchase Order]	Phone	Fax
Dy. General Manager (Projects) [Site address]	Phone	Fax
Dy. General Manager CPCL [concerned Port office]	Phone [Concerned Office Address]	Fax

(For imports)

With copies to the following in case a Procurement Co-ordinator or Project Management Consultant is involved:

(i) Resident Construction Manager Phone Fax
[Site address]

(ii) Dy. General Manager (Inspection) of the Procurement Co- ordinator/Project Management Consultant.

33.0 PART ORDER/REPEAT ORDER

33.1 The Vendor hereby agrees to accept part order at unit prices without limitation and also agrees in case of bulk consumables to accept repeat order(s) during a period of 6 (six) months from the date of original Purchase Order at the same unit price(s) and on the same terms and conditions.

34.0 ARBITRATION AND GOVERNING LAW

34.1 Arbitration: All disputes or differences which may arise out of or in connection with the Contract including any dispute or difference regarding the interpretation of the terms and conditions of any clause thereof which cannot be amicably resolved between the parties may be referred to Arbitration.

The Vendor may select one person out of a panel of three persons nominated by CPCL who shall act as the sole arbitrator.

The Arbitration proceedings shall be governed by and conducted in accordance with the Arbitration and Conciliation Act, 1996 as amended from time to time. The seat of the arbitration shall be New Delhi, India. However, with the consent of CPCL and the Vendor, the arbitrator may agree upon any venue at London or Singapore.

Laws of India only would be applicable.

34.2 Arbitration in case of order on Indian CPSE shall be governed by the prevailing guidelines of Department of Public Enterprises (DPE).

34.3 **CONCILIATION**

At any time prior to or during arbitration of any arbitrable dispute(s) pursuant to CPCL Arbitration Clause if eligible, the Vendor may seek resort to the Conciliation under the CPCL Conciliation Rules, 2014 as amended and / or re-enacted from time to time. The proposal for conciliation shall be made in accordance with said Conciliation Rules. The said Rules are available on the CPCL website.

34.4 Governing Law: The Contract(s) shall be construed in accordance with and governed by the laws of India. CPCL shall warrant that the terms and conditions of the Purchase Order shall be valid under existing Indian laws.

35.0 JURISDICTION

35.1 Notwithstanding any other Court or Courts having jurisdiction to decide the disputed issue, and without prejudice to the provisions or generality of the Arbitration clause, jurisdiction to decide the question(s) arising out of or relative to the Contract in all matters touching or affecting any arbitration, or arising out of or in relation to or under or in accordance with the Arbitration and Conciliation Act, 1996 or otherwise under or with reference to the Contract shall vest exclusively in the court(s) of competent civil jurisdiction at _____ [where the contract(s)/Purchase Order shall be signed on behalf of CPCL] or at New Delhi and only the said Court(s) shall have the jurisdiction to entertain and try any such actions and/or proceedings to the exclusion of all other Courts, provided that nothing herein stated shall be deemed to anywise authorize any party to seek resolution of any dispute(s) otherwise than the recourse to arbitration in accordance with the provisions of the Arbitration clause herein.

Provided always that an award rendered in any arbitration proceedings arising out of or in relation to the Contract may be enforced or executed in any other

country or jurisdiction including without limitation a country in which any party against whom the award is to be enforced or executed is located and a country in which the assets of any such party are located.

36.0 LIMITATION OF LIABILITY

36.1 The maximum liability shall be limited to 100% of total contract value except in case of fraud, deliberate default, negligence or misconduct by the vendor. The vendor shall not be liable for consequential losses on account of production, revenue or profit.

37.0 USE OF WHITE/ ERASING FLUID

37.1 In case the bidder has been asked to submit price bid/ price implication in physical form, the use of white/ erasing fluid for correcting the rates is banned. Wherever the rates are corrected with white/ erasing fluid, the bids will be summarily rejected.

* * * *

Annexure – A

FORM OF BANK GUARANTEE IN LIEU OF EARNEST MONEY DEPOSIT

(On non-judicial stamp paper of appropriate value)

BG NO:_____
DATED:_____
VALID UPTO:_____

To,
Chennai Petroleum Corporation Limited
(Refineries Division)

[Address]Dear

Sirs,

In consideration of the Chennai Petroleum Corporation Limited [Address] ……., (herein after called) which expression shall include its successors and assigns) having agreed interalia to consider the tender of _____(Name)_____, (Constitution)_____(Address)_____ (hereinafter referred to as the "Tenderer" which expression shall wherever the subject or context so permits include its successors and assigns) for supply of materials to be awarded under Tender No._____upon the Tenderer furnishing an undertaking from the Bank as hereinafter appearing in lieu of payment of the Earnest Money.

We _____(name) a body corporate registered/constituted under the laws of _____and having a branch at [Indian branch address with IFSC Code] (hereinafter called "the Bank" which expression shall include its successors and assigns), at the request of the Tenderer and with the intent to bind the Bank and its successors and assigns do hereby unconditionally and irrevocably undertake to pay the Corporation at New Delhi forthwith on first demand without protest or demur or proof or satisfaction or condition and without reference to the Tenderer, all sums payable by the Tenderer as and by way of Earnest Money to the Corporation, for an amount of Rs. _____ (Rupees_____) only or US Dollar_____ (United States Dollars_____)

AND THE BANK DOTH HEREBY FURTHER AGREE AS FOLLOWS:

1. The amount stated by the Corporation in any demand, claim or notice made or given with reference to this Guarantee/Undertaking shall as between the Bank and the Corporation for the purpose of this Guarantee/Undertaking be conclusive of the amount payable by the Bank to the Corporation hereunder.

2. This Guarantee/Undertaking shall not be determined or affected by the liquidation or winding up or dissolution or change of constitution or insolvency of the Tenderer or by any change in the legal constitution of the Bank or the Corporation.

3. Without prejudice to any other mode of service, a demand or claim or other communication may be transmitted by the Corporation to the Bank either by post or by email. If transmitted by email, the transmission shall be complete as soon as the mail is delivered to the mail id of the Bank mentioned in the BG.

4. Notwithstanding anything to the contrary contained herein above:

   i. The Bank's liability under this Guarantee/Undertaking shall not exceed Rs. / US $ _____ (Rupees / United State Dollars …… only).

   ii. This Guarantee/Undertaking shall remain in force upto (fill in date) and any extension(s) thereof; and

   iii. The Bank shall be released and discharged from all liability under this Guarantee/Undertaking unless a written claim or demand is issued to the Bank on or before the midnight of_____and if extended, the date of expiry of the last extension of this Guarantee/Undertaking.

5. Our Guarantee will be governed and construed in accordance with Indian Laws and subject to exclusive jurisdiction of Courts in India.

6. The Bank doth hereby declare that Shri [designation] is authorised to sign this Guarantee/Undertaking on behalf of the Bank and to bind the Bank thereby.

This_____day of_____20

Yours faithfully,
Signature:_____
Name & Designation:_____
Name of the Branch:_____

NOTES:

1. The Bank Guarantee in lieu of Earnest Money Deposit shall be strictly as per above proforma and shall be through a Scheduled Bank operating in India, including the Indian Branch of a foreign bank recognized as a Scheduled bank in India.
2. The Bank Guarantee shall be typed on stamp paper applicable to an agreement in the State in which the Bank Guarantee is issued.

   This Bank Guarantee shall be sent by the Bank directly to:- Dy. General Manager (Materials), CPCL.

Annexure - B

[IRREVOCABLE LETTER OF CREDIT- FORMAT]

To:

This communication is to be considered as our credit instrument and is subject to the Uniform Customs and Practice for Documentary Credit 1993 revision, I.C.C. publication No. 500.

At the request of Chennai Petroleum Corporation Limited. (hereinafter called "the Corporation" which expression shall include its successors and assignees) we hereby establish our irrevocable Letter of Credit No. _____dated _____in favour of :

(Beneficiary)
for a sum not exceeding (Currency in figures and words) which is payable at sight at the counter of [Beneficiary's Bank] in [Address], covering the FOB/CIF/C&F port supply of Material(s) and equipment under Purchase Order No. [ ] for the Unit/Refinery of Chennai Petroleum Corporation Limited.. (CPCL).

This Letter of Credit shall be valid up till [Date]

TERMS OF PAYMENT

Payments under this Letter of Credit shall be made of _____ % of the value of the Invoice(s) drawn by the Beneficiary on CPCL under the said Purchase Order against presentation of the following documents:

ImportedMaterials:

i. Invoice;
ii. Clear Bill of Lading/Airway Bill covering the Materials(s) invoiced;
iii. Packing list for the consignment;
iv. Material(s) Inspector's Certificate covering the invoiced.
v. Test/Composition Certificate;
vi. Certificate of origin;
vii. Drawing(s)/Catalogue(s) covering the Material(s);
viii. Export Certificate, wherever applicable;
ix. Invoice of Inspector's charges, wherever applicable.
x. Freight Memo(s) if the freight particulars are not shown in the Bill of Lading/Airway Bill.
xi. Copy of faxed/e-mail intimation of the shipment to CPCL certified by the Beneficiary to be a true copy of the faxed intimation of shipment to CPCL

IndigenousSupply:

i. Invoice;
ii. Clear Railway Receipt/Truck Receipt/Goods Receipt covering the Materials(s) invoiced;
iii. Packing list for the consignment;

iv. Third Party Inspector's Certificate covering the invoiced Material/Release Note.
v. Test/Composition Certificate;
vi. IBR Certificate/CMRS Certificate;
vii. Drawing(s)/Catalogue(s) covering the Material(s);
viii. Guarantee/Warranty Certificate(s), wherever applicable;
ix. Invoice of Inspector's charges, wherever applicable.
x. Freight Memo(s) if inland freight is not included in the Price and the RR/TR/GR does not give the freight particulars..
xi. Copy of faxed/e-mail intimation of the shipment to CPCL certified by the Beneficiary to be a true copy of the faxed intimation of shipment to CPCL

SPECIAL INSTRUCTIONS:

* All Bank charges and commissions outside India shall be borne by the Beneficiary
* Drawing permitted in not more than 3 (three) tranches.
* Partial shipment is not permitted.
* Trans-shipment not permitted.
* Negotiating Bank to reimburse itself after giving 7 (seven) banking days notice to the Issuing Bank from Issuing Bank's account with ____ (Name of the Issuing Bank's correspondent in _____) at maturity under telex/cable advice to the Issuing Bank indicating the amount claimed.
* CPCL to be intimated the shipping details by fax immediately on shipment.
* Documents to be dispatched to Issuing Bank by courier within 7 banking days of receipt by Negotiating Bank.
* Documents must be presented within _____ days after the date of shipment/dispatch to ensure receipt thereof 7 days prior to the arrival of the vessel.

* * * *

Annexure -C

[PROFORMA OF BANK GUARANTEE (PERFORMANCE)]

(On non-judicial stamp paper of appropriate value)

To
Chennai Petroleum Corporation Ltd.
[Address]

Dear Sirs,

In consideration of the Chennai Petroleum Corporation Ltd. (hereinafter called the "Corporation" which expression shall include its successors and assigns) having awarded to [Name], [Constitution] _____ [Address] _____ (hereinafter referred to as the "Supplier" which expression shall wherever the subject or context so permits include its successors and assigns) a Supply Contract in terms interalia, of the Corporation(s) Purchase Order No. [ ] dated ..... (hereinafter referred to as the "Contract") upon the condition of the Supplier's furnishing security in lieu of cash deposit for the Performance of the Supplier's obligations and/or discharge of the Supplier's liabilities under and/or in connection with the said Supply Contract upto a sum of Rs./US$ [ ] [Rupees/United States Dollars_____] only amounting to __% (__ percent) of the total contract value.

We _____(name) a body corporate registered/constituted under the laws of _____and having a branch at [Indian branch address with IFSC Code] (hereinafter called "the Bank" which expression shall include its successors and assigns) with the intent to bind the Bank and its successors and assigns, hereby undertake to pay the Corporation at New Delhi on first demand in writing without protest or demur or proof or satisfaction or condition and without reference to the Supplier, any and all amounts from time to time demanded by the Corporation from the Bank with reference to this Guarantee/Undertaking upto an aggregate limit of Rs./US$ [ ] only or [Rupees / United States Dollars_____].

AND THE BANK DOTH HEREBY FURTHER AGREE AS FOLLOWS:

1. This Guarantee/Undertaking shall be in addition to any other guarantee or any other security whatsoever that the Corporation may now or at any time anywise have in relation to the Supplier's obligations and/or liabilities under and/or in connection with the Contract and the Corporation shall have full authority to take recourse to or reinforce this Guarantee/Undertaking in preference to the other guarantee(s) or security(ies) at its discretion, and no failure on the part of the Corporation in enforcing or requiring enforcement of any other guarantee or security shall have the effect of releasing the Bank from its full liability hereunder.

2. The Corporation shall be at liberty without reference to the Bank and without affecting the full liability of the Bank hereunder to take any other security in respect of the Supplier's obligation(s) and/or liability(ies) under/or in connection with the Contract and/or to vary the terms vis-a-vis the Supplier of the Contract and/or to grant time and/or indulgence to the Supplier or to reduce or to increase or otherwise vary the prices or the total contract value or to release or to forebear from enforcement of all or any of the obligations of the Supplier under the Contract and/or the remedies of the Corporation under any other guarantee(s) or security(ies) now or hereafter held by the Corporation and no such dealing(s),

variation(s), reduction(s) or other indulgence(s) or arrangement(s) with the Supplier or release or forbearance whatsoever shall have the effect of releasing the Bank from its full liability to the Corporation hereunder or of prejudicing the rights of the Corporation against the Bank.

3. This Guarantee/Undertaking shall not be determined or affected by the liquidation or winding up or dissolution or change of constitution or insolvency of the Supplier or any change in the legal constitution of the Bank or of the Corporation.

4. The Bank hereby waives all rights at any time inconsistent with the terms of this Guarantee/Undertaking and the obligations of the Bank in terms hereof shall not be anywise affected or suspended by reasons of any dispute or disputes having been raised by the Supplier (whether or not pending before any Arbitrator, Officer, Tribunal or Court) or any denial of liability by the Supplier or any other order or communication whatsoever by the Supplier stopping or preventing or purporting to stop or prevent any payment by the Bank to the Corporation in terms hereof.

5. Notwithstanding anything to the contrary contained herein above:

   a) The Bank's liability under this Guarantee/Undertaking shall not exceed (currency in figures) . . . . . (currency in words only) . . . . ;

   b) This Guarantee/Undertaking shall remain in force upto _____ (here put the entire period required for the performance of the Contract and the defect liability period plus 3 (three) months) and any extension(s) thereof; and

   c) The Bank shall be released and discharged from all liability under this Guarantee/Undertaking unless a written claim or demand is issued to the Bank on or before_____ and if extended, the date of expiry of the last extension of this Guarantee/Undertaking.

6. Our Guarantee will be governed and construed in accordance with Indian Laws and subject to exclusive jurisdiction of Courts in India.

7. Without prejudice to any other mode of service, a demand or claim or other communication may be transmitted by the Corporation to the Bank either by post or by email. If transmitted by email, the transmission shall be complete as soon as the mail is delivered to the mail id of the Bank mentioned in the BG.

8. The Bank doth hereby declare that Mr._____[name & designation of the person authorised to sign on behalf of the Bank] is authorised to sign this Guarantee/Undertaking on behalf of the Bank and to bind the Bank thereby.

Yours faithfully,

(Authorised Signature)

NAME & DESIGNATION

NAME OF THE BANK

Annexure D

PROFORMA OF BANK GUARANTEE
(FULL VALUE OF FREE ISSUE MATERIALS)

(On non-judicial stamp paper
Of appropriate value)

To

Chennai Petroleum Corporation Ltd.
[Address]

Dear Sirs,

In consideration of the Chennai Petroleum Corporation Ltd. [address] (hereinafter called the "Corporation" which expression shall included its successors and assigns) having having [Name], _____(Constitution) _____ (Address) awarded to _____ (hereinafter referred to as the "Supplier" which expression shall wherever the subject or context so permits include its successors and assigns) a Supply Contract in terms, interalia, of the Corporation(s) Purchase Order No. [_____] dated [_____] (hereinafter referred to as the Contract") and having agreed to provide certain free issue material(s) hereinafter referred to as "Free Issue Material" to the Supplier for incorporation in the Condition of the supplier furnishing security in lieu of cash for the said Free Issue Material(s) upto a sum of Rs./US$ _____ (Rupees/ United States Dollars _____ only] being the value of the Free Issue Material(s):

We_____[name] a body corporate registered/constituted underthe laws of [Country] and having a branch at [Indian branch address with IFSC code] (hereinafter called "the Bank" which expression shall included its successors and assigns) with the intent to bind the Bank and its successors and assigns, hereby undertake to pay the Corporation at New Delhi on first demand in writing without protest or demur or proof or satisfaction or condition, and without reference to the Supplier, any and all amounts from time to time demanded by the Corporation from the Bank with reference to this Guarantee/Undertaking upto an aggregate limit of Rs./US$ _____United States Dollars _____only]

AND THE BANK DOTH HEREBY FURTHER AGREE AS FOLLOWS:

1. This Guarantee/Undertaking shall be in addition to any other guarantee or any other security whatsoever that the Corporation may now or at any time anywise have in relation to the Supplier's obligations and/or liabilities under and/or in connection with the Contract and/or the Free Issue Material and the Corporation shall have full authority to take recourse to or enforce this Guarantee/Undertaking in preference to the other guarantee(s) or security(ies) at its discretion, and no failure on the part of the Corporation in enforcing or requiring enforcement of any other guarantee or security shall have the effect of releasing the Bank from its full liability hereunder.

2. The Corporation shall be at liberty without reference to the Bank and withoutaffecting the full liability of the Bank hereunder, to take any other security in respect of the Supplier's obligation(s) and/or liability under or in connection with the Contract or the Free Issue Material and to vary the terms vis-a-vis the Supplier of the Contract or the Free Issue Material and/or to grant time and/or indulgence to the Supplier and/or to reduce or to increase or otherwise vary the prices or the total contract value or the value of the Free IssueMaterial or to release or to forebear from enforcement of all or any of the obligations of the Supplier under the Contract and/or the remedies of the Corporation under any other guarantee(s) or security(ies) now or hereafter held by the Corporation and no such dealing(s), variation(s), reduction(s) or other indulgence(s) or arrangement(s) with the Supplier or release or forbearance whatsoever shall have the effect of releasing the Bank from its full liability to the Corporation hereunder or of prejudicing the rights of the Corporation against the Bank.

3. This Guarantee/Undertaking shall not be determined or affected by the liquidation or winding up or dissolution or change of constitution or insolvency of the Supplier or any change in the legal constitution of the Bank or of the Corporation.

4. The Bank hereby waives all rights at any time inconsistent with the terms of this Guarantee/Undertaking and the obligations of the Bank in terms hereof shall not be anywise affected or suspended by reasons of any dispute or disputes having been raised by the Supplier (whether or not pending before any Arbitrator, Officer, Tribunal or Court) or any denial of liability by the Supplier or any other order or communication whatsoever by the Supplier stopping or preventing or purporting to stop or prevent any payment by the Bank to the Corporation in terms hereof.

5. Notwithstanding anything to the contrary contained herein above:

   a. The Bank's liability under this Guarantee/Undertaking shall not exceed [currency in figures _____ [currency in words] _____ only.

   b. This Guarantee/Undertaking shall remain in force upto the midnight of (here put the scheduled date of delivery of the Free Issue Material + 3 months) and any extension thereof.

   c. The Bank shall be released and discharged from all liability under this Guarantee/Undertaking unless a written claim or demand is issued to the Bank on or before [ ] and if extended, the date of expiry ofthe last extension of this Guarantee/Undertaking.

6. The Bank doth hereby declare that Mr._____ (name &designation of the person authorised to sign on behalf of the Bank) is authorised to sign this Guarantee/Undertaking on behalf of the Bank and to bind the Bank thereby.

7. Our Guarantee will be governed and construed in accordance with Indian Laws and subject to exclusive jurisdiction of Courts in India.

8. Without prejudice to any other mode of service, a demand or claim or other communication may be transmitted by the Corporation to the Bank either by post or by email. If transmitted by email, the transmission shall be complete as soon as the mail is delivered to the mail id of the Bank mentioned in the BG.

Yours faithfully,
(Signature)
NAME & DESIGNATION
NAME OF THE BANK

NOTES:

1. The Bank Guarantee shall be for the full value of the Free Issue Material(s) as specified by CPCL and shall be strictly as per above proforma and shall be through a Scheduled Bank operating in India, including the Indian Branch of a foreign bank recognized as a Scheduled Bank in India.

2. The Bank Guarantee shall be typed on stamp paper applicable to an agreement in the State in which the Bank Guarantee is issued.

3. This bank guarantee shall be sent by the Bank directly to:- Dy. General Manager (Materials), CPCL [Address].

Annexure E

PROFORMA OF BANK GUARANTEE
(FULL VALUE OF FREE ISSUE MATERIALS)

(On non-judicial stamp paper
Of appropriate value)

To

Chennai Petroleum Corporation Ltd.
[Address]

Dear Sirs,

In consideration of the Chennai Petroleum Corporation Ltd. [address] (hereinafter called the "Corporation" which expression shall included its successors and assigns) having [Name], _____(Constitution) _____ (Address) awarded to _____ (hereinafter referred to as the "Supplier" which expression shall wherever the subject or context so permits include its successors and assigns) a Supply Contract in terms, interalia, of the Corporation(s) Purchase Order No. [_____] dated [_____] (hereinafter referred to as the Contract") and having agreed to permit the Supplier to remove certain rejected material(s) for which the Corporation has paid the Supplier (hereinafter referred to as the "Rejected Material(s)") upon the condition of the Supplier furnishing security for the satisfactory replacement of the Rejected Material(s) upto a sum of Rs./US$ _____ (Rupees/ United States Dollars _____ only] amounting to the full value of the Rejected Material(s).

We_____[name] a body corporate registered/constituted underthe laws of [Country] and having a branch at [Indian branch address with IFSC code] (hereinafter called "the Bank" which expression shall included its successors and assigns) with the intent to bind the Bank and its successors and assigns, hereby undertake to pay the Corporation at New Delhi on first demand in writing without protest or demur or proof or satisfaction or condition, and without reference to the Supplier, any and all amounts from time to time demanded by the Corporation from the Bank with reference to this Guarantee/Undertaking upto an aggregate limit of Rs./US$ _____United States Dollars _____only]

AND THE BANK DOTH HEREBY FURTHER AGREE AS FOLLOWS:

1. This Guarantee/Undertaking shall be in addition to any other guarantee or any other security whatsoever that the Corporation may now or at any time anywise have in relation to the Supplier's obligations and/or liabilities under and/or in connection with the Contract and/or the Rejected Material(s) and the Corporation shall have full authority to take recourse to or enforce this Guarantee/Undertaking in preference to the other guarantee(s) or security(ies) at its discretion, and no failure on the part of the Corporation in enforcing or requiring enforcement of any other guarantee or security shall have the effect of releasing the Bank from its full liability hereunder.

2. The Corporation shall be at liberty without reference to the Bank and without affecting the full liability of the Bank hereunder, to take any other security in respect of the Supplier's obligation(s) and/or liability under or in connection with the Contract and/or the Rejected Material(s) and to vary the terms vis-a-vis the Supplier of the Contract or Rejected Material(s) and/or to grant time and/or indulgence to the Supplier and/or to reduce or to increase or otherwise vary the prices of the total contract value or the value of the Rejected Material or to release or to forebear from enforcement of all or any of the obligations of the Supplier under the Contract or otherwise in respect of the Rejected Material and/or the remedies of the Corporation under any other guarantee(s) or security(ies) now or hereafter held by the Corporation and no such dealing(s), variation(s), reduction(s) or other indulgence(s) or arrangement(s) with the Supplier or release or forebearance whatsoever shall have the effect of releasing the Bank from its full liability to the Corporation hereunder or of prejudicing rights of the Corporation against the Bank.

3. This Guarantee/Undertaking shall not be determined or affected by the liquidation or winding up or dissolution or change of constitution or insolvency of the Supplier or any change in the legal constitution of the Bank or of the Corporation.

4. The Bank hereby waives all rights at any time inconsistent with the terms of this Guarantee/Undertaking and the obligations of the Bank in terms hereof shall not be anywise affected or suspended by reasons of any dispute or disputes having been raised by the Supplier (whether or not pending before any Arbitrator, Officer, Tribunal or Court) or any denial of liability by the Supplier or any other order or communication whatsoever by the Supplier stopping or preventing or purporting to stop or prevent any payment by the Bank to the Corporation in terms hereof.

5. Notwithstanding anything to the contrary contained herein above:

   d. The Bank's liability under this Guarantee/Undertaking shall not exceed [currency in figures _____ [currency in words] _____ only.

   e. This Guarantee/Undertaking shall remain in force upto (put date fixed for replacement of the Rejected Material + 3 months) and any extension(s) thereof and

   f. The Bank shall be released and discharged from all liability under this Guarantee/Undertaking unless a written claim or demand is issued to the Bank on or before [ ] and if extended, the date of expiry of the last extension of this Guarantee/Undertaking.

6. The Bank doth hereby declare that Mr._____ (name &designation of the person authorised to sign on behalf of the Bank) is authorised to sign this Guarantee/Undertaking on behalf of the Bank and to bind the Bank thereby.

7. Without prejudice to any other mode of service, a demand or claim or other communication may be transmitted by the Corporation to the Bank either by post or by email. If transmitted by email, the transmission shall be complete as soon as the mail is delivered to the mail id of the Bank mentioned in the BG.

8. Our Guarantee will be governed and construed in accordance with Indian Laws and subject to exclusive jurisdiction of Courts in India.

Yours faithfully,
(Signature)
NAME & DESIGNATION
NAME OF THE BANK

NOTES:

1. The Bank Guarantee for full value of the Rejected Material(s) shall be strictly as per above proforma and shall be through a Scheduled Bank operating in India, including the Indian Branch of a foreign bank recognized as a Scheduled Bank in India.

2. The Bank Guarantee shall be typed on stamp paper applicable to an agreement in the State in which the Bank Guarantee is issued.

3. This bank guarantee shall be sent by the Bank directly to:- Dy. General Manager (Materials), CPCL [Address].

Annexure F

PROFORMA OF BANK GUARANTEE

(ADVANCE)

(On non-judicial paper of appropriate value)

To
Chennai Petroleum Corporation Ltd.
……
……
……

Dear Sirs,

WHEREAS CHENNAI PETROLEUM CORPORATION LIMITED (Hereinafter called "The CPCL" which expression shall include its successors and assigns) has awarded M/s _____ (hereinafter called "The Supplier" which expression shall include its successors and assigns) the work of designing, manufacturing, fabricating and supply of in terms of a contract as constituted by Purchase Order No. dated _____ issued by the CPCL to the Supplier (hereinafter called "The Contract" which expression include all the amendments and/or modifications of the Purchase Order).

AND WHEREAS the CPCL has agreed to advance the Supplier a sum of Currency _____ (Currency _____) hereinafter called "The Advance" as financial assistance to the Supplier under the Contract on the condition, inter alia, that the advance shall be secured by a Bank Guarantee as hereinafter appearing.

We _____(name) a body corporate registered/constituted under the laws of _____and having a branch at [Indian branch address with IFSC Code] (hereinafter called "The Bank" which expression shall include our successors and assigns) in consideration of the aforesaid promises and at the request of the Supplier DO HEREBY bind ourselves and our successors and assigns, and irrevocably undertake to pay the CPCL at New Delhi on first demand in writing without protest or demur or proof or condition and without reference to the Supplier any and all amounts at any time and from time to time claimed by you, as due to you under or in respect of the said advance, and demanded by you from us, with reference to this undertaking upto an aggregate limit of Currency (Currency _____).

AND, we, the Bank DO HEREBY further agree as follows:

i. The CPCL shall have the fullest liberty without reference to the Bank and without affecting in any way the liability of the Bank under this Guarantee/Undertaking, at any time and/or from time to time to anywise vary the Contract and/or any of terms and conditions thereof or of or relative to the advance and to extend time for the performance of the Contract and/or repayment of the advance or to postpone for any time or from time to time the obligations of the Supplier and to waive or postpone exercise of any of the rights available to the CPCL against the Supplier or to forebear from enforcing any of the terms or conditions of the Contract and/or the advance or any security(ies) available to the CPCL, AND the liability of the Bank shall remain in full force and effect notwithstanding any exercise by the CPCL of the liberty with reference to any or all the matters aforesaid or by reason of time being given to the Supplier or any forebearance, waiver, act or omission on the part of the CPCL or any indulgence by the CPCL to the Supplier or any other act, matter or thing whatsoever which under the law relating to sureties would have the effect of releasing the Bank from its liability hereunder or any part thereof, AND the BANK DOTH HEREBY waive all rights at any time inconsistent with the terms of this Guarantee/Undertaking.

ii. It shall not be necessary for the CPCL to proceed against the Supplier before proceeding against the Bank and this guarantee/undertaking shall be enforceable against the Bank as principal debtor notwithstanding the existence of any other security for any indebtedness of the Supplier to the CPCL

(including relative to the advance) and notwithstanding that any such security shall at the time when claim is made against the Bank or proceedings taken against the Bank be outstanding or unrealised.

iii. As between the Bank and the CPCL for the purpose of this guarantee/undertaking the amount claimed by the CPCL from the Bank with reference to this guarantee/undertaking shall be final and binding upon the bank as to the amount payable by the Bank to the CPCL hereunder.

iv. The liability of the Bank to the CPCL under this guarantee/undertaking shall remain in full force and effect notwithstanding the existence of any difference or dispute between the Supplier and the CPCL, the Supplier and the Bank and/or the Bank and the CPCL, or otherwise howsoever touching or affecting these presents or the liability of the Supplier to the CPCL, and notwithstanding the existence of any instructions or purported instructions of the Supplier or any other person(s) to the Bank not to pay or for any cause withhold or defer payment to the CPCL under these presents with the intent that notwithstanding the existence of such difference, dispute or instruction, the Bank shall be and remain liable to make payment to the CPCL.

v. This guarantee/undertaking shall not be affected by any change in the constitution of the Bank or that of the Supplier or the CPCL, or any irregularity in the exercise of borrowing powers by or on behalf of the Supplier.

vi. This guarantee/undertaking shall be valid for all claims/demands made by the CPCL to or upon us upto the business hours of _____ provided always that if for any reason, the Supplier is unable to complete supplies under the Contract, the Bank shall at the request of the CPCL and without recourse to the Supplier extend the validity of this guarantee/undertaking for a further period of six months. For the purpose of this clause, the CPCL's statement that the Supplier is unable to complete supplies under the contract shall be conclusive and final binding on us.

vii. Notwithstanding anything to the contrary contained herein above:

a. The Bank's liability under this Guarantee/Undertaking shall not exceed Currency..... (Currency..... only),

b. This Guarantee/Undertaking shall remain in force upto..... (Expiry date) and any extension(s) thereof; and

c. The Bank shall be released and discharged from all liability under this Guarantee/Undertaking unless a written claim or demand is issued to the Bank on or before (Claim Expiry date i.e. ___months from the BG Expiry date quoted in point (b) above) and if extended, the claim expiry date of the last extension of this Guarantee/Undertaking.

viii. Without prejudice to any other mode of service, a demand or claim or other communication may be transmitted by the Corporation to the Bank either by post or by email. If transmitted by email, the transmission shall be complete as soon as the mail is delivered to the mail id of the Bank mentioned in the BG.

ix. Our Guarantee will be governed and construed in accordance with Indian Laws and subject to exclusive jurisdiction of Courts in India.

x. The Bank DOTH HEREBY declare that Mr. _____(name of the person signing on behalf of the Bank) who is _____ (his designation), is authorised to sign this guarantee/undertaking on behalf of the Bank and to bind the Bank thereby.

Dated this _____day of_____

Yours faithfully

For _____

Signature _____

Name & Designation _____
Name of the Branch _____

Valida Print Help

Item Wise BoQ

Tender Inviting Authority: GM (SCM), ENGINEERS INDIA LIMITED, GURUGRAM-122001, HARYANA

Name of Work / Enquiry No.: SUPPLY OF PRESSURE VESSEL-CS/LAS+SS CLAD FOR GROUP-II/III LOBS PROJECT AT MANALI REFINERY, CHENNAI RFQ NO MKP/B649-095-VV-MR-6009/04

Preamble to Price Schedule (BoQ):
i) This BOQ sheet "BoQ1" shall be filled by the BIDDER.
ii) Currency of Quote shall be selected from INR only.
iii) Basic Supply Price shall be on FOT Despatch point basis.
iv) SUPPLY PRICE" should be inclusive of all all taxes & duties (except GST) applicable to the bidder but shall exclude Road Transportation charges (which is sought separately). Any other charges indicated as applicable "EXTRA" elsewhere in the bid shall be considered "Included" in the quoted "SUPPLY PRICE".
Bidder to note that no other charges except those quoted in the Price Schedule (BoQ) shall be considered for evaluation/ordering.
v) Bidders shall quote Road Transportation charges for Main Equipment till Site at Column No. 7.
vi) Bidders are advised not to tamper with this BoQ format.
vii) "SUPPLY PRICE" should be inclusive of all all taxes & duties (except GST) applicable to the bidder but shall exclude Road Transportation charges (which is sought separately). Any other charges indicated as applicable "EXTRA" elsewhere in the bid shall be considered "Included" in the quoted "SUPPLY PRICE".
viii) Bidder to note that no other charges except those quoted in the Price Schedule (BoQ) shall be considered for evaluation/ordering.
ix) Bidders are advised not to tamper with this BoQ format.
x) Unit Rates for additon/deletion shall be applicable as per Pre filled rates as per Annexure-I to BDS.
xi) Evaluation & Ordering shall be done on ITEM WISE BASIS. For details of each item, please refer the MR.
xii) Bidder shall ensure filling of prices in the excel sheet named "2YOM" for 2 Year O&M Spares (as per MR requirements).
xiii) In case of any discrepancy in the Item Description/Tag No./ Group No/ Qty. specified in this Price Schedule vis-à-vis the corresponding details in MR, the Item Description/Tag No./ Group No / Qty.specified in MR shall prevail.
xiv)Bidder confirms that he has noted the contents of this Preamble, RFQ, Material Requisition, etc. and quoted his prices accordingly without any deviation.

Name of the Bidder/ Bidding Firm / Company :	

PRICE SCHEDULE

(This BOQ template must not be modified/replaced by the bidder and the same should be uploaded after filling the relevent columns, else the bidder is liable to be rejected for this tender. Bidders are allowed to enter the Bidder Name and Values only)

Slno	ITEM DESCRIPTION AS PER MR	QTY.	UNIT	Quoted Currency (in INR only)	SUPPLY PRICE BUT EXCL. GST ON FOT DESPATCH POINT BASIS AS PER MR SL. NO. 01.00 (in currency INR) (To be filled by bidder)	ROAD TRANSPORTATION FOR MAIN EQUIPMENT CHARGES EXCL. GST TILL SITE AS PER MR SL. NO. 03.00 (in currency INR) (To be filled by	TOTAL AMOUNT Without GST	TOTAL AMOUNT EXCL.GST In Words (Automatically Populated; Bidder shall not change the Words that appear here)
1	2	3	4	5	6	7	8	9
1.00	MR ITEM SL. NO. 01.001 : **TAG NO 207A-C209N** HCGO HOT LOW PRESSURE SEPARATOR	1	NO	**INR**			**0.00**	INR Zero Only
2.00	MR ITEM SL. NO. 01.002 : **TAG NO 95-C-054** RECYCLE GAS COMPRESSOR KO DRUM	1	NO	**INR**			**0.00**	INR Zero Only

Name of Work / Enquiry No.: SUPPLY OF PRESSURE VESSEL-CS/LAS+SS CLAD FOR GROUP-II/III LOBS PROJECT AT MANALI REFINERY, CHENNAI RFQ NO MKP/B649-095-VV-MR-6009/04

NOTES:
1.Vendor shall furnish an itemised list of recommended spare parts with unit prices for two years normal and continuous operation as per MR, required over and above the mandatory spares.
2. Separate list shall be provided by the Bidder in the format enclosed herewith with itemized cost. Ordering of these spares may be done by the Purchaser separately at a later date and not alongwith main equipment. The price of Two Years Operation and Maintenance Spares shall not be considered for commercial (price) evaluation.
3. In case bidder has quoted for any item but not indicated freight for that particular item, in such case freight for that particular item shall be considered included.

Bidder Name :	

PRICE SCHEDULE (FOR TWO YEARS OPERATION AND MAINTENANCE SPARES)

(This BOQ template must not be modified/replaced by the bidder and the same should be uploaded after filling the relevent columns, else the bidder is liable to be rejected for this tender. Bidders are allowed to enter the Bidder Name and Values only)

Sl. No.	Item Description	UOM	Quantity	Unit Price on FOT Despatch Point Basis (in INR)	Road Transportation Charges (in % of Unit Price)
				0.00	
1				0.00	
2				0.00	
3				0.00	
4				0.00	
5				0.00	
6				0.00	
7				0.00	
8				0.00	
9				0.00	
10				0.00	
				0.00	

Material Requisition (Top Sheet)

Item Description :	PRESSURE VESSEL-CS/LAS+SS CLAD		
Item Code :	01AK	**Destination :**	As per Commercial Documents
Item Category :	I	**Delivery Period :**	As per Commercial Documents

Requisition Number (Always quote this Number given below as reference)

B649	095	VV	MR	6009	A
Job No.	**Unit/ Area**	**Main Cost Centre**	**Doc. Code**	**Sr. No.**	**Rev.**

22/07/2026	80	42
Date	**Divn.**	**Dept.**
	Originator	

Notes :

1. This sheet is a record of all the Revisions of this Requisition.
2. Vendor shall note the item category and shall submit his offer in line with the requirements included in attached 'Instructions to Vendors'.
3. The nature of the Revision is briefly stated in the 'Details' column below, the Requisition in its entirety shall be considered for contractual purposes.

Rev.	Date	Prepared By	Checked By	Approved By	Details
A	22/07/2026	Mohendra Kumar Sharma	Saikat Chakraborty	Tarun Khurana	ISSUED FOR BIDS

This is a system generated approved document and does not require signature.

Print Date : 7/22/2026 10:02:59AM

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Sr. No.	Tag No./Item Code/ [ID.No.]	Description	Quantity	Delivery Location
01.00		Residual Design including MPT Calculations along with Transportation Design Calculations, Detail Engineering, Procurement of materials and bought out components including all internals, assembly, sub-assemblies at shop, fabrication, PWHT, NDT, Third party inspection, Testing at vendor's shop, Certification, Hydrotesting, Pickling & Passivation of SS surfaces, Surface preparation and application of primer and finish paint at shop including necessary internal coating for applicable services (as applicable), packing (Desiccant filling) & delivery of the following including supply of all mandatory, erection & commissioning spares, special tools & tackles, Skirt template (as applicable) and documentation as per the enclosed Datasheets, Job specification, Licensor's/EIL Standards, standard specifications, instructions to bidder / vendors, job sketches, data sheets etc. and other codes and standards attached or referred to.		
▶ **01.001**	**207A-C209N**	**HCGO HOT LOW PRESSURE SEPARATOR**	**1 No**	
▶ **01.002**	**95-C-054**	**RECYCLE GAS COMPRESSOR KO DRUM**	**1 No**	
02.00		Deleted		
03.00		Transportation of equipments in single piece as per instructions to bidder / vendors on transportation saddles with all internals in installed condition except Demister from vendor's work / shop to site (Refer commercial document) for the following as per the enclosed Licensor's / EIL standard specifications, instructions to bidder / vendors, job specification, data sheets etc.		
▶ **03.001**	**{03}207A-C209N**	**For Item SL. No. 01.001**	**1 No**	
▶ **03.002**	**{03}95-C-054**	**For Item SL. No. 01.002**	**1 No**	
04.00		Deleted		
05.00		Two Years Operation and Maintenance Spares, only price list is to be provided by Bidder.		
05.001	**207A-C209N**	**HCGO HOT LOW PRESSURE SEPARATOR**	**1 Lot**	

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Sr. No.	Tag No./Item Code/ [ID.No.]	Description	Quantity	Delivery Location
05.002	**95-C-054**	**RECYCLE GAS COMPRESSOR KO DRUM**	**1 Lot**	
06.00		Unit Rates for Addition / Deletion.		
06.001		**Change in support dimension SA 387 GR 22 CL.2**	**1 kg**	
06.002		**Change in shell dimension SA 387 GR 22 CL.2 (IMPACT TESTED) + 347 WOL**	**1 kg**	
06.003		**Change in size of nozzle flange SA 182 F22 CL.3 (IMPACT TESTED) + 347 WOL**	**1 kg**	
06.004		**Change in size of nozzle neck SA 335 P22 (IMPACT TESTED) + 347 WOL**	**1 kg**	
06.005		**Change in shell dimension SA 516 GR 70 (IMPACT TESTED) + 304L CLAD (NACE COMPLIANT)**	**1 kg**	
06.006		**Change in size of nozzle flange SA 266 GR 4 (IMPACT TESTED) + 304L WOL (NACE COMPLIANT)**	**1 kg**	
06.009		**Change in support dimension SA 516 GR 70**	**1 kg**	
06.010		**Change in support dimension IS 2062 E250BR/BO**	**1 kg**	
06.011		**Change in pipe fitting SA 234 WP 22 CL.3 (IMPACT TESTED) + SS 347 WOL**	**1 kg**	
06.012		**Change in pipe fitting SS 347**	**1 kg**	
06.013		**Change in pipe fitting SA 234 WPB (IMPACT TESTED) + SS 304L WOL (NACE COMPLIANT)**	**1 kg**	
06.014		**Change in pipe fitting SA 403 WP 304L (NACE COMPLIANT)**	**1 kg**	
06.015		**Change in pipe SA 312 TP 304L (NACE COMPLIANT)**	**1 kg**	
06.016		**Change in plate dimension SS 321 (NACE COMPLIANT)**	**1 kg**	
06.017		**Change in pipe SA 106 GR B**	**1 kg**	
06.018		**Change in plate dimension SA 240 GR 304L (NACE COMPLIANT)**	**1 kg**	
06.019		**Change in plate SS 316**	**1 kg**	
06.020		**Change in Plate dimension SS 347**	**1 kg**	

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Sr. No.	Tag No./Item Code/ [ID.No.]	Description	Quantity	Delivery Location
07.00		Deleted		
08.00		Deleted		
09.00		Drawings and documents as per attached Vendor Data requirement for all supplies and services covered above in Sr.Nos.1.00 to Sr.No.8.00	Lot	
10.00		Deleted		
11.00		Deleted		
12.00		Deleted		
13.00		Deleted		
14.00		Deleted		
15.00		Deleted		
16.00		Deleted		
17.00		Deleted		
18.00		Deleted		
19.00		Deleted		
20.00		Deleted		
21.00		Deleted		
22.00		Deleted		
23.00		Deleted		
24.00		Deleted		
25.00		Deleted		
27.00		Deleted		
31.00		Deleted		
32.00		Deleted		

- Bidder shall quote prices in EIL Price Schedule except for Sr.No.09.00. Price for documentation is implied to be included in the prices quoted against Sr.No. 01.00 to Sr.No. 32.00

  Vendor to note that the numbers given in square '[]' and curly '{}' brackets are not for their use and meant for store purpose only. Items shall be tagged as per main equipment Tag No. only.

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Clause Related Remarks	
Supply	
1.	Bidders to include cost of external attachments (Platform / ladder supports cleats and Piping support clips as mentioned in clause (B) 7 of Job specification (Doc. No. B649-80-42-IB-6009) in quoted price.
Unit Rate	
1.	a). Rates shall be applicable for components / Parts / Attachments including Fabrication & Testing. b). The Unit rates shall be applicable for Addition as well for Deletion of the Items. c). Bidder to note that Finish / Fabricated Weights shall be considered for final settlement. d). The Weights indicated on Engineering Drawings do not form the basis for Offer, since these are approximations only. e). For Standard / Non-Standard Nozzle / Shell Flange (WN/LWN/SRWN), the Unit Rates shall be applicable for complete Assembly / Nozzle Assembly comprising Nozzle Pipe / Neck / RF Pad / Insert Plate (as applicable), Nozzle / Shell Flange, Stiffeners (as applicable), Studs, Nuts & Gaskets (when supplied with Blind / Reducing Flange) and associate Fabrication Testing.

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List of Attachments						
Sr. No.	**Document Title**	**Document No.**	**Revision**			
			Rev.	**Rev.**	**Rev.**	**Rev.**
			Date	**Date**	**Date**	**Date**
LIST OF ATTACHMENT						
1.	TABLE OF CONTENTS	B649-095-VV-MR-6009	A			
			22/07/2026			

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INSTRUCTIONS TO BIDDER (STATIC & MACHINERY EQUIPMENT DEPARTMENT)

PROJECT	:	**GROUP II / III LOBS PROJECT**
OWNER	:	**CHENNAI PETROLEUM CORPORATION LTD.**
EPCM/PMC	:	**ENGINEERS INDIA LIMITED**
JOB NO	:	**B649**

Rev. No	Date	Purpose	Prepared By	Checked By	Approved By
A	03.07.2026	ISSUED FOR BIDS	MKS	SC	TKh

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INSTRUCTIONS TO BIDDER

(A) Bidding Instructions (Category-I MR):

1. Bidder to note that no correspondence shall be entered into or entertained after the Bid submission. Bids shall be evaluated based on the details submitted by the Bidder. In view of the same Bidder in his own interest is advised not to submit any deviation.
2. Bidder shall furnish quotations only for those Items/Parts:
   a. for which Bidder is approved and
   b. for which Bidder can supply strictly as per technical specifications
3. No Technical Details of any nature shall be included in the Offer. If the Offer contains any Technical Deviations or Clarifications or stipulates any unsolicited Technical Specifications (even if in line with the MR requirements), the Offer is liable for rejection.
4. The submission of prices by the Bidder shall be construed to mean that he has confirmed compliance with all Technical Specifications of the corresponding item(s).
5. Bidders to note that a pre-bid meeting may be held in which Bidders can seek any clarifications/confirmation if required. Bidder to ensure that this conference is attended by both technical and commercial personnel of the Bidder who should have thoroughly scrutinized the MR beforehand so that all issues are finally resolved in this meeting.
6. The list of sub vendor provided in relevant standard specifications related to static equipment/internals is hereby stand withdrawn. Bidder to refer approved vendor list attached elsewhere in bid document. Bidder to note that name of third-party inspection agency provided in standard specifications related to static equipment is hereby also withdrawn. Bidder shall refer commercial document for list of approved TPA.

(B) It is understood that:

1. The quoted price includes for the “HOLDS” details of which are indicated on the EIL Engineering Drawing.
2. The weights indicated on EIL Engineering Drawing do not form the basis for offer, since these are approximations only.
3. Spares shall be included in the quoted price as described and applicable in Doc. No. B649-80-42-SOW-6009.
4. Bidder to note that quoted price shall include the weight of internals attachments as provided in MR. Any increase/decrease in weight shall be settled as per Material Requisition.
5. The quoted price shall include stiffeners, increased shell / skirt course thickness, insert plates, base ring / top ring of adequate thickness required as per requirement of design calculation for lifting and tailing forces. No price implication shall be paid on this account
6. Any addition/deletion of parts shall be settled on finished items only.
7. Bidder to note that quoted price shall include following weights for structural/platform/ladder clips & piping clips. Any increase/ decrease in weight shall be settled as per Unit rates addition & deletion. Insulation & Fire Proofing Cleats to be provided as per applicable EIL STDs and shall be included in the quoted price.

ITEM No.	IS 2062 E250 QUALITY-BR/B0 (Kg)	SA 516 GR.70 (Kg)	SA387 GR.22 CL.2 (Kg)
95-C-054	500	500	-

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(C) Transportation of equipment shall be governed by commercial conditions.

Transportation of all vessels shall be done in single piece with internals in installed condition except for "Recycle Gas Compressor K.O. Drum (Tag No. 95-C-054) where Demister Pad shall be supplied loose in packed condition from vendor works/shop to site as per commercial documents and shall be carried out by Bidder. Transportation Design load calculation shall be done by Vendor / Bidder as per applicable clauses of the relevant Licensor's specification.

Further, Bidder to note that Equipment transportation / movement during transit & especially inside the Refinery premises may call for the use of SPMT platform vehicle, for easy maneuvering, which should be properly assessed & included in the quoted price by the Bidder as part of his responsibility. Thus, Bidder to select the transportation logistics accordingly. Necessary unloading/loading into the SPMT trailer bed, if required, shall be in the scope of the Bidder.

(i) Transportation of equipment shall be governed by commercial conditions. However, in general, the following shall be strictly complied with.

a) The Package / Equipment with length upto 20-meter, width upto 4-meter, height up to 4 meter and weight upto 32 MT can be transported on Mechanical Trailers. However, additional support is to be given for the length to avoid overhanging.

b) The Equipment / Package where any of the dimension (Length, Width, Height or Weight) exceeds the limits indicated in Point no. a) above, shall compulsorily be transported by Hydraulic Axles in compliance to all statutory norms of the authorities concerned.

c) For the movement of Hydraulic axles, the GPS tracking system shall be installed on the prime movers(s) of the vehicle(s). The operational password for the GPS tracking system shall be provided to EIL/ Owner, to enable all concerned to monitor movement enroute.

d) MORTH approval shall be taken before commencement of transportation by Road & copy of same shall be submitted to EIL. Movement should not commence without MORTH approval.

(ii) All external surfaces shall be properly protected / covered. All open ends of Shell flanges to be protected with Temporary Steel Covers along with Gasket & Bolting. Gasket & Bolting used for transportation shall not be used as Service Gaskets & Bolting.

(iii) After Completion of all Testing and Inspection, the inside of Complete Equipment shall be thoroughly drained and dried out. The Internal Surface then shall be thoroughly cleaned to remove all dirt's, debris etc. Equipment shall be then completely dried by passing hot air for sufficient time until no further increase in relative humidity of outgoing air is observed. Alternatively, Vacuum Drying is also acceptable. For preparation of shipment, Bidder / Vendor to refer relevant clauses i.e., clause B of the applicable Licensor specification. Following to be noted –

a) Steel covers along with temporary gasket & bolting
b) Labelling shall be done as per EIL coating specification only.

Technical Instructions

1. The shape of dished end~~/tori-cone~~, ~~profile of conical reducer, knuckle radius, cone angle and overall height~~ shall be as per engineering drawing. Minimum thickness of shell, ~~cone~~ and dished end shall be achieved taking into account thinning and scaling etc.

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2. Dished ends and hot formed pressure parts shall have 10% or 2 mm (whichever is higher) allowance for scaling and thinning, in order to achieve minimum thickness specified on the drawing.

3. a. Wherever hot forming and subsequent heat treatment is involved, adopted procedure shall not impair the mechanical properties of the material beyond the limits specified in respective material specification.

   b. For ~~low temperature equipment and~~ hot formed parts, number of production coupons and the tests thereon shall be as per requisition.

4. Post weld heat treatment and all non-destructive testing like radiography, ultrasonic testing, magnetic particle/dye penetrant examination etc. shall be conducted as per requisition.

5. Workmanship and materials, in Bidder's scope of supply, shall be guaranteed.

6. Pickling and passivation of internal/external surfaces of stainless-steel clad equipment shall be carried out as per ASTM A-380.

7. **For 95-C-054:** Demister/Coalescer shall be procured through approved supplier list (EIL Doc. No. B649-80-42-SVL-6009) only. Wherever, Demister/Coalescer supplier is providing on equivalent make / model of “YORK MESH 421”, equivalency needs to be established by Bidder before placement of order on Demister/Coalescer supplier. All necessary sizing calculations, process calculations, proven past record shall be furnished for EIL / Owner / Licensor for approval. Any additional document required to establish equivalency shall also be provided without any time & cost implication to the Owner. Selection of Demister/Coalescer supplier without equivalency establishment beforehand will be liable for rejection for the Demister/Coalescer supplier.

8. Transportation of Equipment to Site shall be in a Single Piece with all Internals As-Installed Conditions. However, Demister/ Coalescer Pad Assembly shall be supplied loose along with Equipment.

9. The main equipment shall not be subcontracted; however, the bought-out items/ components can be sub-ordered or sub-contracted to approved vendors / sub-contractors only (refer attached list with the MR).

10. If any discrepancy is found during design and fabrication stages, the fabricator shall report in writing to EIL immediately and shall obtain necessary clarification/approval before proceeding with that portion of the job any further. EIL decision in this regard shall be final & binding.

11. Fabricator to supply Box type template for skirt supported vessels for checking the orientation of anchor bolts on foundation and to ensure matching of base plate that is drilled in advance. The schedule of supply of template at site shall be 8 weeks from LOI.

12. Bidder shall provide permanent sloping steel monkey ladder inside skirt (1 Nos.) to approach bottom dish end for the Vessels. If Vessel ID is less than 1200 mm then ladder rungs shall be provided. In case ladder rungs have been provided, 2 sets of ladder rungs at 180° apart shall be provided. Bidder to also refer MDS for the same.

13. Lifting Lugs / Lifting Trunnions (as applicable) for Vertical Vessels shall be designed by Bidder considering an impact factor of 1.5. Suitable insert plate (wherever applicable) shall be provided. Stresses of shell, pad & lug shall be limited to 0.9 times of Yield Stress. (Refer applicable notes of MDS). Insert Plate shall be min. twice the dia. Of lifting trunnion / twice the width of lifting lug with thickness of Insert Plate shall be twice the Shell thickness (min.) Welds of lifting trunnion shall be mandatorily Groove cum fillet.

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14. Bidder shall ensure that all internals shall pass through nearest man way. Size indicated in Data sheets are minimum to be followed. In case the same is inadequate, man way size shall be increased suitably. Vendor to include the same in Quoted price. The same shall be informed to EIL within 30 days of receipt of FOA.

15. Gaskets used for hydro testing of equipment shall be of the same specification as service gaskets and the Bidder shall supply new gaskets for manholes and nozzles with blind /reducing flanges apart from spares. Gaskets used for hydrotest shall not be used for actual hook up.

16. **Internals:** Design, detail engineering, supply and fabrication of all internals like Outlet baffle, Wear Plate, Feed inlet, sloped inlet shroud ring, donut baffle, strainer and vortex breaker including all their supports (both Bolted and Welded) as per the internal details given in the respective CLG's Standards / Mechanical datasheet shall be in the scope of Bidder. The welded supports for Demisters / Coalescer shall be designed by the Internals supplier & provided by the Bidder taking guidance from EIL Std. 7-12-0036. Minimum requirements specified in EIL-MDS shall be followed. Bidder shall also consider Trial/ Mock-up at shop for all the internals, pickling & passivation of SS surfaces, inspection, testing at Bidder's/Supplier's shop, packing, delivery including supply of all mandatory spares for Internals.

17. While transporting completed equipment/equipment sections, Bidder shall provide suitable type and no. of supports so that no deformation of any part of the equipment takes place.

18. Bidder shall exercise utmost care during loading/unloading of equipment including transportation so that no damage to any part of the equipment, installed internals occurs. Suitable type and no. of supports shall be provided for supporting the equipment and also the pre-installed internals. Any damages in the internals during loading/unloading of equipment including transportation shall be rectified by the bidder/vendor without any time and cost implication. It is completely bidder/vendor responsibility to ensure the complete intactness and integrity of pre-installed internals.

19. Bidder shall also design and provide suitable stiffening arrangement, as necessary, at the base anchor chair so that deformation/buckling of base and bottom zone does not occur at the time of handling and lifting while the vessels/vessel sections attached with skirt are being erected.

20. Erection weight given in equipment data sheet is approximate. Vendor shall calculate the actual equipment weight (including installed ladders & platforms) and furnish calculation. Erection weight furnished in vendor drawing shall be based on fabricator's calculation. Calculation of Centre of Gravity (C.G.) of the equipment shall also be done by vendor and marked on the equipment & in the fabrication GA drawing. Thickness of equipment after fabrication shall be recorded and shall be submitted along with final document.

21. Bidder shall provide transportation steel saddles for transportation of each vertical vessel while suitable temporary wooden sleepers shall be provided for each horizontal vessel below each base plate of saddles. Bidder can take back the transportation saddles only after the erection of equipment. Bidder to provide isolation (like rubber sheet etc.) between steel saddles and vessel. Care shall be taken to prevent rusting of saddles during transportation. Bidder to quote accordingly and the same shall be included in the quoted price.

22. Hardness of forging & plates shall be followed as specified in the applicable CLG specifications/ EIL MDS.

23. All the davit arrangements/ lifting lugs/ any other handling tools must undergo stagewise inspection and to be tested for its holding capacity for a period of at least 8 hours at suppliers shop to ensure that there is no failure of welded joints/ fillets/ links / eye bolts etc All Load bearing weld joints of all types of Davits shall be suitably tested by one of the NDT methods in accordance to the approved ITP in addition to load testing mentioned in applicable EIL standard.

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24. Bidder to provide fire proofing nuts for Angle leg/ Pipe Leg/ Skirt supported vessels.

25. Bidder to provide insulation supports as per EIL Standard as applicable.

26. For HCGO Hot Low-Pressure Separator (Tag No. 207A-C209N)

a) Equipment along with support shall be impact tested. For Pressure containing Parts Impact Test shall be carried out as per clause 6.1.7 & 6.1.12 (b) of CLG specification no. PVM-SU-5290 while for non-pressure parts, Impact Test shall be carried out as per clause 6.2 of CLG specification no. PVM-SU-5290.

b) ISR shall be carried out as per clause 8.2.1 of CLG specification no. PVM-SU-5290. DHT in lieu of ISR shall be done only after obtaining written permission from EIL/Owner only for non-restrained areas. For highly restrained areas, DHT in lieu of ISR is not acceptable.

c) Step Cooling shall be done as per clause 14.1 to 14.3 of CLG specification no. PVM-SU-5290.

27. Internal coating (as applicable) for the equipment shall be carried out as per Table-5 of Licensor specification no. COM-SU-5191 for the applicable services mentioned for new equipments, if applicable, thin film linings shall be used.

28. Earthing lugs, Insulation supports and fire proofing supports shall be as per EIL standard as mentioned in the MDS.

General Instructions

1. All materials including those used for internals shall be procured with stage-wise inspection. Stage wise and final inspection of equipment and its components shall be carried out as follows:

a) Stage wise & final inspection of the equipment at vendor shop/site including all components and internals shall be carried out by Third party inspection agency for which cost incurred shall be borne by Bidder. Bidder to quote accordingly.

b) For acceptable third-party inspection agencies refer commercial documents.

c) Inspection of imported material like pate/forging shall also be done by TPIA and cost of the same shall be included in the quoted base price.

2. Raw material procurement may proceed prior to approval of vendor drawings on the basis of Purchase Requisition. However, for plates & nozzles where there is no Hold in size, thickness, etc., material procurement shall be initiated by successful vendor based on MR drawings immediately after placement of order.

3. Shell rolling/welding and dished - head fabrication shall commence without awaiting fabrication drawing approval.
Inspection shall be carried out in accordance with Purchase Requisition documents till approved fabrication drawings are available.

4. Material for shell, dished ends and skirt shall be considered as major raw material wherever identification of major raw material is one of the payment milestones.

5. Purchase requisition shall be issued within two weeks of Fax of intent (FOI). Final nozzle sizes (without any holds) shall be issued along with purchase requisition or latest within two months of Fax of intent (FOI).

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6. Nozzle orientation of the equipment shall be issued latest by mid of contractual delivery period reckoned from date of issue of Fax of Intent (FOI).

7. Pipe support/structural clip details shall be issued latest by three months before contractual delivery date.

8. Minimum course width shall be 1 meter. Maximum no. of longitudinal seams shall be as follows:

(a)	vessels upto 2 meter diameter	:	1 seam
(b)	vessels from 2 meter upto 4 meter dia.	:	2 seam
(c)	vessels from 4 meter upto 6 meter dia.	:	3 seam
(d)	vessels from 6 metre upto 8 metre dia.	:	4 seam
(e)	vessels from 8 metre upto 12 metre dia.	:	5 seam

In any case, distance between two longitudinal seams shall not be less than 3 meter.

9. If any discrepancy is found during design and fabrication stages, the fabricator shall inform EIL immediately and shall obtain necessary clarification/approval before proceeding with that portion of the job any further.

EIL's review of fabricator's drawings and documents must not be considered as a check and shall not relieve the fabricator of his responsibilities to supply equipment as per requisition. Fabricator shall remain responsible for conflicts between his drawings/documents and EIL drawings/documents.

10. English language and metric units shall be used in all documents. Drawings shall be prepared in prescribed sizes as standardized by bureau of Indian standards (BIS) and shall be preferably in the sizes such as 210 x 297, 297 x 420, 420 x 597, 597 x 841, 841 x 1189 mm.

11. All fabricator's drawings submitted to EIL shall be based on purchase requisition and shall bear reference number and revision of the corresponding EIL drawings. In addition, it shall indicate item number, client's name, project name, fabricator's name, purchase order number, purchase requisition number, drawing number, revision number etc. all in the lower right-hand corner. All revisions shall be clearly marked by encircling with revision marks.

12. Vendor shall submit DCI of those Drawings, which are to be submitted for review to EIL along with the Submission dates for each Drawing within 15 days of Placement of FOI. Vendor shall strictly adhere to this Drawing Submission Schedule. Updated Vendor Drawing / Document Submission Schedule shall be forwarded along with each Submission.

13. Submission of required drawings/documents shall be the responsibility of fabricator. In the event of fabricator's failure to meet this requirement, the supply of equipment shall be considered as incomplete.

14. All drawings shall be thoroughly checked and duly signed by Vendor. Unchecked drawings and drawings without revisions clearly marked shall be returned without review. Successive review of the same fabricator's drawing shall apply only to EIL's latest data sheet/comments on the previous revision. Drawings and documents returned to Vendor for revision shall be resubmitted preferably within ten days of receipt.

15. Dimensions of all flanges shall be as per ANSI B16.5 for nozzle sizes 24"NB and below and as per ANSI B16.47 Series B for sizes greater than 24"NB unless otherwise stated.

16. Fabrication drawings shall be submitted in following stages:

Stage-I

- General arrangement drawing indicating design data, center of gravity, fabricated equipment's weight, general notes, nozzle schedule, detail of shell, heads,

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skirt/supporting arrangement, main weld seams, nozzle orientation plan, ~~cutting layout (in case of free issue material)~~ etc.

- Detail of all nozzles, manholes, accessories, shell flanges (if applicable) etc.
- Bill of material for each item showing part size, quantity, material specification, scope of supply and weight etc.

Stage-II

- Detail of internals such as outlet baffles, feed inlet pipes, wear plate, sloped inlet shroud ring, donut baffle, internals support, strainer & vortex breaker etc. as per MDS / CLG's standards.
- Detail of Demisters, it's supports as per MDS / PDS.
- Detail of external clips for ladder, platform, pipe support, insulation, fire proofing, pipe davit, lifting trunnions / lugs, tailing lugs, etc.
- Shell development drawing incorporating all attachments and weld seams, highlighting all the fouling along with suggestions for resolution.
- Bill of material for each item showing part size, quantity, material specification, scope of supply and weight etc.

17. After receipt of order, fabricator shall submit to EIL, the planning of fabrication which shall indicate the details and scheduled date of:

- Sub orders.
- Purchase of materials such as plates, pipes, forged flanges, bolting, gaskets, etc.
- Start of fabrication of dished ends and shell, welding of shell and heads.
- Welding of nozzles and external attachments such as cleats etc.
- Phased program of shop/site fabrication of equipment section, shop assembly, testing and dispatch to site.

This planning shall be established to extend from purchasing of equipment components to the end of fabrication and delivery of equipment.

18. All drawings shall be drawn in AUTOCAD 2017 or above. No hand drawn drawings shall be accepted. All residual calculations shall be computerized.

19. Documents and letters shall be furnished in electronic format. The software used shall be as follows.

a) MS OFFICE 2007 or above
b) ADOBE ACROBAT 2017 & above

20. ~~In case of owner's free issue plate material, fabricator shall tabulate each plate detail as per following table:~~

~~Plate No./ Plate mark~~	~~Material Specification~~	~~Size of Plate Thk x Width x Length (mm) x (M) x (M)~~	~~Equipment Number~~	~~Ref. Dwg. No.~~	~~Part number and qty.~~

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~~Size of Plate Thk x Width x Length (mm) x (M) x (M)~~	~~Heat/ sample number~~	~~Plate Weight (Kgs.)~~	~~Used plate weight (Kgs.)~~	~~Burnout losses (Kgs.)~~

~~Returnable offcut size with sketch~~	~~Returnable offcut weight (Kgs.)~~	~~Non-returnable scrap weight (Kgs.)~~	~~Remarks~~

~~Fabricator shall be allowed 2% burnout losses against the material consumed in case of pressure vessels and 1% in case of storage tanks and spheres unless otherwise specified. In general purchase conditions all offcuts of width 800mm and perimeter 2400mm in carbon steel materials and 150mm width and perimeter 1500mm in stainless steel materials shall be returned to owner, unless otherwise specified. Returnable offcuts shall have original plate markings transferred and duly stamped by inspector. All other non-returnable scrap shall be sold by fabricator on the basis of owner's instructions and approval.~~

21. **Further to pt. No. 8 of General Instructions; Bidder to note that interference of nozzles with circumferential seams is not acceptable. However, Bidder to take suitable care to avoid fouling of external/internal attachments with C-seams. Bidder to submit shell development drawing in the Ist submission**.

22. Words “Fabricator”, “Vendor”, “Contractor” appearing anywhere in the MR/Specifications/ Standards etc. shall be read as “Bidder”.

23. Bidder to note that name of third-party inspection agency provided in standard specifications related to static equipment is hereby withdrawn. Bidder shall refer commercial document for list of approved TPA.

24. Vendor shall note that the vendor drawing for the following standard components (as applicable) shall not be reviewed by EIL. It shall be solely vendor's responsibility to ensure the correctness of these drawings as per EIL Standards and Drawings. If any discrepancy / mistake is found at a later date the vendor shall rectify the same at his time and cost.

- Manhole davit/ hinge arrangement & Ladder rungs
- EIL Name plate, Manufacturer's Nameplate, Nameplate bracket
- Lifting lugs (Standard Type)
- Fireproofing and Insulation Support
- Earthing lug
- Pipe support cleats and ladder and platform cleats (Fouling of external support cleats (if any), shall be highlighted to EIL along with shell development drawing and suggestion for resolution)

However, all these above listed drawing shall be submitted in a separate drawing for information to EIL ensuring adherence to all the respective EIL Standards and Drawings. Clarifications (if any) shall be obtained by Vendor from EIL before fabrication.

25. Standard Inspection and test plans are attached with requisition for guidance. Vendor shall submit inspection test plans for the approval of visiting inspection engineer. Inspection shall take care of all the requirements of specifications. Inspection engineer has the authority to modify the plan depending on the criticality of the item.

26. **In the event of conflict, requirements specified in Licensor’s specifications / standards shall govern. Requirements not available in Licensor’s specifications shall be followed from the relevant EIL specifications / standards. Below order of precedence shall be followed to resolve the conflict:**

a) Statutory Regulations.

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b) Mechanical Datasheet, PDS (as applicable).
c) This Specification.
d) CLG's Specifications & Standards.
e) EIL specifications & Standards.
f) Applicable codes.

However, in event of any conflict other than one mentioned above, it shall be the duty of the Bidder / Vendor to bring it to EIL's notice for resolution and resolve it before proceeding with the manufacturing. EIL's decision shall be final and binding on Bidder / Vendor without any cost and delivery implications.

27. Equipment tag number shall be engraved/welded on all components (if equipment is being sent in parts) and on all spares. If spares are small enough or if welding/engraving will damage the spares, metallic tag shall be provided with equipment tag number. If welding/engraving on equipment can result in damages, equipment tag number shall be painted on the equipment or its part (minimum three locations).

28. On completion of order, Supplier shall submit the final documents to the Owner/ Consultant, as per "Supplier Data Requirement" with 3 nos. hard copies (1 Original + 2 Duplicate) and 2 nos. soft copies in external HDD in addition to submission through EIL Bidder Portal. The final documents supplied shall be duly endorsed by the TPIA or EIL (which ever applicable) as 'VERIFIED AND CERTIFIED FOR COMPLETENESS OF DOCUMENTATION AS PER SUPPLIER DATA REQUIREMENT'.

29. Surface preparation, application of Primer & Finish painting, internal coating (as applicable) of equipment shall be in Bidder's / Vendor's scope.

30. EIL /OWNER would review the actual job progress at place of fabrication / Manufacturing shop as & when desired or on monthly basis.

Bidder shall have provisions for showing live feed of job status /Work in progress at place of fabrication / shop floor for the ordered equipment by use of high-resolution cameras / helmet cameras or any suitable electronic device during virtual progress review. Recording of the job status (in soft) shall be captured & submitted to EIL / Owner after each review.

Bidder shall ensure that requisite facilities to show virtual job progress/ Work in Progress are available at their manufacturing shop/place of fabrication and in case these are not available same shall be installed within one month of award".

31. MARKING

In addition to General Purchase Conditions, the following requirements shall also be complied with.

a) All Loose Components such as Studs, Nuts, Washers, Gaskets, etc. shall be packed in Crates and shall be marked for the Project, Consignee, Consigner, Job Number, Item Number, Order Number, Gross and Net Weight, Dimensions etc.

b) Additional indications such as North / East / West / South along with Centre of Gravity shall be clearly marked with White Paint.

c) Specific Marking with White Paint for Slinging shall be provided for all Heavy Lifts weighing 5 Tons and above.

A Copy of Packing List shall accompany the Material enclosed in a Water Tight Envelope Fastened inside a Shell Connection with an identifying arrow sign "Documents" applied with Indelible Paint.

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32. PACKING, SHIPMENT & TRANSPORTATION

In addition to General Purchase Conditions, the following requirements shall also be complied with:

a) Vessels shall be provided with suitable Wooden Saddles with Steel Ties and Tension Rods. The Minimum Height of the Saddle shall correspond to the Maximum Projected Length of the connected attachments plus an additional clearance of 45 mm. Saddles spacing shall depend on the length of the Equipment.

b) Fragile or Machined Components shall be adequately protected against nature during handling and transit.

c) Demister/Coalescer Pad Assembly for 95-C-054, Mandatory Spares and Commissioning Spares & Special Tools and Tackles shall be packed separately.

33. GUARANTEE

Unless Specified Otherwise in General Purchase Conditions regarding Guarantee, the following shall govern:

a) Manufacturer shall guarantee that all Materials used in the Equipment are new and have been submitted to regular acceptance procedure and are free from any defect regarding quality, form and appearance.

b) Approval of Work by EIL or Release of Vessels for Shipment shall in no way release or relieve the Manufacturer of any responsibility for carrying out all provisions of this Specification.

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SCOPE OF WORK & SUPPLY (STATIC & MACHINERY EQUIPMENT DEPARTMENT)

PROJECT	:	**GROUP II / III LOBS PROJECT**
OWNER	:	**CHENNAI PETROLEUM CORPORATION LTD.**
EPCM/PMC	:	**ENGINEERS INDIA LIMITED**
JOB NO	:	**B649**

Rev. No	Date	Purpose	Prepared By	Checked By	Approved By
A	03.07.2026	ISSUED FOR BIDS	MKS	SC	TKh

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BIDDER'S SCOPE OF WORK (SHALL INCLUDE BUT NOT TO BE LIMITED TO THE FOLLOWING)

1.1 Residual design, MPT Calculation along with Transportation Design Calculations & Detail Engineering of vessels, internals & all accessories (as per engineering datasheet) viz. Demisters/Coalescer (including their support), outlet baffles, feed inlet pipes, wear plates, sloped shroud rings, donut baffles, stilling wells, Baffles, strainer and vortex breakers, wear plates, all necessary supports/cleats/external attachments for insulation, piping supports, structural supports, fire proofing, pipe davits etc.

1.2 Procurement of all raw materials, bought out components including all internals i.e., Demisters/Coalescers, outlet baffles, internal feed pipes, internal flanges, wear plates, sloped shroud rings, donut baffles, manway in baffles, SS wire mesh demisters through EIL approved sub-vendor list only, including supports, rain shield, monkey ladders / rungs, top manway davit supports, fabrication including assembly at Vendor's shop, necessary installation of internals as per provisions of Requisition, all NDT, PWHT, hydrotesting, inspection, testing at Vendor's work.

1.3 Surface preparation, primer and finish paint application including internal coating (as applicable) for the applicable services as per painting specification attached with the Requisition.

1.4 Pickling & passivation of all Stainless-steel internal surfaces.

1.5 All necessary supports/ cleats/ external attachments for insulation, fabrication of pipe davit, piping support cleats, structural cleats, fire proofing supports etc. detailing shall be done by Bidder based on EIL Standards.

1.6 Design of Lifting Lugs / Trunnions, tailing lugs etc. required as per MDS.

1.7 Testing of all Load bearing weld joints of all types of Davits by one of the NDT methods in accordance to the approved ITP in addition to the load testing specified in applicable EIL / Licensor standards.

1.8 Design and fabrication of transportation & Unloading/Loading saddles for all vertical vessels.

1.9 Transportation of Vessels in single piece along with installed internals excluding wire mesh Demisters which shall be provided in separate Packed condition along with the equipment from Vendor's shop to site as specified in Requisition.

1.10 Establishment of equivalency including necessary approval with respect to make / model of Demisters vis-à-vis Licensor designated Demister is in the scope of Bidder wherever the proposed supplier Demister is different from that of Licensor specified make / model of Demister in respective PDS. Licensor decision on equivalency shall be final and binding.

1.11 All inspection as per clause 1 of general instructions (B649-80-42-IB-6009).

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2.0 BIDDER'S SCOPE OF SUPPLY (TO INCLUDE BUT NOT TO BE LIMITED TO THE FOLLOWING)

2.1 Complete Equipment along with Nozzles, Demisters/Coalescer, feed inlet assembly, outlet strainers & vortex breaker assembly, and all the other Internals (scope as per engineering datasheet), wear plate, sloped shroud ring, including internal flanges with gasket & bolting, all welded and bolted Internals and other accessories as per requirements of Process datasheet / Licensor Standards / Engineering Datasheets and Material requisition except as covered in exclusions in clause 3.0.

2.2 All consumables, tools, tackles, machinery & accessories for completing the equipment in all respect as per Material requisition.

2.3 Gasket / Studs / bolts and nuts for manway / handholes / internal flanges / nozzles with blind flange / shell flange/ pad flanges / reducing flange / companion flange along with piping spool, including jack bolts for manways, shell flanges and reducing flanges, whichever is applicable. Jack Bolts to be provided for all shell flanges.

2.4 Pipe Davits with all accessories shall be supplied by Bidder.

2.5 **Spares:**

a) Mandatory Spares (to be included in base price)

i. 10% (min 2 nos. in each size) of installed fasteners and two set of installed gaskets for each nozzle shown with blind flange / pad flanges / reducing flange / shell flange / companion flange / internal flanges, whichever is applicable.

ii. 10% (min 2 nos. in each size) of fasteners for all internal bolting, clamps etc.

b) Commissioning Spares and Special Tools & Tackles (to be included in base price)

Commissioning Spares are to be recommended and supplied by the Bidder.

Bidder shall ensure that all spares required over and above the recommendation are procured in time and are supplied with equipment.

Any leftover (unused) spares after commissioning, out of those included by Bidder in their offer, shall be handed over to the Owner.

c) Two years normal operational spares recommended by vendor, if any (provide separate price)

List of additional spares required over and above mandatory spares for 2 years operation and maintenance shall be recommended by the Bidder along with the offer. Item wise Price for same shall not be included in base price and to be quoted separately.

The price for 2 years operation and maintenance spares shall be valid for at least 2 years from the date of main order.

d) Mandatory & Commissioning spares for Demister/ Coalescer shall be as per EIL Standard specification no. 6-44-0017 (to be included in base price).

2.6 All the drawings and documents (including design calculations) as specified in the requisition.

2.7 All tools, tackles, consumables, manpower required for Surface preparation & application of Primer, supply of primer at shop.

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2.8 Supply of all Lifting Trunnions / Lugs & tailing lugs with insert plates as applicable as per MDS.

2.9 Box type Template for foundation for skirt supported vessels as required.

2.10 All necessary supports/ cleats/ external attachments for insulation, fireproofing, pipe davit supports, piping support clips, structural cleats etc. shall be provided by Bidder.

3.0 EXCLUSIONS

3.1 Supply & installation of Permanent instruments

3.2 Foundations

3.3 Anchor bolts

3.4 Unloading of vessels at site.

3.5 Supply and installation of platform and ladders. (However, welded cleats shall be in Vendor's scope).

3.6 Supply & installation of insulation and fireproofing. (However, insulation cleats and fire proofing cleats are in vendor's scope).

3.7 Erection of vessels.

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VENDOR DATA REQUIREMENTS
FOR
CS/LAS + SS CLAD VESSELS

Rev. No.	Date	Purpose	Prepared by	Checked by	Approved by
A	03-JUL-2026	ISSUED FOR BIDS	MKS	SC	TKh

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VENDOR DATA REQUIREMENTS

The following drawings/documents marked "✔" shall be furnished by the bidder.

PRESSURE VESSEL

S. NO.	DESCRIPTION	WITH BID	POST ORDER			REMARKS
			FOR REVIEW	FOR RECORD	WITH DATA BOOK (FINAL)	
1.	GENERAL ARRANGEMENT DRAWING INDICATING DESIGN DATA FABRICATED EQUIPMENT WEIGHT, GENERAL NOTES, NOZZLE SCHEDULE, DETAIL OF SHELL, HEADS, BOOT DETAILS, SUPPORTING ARRANGEMENT, MAIN WELD SEAMS, NOZZLE ORIENTATION PLAN.		✔		✔	
2.	DETAIL OF NOZZLES, MANHOLES, ACCESSORIES ETC.		✔		✔	
3.	DETAIL OF INTERNALS LIKE FEED INLET PIPES, WEAR PLATE, SLOPED SHROUD RING, DONUT BAFFLE, OUTLET BAFFLE, STRAINER & VORTEX BREAKER, ETC. ALONG WITH THEIR NECESSARY SUPPORTS.		✔		✔	
4.	DESIGN CALCULATIONS & DETAIL DRAWING OF DEMISTERS / COALESCER ALONG WITH SUPPORT DETAILS ETC.		✔		✔	
5.	DETAIL OF EXTERNAL ATTACHMENTS LIKE CLIPS FOR LADDER, PLATFORM, PIPE SUPPORTS, INSULATION CLEATS, FIREPROOFING CLEATS, PIPE DAVIT ETC. AS APPLICABLE.			✔	✔	
6.	CALCULATION & DETAIL FOR LIFTING LUGS AND TAILING LUG AS APPLICABLE ETC.		✔		✔	
7.	SHELL DEVELOPMENT DRAWINGS INCORPORATING ALL ATTACHMENTS AND WELD SEAMS.			✔	✔	
8.	BILL OF MATERIAL FOR EACH ITEM SHOWING PART SIZE, QUANTITY,MATERIAL SPECIFICATION, SCOPE OF SUPPLY AND WEIGHT ETC.			✔	✔	*WITH EACH DWG.*
9.	DETAILS OF TEMPLATE FOR FOUNDATION AS APPLICABLE.			✔	✔	
10.	MATERIAL TEST CERTIFICATES @				✔	
11.	WELDING PROCEDURES AND QUALIFICATION TEST REPORTS @				✔	
12.	DESTRUCTIVE AND NON-DESTRUCTIVE TEST REPORTS @				✔	
13.	RADIOGRAPHIC EXAMINATION REPORTS WITH FILMS @				✔	
14.	HEAT TREATMENT PROCEDURE AND TIME TEMPERATURE CHARTS @				✔	

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S. NO.	DESCRIPTION	WITH BID	POST ORDER			REMARKS
			FOR REVIEW	FOR RECORD	WITH DATA BOOK (FINAL)	
15.	CODE CERTIFICATES (INCLUDING INSPECTION CERTIFICATE, HYDROSTATIC TEST CERTIFICATE, LOCAL CODE REQUIREMENTS, NAME PLATE ETC.) @				✔	
16.	FABRICATION ASSEMBLY DETAILS @				✔	
17.	HYDROTEST PROCEDURE.			✔	✔	
18.	CERTIFIED 'AS BUILT' DRAWINGS INCORPORATING ACTUAL DIMENSIONS AND MATERIAL USED, DULY CERTIFIED BY THE INSPECTOR @				✔	
19.	DATA FOLDER AS PER SPECIFICATION @				✔	
20.	INSPECTION AND TESTING PLAN @				✔	
21.	PACKING AND FORWARDING INSTRUCTIONS @				✔	
22.	MPT Calculations			✔	✔	
23.	TRANSPORTATION DESIGN CALCULATION AND TRANSPORTATION DRAWING			✔	✔	
24.	LIST OF MANDATORY & COMMISSIONING SPARES INCLUDING SPECIAL TOOLS & TACKLES.			✔	✔	

Notes :

1. "TICK" denotes applicability.
2. Post order, drawing / document review shall commence only after approval of Document Control Index (DCI).
3. All post order documents shall be submitted / approved through EIL eDMS portal.
4. Refer - 6-78-0001: Specification for quality management system from Bidders.
5. Refer - 6-78-0003: Specification for documentation requirement from Suppliers.
6. All drawings & documents shall be submitted in A4 or A3 paper sizes. Documents in higher paper size shall be submitted in exceptional circumstances or as indicated in the MR/Tender.
7. Post order- The schedule of drawing / data submission shall be mutually agreed between EIL & the bidder / contractor / supplier during finalization of Document Control Index (DCI).
8. "@" indicates submission of documents to Inspection Agency.
9. Bill of Material shall form part of the respective drawing.
10. ALL THE DATA FOLDERS SHALL BE MARKED ON BOTH FRONT AND SPINE WITH THE FOLLOWING DATA:

- PROJECT TITLE / PURCHASE ORDER NO.
- EQUIPMENT / MATERIAL DESCRIPTION.
- TAG NUMBER(S) IF APPLICABLE
- DATA BOOK TITLE

11. Submission of documents to EIL for information shall not be a HOLD point for fabrication. Vendor can proceed with fabrication based on requirements of requisition. Any clarification required from EIL shall be obtained prior to proceeding with fabrication of the equipment. However, it is EIL¿s prerogative to review fabrication drawings at its discretion and EIL comments in line with requirements of requisition shall be

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incorporated by vendor without any cost and delivery implications.

12. For all Equipment, Contractor Quality Assurance Plan, Weld procedures, Inspection Records/Data and Material Test Certificates shall be reviewed by Authorized Inspector.

13. Each Vendor data item must be identified with Purchase Order Number and Equipment tag Number located on the cover sheet of the first page of the said item and in case of Engineering Drawing on each and every drawing.

14. SUCCESSFUL BIDDER TO SUBMIT DCI(DOCUMENT CONTROL INDEX) AND ALL THE VENDOR DATA THROUGH VENDOR PORTAL(HTTP://EDOCX.EIL.CO.IN/VPORTAL) ONLY. THE DETAIL GUIDELINES FOR UPLOADING THE DOCUMENT ON VENDOR PORTAL ARE AVAILABLE ON THE SITE ITSELF. SUCCESSFUL BIDDER TO SUBMIT ALL THE DOCUMENT AS SCHEDULED DOCUMENT ONLY.

15. SOFT COPIES OF VENDOR DOCUMENTS SHALL BE SUBMITTED IN AUTOCAD AS WELL AS PDF FORMAT

16. On completion of order, Vendor shall submit the final documents to the Owner/ Consultant, as per ¿Vendor Data Requirement¿ with 3 nos. hard copies (1 Original + 2 Duplicate) and 2 nos. soft copies in external HDD in addition to submission through EIL Bidder Portal. The final documents supplied shall be duly endorsed by the TPIA or EIL (which ever applicable) as `VERIFIED AND CERTIFIED FOR COMPLETENESS OF DOCUMENTATION AS PER SUPPLIER DATA REQUIREMENT¿

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1. EIL APPROVED SUPPLIER LIST FOR WIRE MESH TYPE DEMISTER/ MIST ELIMINATOR **:

Sl. No.	SUPPLIER CODE	SUPPLIER NAME	REMARKS
1.	C121	CONTINENTAL PROFILES LTD [FARIDABD]	
2.	C507	COSTACURTA SPA-VICO	OFFER IS ACCEPTABLE EITHER FROM C507 OR C507A
3.	C507A	COSTACURTA SPA-VICO [LECCO]	
4.	E020	EVERGREEN TECHNOLOGIES PVT. LTD. [KILLA PARDI]	
5.	H082	STANMESH PRIVATE LIMITED [DISTT. VADODRA]	
6.	K030	MUNTERS INDIA HUMIDITY CONTROL PRIVATE LIMITED [THANE]	
7.	K159	KOCH ENGINEERED SOLUTIONS INDIA PVT. LTD. [BARODA]	
8.	M721	MUNTERS EUROFORM GMBH	
9.	P073	PACE ENGINEERING INDUSTRIES PVT. LTD. [PALGHAR]	
10.	4325	SULZER INDIA PRIVATE LIMITED [PUNE]	FOR VACUUM COLUMNS: MODELS 4540, 9001, 9032, 9059. FOR OTHER COLUMNS: ALL RANGE

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JOB SPECIFICATION FOR SURFACE PREPARATION AND PROTECTIVE COATING

PROJECT: GROUP-II/III LOBS Project

CLIENT: M/s CPCL

JOB NO: B649

Rev. No	Date	Purpose	Prepared by	Checked by	Approved by
3	19-06-2026	ISSUED FOR JOB SPECIFICATION	JATIN	SAHIL	SRIDHAR
2	30-04-2023	ISSUED FOR JOB SPECIFICATION	JATIN	SAHIL	SRIDHAR
1	28-06-2026	ISSUED FOR JOB SPECIFICATION	JATIN	SAHIL	SRIDHAR
0	18-04-2023	ISSUED FOR JOB SPECIFICATION	UTTAM	SAHIL	SRIDHAR

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Abbreviations:

AS	:	Alloy Steel
ASTM	:	American Society for Testing and Materials
AWWA	:	American Water Works Association
CS	:	Carbon Steel
DFT	:	Dry Film Thickness
GI	:	Galvanized Iron
ID	:	Internal Diameter
ISO	:	International Organization for Standardization
LTCS	:	Low temperature Carbon Steel
MS	:	Mild Steel
MR	:	material requisition
NB	:	Nominal Bore
NA	:	Not applicable
OD	:	Outside Diameter
OSHA	:	Occupational Safety and Health Act
RCC	:	Reinforced Cement Concrete
RH	:	Relative humidity
SS	:	Stainless Steel
SOR	:	Schedule of Rate
SSPC	:	Steel Structure Painting Council
WFT	:	Wet Film Thickness

REVISION HISTORY:

1. Painting system for equipment located within 100 m of cooling towers has been updated. Refer Table 6.0.
2. 3 LPE coating system has been included for underground piping. Refer Table 10.0.
3. SS foils are included as protective measure along with protective coating for SS insulated items – Refer Table 11.0
4. Coating system for manhole covers has been incorporated in Table 14.0.
5. CPCL color coding requirements have been incorporated in Annexure-1.

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Sl. No.	CONTENT
1.0	General
2.0	Scope
3.0	Reference codes & standards
4.0	Equipment
5.0	Surface preparation, shop primer, coating application, repair and documentation
6.0	Paint materials
7.0	Coating systems
8.0	Storage
9.0	Color code
10.0	Identification of vessels, piping
11.0	Painting for civil defense requirements
12.0	Quality control, inspection and testing
13.0	Guarantee
14.0	Qualification criteria of painting CONTRACTOR/sub-CONTRACTOR
15.0	Qualification/acceptance criteria of paint coating system
16.0	Method of sampling & dispatch for laboratory testing
17.0	ANNEXURE-1: COLOUR CODE

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1.0 GENERAL

1.1 This technical specification shall be applicable for the work covered by the contract and without prejudice to the provisions of various international codes of practice, standard specifications etc. It is understood that the CONTRACTOR shall carry out the work in all respects with the best quality of materials and workmanship and in accordance with the best engineering practice and instructions of Engineer-In-Charge.

1.2 Wherever it is stated in the specification that a specific material is to be supplied or a specific work is to be done, it shall be deemed that the same shall be supplied or carried out by the CONTRACTOR.

Any deviation from this standard without written deviation permit from appropriate authority will result in rejection of the job.

1.3 This specification covers the requirement for protective coating for new construction.

2.0 SCOPE

2.1 Scope of work covered in the specification shall include, without being limited to, the following:

This specification defines the requirements for surface preparation, selection and application of primers and paints on external surfaces of equipment, vessels, machinery, piping, ducts, steel structures, external & internal protection of storage tanks for all services and chimneys, if any. The items listed in the heading of tables of paint systems are indicative only. However, the CONTRACTOR is fully responsible for carrying out all the necessary painting, coating and lining job on external and internal surfaces as per the tender requirement.

2.2 Extent of work

2.2.1 The following surfaces and materials shall require shop, pre-erection and field painting:

- All un-insulated carbon steel & alloy steel equipment like vessels, columns, storage tanks, exchangers, parts of boilers etc.
- All un-insulated carbon steel and low alloy plant and related piping, fittings and valves (including painting of identification marks), furnace ducts and stacks.
- All insulated parts of vessels, boilers, chimneys, stacks, piping and steam piping and other insulated items present, if any.
- All items contained in a package unit.
- All structural steel work, pipes, structural steel supports, walkways, handrails, ladders, platforms etc.
- Flare lines, external surfaces of MS chimney with or without refractory lining and internal surfaces of MS chimney without refractory lining.

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- Identification of colour bands on all piping, as required, including insulated aluminium clad, galvanized, SS and nonferrous piping.

- Identification lettering/ numbering on all painted surfaces of equipment/piping insulated aluminium clad, galvanized, SS and non-ferrous piping.

- Marking / identification signs on painted surfaces of equipment and piping including hazardous service.

- Supply of all primers, paints and all other materials required for painting (other than OWNER supplied materials)

- Application of pre-erection/fabrication and shop primer.

- Repair work of damaged pre-erection/ fabrication and shop primer and weld joints in the field/site before and after erection.

- All CS piping, equipment, storage tanks and internal surfaces of RCC tanks in ETP plant.

- Quality control, testing and inspection during all stages of work (surface preparation, application of coating and testing of furnished coating).

2.2.2 The following surfaces and materials shall not require painting in general. However, if there is any specific requirement by the OWNER, the same shall be painted as per the relevant specifications:

a. Un-insulated austenitic stainless steel.
b. Plastic and/or plastic coated materials.
c. Non-ferrous materials like aluminum, Cu-Ni alloys, galvanized steel.

In general Galvanized steel doesn't require painting. However if painting is required due to OWNERs instructions, contractual or for colour coding requirement then coating system in Table 4.0 shall be followed.

2.3 Unless otherwise instructed, final paint coating (i.e. application of field primer, intermediate and top coats) on pre-erection/ shop primed equipment shall be applied at site, only after all welding, testing on systems are completed as well as after completion of steam purging.

3.0 REFERENCE CODES & STANDARDS

3.1 Without prejudice to the provision of clause 1.1 above and the detailed specifications of the contract, latest editions of the following codes and standards are applicable for the work covered by this contract:

ISO-12944	:	Corrosion protection of steel structures by protective paint system
ASTM-Vol. 6.01&6.03	:	American standard test methods for paints and coatings
IS-101	:	Methods of sampling and test for paints, varnishes and related products
RAL CLASSIC	:	International standard for colour shade

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IS-2379 : Indian standard for pipe line identification-colour code

ISO 8501-1/SIS-05 5900 : ISO standard for preparation of steel substrates before application of paints and related products.

- Steel Structures Painting Council, U.S.A. (surface preparation specifications SSPC-SP).

- National Association of Corrosion Engineers (NACE), U.S.A.

3.2 Surface preparation standards

The latest editions of any of the following standards shall be followed for surface preparation:

3.2.1 ISO 8501-1 / SIS-05 59 00: ISO standard for preparation of steel substrates before application of paints and related products. This standard contains photographs of the various standards on four different degrees of rusted steel and as such is preferable for inspection purpose by the Engineer-In-Charge.

3.2.2 Steel Structures Painting Council, U.S.A. (surface preparation specifications SSPC-SP).

3.2.3 National Association of Corrosion Engineers (NACE), U.S.A.

3.2.4 Various international standards equivalent to Swedish standard for surface preparations are given in Table-1.

3.3 The CONTRACTOR shall arrange, at his own cost, to keep a set of latest edition of above mentioned standards and codes at site.

3.4 The paint manufacturers' instructions shall be followed as far as practicable at all times for the best results. Particular attention shall be paid to the following:

a. Instructions for storage to avoid exposure as well as extremes of temperature.
b. Surface preparation prior to painting shall be followed as per Table-4 to 13 of this standards.
c. Mixing and thinning.
d. Application, recommended limit on time intervals in between coats & DFT.

4.0 EQUIPMENT

4.1 All tools, brushes, rollers, spray guns, blast material, hand power tools for cleaning, all equipment, scaffolding materials, shot & grit blasting equipment , air compressors etc. required shall be suitable for the work and shall be arranged by the CONTRACTOR in sufficient quantity at the site. The manufacturers test certificates / data sheets for all the above mentioned items shall be reviewed by Engineer-In-Charge at site before start of the work.

4.2 Mechanical mixer shall be used for paint mixing operations in case of two pack systems except in case of specific requirement; the Engineer-In-Charge may allow the hand mixing of small quantities at his discretion for touch up work only.

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5.0 SURFACE PREPARATION, SHOP PRIMER / COATING APPLICATION, REPAIR AND DOCUMENTATION

5.1 General

5.1.1 Adhesion of the paint film to the surface depends largely on the degree of cleanliness of the metal surface. Proper surface preparation contributes more to the success of the paint protective system. In order to achieve the maximum durability, one or more of the following methods of surface preparation shall be followed, depending on condition of surface to be painted and as instructed by the Engineer-In-Charge.

a. Abrasive blast cleaning

b. Mechanical or power tool cleaning

5.1.2 Mill scale, rust, rust scale and foreign matter shall be removed fully to ensure that a clean and dry surface is obtained. Unless otherwise specified, surface preparation shall be done as per provisions of relevant tables given elsewhere in this specification. The minimum acceptable standard, in case of thermally sprayed metal coatings, in case of mechanical or power tool cleaning shall be St. 3 or equivalent. In case of blast cleaning, it shall be Sa 2-1/2 as per Swedish standard SIS-055900 (latest edition) or SSPC-SP or ISO 8501-01. Blast cleaning shall be Sa 3 as per Swedish standard in case of thermally sprayed metal coatings.

Before surface preparation by blast cleaning, the surface shall be degreased by aromatic solvent to remove all grease, oil etc as per SSPC-SP-1.

5.1.3 Irrespective of whether external or internal surface to be coated, blast cleaning shall not be performed where dust can contaminate surfaces undergoing such cleaning or during humid weather conditions having humidity >85%. In case of internal coating of storage tanks, dehumidifier shall be used to control humidity level below 60%. Dehumidifier should depress the dew point of air in the enclosed space, sufficient enough so as to maintain it 3ºC below the metal substrate temperature during entire period of blasting and coating application. During the interval time between application of primer coat and subsequent intermediate and top coats or between blast cleaning completion and start of application of primer coat, dehumidifier unit should be in continuous operation to ensure that no condensation occurs on the substrate.

Dehumidifier should be able to maintain grain drop (moisture removal) at the rate of 25 grains per pound of air per hour. Dehumidifier should have capacity of at least 2 air changes per hour of the enclosed space. All necessary psychometric data should be collected by the CONTRACTOR for the given site conditions, before starting operation of dehumidifier, to ensure that desired values of dew point and moisture content in enclosed space is achieved.

Dehumidification to be maintained round the clock for surface preparation and painting till the total coating application is over.

Dehumidifier shall not be stopped under any condition till the entire blasted surface is primed to the satisfaction of the technical representative of the paint

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manufacturer. In case the dehumidifier breaks down in middle of the job, the same shall be replaced at the risk and cost of the CONTRACTOR.

5.1.4 The Engineer in-Charge shall have the right to disallow usage of dehumidifier if the performance is not meeting the specified requirements. Under such circumstances, the CONTRACTOR shall remove the equipment and replace the same with another equipment to provide satisfactory results without any additional cost to the OWNER.

5.1.5 Irrespective of the method of surface preparation, the first coat of primer must be applied by airless spray/ air assisted conventional spray, if recommended by the paint manufacturer, on dry surface. This should be done immediately and in any case within 4 hours of cleaning of the surface. However, at times of unfavorable weather conditions, the Engineer-In-Charge shall have the liberty to control the time period, at his sole discretion and/or to insist on re-cleaning before primer application is taken up. In general, during unfavorable weather conditions, blasting and painting shall be avoided as far as practicable.

5.1.6 The external surface of RCC chimney to be painted shall be dry and clean. Any loose particle of sand, cement, aggregate etc. shall be removed by scrubbing with soft wire brush. Acid etching with 10-15% HCl solution for about 15 minutes shall be carried out. The surface must be thoroughly washed with water to remove acid and loose particles and then dried completely before application of paint.

5.2 Procedure for surface preparation

5.2.1 Air blast cleaning with abrasives

The surfaces shall be blast cleaned using one of the abrasives such as Al_2O_3 particles, chilled cast iron or steel grit, copper slag or nickel slag at a pressure of 7.0 kg/cm^2 and at an appropriate distance & angle depending on nozzle size maintaining constant velocity and pressure. Chilled cast iron or steel shall be in the form of shot or grit of size in the range of G16 – G42 conforming to SSPC AB1 and S250 grade size of steel shots (maximum) to obtain a desired surface profile of 35-50 microns trough to peak. For all other abrasives, size shall be in the range of G16 – G24. The combination of steel grits and shots shall be normally in the ratio of 3:1. The quality of abrasives shall be free from contaminants and impurities and shall meet the requirements of SSPC AB1. The compressed air shall be free from moisture and oil. The blasting nozzles should be venturi style with tungsten carbide or boron carbide as the material for liners. Nozzle orifice may vary from 3/16” to 3/4”. On completion of blasting operation, the blasted surface shall be clean and free from any scale or rust and must show a grey-white metallic luster. Primer or first coat of paint shall be applied within 4 hours of surface preparation. Blast cleaning shall not be done outdoors in bad weather without adequate protection. If there is dew on the metal surface, it shall be cleaned. The surface profile shall be uniform to provide good adhesion (i.e. 35 to 50 microns) to the paint. If possible, a vacuum collector shall be installed to collect and recycle the abrasives.

5.2.2 Mechanical or power tool cleaning

Power tool cleaning shall be done by mechanical striking tools, chipping hammers, grinding wheels or rotating steel wire- brushes. Excessive burnish of the surface

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shall be avoided as it can reduce paint adhesion. On completion of cleaning, the detached rust, mill scale etc. shall be removed by clean rags and /or washed by water/steam and thoroughly dried with compressed air jet before application of paint.

5.3 Non-compatible shop coat primer

For equipment on which application of total protective coating (primer + intermediate + top coat) is carried out at shop, compatibility of finish coat with primer should be checked with the paint manufacturer. If the shop coat is in satisfactory condition showing no major defect upon arrival at site, the shop coat shall not be removed.

5.4 Shop coated equipment (coated with primer & finishing coat) should not be repainted unless paint is damaged. Repair shall be carried out as per Table-3 of paint systems depending upon the compatibility of paint.

5.5 Shop primed equipment and surfaces will only be 'spot cleaned' in damaged areas by means of power tool brush cleaning or hand tool cleaning and then spot primed before applying one coat of field primer, unless otherwise specified. If shop primer is not compatible with field primer, then shop coated primer should be completely removed before application of selected paint system for a particular environment.

5.6 For package units/equipment, shop primer should be as per the paint system given in this specification. However, manufacturers' standard can be followed after review.

5.7 Coating procedure and application

All coatings shall be applied by airless spray except for the following special cases, where application can be carried out by brush subject to suitability of the application of the paint product by brush.

- Spot repair
- Stripe coating on edges
- Small bore parts not suitable for spray application

Irregular surfaces such as sharp edges, welds, small brackets, and interstices may stripe coated to ensure specified DFT is achieved. Paint manufacturer recommendation should be followed before deciding for brush application.

5.7.1 Surface shall not be coated in rain, wind or in an environment where injurious airborne elements exist, when the steel surface temperature is less than 5 ^{0}F above dew point, when the relative humidity is greater than 85%, when the temperature is below 40^{o}F and when the ambient/substrate temperature is below the paint manufacturers recommended temperature of application and curing. De-humidifier equipment shall be used to control RH and Dew point. The paint application shall not be done when the wind speed exceeds 20 km per hour.

5.7.2 Blast cleaned surface shall be coated with complete application of primer as soon as practicable but in no case later than 4 hours the same day.

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5.7.3 To the maximum extent practicable, each coat of paint shall be applied as a continuous film with uniform thickness and free of probes. Any spots or areas missed in application shall be re-coated and permitted to dry before the next coat is applied. Applied paint should have the desired wet film thickness.

5.7.4 Each coat shall be in proper state of cure or dryness before the application of succeeding coat. Material shall be considered dry for re-coating when an additional coat can be applied without the development of any detrimental film irregularities such as lifting or loss of adhesion of the under coat. Manufacturer's instructions shall be followed for inter coat interval.

5.7.5 When the successive coat of the same colour have been specified, alternate coat shall be tinted, when practical, sufficiently to produce enough contrast to indicate the complete coverage of the surface. The tinting material shall be compatible with the material underneath and shall not be detrimental to its service life and shall be recommended by the original paint manufacturer.

5.7.6 Airless spray application shall be in accordance with the following procedure:
As per steel structure paint manual Vol.1 & Vol.2 by SSPC, USA. Airless spray relies on hydraulic pressure rather than air atomization to produce the desired spray. An air compressor or electric motor, issued to operate a pump to produce pressures of 1000 to 6000 psi paint, is delivered to the spray gun at this pressure through a single hose within the gun. A single paint stream is divided into separate streams, which are forced through a small orifice resulting in atomization of paint without the use of air. This results in more rapid coverage with less over spray. Airless spray usually is faster, cleaner, more economical and easier to use than conventional air spray.

Airless spray equipment is mounted on wheels, and paint is aspirated in a hose that sucks paint from any container, including drums. The unit shall have in-built agitator that keeps the paint uniformly mixed during the spraying. The unit shall consist of in-built strainer. Usually, a very small quantity of thinning is required before spray. In case of high build epoxy coating (two packs), 45:1 pump ratio and 0.020-0.023" tip size will provide a good spray pattern. Ideally, fluid hoses should not have ID less than 3/8" and not longer than 50 ft. to obtain optimum results.

In case of gun choking, de-choking steps shall be followed immediately.

5.7.7 Brush application of paint shall be in accordance with the following:

a. Brushes shall be of a style and quality that will enable proper application of paint.

b. Round or oval brushes are most suitable for rivets, bolts, irregular surfaces, and rough/ pitted steel. Wide flat brushes are suitable for large flat areas but they shall not have width over 5 inches.

c. Paint shall be applied into all corners.

d. Any runs or sags shall be brushed out.

e. There shall be a minimum of brush marks left in the applied paint.

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f. Surfaces not accessible to brushes shall be painted by spray, daubers, or sheepskin.

5.7.8 For each coat, the painter should know the WFT corresponding to the specified DFT and standardize the paint application technique to achieve the desired WFT. This has to be ensured in the qualification trial.

5.8 Drying of coated surfaces

5.8.1 No coat shall be applied until the preceding coat has dried. The material shall be considered dry for re-coating when another coat can be applied without the development of any film irregularities such as lifting or loss of adhesion of undercoats. Drying time of the applied coat should not exceed maximum specified for it as a first coat. If this exceeds, the paint material has possibly deteriorated or mixing is faulty.

5.8.2 No paint shall be force dried under conditions which will cause chalking, wrinkling, blistering formation of pores, or detrimentally affect the conditions of the paint.

5.8.3 No drier shall be added to paint on the job unless specifically called for in the manufacturers' specification for the paint.

5.8.4 Paint shall be protected from rain, condensation, contamination, snow and freezing until dried to the fullest extent practicable.

5.9 Spot repair of damaged primer

5.9.1 Where pre-erection/shop primer has been damaged at isolated localized spots during handling and transportation or after erection / welding, its repair shall be done as given below and as per the Table-3 of this specification.

5.9.2 Surface preparation: Quickly remove the primer from damaged area by mechanical scraping and emery paper conforming to SSPC-SP-3 to expose the white metal. Blast clean the surface, if possible. Feather the primed surface, over the intact adjacent surface surrounding the damaged area, by emery paper.

Primer coating: One coat of F-9 shall be applied wherever damage was observed on pre-erection / pre fabrication or shop primer of inorganic zinc silicate coating (F-9). Similarly one coat of F-12 shall be applied wherever damage observed on pre-erection ***/*** pre-fabrication/shop primer of silicone aluminium (F-12).

5.9.3 If damaged areas are found to be extensive and spread over large areas, then entire pre-erection/pre-fabrication/shop primer shall be removed by blasting to achieve SSPC-SP-10 and entire blasted surface shall be primed again with F-9 or F-12, as applicable, for the intended design temperature. (See note of Table-3).

5.10 Assessment of painting requirement

The paint system to be applied for a specific job shall be arrived at sequentially as given below:

- Identify the environment from area classification details and choose the appropriate table.
- Identify the design temperature from the technical documents.

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- Identify the specific field paint system and surface preparation requirement from the above identified table and temperature range.
- Identify the shop priming requirement from based on compatibility of the above paint system.
- Identify the need of repair of shop primer and execute as per Table-3.

5.11 Documentation / records

5.11.1 A written quality plan with procedure for qualification trials and for the actual work including test and inspection plan & procedure for approval before start of work.

5.11.2 Daily progress report with details of weather conditions, particular of applications, no. of coats and type of materials applied, anomalies, progress of work versus program.

5.11.3 Results of measurement of temperatures, relative humidity, surface profile, film thickness, holiday detection, adhesion tests with signature of appropriate authority.

5.11.4 Particulars of surface preparation and paint application during trials and during the work.

5.11.5 Details of non-compliance, rejects and repairs.

5.11.6 Type of testing equipment and calibration.

5.11.7 Code and batch numbers of paint materials used.

The coating applicator must maintain a job record consisting of all the information as per 5.11.2 -5.11.7 above as well as the approved procedure of work (5.11.1 above). The job record consisting of information in accordance to 5.11.2 – 5.11.7 shall be entered on daily basis and should be daily signed by Engineer-in-charge.

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TABLE-1: SURFACE PREPARATION STANDARDS

Sl. No.	DESCRIPTION	VARIOUS INTERNATIONAL STANDARDS (EQUIVALENT)			REMARKS
		ISO 8501-1/ SIS-05 59 00	SSPC-SP, USA	NACE, USA	
1	Manual or hand tool cleaning Removal of loose rust, loose mill scale and loose paint, chipping, scrapping, standing and wire brushing. Surface should have a faint metallic sheen.	St 2	SSPC-SP-2	-	This method is applied when the surface is exposed to normal atmospheric conditions when other methods cannot be adopted and also for spot cleaning during maintenance painting.
2	Mechanical or power tool cleaning Removal of loose rust loose mill scale and loose paint to degree specified by power tool chipping, de-scaling, sanding, wire brushing and grinding, after removal of dust, surface should have a pronounced metallic sheen.	St 3	SSPC-SP-3	-	
3	Dry abrasive Blast cleaning There are four common grades of blast cleaning:				
3.1	White metal Blast cleaning to white metal cleanliness. Removal of all visible rust. Mill scale, paint & foreign matter 100% cleanliness with desired surface profile.	Sa 3	SSPC-SP-5	NACE No.1	Extremely clean surface can be expected to have prolonged life of paint system.
3.2	Near white metal Blast cleaning to near white metal cleanliness, until at least 95% of each element of surface area is free of all visible residues with desired surface profile.	Sa 2½	SSPC-SP-10	NACE No.2	The minimum requirement for chemically resistant paint systems such as epoxy, vinyl, polyurethane based and inorganic zinc silicate paints. Also for

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Sl. No.	DESCRIPTION	VARIOUS INTERNATIONAL STANDARDS (EQUIVALENT)			REMARKS
		ISO 8501-1/ SIS-05 59 00	SSPC-SP, USA	NACE, USA	
					conventional paint systems used under fairly corrosive conditions to obtain desired life of paint system.
3.3	Commercial Blast Blast cleaning until at least two-third of each element of surface area is free of all visible residues with desired surface profile.	Sa 2	SSPC-SP-6	NACE No.3	For steel required to be painted with conventional paints, for exposure to mildly corrosive atmosphere and for longer life of the paint systems.
3.4	Brush-off Blast Blast cleaning to white metal cleanliness, removal of all visible rust, mill scale, paint & foreign matter. Surface profile is not so important.	Sa 1	SSPC-SP-7	NACE No.4	-

6.0 PAINT MATERIALS

Typical characteristics and codes of various paint materials used in this specification are as follows.

TABLE-2: PAINT MATERIALS
(Refer to *general notes* at the end of this table)

DESCRIPTION	P-6	P-7	P-4
Technical name	Epoxy Zinc Phosphate Primer	Cold Galvanizing product	Etch Primer/Wash Primer
Type and composition	Two component polyamine cured epoxy resin medium, pigmented with zinc phosphate.	One pack synthetic resin based zinc galvanizing containing min 92% of electrolytic zinc dust of 99.95% purity.	Two pack polyvinyl butyral resin medium cured with phosphoric acid solution pigmented with zinc tetroxy chromate.

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Volume solids % Minimum	49	37	9
DFT per coat, µ	40-50	40-50	8-10
Theoretical covering capacity in m^2/coat/ litre	9.8-12.2	3.4-4.4 m²/kg	9.0-11.2
Weight per litre in Kg/litre	1.4±0.05	2.67 kg at 15°C	1.2±0.05
Touch dry at 30°C (max.)	30 min.	10 min.	2 hrs.
Hard dry at 30°C (max.), hrs	8	24	24
Over-coating interval, hrs	Min. 8	Min. 4	Min. 4-6
Pot life at 30 ° C for two component paints, hrs	6 - 8	NA	NA
Adhesion (ASTM D 4541)	>7	NA	NA
Temperature resistance (min) ° C (ASTM D 2485) *Note 8	80 (Method A)	50 (Method A)	NA

TABLE- 2: PAINT MATERIALS (Contd.)

DESCRIPTION	F-2	F-3	F-6A/B	F-6C
Technical name	Acrylic Polyurethane finish paint	Chlorinated rubber based finish paint	Epoxy-High Build coating	Solvent less epoxy coating
Type and composition	Two-pack aliphatic isocynate cured acrylic finish paint (free of alkyd/polyester resins).	Single pack plasticized chlorinated rubber based medium with chemical and weather resistant pigments.	F-6A Two-Pack Aromatic polyamine cured epoxy resin medium suitably pigmented. F-6B: polyamide cured epoxy resin medium suitably pigmented with MIO.	Two pack, cured with Amine Adduct; catalyzed epoxy resin suitably pigmented

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Volume Solids % Minimum.	40	36	57	98
DFT per coat, µ	30-40	30-40	100-125	250-500
Theoretical covering capacity in m^2/coat/litre	10-13.3	9-12	4.6-5.7	2-3.9
Weight per liter in Kg/litre	1.15±0.03	1.15±0.03	1.42±0.03	1.40±0.03
Touch dry at 30 ^{o}C (max)	30 min.	30 min.	3 hrs	3 hrs
Hard dry at 30 ^{o}C (max.), hrs	8	8	16	16
Full cure at 30 ^{o}C (for immersion/ high temperature service)	NA	NA	5 days	5 days
Over-coating Interval, hrs	Min.12.	Min. Overnight	Min. Overnight Max. 5 days	Min. 8. Max. 48
Pot life (approx.) at 30 ^{o}C for two component paints, hrs	5-8	NA	3-6	0.5
Adhesion (ASTM D 4541)	>5	>4	>7	>8
Abrasion Resistance (ASTM D4060) For 1000 g load	<300 mg /1000 cycles/CS17 or <100 mg /1000 cycles/CS10	NA	NA	NA
Temperature resistance (min.) ^{o} C (ASTM D 2485) *Note 8	80 (Method A)	60 (Method A)	80 (Method A)	120 (Method A)

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TABLE- 2: PAINT MATERIALS (Contd.)

DESCRIPTION	F-7	F-8	F-9 (primer)	F-11	F-12
Technical name	High build coal tar epoxy coating.	Self priming type surface tolerant high build epoxy coating (complete rust control Coating)	Inorganic zinc silicate Coating	Heat resistant synthetic medium based two pack Aluminium Paint suitable up to 250°C dry temp.	Heat resistant silicone Aluminium Paint suitable up to 540°C dry temp.
Type & composition	Two pack polyamide cured epoxy resin blended with coal tar medium, suitably pigmented	Two pack epoxy resin based suitable pigmented and capable of adhering to manually prepared surface and old coating.	A two pack air drying self curing solvent based inorganic zinc silicate coating with minimum 80% zinc content on dry film. The final cure of the dry film shall pass the MEK rub test as per ASTM D4752.	Heat resistant synthetic medium based two pack Aluminium paint suitable up to 250°C.	Single pack silicone resin based medium with Aluminium flakes.
Volume Solids % Minimum.	62	75	57	35	18
DFT per coat in µ	100-125	100-125	65-75	15-20	15-20
Theoretical covering capacity in M^2/coat/ litre	5-6.2	6-7.5	7.6-8.8	17.5-23.3	9-12
Weight per liter in Kg/litre	1.40±0.03	1.41±0.03	2.3±0.03	0.95±0.03	1.00±0.03
Touch dry at 30°C (maximum), hrs	4	3	0.5	3	0.5
Hard dry at 30°C (maximum), hrs	48	24	12	12	24
Full cure 30°C (for immersion /high temperature service)	5 days	5 days	NA	NA	NA
Over-coating interval, hrs	Min. 24 hrs max.5days	Min. 10	Min. 12 hrs at 20°C & 50% RH	Min. 24	Min. 24

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Pot life at 30°C for two component Paints, hrs	4-6	1.5	4-6	NA	NA
Adhesion MPa (ASTM D 4541)	>5	>5	>5	NA	NA
Temperature resistance (min.) ° C (ASTM D 2485) *Note 8	80 (Method A)	80 (Method A)	400 (Method B)	250 (Method A)	540 (Method B)

TABLE- 2: PAINT MATERIALS (Contd.)

DESCRIPTION	F-14	F-15	F-16	F-17	F-20
Technical name	Polyamine cured coal tar epoxy	Two-component Epoxy phenolic coating cured with Polyamine adduct hardener	Inert polymeric matrix coating suitable for under insulation for CS and SS.	Novolac epoxy phenolic coating cured with Polyamine adduct hardener	Glass flake reinforced vinyl ester coating.
Type & composition	Specially formulated polyamine cured coal tar epoxy suitable for application under insulation	Two pack ambient temperature curing epoxy phenolic coating system suitable for application under insulation	Suitable for high temperature service and under insulation coating for CS, alloy steel and SS	Novolac epoxy phenolic coating cured with Polyamine adduct hardener	Two component glass flake filled vinyl ester lining for under immersion services up to 90 deg. C.
Volume Solids % Minimum.	67	67	50	98	98
DFT per coat in microns	100-125	75-100	100-125	300-450	500-600
Theoretical covering capacity in M^2/coat/ litre	5.4-6.7	6.7-8.9	4-5	2.2-3.3	1.6-2
Weight per liter in Kg/litre (mix paint)	1.45±0.03	1.65±0.03	>1.3	1.7	>1.2
Touch dry at 30°C (max) , hrs	4	3	1	2	2

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Hard dry at 30°C (max), hrs	24	24	16	24	4
Full cure 30°C (for immersion / high temp service)	168 hrs (7 days)	168 hrs (7 days)	NA	168 hrs (7 days)	96 hrs (4 days)
Over-coating interval	Min. 6 hrs Max.5 days	Min. 36 hrs Max.21 days	Min.6 hrs Max. Not applicable	Min. 16 hrs Max.21 days	Min. 4 hrs Max.3 days
Pot life at 30°C, hrs (for two component paints)	4	4-6	1	1	50 min-1 hr
Adhesion, MPa (ASTM D 4541)	>6	>7	NA	>8	>7, Tensile strength >20N/mm2 (ASTM D 638)
Temperature resistance (min.) ° C (ASTM D 2485) *Note 8 & 9	125 (Method A)	150 (Method A)	650 (Method B)	150 (Method A)	90 (Method A)

General notes for TABLE-2:

1. Covering capacity and DFT achieved per coat depends on method of application. Covering capacity specified above is theoretical. For estimation of actual quantity of paints required, include the losses during application. Minimum specified DFT should be maintained in any case.
2. All primers and finish coats should be ambient temperature curing and air drying unless otherwise specified.
3. All paints shall be applied in accordance with manufacturer's instructions for surface preparation, intervals, curing and application. The surface preparation, quality and workmanship should be ensured. Wherever a deviation is noticed from the specification in manufacturer data sheet, more stringent one between the data sheet and the specification shall prevail e.g. if this specification recommends Sa 2 ½ and the manufacturer data sheet requires Sa3, the surface preparation shall be done as per Sa 3. However in another case if this specification requires the surface preparation of Sa 2 ½ and the manufacturer data sheet recommends only Sa 2 as minimum, the surface preparation shall be done as per Sa 2 ½.
4. Technical data sheets for all paints shall be supplied at the time of submission of quotations.
5. Higher specific gravity of F-9 is also acceptable.
6. Internationally recognized & acceptable testing method shall be used for lab testing wherever testing standards are not mentioned.
7. Touch dry, hard dry, pot life, full cure period, & over coating interval shall be as per manufacturer's data sheets and no testing is required. Slight variation in the values of these

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parameters along with weight per liter may be permissible with the discretion of engineer-in-charge only.

8. Temperature resistance tests (ASTM D2485) shall be carried out for minimum required temperature resistance indicated.
9. Wherever ASTM D 2485 method B is applicable, test results of the panels subjected to salt spray (ASTM B 117) after muffle furnace exposure shall be submitted for pre-qualification purpose.
10. F-6A shall be suitable for immersion services in line with recommendations in tables-7 to 9.
11. F-6C shall be suitable for immersion services of hydrocarbons and DM water (in line with recommendations in table-9)
12. F-7 shall be suitable for immersion service of hydrocarbons & underground service up to 80 Deg C minimum.
13. F-14 shall be suitable for under insulation service up to 125 deg C.
14. F-15 shall be suitable for high temperature immersion & under insulation services in line with recommendations from table-8 to 13.
15. F-17 shall be suitable for high temperature immersion service and underground services as recommended in table-9 to 10.

7.0 COATING SYSTEMS

The coating system should be selected based on the plant location as given below:

Classification based on plant location:

a) Plant located in inland area (more than 50 km from coast).
Environment classification – Industrial

- For offsite areas: Table-5 to be followed.
- For all process unit areas including DM, CPP and Cooling tower: Table-6 to be followed.

b) Plant located on sea coast or within 50 km from sea coast.
Environment classification- Industrial marine

- For offsite areas, process unit areas including DM, CPP, Cooling tower etc.: Table-6 to be followed.

c) For external surface of above ground tanks, Table-8 to be followed for all locations (inland or coastal)

General notes for clause 7.0:

1. Coating systems (primers, finish paints etc.) based on area classification/environments/applications are tabulated in Table-4 to Table-13
2. Repair of pre-erection/pre-fabrication & shop priming after erection/ welding shall be done as per Table-3.

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TABLE-3: REPAIR OF PRE-ERECTION/PRE-FABRICATION OR SHOP PRIMER AFTER ERECTION/WELDING

(For all un-insulated CS, LTCS & low allow steel items in all environments)

Sl. No.	Design Temp. in °C	Surface Preparation	Coating System	Total DFT in Microns (min.)	Remarks
3.1	-45 to 80	SSPC-SP-3	2 coats of F-8 @ 125 µ DFT/coat + 1 coat of F-2 @ 40µ DFT/coat	290	See Note-1 below and clause 5.9.3
3.2	81 to 400	SSPC-SP-3	1 coat of F-9 @ 65-75 µ DFT/coat + 2 coats of F-12 @ 20 µ DFT/coat	105-115	
3.3	401 to 550	SSPC-SP-3	3 coats of F-12 @ 20 µ DFT/coat	60	

Notes for Table-3:

Note-1: The application and repair of pre-erection/pre-fabrication or shop primer given in above tables shall be done for all the items to be painted. In case the damages of primer are severe and spread over large area, entire primer shall be removed by blasting to achieve SSPC-SP-10 and surfaces to be primed again with F-9 or F-12 as applicable.

TABLE-4: COATING SYSTEM FOR GI HAND RAILS & GI ITEMS

(Refer clause 2.2.2)

Sl. No.	Design Temp. in °C	Coating System	Total DFT in Microns (min.)	Remarks
4.1	Up to 60	Hot Dip Galvanizing to 80-85 microns (600-610 gm/m^2)as per IS 4759, 2629, 4736, 2633 + 1 coat of P-6 @ 40µ DFT/coat + 1 coat of F-2 @ 40 microns DFT/coat	80 of finish coat (excluding the thickness of galvanizing)	-

General notes for Table-4:

1. No galvanized specimen shall have thickness less than 80 microns.
2. Repair of the damaged areas of galvanized coatings due to welding during erection shall be carried out as per recommended practice IS: 11759, using cold galvanizing spray process. Organic paint systems are not acceptable for the repair.
3. After repair of damaged galvanized coating by cold galvanization (P-7), the repaired area shall be top coated with paint system as given in table-4 above (i.e. 1 coat of P-6 @ 40µ DFT/coat + 1 coat of F-2 @ 40µ DFT/coat).
4. Suggested cold galvanizing manufacturers are ZINGA, LOCTITE or Z.R.C.
5. Galvanized gratings don't require painting in general until otherwise specified elsewhere or as per the requirement of the OWNER. Ladders, stairways & hand rails

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require painting to meet the colour coding requirement of the OWNER. Contractor has to ensure the applicable colour coding prior to application of coating as per this clause.

TABLE-5: COATING SYSTEM FOR NORMAL CORROSIVE AREAS (OFFSITES)
(For all un-insulated above ground CS, LTCS & low allow steel piping, equipment, structures, valves, vessels & columns etc.)

Sl. No.	Design Temp. in °C	Surface Preparation & Pre-erection/Shop Primer	Coating System		Total Final DFT in Microns (min.)	Remarks
			Primer	Finish Coat		
5.1	-45 to -15	SSPC-SP-10; 1coat of F-9 @ 65-75µ DFT/coat	-	None	65-75	No over-coating shall be done on F-9 F-12 shall be ambient temperature curing type
5.2	-16 to 80	SSPC-SP-10; 1coat of F-9 @ 65-75µ DFT/coat	1 coat of P-6 @ 40 µ DFT/coat	1coat of F-6B @ 100 µ DFT/coat+ 1 coat of F-2 @ 40µ DFT/coat	245-255	
5.3	81 to 250	SSPC-SP-10; 1coat of F-9 @ 65-75µ DFT/coat	-	2 coats of F-11 @ 20µ DFT/coat (2x20=40)	105-115	
5.4	251 to 400	SSPC-SP-10; 1coat of F-9 @ 65-75µ DFT/coat	-	2 coats of F-12 @ 20µ DFT/coat (2x20=40)	105-115	
5.5	>400	SSPC-SP-10; 1coat of F-12 @ 20µ DFT/coat	-	2 coats of F-12 @20µ DFT/coat (2x20=40)	60	This system is suitable up to 540^0 C

General notes for Table-5:

1. The list of items given in the heading of the above table is not exhaustive. There may be more items for a particular contract where these specifications are used. The CONTRACTOR is fully responsible for completing painting including prefabrication primer for all the items supplied and fabricated through his scope of work as per tender document.
2. Flare line within unit or offsite areas shall be coated as per clause 6.3 of Table-6.
3. If the application of pre-erection/pre-fabrication/shop primer has already been completed, the same shall not be repeated in the field. In case the damages to the primer coat are severe and are spread over large areas, the Engineer-In-Charge may decide & advise re-blasting and re-application of the primer coat. Repair of pre-fabrication/pre-erection primer, if required, shall be done as per Table-3**.**
4. In case of paint systems as per sl. nos. 5.4 and 5.5, the colour bands shall be applied over the aluminum paint as per the colour coding requirement for specific service of the piping.
5. All coating system including surface preparation, primer, intermediate and finish coat for piping is recommended to be done at field only until otherwise specified.
6. For equipment, vessels, columns & valves etc scope of surface preparation, application of primers to finish coat including repair shall be as per applicable contractual documents like SOR, MR and PR etc.

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TABLE-6: COATING SYSTEM FOR
(For all un-insulated above ground CS, LTCS & low alloy steel piping, equipment, structures, valves, vessels & columns, furnace stacks etc.)

CORROSIVE AREAS (PROCESS UNITS, DM AND CPP)						
Sl. No.	**Design Temp. °C**	**Surface Preparation & Pre-erection/Shop Primer**	**Coating System**		**Total DFT in Microns (min.)**	**Remarks**
			Primer	**Finish Coat**		
6.1	-45 to -15	SSPC-SP-10; 1coat of F-9 @ 65-75µ DFT/ coat	-	-	65-75	a)No over coating on F-9 is allowed b) F-12 shall be ambient temperature curing type
6.2	-16 to 80	SSPC-SP-10; 1coat of F-9 @ 65-75µ DFT/ coat	1 coat of P-6 @ 40 µ DFT/ coat	2 coats of F-6B @ 100 µ DFT/coat + 1 coat of F-2 @ 40µ DFT/coat (2x100 + 40= 240)	345-355	
6.3	81 to 400	SSPC-SP-10; 1coat of F-9 @ 65-75µ DFT/ coat	-	2 coats of F-12 @ 20µ DFT/coat 2x20=40	105-115	
6.4	>400	SSPC-SP-10; 1coat of F-12 @ 20-25µ DFT/coat	-	2 coats of F-12 @ 20-25µ DFT/coat (2x20=40)	60	This system is suitable up to 540^{O} C
CORROSIVE AREAS (LOCATED WITHIN A 100-METER RADIUS OF THE COOLING TOWER)						
Sl. No.	**Design Temp.°C**	**Surface Preparation & Pre-erection/Shop Primer**	**Coating System**		**Total DFT in Microns (min.)**	**Remarks**
			Primer	**Finish Coat**		

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6.5	-16 to 80	SSPC-SP-10; 1coat of F-9 @ 65-75µ DFT/ coat	1 coat of surface moisture tolerant Aluminum-Filled Phenylamine /Polyamine Epoxy Mastic (Vol. of solid: 80% min.) @ 150 µ DFT/ coat	1 coat of moisture tolerant Phenalkamine/ Polyamine epoxy (Vol. of solid: 80% min.) @ 150 µ DFT/coat + 1 coat of 2-Pack Aliphatic Polyurethane Finish Coat @ 50µ DFT/coat	350	Primer & Intermediate coat shall have compliance to Norsok M-501 Edition 6 (Coating system 7) standard

General notes for TABLE-6:

1. The list of items given in the heading of the above table is not exhaustive. There may be more items for a particular contract where these specifications are used. The CONTRACTOR is fully responsible for completing painting including prefabrication primer for all the items supplied and fabricated as per tender document and scope of work.
2. If the application of pre-erection/pre-fabrication/shop primer has already been completed, the same shall not be repeated in the field. In case the damages to the primer coat are severe and spread over large areas, the Engineer-In-Charge may decide & advise re-blasting and re-application of the primer coat. Repair of pre-fabrication/pre-erection primer, if required, shall be done as per Table-3.
3. For external surface of RCC Chimney, 2 coats of F-6 B@ 100µ DFT/coat to obtain a total DFT of 200 µ shall be applied after proper surface preparation as per guidelines in 5.1.6.
4. In case of paint systems as per sl. Nos. 6.3 and 6.4, the colour bands shall be applied over the aluminum paint as per the colour coding requirement for specific service of piping given in clause 9.0.
5. For piping, equipment, vessels, columns & valves etc, scope of surface preparation, application of primers to finish coat including repair shall be as per applicable contractual documents like SOR, MR, and PR etc.

TABLE-7: COATING SYSTEMS FOR EFFLUENT TREATMENT PLANT (ETP)

Sl. No.	Design Temp. in °C	Surface Preparation	Coating system		Total DFT in Microns (min.)	Remarks
			Primer	Finish Coat		
7.1	For external surfaces of C.S./M.S. items: screens, walk way bridges, baffles, dual media filters, vertical pumps, piping in treated effluent sump, bio sludge pump, screw pump and pump house, CS tanks, sumps and vessels.					

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	-14 to 80	SSPS-SP-10	1 coat of F-9 @ 65-75µ DFT/coat	2 coats of F-6A @100µ DFT/coat + 1 coat of F-2 @ 40µ DFT/coat (2x100+40 =240)	305-315	-
7.2	For internal surfaces of CS/MS Items: bio-sludge sump, filter feed sump, process sump, sanitary sump, transfer sump, sludge, slop oil tank, scrapping mechanism in clarifier.					
	-14 to 80	SSPS-SP-10	1 coat of F-6A @ 100µ DFT/coat.	2 coats of F-6A @100µ DFT/coat (2x100=200)	300	-
7.3	All R.C.C./concrete surfaces exposed to effluent water / liquid such as tanks, structures, drains etc. in process sump, TPI separator (process and oil), aeration tank and transfer sump etc.					
	-14 to 80	Blast cleaning to SSPC-SP guide lines and acid etching with 10-15% HCl acid followed by thorough water washing.	Epoxy screed lining		3000	Epoxy screed lining shall be applied as per manufacturer and Engineer-In-Charge instructions
7.4	CS dual media filters, chemical dosing tanks – internal surface					
	Up to 60	SSPC-SP-10	1 coat of clear two component solvent free vinyl ester primer @ 100µ DFT/ Coat + 2 coats of F-20 @ 500µ DFT/ Coat		1100	-

TABLE-8: EXTERNAL COATING SYSTEMS FOR CARBON STEEL AND LOW ALLOY STEEL STORAGE TANKS

Sl. No.	Design Temp. in °C	Surface Preparation	Coating system (Note-1)		Total DFT in Microns (min.)	Remarks
			Primer	Finish Coat		
8.1	All external surfaces of shell, wind girders, appurtenances, roof tops of all above ground tank including top side of external and internal floating roof and associated external structural works.					

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8.1.1	-14 to 80	SSPC-SP-10	1coat of F-9 @ 65-75µ DFT/coat + 1coat of P-6 @ 40µ DFT/ coat	2 coats of F-6B @ 100µ DFT /coat + 1 coat of F-2 @ 40µ DFT/ coat	345-355	F-6 should be suitable for occasional water immersion
8.1.2	81 to 150	SSPC-SP-10	1 coat of F-15 primer @ 80µ DFT/ coat + 1 coat of F-15 intermediate coat @ 80µ DFT/coat	1 coat of F-15 finish coat @80µ DFT/ coat + 1coat of F-2 @ 40µ DFT/ coat	280	-
8.1.3	151 to 400	SSPC-SP-10	1coat of F-9 @ 65-75µ DFT/coat	2 coats of F-12 @20 µ DFT/ coat or 1 coat of F-16 @ 125 µ DFT / coat	105-115 or 190-200	
8.2	External surfaces of bottom plate (in contact with soil) for all storage tanks.					
8.2.1	-14 to 80	SSPC-SP-10	1 coat of F-9 @ 65-75µ DFT/ coat	3 coats of F-7@ 100µ DFT/coat (3x100=300)	365-375	F-7 should be suitable for immersion service
8.2.2	81 to 150	SSPC-SP-10	1 coat of F-17 primer @ 400µ DFT/ coat	1 coat of F-17 finish coat @ 400µ DFT/ coat	800-825	-
8.3	For underside of the bottom plate (not in contact with soil) (in case tank is not lifted during PWHT) (Note- 2)					
8.3.1	-29 to 400	SSPC SP-10	1 coat of inert polymeric matrix coating @ 125 µ	1 coat of inert polymeric matrix coating @ 125 µ	250-300	-

Notes for TABLE-8:

1) All paint coating application including primer for tanks shall be carried out at field after erection and completion of entire welding.
2) For underside of bottom plate :
   a) Painting shall be carried out before laying the bottom plate for tanks with non-Post Weld Heat Treatment (PWHT).
   b) For tanks with PWHT, painting shall be carried out after PWHT.
   c) In case tank is not lifted during PWHT then painting shall be applied before laying the bottom plate, clause no.8.3.1 shall be followed.

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3) If any tank or vessel with different temperature or service is not covered in the above table, then contractor shall develop a suitable system. The same may be submitted for PMC/OWNER review.

TABLE-9: INTERNAL COATING SYSTEMS FOR CARBON STEEL AND LOW ALLOY STORAGE TANKS

Sl. No.	Design Temp. in °C	Surface Preparation	Coating system		Total DFT in Microns (min.)	Remarks
			Primer	Finish Coat		
9.1	Potable, Raw & Fire water All internal surfaces but not limited to internal surfaces of shell for full height, bottom plate, accessories, roof and roof structures of cone and dome roof tanks etc.					
9.1.1	-14 to 60	SSPC-SP-10	1 Coat of F-6A @ 100 µ DFT/coat	2 Coats of F-6A @ 100µ DFT/ Coat (2x100=200)	300	-
9.2	De-mineralized (DM) water & Condensates etc.: All internal surfaces but not limited to internal surfaces of shell for full height, bottom plate, accessories, roof and roof structures of cone and dome roof tanks etc.					
9.2.1	-14 to 60	SSPC-SP-10	1 Coat of F-6A @ 100µ DFT/coat	2 coats of F-6C @ 200µ DFT/ coat (2x200=400)	500	Single coat of F-6C @ 400 µ is also acceptable
9.2.2	61 to150	SSPC-SP-10	1 coat of F-15 primer @ 80µ DFT/ coat	1 coat of F-15 intermediate coat @ 80µ DFT/coat + 1 coat of F-15 finish coat @ 80µ DFT/ coat (80+80=160)	240	-
9.3	Inside pontoon and inside of double deck of all floating roofs					
9.3.1	-14 to 80	SSPC-SP-3	1 coat of F-8 @ 100µ DFT/coat	1 coat of F-8 @ 100µ DFT/coat	200	-
9.3.2	81 to150	SSPC-SP-10	1 coat of F-15 primer @ 80µ DFT/ coat	1 coat of F-15 intermediate coat @ 80µ DFT/coat + 1 coat of F-15 finish coat @ 80µ DFT/ coat (80+80=160)	240	-
9.4	Other Corrosive liquids and high temperature hydrocarbons All internal surfaces but not limited to internal surfaces of shell for full height, bottom plate, accessories, roof and roof structures of cone and dome roof tanks etc.					
9.4.1	Up to 150oC	SSPC-SP-10	-	1 coat of F-17 @ 375µ DFT/coat	375-425	-

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Notes:

1) Contractor shall develop a suitable internal coating system for any different temperature or service which is not covered in the above table however needs to be internally coated as per project/ data sheets/tender etc. requirement. The same may be submitted for PMC/OWNER review.

TABLE-10: COATING SYSTEMS FOR EXTERNAL SIDE OF UNDERGROUND CARBON STEEL PLANT PIPING AND VESSELS

Sl. No.	Design Temp. in °C	Surface Preparation & Shop Primer	Coating system		Total DFT in Microns (min.)	Remarks
			Surface Preparation & Primer	Finish Coat		
10.1	Underground carbon steel plant piping					
10.1.1	25 to 80	-	3-layer polyethylene (3LPE) coating as per document B649-000-06-42-PLS-04		--	--
10.1.2	81 to 150	-	SSPC-SP-10; 1 coat of F-17 primer @400 µ DFT/ coat	1 coat of F-17 @ 400 DFT/coat	800	Note-2
10.2	External side of un-insulated underground storage vessels					
10.2.1	-40 to 80	SSPC-SP -10; 1 coat of F-9 @ 65-75µ DFT/ coat	-	3 coats of F-7 @ 100µ DFT/coat	365-375	-
10.2.2	81 to 150	SSPC-SP -10	1 coat of F-17 primer @400 µ DFT/ coat	1 coat of F-17 @400µ DFT/ coat	800	Note-2

Notes for TABLE-10:

1 For underground vessels, scope of surface preparation, application of primers to finish coat including repair shall be as per applicable contractual documents like SOR, MR, PR etc.
2 This coating system can be used for the lines/equipment/vessels which have operating temperature between ambient and 150 °C (normal operating temperature) with occasional rise in the temperature up to 200 °C (design temperature).
3 If any piping or vessels with different temperature or service is not covered in the above table, then contractor shall develop a suitable system. The same may be submitted for PMC/OWNER review

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TABLE-11: COATING SYSTEMS FOR UNDER INSULATION (ALL UNIT AREAS & OFFSITES)

(For insulated piping, equipment, storage vessels, tanks, columns etc. of CS, LTCS, Low alloy steel & stainless steels in all environments.)

Sl. No.	Design Temp. °C	Surface Preparation & Pre-erection/Shop Primer	Coating system		Total DFT Microns (min.)	Remarks
			Primer	Finish paint		
11.1	Carbon steel, LTCS and Low alloy steel Piping, Storage tanks, Vessels, Equipment etc.					
11.1.1	-45 to 125	SSPC-SP-10; 1coat of F-15 @ 75µ DFT/coat	None	2 coats of F-15 @75µ DFT/coat	225	
11.1.2	126 to 450	SSPC-SP-10; 1 coat of F-16 @ 125µ DFT/coat	None	1 coat F-16 @ 125µ DFT/coat	250	
11.2	Stainless Steel, Alloy Steel, Duplex Stainless steel, Super Duplex stainless steel & Alloy-20 Piping, Tanks, Vessels & Equipment etc.					
11.2.1	-45 to 125	SSPC-SP-7; (15-25µ surface profile) 1 coat of F-15 @75 µ DFT/coat	None	2 coats of F-15 @75µ DFT/coat	225	Note-2
11.2.2	126 to 550	SSPC-SP-7; (15-25µ surface profile) 1 coat of F-16 @125µ DFT/coat	None	1 coat of F-16 @125µ DFT/coat	250	Note-2
11.3	Cyclic service of CS, LTCS, SS, & Alloy Steels (Note-1)					
11.3.1	-45 to 150 (Note-1)	SSPC-SP-10 For CS, LTCS & low Alloy steel. SSPC-SP-7 for SS; (15-25µ surface profile) 1 coat of F-15 @75 µ DFT/coat	None	2 coats of F-15 @75µ DFT/coat	225	Note-2 & 3
11.3.2	-29 to 400 (CS) -180 to 400 (SS) -29 to 650 (LAS)	SSPC-SP-10 for CS, LTCS & Alloy steel. SSPC-SP-7 for SS (15-25µ surface profile) 1 coat of F-16 @ 125µ DFT/coat	None	1 coat of F-16 @ 125µ DFT/coat	250	Note-2 & 4

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Notes

1. In case of overlapping of cyclic temperature ranges as mentioned in 11.3.1 and 11.3.2 then clause 11.3.1 shall be followed.
2. For all insulated stainless steel (SS) items, stainless steel foil with a minimum thickness of 0.1 mm shall be provided along with the protective coating as a corrosion barrier. And for insulated stainless steel (SS) items with a design temperature exceeding 650°C, stainless steel (SS) foil having a minimum thickness of 0.1 mm shall be used for corrosion protection.
3. Alternatively, for this temperature range, 1 coat of F-17 @ 400 µ is also acceptable.
4. The coating system applicable for any other temperature range shall be reviewed if it is encountered.
5. Contractor shall develop a suitable coating system for any different temperature or service which is not covered in the above table however needs to be coated as per project/ data sheets/tender etc. requirement. The same may be submitted for PMC/OWNER review.

General notes for TABLE-11:

1. "Cyclic Service" is characterized by rapid or periodical temperature fluctuation or temperature cycles.
2. The blasting abrasives for SS and alloy steels shall be aluminum oxide or garnet only.
3. For insulated vessels & equipment scope of surface preparation, application of primers to finish coat including repair shall be as per applicable contractual documents like SOR, MR, and PR etc.

TABLE-12: COATING SYSTEMS (INTERNAL PROTECTION) FOR CARBON STEEL COMPONENTS OF COOLERS/CONDENSERS WITH FRESH COOLING WATER SERVICE

Sl. No.	Design Temp. in °C	Surface Preparation & Pre-erection/Shop Primer	Coating System		Total Final DFT in Microns (min.)	Rema-rks
			Primer	Finish paint		
12.1	Up to 80	SSPC-SP-10	1 coat of F-15 @ 80 µ DFT/coat	2 coats of F-15 @ 80 80 µ DFT/coat	240	-
12.2	81 to 150	SSPC-SP-10	---	2 coats of F-17@ 400µ/coat	800	-

TABLE-13: COATING SYSTEM (INTERNAL PROTECTION) FOR GALVANIZED OR NON FERROUS OR STAINLESS STEEL/ DUPLEX STAINLESS STEEL COMPONENTS OF COOLERS/ CONDENSERS WITH FRESH COOLING WATER SERVICE

Sl. No.	Design Temp. in °C	Surface Preparation & Pre-erection/Shop Primer	Coating System		Total Final DFT in Microns (min.)	Rema-rks
			Primer	Finish paint		

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13.1	Up to 80	Sweep Blasting	1 coat of F-15 @ 80µ DFT/coat;	2 coats of F-15 @ 80µ DFT/coat;	240	-
13.2	81 to 150	Sweep Blasting	-	2 coat of F-17@ 400µ/coat	800	-

Note:
Coating systems in Table – 12 & 13 are applicable only as per the requirement specified in the data sheets of the respective exchangers/equipment.

TABLE-14: COATING SYSTEM FOR MANHOLES COVERS (SEWER/OWS/CRWS)

S.No	Contractor/Vendor shall submit the suitable proposed coating system for manhole covers to EIL for review and approval. The approved coating system shall be adopted for supply. Color coding shall conform to the requirements of Annexure-1.

8.0 STORAGE

8.1 All paints and painting materials shall be stored in rooms only which are to be arranged by CONTRACTOR and approved by Engineer-In-Charge for the purpose. All necessary precautions shall be taken to prevent fire. The storage building shall preferably be separate from adjacent building. A signboard bearing the word "PAINT STORAGE- NO NAKED LIGHT-HIGHLY INFLAMMABLE" shall be clearly displayed outside. Manufacturers' recommendation shall be followed for storage of paint materials.

9.0 COLOUR CODE

The color-coding system as per Annexure-1 of this specification shall be followed.

10.0 IDENTIFICATION OF VESSELS, PIPING ETC.

10.1 Equipment number shall be stenciled in black or white on each vessel, column, equipment & machinery (insulated or un-insulated) after painting. Line number in black or white shall be stenciled on all the piping of more than one location as directed by Engineer-In-Charge. Size of letter printed shall be as below:

Column & vessels	:	150 mm (high)
Pump, compressor & other machinery	:	50 mm (high)
Piping	:	40-150 mm

10.2 Identification of storage tanks

The storage tanks shall be marked as detailed in the drawing.

11.0 PAINTING FOR CIVIL DEFENCE REQUIREMENTS

11.1 The following items shall be painted for camouflaging, if required by the client:

a. All columns

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b. All tanks in offsite
c. Large vessels
d. Spheres

11.2 Two coats of selected finishing paint as per defence requirement shall be applied in a particular pattern as per clause 11.3 and as per the instructions of Engineer-In-Charge.

11.3 Method of camouflaging

11.3.1 Disruptive painting for camouflaging shall be done in three colours in the ratio of 5:3:2 (all matte finish).

Dark Green		Light Green		Dark Medium Brown
5	:	3	:	2

11.3.2 The patches should be asymmetrical and irregular.
11.3.3 The patches should be inclined at 30º to 60º to the horizontal.
11.3.4 The patches should be continuous where two surfaces meet at an angle.
11.3.5 The patches should not coincide with the corners.
11.3.6 Slits and holes shall be painted in dark shades.
11.3.7 Width of patches should be 1 to 2 m.

12.0 QUALITY CONTROL, INSPECTION AND TESTING

12.1 All painting materials including primers and thinners brought to site by CONTRACTOR for application shall be procured directly from manufactures as per specifications and shall be accompanied by manufacturers' test certificates. Paint formulations without certificates are not acceptable (see section 14.0 & 15.0).

12.2 The CONTRACTOR must produce test certificate from pre qualified paint manufacturer for various tests as detailed out in section 15.0 of this document. The Engineer-In-Charge shall have the right to test wet samples of paint at random for verifying quality of paint supplied. CONTRACTOR shall arrange to have such tests, when required by Engineer-in-Charge, performed at his cost at any one of the NABL accredited laboratories.

Samples for the test will be drawn at random in presence Engineer-In-Charge or his representations. Following tests to be carried out if called for by Engineer-in-Charge:

- Specific Gravity
- % solids by weight (% zinc content in case of inorganic or organic zinc primer)
- Drying time (touch dry & full curing)
- Adhesion
- Storage stability (pot life)

Test methods for above tests shall be as per relevant ASTM or ISO Standard.

12.3 The painting work shall be subject to inspection by Engineer-In-Charge at all times. In particular, following stage-wise inspection will be performed and CONTRACTOR shall offer the work for inspection and approval of every stage before proceeding with the

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next stage. The record of inspection shall be maintained in the registers. Stages of inspection are as follows:

a) Surface preparation
b) Primer application
c) Each coat of paint

Following tests are to be carried out during surface preparation:

- Test for presence of oil/grease and contamination

  The steel substrate after degreasing as per SSPC-SP-1 shall be tested as per following procedure to validate absence of oil and grease contamination:

a) Visual inspection - continue degreasing until all visible signs of contamination are removed.
b) Conduct a solvent evaporation test by applying several drops or a small splash of residue-free tri-chloromethane on the suspect area especially pitting, crevice corrosion areas or depressed areas. An evaporation ring formation is indicative of oil and grease contamination. Continue degreasing and inspection till test is passed.
   - Tests for surface finish of blasted surface shall be done by visual inspection using SSPC-VIS1. Clear cellophane tape test as per ISO 8502-3 shall be used to confirm absence of dust on blasted surface. Checks shall be done on each component at least once per 200 m^2 of blasted surface and a minimum of 3 checks per shift.
   - Test for presence of soluble salt as per method ISO 8502-9. Maximum allowable salt content shall be considered 20 mg/m^2.Checks shall be done on each component at least once per 200 m^2 of blasted surface and minimum of 3 checks per shift. In case salt exceeds specified limit, the contaminated surface shall be cleaned by method as per Annexure-C of IS 12944-4 (water cleaning). After cleaning, surface shall be retested for salt after drying.

Blast Profile Measurement: (In-Process testing during actual production before application coating)

The angular profile depth measurement shall be done by profile tape as per method NACE Standard RP 0287 or ASTM D 4417 method B (Profile depth gauge micrometer). Spot measurement shall be carried out every 15m^2 of blasted surface. At each spot three measurements shall be taken over an area of 10 cm^2 and average of measurements to be recorded and reported.
If profile is <65 microns blasting shall continue till greater than 65 microns depth profile is achieved.

-

Tests for blasting media, blasting air

Blasting Media (For every fresh batch of media and one random test during blasting)

Blasting Media shall be visually inspected for absence of contamination and debris using 10 X magnification.

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a) Inspection for the absence of oil contamination shall be conducted using following procedure :

- Fill a small clean 200 ml bottle half full of abrasive.
- Fill the bottle with potable water, cap and shake the bottle.
- Inspect water for oil film/slick. If present, the blasting media is not to be used.

b) Soluble salt contamination if suspected shall be verified by method ASTM D 4940. If present, media to be replaced.

c) Clean blasting equipment, especially pot and hoses, and then replace blasting media and retest.

Test for Blasting Air (Once Daily before start of blasting and once at random during blasting)

The air for blasting shall be free from moisture and oil. The compressor air shall be checked for oil and water contamination per ASTM D 4285.

In addition to above, record should include type of shop primer already applied on equipment e.g. zinc silicate or zinc rich epoxy or zinc phosphate. Any defect noticed during the various stages of inspection shall be rectified by the CONTRACTOR to the entire satisfaction of Engineer-In-Charge before proceeding further. Irrespective of the inspection, repair and approval at intermediate stages of work, CONTRACTOR shall be responsible for rectifying any defects found during final inspection/guarantee period/defect liability period as defined in general conditions of the contract. Dry film thickness (DFT) shall be checked and recorded after application of each coat and extra coat of paint should be applied to make-up the DFT specified without any extra cost to OWNER, the extra coat should have prior approval of Engineer-in-charge.

12.4 Final inspection of finished coating shall consist of the following:

1) **Coating dry film thickness check:** DFT measurement shall be as per ISO 2808. Type II electromagnetic gauges should be used for ferrous substrates. DFT gauge calibration, number of measurement shall be as per SSPC-PA-2. Measured DFT shall be within + 10% of the dry film thickness, specified in the specifications.

2) **Adhesion testing:** Adhesion of the primer to the steel substrate and inter-coat adhesion of the subsequent coat(s) after curing for at least a week after application of the topcoat shall be examined by a knife test in accordance with ASTM D6677. For the knife test, if the rating is better than 8, the adhesion is considered acceptable. The adhesion is destructive and tested areas shall be repaired afterward using the spot repair procedure. Alternatively, the applicator may perform the adhesion test on a steel panel coated using the same surface preparation and coating application procedure as the work piece. Adhesion testing shall be carried out for each component at least once per 200 m2 of coated surface.

3) **Holiday detection check:** Holiday testing shall be conducted in accordance with NACE SP0188. For immersion services, 100% of coated area shall be inspected for holidays. For atmospheric exposure, 10% of coated area which must include weld seams, corners and edges to be holiday tested. Voltage at which test is to be carried out will depend upon DFT of coating being tested and shall be as per NACE

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SP0188. Any holiday is unacceptable and should be marked and repaired immediately**.**

The CONTRACTOR shall arrange for spot checking of paint materials for specific gravity, glow time (ford cup) and spreading rate.

13.0 GUARANTEE

The CONTRACTOR shall guarantee that the chemical and physical properties of paint materials used are in accordance with the specifications contained herein/to be provided during execution of work.

14.0 QUALIFICATION CRITERIA OF PAINTING CONTRACTOR/SUB-CONTRACTOR

Painting CONTRACTOR who is awarded any job for EIL, Projects under this standard must have necessary equipment, machinery, tools and tackles for surface preparation, paint application and inspection. The CONTRACTOR must have qualified, trained and experienced surface preparator, paint applicator, inspector and supervisors. The CONTRACTOR supervisor, inspector, surface preparator and paint applicator must be conversant with the standards referred in this specification.

15.0 QUALIFICATION/ACCEPTANCE CRITERIA FOR PAINT COATING SYSTEM

15.1 Pre-qualification of paint coating manufacturer and his products

Paint manufacturer meeting the following requirements shall be considered by the CONTRACTOR for supply of the paint products.

- Manufacturer should have been in continuous business of paint coating formulation and manufacturer for at least past 5 years.
- Manufacturer should posses past experience of supplying his products to hydrocarbon, petrochemical, fertilizer. Chemical processing industry or offshore platforms in the past 5 years.
- Coating manufacturer should have supplied at least 10000 litre of an individual product to hydrocarbon, petrochemical, fertilizer. Chemical processing industry or offshore platforms.
- The manufacturer's manufacturing procedure & QA/QC system shall meet ISO 9001 requirements and preferably should posses ISO 14000 certificate.
- The Quality control set up should be manned by qualified paint technologists whose bio data should be sent along with quality control organization chart.

CONTRACTOR shall procure the paint materials from the qualified manufacturer meeting above requirements and after obtaining prequalification testing approval as per requirements mentioned in clause 15.2 below.

15.2 Pre-Qualification Testing procedure:

The paint manufacturer engaged by the mechanical contractor shall carry out the tests in NABL accredited government laboratories like national test house (NTH), as a part of qualification. Paint manufacturer shall provide the paint samples to laboratory for testing of the parameters mentioned in Table-2 (typical characteristics) and Table-15 (tests on coating systems) of this specification. The testing laboratory will confirm the compliance of the paint material with respect to the acceptance

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criteria mentioned in the respective tables. Contractor shall furnish these test certificates along with all necessary supporting documents/information to EIL site for approval/ acceptance. The paint manufacturer will be qualified and approved by EIL site for supply of paints after review/assessment of the submissions made by the contractor. Test certificates which are more than 3 years old will not be considered. Paint manufacturers are advised to carryout pre-qualification testing accordingly for paints supply to EIL projects.

TABLE-15: PRE-QUALIFICATION TESTING

SYSTEM No.	COATING SYSTEM	REFERENCE CLAUSE (from table-3 to 12)	TOTAL DFT µ (min)
1.	F-9+P6+F6B+F2	6.2	345
2.	F12+F12+F12	6.4	60
3.	F15+F15+F15	9.2.2	240
4.	F16+F16	8.3.1	250
5.	F17	9.4.1	375
6.	F8+F8	9.3.1	200
7.	F20+F20	7.4	1100
8.	F6A+F6C+F6C	9.2.1	500
9.	F6A+F6A+F6A	9.1.1	300

S. No.	TEST	FOR SYSTEM NUMBER	DURATION	ACCEPTANCE CRITERIA
1.	Cyclic Test Salt Spray : 72 hrs. Drying in air: 16 hrs. UV-A340 nm weather meter: 80 hrs. One cycle: 168 hrs. (25 cycles at 168 hrs. each cycle) (ASTM D5894)	1	4200 hrs	Shall pass. No chalking, cracking, flaking, blistering or peeling shall be observed.
2.	Chemical Resistance Test (ASTM D543)			
2a.	10% & 40% NaOH	3,5,7 & 9* *H_2SO_4 solution pH = 5.0 to 5.5 for system 9	1000 hrs	Shall pass. No cracking, discoloration, blistering, peeling or softening of film shall be observed.
2b.	5% H2SO4		168 hrs	
2c.	Xylene		4 weeks	
2d.	Acetone		4 weeks	
2e.	Ethanol		4 weeks	
2f.	Kerosene		4 weeks	
2g.	Sea water		2000 hrs	

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3.	Immersion in DM/DI water @90^0 (ASTM C868)	3,5 & 7	30 days	No softening, blistering or film damage.
4.	Resistance to DM water using water immersion. (ASTM D870)	8	2000 hrs	Shall pass. No chalking, cracking, flaking, blistering or peeling.
5.	100% Humidity Test (ASTM D2247)	1 to 9 (except system-2)	1440 hrs	Shall pass
6.	Thermal Shock Resistance Test; 5 cycles @ 30 minutes in furnace at 120 º C and 15 minutes in water after quenching in water for each cycle. (ASTM D2485 method A)	2,3 & 4 (For system-2, testing to be done after heating the panels at 175°C for 2 hrs.)	-	Shall pass
7.	Cathodic Disbondment Test (ASTM G8 @60°C)	3&5	-	Shall pass

Each coating product to be qualified shall be identified by the Specific gravity of base and curing agent (ISO 2811). The identification shall be carried out on the batch, which is used for the pre-qualification testing.

15.3 Information to be furnished during delivery of paint materials:

CONTRACTOR along with delivery of paint material has to furnish following information from paint manufacturer to EIL for acceptance/approval of products:

a) Batch test certificates:

Along with paint products delivery to site from the pre-qualified coating manufacturer, CONTRACTOR has to produce test certificate (from paint manufacturer) for each category of product for the following test items. All test results must mention clearly the batch no. and category of product tested. Tests to be conducted for following properties:

- Specific Gravity
- % solids by weight (% zinc content in case of inorganic or organic zinc primer)

b) Product information sheet/ technical data sheet for each category of product.

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The contractor shall be fully responsible for the quality of the paints products as per prequalification testing. After the paint materials are supplied to site, the supplier shall organize random sampling and testing in a NABL laboratory as per discretion of the Engineer-in-charge. Failing to meet the specified quality requirements may cause rejection of the paint products.

16.0 METHOD OF SAMPLING & DISPATCH FOR LABORATORY TESTING

(Pre-Qualification tests (sec. 15.2), Batch testing (sec. 15.3) and Inspection testing (sec. 12.0))

16.1 Samples of coating materials should be submitted to the laboratory in sealed containers with batch no. and test certificate on regular format of manufacturer's testing laboratory.

16.2 All test panels should be prepared by testing laboratory. Surface preparation for a system shall be done in accordance with this specification. For individual products testing, minimum shall be Sa 2.5. Colour photographs of test panels should be taken before and after the test and should be enclosed along with test report.
Sample batch no. and manufacturer's test certificate should be enclosed along with the report. Test report must contain details of observation and rusting if any, as per the testing code.

16.3 Manufacturers should intimate EIL, details of sample submitted for testing, name of testing agency, date, and contact personnel of the testing agency.

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ANNEXURE-1: COLOUR CODE

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1.0 SCOPE

The specification covers the requirement of color coding for the identification of the material of construction in CPCL refinery. The intent of this standard is to provide a system for easy identification of piping materials by general material classifications.

2.0 REQUIREMENT

The purpose of color coding in piping systems is to provide visual identification and communicate important information such as material of construction & specific services (H2, IBR, welded & HIC). This helps to prevents accidental mix up of material of construction whose appearance is similar to each other.

3.0 GUIDELINES

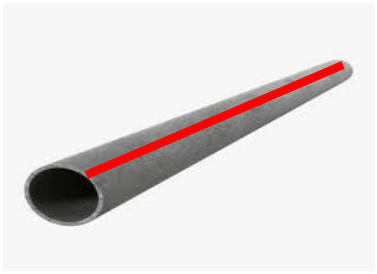
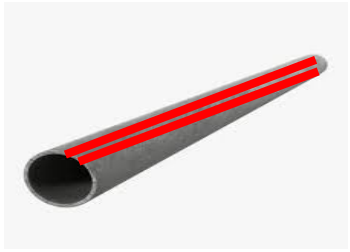
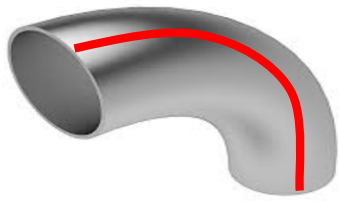
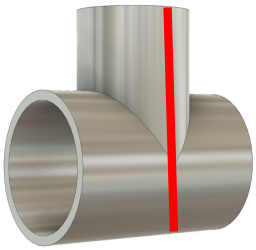
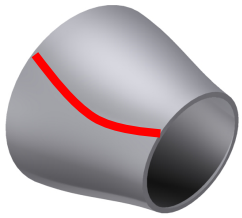
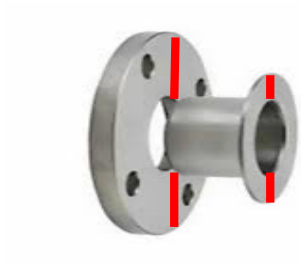
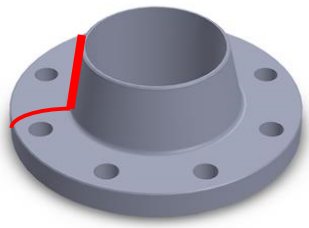
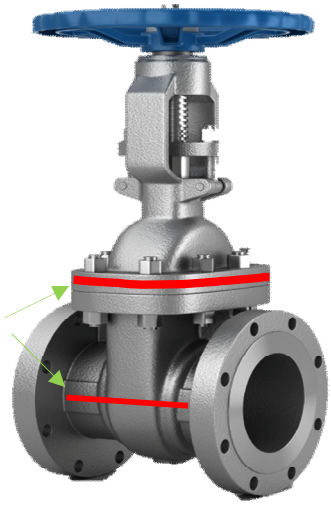
- Colour Coding for piping & fittings shall be done as per Metallurgy wise & Service wise as specified.
- Pipes & pipe materials shall be painted with one color or a combination of colors, as specified; whenever combination of two or more colors are required / specified, materials shall be painted in parallel color bands as close to each other as possible.
- Additional colour band shall be painted as per Service Color identification is not required on Regular Carbon Steel metallurgy, Galvanized material, non-ferrous material such as Copper, Aluminum, lead etc. because they are easily distinguished by their specific colour & character.
- Width of the color band:

Less than 3 IN size	Min. 12mm
Greater than or equal to 3 IN size	25mm

- The paint shall not cover welding surfaces and end of the materials which are to be welded.
- Surface Preparation & Application of the Colour Coding shall be as below:
  - ✓ Surfaces to be color-coded or marked shall be free from oil, grease, dirt, and other surface contaminants that might be detrimental to the adhesion of the paint used for coding and marking.
  - ✓ Whenever color coding or marking paint is to be applied to a primed surface, the primer shall be dried completely before the color coding or marking paint is applied.
  - ✓ Color coding, marking, and identification paint shall be applied to dry, clean surfaces in accordance with the manufacturer's instructions.
  - ✓ Unless otherwise recommended by the manufacturer, color coding and marking shall not be undertaken when the ambient temperature is less than 10 °C (50 °F), or the relative humidity > 90%
  - ✓ Color coding and marking paint shall be applied in 1 coat.
- Paints containing chlorine, chlorides, Sulphur, lead, zinc or any contents detrimental to materials are not acceptable. The bidder shall satisfy the same & undertaking has to be given during bid submission.
- Identification by this method is not a substitute for permanent manufacturer's marking, as is required by applicable ASTM or other piping specification

4.0 LOCATION OF MARKING

- Studs / Bolts and Nuts shall be painted only on head.
- Identification color on gaskets shall be painted on the peripheral face of each gasket.
- Pipe shall be marked with paint, dyes, tapes, etc. For its full length.
- Flanges shall be banded on the back of the flange at the intersection of the back face and the hub.
- Fittings shall be stripped from bevel to bevel.
- The paint shall not cover welding surfaces. Typical markings on components are shown below.

		
Single Band colored pipes	Double band-colored pipes	90-degree Elbow
		
Tee	Reducer	Flange
		
Cap	Coupling	

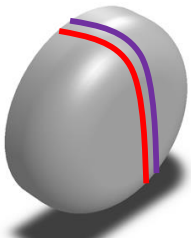
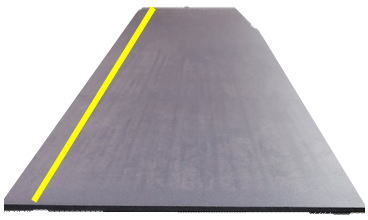
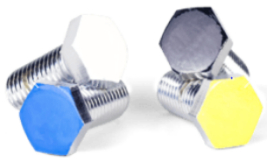
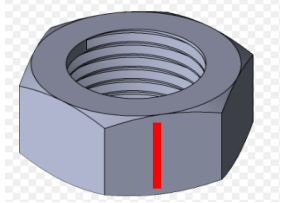

	

Plate	Plate	Valve
	 	
Gasket	Stud, Nut, Bolt	

5.0 COLOUR IDENTIFICATION – MATERIAL WISE

Material	Colour	Pipe	Tube	Wrought fitting	Forging	Plate	Cast Valve
LTCS	1 Chocolate brown	A 333 Gr.1 or 6	A 334 Gr.1 or 6	A 420 WPL6	A 350 LF1	SA 537 CL 1	A 352
P11	1 Sky Blue	A 335 Gr. P11	A 213 Gr. T11	A 234 Gr. WP11	A 182 Gr. F11	A 387 Gr. 11	A217 WC6
P22	2 Sky Blue	A 335 Gr. P22	A 213 Gr. T22	A 234 Gr. WP22	A 182 Gr. F22	A 387 Gr. 22	A217 WC9
P5	1 Sky Blue & 1 Signal grey	A 335 Gr. P5	A 213 Gr. T5	A 234 Gr. WP5	A 182 Gr. F5	A 387 Gr. 5	A217 C5
P9	1 Sky Blue & 1 Deep Orange	A 335 Gr. P9	A 213 Gr. T9	A 234 Gr. WP9	A 182 Gr. F9	A 387 Gr. 9	A217 C12
SS 304	1 Signal black	A 312 TP 304	A 213 TP 304	A 403 WP 304	A 182 F 304	A 240 304	A351-Gr.CF8 CF8a
SS 304 L	2 Signal black	A 312 TP 304L	A 213 TP 304L	A 403 WP 304L	A 182 F 304L	A 240 304L	A351-Gr.CF3 CF3a
SS 304 H	1 Signal black & 1 Light Pink	A 312 TP 304H	A 213 TP 304H	A 403 WP 304H	A 182 F 304H	A 240 304H	A351.Gr.CF10
SS 310	1 Signal Black & 1 Sky Blue	A 312 TP 310	A 213 TP 310	A 403 WP 310	A 182 F 310	A 240 310	NA
SS 316	1 Signal Grey	A 312 TP 316	A 213 TP 316	A 403 WP 316	A 182 F 316	A240 316	A351-Gr.CF8M
SS 316 L	2 Signal Grey	A 312 TP 316 L	A 213 TP 316 L	A 403 WP 316 L	A 182 F 316 L	A240 316 L	A351-Gr.CF3M
SS 321	1 Light Pink	A 312 TP 321	A 213 TP 321	A 403 WP 321	A 182 F 321	A 240 321	NA
SS 321 H	2 Light Pink	A 312 TP 321H	A 213 TP 321H	A 403 WP 321H	A 182 F 321H	A 240 321H	NA
SS 347	1 Copper brown	A 312 TP 347	A 213 TP 347	A 403 WP 347	A 182 F 347	A 240 347	A351-Gr.CF8C
SS 347 H	2 Copper brown	A 312 TP 347H	A 213 TP 347H	A 403 WP 347H	A 182 F 347H	A 240 347H	A351-Gr.CF8C
SA 405	1 Traffic Green & 1 Traffic black					A 240 405 S	
SS 410	1 Traffic Green & 1 Sky Blue					A 240 410 S	
Standard DSS 32205 / SS 31803	1 Signal Grey & 1 Traffic Green	A 790 UNS 31803/32205	A 789 UNS 31803/32205	SA-815 UNS 31803/32205	SA-182 Gr. F51	A-240 UNS 31803	NA
DSS 32750 (Super Duplex)	1 Signal Grey & 1 Sky Blue	A790 UNS 32750	A 789 UNS 32750	SA-815 UNS 31803/32205	SA-182 Gr. F53	A-240 UNS 32750	NA
Incolloy 800/800H & 825	1 signal black & 1 Deep Orange	B 407, B 514	B 407, B 515	B 366 WPN1C	B 408	B 409	ASTM A743
Inconel 625	1 Sky blue & 1 Light pink	B 167, B 517	B 516, B 163	B 366 WPNC1	B 166	B 168	ASTM A743 Grade CY-40
Monel	1 signal black & 1 Pearl Violet	B 165	B 165, B 163	B 366 WPNC	B 164	B 127	ASTM 743 Grade M3-35-1

6.0 COLOUR CODE – PIPE & PIPE FITTINGS

S.No	Material	Colour	Colour identification	RAL No	IS No
1	LTCS	1 Chocolate brown		8017	412
2	P11	1 Sky Blue		5015	101
3	P22	2 Sky Blue		5015	101
4	P5	1 Sky Blue		5015	101
		1 Signal grey		7004	628
5	P9	1 Sky Blue		5015	101
		1 Deep orange		2011	591
6	SS 304	1 Signal black		9004	
7	SS 304 L	2 Signal black		9004	
8	SS 304 H	1 Signal black		9004	
		1 Light Pink		3015	442
9	SS 310	1 Signal black		9004	
		1 Sky Blue		5015	101
10	SS 316	1 Signal Grey		7004	628
11	SS 316 L	2 Signal Grey		7004	628
12	SS 321	1 Light Pink		3015	442
13	SS 321 H	2 Light Pink		3015	442
14	SS 347	1 Copper brown		8004	414
15	SS 347 H	2 Copper brown		8004	414

16	SA 405	1 Traffic Green		6024	267
		1 signal black		9004	
17	SS 410	1 Traffic Green		6024	267
		1 Traffic Blue		5017	169
18	DSS 32205 / SS 31803 (Standard Duplex)	1 Signal Grey		7004	628
		1 Traffic Green		6024	267
19	DSS 32750 (Super Duplex)	1 Signal Grey		7004	628
		1 Traffic Blue		5017	169
20	Incolloy 800/800H & 825	1 signal black		9004	
		1 Deep orange		2011	591
21	Inconel 625	1 Traffic Blue		5017	169
		1 Light Pink		3015	442
22	Monel	1 signal black		9004	
		1 Pearl Violet		4011	796

7.0 COLOUR CODE – STUD, NUTS & BOLTS

S.No	Material	Colour	Colour identification	RAL No	IS No
1	Bolt - A 193 Gr B7	Sky Blue		5015	101
2	Nut - A 194 GR 2 H				
3	Bolt - A 193 Gr B7M	Traffic Blue		5017	169
4	Nut - A 194 GR 2 HM				
5	Bolt - A 193 GR B 16	Traffic Green		6024	267
6	Nut - A 194 GR 4				
7	Bolt - A 320 GR L 7	Chocolate brown		8017	412
8	Bolt – A 320 Gr B8	Signal black		9004	
9	Nut – A 194 gr 8				

8.0 COLOUR CODE – GASKETS

S.No	Material	Colour	Colour identification	RAL No	IS No
1	Compressed Fibre (Ring & Full Face) - BS753I GR.X (FULL FACE)	Dove Grey		7005	694
2	Compressed Fibre (Ring & Full Face) - BS753I GR.X (RING)	-		-	
3	Spiral Wound gasket – SS 304 SPR WND + Grafoil Filler	Satin Blue		5012	177
4	Spiral Wound gasket – SS 304 H + Grafoil Filler	Salmon Pink		3012	443
5	Spiral Wound gasket – SS 304 + Grafoil Filler	Pale brown		8025	368
6	Spiral Wound gasket – SS 304 + Teflon Filler	Indian Brown		8016	415
7	Spiral Wound gasket – SS 304 L SPR WND + Grafoil Filler	Leaf Brown		8003	489
8	Spiral Wound gasket – SS 316 SPRWND + Grafoil Filler	Deep Orange		2011	591
9	Spiral Wound gasket – SS 316 L SPR. WND + Grafoil Filler	Traffic Yellow		1023	368
10	Spiral Wound gasket – SS 316 H SPR. WND + Grafoil Filler	Canary Yellow		1018	309
11	Spiral Wound gasket – Teflon Jacketed SPR. WND	Dark Violet		4006	796
12	Spiral Wound gasket – SS321 Spr WND + Grafoil filler	Sea Green		6018	217
13	Spiral Wound gasket – SS347 Spr WND + Grafoil FIller	Sky Blue		5021	101
14	Oct. Ring Gasket – 5 Cr,1/2 Mo (Max. 120 BHN)	Fire Green		6009	218
15	Soft Iron (Max. 90 BHN)	Dove Grey		7005	694

9.0 ADDITIONAL COLOUR IDENTIFICATION – SERVICE WISE

S.No	Service	Colour	Colour identification	RAL No	IS No
1	H2 Service (All)	Luminous Yellow		1026	356
2	IBR service (All)	Traffic Red		3020	570
3	Welded connection (Incl. EFSW or Welded Pipe)	Traffic White		9016	
4	HIC (All)	Luminous Green		6038	219

1.0 SCOPE

The specification covers the requirement of colour scheme for the identification content of the pipeline, vessel, tanks carrying hydrocarbons, water, air, gases and chemicals, etc in our Refinery.

2.0 IDENTIFICATION

The system of colour coding consists of a ground colour and secondary colour bands superimposed over the ground colour.

2.1 Insulated Lines –

On metal cladded insulated lines, minimum 2-meter-long portion should be painted in ground colour.

2.2 Un-Insulated Lines –

On Un-insulated pipes, the entire pipe length has to be painted in ground colour.

3.0 PRIMARY IDENTIFICATION - GROUND COLOUR

The ground colour identifies the basic nature of the service and the secondary colour band over the ground colour distinguishes the particular service. For insulated pipelines ground colour shall be provided as per the specified length and interval to identify the basic nature of service and secondary colour bands to be painted on these specified lengths to identify the particular service. The colour coding system is applicable for both process units and offsite pipelines. All drains & Vents shall be painted in main line color.

The following ground colour designation for identification of basis classification of various important services shall be as follows:

S.NO	SERVICE	GROUND COLOUR	SHADE	RAL No	IS NO
1	Hydrocarbons / Combustible liquids	Golden Brown			414
2	Fuel gas, Hydrogen, Hydrogen Sulphide, Nitrogen, Inert gas, Carbon monoxide, Carbon dioxide and Amine	Golden Yellow		1004	356
3	Acids / Alkalis, Ammonia & Chlorine	Wild Lilac			
4	Raw Water, Cooling Water, Service Water, DM Water (e.g. **OTHER THAN FIRE WATER**)	Aquamarine			
5	Fire Water	Post Office Red		3002	538

6	Air	Sky Blue		5015	101
7	Flare	Heat Resistant Aluminum		9006	
8	RLNG	Deep Orange		2011	591
9	General Structures	Opaline Green		6026	275
10	Heater Structures	Heat Resistant Aluminum		9006	
11	Hand Rails, Ladders & Walkways	Golden Yellow		1004	356
		Jet Black		9005	

4.0 SECONDARY IDENTIFICATION - BAND COLORS

The narrow bands presenting the secondary colour which identifies the specific service shall be applied by painting. For some of the products, two bands with different colors are specified in order to distinguish the content of the pipeline more specifically.

S.NO	SERVICE (PRODUCT)	RECOMMENDED COLOUR BAND	SHADE	RAL No	IS NO
Category 1: HYDROCARBON - LIQUID GROUND COLOUR: GOLDEN BROWN					
1	*Crude Oil*	Golden Yellow		1004	
2	*LPG*	Traffic White		9016	
		Post Office Red		3002	538
3	*Propylene*	Traffic White		9016	
		Sky Blue		5015	101
4	*Butylene*	Traffic White		*9016*	
		Opaline Green		*6026*	275
5	*PBFS*	Traffic White		*9016*	
		Traffic Purple		*4006*	
6	*Naphtha*	Opaline Green		*6026*	*275*

7	*Gasoline*	Post Office red		*3002*	*538*
8	*Diesel / Gas oil*	Sky blue		*5015*	*101*
9	*ATF*	Sky Blue		*5015*	*101*
		Post Office red		3002	538
10	*Kerosene*	Sky Blue		*5015*	*101*
		Traffic white		9016	
11	*Fuel oil*	Jet Black		*9005*	
12	*Asphalt*	Jet Black		*9005*	
		Post Office red		3002	538
13	*Flushing oil*	Traffic Purple		*4006*	
		Antique pink		3014	
14	*Slop*	Golden Yellow		*1004*	
		Post Office Red		3002	538
15	*Wax*	Traffic Purple		*4006*	
		Opaline Green		6026	275
16	Lube oil	Traffic Purple		*4006*	
Category 2: HYDROCARBON - GASEOUS GROUND COLOUR: GOLDEN YELLOW					
17	Fuel gas	Antique Pink		*3014*	
18	Hydrogen	Post Office Red		*3002*	*538*
19	Nitrogen	Sky Blue		*5015*	*101*
20	Amine service	Opaline Green		*6026*	275
21	CO2	Traffic White		*9016*	

Category 3: CHEMICALS/ACID GROUND COLOUR: WILD LILAC					
22	*Ammonia - Gas / Liquid NH3*	*Jet Black*		*9005*	
23	*Chlorine*	*Golden Yellow*		*1004*	
24	*Caustic*	*Traffic White*		*9016*	
25	*Hydrochloric acid*	*Sky Blue*		*5015*	*101*
26	*Sulphuric acid*	*Opaline Green*		*6026*	*275*

5.0 COLOUR BANDS AND IDENTIFICATION LETTERING:

5.1 The colour bands as specified in the attachment shall be followed for identifying the piping contents. The line number and the flow direction shall be stenciled on each pipe section and pipeline together with the colour coding, to provide the pipeline with unique traceability.

5.2 The colour band width requirements are;

2" to 6" inclusive : 100 mm
Greater than 6": 150 mm

In case of two band colors, both the bands shall be applied adjacently and the width of each band shall also be the same as mentioned above.

5.3 The interval of colour bands application shall be as below:

Plant area : at an interval of 15.0 meter
Offsite area : at an interval of 30.0 meter

5.4 Colour Coding Bands shall be provided at the following locations in addition to the interval specified above:

- On both sides of and adjacent to valves, tees, important fittings,
- Wall entries, Road crossings and both the sides of culverts
- Outboard of flanges to and from equipment such as vessels, tanks, exchangers, compressors, pumps and Unit battery limits.

5.5 Size of Lettering and Direction Arrow shall be as follows:

Lettering

Pipe Diameter	Width x Height - Spacing
2" to 6" inclusive	30 x 30mm - 10 mm min.
6" to 50" inclusive	80 x 80mm - 25 mm min.
greater than 50"	100 x 100mm - 30 mm min.

Direction Arrow

Pipe Diameter	W1	W2	L1	L2 (in mm)

2” to 6” inclusive	10	30	45	135
6” to 50” inclusive	40	80	90	270
greater than 50”	50	100	150	450

The lettering and flow direction arrows shall be carried out with BLACK colour

5.6 Two finish coat of relevant painting system shall be applied for banding.

5.7 The areas of the painted surfaces to which the colour coding bands, pipeline identification numbers and flow direction arrows are to be applied shall be degreased, water washed and lightly abraded, in order to ensure good adhesion of the applied paint to the existing topcoat paint.

6.0 INSPECTION:

The contractor shall depute NACE certified Level -1 coating inspectors for the inspection of protective coating application jobs. It is the primary responsibility of the painting contractor to ensure proper surface preparation, using of correct paint material, ensuring proper mixing ratio and application methods. Any deviation from the standard procedure shall be strictly avoided and proper control over the painting jobs shall be ensured by the Inspector at all times.

7.0 GUARANTEE:

The painting contractor shall stencil the date of painting in a legible manner in each area after completion of painting and provide a guarantee for a period of 3 years. Any deterioration noticed in the painting within 3 years shall be repainted as per the painting system at free of cost.

CPCL	CHENNAI PETROLEUM CORPORATION LTD (A group company of IndianOil) चेन्नै पेट्रोलियम कॉर्पोरेशन लिमिटेड					STANDARDISATION OF COLOUR CODING FOR MANHOLE COVER IN OWS & CRWS	
0	08.02.23	FOR CONSTRUCTION	RK	SD	SS	DRAWING NO.	00-E-11649
REV.	DATE	DESCRIPTION	DRN	CHD	APP.	JOB NO	–

STANDARDISATION OF COLOUR CODING FOR MANHOLE COVER IN OWS & CRWS

Preface :

Standardization of Color Coding for of OWS (Oily Water Sewer) &

CRWS (Contaminated Rain Water Sewer) as per OISD – 109

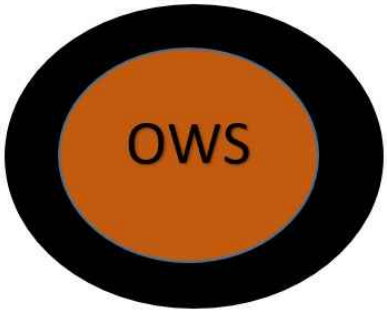
Recommendations

The system of colour coding consist of Outer Core Ground Colour & Inner Core Band Colour.

OWS (Oily Water Sewer)

Outer Core Ground Colour : Black

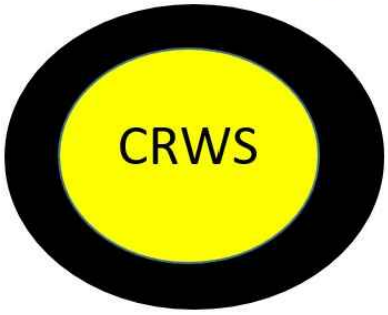
Inner Core Band Colour : Brown

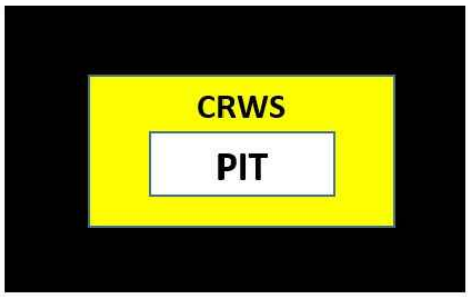


CRWS (Contaminated Rain Water Sewer)

Outer Core Ground Colour : Black

Inner Core Band Colour : Yellow





Lettering shall be made in the Black Colour in the Inner Code.

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NOZZLES AND CONNECTIONS (नोजल व कनैक्शन)

MARK	QTY	NOM. DIA	SCH./THK.	FLANGES			PROJECTION NOTE-4	PAD W x T	SERVICE
		नामिनल	शड्यूल व	CLASS	TYPE	FACING			
मार्क	क्वान्टिटी	डांया	थिकनैस	क्लास	टाईप	फेसिंग	प्रोजेशन	पैड	सार्विस
M	1	460 I.D. BARE	$	1500	SRWN	RTJ	$	-	MANWAY + BF + GASKET + BOLTING + DAVIT
VI	1	100	$	1500	SRWN	RTJ	$	-	VAPOR INLET
VO	1	100	$	1500	SRWN	RTJ	$	-	VAPOR OUTLET
LO	1	80	$	1500	SRWN	RTJ	$	-	LIQUID OUTLET
LT 1-4	4(H)	80(H)	$	1500	SRWN	RTJ	$	-	LEVEL TAPS (NOTE-25) △B
AO	1	450 I.D.	14 THK.	AS PER EIL STD.					ACCESS OPENING
SKV 1-3	3	100	Sch 40	AS PER EIL STD.					SKIRT VENT

T' DENOTES CORRESPONDING SHELL/CONE/DISH END/SKIRT NOMINAL THICKNESS & (H) DENOTES HOLD

$' REFER SHEET-3 FOR DETAILS

LICENSOR'S SPECIFICATION
लाईसेन्सर स्पेसिफिकेशन्स

PVM-SU-4750
PVM-SU-1322
GC-C31554
RC-940132
PVM-GD-C99663
RF-224147 (GN-11) △B

FOUNDATION LOADING DATA (OPERATING CONDITION)
फॉऊडेशन लाडिगं डाटा (आपरेटिंग कनडिशन)

TYPE	MAX MOMENT AT BASE (M) (kgm)	MAX. SHEAR FORCE AT BASE (H) (kg)
SEISMIC	31000	6300
	-	-
WIND	14000	4300

GENERAL NOTES (जनरल नोट्स)

UNLESS STATED OTHER WISE

1 ALL DIMENSIONS ARE IN MM UNLESS OTHERWISE STATED.
2 ALL ANCHOR BOLT HOLES TO STRADDLE N/S CENTRE LINE.
3 NORTH DIRECTION WHEREVER SHOWN IS WITH RESPECT TO PLAN VIEW.
4 (A) FOR NOZZLES ON SHELL PROJECTIONS ARE REFERRED FROM VESSEL CENTER LINE TO FLANGE CONTACT FACE (BARE, WITHOUT CLADDING/WELDOVERLAY).
4 (B) FOR NOZZLES ON HEAD PROJECTIONS ARE REFERRED FROM HEAD T.L TO FLANGE CONTACT FACE (BARE, WITHOUT CLADDING/WELDOVERLAY).
5 THE INDICATED THICKNESS IS THE MINIMUM ACCEPTABLE AFTER CONSTRUCTION.
6 (A) FLANGE GASKET FACE SHALL HAVE 125 AARH FINISH AND SIDE WALL FACE OF GASKET GROOVE SHALL HAVE 63 AARH FINISH.
6 (B) FOR RTJ FACING FLANGE, SURFACE HARDNESS DIFFERENCE BETWEEN FLANGE ANG GASKET SHALL BE 20 BHN. MAXIMUM HARDNESS OF RTJ GASKET SHALL BE AS PER MATERIAL OF CONSTRUCTION.
7 DIMENSIONS OF FLANGES FOR NOZZLES UPTO 600NB SHALL BE AS PER ASME B16.5 AND FOR NOZZLE ABOVE 600NB SHALL BE AS PER ASME B16.47 SERIES 'B' UNLESS SPECIFIED OTHERWISE.
8 ID OF WELD NECK FLANGES SHALL MATCH WITH CORRESPONDING ID OF NOZZLE PIPE/SHELL.
9 AFTER COMPLETION OF FABRICATION AND TESTING ALL SS CLAD SURFACES ALONG WITH SS INTERNALS SHALL BE PICKLED AND PASSIVATED AS PER ASTM A380.
10 ALL BUTT WELD(CIRCUFERENTIAL AND LONGITUDINAL SEAM) IN PRESSURE PART INCLUDING NOZZLE NECK TO FLANGE, PIPE TO PIPE, PIPE TO PIPE FITTING, NOZZLE TO SHELL SHALL BE 100% RADIOGRAPHED.
11 ALL INTERNALS INCLUDING SHROUD RING, STRAINER & VORTEX BREAKER, DONUT BAFFLE, DEMISTER WITH SUPPORTS AS PER LICENSOR & EIL SPEC. ARE IN SCOPE OF VENDOR. INTERNALS TO BE SUPPLIED IN AS INSTALED CONDITION EXCEPT DEMISTER WHICH SHALL BE LOOSE SUPPLIED.
12 IT SHALL BE FABRICATORS RESPONSIBILITY TO SELECT NOZZLE SCHEDULES/THK. SO THAT NO SHRINKAGE OR WARPING OCCURS DURING WELD OVERLAY.
13 NOZZLES SHOWN WITH BLIND FLANGES SHALL BE PROVIDED WITH BOLTS/NUTS AND GASKETS AND SPARES AS PER MR REQUIREMENTS.
14 SURFCAE PREPERATION (INSIDE & OUTSIDE) ALONG WITH PRIMER & FINISH PAINT IS IN THE SCOPE OF VENDOR. REFER JOB SPEC. B649-000-06-41-PLS-01.
15 ALL REMOVABLE INTERNALS SHALL BE SIZED SO AS TO PASS THROUGH NEAREST MANHOLE.
16 COMPLETE EQUIPMENT SHALL BE POST WELD TREATMENT. HARDNESS OF BASE METALS, WELDS AND HAZ AFTER PWHT SHALL BE LIMITED AS PER PVM-SU-4750.
17 VENDOR SHALL PROVIDE 1 NOS. PERMANENT SLOPING STEEL MONKEY LADDERS INSIDE THE SKIRT TO ACCESS THE BOTTOM HEAD FOR INSPECTION.
18 CORROSION ALLOWANCE (mm)
SHELL & HEADS - 3mm OF SS 304L CLAD
INTERNALS - 1.0
BASE PLATE EXTERNAL - 1.5
19 UNLESS OTHERWISE SPECIFIED MINIMUM THICKNESS OF WELDED INTERNAL & BOLTED INTERNAL SHALL BE 10mm. IN CASE OF PIPE AND PIPE FITTINGS SCHEDULE TO BE SELECTED ACCORDINGLY.
20 ALL PLATES AND WELDS INCLUDING HAZ SHALL BE MEET THE IMPACT TEST REQUIREMENTS AT 0^0C AS PER CODE/SPEC. 6-12-0011.
21 SKIRT TO BOTTOM HEAD JOINT SHALL BE DONE CONSIDERING WELD METAL BUILD UP HAVING SAME NOMINAL COMPOSITION AS THAT OF THE BASE METAL I.E. BOTTOM HEAD AND TOP SKIRT SECTION. THE JOINT BETWEEN SUPPORT SKIRT AND BOTTOM HEAD SHALL HAVE A SMOOTH STREAMLINE GEOMETRY AS SHOWN IN DETAIL–B (SHEET-2). JOINT DETAIL SHALL BE MADE BY WELD METAL BUILD-UP. WELD SHALL BE GROUND /MACHINED TO A SMOOTH FLUSH CONTOUR AND SHALL BE PWHT. 100% UT EXAMINATION SHALL BE CARRIED OUT BEFORE AND AFTER PWHT.
22 DEMISTER SHALL BE SUPPLIED BY VENDOR THROUGH EIL APPROVED SUB-SUPPLIER ONLY. EQUIVALENCY SHALL BE ESTABLISHED BY VENDOR. VENDOR TO FOLLOW EIL SPEC. 6-14-0017 & EIL STD. 7-12-0036.
23 FOR WELD OVERLAID FLANGES, MINIMUM THICKNESS OF FLANGES SHALL BE AS PER ASME B16.5 / B16.47 UNLESS SPECIFIED OTHERWISE. VENDOR TO CONSIDER EXTRA THICKNESS TO MEET UNDERCUT REQUIREMENT FOR WELD OVERLAY.
24 ALL MOC AND MANUFACTURING REQUIREMENT SHALL BE NACE MR0103 COMPLIANT.
25 △B LEVEL TAPS TO BE ORIENTED AT APPROXIMATELY 270^0-330^0 DOWNSTREAM FROM INLET.
26 △B EQUIPMENT SHALL MEET PVM-SU-4750 SUPPLIMENT `B' REQUIRMENTS.

SPECIFICATIONS (स्पेसिफिकेशन्स)

X	DENOTES APPLICABLITY	
X	GENERAL SPEC. FOR PRESSURE VESSELS	6-12-0001
X	SUPP. SPEC. FOR CS VESSELS	6-12-0002
X	SUPP. SPEC. FOR SS CLAD VESSELS	6-12-0007
X	STD. SPEC. FOR BQCS PLATE	6-12-0011
X	~~STD.~~ SPEC. FOR STRUCTRAL QUALITY PLATES	6-12-0014
X	STD. SPEC. FOR CLAD PLATES	6-12-0015
X	~~STD.~~ SPEC. FOR STAINLESS STEEL PLATES	6-12-0020
X	~~GEN.~~ SPEC. FOR WIRE MESH DEMISTER	6-14-0017
X	SPEC. FOR SURFACE PREPARATION & PROTECTIVE COATING	B649-000-06-41-PLS-01

STANDARDS (स्टैण्डर्डस)

X	VESSEL TOLERANCES	7-12-0001
	SUPPORT FOR HORIZONTAL VESSEL	7-12-0002
	WOODEN PILLOWS FOR SADDLE SUPPORT	7-12-0003
X	SKIRT BASE DETAILS	7-12-0004
X	SKIRT OPENING DETAILS	7-12-0005
	ANGLE LEG SUPPORT	7-12-0006
	PIPE LEG SUPPORT	7-12-0007
	BRACKET SUPPORT FOR VERTICAL VESSEL	7-12-0008
	MANHOLE WITH HINGED COVER	7-12-0009
X	MANHOLE WITH DAVIT	7-12-0010
X	LADDER RUNGS FOR MANHOLE/DEMISTER	7-12-0011
	RETAINING PLATE	7-12-0012
X	NOZZLE REINFORCEMENT AND PROJECTION	7-12-0013
	PAD NOZZLES FOR VESSELS	7-12-0014
X	STANDARD BOLT HOLE ORIENTATION	7-12-0015
X	ALLOY LINER DETAILS	7-12-0016
	SIGHT GLASSES FOR PRESSURE VESSELS	7-12-0017
	INTERNAL FLANGES	7-12-0018
	VORTEX BREAKERS	7-12-0019
	INLET DEFLECTOR BAFFLE	7-12-0020
	SUPPORT RING AND BOLTING BAR	7-12-0021
	SUPPORT RING SIZES FOR PACKED TOWERS	7-12-0022
X	PIPE DAVIT	7-12-0023
X	LIFTING LUG TOP HEAD TYPE	7-12-0024
X	FIRE PROOFING AND INSULATION SUPPORTS	7-12-0025
X	EARTHING LUG	7-12-0026
X	NAME PLATE	7-12-0027
X	MANUFACTURER NAME PLATE	7-12-0028
X	BRACKET FOR NAME PLATE	7-12-0029
	NAME PLATE FOR SMALL EQUIPMENT	7-12-0030
	DETAILS OF FORGED NOZZLES	7-12-0031
X	SUPPORTS FOR INTERNAL FEED PIPE	7-12-0032
	HOT INSULATION SUPPORT FOR HORIZONTAL VESSEL	7-12-0033
	PIPE DAVIT SUPPORT FOR COLD INS. VESSELS	7-12-0034
X	TYP. DETAILS OF WIRE MESH DEMISTER SUPPORTS	7-12-0036
X	S.R NOZZLE NECK	7-12-0037
X	ALLOWABLE NOZZLE LOADS	7-12-0038

△B REFERENCE DRAWINGS(रेफरेन्स डाईग)

	NOZZLE ORIENTATIONS	
	LADDERS/PLATFORM CLEATS	
	PIPE SUPPORT CLEATS	
	TRAY SUPPORT AND BOLTING BARS	
	DEMISTER DATA SHEET	
	PROCESS DATASHEET	
X	SKETCH FOR MONLEY LADDER	B649-000-80-42-SK-0002-0-1

HOLD UPS (होल्ड अप्स)

X	NOZZLE ORIENTATIONS	X	SIZE OF NOZZLES AS MARKED
X	NOZZLE ELEVATIONS AS MARKED	X	PIPE SUPPORT CLEATS
X	SUPPORT HEIGHT	X	LADDER/PLATFORM CLEATS
X	DETAILS OF INTERNALS		TRAY SUPPORT/BOLTING BARS
		X	PIPE DAVIT
			PACKING SUPPORT

DESIGN DATA (डिजाइन डाटा)

CODE	ASME SECTION VIII DIV-1 EDITION LATEST
WORKING PRESSURE (kg/cm^2g)	157.3
DESIGN PRESSURE (kg/cm^2g)	INT. 165.3 EXT. HV
WORKING TEMPERATURE (°C)	55
DESIGN TEMPERATURE (°C)	139
MDMT (°C)	18
CORROSION ALLOWANCE (mm)	(NOTE-18)
TYPE OF HEAD	HEMISPHERICAL
JOINT EFFICIENCY	SHELL 1 HEAD 1
RADIOGRAPHY	SHELL 100% HEAD 100%
POST WELD HEAT TREATMENT	YES (REFER NOTE-16)
HEAT TREATMENT	HEAD:AS PER CODE / SPEC.
OPERATING MEDIUM	HYDROCARBON / HYDROGEN / H2S
SP. GRAVITY	0.978
WIND SPECIFICATION	IS-875 (PART-3)
SEISMIC	SITE SPECTRA + IS-1893 (PART-4)
CAPACITY (M^3)	2.13
PAINTING/CLEANING	AS PER JOB SPECIFICATION (REFER NOTE-14)
INSULATION THICKNESS (BY OTHERS) (mm) (IH)	NIL ☐ HOT ☐ COLD
FIRE PROOFING CLEATS (BY OTHERS) (mm)	50 (BOTH SIDE) [X] YES ☐ NO
HYDROSTATIC TEST	~~HORIZONTAL~~ / VERTICAL (AT TOP)
PRESSURE (NEW & COLD) (kg/cm2g)	223.943
INSPECTION BY	☐ EIL ☐ CIB [X] TPIA

MATERIAL OF CONSTRUCTION (मैटिरियल ऑफ कन्सट्रक्शन)
(AS PER ASME / IS ~~OR EQUIVALENT~~) (NOTE-24)

SHELL	SA 516 GR.70 # + SA 240 GR.304L CLAD
~~REINFORCEMENT PAD~~/INSERT PLATE	SA 516 GR.70 # + SA 240 GR.304L CLAD
HEADS	SA 516 GR.70 # + SA 240 GR.304L CLAD
SHELL FLANGES	-
LWN / SRWN NOZZLES	SA 266 Gr.4 # + SA 240 GR.304L W.O.
NOZZLE FLANGES	SA 266 Gr.4 # + SA 240 GR.304L W.O.
NOZZLE NECK UPTO 250 NB	SA 266 Gr.4 # + SA 240 GR.304L W.O.
NOZZLE NECK ABOVE 250 NB	SA 266 Gr.4 # + SA 240 GR.304L W.O.
PIPE FITTINGS	SA 234 GR.WPB # + SA 240 GR.304L W.O.
GASKET EXTERNAL	FOR NOZZLE 'M' OCTAGONAL RING , 1500 RATING SOFT IRON (90 BHN) AS PER ASME B16.20
GASKET INTERNAL	-
EXTERNAL STUDS/BOLTS/NUTS	SA 193 GR.B7/SA 194 GR.2H
INTERNAL STUDS/BOLTS/NUTS	SS 304L
SKIRT~~/ SUPPORT~~	SA 516 GR.70 / IS 2062 GR.B E250 QUALITY BR/B0
SKIRT~~/ SUPPORT~~ BASE	IS 2062 GR.B E250 QUALITY BR/B0
INTERNAL PARTS (WELDED)	SA 240 TP 304L/SA 312 GR.304L
INTERNAL PARTS (BOLTED)	SA 240 TP 304L/SA 312 GR.304L/SA 403 Gr.WP304L
CLIPS & ATTACHMENTS DIRECTLY WELDED (EXTERNAL)	SA 516 GR.70
CLIPS & ATTACHMENTS WELDED OVER PAD (EXTERNAL)	IS 2062 GR.B E250 QUALITY BR/B0
TAILING LUG ON SKIRT	-
DEMISTER	UNS S31600 (SS 316)
WEAR PLATE	SS 321
PIPE DAVIT	SA 106 GR.B
LIFTING LUGS	SA 516 GR.70

IMPACT TESTED MATERIAL AT 0^0C AS PER CODE (NOTE-20).

SPECIAL SERVICE : HYDROGEN & SOUR SERVICE

STATUTORY REGULATIONS (स्टैचुअरी रेगुलेशन्स)

INDIAN BOILER REGULATIONS (IBR)
DEPARTMENT OF EXPLOSIVES, NAGPUR (CCOE)

APPROXIMATE WEIGHT (kgs) (PER ITEM) (एपरौक्सिमेंट वजन)

ERECTION	12000	OPERATING	21000
HYDROTEST(SHOP)	15000	HYDROTEST (FIELD)	23000

NUMBER OF ITEMS : ONE

ENGINEERS INDIA LIMITED
NEW DELHI
इंजीनियर्स इंडिया लिमिटेड
(भारत सरकार का उपक्रम)

CHENNAI PETROLEUM CORPORATION LTD.
GROUP II/III LOBS PROJECT,
AT MANALI, CHENNAI

REV	DATE	REVISION	BY	CHECKED	APPROVED
B	21.07.2026	RE-ISSUED FOR BIDS / COMMENTS	J.SINGH	HS/MKS	TKh
A	07.07.2026	ISSUED FOR BIDS / COMMENTS	J.SINGH	HS/MKS	TKh

DESIGN DATA
RECYCLE GAS COMPRESSOR KO DRUM
ITEM NO. 95-C-054

डिजाइन डाटा
रिसाईकल गैस काम्प्रैशर केओ ड्रम
आइटम नम्बर 95–सी–054

DRAWING NUMBER / डाईग नम्बर: **B649-095-80-42-DS-3010** REV: B
SHEET 1 OF 3

FORMAT NO.3-8042-0501 REV.0 (28.02.2024) A3-420 X 297

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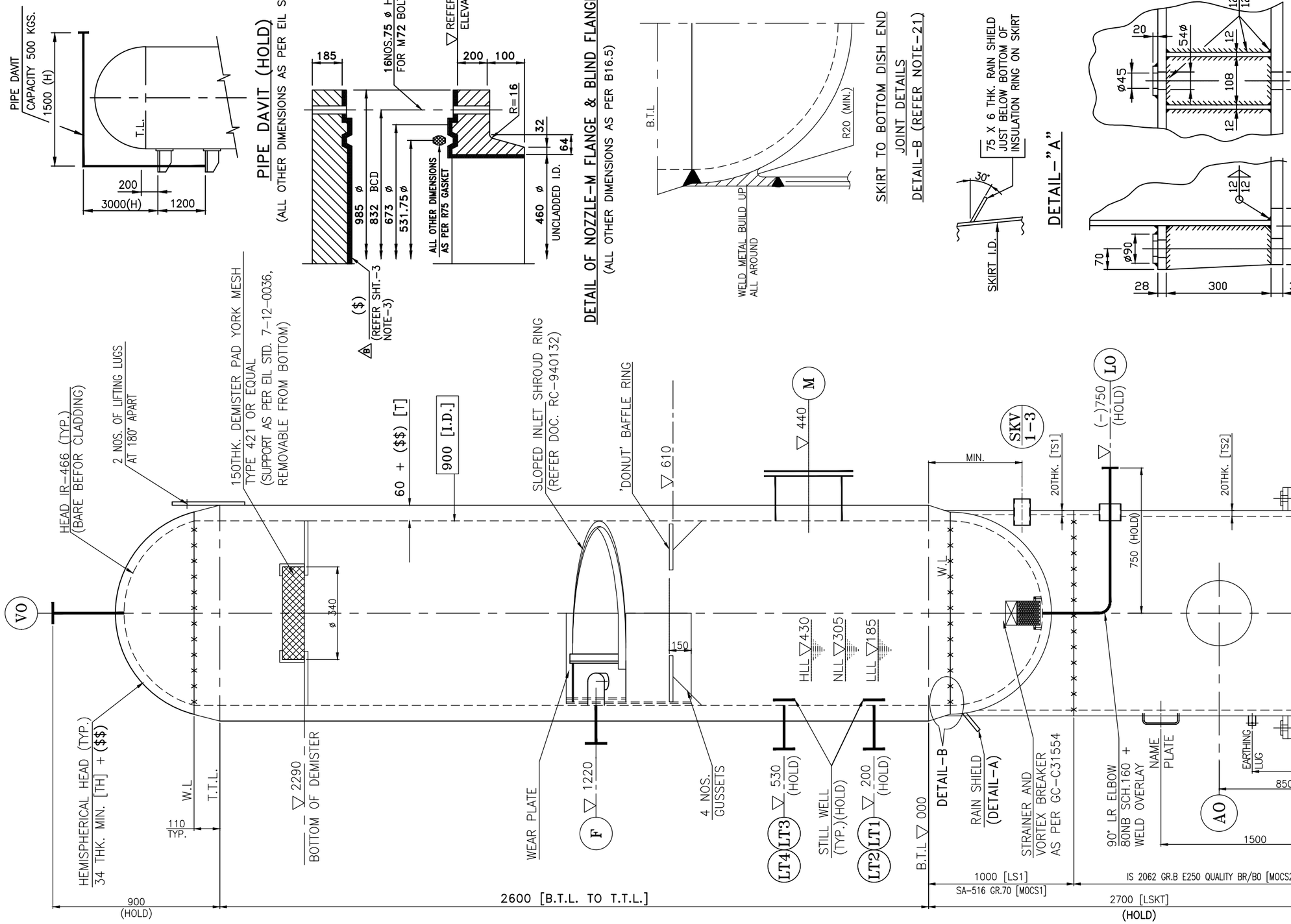


PIPE DAVIT (HOLD)

(ALL OTHER DIMENSIONS AS PER EIL STD.)

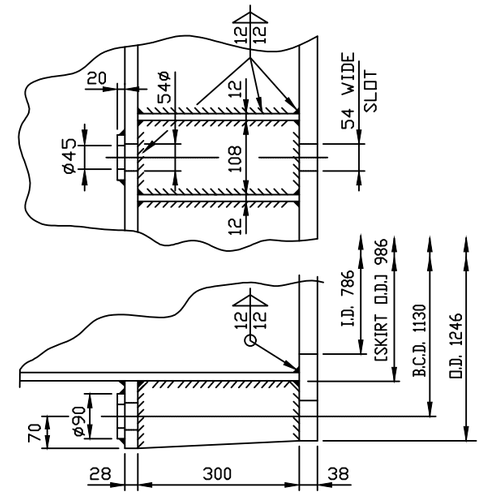
DETAIL OF NOZZLE-M FLANGE & BLIND FLANGE

(ALL OTHER DIMENSIONS AS PER B16.5)

SKIRT TO BOTTOM DISH END JOINT DETAILS

DETAIL-B (REFER NOTE-21)

DETAIL-"A"



SKIRT BASE DETAIL

(ALL OTHER DIMENSIONS AS PER EIL STD.)

B	21.07.2026	RE-ISSUED FOR BIDS/COMMENTS	J.SINGH	HS/MKS	TKh	
A	07.07.2026	ISSUED FOR BIDS/COMMENTS	J.SINGH	HS/MKS	TKh	
REV.	DATE	REVISION	BY	CHK	APPROVED	APPROVED

ENGINEERS INDIA LIMITED
NEW DELHI

CHENNAI PETROLEUM CORPORATION LTD.
GROUP II/III LOBS PROJECT, AT MANALI, CHENNAI

VESSEL DESIGN DATA
RECYCLE GAS COMPRESSOR KO DRUM
ITEM NO. 95-C-054

वेसेल डिजाइन डाटा
रिसाईकल गैस काम्प्रैशर केओ ड्रम
आइटम नम्बर 95-सी-054

DRAWING NO. B649-207-80-42-DS-3010
REV. B
SHEET 2 OF 3

1-1641-0503 REV.0 A3-420x297

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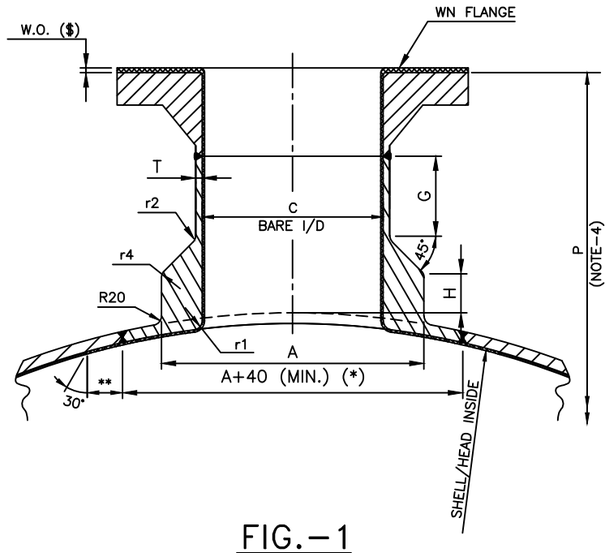

FIG.–1
(RADIAL NOZZLE)

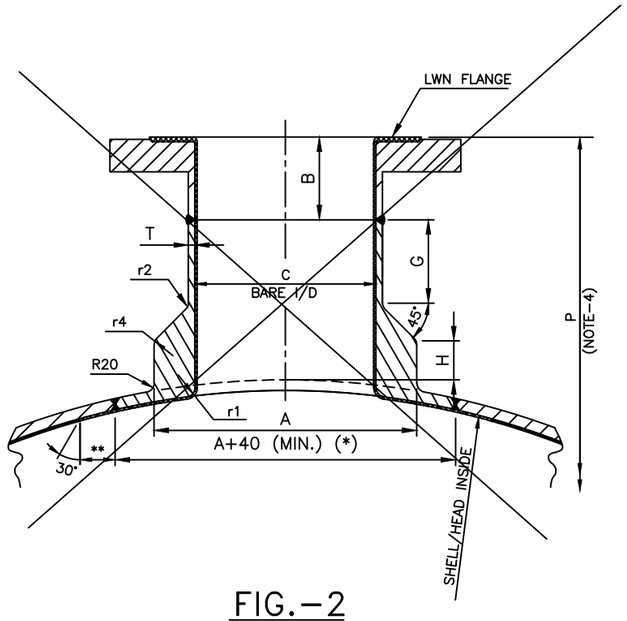

FIG.–2
(RADIAL NOZZLE)

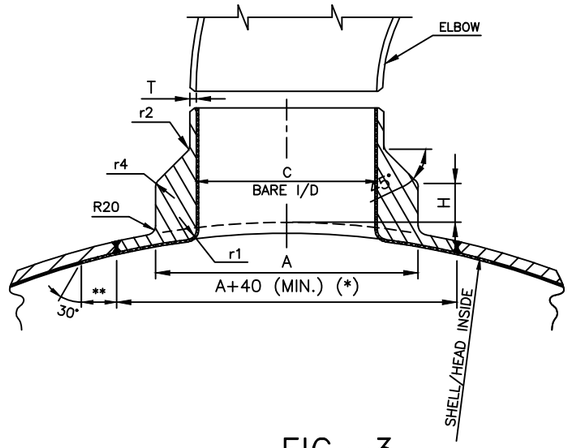

FIG.–3
(BEVEL RADIAL NOZZLE)

** CHIPBACK CLADDING BEFORE WELD METAL DEPOSIT AS PER EIL STANDARD

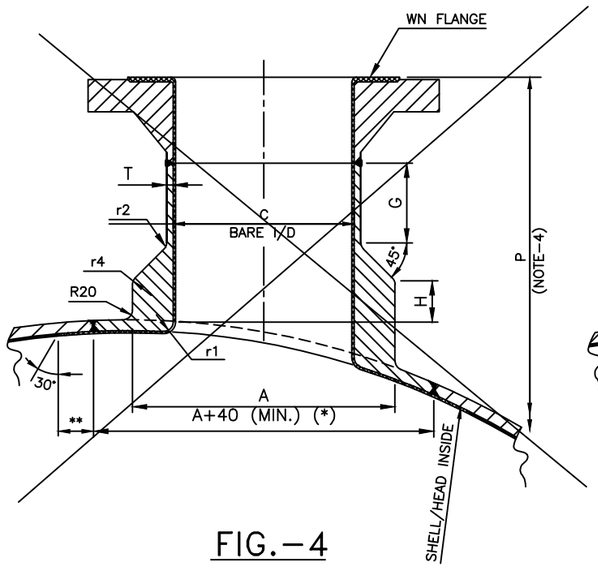

FIG.–4
(OFFSET NOZZLE)

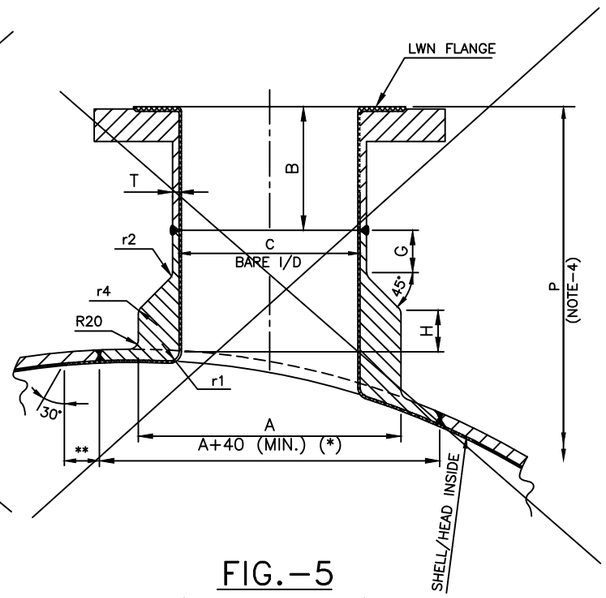

FIG.–5
(OFFSET NOZZLE)

NOZZLE MARK	SIZE (NB) NOTE–(1,2)	P (MM)	A (MM)	C (MM) (NOTE–5)	H (MM) (MIN.)	T (MM)	G (MM)(MIN.) NOTE–(3)	FIG.
M	460 BARE I.D.	1100	730	460	100	32+@	70	1
VI	100	820	222.04	92.04	50	SCH.120+@	50	1
VO	100	SHT–2	184.04	92.04	40	SCH.120+@	50	1
LO	80	SHT–2	146.64	66.64	35	SCH.160+@	50	3
LT1–4	80	790	176.64	66.64	40	SCH.160+@	50	1

DETAIL OF SELF REINFORCED NOZZLE

NOTE:

1. ALL NOZZLE TO SHELL JOINTS ARE TO BE FULLY RADIOGRAPHED.
2. RADIUS 'r1','r2','r4' TO BE FINALISED IN ACCORDANCE WITH DESIGN CODE.
3. PROJECTION 'P' SHALL BE MAINTAINED. 'G' VALUE FURNISHED ARE MINIMUM AND SHALL BE SUITABLY INCREASED TO MAINTAIN PROJECTION 'P'.

(*) REQUIREMENTS OF ASME SEC.VIII DIV–I FIG. UW–16.1F(4) SHALL BE FOLLOWED.
A LIP OF 20MM IS MINIMUM TO BE PROVIDED, VENDOR TO INCREASE THE SAME IF REQUIRED.

(@) WELD OVERLAY (MAINTAIN 3MM UNDILUTED CHEMISTRY).

($) WELD OVERLAY (MAINTAIN 5MM UNDILUTED CHEMISTRY).

4. NOZZLE NECK OD SHALL BE SAME AS PIPING OD.
5. 'C' IS BARE I.D. OF NOZZLE WITHOUT WELD OVERLAY.

ENGINEERS INDIA LIMITED
NEW DELHI

CHENNAI PETROLEUM CORPORATION LTD.
GROUP II/III LOBS PROJECT, AT MANALI, CHENNAI

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A	07.07.2026	ISSUED FOR BIDS/COMMENTS	J.SINGH	HS/MKS	TKh	

VESSEL DESIGN DATA
RECYCLE GAS COMPRESSOR KO DRUM
ITEM NO. 95–C–054

वेसेल डिजाइन डाटा
रिसाईकल गैस काम्प्रैशर केओ ड्रम
आइटम नम्बर 95–सी–054

DRAWING NO. B649–095–80–42–DS–3010
REV. B
SHEET 3 OF 3

1–1641–0503 REV.0 A3–420x297

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NOZZLES AND CONNECTIONS (नोजल व कनैक्शन)

MARK	QTY	NOM. DIA	SCH./THK.	FLANGES			PROJECTION NOTE-4	PAD W x T	SERVICE
		नामिनल	शड्यूल व	CLASS	TYPE	FACING			
मार्क	क्वान्टिटी	डांया	थिकनैस	क्लास	टाईप	फेसिग	प्रोजेशन	पैड	सर्विस
M	1	600	14+@	600	SRWN	RF	1120	-	MANHOLE + BF + GASKET + BOLTING (NOTE-12)
F	1	100 I.D.	12+@	$	SRWN	RF	920	-	FEED INLET (NOTE-22)
VO	1	50.8 ID	16.6+@	600	LWN	RF	SEE DWG.	-	VAPOR OUTLET
LO	1	101.6 ID	25.2+@	600	LWN	RF	SEE DWG.	-	LIQUID OUTLET
PR (H)	1	101.6 ID	25.2+@	600	LWN	RF	SEE DWG.	-	PRESSURE RELIEF
V	1	76.2 ID	20.4+@	600	LWN	RF	SEE DWG.	-	VENT
SO	1	76.2 ID	20.4+@	600	LWN	RF	850	-	STEAM OUT
PC	1	76.2 ID	20.4+@	600	LWN	RF	SEE DWG.	-	PRESSURE CONNECTION
LT 1-4(H)	4	76.2 ID	20.4+@	600	LWN	RF	900	-	LEVEL TAPS

T' DENOTES CORRESPONDING SHELL/CONE/DISH END/SKIRT NOMINAL THICKNESS

@ 6-8mm WELD OVERLAY (FOR MIN. 3mm UNDILUTED CHEMISTRY).

'H' DENOTES H0LD

$ REFER DETAIL ON SHEET-4

NO PIPING, STRUCTURE, LADDER, PLATFORM SHALL BE SUPPORTED FROM THIS EQUIPMENT.

LICENSOR'S SPECIFICATION
लाईसेन्सर स्पेसिफिकश्सन्स

PVM-SU-5290
PVM-SU-1322
GC-C31554
RC-940132
PVM-GD-C99663

FOUNDATION LOADING DATA (OPERATING CONDITION)
फॉऊडशन लोडिगं डाटा (आपरेटिंग कनडिशन)

TYPE	MAX MOMENT AT BASE (M) (kgm)	MAX. SHEAR FORCE AT BASE (H) (kg)
SEISMIC (DB)	-	-
SEISMIC (MC)	-	-
WIND	-	-

GENERAL NOTES (जनरल नोट्स)

UNLESS STATED OTHER WISE

1 ALL DIMENSIONS ARE IN MM UNLESS OTHERWISE STATED.
2 ALL ANCHOR BOLT HOLES TO STRADDLE N/S CENTRE LINE.
3 NORTH DIRECTION WHEREVER SHOWN IS WITH RESPECT TO PLAN VIEW.
4 (A) FOR NOZZLES ON SHELL PROJECTIONS ARE REFERRED FROM VESSEL CENTER LINE TO FLANGE CONTACT FACE.
4 (B) FOR NOZZLES ON HEAD PROJECTIONS ARE REFERRED FROM HEAD T.L TO FLANGE CONTACT FACE.
5 THE INDICATED THICKNESS IS THE MINIMUM ACCEPTABLE AFTER CONSTRUCTION.
6 FLANGE GASKET FACE SHALL HAVE 125-250 AARH FINISH.
7 DIMENSIONS OF FLANGES FOR NOZZLES UPTO 600NB SHALL BE AS PER ASME B16.5 AND FOR NOZZLE ABOVE 600NB SHALL BE AS PER ASME B16.47 SERIES 'B' UNLESS SPECIFIED OTHERWISE.
8 ID OF WELD NECK FLANGES SHALL MATCH WITH CORRESPONDING ID OF NOZZLE PIPE/SHELL.
9 NOZZLES 50NB AND BELOW SHALL BE STIFFENED WITH 2 NOS. 40 x 6 THK FLATS 90^0 APART.
10 ALL FABRICATED NOZZLES SHALL BE 100% RADIOGRAPHED.
11 REQUIREMENT OF INTERNAL GASKETS SHALL BE AS PER SPEC./STD. IN CASE INTERNAL GASKET ARE REQUIRED SAME SHALL BE FULL FACED. VENDOR TO ENSURE THAT MOC FOR INTERNAL GASKET SHALL BE SUITABLE FOR DESIGN TEMPERATURE & OPERATING FLUID.
12 NOZZLES SHOWN WITH BLIND FLANGES SHALL BE PROVIDED WITH STUDS/NUTS AND GASKETS.
13 SURFACE PREPARATION, APPLICATION OF SHOP PRIMER & FINISH PAINTING AS PER JOB SPECIFICATION FOR SURFACE PREPARATION AND PROTECTIVE COATING IS IN VENDOR'S SCOPE. SAME SHALL BE CARRIED OUT BY VESSEL VENDOR AT SHOP BEFORE DISPATCH TO SITE.
14 ALL REMOVABLE INTERNALS SHALL BE SIZED SO AS TO PASS THROUGH NEAREST MANHOLE.
15 AFTER COMPLETION OF FABRICATION AND TESTING ALL SS SURFACES WITH SS INTERNALS SHALL BE PICKLED AND PASSIVATED.
16 EQUIPMENT VENDOR SHALL PERFORM LOCAL LOADS ANALYSIS AS PER WRC (FOR OTHER REQUIREMENTS REFER REQUISITION) AND CHECK THE ADEQUACY OF EQUIPMENT AT THE LOCATION OF CLEATS (PIPING AND STRUCTURAL) FOR FURNISHED LOADS AND INCREASE THE PAD THICKNESS / PROVIDE INSERT PLATES (IF REQUIRED). QUOTED PRICE TO INCLUDE THE SAME.
17 UNLESS OTHERWISE SPECIFIED MINIMUM THICKNESS OF WELDED INTERNAL & BOLTED INTERNAL SHALL BE 12mm. IN CASE OF PIPE AND PIPE FITTINGS SCHEDULE TO BE SELECTED ACCORDINGLY.
18 ALL SS SURFACES SHALL BE PICKLED AND PASSIVATED AS PER ASTM A-380.
19 CORROSION ALLOWANCE (mm)
SHELL & HEADS- 3mm OF SS 347 WOL
PERMANENT INTERNALS- 3.0
REMOVABLE INTERNALS- 1.5
20 NOZZLE SCHEDULES INDICATED ARE MINIMUM REQUIRED. IT SHALL BE FABRICATORS RESPONSIBILITY TO SELECT NOZZLE SCHEDULES SO THAT NO SHRINKAGE OR WARPING OCCURS DURING WELD OVERLAY.
21 FOR WELD OVERLAID FLANGES, MINIMUM THICKNESS OF FLANGES SHALL BE AS PER ASME B16.5 / B16.47 UNLESS SPECIFIED OTHERWISE. VENDOR TO CONSIDER EXTRA THICKNESS TO MEET UNDERCUT REQUIREMENT FOR WELD OVERLAY.
22 NOZZLE FLANGE SHALL MATCH WITH 100NB, #600 WITH ID OF NOZZLE 100mm. 90^0 SR ELBOW OF SS347 SHALL BE Sch.40s, IN ORDER TO BRING I.D. 100mm, WELD BUILD-UP WITH SUBSEQUENT MACHINING TO BE DONE INSIDE.
23 EQUIPMENT SHALL BE FABRICATED, SUPPLIED & TRANSPORTED IN SINGLE PIECE TO SITE.
24 COMPLETE EQUIPMENT SHALL BE POST WELD HEAT TREATED (PWHT). HARDNESS OF BASE METAL, WELDS & HAZ AFTER PWHT SHALL BE MEASURED AND LIMITED TO AS PER CLAUSE 7.7 & 8.3 OF CLG SPECIFICATION No. PVM-SU-5290
25 INSIDE SURFACE OF THE EQUIPMENT INCLUDING NOZZLES SHALL BE WELD OVERLAID WITH SS347. WELD OVERLAY SHALL BE AS PER CLAUSE 7 OF CLG SPECIFICATION No. PVM-SU-1322. 6 TO 8 MM SS347 WELD OVERLAY TO ACHIEVE 3MM OF UNDILUTED CHEMISTRY FROM TOP. DETAILS OF ELECTRODES MENTIONED IN 7-12-0016 SHALL NOT BE CONSIDERED.
26 ALL LAS MATERIAL FOR PRESSURE PARTS SHALL BE IMPACT TESTED AS PER CLAUSE 6.1.(7) & 6.1.(12) OF CLG SPECIFICATION NO. PVM-SU-5290 WHILE FOR NON-PRESSURE PARTS INCLUDING SUPPORTS, IMPACT TEST SHALL BE CARRIED OUT AS PER CLAUSE 6.2 OF CLG SPECIFICATION NO. PVM-SU-5290.
27 ISR SHALL BE CARRIED OUT AS PER CLAUSE 8.2 OF CLG SPECIFICATION NO. PVM-SU-5290.
28 STEP COOLING SHALL BE DONE AS PER CLAUSE 14.1 TO 14.3 OF CLG SPECIFICATION NO. PVM-SU-5290.
29 INTERNAL PROJECTION (MIN.) FOR PIPE OF NOZZLE "F" SHALL BE MIN. WITH WELD OVERLAY ON BOTH INSIDE AND OUTSIDE. WELD EDGE OF NOZZLE "F" (TO BE WELDED WITH SR ELBOW) SHALL BE BUTTERED WITH SS347 WELD OVERLAY, EDGE PREPARED BEFORE PWHT. AFTER PWHT, SLOPED SR ELBOW SHALL BE WELDED. DETAILS TO BE DEVELOPED BY VENDOR AND SHALL BE SUBMITTED FOR EIL APPROVAL BEFORE FABRICATION.

SPECIFICATIONS (स्पेसिफिकेशन्स)

X	DENOTES APPLICABLITY	
X	GENERAL SPEC. FOR PRESSURE VESSELS	6-12-0001
X	SUPP. SPEC. FOR LAS VESSELS	6-12-0003
X	SUPP. SPEC. FOR SS CLAD VESSELS	6-12-0007
X	STD. SPEC. FOR CLAD PLATES	6-12-0015
X	STD. SPEC. FOR 2.25 Cr - 1 Mo STEEL PLATES	6-12-0018
X	SPEC. FOR SURFACE PREPARATION & PROTECTIVE COATING	B649-000-06-41-PLS-01

STANDARDS (स्टैण्डर्डस)

X	VESSEL TOLERANCES	7-12-0001
	SUPPORT FOR HORIZONTAL VESSEL	7-12-0002
	WOODEN PILLOWS FOR SADDLE SUPPORT	7-12-0003
	SKIRT BASE DETAILS	7-12-0004
	SKIRT OPENING DETAILS	7-12-0005
	ANGLE LEG SUPPORT	7-12-0006
	PIPE LEG SUPPORT	7-12-0007
X	BRACKET SUPPORT FOR VERTICAL VESSEL	7-12-0008
	MANHOLE WITH HINGED COVER	7-12-0009
X	MANHOLE WITH DAVIT	7-12-0010
X	LADDER RUNGS FOR MANHOLE/DEMISTER	7-12-0011
	RETAINING PLATE	7-12-0012
X	NOZZLE REINFORCEMENT AND PROJECTION	7-12-0013
	PAD NOZZLES FOR VESSELS	7-12-0014
X	STANDARD BOLT HOLE ORIENTATION	7-12-0015
X	ALLOY LINER DETAILS	7-12-0016
	SIGHT GLASSES FOR PRESSURE VESSELS	7-12-0017
	INTERNAL FLANGES	7-12-0018
	VORTEX BREAKERS	7-12-0019
	INLET DEFLECTOR BAFFLE	7-12-0020
	SUPPORT RING AND BOLTING BAR	7-12-0021
	SUPPORT RING SIZES FOR PACKED TOWERS	7-12-0022
	PIPE DAVIT	7-12-0023
X	LIFTING LUG TOP HEAD TYPE	7-12-0024
X	FIRE PROOFING AND INSULATION SUPPORTS	7-12-0025
X	EARTHING LUG	7-12-0026
X	NAME PLATE	7-12-0027
X	MANUFACTURER NAME PLATE	7-12-0028
X	BRACKET FOR NAME PLATE	7-12-0029
	NAME PLATE FOR SMALL EQUIPMENT	7-12-0030
	DETAILS OF FORGED NOZZLES	7-12-0031
	SUPPORTS FOR INTERNAL FEED PIPE	7-12-0032
	HOT INSULATION SUPPORT FOR HORIZONTAL VESSEL	7-12-0033
	PIPE DAVIT SUPPORT FOR COLD INS. VESSELS	7-12-0034
	TYP. DETAILS OF WIRE MESH DEMISTER SUPPORTS	7-12-0036
	S.R NOZZLE NECK	7-12-0037
X	ALLOWABLE NOZZLE LOADS	7-12-0038

REFERENCE DRAWINGS(रेफरेन्स डाईग)

X	NOZZLE ORIENTATIONS	
	LADDERS/PLATFORM CLEATS	
	PIPE SUPPORT CLEATS	
	TRAY SUPPORT AND BOLTING BARS	
	DEMISTER DATA SHEET	
X	PROCESS DATA SHEET	RD-282191

HOLD UPS (होल्ड अप्स)

X	NOZZLE ORIENTATIONS	X	SIZE OF NOZZLES AS MARKED
X	NOZZLE ELEVATIONS AS MARKED		PIPE SUPPORT CLEATS
X	SUPPORT ELEV. & DETAILS		LADDER/PLATFORM CLEATS
	DETAILS OF INTERNALS		TRAY SUPPORT/BOLTING BARS
			PIPE DAVIT
			PACKING SUPPORT

DESIGN DATA (डिजाइन डाटा)

CODE	ASME SECTION VIII DIV-1 EDITION LATEST
WORKING PRESSURE (kg/cm^2g)	39.3
DESIGN PRESSURE (kg/cm^2g)	INT. 43.2 EXT. HV
WORKING TEMPERATURE (°C)	260
DESIGN TEMPERATURE (°C)	343
MDMT (°C)	18
CORROSION ALLOWANCE (mm)	NOTE-19
TYPE OF HEAD	2:1 ELLIPSOIDAL
JOINT EFFICIENCY	SHELL 1 HEAD 1
RADIOGRAPHY	SHELL FULL HEAD FULL
POST WELD HEAT TREATMENT	COMPLETE EQUIPMENT (NOTE-24)
HEAT TREATMENT	AS PER CODE / SPEC.
OPERATING MEDIUM	HYDROCARBON
SP. GRAVITY	0.718
WIND SPECIFICATION	IS-875 (PART-3)
SEISMIC	SITE SPECTRA + IS-1893 (PART-4)
CAPACITY (M^3)	8.03
PAINTING/CLEANING	AS PER JOB SPECIFICATION (REFER NOTE-13)
INSULATION THICKNESS (mm) (IH)	75(HOLD) [X] HOT [ ] COLD
FIRE PROOFING CLEATS	[ ] YES [X] NO
HYDROSTATIC TEST	~~HORIZONTAL~~ / VERTICAL (AT TOP)
PRESSURE (NEW & COLD) (kg/cm2g)	74.1
INSPECTION BY	[ ] EIL [ ] CIB [X] TPIA

MATERIAL OF CONSTRUCTION (मैटिरियल ऑफ कन्सट्रक्शन)
(AS PER ASME / IS ~~OR EQUIVALENT~~)

SHELL	SA 387 GR.22 CL.2 # + SS 347 WOL
REINFORCEMENT PAD/INSERT PLATE	SA 387 GR.22 CL.2 # + SS 347 WOL
HEADS	SA 387 GR.22 CL.2 # + SS 347 WOL
SHELL FLANGES	-
LWN NOZZLES	SA 182 GR.F22 CL.3 # + SS 347 WOL
NOZZLE FLANGES	SA 182 GR.F22 CL.3 # + SS 347 WOL
NOZZLE NECK UPTO 250 NB	SA 335 GR.P22 # + SS 347 WOL (SEAMLESS)
NOZZLE NECK ABOVE 250 NB	SA 387 GR.22 CL.2 # + SS 347 WOL
PIPE FITTINGS	SA 234 WP 22 CL.3 # + SS 347 WOL/SS 347 (INTERNAL)
GASKET EXTERNAL	SPIRAL WOUND GASKET WITH SS 304 METAL WINDING AND GRAFOIL FILLER MATERIAL WITH SS INNER & OUTER RING AS PER ASME B 16.20.
GASKET INTERNAL	COMPRESSED FIBRE ASBESTOS FREE (NOTE-11)
EXTERNAL STUDS/BOLTS/NUTS	SA 193 GR.B7/SA 194 GR.2H
INTERNAL STUDS/BOLTS/NUTS	SS 347
~~SKIRT /~~ SUPPORT GUSSETS	SA 387 GR.22 CL.2 #
~~SKIRT /~~ SUPPORT BASE	SA 387 GR.22 CL.2 #
INTERNAL PARTS (WELDED)	SS 347
INTERNAL PARTS (BOLTED)	SS 347
CLIPS & ATTACHMENTS DIRECTLY WELDED (EXTERNAL)	SA 387 GR.22 CL.2 #
CLIPS & ATTACHMENTS WELDED OVER PAD (EXTERNAL)	SA 387 GR.22 CL.2 #
TAILING LUG ON SKIRT/SHELL	-
DEMISTER/GRID	-
PAD FOR EXTERNAL ATTACHMENT	SA 387 GR.22 CL.2 #
PIPE DAVIT	-
LIFTING LUGS	SA 387 GR.22 CL.2 #

 # - Impact Tested

SPECIAL SERVICE : NIL

STATUTORY REGULATIONS (स्टैचुअरी रेगुलेशन्स)

INDIAN BOILER REGULATIONS (IBR)
DEPARTMENT OF EXPLOSIVES, NAGPUR (CCOE)

APPROXIMATE WEIGHT (kgs) (PER ITEM) (एपरौक्सिमेंट वजन)

ERECTION	12200	OPERATING	20600
HYDROTEST(SHOP)	20000	HYDROTEST (FIELD)	21000
NUMBER OF ITEMS :	ONE		

REV	DATE	REVISION	BY	CHECKED	APPROVED
A	06.07.2026	ISSUED FOR BIDS / COMMENTS	J.SINGH	MKS/SC	TKh

ENGINEERS INDIA LIMITED
NEW DELHI
इंजीनियर्स इंडिया लिमिटेड
(भारत सरकार का उपक्रम)

CHENNAI PETROLEUM CORPORATION LTD.
GROUP II/III LOBS PROJECT,
AT MANALI, CHENNAI

DESIGN DATA
HCGO HOT LOW PRESSURE SEPARATOR
ITEM NO. 207A-C209N

डिजाइन डाटा
एचसीजीओ हॉट लो प्रेशर सेपारेटर
आइटम नम्बर 207ए-सी209एन

DRAWING NUMBER
डाईग नम्बर
B649-207-80-42-DS-3019
REV A
SHEET 1 OF 4

FORMAT NO.3-8042-0501 REV.0 (28.02.2024) A3-420 X 297

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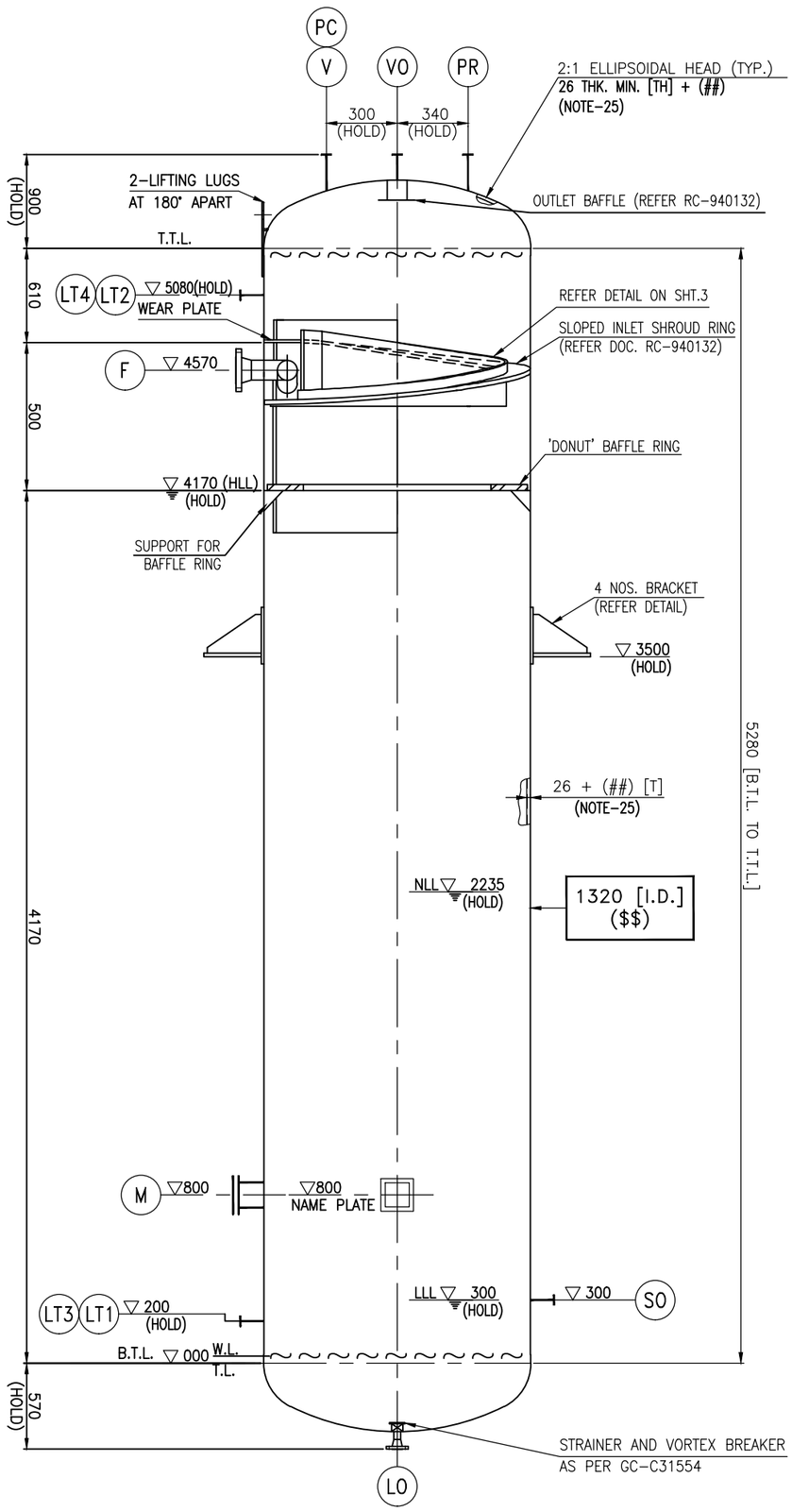
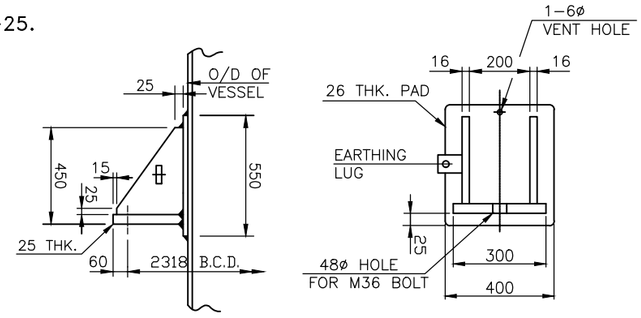


(##) – INCLUDING 3THK. SS 347 WOL UNDILUTED CHEMISTRY

($$) – AFTER WELD OVERLAY OF SS347 REFER NOTE-25.



BRACKET SUPPORT DETAIL (HOLD)

(ALL OTHER DIMENSIONS AS PER EIL STD.)

ENGINEERS INDIA LIMITED
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CHENNAI PETROLEUM CORPORATION LTD.
GROUP II/III LOBS PROJECT, AT MANALI, CHENNAI

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HCGO HOT LOW PRESSURE SEPARATOR
ITEM NO. 207A-C209N

वेसेल डिजाइन डाटा
एचसीजीओ हॉट लो प्रेशर सेपरेटर
आइटम नम्बर 207ए-सी209एन

DRAWING NO. B649-207-80-42-DS-3019

REV. A

SHEET 2 OF 4

1-1641-0503 REV.0 A3-420x297

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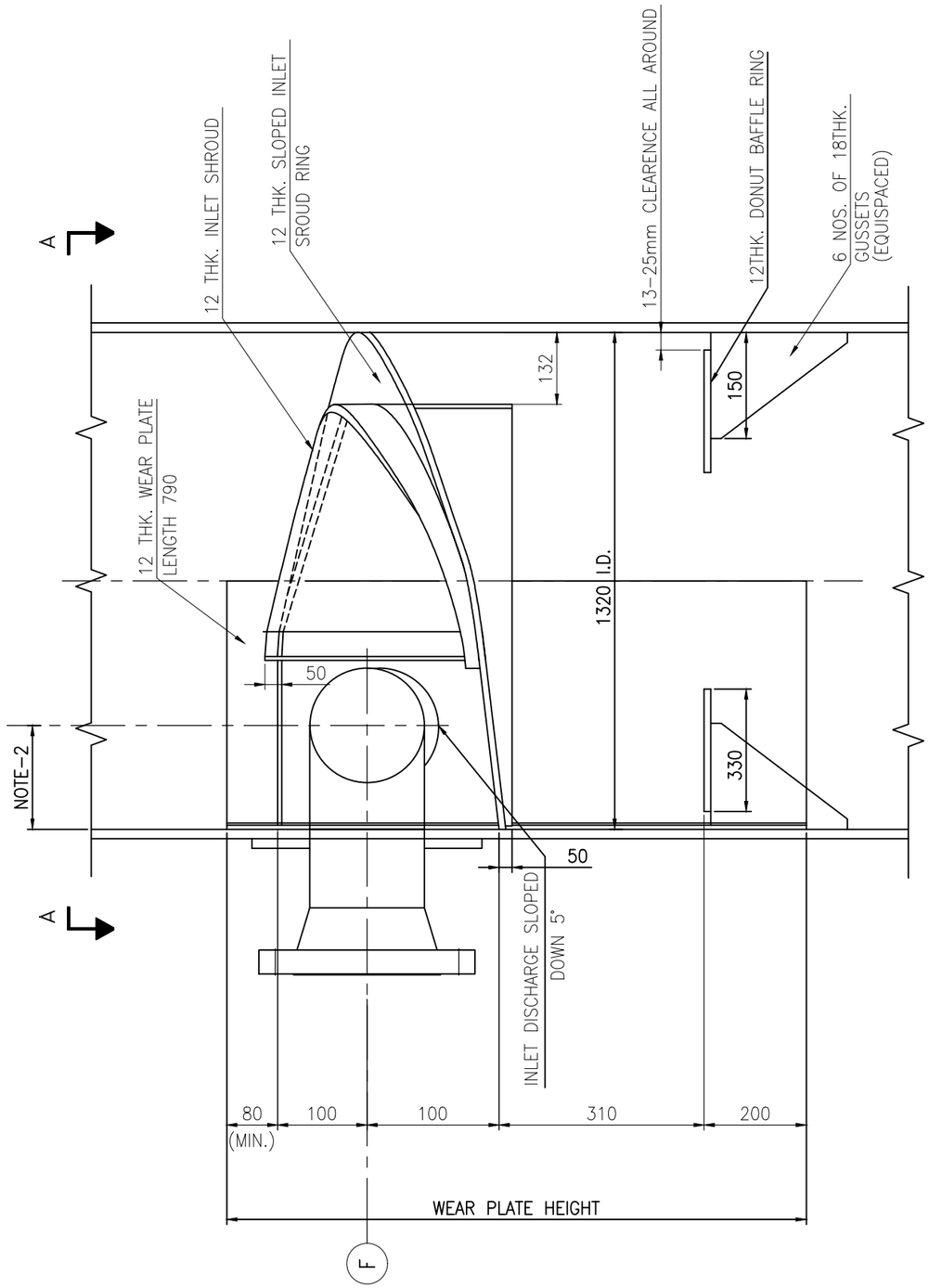
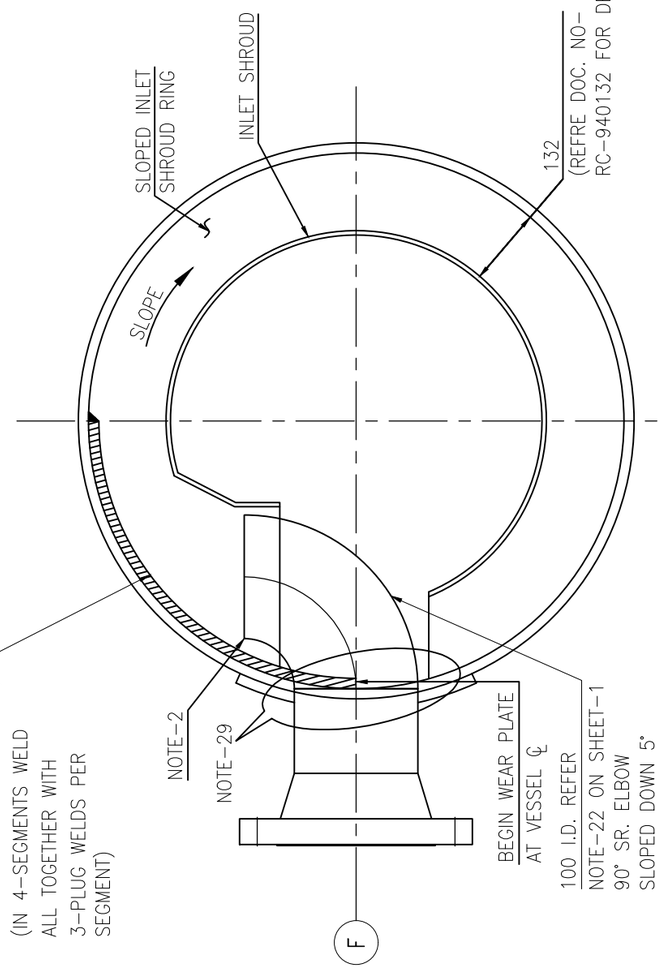


ELEVATION



PLAN VIEW

NOTE:–
1. REFER LICENSOR STANDARD RC–940132 FOR COMPLETE DETAIL.
2. MIN. REFER NOTE–6 OF RC–940132

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	A	06.07.2026	ISSUED FOR BIDS/COMMENTS	J.SINGH	MKS/SC	TKh	
	REV.	DATE	REVISION	BY	CHK	APPROVED	APPROVED

VESSEL DESIGN DATA HCGO HOT LOW PRESSURE SEPARATOR ITEM NO. 207A–C209N	वेसेल डिजाइन डाटा एचसीजीओ हॉट लो प्रेशर सेपारेटर आइटम नम्बर 207ए–सी209एन	DRAWING NO. B649–207–80–42–DS–3019	REV. A
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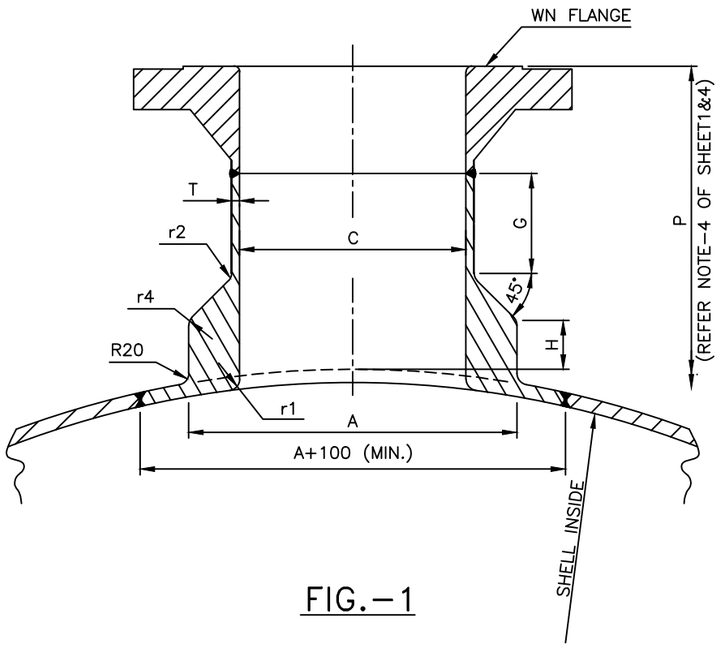


FIG.–1

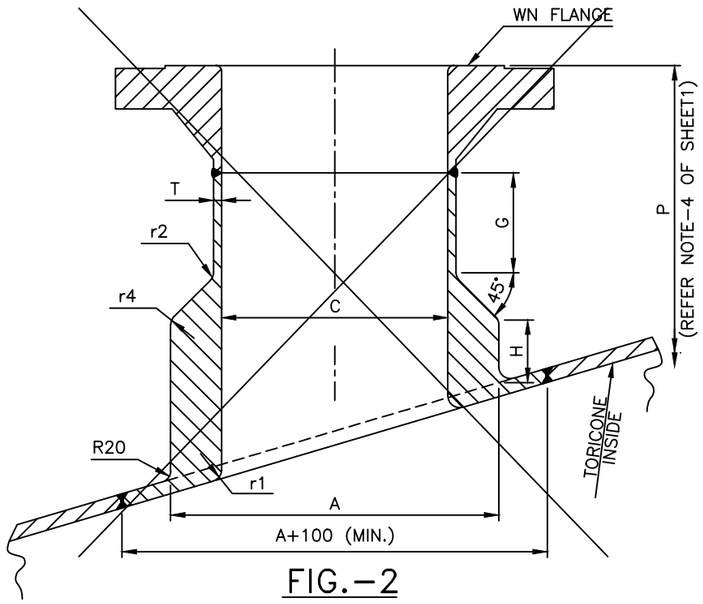


FIG.–2

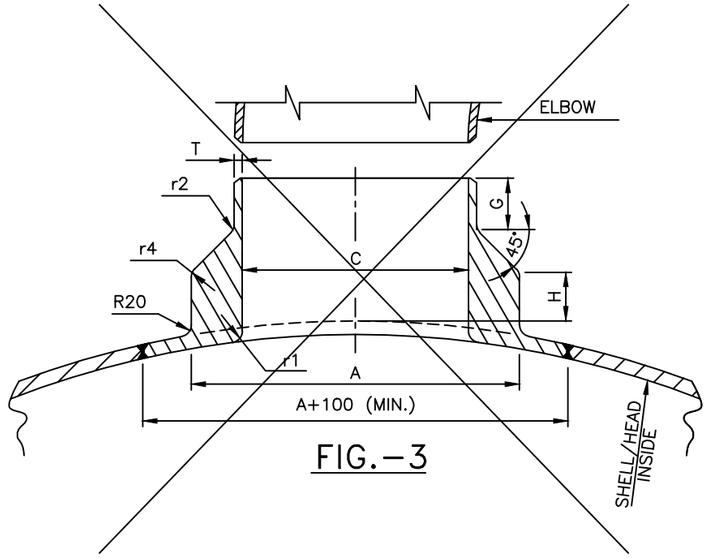
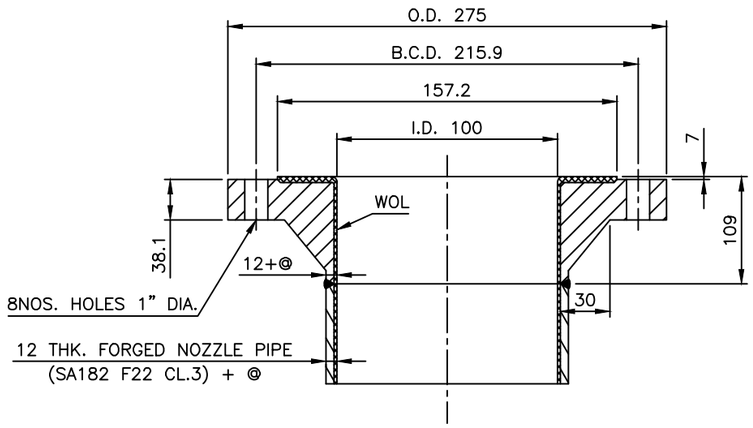


FIG.–3

NOZZLE MARK	SIZE (NB) (NOTE–1,2)	P (MM)	A (MM)	C (MM)	H (MM)	T (MM)	G (MM) (MIN.) (NOTE–4)	FIG.
F (@)	100 I.D.	SEE DWG.	160	100	50	12THK.+@	50	1
M (#)	600	SEE DWG.	842	582	50	14THK.+@	50	1

DETAIL OF SELF–REINFORCED NOZZLE



NOZZLE DETAIL FOR NOZZLE–'F' (@@)

NOTE:

1. ALL NOZZLE TO SHELL JOINTS ARE TO BE FULLY RADIOGRAPHED.
2. RADIUS 'r1','r2','r4' TO BE FINALISED IN ACCORDANCE WITH DESIGN CODE.
3. REQUIREMENTS OF ASME SEC.VIII DIV–1 Fig.UW–16.1 f(1) & f(4) SHALL BE FOLLOWED. A LIP OF 100MM i.e. 50mm (MIN.) BOTH SIDE IS MINIMUM TO BE PROVIDED, VENDOR TO INCREASE THE SAME IF REQUIRED.
4. PROJECTION 'P' SHALL BE MAINTAINED. 'G' VALUE FURNISHED ARE MINIMUM AND SHALL BE SUITABLY INCREASED TO MAINTAIN PROJECTION 'P'.

\# BEFORE WELD OVERLAY
@ SS 347 WELD OVERLAY FOR MIN. 3MM UNDILUTED CHEMISTRY, THK. PROVIDED IS EXCLUDED 3mm THK. WOL FOR UNDILUTED CHEMISTRY.
@@ THICKNESS/DIMENSIONS SHOWN ARE FOR BARE MATERIALS WITHOUT CONSIDERING 3MM WOL, 3MM UNDILUTED CHEMISTRY OF SS347 WELD OVERLAY TO BE CONSIDERED BY VENDOR & DERIVE THE DIMENSIONS OF FLANGE & FORGED PIPE ACCORDINGLY. I.D. OF FLANGE & FORGED PIPE SHALL BE MENTIONED 100MM AFTER 3MM UNDILUTED CHEMISTRY SS347 WOL.

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GROUP II/III LOBS PROJECT, AT MANALI, CHENNAI

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PVM-SU-1322-I

Pressure Vessels with High-Alloy Cladding or Overlay

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Rev.	Date	Description	Author	Technology Leader
F	11/05	General revision.	K. Leung	D. Bosi
G	07/10	General revision.	R. Shockley	D. Bosi
H	06/14	Major revision; greater than 50% rewrite.	C. Boyak	J. Taagepera
I	10/19	General revision	R. Darby	J. Taagepera

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Summary of Changes

Revised technical content may be marked with change bars in the right margin as directed by Author. The Author denotes the following technical changes to this revision as detailed below:

	Author notations regarding this revision
	Section 4.1 – Minimum vessel diameter indicated
	Section 4.5 – Internal Structural Supports major revision
	Section 5.4 – PMI requirements clarified
	Section 7.3 – Weld procedure qualification PMI requirement added
	Safety Critical Changes (if applicable) are noted below
	n/a

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1.0 Scope

1. This specification is to be used when corrosion-resistant alloy is required to protect the carbon steel or low-alloy steel shell from severe corrosive attack by the process environment.
2. This specification addresses cladding by high-alloy sheet bonded to the plate or weld overlay on the plate.
3. This specification is intended to be used in conjunction with the appropriate pressure vessel specification that applies to the chosen base metal.

2.0 References

2.1 Owner Documents

CPM-PU-5074	Material Verification Program
PVM-SU-4750	Carbon Steel Pressure Vessels
PVM-DS-4750	Data Sheet for Carbon Steel Pressure Vessels
PVM-SU-5290	Conventional 2-1/4 Cr-1 Mo Steel Pressure Vessels
PVM-SU-5291	1-1/4 Cr- 1/2 Mo Steel Pressure Vessels Operating Below 825°F (440C°)
[PVM] GD-C99663	Standard Connections and Welding Details for Clad Steel Vessels

2.2 Industry Codes and Standards

Unless specified herein, use the latest edition of the referenced documents.

American Petroleum Institute (API)

RP 582	Welding Guidelines for the Chemical, Oil, and Gas Industries

ASME International (ASME)

Boiler and Pressure Vessel Code (*BPVC*)

Section II: Materials – Ferrous Material Specification

SA-240/SA-240M	Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
SA-263	Specification for Stainless Chromium Steel-Clad Plate
SA-264	Specification for Stainless Chromium-Nickel Steel-Clad Plate
SA-265	Specification for Nickel and Nickel-Base Alloy-Clad Steel Plate
SA-578/SA-578M	Specification for Straight-Beam Ultrasonic Examination of Rolled Steel Plates for Special Applications

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Section VIII, Division 1: Rules for Construction of Pressure Vessels

Section VIII, Division 2: Rules for Construction of Pressure Vessels– Alternative Rules

Section IX: Welding, Brazing and Fusing Qualifications

ASTM International (ASTM)

A265	Standard Specification for Nickel and Nickel-Base Alloy-Clad Steel Plate

American Welding Society (AWS)

A4.2	Standard Procedures for Calibrating Magnetic Instruments to Measure the Delta Ferrite Content of Austenitic and Duplex Ferritic-Austenitic Stainless Steel Weld Metal

Society for Protective Coatings (SSPC)

SP 5	White Metal Blast Cleaning
SP 10	Near-White Metal Blast Cleaning

2.3 Conflict Resolution

Supplier shall submit any conflicts among the inquiry documents in writing to Owner for resolution.

2.4 Units of Measurement

This document contains both U.S. Customary and International System (SI) units of measure, with U.S. units listed first.

3.0 Acronyms

ESW	Electroslag Welding
FCAW	Flux-Cored Arc Welding
FN	Ferrite Number
ID	Inside Diameter
PMI	Positive Material Identification
PT	Liquid Penetrant Test (or Examination)
PWHT	Post Weld Heat Treatment
SAW	Submerged Arc Welding
SMAW	Shielded Metal Arc Welding
UNS	Unified Numbering System
UT	Ultrasonic Testing (or Examination)

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4.0 Design

1. [PVM] GD-C99663 shall be considered part of this specification.
2. Vessel drawings that are part of PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291 shall be included with purchase documents for high-alloy clad pressure vessels.
3. Only those alloys listed in the ASME Code, Section II, SA-263, SA-264, or SA-265 shall be used for clad or weld overlaid vessels or vessel parts.
   a. Duplex and super duplex stainless steels shall be excluded.
   b. Other alloys that may be used shall be subject to review and authorization by Purchaser.

4.1 Pressure Design

1. Thickness of the stainless steel clad or weld overlay shall not be included in the pressure design unless specifically authorized by Purchaser in writing.
2. The thickness of the base metal shall include a minimum external corrosion allowance of 1/16 inch (1.5 mm).
3. The minimum inside vessel diameter shall be 24 inches (610 mm).

4.2 Solid Alloy Components

Solid alloy materials shall not be used for pressure-containing parts unless specifically detailed in Purchaser vessel drawing or authorized by Purchaser in writing.

4.3 Openings

1. For clad vessels with a shell thickness of less than 2 inches (50 mm), nozzles and manways shall conform to [PVM] GD-C99663.
2. Nozzle liners shall not be used unless specified in the purchase order vessel drawings.

4.4 Weld Details

1. Weld details shall show the clad removal distance (strip back) from the pressure boundary butt weld preparation and the bevel angle to allow the clad restoration weld deposit to tie easily into the clad. The removal distance shall be a minimum of 1/4-inch (6.35 mm) from the butt weld preparation.
2. Clad restoration weld overlay details shall show the following:
   a. Required weld metal alloy.
   b. Weld process.
   c. Number of weld overlay layers.
   d. Minimum undiluted thickness.

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3. Weld details for nozzles and manways shall show generous radius at the intersection of the nozzle neck and shell or head inside diameter (ID) surfaces to allow for full minimum thickness weld overlay at all locations. As a minimum, all base metal radii shall meet double ASME Code minimums.

4. Weld details shall show flange face weld overlay to flange bore intersection bevel to allow for full minimum thickness weld overlay at all locations on the nozzle bore.

5. Alternatively, the flange face may be weld overlaid first and the bore second, including the full thickness of the flange face overlay thickness. If this alternative is chosen, instructions shall be placed in the drawings.

4.5 Internal Structural Supports

1. Supports (excluding weld build-up supports, shown in item 4 below) shall be constructed of the same material as the weld overlay or cladding.

2. In weld overlay applications, internal attachments shall be welded directly to the overlay.

3. In roll-bond and explosion bond cladding applications the following shall apply:

   Note: *Heavy loads are defined as those that impose greater than 5000 psi (35 MPa) in shear and 10,000 psi (70 MPa) in tensile.*

   a. Lightly loaded internal attachments may be welded directly to the clad.

   b. Heavily loaded attachments shall be welded to the base metal of the vessel wall and overlaid with the same alloy as used for the cladding except as allowed in item c below.

   c. Heavily loaded supports may be welded directly to the clad with Purchaser acceptance when the following material testing requirements have been met and submitted Purchaser for review:

      1) Bond shear test shall be performed per ASTM A265.

      2) Through thickness tensile test per manufacturer standard with Purchaser acceptance as an alternative to bond shear test.

      3) Tests shall be conducted using specimens from the same lot as used in the vessel and shall be in the “as delivered” condition, unless otherwise requested by Purchaser.

4. Supports designed as weld build-up shall be the same metallurgy as the vessel base metal and welded directly to the base metal. The support shall then be overlaid with the same alloy as the vessel overlay or cladding.

5.0 Materials

5.1 Base Materials

Base materials as shown in vessel drawings shall be furnished according to the requirements of PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291, whichever applies.

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5.2 Welding Consumables

1. Welding consumables shall comply with the SFA designation listed in Section 2.2 for the following:
   a. Alloy(s) specified on vessel drawings for clad matching chemistry weld deposits.
   b. Alloy(s) specified on vessel drawings for clad non-matching chemistry weld deposits.
   c. Alloy chosen to overmatch the clad chemistry.
2. The non-matching alloy chosen shall be compatible with the clad chemistry for the vessel service.
3. In addition to SFA specifications, vessel manufacturer may use the appropriate alloy electrode and flux combinations for the submerged arc welding (SAW) process and electroslag welding (ESW) process for weld overlay or clad restoration welding.
4. The electrode alloy, diameter or strip size, and flux combinations are manufacturer and brand name specific and no intermixing of different brands shall be used in production without requalification.
5. For Type 300 series stainless steels and Alloy 400 (Monel), the welding consumables listed in API 582 may be used for weld overlay and clad restoration.
6. For Types 300/400 series stainless steels, copper-nickel, and nickel-base alloys, refer to API 582 for typical consumables that may be used for combinations of these alloys (clad restoration) to carbon steel or low-alloy steels.
7. Weld consumables and their applications shall be subject to Purchaser review and acceptance. See Table 1 (at the end of this specification) for typical weld consumables for each vessel alloy specified as a guide for weld consumable selection.

5.3 Clad and Weld Overlay

5.3.1 Applications

1. Weld overlay shall be used for the following:
   a. Base metal groove welds in cladding (clad restoration welding).
   b. Base materials such as nozzles, flange faces, and other small area weld overlays.
2. Weld repair areas shall be composed of clad plate or weld overlays.
3. Weld overlay shall be applied to plate after it has been formed into the shapes required for the components of a pressure vessel.

5.3.2 Thickness

1. Roll bond clad materials shall not be used for base metal thickness greater than 4 inches (100 mm).
2. Explosive bond clad material shall be used only for base metal thickness of 1/2 inch (13mm) to 16 inches (400 mm).
3. Minimum clad or weld overlay thickness shall be 1/8 inch (3 mm) unless otherwise authorized by Purchaser.

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4. Maximum clad or weld overlay thickness shall not exceed 5/16 inch (8 mm) without Purchaser's written authorization.

5.4 Roll Bond or Explosive Bond Clad

1. Clad plate shall comply with the requirements of ASME Code, Section II, SA-263, SA-264, or SA-265.
2. Stainless-clad plate material shall be as specified in the vessel drawing and shall comply with the ASME Code, Section II, SA-240/SA240M, including unified numbering system (UNS) if specified.
3. Nickel and nickel-alloy/copper-nickel-alloy clad shall be as specified in the vessel drawing and shall comply with the applicable specified alloy listed in the ASME Code, Section II, SA-265, Section 2, including UNS number if specified.
4. Clad plate of any alloy produced by the explosion bond process shall not have weld seams without the vessel manufacturer's knowledge.
5. Weld seams in clad alloys shall be as follows:
    a. Full fusion, full penetration welds.
    b. Before explosive bonding, welds shall be as follows:
        1) Tested for ferrite content, where applicable.
        2) Positive material identification (PMI) performed per CPM-PU-5074 to verify elemental analysis of the alloy.
    c. Ground flush and smooth, and not readily apparent to the naked eye.
    d. Permanently marked.
    e. After explosive bonding, welds shall be as follows:
        1) Weld seams shall be liquid-penetrant examined.
        2) PMI performed per CPM-PU-5074 to verify an alloy match. Full elemental analysis is not required for this check.
6. The explosion bond manufacturer shall supply as-built locations of all weld joints in the clad.
7. The explosion bond manufacturer shall supply as-built locations of all clad weld repairs.
8. Clad plate of any alloy produced by roll bond process shall supply the vessel manufacturer with as-built locations of all repairs.
9. Weld repairs made by roll bond or explosive bond clad manufacturers shall meet the weld repair requirements of the ASME Code, Section II, SA-263, SA-264, or SA-265, as applicable. Weld repair welding consumables shall match the welding consumables that the vessel manufacturer plans to use for clad restoration or clad repairs.
10. Vessel manufacturer shall specify the welding consumable alloy in the purchase order to the clad manufacturer.

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11. Clad materials produced by roll bond or explosion bond process shall meet the bond quality level class 1 required by the applicable ASME Code, Section II, SA-263, SA-264, or SA-265 specification, whichever applies.

12. Ultrasonic examination (UT) shall meet the ASME Code, Section II, SA-578 with 100 percent coverage per supplemental requirement S1.

13. Type 300 series austenitic stainless steel clad shall be supplied to the vessel manufacturer in the solution-annealed condition as this is the most corrosion-resistant condition.

    a. Roll bond-produced clad materials shall be heat treated at 1675°F to 1725°F (915°C to 940°C) for 1 hour per inch (25 mm) of thickness followed by individual air cooling or by alternate heat treatment, subject to authorization by Purchaser.

    b. Explosion bond-produced clad shall not need to be heat treated if supplied in the solution annealed condition. However, if the explosion bond clad plate manufacturer applies any heat treatment above 800°F (425°C); the procedure shall be submitted to Purchaser for authorization.

    c. Copper-nickel alloys and nickel alloys shall be supplied in a solution-annealed condition or as specified in the ASME Code, Section II, SA-265.

14. 405 and 410S ferritic stainless steel shall be supplied to vessel manufacturer in the normalized and tempered condition.

    a. Roll bond-produced clad materials shall be normalized at 1650°F ±25°F (900°C ±15°C) and tempered at 1250°F (675°C) for one hour per 1 inch (25 mm) of thickness.

    b. Explosion bond-produced clad typically does not need to be heat treated to be supplied in the normalized and tempered condition. However, if the explosion bond clad manufacturer applies any heat treatment above 800°F (425°C), the procedure shall be submitted to Purchaser for authorization.

5.5 Weld Overlays

Weld overlays shall comply with the requirements of Sections 6.0, 7.0, and 8.0 of this specification.

6.0 Fabrication

6.1 Material Handling and Protection

1. Clad materials shall be handled with care and protected during vessel manufacturing activities.

2. Clad surfaces and the as-welded surfaces of weld overlay shall be protected from the following activities:

    a. Arc gouging and grinding of carbon or low-alloy base metals that allow the splatter or grinding debris to impact the clad surface.

    b. Lifting or moving materials in a manner that causes carbon steel clamps or lifting chains to contact and abrade the cladding.

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c. Personnel walking on the clad surface.

d. Use of carbon steel fitting tools that come in contact with the clad surfaces.

3. Weld overlaid surfaces that have been machined shall be protected from carbon steel or low alloy steel contact by covering and taping.

4. Clad and weld overlays shall also be protected from low-temperature melting elements (e.g., lead, bismuth, zinc).

5. The procedure for material handling and protection of high-alloy clad or weld-overlaid vessel parts and the completed vessel shall be submitted by the vessel manufacturer. This procedure shall include steps to clean objectionable embedded carbon or low alloy steel, dirt, grease, oil, or other surface contaminants.

6. Grinding disks shall be suitable for use with alloy clad or weld overlay deposit.

   a. Grinding disks shall be segregated from disks used for carbon steel and shall be marked for easy identification.

   b. Care shall be taken to prevent contamination of the grinding disks from carbon or low-alloy steel when repairs are made to clad or weld overlays.

   c. Disks contaminated with carbon or low alloy steels shall be discarded.

6.2 Material Marking

1. Clad material and weld overlays shall be marked only with compatible marking pens that are low in the following:

   a. Sulfur

   b. Chlorides.

   c. Fluorides.

   d. Low-temperature melting elements such as lead, bismuth, or zinc.

2. Marking pens to be used in production shall have manufacturer's certification of compatibility with the high-alloy clad material and/or weld overlay deposit.

6.3 Material Cutting and Clad Strip Back

1. Non-clad materials shall be cut to shape by oxy-fuel, plasma, laser, water-jet, shearing, and machining methods as limited by the ASME Code and PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291.

2. Clad materials may be cut to shape by thermal means provided that the thermal cutting process does not embrittle or otherwise degrade the clad alloy's ability to perform in service.

3. Sufficient clad alloy material, 1/8 inch (3 mm) minimum, shall be machined or ground to remove the effects of thermal cutting.

4. Clad shall be stripped back from pressure-containing joints a minimum of 1/4 inch (6 mm) on either side of the pressure-containing joint.

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5. If strip back is required for heavily loaded attachments, the alloy clad shall be stripped back a minimum of 3/8 inch (10 mm) from the toe of the attachment weld, including any fillet weld required.

6. Clad strip back shall be done by machining, grinding, plasma arc gouging, or carbon arc gouging.

   a. If either plasma or carbon arc gouging is used, the exposed clad shall be protected from splatter and gouging debris.

   b. If carbon arc gouging is used, the cut edges of the clad shall be ground to remove 1/8 inch (3 mm) minimum of material in order to remove the thermal and carbon contamination effects on the alloy material.

7. Clad removal (strip back or repairs) shall be verified using copper sulfate to ensure complete removal of the clad before alloy welding begins.

8. Carbon steel tools shall not be used on clad or weld overlay surfaces unless protected by alloy sheet or weld metal to limit exposure of the clad or weld overlay to embedded iron or surface iron contamination.

6.4 Material Forming

1. Non-clad materials may be formed using methods as limited by ASME Code and PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291.

2. Weld overlays shall not be formed in any manner. Weld overlays shall be applied after base metal forming and heat treatment, if required.

3. Clad materials may be roll formed, press formed, spun formed, or press brake formed provided that the alloy clad is protected from roll or press/spinning, die indentations, and iron contamination.

4. Forming operations shall take into account thinning of the base metal and/or alloy clad materials. Formed parts shall be checked for thickness after forming to ensure that minimum thickness has been met for base metal and alloy clad.

5. Forming may be done hot or cold. If done hot (above 800°F [427°C]), clad material shall be either post-formed heat treated to a condition equal to base material mill flat plate supply condition or hot formed in a temperature range and cooled to a condition equal to base material mill flat plate supply condition.

6. If formed, the vessel parts shall be heat treated per the ASME Code after forming when extreme fiber of the base material exceeds the limit identified in PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291.

   a. The ASME formulas for determining extreme fiber shall be used for all forming methods except spin forming.

   b. Spin forming of head percent extreme fiber elongation shall be determined by the following formula:

$$Strain\ (\%) = 100\ ln\ (D_b/D_e\text{-}2t)$$

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Equation 1

Where:

T = Nominal thickness of the base metal in inches (or mm)

D_b = Diameter of the starting blank in inches (or mm)

D_e = External diameter of the formed head in inches (or mm)

Ln = Natural logarithm

7.0 Welding

7.1 Base Metal Welding Procedure Qualification

Welding procedures and procedure qualifications for welded joints in the base metal shall be in accordance with PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291.

7.2 Clad Restoration and Repair Welding Procedure Qualification

Welding procedures for clad restoration covering base metal weld joints and clad repairs shall be in accordance with [PVM] GD-C99663 for clad material, and in accordance with Section 7.3 for weld overlaid base material.

7.3 Weld Overlay Procedure Qualification

1. Weld overlay procedures shall be qualified in accordance with the ASME Code, Section IX and applicable requirements of Section VIII, Division 1 Division 2.
2. Details of the weld overlay procedure shall be established by fabricator, submitted for acceptance by Purchaser, and shall include the following:
   a. Weld process.
   b. Wire.
   c. Flux.
   d. Number of layers.
   e. Welding parameters.
3. Single pass procedures, whereby the filler metal-base metal mix results in an equivalent chemical analysis at the specified minimum thickness, shall be considered, provided that fabricator can demonstrate the reliability of the proposed procedure and freedom from interface micro-cracking.
4. SAW and ESW procedure qualifications shall be made using the specified base material and the same type and brand of flux and weld wire/strip to be used in production overlay.
5. Weld qualification procedures shall verify by sample removal and testing that the specified chemical composition of the alloy filler metal is met at a depth of 1/8 inch (3 mm) from the top finished surface, or as indicated on the vessel drawing.

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6. Before bend and liquid penetrant testing (PT), the weld overlay procedure qualification test plate and weld overlay shall be subjected to the total thermal history planned for the completed vessel.
7. After heat treatment, the weld overlay procedure qualification test plate shall be examined by liquid penetrant in accordance with the ASME Code, Section IX, as a minimum, and Section VIII, Division 1 or Division 2, as applicable.
8. After PT of the weld overlay procedure qualification test plate has been completed and accepted, bend specimens shall be examined as follows:
   a. Removed from the test plate and tested per the ASME, Code Section IX.
   b. PT examined on the convex surface in addition to the ASME Code required visual exam.
9. PT results shall meet a “snow white” condition when viewed within the timeframe indicated in the supplied non-destructive examination (NDE) procedure.

7.4 Welders and Welding Operator Qualifications

1. Welders and welding operators shall be qualified per the requirements listed in the ASME Code, Section IX, and Section VIII, Division 1 or Division 2, as applicable.
2. Before ASME Code qualification and production welding, welders and welding operators shall be trained in the application of welding techniques that minimize the following:
   a. Base metal dilution of weld overlay deposits, including bead-to-bead overlap percentage.
   b. Overlap distances.
   c. Arc impingement on the base metal.

7.5 Production Weld Overlay, Clad Restoration, and Clad Repair

1. Per the weld procedure qualification (Section 7.3, item 5), weld overlay shall be designed to meet the specified chemical composition of the alloy filler metal at a depth of 1/8 inch (3 mm) from the top finished surface, or as indicated on the vessel drawing.
2. Maximum total weld overlay thickness shall not exceed 5/16 inch (8 mm) without written Purchaser authorization.
3. Composition of the finished weld overlay deposit from the surface to the specified minimum overlay depth shall conform to requirements of the applicable ASME SFA specifications for the high alloy as listed in Section 2.2.
4. These requirements shall be specified in the drawings for matching or overmatching the chemistry of base metal clad or equivalent weld metal alloy.
5. For the SAW and ESW processes, the finished weld deposit composition shall meet the equivalent chemistry for the specified or vessel manufacturer’s chosen alloy and Purchaser authorization, as required for vessel service.

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6. Ferrite content of 300 series stainless steel deposited weld metal shall be between 3 and 10 ferrite number (FN) (8 FN is preferred for PWHT'd vessels) as determined by a magnetic instrument calibrated according to AWS A4.2 standard procedure.
7. Ferrite content shall be measured prior to any heat treatment.
8. Weld overlays shall be applied to base metal, which has been machined, ground, or blast cleaned to SSPC SP 10 near-white metal finish, except that a SSPC SP 5 white metal finish shall be required for weld overlay processes that do not use a flux (SAW and ESW).
9. Weld overlays shall be applied after forming and heat treatment of the base metal with the exception of post weld heat treatment (PWHT).
10. Weld overlay shall be applied circumferentially to shell, heads, and nozzles. For long or small nozzles where it is impractical to apply the weld overlay circumferentially, it may be applied longitudinally with Purchaser acceptance.

7.6 Weld Overlay, Clad Restoration, and Clad Repair Quality Control

1. Production high-alloy welding shall strictly adhere to welding procedure essential and non-essential variables, including the following:
   a. Bead-to-bead overlap.
   b. Preheat and interpass temperatures.
   c. Inter-layer or individual weld bead cleaning.
   d. Travel speed.
2. Before starting to weld overlay base materials, the base metal shall be machined, ground, or blast cleaned to SSPC-SP 10 near-white metal finish, except that an SSPC-SP 5 white metal finish shall be required for weld overlay processes that do not use a flux (SAW and ESW).
3. Welding consumables shall be verified to be acceptable and to meet manufacturer's brand name and applicable SFA specifications for the required alloy, including UNS number.
4. Electrodes, strip, fluxes, and shielding gases shall be controlled in accordance with the weld consumable manufacturer's recommendations.
5. Welders and welding operators' qualifications shall be verified as to acceptability and to ensure that they are up to date for the welding procedures, processes, and techniques to be used.
   a. Periodic checks shall be made to verify welders' and welding operators' compliance with the welding procedures being used.
   b. Frequency of the checks and verifications shall be determined by the vessel manufacturer and Purchaser.
6. Production welding equipment shall be the same or higher quality than the equipment used to qualify the welding procedures. Welding equipment, including welding ground and cabling, shall be in good repair during all production welding.

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7. Periodic layer thickness measurements shall be made to control minimum and maximum weld overlay limits as the weld deposit is being performed. Frequency of the measurements shall be determined by the vessel manufacturer and Purchaser.
8. For type 300 series weld overlays and clad restoration/repairs, periodic ferrite checks shall be made as the weld overlay is being completed to control final deposit ferrite before PWHT.
9. Ferrite check shall meet the required 3 to 10 FN (8 FN is preferred for PWHT vessels).
10. Periodic spot PT exams shall be performed to confirm that cracking is not occurring as the weld deposit is being completed. The frequency of the checks shall be determined by the vessel manufacturer and Purchaser.

8.0 Preheat and Post Weld Heat Treatment

8.1 Preheat

1. Base metal PWHT requirements of PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291 shall apply to the application of weld overlay, clad restoration, and clad repair welding.
2. For multiple-layer overlays using shielded metal arc welding (SMAW), gas metal arc welding (GMAW), flux-cored arc welding (FCAW), or gas tungsten arc welding (GTAW) processes, preheat shall not be required for layers after the first layer, provided that vessel manufacturer demonstrates to Purchaser's satisfaction that the initial layer is sufficiently thick to prevent hardening of the underlying base steel.

8.2 Post Weld Heat Treatment

The base metal PWHT requirements of PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291 shall apply to clad and weld overlaid vessels.

9.0 Inspection

9.1 Pressure-Containing Welds

1. Category A, B, C, and D pressure-containing welds shall be inspected according to the requirements of PVM-SU-4750, PVM-SU-5290, and PVM-SU-5291.
2. Magnetic particle examination shall be made of the surfaces of Category A, B, C, and D welds that are to be covered by weld overlay at the last stage of fabrication prior to applying the overlay.

9.2 Clad Vessel Parts

9.2.1 Positive Material Identification

Clad materials shall be positively identified upon receipt by the vessel manufacturer in accordance to CPM-PU-5074.

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9.2.2 Ultrasonic Examination

1. UT according to the ASME Code, Section II, SA-578, shall be made of clad plates within 1T (1 × total clad thickness + base material plate thickness) of the edges to be welded, and covering an area extending 1T from load-bearing internal attachments designated in the vessel drawing.
2. Clad shall be removed from areas where lack of bond is detected exceeding quality level Class 1 of the applicable ASME Code, Section II, SA-263, SA-264, or SA-265 and repaired by weld overlay.

9.3 Weld Overlays, Clad Restoration, and Clad Repairs

9.3.1 Positive Material Identification and Chemical Analysis

1. Composition of the top layer of each welding process and each weld metal alloy and weld overlay deposit shall be positively identified in accordance with CPM-PU-5074. Weld overlay sampling for chemical content shall be as follows:
   a. Two locations for each cylindrical shell course and head.
   b. One location for each nozzle and for each Category A, B, C, and D weld seam.
   c. All manual repair areas to machine-applied weld overlays.
2. When it is necessary to remove a sample from the overlay or cladding to make the analysis, the following shall apply:
   a. The sample to be analyzed shall be removed from the top surface to a depth of 1/16 inch (1.6 mm) below the top surface.
   b. The sample may be taken by drilling, filing, or machining, provided that the sample is not contaminated by the sample removal method.
   c. After removal, the sample area shall be weld repaired.
3. Check analysis of carbon content shall be furnished for all austenitic stainless steels, clad, solid, and/or weld metal, in accordance with the applicable ASME Code requirements and the following additional requirements:
   a. Carbon check analysis shall be required only for alloy pressure-containing parts such as nozzles and bosses and for clad in contact with the contents of the vessel.
   b. Carbon check analysis shall be required for each individual clad plate in contact with the contents of the vessel.
   c. Carbon check analysis shall be required for each heat and each batch of electrode flux combinations and a minimum of one check for each shell ring head and nozzle.
   d. For other weld processes, a minimum of one check per heat of electrodes (filler metal) and one check for each longitudinal and girth seam and nozzle weld shall be performed.

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9.3.2 Ferrite Determination

1. The ferrite content of the top layer of type 300 series stainless steel weld overlay shall be determined before PWHT by a magnetic instrument calibrated according to AWS A4.2.
2. The minimum number of locations to be tested shall be as follows:
   a. Ten locations per shell course, selected at random.
   b. Ten locations on each vessel head, selected at random.
   c. One location per foot (300 mm) of circumference at both ends of nozzles and manways.
   d. One location at each Category A, B, C, and D weld.
3. If the ferrite content does not meet 3 to 10 FN, the extent of nonconformity shall be determined by additional ferrite measurements, and nonconforming weld overlays shall be removed, replaced. or repaired, and retested.

9.3.3 Ultrasonic Examination

1. The completed vessel bulk weld overlay (primary method of applying the internal corrosion-resistant lining) shall be checked for bond flaw indications after PWHT by UT in accordance with the ASME Code, Section II, SA-578.
   a. If indications of flaws are detected, the size of the flaws shall be determined by additional UT.
   b. Flaws that exceed acceptance Level C shall be removed and repaired.
2. Unless alternate UT is selected in PVM-DS-4750, the coverage of the UT shall be as follows:
   a. One location 1 inch × 2 inches (25 mm × 50 mm) per linear foot (300 mm) along the vessel length.
   b. One location 1 inch × 2 inches (25 mm × 50 mm) per foot (300 mm) along a meridian of each vessel head
   c. One location 1 inch × 2 inches (25 mm × 50 mm) per foot (300 mm) of circumference at both ends of nozzles and manways.
   d. One location 1 inch × 2 inches (25 mm × 50 mm) per foot (300 mm) of length of clad restoration of Category A, B, C, and D weld seams.
3. When the alternate UT is selected in PVM-DS-4750, coverage of the UT shall be as follows:
   a. One location 1 inch × 2 inches (25 mm × 50 mm) per linear foot (300 mm) along the vessel length in each of four quadrants.
   b. One location 1 inch × 2 inches (25 mm × 50 mm) per foot (300 mm) along a meridian of each vessel head in each of four quadrants.
   c. One location 1 inch × 2 inches (25 mm × 50 mm) per foot (300 mm) of circumference at both ends of nozzles and manways.
   d. One location 1 inch × 2 inches (25 mm × 50 mm) per foot (300 mm) of length of clad restoration of Category A, B, C, and D welds.

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9.3.4 Liquid Penetrant Examination

1. The entire surface of weld overlays and clad restoration welds, and clad or weld overlay repairs, shall be sufficiently smooth and clean to permit PT for fissures and cracks.
2. Prior to any heat treatment including PWHT, PT of the entire weld surface shall be performed in accordance with the ASME Code, Section VIII, Division 2, Part 7.5.7.
3. Relevant indications of surface fissures and cracks shall be removed, and the area shall be repaired and re-inspected.

10.0 Hydrostatic Test

1. The hydrostatic test of the completed vessel shall be conducted according to PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291.
2. Hydrotest water quality shall be as follows:
   a. Potable.
   b. Chloride content shall not exceed 50 ppm.
   c. pH between 6 and 8.3.
   d. Mercury less than 0.005 ppm.
   e. Copper less than 0.02 ppm.
3. Chemical analysis or certification from water supply (not older than 30 days) shall be required.
4. After hydrostatic testing, the vessel shall be promptly and carefully drained and completely dried, especially around trays and internals.
   a. Vessel shall be internally inspected to ensure that no residual moisture remains.
   b. Critical vessels or vessels shipped by sea may require nitrogen blanketing.
   c. These requirements shall be specified on the vessel purchase drawings.

11.0 Records

Records listed in this section shall supplement the records required by PVM-SU-4750, PVM-SU-5290, or PVM-SU-5291.

11.1 Explosion Bond Clad Alloys

The following minimum records shall be supplied upon completion of the vessel:

1. Base metal mill certificates.
2. Alloy sheet or plate mill test certificates.
3. Filler metal certificates for butt welds in alloy sheet or plate.
4. UT reports.
5. PT reports.
6. Weld maps of all butt welds and repairs.

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7. Heat treatment procedures (over 800°F [427°C]) and heat treatment records.
8. PMI reports.

11.2 Roll Bond Clad Alloys

The following minimum records shall be supplied upon completion of the vessel:

1. Base metal/alloy mill certificates.
2. Filler metal certificates for weld repairs to clad.
3. Weld maps of clad repairs.
4. PMI reports.

11.3 Weld Overlays, Clad Restoration Welds, and Clad Repair Welds

The following minimum records shall be supplied upon completion of the vessel:

1. Weld consumable test certificates.
2. Weld procedures and supporting procedure qualification records.
3. PMI and chemical analysis records including portable analyzer calibration certification.
4. Ferrite reports including instrument calibrations (as applicable).
5. UT reports.
6. PT reports.
7. Weld maps of clad restoration welds and repairs to clad or weld overlay.
8. Heat treatment (over 800°F [427°C]) records.

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Table 1: Clad and Weld Deposit Alloy Chemical Element Verification Requirements

Product Form		Clad Sheet and Solid Materials (1)						Acceptable Filler Metals (2)		Clad Restoration and Repair Weld Deposits (3)						Weld Overlay Deposits (3)					
Chemical Elements		C	Cr	Ni	Mo	Cu	Fe			C	Cr	Ni	Mo	Cu	Fe	C	Cr	Ni	Mo	Cu	Fe
Solid Alloy	UNS Number							Filler Alloy	UNS Number												
405	S40500		X	X				309L 308L NiCrFe-3	W30913 W30813 N06082 W86182		X	X					X	X			
410S	S41008		X	X				309L 308L NiCrFe-3	W30913 W30813 N06082 W86182		X	X					X	X			
304L	S30403	X	X	X				309L 308L	W30913 W30813	X	X	X				X	X	X			
310S	S31008	X	X	X				309L 310	W30913 W31010	X	X	X				X	X	X			
316L	S316003	X	X	X	X			309LMo 316L	W30923 W31610	X	X	X	X			X	X	X	X		
317L	S31703	X	X	X	X			309LMo 317L	W30923 W31710	X	X	X	X			X	X	X	X		
321 (4)	S32100	X	X	X				309L 347	W30913 W34710	X	X	X				X	X	X			
347 (4)	S34700	X	X	X				309L 347	W30913 W34710	X	X	X				X	X	X			
Inco Monel	N04400			X		X	X	NiCu-7	N04060 W84190			X		X	X (5)			X		X	X (5)

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Product Form		Clad Sheet and Solid Materials (1)						Acceptable Filler Metals (2)		Clad Restoration and Repair Weld Deposits (3)						Weld Overlay Deposits (3)					
Chemical Elements		C	Cr	Ni	Mo	Cu	Fe			C	Cr	Ni	Mo	Cu	Fe	C	Cr	Ni	Mo	Cu	Fe
Solid Alloy	UNS Number							Filler Alloy	UNS Number												
Inco 600	N06600		X	X			X	NiCrFe-1 NiCrFe-3 NiCrFe-5	N06132 W86132 N06082 W86182 N06062		X	X			X	X	X				X
Inco 625	N06625		X	X	X	(6)	X	NiCrMo-3	N06625 W8612		X	X	X		X		X	X	X		X
Inco 825	N08825		X	X	X		X	NiCrMo-3	N06625 W8612		X	X	X		X		X	X	X		X
Hast C276	N10276	X	X	X	X	(7)		NiCrMo-10 NiCrMo-14	N0622 W86022 No6686 W86026	X	X	X	X	(7)	X	X	X	X	X	(7)	X
Hast C22	N06022	X	X	X	X	(7)		NiCrMo-10	N0622 W86022	X	X	X	X	(7)	X	X	X	X	X	(7)	X
Nickel	N02201			X				Ni-1	N02061 W82141			X					X				
20 Cb-3 (4)	N08020	X	X	X	X	(7)	X	320LR	W88022	X	X	X	X	(7)	X	X	X	X	X	(7)	X

Notes:

1. *Chemical elements shall meet the applicable ASME SA or SB material specification chemistry limits.*
2. *Limited example SMAW and bare filler metals listed. Other equivalent filler metals for SAW, ESW, and FCAW may be used.*
3. *Chemical elements shall meet the applicable ASME SFA filler metal specification chemistry limits. For those equivalent filler metals and flux combinations not covered by SFA specs, chemical elements shall meet the solid alloy requirements specified for the vessel or by agreement with Purchaser for non-matching chemistry weld deposits.*
4. *Stabilized alloys shall be analyzed for carbon and titanium or niobium (Nb or Cb) and the carbon to stabilizing element ratio determined to be within applicable ASME specification for solid material or weld deposits.*
5. *For Monel (N04400) weld deposits, the iron (Fe) content shall be limited to 5.0%.*
6. *Aluminum shall be determined for Alloy 625.*
7. *Cobalt shall be determined for these alloys.*

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PVM-SU-4750-I

Carbon Steel Pressure Vessels

CTC-PVM-SU4750-I

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Rev	Date	Description	Author	Technology Leader
E	07/2008	General revision.	K. Leung	D. Bosi
F	12/2011	General revision; scope change to include all thicknesses.	J. Taagepera	D. Bosi
G	11/2013	General revision. Replaced PVM-4748 replace PVM 8.00.	J. Taagepera	C. Boyak
H	01/2017	General revision. Replaces PVM-SU-997.	R. Darby	J. Taagepera
I	10/2020	General revision.	R. Darby	J. Taagepera

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Summary of Changes

	Author notations regarding this revision
	Update with changes throughout to align with IOGP S-619.
	Improved MACA verbiage, incorporation of Changes to Section VIII including Mandatory Appendix 46, relaxed requirements for use of MT, limited use of “set-on” nozzles now permitted, improved verbiage in Section 9.
	Section 18 added for design of girth flanges.

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1 Scope

1. This specification covers general requirements for carbon steel pressure vessels designed and fabricated in accordance with the ASME Boiler and Pressure Vessel Code (BPVC), Section VIII, Division 1, or Division 2. This specification is limited to design temperatures at or below 800°F (425°C).

2. PVM-SU-4750 Section 14 through Section 17 contain supplemental requirements for specified conditions.

> ***Note:*** *Design information and supplemental requirements for the pressure vessel are specified on the individual vessel drawing or data sheet. The supplemental requirements are essential to vessels intended for certain services. Be sure to review the data sheet guide PVM-DG-4750 and to select any applicable requirements for completing the data sheet PVM-DS-4750-US (SI). Consult a specialist for further guidance. Where numerous vessels are purchased, prepare a separate data sheet for each, as appropriate.*
>
> *The data sheet PVM-DS-4750-US (SI) need not be used if an Owner-accepted alternative is provided that contains the same information.*

1.1 Conflict Resolution

1. In cases of conflict between this specification and other drawings or specifications, the vessel manufacturer (hereafter referred to as the fabricator) shall call attention to the conflict and request an interpretation by Purchaser.
2. Fabricator shall not assume which instructions govern.

1.2 Units of Measurement

When applicable, both U.S. Customary and International System of Units (SI) units of measure have been provided, with U.S. Customary listed first. In cases where a dual unit of measurement has not been provided, an exact conversion is not available or applicable.

2 References

Unless otherwise specified herein, use the latest edition of the referenced documents.

2.1 Owner Documents

CIV-SU-398 Fabrication of Structural and Miscellaneous Steel

CIV-DU-5009 Structural Design Criteria

CIV GD-M99971 Design Guide For Cantilever Brackets For Vertical Vessel Platforms

COM-SU-5014 Internal Linings for Vessels in Wet H_2S Service

COM-SU-5191 Coating Systems

EXH-SU-5150 Graphite-Covered Corrugated Metal (CMG)-Style Gaskets

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EXH-SU-5151 Kammprofile (KAM)-Style Gaskets

FPM-DU-5091 Onshore Passive Fire Protection Materials and Application

FPM-SU-5141 Design of Firewater Systems for Fire Protection including Foam System

PIM-SU-5209 Flange Gaskets and Bolting

PVM-SU-1322 Pressure Vessels with High-Alloy Cladding or Overlay

PVM-DG-4750 Data Sheet Guide for Carbon Steel Pressure Vessels

PVM-DS-4750-SI Data Sheet for Carbon Steel Pressure Vessels (SI)

PVM-DS-4750-US Data Sheet for Carbon Steel Pressure Vessels (US)

PVM-DU-6158 Design by Analysis of Pressure Vessels and Similar Components

PVM-GF-C14311 Pressure Vessel Nozzles

PVM-GD-C78876 Standard Skirt and Base Details for Vertical Vessels

PVM-GF-C87280 Standard Manway Cover Davits and Hinges

PVM-GF-C99694 Standard Details of Saddles for Horizontal Vessels

PVM-GB-C99772 Standard Ladder Rungs Inside Columns and Vessels

PVM-GA-C99913 Vortex Breaker for Vessel Liquid Outlet

PVM-GB-C99945 Standard Support Legs for Vertical Vessels

QAM-SU-6053 Quality Assurance Requirements for Equipment and Material Manufacturers

QAM-SU-6163 Performance Demonstration Qualification Program for NDE Contractors Performing Ultrasonic Inspection

SID-SU-5106 Safety In Designs

2.2 Industry Codes and Standards

American Petroleum Institute (API)

API Std 660 Shell-and-Tube Heat Exchangers

API Std 2510 Design and Construction of LPG Installations

API Publ. 2510A Fire-Protection Considerations for the Design and Operation of Liquefied Petroleum Gas (LPG) Storage Facilities

American Society of Civil Engineers (ASCE)

ASCE 7 Minimum Design Loads for Buildings and Other Structures

American Welding Society (AWS)

AWS A4.3 Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and

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Ferritic Steel Weld Metal Produced by Arc Welding

AWS D1.1/D1.1M Structural Welding Code—Steel

ASME International (ASME)

ASME B16.47 Large Diameter Steel Flanges – NPS 26 Through NPS 60 Metric/Inch Standard

ASME B16.5 Pipe Flanges and Flanged Fittings NPS 1/2 through NPS 24 Metric/Inch Standard

ASME B31.3 Process Piping

ASME PCC-2 Repair of Pressure Equipment and Piping

ASME STS-1 Steel Stacks

ASME Boiler and Pressure Vessel Code (BPVC, also referred to as 'the ASME Code')
Section II, Materials, Part C, Specifications for Welding Rods, Electrodes, and Filler Metals

ASME Boiler and Pressure Vessel Code (BPVC, also referred to as 'the ASME Code')
Section II, Materials, Part D, Properties

ASME Boiler and Pressure Vessel Code (BPVC, also referred to as 'the ASME Code')
Section V, Nondestructive Examination

ASME Boiler and Pressure Vessel Code (BPVC, also referred to as 'the ASME Code')
Section VIII, Division 1, Rules for Construction of Pressure Vessels

ASME Boiler and Pressure Vessel Code (BPVC, also referred to as 'the ASME Code')
Section VIII, Division 2, Alternative Rules – Rules for Construction of Pressure Vessels

ASME Boiler and Pressure Vessel Code (BPVC, also referred to as 'the ASME Code')
Section IX, Qualification Standard for Welding, Brazing, and Fusing Procedures, Welders, Brazers, and Welding and Fusing Operators

ASTM International (ASTM)

ASTM A578/A578M Standard Specification for Straight-Beam Ultrasonic Examination of Rolled Steel Plates for Special Applications

International Association of Oil & Gas Producers (IOGP)

Note: *IOGP JIP33 specifications are available at https://www.iogp.org/bookstore/.*

IOGP S-619 Pressure Vessels - Unfired, Fusion Welded

NACE International (NACE)

NACE MR0103 / ISO 17945 Petroleum, Petrochemical, and Natural Gas Industries - Metallic Materials Resistant to Sulfide Stress Cracking in Corrosive Petroleum Refining Environments

NACE MR0175 / ISO 15156 Petroleum and Natural Gas Industries - Materials for use in H_2S-Containing

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Environments in Oil and Gas Production

Process Industry Practices (PIP)

PIP VESV1003 Special Fabrication Requirements for Welded Vessels and Tanks to Be Lined

PIP VEFV1103 Vessel Grounding Lug

PIP VEFV1122 Horizontal Vessel Insulation Supports

PIP VEFV1123 Vertical Vessel Insulation and Fireproofing Supports

Welding Journal Research Supplement

Zick, L.P., "Stresses in Large Horizontal Cylindrical Pressure Vessels on Two Saddle Supports"

The Welding Research Council (WRC)

WRC Bulletin 297 Local Stresses in Cylindrical Shells Due to External Loadings

WRC Bulletin 537 Precision Equations and Enhanced Diagrams for Local Stresses in Spherical and Cylindrical Shells due to External Loadings for Implementation of WRC Bulletin 107

Note: *WRC 537 supersedes WRC Bulletin 107.*

3 Acronyms

ET Eddy Current Surface Examination

FEA Finite Element Analysis

GMAW Gas Metal Arc Welding

GTAW Gas Tungsten Arc Welding

HAZ Heat Affected Zone

MACA Maximum Allowable Corrosion Allowance

MAEP Maximum Allowable External Pressure

MAP Maximum Allowable Pressure

MAWP Maximum Allowable Working Pressure

MDMT Minimum Design Metal Temperature

MT / MPI / MPT Magnetic Particle Examination Testing

NDE/NDI/NDT Nondestructive Examination/Inspection/Testing

PQR Procedure Qualification Record

PWHT Post Weld Heat Treatment

RT Radiographic Examination (or Inspection or Testing)

SAW Submerged Arc Welded/Welding

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SMAW Shielded Metal Arc Welding

UT Ultrasonic Test or Examination

UTSW Ultrasonic Shear Wave Examination

WFMT Wet Fluorescent Magnetic Particle Examination

WPS Welding Procedure Specification

4 ASME Code Conformance

4-1 - Design and fabrication shall conform to the ASME BPVC, Section VIII, Division 1 or 2. Hereinafter, ASME BPVC, Section VIII will be referred to as the ASME Code.

4-2 - All vessels within the scope of the ASME Code shall be Code stamped. Stamping and inspection for other vessels shall be a matter of agreement with the Owner.

4-3 - The latest addenda issued through the date of the purchase order or contract shall be used.

4-4 - When the ASME Code, Division 2 is specified, requirements from Division 2 shall be met as a minimum. Applicable corresponding Division 2 paragraphs shall be substituted for Division 1 paragraphs referenced by this specification.

4-5 - Division 1, Appendix 32, and Division 2, paragraphs 4.3.8, 4.4.10, and 4.14.2 of the ASME Code (addressing local thin areas) shall not be used without written authorization from Owner.

4-6 - Calculations covered by U-2(g) of Division 1 or Part 5 of Division 2 of the ASME Code shall be supplied to the Owner for review and acceptance. When finite element analysis (FEA) is used, it shall comply with PVM-DU-6158.

5 Design

5.1 Design Conditions

5.1.1 Structural Loads

5.1.1-1 - Seismic and wind loading shall be in accordance with ASCE 7-16.

5.1.1-2 - Allowable stresses shall be in accordance with the ASME Code.

5.1.1-3 - Local jurisdiction or regulatory standards shall govern only if they produce a more stringent design.

5.1.1-4 - Alternative structural codes may be used only with Owner’s acceptance.

5.1.1-5 - Unless otherwise specified by Owner, transportation design loads shall be based on a lateral acceleration of 0.65 g acting along both the longitudinal axis and the transverse axis, and a vertical acceleration of 1.5 g as a minimum. The allowable stress may be increased by 33 percent for the transportation design condition.

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5.1.2 Design Pressure and Temperature

5.1.2-1 - Vessel design conditions shall be established by the process engineer.

5.1.2-1.1 - These conditions shall consider startup, normal operation, shutdown, upset, and any other conditions reasonably anticipated.

5.1.2-1.2 - Multiple combinations of pressure/temperature conditions shall be acceptable and are often necessary.

5.1.2-1.3 - Mechanical design temperature for vessels with internal insulating lining shall not be less than metal temperature under the lining.

5.1.2-1.3.1 - Calculation shall be based on maximum operating temperature, atmospheric conditions of no wind, and ambient temperature of 100ºF (40ºC).

5.1.2-1.3.2 - Credit shall not be taken for radiative heat loss.

5.1.2-2 - Vessels subject to steamout for maintenance shall be designed for 7.5 psi (0.5 bar) external pressure at 350ºF (180°C).

5.1.3 Maximum Allowable Pressures and Corrosion Allowance

5.1.3-1 - The vessel data sheet shall state the design pressure and coincident design temperature, and design corrosion allowance (CA).

5.1.3-2 - Minimum design corrosion allowance shall be 0.06 inch (1.5 mm) unless the vessel is clad or overlaid. For corrosion allowance on clad or overlaid vessels, refer to PVM-SU-1322.

5.1.3-3 - Pressure vessel shall meet or exceed requirements specified in the equipment data sheet and in the Purchaser's documents.

5.1.3-4 - Maximum allowable working pressure (MAWP) and temperature, maximum allowable corrosion allowance (MACA), and field and shop hydrostatic test pressure of a vessel shall not be limited by the mechanical design of minor components such as nozzles (including nozzle flange ratings), manways, opening reinforcement, or bolting, unless Owner's written acceptance is obtained.

5.1.3-5 - Maximum shall be designed for the design pressure(s) at the coincident temperature(s) as indicated on the vessel data sheet.

Note: *Calculation of MAWP(s) is not required for Section VIII, Division 1 vessels (see the ASME Code, Section VIII, Division 1, End Note 35). At the fabricator's option, the MAWP may be calculated. This calculated MAWP may be slightly higher than design pressure.*

5.1.3-6 - MACA methodology shall be used to optimize the design for corrosion allowance.

5.1.3-6.1 - The MACA for pressure components is the difference between the nominal thickness and the calculated required (retirement) thickness as per the design code. MACA is the design minimum corrosion allowance (internal, external, or both) plus the thickness added to obtain a commercially available nominal thickness.

5.1.3-6.2 - Excess component thickness (round up thickness) beyond what is required for governing minimum

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thickness plus design corrosion allowance shall be added to the design corrosion allowance in order to determine the MACA.

5.1.3-6.3 - The MACA shall first be determined individually for each of the major components (individual shell courses, head components, conical transitions, and body flanges). For vessels with more than one shell course, the MACA shall be calculated separately for each course.

5.1.3-6.4 - Nozzle reinforcement calculations per UG-37 of the ASME Code, Division 1 or Part 4.5 of Division 2 shall be performed using $A_1 = 0$. Additional thickness that is added to the component for the sole purpose of nozzle loading and reinforcement may be used for A_1.

5.1.3-6.5 - If the Pressure Area Method is used for nozzle reinforcement design, and a plate thicker than the least nominal thickness necessary is used, the additional thickness added solely for nozzle loading and reinforcement shall not be required to be counted towards MACA.

Note: *As an example of item 5.1.3-6.4 and item 5.1.3-6.5, a shell course may have a minimum required (corroded) thickness of 0.60-inch, while the design corrosion allowance is 0.125-inch. These add up to 0.725-inch while the next available nominal plate thickness is 0.75-inch. The MACA then is 0.75-inch – 0.60-inch = 0.15-inch. If the fabricator chooses to use 1-inch plate solely to reduce other added nozzle reinforcement, the MACA remains 0.15-inch and the extra 0.25-inch may be credited as additional thickness available in the shell.*

5.1.3-6.6 - When attached to cylindrical shell components or flat heads, minor components shall inherit the MACA of the cylindrical shell or flat head component to which they are attached.

5.1.3-6.7 - Formed head and formed transition component thickness shall be measured after forming. The measured minimum thickness shall be used in the as-built calculations and drawings, and the as-built MACA shall be calculated.

5.1.3-6.8 - When attached to a formed head or formed transition component, the corrosion allowance of minor components shall be 150 percent of the vessel's design corrosion allowance.

5.1.3-6.9 - The as-built MACA for each major component (each shell course, each head, etc.) shall be included on the vessel drawing and data report.

5.1.3-6.10 - MACA for each major component shall be determined at least to the nearest 0.01 inch (0.2 mm).

5.1.3-6.11 - MACA shall be calculated regardless of whether the corrosion allowance is internal or external.

Note: *MACA need not exceed twice the corrosion allowance.*

Note*: When calculating required thicknesses using the design formulas in UG-27 of the ASME Code and with an internal corrosion allowance, the calculations must be iterated to arrive at a true component MACA value. This is due to the dimensions being based on the corroded condition per UG-16(e). In order to do this, start with an initial corrosion allowance (e.g., 1/8-inch), calculate the corroded radius, calculate minimum required thickness, and calculate MACA (nominal thickness less minimum required thickness). Next, plug that MACA back into the formulas as the corrosion allowance and re-calculate the corroded radius, minimum required thickness, and another MACA. Repeat until consecutive MACA values are the same within a specified tolerance (usually 0.01-inch). This can take 2–5 iterations to converge, depending how close the initial corrosion allowance is to the true*

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MACA.

The alternative is to use the formulas per the ASME Code, Mandatory Appendix 1, which are based on outside dimensions and therefore are not affected by internal corrosion allowance. This option provides for an explicit, non-iterative solution.

***Note**: Corrosion allowance is not required by the ASME Code or this specification to be displayed on the nameplate. This information is best shown on the manufacturer's data report that has room for multiple, unique component corrosion allowances.*

5.1.3-7 - Manufacturer's calculations shall also include the maximum allowable pressure (MAP) in the new (uncorroded) and cold condition.

5.1.3-8 - Vessels designed for internal pressure only shall also be stamped for the calculated maximum allowable external pressures (MAEP) at the internal pressure design temperature and steamout temperature listed in item 5.1.2-2.

5.1.3-9 - Manufacturer's calculations shall clearly state the minimum required thickness for all major components of the vessel. In addition to being stated in the calculations, these thicknesses shall be included in a table on the General Arrangement drawing.

5.1.3-10 - Minimum corroded thickness of shell and heads shall be 0.12 inch (3.0 mm).

5.1.4 Minimum Design Metal Temperature

5.1.4-1 - Minimum design metal temperature (MDMT) shall be selected in consideration of the operating cases and of ASME Code rules in Division 1, paragraphs UG-20(b). For Division 2 vessels, paragraph 4.1.5.2(e) shall be referenced.

5.1.4-2 - Impact test exemption per the ASME Code, Division 1, paragraph UG-20(f), shall not be allowed except for P1, Group 1 material up to 1/2 inch (13 mm) thick.

5.1.4-2.1 - Impact test exemption for forged materials per the ASME Code, Division 1, paragraph UCS-66(c) and Division 2, paragraph 3.11.2.4(a) shall not be used.

5.1.4-3 - The following MDMT and coincident pressures shall be indicated on both the vessel nameplate and the manufacturer's data report.

5.1.4-3.1 - MDMT at MAWP.

5.1.4-3.2 - Lowest allowable MDMT and coincident maximum pressure.

5.1.4-4 - All attachments to the pressure boundary with aggregated primary stress in the attachment weld exceeding 8,000 psi (55 MPa) shall be considered essential to structural integrity for the purposes of establishing the MDMT of the vessel.

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5.1.5 Cyclic Service

5.1.5-1 - Vessels designated as subject to cyclic or transient loading of pressure or temperature shall be screened in accordance with the ASME Code, Division 2, paragraph 5.5.2.

5.1.5-2 - If this screening determines that fatigue analysis is required, design rules and details shall be per Division 2, paragraph 5.5 of the ASME Code. The use of paragraph 5.5.5 for the fatigue assessment of welds shall not be permitted without Owner's acceptance.

5.1.5-3 - Welds shall be blend-ground to form a smooth transition with the parent metal.

5.1.5-3.1 - Welds need not be ground flat.

5.1.5-3.2 - Welds shall be sized to meet the ASME Code requirements after grinding.

5.1.6 Load Combinations

5.1.6-1 - Vessels shall be designed with consideration given to the following loads and load combinations:

5.1.6-1.1 - Erect load with full wind—Dead load shall include installed weight of the vessel including internals, platforms, insulation, and other permanent attachments.

5.1.6-1.2 - Operating load with full wind or seismic loads—Operating load shall include all dead and live loads, high-high liquid level and zero process liquid levels, and full and zero pressures or partial or full vacuum (FV) when specified as a design case. Thrust loads from attached piping shall also be considered.

5.1.6-1.3 - Field hydrostatic pressure test load with partial wind—When a field hydrostatic pressure test is required, the vessel shall be assumed to be full of water, and support shall be designed for 50 percent of design wind speed (or 25 percent of wind load).

5.2 Welded Joints

5.2-1 - Category A and B joints shall be double-butt-welded (Type 1), or an acceptable equivalent agreed to by Purchaser.

5.2-2 - Category C and D joints shall be full penetration welds in accordance with PVM-GF-C14311 or an acceptable equivalent agreed to by Owner.

5.2-3 - An acceptable equivalent for Category A, B, C and D joints in nozzles, elbows, and single-sided closure seams shall be a single-welded butt joint with gas tungsten arc welding (GTAW) root bead.

5.2-4 - Permanent backup rings or strips shall not be used.

5.2-5 - Plate and sheet cut by shearing shall be ground such that shear-damaged material is removed.

5.2-6 - On horizontal vessels, no longitudinal seam welds shall be located more than 30 degrees below the horizontal centerline.

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5.2-7 - Weld joints between the support skirt and the vessel shall be as shown on the vessel drawing or data sheet. Other longitudinal and circumferential weld joints shall be in accordance with item 5.2-1.

5.3 Heads and Transitions

5.3-1 - Ellipsoidal or hemispherical heads shall be preferred; however, torispherical heads may be permitted when designed per the ASME Code, Division 2, Sections 4.3.6 and 4.4.8, and reviewed and accepted by Owner.

5.3-2 - Toriconical transition sections shall be preferred; however, conical transitions may be permitted when reviewed and accepted by Owner.

5.3-3 - The following shall apply when straight conical sections are used:

5.3-3.1 - Design shall be per the ASME Code, Division 2, Sections 4.3.11 and 4.3.13.

5.3-3.2 - Joint efficiency of 1.0 shall not be permitted for conical transitions without a knuckle due to the difficulty in examining the joints.

5.3-3.3 - Checks shall be made for all design conditions and hydrotest conditions, including shop and field.

5.3-4 - Stiffening rings, if used, shall be sufficiently away from butt joints to allow easy nondestructive examination (NDE) (ultrasonic examination [UT] and magnetic particle examination [MT]) to occur during vessel manufacturing and while in service.

5.3-5 - Ellipsoidal heads, torispherical heads, toriconical heads, and toriconical transition sections shall have straight flanges at least 1-1/2 inches (38 mm) long.

5.3-6 - Cold-formed heads shall be stress relieved. Certified documentation from head Supplier shall state if heads were cold formed (and, if cold formed, stress relieving temperature) or hot formed (and, if hot formed, the forming temperature).

5.3-6.1 - Certified documentation from head Supplier shall state if heads were cold formed or hot formed.

5.3-6.2 - The documentation shall include stress relieving temperature if cold formed and forming temperature if hot formed.

5.3-6.3 - If the completed vessel is to be stress relieved, then head may be stress relieved at this time, provided the entire head is stress relieved and not just its weld seams.

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5.4 Nozzles, Manways, Bosses, and Other Openings

5.4-1 - Nozzles for NPS size 1-1/2 (DN 40) and larger shall be in accordance with PVM-GF-C14311.

5.4-2 - Smaller nozzles shall not be used except as required to meet other standards or specifications such as in item 5.4-8.2.

***Note**: It is preferable for nozzles to be integrally reinforced (i.e., no reinforcing pads), but this type of design is only required as noted. Integral reinforcement can be achieved with any combination of extra thick shell (or other component) plates, local weld buildup of the shell, extra large fillet welds, and extra thick nozzle necks (i.e., long weld neck or heavy wall forging).*

5.4-3 - Nozzles in shells or heads with a nominal thickness 2 inches (50 mm) or greater shall be integrally reinforced (i.e., reinforcing pads shall not be permitted).

5.4-4 - Nozzles shall be integrally reinforced in vessel services as follows:

5.4-4.1 - Design temperature exceeding 650°F (345°C).

5.4-4.2 - Cyclic service.

5.4-4.3 - Severe process environments having stress-corrosion-cracking potential.

5.4-4.4 - Vessels in mild or benign services, but having defined process upsets, which may carry over stress-corrosion-cracking agents.

5.4-4.5 - Field-erected LPG storage spheres.

5.4-5 - Flanged nozzles NPS 2 (DN 50) and smaller shall be seamless, long weld neck flanges, or fabricated from Schedule 160 or XXS seamless pipe and weld neck flanges.

5.4-6 - When shell thickness exceeds 3 inches (75 mm) and nozzle thickness is significantly less than the shell thickness, ‘set-on’ nozzle designs shall be permitted only with written acceptance by Owner. When used, the following shall apply:

5.4-6.1 - The use of set-on nozzles shall be prohibited in the following services:

1. Sour service.
2. Hydrogen service.
3. Cyclic service.

5.4-6.2 - The following supplemental NDE shall be required for installation of set-on nozzles:

1. The base material shall receive UT per SA-578, 100 percent coverage of the indicated area, acceptance Level C on base material in a 4 inch radius around opening prior to nozzle fit-up.
2. MT of the through-wall cut shall be required.
3. Completed weld shall receive UT per Section 9.10.

5.4-7 - When shell thickness exceeds 3 inches (75 mm) and nozzle thickness is significantly less than the shell

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thickness, a hybrid set-in/set-on design per the ASME Code, Division 2, Table 4.2.10, Detail 7 shall be acceptable with Owner agreement. When used, the following shall apply:

5.4-7.1 - When installing straight hub welding flanges of non-uniform thickness (commonly referred to as "heavy barrel nozzles"), the dimension "t_n" in the ASME Code, Division 2, Table 4.2.10, Detail 7 shall be defined as the minimum nominal thickness of the nozzle along the hub's length.

5.4-7.2 - Weld shall be back-gouged and rewelded as necessary to achieve full penetration.

5.4-7.3 - The following supplemental NDE shall be required for installation of hybrid set-in/set-on nozzles:

5.4-7.3.1 - UT inspection of base material per SA-578, 100 percent coverage of the indicated area, acceptance Level C on base material in a 4 inch radius around opening prior to nozzle fit-up.

5.4-7.3.2 - MT of through-wall cut.

5.4-7.3.3 - UT inspection of completed weld per Section 9.10.

5.4-8 - Manways shall be in accordance with PVM-GF-C14311 (as defined in item 5.4-1) and PVM-GF-C87280.

5.4-8.1 - For vessels with a nominal shell inside diameter (ID) less than 48 inches (1.2 m), manways shall have a minimum uncorroded ID of 17.9 inches (454 mm). This corresponds to the nominal ID of an NPS 20 Schedule 80 (DN 500).

5.4-8.2 - For vessels with a nominal shell ID 48 inches (1.2 m) and larger, manways shall have a minimum uncorroded ID of 21.5 inches (546 mm). This corresponds to the nominal ID of an NPS 24 Schedule 80 (DN 600).

***Note**: Forgings such as LWN and other heavier wall nozzles that typically have an ID of the nominal pipe diameter, 18 inches (457mm) and 24 inches (609mm), are acceptable alternatives to nozzles built up with pipe.*

5.4-9 - Threaded connections may be used only if required by another code or standard such as Tubular Exchanger Manufacturers Association (TEMA).

5.4-9.1 - Threads shall be chased after installation or post weld heat treatment (PWHT).

5.4-9.2 - Threaded connections shall not be acceptable for use on internally coated or lined vessels.

5.4-10 - The following items shall be prohibited for use as vessel connections:

5.4-10.1 - Threaded connections greater than NPS 1 (DN 25).

5.4-10.2 - Tongue-and groove flange facing.

5.4-10.3 - Integrally reinforced welded branch fittings (such as Weldolets™, Thredolets™, and Sockolets™).

5.4-10.4 - Pipe fittings.

5.4-11 - Threaded connections NPS 1 (DN 25) and smaller, when shown on the vessel drawing, shall be full coupling, ASME Class 6000, 'set through' the vessel shell with a full penetration weld, per item 5.2-2. This application shall be limited to shell thickness less than 3/4 inch (19 mm) and design temperature less than 450°F (230°C).

5.4-12 - Seamless long weld neck and integrally-reinforced nozzle forgings shall be preferred for nozzles and manways.

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5.4-13 - Openings should not penetrate and reinforcing pads (when permitted) and should not cover welded joints. When unavoidable, the weld seam shall be ground flush and 100 percent radiographed to a minimum of 2 inches [50 mm] beyond each side of the opening or repad.

5.4-14 - For vessels supported by a skirt, the following shall apply:

5.4-14.1 - The first piping flange welded to any nozzle in the bottom head shall be located outside the skirt.

5.4-14.2 - No branch connections shall be allowed within the skirt.

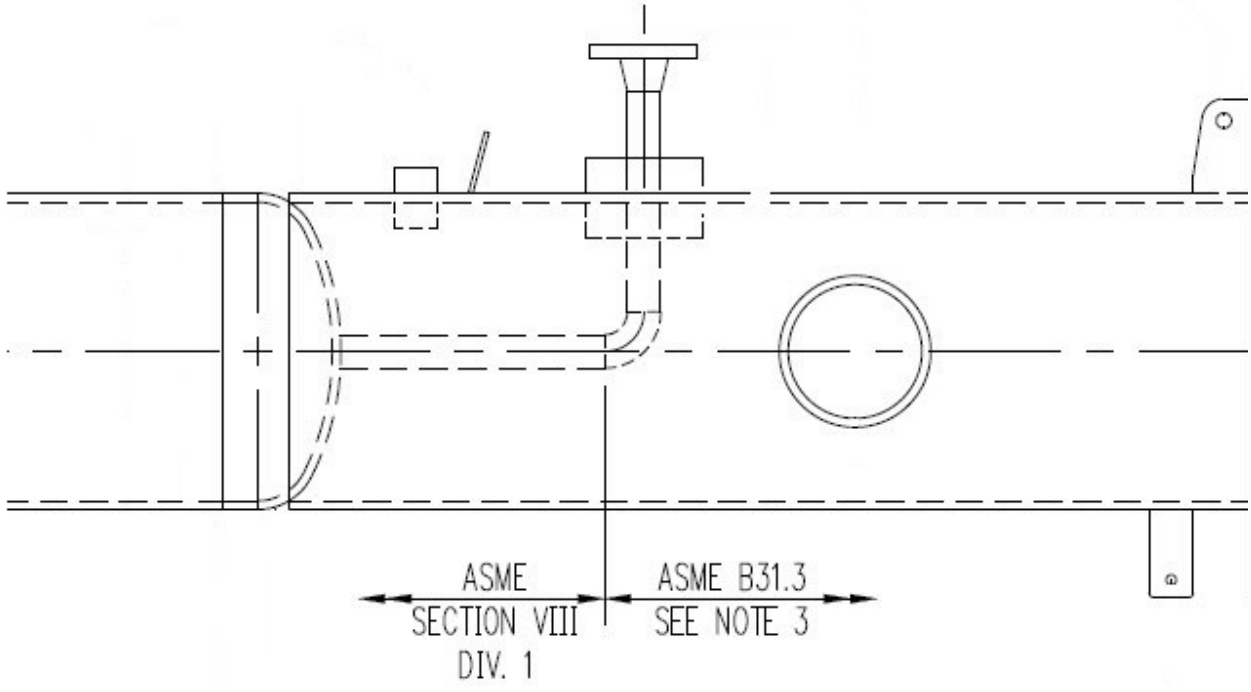
5.4-14.3 - No piping shall be routed through skirt manway.

5.4-14.4 - Vertical nozzles within the skirt shall be designed and fabricated in accordance with the requirements of ASME Code, Section VIII and this specification.

5.4-14.5 - Vertical piping (if any), elbow, horizontal piping, and flange shall be designed and fabricated in accordance with the requirements of ASME B31.3.

5.4-14.6 - The code specification break between the ASME Code, Section VIII and ASME B31.3 shall be clearly documented on the drawings and manufacturer's data report. See the example sketch and example U-1 verbiage in Figure 1 and Figure 2 below.

PVM-SU-47540, Figure 1: Code Spec Break Example



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PVM-SU-4750, Figure 2: Example of Spec Break Verbiage from a U-1 Form

19. Nozzles, inspection, and safety valve openings:

Purpose (Inlet, Outlet, Drain, etc.)	No.	Diameter or Size	Type	Material		Nozzle Thickness		Reinforcement Material	Attachment Details		Location Open
				Nozzle	Flange	Nom.	Corr.		Nozzle	Flange	
(A) VAPOR OUTLET	1	6"	300# RF	SA-105	SA-105	1.06"	0.135"	SA-516-70	WELDED	INTEGRAL	TOP
(B) FEED INLET	1	6"	300# RF	SA-105	SA-105	1.06"	0.196"	SA-516-70	WELDED	INTEGRAL	SIDE
(C) STEAM INLET	1	4"	300# RF	SA-105	SA-105	0.875"	0.191"	SA-516-70	WELDED	INTEGRAL	SIDE
(D) BOTT OUTLET	1	4"	STUB	SA-106B		0.438"	0.125"	SA-516-70	WELDED	WELD TO B31.3 PIPING	BOTT
(E) SAMPLE	1	2"	300# RF	SA-105	SA-105	0.655"	0.191"		WELDED	INTEGRAL	SIDE
(F) SAMPLE	1	2"	300# RF	SA-105	SA-105	0.655"	0.191"		WELDED	INTEGRAL	SIDE
(G) SAMPLE	1	2"	300# RF	SA-105	SA-105	0.655"	0.191"		WELDED	INTEGRAL	SIDE
(J) STEAM INLET	1	1 1/2"	300# RF	SA-105	SA-105	0.625"	0.196"		WELDED	INTEGRAL	SIDE

Additional Nozzles - See Attached U-4...

20. Supports: Skirt YES (Yes or no) Lugs N/A (Number) Legs N/A (Number) Others N/A (Describe) Attached WELDED TO BOTT HEAD (Where and how)

21. Manufacturer's Partial Data Reports properly identified and signed by Commissioned Inspectors have been furnished for the following items of the report (list the name of part, item num Manufacturer's name, and identifying number):

N/A

22. Remarks

(1) NEW 126" I.D. X 24'-0" TAN/TAN VERTICAL VESSEL.
ITEM NO.: 123-C
P.O. NO.: 0010553687
BOTTOM OUTLET (D) & DRAIN (H) ARE B31.3 PIPING CODE, WELDED TO NOZZLE "D" STUB.
OPENING ATTACHMENTS PER UW-16.1
SAFETY RELIEF DEVICES LOCATED ELSEWHERE IN THE SYSTEM.

5.4-15 - Process nozzle connections and bridle piping nozzles shall be designed for local stresses due to external loads from the attached piping. The external loads shall be consistent with the loads provided by API 660, Table 2, unless otherwise stated by Purchaser.

5.4-16 - Local stresses on the vessel-to-nozzle connection imposed by external piping or attachments shall be evaluated. The use of WRC Bulletin 537 or Bulletin 297 shall be acceptable to meet this requirement, when applicable.

5.4-16.1 - Extrapolation of geometric parameter curves in WRC Bulletin 537 and Bulletin 297 shall not be permitted.

5.4-16.2 - If WRC Bulletin 537 and Bulletin 297 are not applicable or not desired, more rigorous analysis methods such as FEA shall be used with results evaluated per ASME Code, Division 2, Part 5.

Note: *WRC Bulletin 537 is an updated version of WRC Bulletin 107, which has been withdrawn.*

5.4-17 - Nozzle flanges, including ASME B16.5 and ASME B16.47, that fall within the ASME Code, Section VIII scope shall be evaluated for external loads.

5.4-18 - Nozzles for vessels identified as being in cyclic or hydrogen service shall be designed to accommodate external inspection by UT and MT.

5.4-19 - Manways shall have no internal projection.

Note*: If a vessel has internal ladder rungs above or below the manway as in PVM-GB-C99772, then finishing flush or modifications to the ladder rung design may be necessary in order to achieve required rung clearance per SID-SU-5106.*

5.4-20 - The following connections shall not have internal projections but shall be finished flush with the inside surface

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of the vessel:

5.4-20.1 - Vents and drains.

5.4-20.2 - Nozzles for two-phase flow.

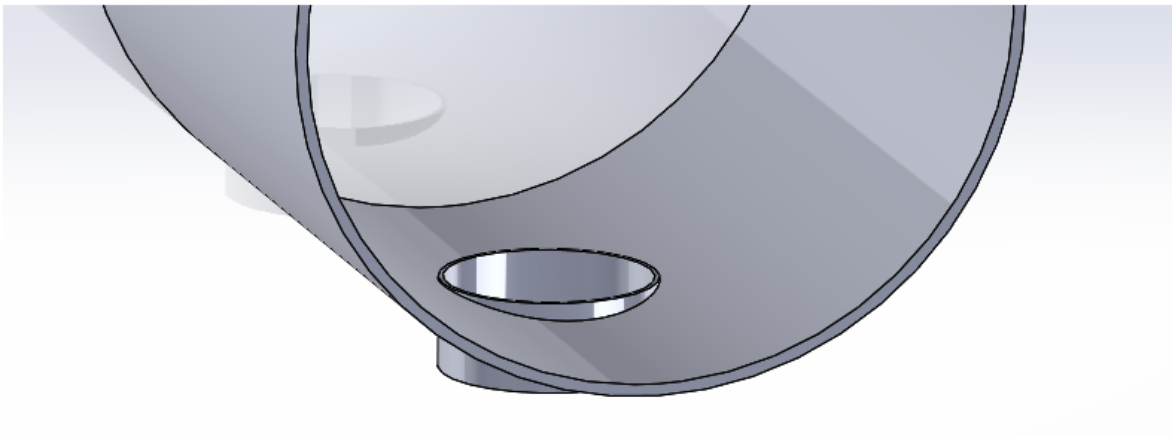
5.4-20.3 - Top vapor outlets.

5.4-20.4 - Bottom liquid outlets except when a vortex breaker compliant with PVM-GA-C99913 is used.

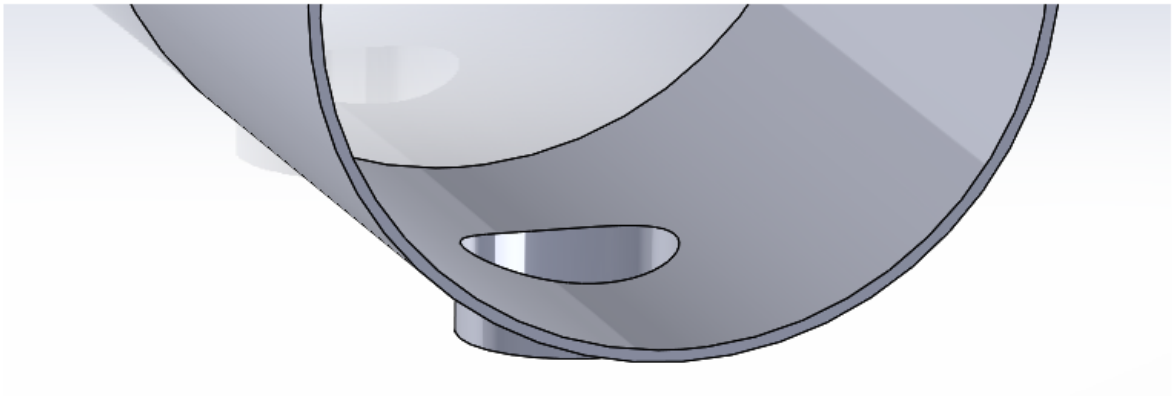
5.4-20.5 - Nozzles in the 6 o'clock position on horizontal vessels.

***Note**: There is a distinction between "no internal projection" and "finished flush." See Figure 3 and Figure 4.*

PVM-SU-4750, Figure 3: No Internal Projection



PVM-SU-4750, Figure 4: Finished Flush



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5.5 Attachments

5.5-1 - External attachments (such as saddles, pads, insulation support rings, stiffening rings, lifting lugs, platform and ladder clips, and nameplate brackets) shall be continuously welded to the vessel unless otherwise agreed to by Owner.

5.5-1.1 - Pads and saddles shall be circular, or shall have a corner radius of at least five times the pad thickness or 2 inches (50 mm), whichever is smaller.

5.5-1.2 - Vents shall be provided in all pads and similar attachments that create a void space.

5.5-1.2.1 - The vent shall be drilled and tapped NPT 1/4 (DN 8) prior to installation on the vessel.

5.5-1.2.2 - The vent shall be located in the lowest practical point of the pad or attachment.

5.5-1.2.3 - Split or two-piece reinforcement pads shall comply with Division 1, paragraph UG-37(h) of the ASME Code.

5.5-2 - Lifting lugs, tailing lugs, and lifting trunnions shall be analyzed for loads transmitted to the shell, heads, or support skirt during lifting.

5.5-2.1 - Trunnions shall be welded directly to the shell (set-on). Reinforcing pads shall be welded to the shell and trunnion.

5.5-2.2 - Trunnions shall have a 1/2-inch (12-mm) weep hole drilled into the bottom.

5.5-2.3 - Lifting lugs shall be seal welded to vessel or reinforcement pad on four sides to prevent corrosion under the lugs. A gap in the weld on the bottom side for drainage shall be acceptable provided that the remaining weld length is of adequate strength for the design.

5.5-3 - Fillet welds shall be at least 3/16 inch (5 mm) and shall be welded with multiple passes.

5.5-3.1 - Fillet welds shall have a smooth flat or slightly concave profile at the specified design throat thickness.

5.5-3.2 - Undercut shall not be permitted.

5.5-4 - When external or internal attachments cover welds in the vessel, the covered area of the weld shall be ground flush and 100-percent inspected by radiography or ultrasonic shear wave examination (UTSW) 2 inches (50 mm) beyond each side of the attachments.

5.5-5 - For internal attachments, efforts shall be made to minimize covering welds in the vessel.

5.5-6 - Internal attachments shall be continuously welded to the vessel, except that a 1-inch (25-mm) skip in the bottom of an internal pad or other attachment detail that creates a void space shall be made for a vent.

5.5-7 - A grounding lug shall be supplied in accordance with PIP VEFV1103.

5.5-8 - Insulation and fireproofing supports shall conform to PIP VEFV1122 and PIP VEFV1123.

5.5-9 - Railings, stairways, walkways, platforms, and ladders shall conform to SID-SU-5106.

5.5-10 - All materials welded directly to the pressure-retaining envelope shall be material of the same P number and

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group number in the applicable table in ASME Code, Section II, Part D.

5.5-11 - Galvanized clips or attachments shall be welded to the vessel only after galvanizing has been removed to a minimum of 1/2-inch (12-mm) beyond the extent of welding.

5.5-12 - Cantilever supports to be attached to vertical vessels shall meet the minimum requirements of CIV-GD-M99971.

5.6 Vessel Supports

5.6-1 - Localized shell stresses at support-to-shell locations shall be considered, as applicable, for wind load, earthquake, and all other loadings described in the ASME Code, Division 1, paragraph UG-22 or, for Division 2 vessels, in the User's Design Specification.

5.6-2 - Support steel shall be no thinner than the smaller of 1/4 inch (6 mm) thick including corrosion allowance (new) or the nominal thickness of the vessel component to which it is attached.

5.6-3 - Support welding shall use continuous seal-welded construction, except as allowed by applicable standard drawing details.

5.6-4 - Design of support skirts for vertical vessels shall conform to PVM-GD-C78876.

5.6-4.1 - Skirt openings shall be provided with rings or collars, sized to provide 100 percent area replacement of the opening.

5.6-4.2 - Provision shall be made for venting of the skirt. Location of the vent shall take into account any required insulation of the bottom head.

5.6-5 - For vertical vessels on legs, design of support legs shall conform to PVM-GB-C99945.

5.6-6 - Generally, leg supports should not be used under the following conditions:

5.6-6.1 - Operating temperature exceeds 450ºF (230ºC).

5.6-6.2 - Service is cyclic.

5.6-6.3 - Vessel height-to-diameter ratio exceeds 5, where height is measured from the base of the support to the top tangent line of the vessel.

5.6-7 - Design of support saddles for horizontal vessels shall conform to PVM-GF-C99694.

5.6-7.1 - Calculation of localized shell stress can be determined using the L.P. Zick method. This method is valid only for vessels with two saddles.

5.6-7.2 - The distance between head tangent line and the saddle centerline shall not be greater than 25 percent of the tangent-to-tangent length of the vessel.

5.6-8 - Horizontal vessels subject to partial or full vacuum and requiring stiffening rings shall have rings located over and integral with each saddle. The rings shall be designed and sized to accommodate the circumferential bending from

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the saddle reaction and the inward radial load due to external pressure.

5.6-9 - Vertical vessels directly supporting heavy equipment, such as thermosyphon reboilers, may require circumferential ring reinforcement to accommodate the radial thrust and weight from each reboiler on the shell. If the vessel is subject to any vacuum condition, the resulting vacuum radial load shall also be accommodated in the ring design.

5.7 Internals

5.7-1 - Dimensional changes or alterations to details shall be made only with prior written acceptance by Purchaser.

5.7-2 - Fabricator shall be responsible for clarification of any questionable points when preparing detailed fabrication drawings.

5.7-3 - Fabricator shall be solely responsible for the effect of the internals upon the pressure-containing components of the vessel.

5.7-4 - Unless specified otherwise, for carbon steel internal components and supports welded to the shell or not easily removable, the corrosion allowance per surface shall be 75 percent of the single side internal design corrosion allowance for the shell as stated in the data sheet.

5.7-5 - Unless specified otherwise for bolted and easily removable carbon steel internal components and supports, the corrosion allowance per surface shall be 50 percent of the single side internal design corrosion allowance for the shell as stated in the data sheet.

5.7-6 - Unless otherwise stated, Supplier shall provide all internals in accordance with the following requirements:

5.7-6.1 - Internals furnished by others shall be installed by the fabricator.

5.7-6.2 - Baffles, partitions, and other parts shall be designed to accommodate differential thermal expansion between vessel shell and internal parts.

5.7-6.3 - All components subject to differential pressure shall be designed to accommodate such loads.

5.7-6.4 - Baffles, partitions, and other parts shall be designed to avoid trapping water or flammable liquids that cannot be drained or purged.

5.7-6.5 - Internal ladders shall comply with PVM-GB-C99772.

5.7-6.6 - Internal non-pressure-containing parts shall be designed using appropriate stress for material and temperature as specified in the ASME Code, Section II, Part D.

Note: *Designs for internals are shown on Purchaser drawings in a manner intended to permit the fabricator to prepare detailed shop fabrication drawings that comply with Purchaser's design with regard to process requirements and corrosion allowance.*

Note: *This section does not apply to heat exchanger internals such as tubesheets, tubes, baffles, and partition plates. These components are covered separately in EXH-SU-2583.*

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5.8 Lifting Requirements

5.8-1 - A minimum impact factor of 1.5 shall be applied to the lift weight for designing lifting devices. Design shall accommodate a sling angle of 30 degrees from vertical.

5.8-2 - Lift weight shall include complete fabricated weight of the vessel plus all components to be included in the lift (e.g., trays, ladders/platforms, insulation, additional piping with insulation, lifting slings, shackles).

5.8-3 - Lifting shall be done when wind speed is below 25 mph (40 km/h or 11 m/s). This wind load shall be included in the consideration of the lift.

5.8-4 - Stresses shall be checked for the lifting condition.

5.8-4.1 - Calculated combined stress excluding peak stresses shall not exceed 80 percent of yield strength at ambient temperature.

5.8-4.2 - Compressive stress shall not exceed 1.2 times the B factor for the ASME Code.

5.9 Wind Requirements for Flexible Columns or Tall Vessels

5.9-1 - Vibration analysis for wind-induced vortex-excited resonance shall be performed on the following:

5.9-1.1 - Vertical vessels with $5 \leq h/d \leq 15$ and natural frequency of vessel less than 2 Hz.

5.9-1.2 - All vertical vessels with $h/d > 15$, irrespective of natural frequency.

5.9-2 - The effective diameter of the vessel shall be used in wind load calculations.

5.9-3 - Critical wind velocity of the vessel shall be greater than 3 times the maximum mean hourly wind speed at the top of the vessel, determined in accordance with ASCE 7-16. (See Section 16 for offshore/topsides vessels.)

Note: *For details and corrective actions to prevent excessive vibrations, refer to ASME STS-1.*

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5.10 Details for Clad or Lined Pressure Vessels

5.10-1 - For pressure vessels that are to be clad or weld overlayed, fabrication details shall comply with PVM-SU-1322. Strip lining shall not be used.

5.10-2 - For pressure vessels that are to be internally lined with a non-metallic lining or coating, fabrication details shall comply with PIP VESV1003.

5.10-3 - Coating systems shall comply with COM-SU-5191.

6 Materials

6.1 Pressure-Containing Parts

6.1-1 - When specified by Purchaser, materials shall be furnished to the specification and grade shown on the vessel drawing, data sheet, or list sheets.

6.1-2 - Fabricator may propose alternate materials in a separate quotation. Any alternate materials shall comply with requirements of this specification.

6.1-3 - All plate, forging, and pipe grades of material shall meet the requirements of Division 1, paragraphs UCS-66 and 67, or Division 2, paragraph 3.11 of the ASME Code for the MDMT shown on the vessel drawing or data sheet.

6.1-4 - For vessels with a shell or head thickness greater than 4 inches (100 mm) or that use high-strength steels (see Section 15), the tensile and Charpy V-notch impact tests required for certification of all plate, forging, and pipe grades of material shall represent the vessel in the completed condition and thermal history. This requirement also applies to tubesheets greater than 4 inches (100 mm) or constructed of high strength materials, that will be subjected to PWHT.

6.1-4.1 - For Division 2 vessels, compliance with Division 2, paragraph 3.11 of the ASME Code shall be verified by subjecting specimens for tensile and Charpy V-notch impact tests to the thermal history required for fabrication of the vessel.

6.1-4.2 - Thermal history shall include heat treatments (hot forming, heat treatment after hot forming, PWHT) and at least two additional PWHT cycles to allow for shop or field repairs.

6.1-4.3 - One extended heat treatment cycle of equivalent hold (soak) time (instead of three consecutive cycles) shall be acceptable for the PWHT portion of the thermal history.

Note: *Compliance with these tensile and impact requirements can be met by including additional testing of base metals in the weld procedure qualification (Section 7.3) or vessel production test plates (Section 7.8) requirements of this specification, with authorization of Owner.*

6.1-5 - Castings shall be prohibited, unless accepted in writing by Owner.

6.1-6 - When normalized materials are used, the normalizing temperature shall not exceed 1750°F (955°C). Normalizing temperature shall be shown on material test reports.

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6.1-7 - Hot-forming temperature for normalized material shall not exceed the normalizing temperature unless the material is renormalized after hot forming. Hot-formed components shall be identified and hot-forming temperature shall be provided on certified documentation by the company performing the hot forming.

6.1-8 - For materials with specified minimum yield strength of 40 ksi (275 MPa) or greater, supplemental requirements in Section 15 shall apply.

6.1-9 - Materials for reinforcing pads (where allowed) for openings and attachments shall be the same (specification, grade, class, etc.) as those of the shell or head to which the pad is to be attached.

6.1-10 - For vessels greater than 4 inches (100 mm) thick (see item 6.1-4) or for vessels that use high-strength steels (see Section 15, fabricator shall prepare a flowchart or list showing the sequence of forming, welding, heat treating, and inspection operations for review and acceptance by Purchaser prior to the start of production.

6.1-11 - For vessels in hydrogen service, material selection shall be based on a margin of 50 psi (3.4 bar) (hydrogen partial pressure) and 50°F (30°C) below the Nelson curve for the particular steel.

6.2 Non-Pressure-Containing Parts

6.2-1 - Skirts and other vessel supports shall be suitable, per paragraph UCS-66 of Division 1, or Section 3.11 of Division 2 of the ASME Code, for the MDMT for the vessel.

6.2-2 - Except for the nameplate, all materials welded to any part of the vessel shall be ASME Code materials.

6.2-3 - Clips on low-temperature pressure vessels (cryogenic) shall be designed with extended insulation to prevent icing.

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6.3 Bolting

6.3-1 - Studs conforming to SA-193, Grade B7, and nuts conforming to SA-194, Grade 2 or 2H shall be used for external bolting on vessels warmer than −55°F (−48°C) when using stud diameters up to 2-1/2-inch (64-mm) and warmer than −40°F (−40°C) when using stud diameters greater than 2-1/2-inch (64-mm).

6.3-2 - For studs in blind holes, either the end of the stud or the bottom of the hole, shall be relieved to prevent galling. The approach described in ASME PCC-2, Article 303, Figure 303-I-1, except that the 'C' dimension does not need to exceed 0.25-inch (6.4-mm). Minimum thread depth shall be determined by the ASME Code, paragraph UG-43.

7 Welding

7.1 General

7.1-1 - All welding, qualification of welding procedures, and qualification of welders and welding operators shall be performed in accordance with the ASME Code, Section IX, except as restricted herein.

7.1-2 - Vessel fabricator shall be responsible for the following, with referenced documents made available for use by welders or welding operators and for reference by the ASME Code-authorized inspector and Purchaser:

7.1-2.1 - Selecting and preparing the welding procedure specification (WPS) for each type of joint, material, and thickness.

7.1-2.2 - Preparing and providing a procedure qualification record (PQR) documenting proof of weldability of the variables described in the WPS.

7.1-2.3 - Preparing and providing a welder performance qualification (WPQ) documenting proof of the welder's ability to deposit sound weld metal within the range of WPS parameters being used.

7.1-2.4 - Submitting weld procedures to Purchaser for acceptance prior to beginning fabrication of the vessel.

7.1-3 - Structural welding involving rolled shapes and plates shall, as a minimum, conform to the requirements of general welding specifications of AWS D1.1/D1.1M.

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7.2 Welding Procedure Specifications

7.2-1 - The welding procedure shall clearly identify the essential variables, materials, specifications, and consumables by name and specification, as well as the standard ASME Code requirements.

7.2-2 - The application for each WPS shall be shown on a separate weld map or welding and testing plan, which shall clearly indicate the WPS for each main type of joint and for attachments. The map or plan shall be submitted prior to the start of welding.

7.3 Welding Procedure Qualification

7.3-1 - The weld procedure shall be qualified for all welding to be used for fabrication of the vessel, including repair welding.

7.3-2 - When Charpy V-notch impact tests are required, the size of consumable and heat inputs used shall be considered supplementary essential variables.

7.3-2.1 - Stringer bead welding technique shall be preferred; however, weave beads shall be allowed for out-of-position welding if the maximum width of weave does not exceed three times the electrode core wire diameter.

7.3-2.2 - The heat input specified on the WPS shall not exceed the highest qualified value on the relevant PQR.

7.3-3 - Welding procedures shall be qualified with PWHT as applicable for each vessel or vessel part.

7.3-4 - The heat treatment total time at (soak) temperature shall be three times the ASME Code minimum required for the thickness in order to qualify the weld procedure for repair welding and possible further heat treatments.

7.4 Acceptable Methods

7.4-1 - Acceptable methods of welding shall be the following:

7.4-1.1 - Shielded metal arc welding (SMAW).

7.4-1.2 - Submerged arc welding (SAW).

7.4-1.3 - Gas metal arc welding (GMAW).

7.4-1.4 - GTAW.

7.4-1.5 - Flux core arc welding – gas shielded (FCAW-G).

7.4-2 - Other welding processes may be used only with Owner's written acceptance.

7.4-3 - The interrupted-arc (short-circuit transfer) GMAW process may be used only for the following applications:

7.4-3.1 - Root passes on circumferential, longitudinal, or nozzle-to-shell welds if back-gouged and back-welded.

7.4-3.2 - Root passes on circumferential piping welds for fabricated nozzles or internal piping.

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7.4-3.3 - Root passes on non-pressure-containing vessel internals.

7.4-4 - Joints welded from two sides shall be ground, back-chipped, or back-gouged to sound metal before continuing welding on the back side of the joint.

7.4-5 - Before welding or hot cutting, weld surfaces and heat-affected zones (HAZ) shall be clean and free of lead, sulfur, dirt, grease, and any other deleterious material.

7.5 Selection of Consumables

7.5-1 - Selection of consumables shall be the responsibility of the fabricator.

7.5-2 - Consumables shall be selected to meet the same strength and toughness requirements as the base metal.

7.5-3 - Weld procedures shall be submitted to Purchaser for acceptance prior to beginning of vessel fabrication.

7.5-4 - Dissimilar metal welds shall be prohibited for Category A, B, C, and D joints unless accepted in writing by Owner.

7.5-5 - SMAW consumables shall be a low-hydrogen type conforming to the ASME Code, Section II, Part C, SFA 5.1.

7.5-6 - All consumables shall be of a low-hydrogen type that is certified by the consumable Supplier, or tested by the fabricator, to result in less than 8 ml of diffusible hydrogen per 100 gm (H-8 designation) of deposited weld metal.

7.5-6.1 - Diffusible hydrogen content shall be measured by the mercury displacement or gas chromatograph methods per AWS A4.3.

7.5-6.2 - Vessel fabricator shall provide written evidence of compliance with this paragraph with welding procedure submittal.

7.5-6.3 - A specification sheet for the proposed brand of filler metal from the consumable Supplier shall be sufficient.

7.5-6.4 - Consumables not of the low hydrogen type may be accepted by the Owner in writing.

7.5-7 - Consumables shall be stored, handled, and used in accordance with the electrode manufacturer’s instructions.

7.5-7.1 - Submerged arc flux shall be kept and stored in a low-humidity (60 percent maximum), warm (minimum 70°F [20°C]), controlled storage area.

7.5-7.2 - Flux shall be dried per flux manufacturer requirements prior to being used on the shop floor.

7.5-7.3 - Flux may be recovered and recycled only twice before it shall be discarded.

7.5-8 - For vessels requiring impact testing, the same brand and manufacturer source consumables used for procedure qualification shall be used for production welding. Any change or variation shall require re-qualification.

7.5-9 - For vessels designed for severe low-temperature applications (below −20°F [−30°C]) or where high-impact strength is required (greater than 35 ft-lb [47 Joules]), the following shall apply:

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7.5-9.1 - All batches of consumables shall be proven suitable by impact testing and certified to be in accordance with the specification before use on production welding.

7.5-9.2 - Batch impact testing may be performed by the consumable Supplier.

7.5-9.3 - Batch impact testing of SAW consumables shall be done by combining the heat number for the electrode and lot number for the flux, and performing the required mechanical tests per the appropriate AWS designation.

7.6 Tensile Strength

7.6-1 - Tensile strength of the weld metal shall be limited to 100 ksi (690 MPa) maximum.

7.6-2 - Tensile strength shall be verified by a hardness test of the weld metal in the weld procedure qualification and shown on the PQR and a hardness test of each Category A and B weld seam.

7.7 Hardness

7.7-1 - Weld metal hardness shall not exceed Brinell hardness number (BHN) 200 for carbon steel.

7.7-2 - Hardness testing shall be performed using Telebrineller™, Eitzen™, or Owner-accepted equal.

7.7-2.1 - Use of Equotip® for hardness testing shall not be permitted.

7.7-2.2 - Other portable hardness test instruments shall be subject to Owner acceptance.

7.7-3 - One hardness test shall be performed on each weld or each 10 linear feet (3 m) of weld, whichever is more frequent.

7.8 Vessel Production Test Plates

7.8-1 - When Charpy V-notch impact testing is required by item 6.1-4, one test plate shall be made for each vessel and for each welding procedure used for Category A and B joints.

7.8-2 - Production (vessel) test plates shall be subjected to the same thermal history expected for the finished vessel, including hot forming, preheat, and PWHT.

7.8-2.1 - Production test plates shall be taken from hot-formed components or shall require a separate thermal cycle to mimic hot-forming thermal history.

7.8-2.2 - Production test plates shall be heat treated with the vessel unless accepted otherwise by Owner.

7.8-3 - When vessel thickness is 2 inches (50 mm) or greater, Charpy V-notch impact tests shall be made for the base metals, weld metals, and HAZs when required by item 6.1-4.

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7.9 Welding After Post Weld Heat Treatment or Tests

7.9-1 - No welding shall be performed on the vessel after PWHT or hydrostatic test.

8 Heat Treatment

8.1 Preheat

8.1-1 - Minimum required preheat shall be as follows:

8.1-1.1 - 100°F (38°C) for carbon steels up to and including 1-1/4 inches (32 mm) thick.

8.1-1.2 - 200°F (90°C) for carbon steels greater than 1-1/4 inches (32 mm) thick.

8.1-1.3 - Interpass temperature, measured inside the weld groove, shall be limited to 500°F (260°C) maximum.

8.1-2 - Preheat shall extend for a minimum distance equal to the material thickness but not less than 3 inches (75 mm) on each side of the weld.

8.2 Post Weld Heat Treatment

8.2-1 - PWHT shall be in accordance with the ASME Code, supplements to this specification, or as specified by Purchaser on the vessel drawing or data sheet for the service conditions.

8.2-2 - The process of heating for PWHT shall be at fabricator's option, except:

8.2-2.1 - Direct flame heating shall not be permitted.

8.2-2.2 - Exothermic methods shall not be permitted.

8.2-2.3 - Vessel shall be PWHT as a whole unless it is not feasible.

8.2-3 - When required, PWHT shall be performed in the temperature range of 1100°F–1200°F (595°C–620°C) and meet or exceed the minimum time required by the ASME Code, Division 1, Table UCS-56, or Division 2, Table 6.8, as applicable, but with a minimum hold time of 1 hour.

8.2-4 - The use of the ASME Code, Division 1, Table UCS-56.1, or Division 2, Table 6.16 for applying PWHT at a lower temperature for an increased length of time shall not be acceptable.

8.2-5 - A detailed procedure for PWHT shall be required for Purchaser review and acceptance, and shall include a vessel sketch of the following:

8.2-5.1 - Locations and types of thermocouples.

8.2-5.2 - Method of thermocouple attachment.

8.2-5.3 - Insulation details for local PWHT.

8.2-5.4 - Methods used to support the vessel or part thereof.

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8.2-5.5 - Bracing to be used to prevent distortion.

8.2-5.6 - Heating control system details.

8.2-6 - A sufficient number of thermocouples in contact with both the internal and external surfaces of the vessel shall be used to continuously and automatically record the PWHT temperature on a chart from the start of controlled heating until the end of the controlled cooldown.

8.2-6.1 - The thermocouples shall be insulated from the heat source.

8.2-6.2 - Recorded temperature data shall be made available to Purchaser's inspector to verify that the required temperature and holding time have been obtained.

8.2-7 - Details of any methods used to prevent scaling on machined surfaces shall be included.

8.2-8 - During PWHT, vessels shall be blocked and supported as necessary to avoid deformation or damage.

8.2-9 - Lead- or sulfur-containing substances such as oil, grease, thread lubricants, and paint shall be removed from vessel surfaces prior to heat treatment.

8.2-10 - Any internal parts installed in the vessel prior to heat treatment shall be assessed to ensure that their corrosion or material properties are not affected, and that their thermal expansion will not negatively affect the free thermal expansion of the vessel.

8.2-11 - The following shall be done after heat treatment:

8.2-11.1 - Parts shall be realigned.

8.2-11.2 - Warpage shall be corrected.

8.2-11.3 - Bolts shall be re-tightened to comply with the vessel drawing.

9 Examination

9-1 - Quality assurance and quality control throughout procurement and fabrication shall be performed in accordance with QAM-SU-6053 and QAM-SU-6163.

9.1 Dimensional Tolerances

9.1-1 - Tolerances for vessel dimensions, nozzle and manway locations and orientations, and other design details shall be in accordance with IOGP S-619 Annex E.

Note: *See Section 17.12 for field-erected carbon steel storage spheres.*

9.1-2 - Vessel fabricator shall calculate maximum permissible deviation from a perfect circle for all shell sections and skirt (if one is used), per the ASME Code, Division 1, UG-80(b) or Division 2, Paragraph 4.4.4. These allowable deviations shall be clearly shown on vessel drawings.

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9.2 Minimum Examination Requirements Common to All Weld Joints

9.2-1 - Weld bevels with a shell component thickness of 2 inches (50 mm) or greater shall receive MT prior to welding.

9.2-1.1 - All indications of linear flaws shall be removed by grinding or chipping, and the area shall be re-examined to ensure satisfactory removal of the flaw.

9.2-1.2 - Repairs shall be made with a welding procedure qualified in accordance with Section 7.3.

9.2-2 - Category A, B, C, and D butt-welded root areas shall receive MT of the weld preparation following back-gouging and prior to back welding. This examination shall be randomly witnessed by a Purchaser inspector for Category C and D joints.

9.2-3 - When specified in the data sheet, Category A, B, C, and D weld joints in shell components with a thickness of 3 inches (75 mm) or greater shall receive full coverage UT after final heat treatment.

9.3 Minimum Examination Requirements for Specific Joint Categories

9.3-1 - As a minimum, Category A and B butt welds shall be spot radiographed in accordance with ASME Code.

9.3-2 - Vessels with thicknesses greater than 1-1/2 inches (38 mm) shall have 100 percent inspection of Category A and B welds.

9.3-2.1 - Inspection shall be either UT or radiographic examination (RT).

9.3-2.2 - If UT is used, it shall be considered ‘in lieu of RT’ per item 9.10-2.

9.3-2.3 - The additional examination requirements in Section 9.4 shall apply.

9.3-3 - Purchaser shall be given three working days’ notice before spot radiographs are made so that Purchaser’s inspector can designate the locations at which the spot radiographs will be taken.

9.4 Additional Examination Requirements for Specific Joint Categories When Full Inspection Is Specified by Purchaser

9.4-1 - When specified by Purchaser in the data sheet, full inspection by either UT or RT shall be required.

9.4-2 - If UT is used, it shall be considered ‘in lieu of RT’ per item 9.10-2.

9.4-3 - When full UT or RT is required by the Purchaser, the following additional requirements shall apply. These requirements shall not apply when the fabricator elects to perform full inspection.

9.4-3.1 - Category B welds, including those on nozzles, shall receive 100 percent RT or UT.

9.4-3.2 - Category C and D welds shall receive the following inspections:

9.4-3.2.1 - NDE shall be performed per ASME Code, Division 2, Table 7.2, for the specific weld configuration and joint category at a 1.0 joint efficiency.

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9.4-3.2.2 - The NDE extent shall be 100 percent if a lower value is allowed per ASME Code, Division 2, Table 7.2.

9.4-3.2.3 - Unless otherwise specified, MT in item 9.4-3.2.1 shall be randomly witnessed by an Owner inspector or representative.

9.4-3.2.4 - UT shall be witnessed as per item 9.10-3.

9.5 Examination Requirements for Structural Attachments and Miscellaneous Joints

9.5-1 - Skirt-to-vessel welds shall receive UT unless another type of examination is accepted by Owner.

9.5-1.1 - Where possible, skirt-to-vessel welds shall be designed to permit UT.

9.5-1.2 - If impossible or impractical to design so that UT can be applied, skirt-to-vessel welds shall receive MT after the root pass is completed and every third layer thereafter.

9.5-2 - Internal and external attachment welds, including those for lifting attachments when supplied, shall receive MT after grinding, if grinding is required. If vessel is subject to heat treatment, MT shall be performed after heat treatment.

9.5-3 - Surfaces from which temporary attachments have been removed shall receive MT after grinding.

9.5-4 - Weld pad buildups for attachment of internal or external components and supports shall receive UT.

9.6 Visual Examination

9.6-1 - Purchaser's inspector may be present during fabrication of the vessel, but the fabricator shall provide shop inspection to verify quality of workmanship and that the work is being performed in accordance with this specification and the ASME Code.

9.6-2 - Any evidence of poor workmanship or damage due to mishandling shall be repaired at the fabricator's expense before the vessel is accepted.

9.7 Magnetic Particle Examination

9.7-1 - Magnetic particle examination (MT) shall comply with ASME Code, Division 2, paragraph 7.5.6.1.

9.7-2 - Acceptance criteria shall comply with ASME Code, Division 2, paragraph 7.5.6.2.

9.7-3 - Prod burns on vessels that receive a PWHT shall be removed by grinding.

9.7-4 - Wet fluorescent magnetic particle examination (WFMT) shall be used for all accessible internal welds on vessels 40 inches (1000 mm) ID and larger, except as noted in item 6.

9.7-5 - MT for external welds shall be of a wet visible type.

9.7-6 - Dry powder MT techniques may be used in the following situations:

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9.7-6.1 - On welds that receive a preheat in excess of 125°F (52°C).

9.7-6.2 - On internal welds that will also receive 100 percent Eddy current surface examination (ET).

9.7-6.3 - For in-process welding checks, such as after back-gouging and weld prep.

9.7-7 - MT procedures shall be subject to Owner review and acceptance.

9.8 Eddy Current Surface Examination

9.8-1 - Eddy current surface examination (ET), when performed, shall be per and acceptance criteria shall comply with the ASME Code, Division 2, paragraph 7.5.8.

9.8-2 - ET procedures shall be subject to Owner review and acceptance.

9.8-3 - Personnel performing and evaluating ET shall be qualified by the Owner's personal demonstration qualification process.

9.9 Radiographic Examination

9.9-1 - Radiographic examination (RT) shall be performed per the ASME Code, Division 2, paragraph 7.5.3.

9.9-2 - Acceptance criteria shall be per the ASME Code, Division 2, paragraph 7.5.3.2.

9.9-3 - Radiographs shall be retained by the fabricator for at least one year after the date of shipment, and shall be made available for review upon request by Owner.

9.9-4 - RT shall be performed after heat treatment, if applicable.

9.10 Ultrasonic Examination

9.10-1 - Ultrasonic examination (UT) shall be made in accordance with ASME Code, Division 2, paragraph 7.5.4.1, and the acceptance criteria of paragraph 7.5.4.2 shall be applied unless otherwise required in this specification.

9.10-2 - UT on Category A and B joints shall be considered "in lieu of RT" per the ASME Code, Division 1, UW-51(a)(4) or Division 2, paragraph 7.5.5 and shall use a recordable UT method.

9.10-3 - UT of welds shall be witnessed by Purchaser's inspector.

9.10-4 - UT methodology shall require prior acceptance by Owner of detailed UT procedures including weld and nozzle configuration, mode of the UT wave, reflection angle, scanning position, and calibration blocks.

9.10-5 - The selected UT method shall be one that creates a permanent record such as phase array ultrasonic testing (PAUT), time of flight diffraction (ToFD), or other equivalent method accepted by the Purchaser.

9.10-6 - UT shall be performed after PWHT, if applicable.

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10 Hydrostatic Testing

10.1 Preparation

10.1-1 - Fabricator shall furnish facilities and test materials, including blinds, bolts, nuts, and gaskets.

10.1-2 - Vessel shall be cleaned internally and externally of dirt, debris, welding slag, and weld spatter before the hydrostatic test is performed.

10.1-3 - No paint or primer shall be applied to the vessel prior to hydrostatic testing.

10.1-4 - Purchaser shall be notified a minimum of 3 working days in advance so that the hydrostatic test can be witnessed.

10.1-5 - Reinforcing pads shall be pneumatically tested per the ASME Code, Section V, Article 10, Appendix I, with 15 psig (1 bar gauge) dry air or nitrogen and a bubble-forming solution.

10.2 Required Stress Levels

10.2-1 - Hydrostatic test pressure for ASME Code, Division 1 vessels shall be based on the MAP (new and cold) as described in UG-99(c) of the ASME Code.

10.2-2 - Hydrostatic test pressure for Division 2 vessels shall be determined in accordance with ASME Code, Division 2, paragraph 8.2.1(e) of the ASME Code.

10.2-3 - Test pressures as determined by item 10.2-1 or item 10.2-2 shall be reduced as necessary to comply with the limits of, Division 2, paragraph 4.1.6.2 of the ASME Code.

10.2-3.1 - Consideration may be given to tolerance in hydrotest pump and gauge setups.

10.2-3.2 - Reductions in hydrotest pressure to reduce induced stresses by greater than 0.03 of the factor multiplied against yield stress (S_y) and primary membrane stress (P_m) shall be reviewed and accepted by Owner.

10.3 Other Requirements

10.3-1 - Hydrostatic testing shall be performed after all welding and required heat treatment is completed.

10.3-2 - Hydrotest shall be 1 hour minimum unless stated otherwise in the purchase order.

10.3-3 - Any suitable sheet gaskets may be used on ASME raised-face flanges for the hydrostatic test, with the following exceptions:

10.3-3.1 - Elastomeric gaskets shall not be used.

10.3-3.2 - ASME raised-face openings with hinged covers shall require service-type gaskets.

10.3-4 - All other gasketed joints shall require service-type gaskets.

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10.3-5 - After hydrostatic testing, the vessel shall be promptly and carefully drained and completely dried, especially around trays and internals.

10.3-6 - Vessel shall be internally inspected to ensure that no residual moisture or foreign matter remains.

11 Painting and Coating

11.1 Surface Preparation

11.1-1 - Exterior surfaces that require painting shall be prepared in accordance with the coating system as specified in the vessel data sheet, PVM-DS-4750-US (SI).

11.2 Priming

11.2-1 - When priming is specified on the vessel drawing, it shall be done after hydrostatic testing and in accordance with the coating system as specified in the vessel data sheet, PVM-DS-4750-US (SI).

11.2-2 - Coating systems shall comply with COM-SU-5191.

11.3 Fireproofing

11.3-1 - Fireproofing shall comply with FPM-DU-5091.

11.3-2 - Intumescent (epoxy) fireproofing coatings shall be strongly preferred over cementitious fireproofing.

12 Nameplate

12-1 - Each vessel shall be identified by a bracket-mounted austenitic stainless steel nameplate.

12-2 - Bracket shall be continuously welded to the vessel prior to heat treatment, and the nameplate continuously welded to the bracket after heat treatment.

12-3 - Bracket shall be mounted so that the nameplate is not inadvertently covered by insulation.

12-4 - In addition to the stamping required by the ASME Code, the nameplate shall show the purchase order number and item number, and MAEP (at design temperature and steamout temperature if applicable) per item 5.1.3-8.

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13 Preparation for Shipment

13.1 Flanges

13.1-1 - Flange faces shall be protected against corrosion and damage by applying stiff grease and by bolting on wooden or steel covers with a minimum of four bolts.

13.1-2 - A plastic sheet shall be inserted between a wooden cover and the greased flange surface.

13.1-3 - Nozzles fitted with blind flanges (including manways and handholes) shall be bolted with specified studs, nuts, and gaskets in place, ready for service. Nuts shall be torqued per PIM-SU-5209.

13.1-4 - Ring gaskets shall be coated with rust preventive grease before installation. Using tape to hold ring gaskets in place during assembly shall not be acceptable.

13.2 Threaded Connections

13.2-1 - Female threaded connections where allowed in item 5.4-1 shall be fitted with solid round steel bar stock plugs of the same material as the connection.

13.2-2 - Plugs shall have a length of shank equal to at least twice the diameter and shall not be free-machining steels.

13.2-3 - Plugs other than carbon steel shall be marked by stamping to designate the material used.

13.3 Vent Holes

13.3-1 - Vent holes and skips in welds provided for venting in reinforcing pads, saddles, etc., shall be plugged with stiff grease.

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13.4 Thread Lubricant

13.4-1 - Bolting shall be lubricated with a recognized brand of high-temperature molybdenum disulfide (MoS_2)-based thread lubricant per PIM-SU-5209.

13.5 Security

13.5-1 - Fabricator shall load and block the vessel in a manner to prevent damage during shipment, including damage from exposure to marine environment during ocean shipping.

13.6 Atmosphere

13.6-1 - Fabricator shall use dry air and desiccant for maintaining a non-corrosive environment in the vessel during shipment.

13.6-2 - Fabricator shall use stainless steel wire to hang stainless steel tags on the vessel manways or handholes to indicate that they contain desiccant.

13.6-3 - The use of nitrogen or other means of creating an oxygen-deficient atmosphere shall not be used without prior Owner acceptance.

13.7 Labeling

13.7-1 - Vessel tag number, purchase order number, and fabricator's name shall be written on the skirt or near a saddle.

13.7-2 - If vessel is heat treated, vessel shall be labeled "Vessel Is Heat Treated – Do Not Weld."

13.7-3 - Character size shall be as follows:

13.7-3.1 - For vessels with ID greater than 40 inches (1000 mm), characters shall be at least 4 inches (100 mm) high.

13.7-3.2 - For vessels with ID greater than 40 inches (1000 mm), characters shall be at least 4 inches (100 mm) high.

13.7-3.3 - Markings other than those required in this section shall have 1 inch (25 mm) minimum high characters regardless of diameter.

13.7-4 - Text shall be stenciled or neatly written in characters at least 4 inches (100 mm) high using coating requirements described in Section 11. A contrasting color of paint shall be used for the text.

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14 Supplemental Requirements for Specific Service Conditions

14-1 - The following supplemental requirements shall apply if so checked in the data sheet.

14.1 Supplemental Requirement A (Stress Corrosion Cracking Services)

14.1-1 - All welds and cold formed parts shall be thermally heat treated at 1150°F (620°C) to 1200°F (650°C) in accordance with Table UCS-56 of the ASME Code, Division 1 but for 1 hour minimum.

14.1-2 - All internal attachment welds shall be a minimum of two pass welds with filler metal added.

14.2 Supplemental Requirement B (Downstream Mild Sour Service)

14.2-1 - All vessels shall meet the materials and manufacturing requirements of NACE MR0103 or NACE MR0175, as appropriate.

14.2-2 - All welds and cold formed parts shall be thermally heat treated at 1150°F (620°C) to 1200°F (650°C) in accordance with Table UCS-56 of the ASME Code, Division 1 but for 1 hour minimum.

14.2-3 - All internal attachment welds shall be a minimum of two pass welds with filler metal added.

14.2-4 - Following heat treatment, internal welds shall be inspected by 100 percent ET for at least 1 inch (25 mm) on each side of the toe of the weld, according to ASME Code, Section V, Article 8.

14.2-5 - If qualified ET personnel are not readily available, then alternatively and with Owner approval, 100 percent WFMT in accordance with ASME Code, Section V, Article 7shall be performed for at least 1 inch (25 mm) on each side of the toe of the weld.

14.2-6 - Reinforcing pads shall not be permitted on vessel nozzles; nozzles shall be integrally reinforced in accordance with PVM-GF-C14311.

14.2-7 - Interior surfaces that require coatings to protect against wet H_2S cracking shall be applied in accordance with COM-SU-5014 using the product as specified by the Purchaser in the vessel data sheet.

14.3 Supplemental Requirement C (Downstream Severe Sour Service and All Upstream Sour Service)

14.3-1 - Steel plates, fabrication, and NDE requirements in this section shall be used for sour services where clad construction is not required and where NACE MR0175 or NACE MR0103 is specified in the vessel data sheet.

14.3-2 - All vessels shall meet the materials and manufacturing requirements of NACE MR0103 or NACE MR0175 as appropriate.

14.3-3 - Reinforcing pads shall not be permitted on vessel nozzles; nozzles shall be integrally reinforced.

14.3-4 - Steel plates used for sour services shall be SA-516 Grade 60, 65, or 70, with the following additional

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requirements:

14.3-4.1 - Plate shall conform to the following chemistry limits:

14.3-4.1.1 - CEq ≤ 0.42 [CEq = C + Mn / 6 + (Cu + Ni) / 15 + (Cr + Mo + V) / 5]

14.3-4.1.2 - S ≤ 0.003 wt %

14.3-4.1.3 - P ≤ 0.010 wt %

14.3-4.1.4 - Nb(Cb) + V ≤ 0.020 wt %

14.3-4.2 - The steel shall be in the normalized heat treatment condition.

14.3-4.3 - One hundred percent of the surface of each plate shall be examined ultrasonically per ASTM A578/A578M with supplemental requirement S1 and acceptance level C.

14.3-5 - Forgings, flanges, fittings, and seamless piping shall meet the following chemistry requirements in addition to the requirements of the respective material specification:

14.3-5.1 - CEq ≤ 0.42 [CEq = C + Mn / 6 + (Cu + Ni) / 15 + (Cr + Mo + V) / 5]

14.3-5.2 - C ≤ 0.23%

14.3-5.3 - S ≤ 0.01%

14.3-5.4 - P ≤ 0.02%

14.3-5.5 - V ≤ 0.02%

14.3-5.6 - Nb ≤ 0.005%

14.3-6 - For upstream services, forgings and flanges shall be SA-350 LF2, seamless pipe shall be SA-333 Grade 6, and fittings shall be SA-420 Grade WPL6.

14.3-7 - Weld procedure qualification shall be as follows:

14.3-7.1 - A supplemental PQR tests shall be performed on the heat of steel with the highest carbon equivalent in each lot or mill run.

14.3-7.2 - A supplemental PQR shall be made with the lowest weld heat input to be used in the vessel construction, including fillet attachments.

14.3-7.3 - For vessels manufactured in accordance with NACE MR0175, hardness testing of PQRs shall be performed in accordance with those requirements and included as part of the PQR documentation.

14.3-7.4 - For vessels manufactured in accordance with NACE MR0103, welding procedure hardness tests shall be performed in accordance with that document and the following requirements:

14.3-7.4.1 - Each weld procedure test shall include two micro-hardness traverses of the weld and weld-HAZs made on a polished cross-section of the weldment.

14.3-7.4.2 - Each micro-hardness traverse shall be located 0.06 inch (1.5 mm) below the surface of the weld toe.

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14.3-7.4.3 - No micro-hardness reading shall exceed HV 248.

14.3-8 - All welds and cold formed parts shall be stress relieved at 1150°F to 1200°F (620°C to 650°C) in accordance with Table UCS-56 of the ASME Code, Division 1 but for 1 hour minimum.

14.3-9 - Internal welds and HAZs shall be inspected after PWHT as follows:

14.3-9.1 - Internal welds shall undergo eddy current surface examination for at least 1 inch (25 mm) on each side of the toe of the weld in accordance with ASME Code, Section V, Article 8.

14.3-9.2 - With Owners acceptance, in lieu of eddy current surface examination, internal welds shall undergo WFMT for at least 1 inch (25 mm) on each side of the toe of the weld in accordance with ASME Code, Section V, Article 7.

14.3-9.3 - Acceptance criteria shall be per ASME Code, Division 1, Appendix 6 (Division 2, paragraph 7.5.6).

14.3-9.4 - Fluid-wetted pressure boundary and attachment welds shall be hardness tested and shall not exceed 200 BHN.

14.3-9.5 - One hardness test shall be performed on each weld or each 10 linear feet (3 m) of weld, whichever is more frequent.

14.3-10 - Interior surfaces that require coatings to protect against wet H_2S cracking shall be applied in accordance with COM-SU-5014 using the product as specified by the Purchaser in the vessel data sheet.

15 Supplemental Requirements for High-Strength Steels

15-1 - The following supplementary requirements shall apply for carbon steel materials with a specified minimum yield strength of 40 ksi (275 MPa) or greater.

15-1.1 - Hot forming temperature for heat-treated (normalized or quenched and tempered) material shall not exceed its heat treatment temperature, unless the material is reheat-treated after hot forming per the material specification.

15-1.2 - Carbon shall be limited to 0.20 percent maximum.

15-1.3 - Carbon equivalent shall be limited to 0.45 percent maximum.

15-1.4 - Niobium shall be limited to 0.03 percent maximum.

15-1.5 - Vanadium shall be limited to 0.05 percent maximum.

15-1.6 - Niobium plus vanadium shall be limited to 0.05 percent maximum.

15-1.7 - Yield strength shall be limited to 75 ksi (517 MPa) maximum.

15-1.8 - Supplier’s experience in welding of these or similar materials shall be submitted with bid for Owner evaluation.

15-1.9 - Three samples of micro-hardness traverses of production welds shall be submitted for Owner review.

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Note: *Compliance with the micro-hardness traverse requirements can be met by including additional testing of the base metals in the weld procedure qualification (Section 7.3) or vessel production test plates (Section 7.8)* requirements with the acceptance of Owner.

16 Supplemental Requirements for Topsides Vessels

16-1 - The following supplemental requirements shall apply if checked in the data sheet:

16-1.1 - The minimum acceptable internal corrosion allowance for the pressure boundary shall be 1/8 inch (3 mm) for all 'wetted' components including but not limited to, the vessel shell, heads, nozzles, internals, etc., unless specified otherwise.

16-1.2 - MDMT shall not be warmer than 50ºF (10ºC).

16-1.3 - Basic wind speed shall be per project technical requirement.

16-1.3.1 - As a minimum, the design wind velocity shall be 100 mph (160 km/hr) (3-second gust).

16-1.3.2 - Unless otherwise stated, Exposure Category D (wind from open body of water) shall be assumed.

16-1.3.3 - For wind-induced vibration, the critical wind velocity of the vessel shall be greater than 3 times the maximum 10-minute sustained wind velocity corrected at the top of the vessel.

16-1.4 - When specified in the data sheet, the vessel shell and heads shall be considered filled with sand (specific gravity = 1.8).

16-1.5 - The minimum impact lifting factor shall be 2.

17 Supplemental Requirements for Field Erected Carbon Steel Storage Spheres

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17.1 General

17.1-1 - Spheres shall be designed in accordance with API 2510 and API 2510A.

17.1-2 - In addition to the nameplate requirements listed in Section 12, the nameplate shall also list the sphere's storage capacity.

17.2 Nozzles

17.2-1 - Flanges for external nozzles or other flanges NPS 26 through NPS 60 (DN 650 through DN 1500), when used, shall be in accordance with ASME B16.47, Series A.

17.2-2 - Details of vortex breakers shall be per PVM-GA-C99913.

17.3 Appurtenances

17.3-1 - Spheres shall be supplied and installed complete with all appurtenances as specified.

17.3-2 - Structural steel supports and appurtenances shall be fabricated in accordance with CIV-SU-398.

17.3-3 - Materials for appurtenances shall meet the requirements of this specification, codes, and the standards referenced therein.

17.3-4 - Spheres and appurtenances shall be designed with sufficient clearances to permit cleaning, painting, and maintenance.

17.3-5 - Appurtenances and other details not covered by Purchaser standard drawings may be of Supplier design, provided such designs comply with applicable provisions of this specification.

17.4 Design Loads

17.4-1 - Structural supports and localized shell stresses at each support-to-shell location shall be designed for the greatest loads resulting from the following conditions:

17.4-1.1 - Load combinations in CIV-DU-5009.

17.4-1.2 - Static loads during erection plus expected wind load during erection.

17.4-1.3 - Static and dynamic loads during maintenance and operations.

17.4-2 - Seismic design shall be per ASCE 7-16 and shall include loads due to liquid sloshing.

17.4-3 - Wind design shall include loads from applicable wind pressures applied to the projected area of the sphere on a vertical plane perpendicular to a horizontal wind from any direction.

17.4-3.1 - The minimum elevation of the bottom of the spheres shall be assumed to be 12 ft. (3.7 m) above the bottom of the support legs base plate.

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17.4-3.2 - Actual clearances shall be indicated after finalizing the layout details.

17.5 Additional Material Requirements

17.5-1 - Materials of construction shall comply with Table 1.

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PVM-SU-4750, Table 1: Material Requirements (Unless Otherwise Specified)

Component	Material (see note 1)
Shells	• SA-516 (normalized) (see note 2) • SA-537 Class 1 (normalized) (see note 2) • SA-537 Class 2 (quenched and tempered) • SA-738 Grade B (quenched and tempered)
Supports (see note 3)	• SA-106 Grade B • SA-671 Grade CC60, CC65, CC70 • SA-516
Flanges	• SA-350 Grade LF2 • SA-350 Grade LF3 for MDMT colder than −50°F(−45°C)
Piping	Attachments welded to shell • SA-333 Grade 6 • SA-333 Grade 3 for MDMT colder than −50°F (−45°C) • Remainder SA 106 Grade B
Weld Fittings	• SA-234 Grade WPB • SA-420-WPL6 or WPL6W for MDMT colder than −20°F (−29°C) • SA-420-WPL3 or WPL3W for MDMT colder than −50°F(−45°C)
Forgings	• SA-350 Grade LF2 • SA-350 Grade LF3 for MDMT colder than −50°F (−45°C) • SA-765 Grade II (see note 4) • SA-765 Grade IV (see note 4)
Internals (clips, braces and other non-pressure parts)	Attachments welded to shell shall be same material as shell. Remainder as follows: • SA-106 Grade B for pipe, or as specified on drawings • SA-516 for plate or as specified on drawings • SA-36 for angles and bars
External Attachments	Attachments welded to shell to be same material as shell. Remainder as follows: • SA-36 for angles and bars • SA-516 for base rings, lugs/clips for stairs, platforms and pipe-support pads
Bolting	SA-193 Grade B-7 for Studs, SA-194 Class 2 fit for studs/nuts
Gasketing	Spiral-wound non-asbestos with 304 SS windings

Notes:

1. Carbon content shall be 0.23 percent max for plate and pipe and, 0.25 percent max for forgings. Carbon content for support bases, etc., may be higher than the shell as hardness is of secondary concern when not in contact with the product. The 0.20 percent is required for all shell plate material, including SA-516 Grade 70 and SA-537 Cl. 1. For forgings, the 0.25 percent is acceptable, as past experience indicates that no hydrogen-induced cracking (HIC) occurs in forgings even with hard HAZs.
2. When agreed to by the Purchaser, SA-516, Grade 70 and SA-537, Cl. 1 may be supplied in the quenched and tempered condition for improved notch toughness or to meet the specified tensile properties in thick plates, or both.
3. Support material attached to shell and extending 39.4 in. (1000 mm) minimum from shell surface shall be of same material as shell.
4. The SA-765 forging specification is intended for larger and heavier forgings than generally available to specification SA-350. SA-765 Class IV forgings have a specified minimum tensile strength of 80 ksi (550 MPa)and 105 ksi (725 MPa) maximum tensile strength. The Class IV forgings are suitable for use with SA-537 Cl. 2 and SA-738 Grade B plates but should not exceed a maximum tensile strength of 100 ksi (690 MPa).

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17.5-2 - If SA 537 Class 2 or SA 738 Grade B materials are to be used, the Owner's materials engineering subject matter expert (SME) shall be consulted for requirements in addition to the requirements of Section 15.

17.5-3 - Forgings shall be worked in all directions and shall preferably have at least 3:1 reduction in all directions from the billet.

17.5-3.1 - Tension and impact testing shall preferably be performed on test specimens removed from the actual forging or forging prolongation. The test coupons shall be removed from the forging after heat treatment.

17.5-3.2 - If separate test blanks are used, they shall be of the same heat of steel, be subjected to substantially the same reduction and working as the production forging it represents, be heat treated in the same furnace charge under the same conditions as the production forging, and be of the same nominal thickness as the maximum heat treated thickness of the production forging.

17.5-4 - With Purchaser review, higher-grade material with better notch toughness and/or stress relieve may be used in lieu of impact testing.

17.5-5 - The test coupons of quenched and tempered plates and forgings, and other plates and forgings that are required to be impact tested, shall be subjected to the same total heat treatment history (hot forming, intermediate stress relief, and PWHT including an additional PWHT cycle for repairs).

17.6 Sphere Supports

17.6-1 - Corrosion allowance of 1/8 in. (3 mm) shall be added to the calculated thickness of carbon steel support legs.

17.6-2 - Diagonal members, such as those used for bracing vertical columns, shall not be attached directly to a vessel unless adequate provisions are made in the design of the vessel for the resulting loads.

17.6-3 - Diagonal bracing shall not be used under stair treads.

17.6-4 - Pressure retaining portions of storage spheres should not contact concrete or masonry supports or concrete or masonry fireproofing, because these contact points may be sites for external corrosion. If such contact points are unavoidable, points should be identified for routine inspection.

17.6-5 - Two grounding lugs shall be welded to each support leg in accordance with PIP VEFV1103.

17.6-6 - MT shall be carried out on the root pass on the leg-to-shell attachment.

17.6-7 - Base plates shall be sized so that the bearing pressure on the concrete foundation does not exceed 725 psi (5 N/mm^2).

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17.7 Stairways and Platforms

17.7-1 - Field erected carbon steel storage spheres shall be built with a circular staircase and platform in accordance with SID-SU-5106.

17.7-2 - Loads for stairs and platforms shall comply with CIV-DU-5009.

17.7-3 - Bracing and supports should be minimized.

17.7-4 - Stairs and platforms shall be designed to accommodate thermal expansion of the equipment on which the stairs are supported.

17.7-5 - Support clips for bolted attachment of platforms and stairs shall be supplied and welded to the sphere.

17.8 Fire Protection Water Deluge System

17.8-1 - Fire protection water spray system shall be in accordance with FPM-SU-5141.

17.8-2 - Exposure protection for spheres shall follow the requirements for ‘Pressure Vessels’ in FPM-SU-5141.

17.9 Field Fabrication

17.9-1 - All necessary parts, personnel, equipment, documents, buildings, facilities, and services required for completion of spheres shall be provided by the fabricator.

17.9-2 - During erection, guy ropes or other methods shall be provided to secure incomplete spheres against damage during windy conditions.

17.9-3 - Protection shall be provided for spheres to ensure coating and painting are not affected by adverse weather.

17.9-4 - Flange faces shall be protected against corrosion and damage by applying stiff grease and by bolting on wooden or steel covers with a minimum of four bolts.

17.9-5 - A plastic sheet shall be inserted between a wooden cover and the greased flange surface.

17.9-6 - Spheres shall be calibrated and calibration charts and data shall be provided to the Purchaser by the fabricator.

17.9-7 - Unless adequate protection is provided, welding shall not be permitted if any of the following conditions are present:

17.9-7.1 - Ambient temperature is less than 40°F(5°C).

17.9-7.2 - Wind speed is greater than 10 miles/hr (5 m/s).

17.9-7.3 - Any inclement weather.

17.9-8 - Plate edges may be sheared, machined, or cut by oxy-fuel or electric arc.

17.9-8.1 - Shearing is limited to plates 3/8 in. (9.5 mm) or less in thickness.

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17.9-8.2 - Edges shall be ground smooth and shall be free of scale and any slag.

17.9-9 - Lifting requirements shall be as follows:

17.9-9.1 - The lifting requirements of Section 5.8 shall apply for lifting the spherical caps except that a minimum impact factor of 2.0 shall be used.

17.9-9.2 - For rigging condition, allowable stresses where S is the ASME Code allowable stress, shall be as follows:

17.9-9.2.1 - 1.5S for local primary membrane stress.

17.9-9.2.2 - 3S for primary membrane plus secondary bending stress at the location of the lifting devices.

17.9-9.3 - Shear stresses for fillet welds on the lifting attachments shall not exceed 0.55 times the ASME Code allowable stress at 100°F (38°C) for the material selected at the following locations:

17.9-9.3.1 - Sphere shell.

17.9-9.3.2 - Head caps.

17.9-9.3.3 - Support legs.

17.9-10 - Protection shall be provided around sphere location to ensure safety of jobsite, equipment, and personnel working in the vicinity.

17.10 Additional Examination

17.10-1 - Full RT or UT shall be required along with the additional requirements as described in Section 9.4.

17.10-2 - Weld deposits subjected to severe forming (ratio of thickness to local radius greater than 5 percent) shall be 100 percent inspected by RT or UT after completion.

17.10-3 - One hundred percent UT shall be required for all category D welds.

17.10-4 - ET or MT for the proposed areas of the sphere shall be per Table 2.

PVM-SU-4750, Table 2: Magnetic Particle Examination Methods

Area of Inspection	Required Method of Magnetic Particle Examination	Remarks
Inside surfaces of all sphere welds after hydrotest	Eddy Current Surface Examination (ET)	This method is required for inside surfaces to set up a baseline of the sphere for potential wet H_2S cracking later in service. WFMT is permitted as an alternative with Owner acceptance.
Outside surfaces of all sphere welds Back gouged joints Root passes on attachments	MT	Dry powder technique per Section 9.7.

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17.11 Hydrostatic Testing

17.11-1 - A hydrotest plan shall be provided, detailing fill rates, settlement checks at 0, 25, 50, 75, and 100 percent liquid fill levels during fill and drain.

17.11-2 - Hydrotest plan shall address the maximum allowable differential settlement.

17.11-3 - Settlements shall be checked and recorded during filling for the following points of hydrotest:

17.11-3.1 - Holding at 25, 50, 75, and 100 percent of liquid fill level.

17.11-3.2 - Emptying at 25, 50, 75, and 100 percent of liquid fill level.

17.11-4 - Settlement measurements shall be taken at all sphere support leg locations at each of these fill levels.

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17.12 Tolerances

17.12-1 - In addition to those tolerances required by ASME Code, fabrication tolerances for field-erected carbon steel storage spheres shall be per Figure 5.

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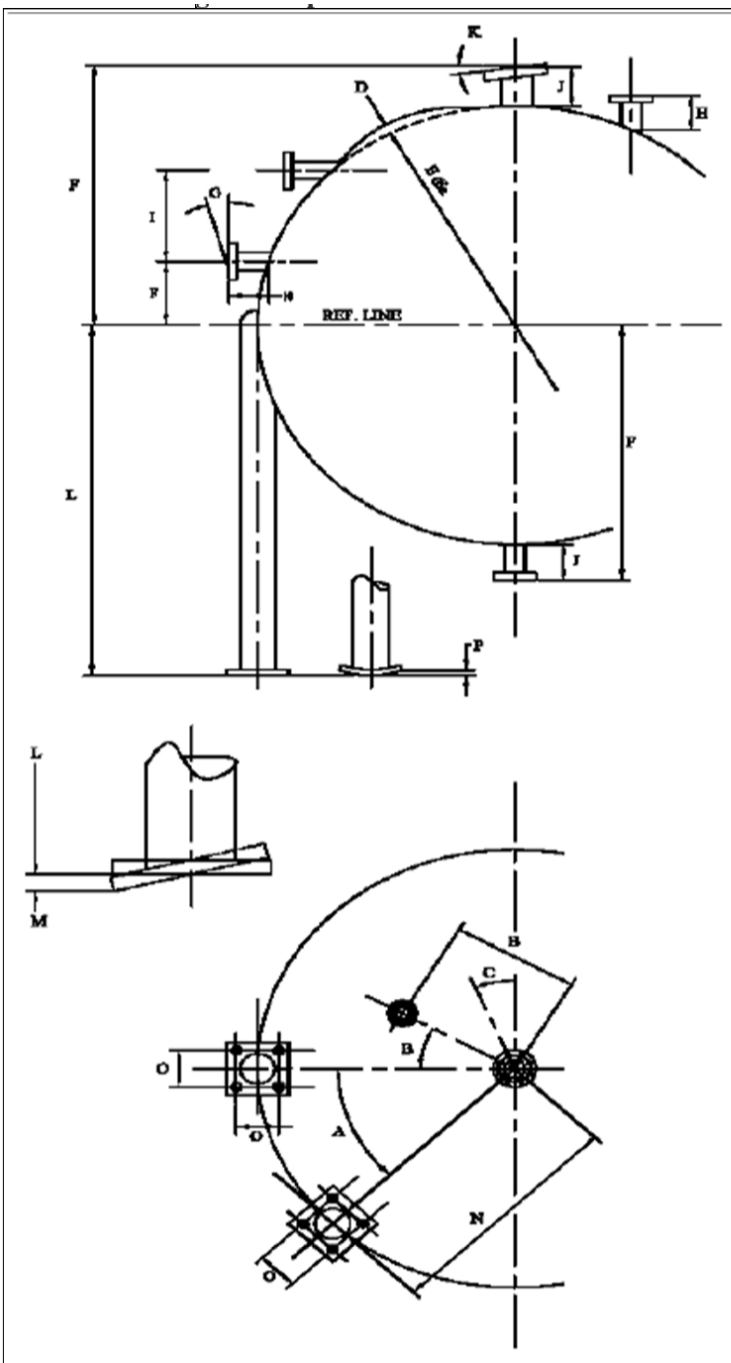
PVM-SU-4750, Figure 5: Sphere Fabrication and Construction Tolerances



LEGEND:

A. ± 9/32-inch (± 7 mm) anchor bolt hole orientation at bolt circle (chord length).

B. Nozzle Orientation: 1/4 inch (6 mm) maximum deviation from reference axis on shell (linear or chord length).

C. Nozzle Bolt Hole Orientation: 1/16-inch (1.5 mm) maximum deviation at bolt circle (chord length).

D. Shell deviation from spherical form shall not me more than the lesser of 1/16 inch per 3 ft (1.5 mm per 1000 mm) with maximum of 3/4-inch (20 mm).

E. Deviation from Nominal Outside Diameter as Determined by Strapping: ±1/8 inch per 3 ft (± 3 mm per 1000 mm) of circumference with maximum of ±3/4 inch (±20 mm).

F. ± 9/32-inch (± 7 mm) nozzle elevation from reference line.

G. ±1/2º nozzle tilt measured across gasket face.

H. ±1/8 inch (± 3 mm) nozzle projection from shell outer surface.

I. 1/16 inch (1.5 mm) centerline-to-centerline liquid level and gauge class connections.

J. ±3/8 inch (± 10 mm) manhole projection.

K. ±1º manhole tilt measured across gasket face.

L. ±3/4 inch (± 20 mm) from reference line to underside of vessel supports.

M.3/16 inch (5 mm) base plate out of perpendicular to longitudinal axis.

N. Anchor Bolt Circle: ±3/16 inch (± 5 mm) radius.

O. Distance Between Bolt Holes in Base Plate: ± 3/16 inch (± 5 mm).

P. Base Plate Flatness: ±5/32 inch (± 4 mm).

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18 Supplemental Requirements for Girth Flange Design and Assembly

18-1 - This specification does not exclude the use of proprietary closures or proprietary devices for loading the gasket. Such devices shall conform to the ASME Code and shall be subject to acceptance in writing by the Purchaser.

***Note**: These requirements are loosely based on those of Sections 7.7 and 7.8 of API 660 and EXH-SU-2583, which are for shell-and-tube heat exchanger body flanges. Much of the information is appropriate for other pressure vessel body flanges.*

18-2 - External girth joints shall be of through-bolted construction.

18-2.1 - Studded-in bolts may be used when specified or accepted by the Purchaser.

18-2.2 - The design of studded-in bolts and flanges shall comply with ASME PCC-2, Appendix 303-I.

18-3 - Flanges for external girth joints shall be of the forged welding-neck type, unless otherwise specified or accepted by the Purchaser.

18-4 - Nubbins shall not be used.

18-5 - The nominal clearance between flanges after assembly shall not be less than 0.1875 inch (5 mm) at the periphery of the flanged joint.

18-6 - Spot facing or back facing of the bolting bearing surfaces shall be in accordance with ASME B16.5 and the ASME Code, Section VIII.

18-7 - Hydraulic bolt tensioning shall be applied for bolting equal to or greater than 2 inch (50 mm) diameter, unless otherwise specified.

18-8 - When bolt tensioning is used, the bolting shall have additional thread length equivalent to one bolt diameter, extending from the nut at one end, to allow attachment of the bolt-tightening device.

18-9 - When bolt tensioning is used, Purchaser shall specify any special requirements necessary to allow for adequate clearance for the bolt-tightening device. The required axial clearance for tensioning equipment shall be provided when tensioning is to be used for studs.

18-10 - When either manual or hydraulic torqueing is applied, through-hardened washers shall be provided under nuts for all bolts to improve the translation of torque into bolt preload by providing a smooth and low friction bearing surface for the nut, and to protect the contact surface of the flange from damage caused by a turning nut.

***Note:** Hardened washers should not be installed on studs that have been designated to be tensioned.*

18-11 - Washers shall conform to the requirements of ASME PCC-1, Appendix M. The washers shall be suitable for re-use as defined in ASME PCC-1, Appendix M, unless otherwise specified by Purchaser.

18-12 - Allowable stresses that have been established on the basis of short-term tensile strength shall not be used for the design of girth flanges. Allowable stresses for the design of girth flanges shall not exceed the allowable stresses for the pre-1999 addenda of the ASME Code, Section II, Part D.

***Note**: In the ASME Code, Section II, the allowable stresses of some high nickel alloys have been established based*

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on short term tensile strength. These stress values may result in dimensional changes due to permanent strain and should not be used for flanges for gasketed joints where slight distortion can cause leakage.

18-13 - Flange design shall be in accordance with the pressure design code.

18-14 - The MAWP shall not be limited by flange bolting.

18-15 - All girth flanges shall be provided with 1/8 inch (3 mm) future machining allowance on the gasket contact seating surfaces.

18-15.1 - This additional thickness shall not be used in the calculation for MAP or MACA.

18-15.2 - This additional thickness shall be shown as machining allowance on the part drawing.

18-15.3 - This requirement shall apply to clad or weld overlay construction.

18-16 - Tongue and groove type configuration shall not be used, unless specified or agreed by the Purchaser.

18-17 - The joint component approach, as defined in ASME PCC-1, Appendix O, and the requirements contained within this section, shall be applied to the girth flange when specified by the Purchaser.

18-18 - Definition of terms and symbology contained within this section are consistent with ASME PCC-1. See A.4.5 for additional guidance. The following equations from ASME PCC-1, Appendix O do not need to be calculated:

18-18.1 - O-6.

18-18.2 - O-10.

Note: *Equations O-6 and O-10 are intended to check the adequacy of existing flanges. Girth flanges for this specification will be designed to the ASME Code, Section VIII, Division 1, Appendix 2 or Division 2, Part 4.16 or Part 5 and these additional requirements after the bolting requirements have been determined.*

18-19 - The minimum permissible design bolt stress, Sb_{min}, shall be 50 percent of the listed yield stress values in the ASME Code, Section II, Part D at assembly (ambient) conditions.

18-19.1 - If studs greater than 2 inch (50 mm) diameter are required, the Sb_{min} may be lower with Purchaser acceptance.

18-19.2 - The maximum permissible bolt stress, Sb_{max}, shall be 70 percent of the listed yield values in the pressure design code at assembly (ambient) conditions.

18-19.3 - The selected assembly bolt stress, Sb_{sel} , shall be established to ensure that the gasket stresses at assembly and the operating conditions comply with the values shown in Table 3. Gasket selection shall be limited to the following:

18-19.3.1 - Corrugated metal graphite coated (corrugated metal graphite faced gasket [CMG] per EXH-SU-5150.

18-19.3.2 - Grooved metal graphite faced Kammprofile [KAM] per EXH-SU-5151) for girth flange construction.

18-19.3.3 - Gaskets indicated in item 1) and item 2) above shall be designed with the following values:

i. M = 3.75
ii. Y = 7600

Note: *In Table 3, a new variable is introduced, Sg $_{T\text{-}O}$, which is the target operating stress at operating*

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conditions.

PVM-SU-4750, Table 3: Assembly Gasket Stress

Gasket Type	Maximum Permissible Gasket Stress, Sgmax psi (MPa)	Minimum Gasket Seating & Operating Stress, Sgmin-S and Sgmin-O psi (MPa)	Target Gasket Operating Stress, Sg T-O psi (MPa)
KAM	55,000 (380)	20,000 (138)	25,000 (172) [1]
CMG	40,000 (275)	20,000 (138)	25,000 (172) [1]
Note: 1. Target gasket stress Sg T-O is increased to 30,000 (207) for gasket with widths over 1 inch (25.4 mm).			

18-20 - Sg $_{T\text{-}O}$ shall be substituted into equation (O-8) of PCC-1, Appendix O, for Sg min-$_O$ for the determination of $Sb_{sel.}$ The selected value of $Sb_{sel.}$ shall not exceed the calculated minimum value by more than 10 percent. The “bolt up” (P=0) gasket stress shall also be checked and shall not exceed Sg_{max} in Table 3.

18-20.1 - The fraction of gasket load remaining after relaxation, Φ, shall be set to 1 for design purposes.

18-20.2 - To minimize potential for flange rotation, the rigidity index as calculated in the ASME Code, Section VII, Division 1, Mandatory Appendix 2, and Division 2, Part 4.16, shall not exceed 1.0.

18-20.3 - If the requirements contained within this section require the bolt area to be increased or gasket geometry revised, then the ASME Code, Section VIII calculations shall be updated and the required flange thickness re-calculated accordingly.

18-20.4 - Controlled bolt-tightening procedures shall be applied and records provided to demonstrate that the selected assembly bolt stress, Sb_{sel}, has been achieved.

18-21 - Permanent match marks (i.e., punch marks or letter/number metal stencils) shall be provided to facilitate proper alignment of the bolted joint (dowels shall not be used).

18-22 - All studs, including studs with zinc plating and Xylan coating, shall be coated with a high temperature molybdenum disulfide-based anti-seize thread lubricant to lubricate studs and nuts, such as Jet-Lube-550 Extreme™, or equal accepted by the Purchaser.

18-22.1 - Lubricant shall be applied generously so that the threads that engage the nuts are filled with lubricant.

18-22.2 - Lubricant shall also be used between the nut and washer and on both ends of the studs to prevent galling and to facilitate consistent torqueing.

18-22.3 - Jet-Lube-550 Extreme or Purchaser-accepted equal shall be used for blind holes or for studs in 371°C (700°F) or higher service.

18-23 - To facilitate the installation of corrugated metal gaskets, a light coat of 3M Super 77™ spray adhesive may be applied on the flange.

18-23.1 - As an alternative, heavy grease may be used to hold the gasket in place while assembling the girth flange.

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18-23.2 - Tape shall not be used to hold gaskets in place during assembly.

18-24 - Stud length shall be sufficient to allow for full thread engagement through the nuts, but shall not extend more than two threads beyond the nut.

18-25 - Girth flange studs shall be torqued using a calibrated clicker-type torque wrench or a hydraulic torque wrench.

18-25.1 - As an alternative to torqueing, girth flange bolting may be tensioned to final loads. Tension requirements will be specified in purchasing documents.

18-25.2 - Air impact guns or RAD^{TM} guns shall not be used to set body or floating head flange stud loads.

18-26 - Studs in girth flange joints shall be loaded to the appropriate torque.

18-26.1 - The formula for calculating stud torque (T, in foot-pounds) shall be as follows:

Equation 1

$$T = K \times \frac{S_{nd}}{12} \times S_{ra} \times Sb_{sel}$$

where:

K	=	Torque coefficient (nut factor or friction factor)
S_{nd}	=	Stud nominal diameter, inches
S_{ra}	=	Stud root area, square inches
Sb_{sel}	=	Assembly stud stress, psi

18-26.2 - Refer to item 10 of this section for calculation of the Sb_{sel} value.

18-26.3 - A friction factor (K) of 0.17 shall be used for molybdenum disulfide-based lubricants.

18-26.4 - A friction factor (K) of 0.12 shall be used for Xylan coated studs with molybdenum disulfide-based lubricant applied to the coated ends.

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PVM-SU-5290-B.1

Conventional 2¼ Cr–1 Mo Steel Pressure Vessels

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Rev.	Date	Description	Author	Technology Leader
—	12/12	Initial release.	J. Taagepera	J. Taagepera
A	09/17	General revision.	R. Darby	J. Taagepera
B	10/20	General revision.	R. Darby	J. Taagepera
B.1	10/21	Technical modification. Section 14.0 item 2.	R. Darby	—

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Summary of Changes

The Author denotes the following technical changes to this revision as detailed below:

	Author notations regarding this revision
	Revision to Examination Section (9.0).
B.1	Replaced MDMT in Section 14.0 Item 2 with maximum design metal temperature.
	Safety Critical Changes (if applicable) are noted below
	N/A

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1.0 Scope

1. This specification covers general requirements for pressure vessels fabricated of 2¼ Cr-1 Mo steel, in accordance with the ASME *Boiler and Pressure Vessel Code* (hereafter referred to as the 'ASME Code'), Section VIII, Division 1 or Division 2.
2. This specification is limited to design temperatures at or below 850°F (455°C).

Note: *Design information and supplemental requirements for the pressure vessel are specified on the individual vessel drawing or data sheet. See PVM-DS-5290 and PVM-DG-5290.*

1.1 Conflict Resolution

1. In cases of conflict between this specification and other drawings or specifications, the vessel manufacturer (hereafter referred to as 'fabricator') shall call attention to the conflict and request an interpretation by Purchaser.
2. Fabricator shall not assume which instructions govern.

1.2 Units of Measurement

When applicable, both U.S. Customary and International System of Units (SI) units of measure have been provided, with U.S. Customary listed first. In cases where a dual unit of measurement has not been provided, an exact conversion is not available or applicable.

2.0 References

Note: *Some of these references may not apply to all projects.*

2.1 Owner Documents

COM-SU-4743	External Coatings
COM-SU-5191	Coating Systems
CPM-PU-5074	Material Verification Program
FPM-DU-1508	Offshore Passive Fire Protection Materials and Application
FPM-DU-5091	Onshore Passive Fire Protection Materials and Application
PIM-DU-5155	Piping Flexibility and Stress Analysis
PVM-SU-1322	Pressure Vessels with High-Alloy Cladding or Overlay
PVM-DG-5290	Data Sheet Guide for Conventional 2¼ Cr–1 Mo Steel Pressure Vessels
PVM-DS-5290	Data Sheet for Conventional 2¼ Cr-1 Mo Steel Pressure Vessels
PVM-DU-6158	Pressure Vessel Design By Analysis
[PVM] GF-C14311	Pressure Vessel Nozzles
[PVM] GD-C78876	Standard Skirt and Base Details for Vertical Vessels
[PVM] GF-C87280	Standard Manway Cover Davits and Hinges
[PVM] GF-C99694	Standard Details of Saddles for Horizontal Vessels
[PVM] GB-C99772	Standard Ladder Rungs Inside Columns and Vessels

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[PVM] GA-C99913	Vortex Breaker for Vessel Liquid Outlet
[PVM] GB-C99945	Standard Support Legs for Vertical Vessels
QAM-SU-6053	Quality Assurance Requirements for Equipment and Material Manufacturers
QAM-SU-6163	Performance Demonstration Qualification Program for NDE Contractors Performing Ultrasonic Weld Inspection
SID-SU-5106	Safety in Designs

2.2 Industry Codes and Standards

Unless specified herein, use the latest edition of the referenced documents.

American Society of Civil Engineers (ASCE)

7-16	Minimum Design Loads for Buildings and Other Structures

American Society of Mechanical Engineers (ASME)

Boiler and Pressure Vessel Code (BPVC, also referred to as 'the ASME Code')

	Section II, Material Specifications, Part D
	Section V, Nondestructive Examination
	Section VIII, Division 1, Rules for Construction of Pressure Vessels
	Section VIII, Division 2, Alternative Rules – Rules for Construction of Pressure Vessels
	Section IX, Welding and Brazing Qualifications
	Code Cases
B31.3	Process Piping
PCC-2	Repair of Pressure Equipment and Piping
STS-1	Steel Stacks

American Welding Society (AWS)

A4.3	Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and Ferritic Steel Weld Metal Produced by Arc Welding
D1.1/D1.1M	Structural Welding Code—Steel

International Association of Oil & Gas Producers (IOGP)

S-619	Specification for Unfired, Fusion Welded Pressure Vessels

Note: *This specification is available for free at https://www.iogp.org/bookstore/*

Process Industry Practices (PIP)

VEFV1100	Vessel/S&T Heat Exchanger Standard Details, including drawings: VEFV1103 Vessel Grounding Lug VEFV1122 Horizontal Vessel Insulation Supports VEFV1123 Vertical Vessel Insulation and Fireproofing Supports
VESV1003	Special Fabrication Requirements for Welded Vessels and Tanks to Be Lined

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Welding Research Council (WRC)

Bulletin 297	Local Stresses in Cylindrical Shells Due to External Loadings on Nozzles
Bulletin 537	Precision Equations and Enhanced Diagrams for Local Stresses in Spherical and Cylindrical Shells Due to External Loadings for Implementation of WRC Bulletin 107

Welding Journal Research Supplement

Zick, L.P., "Stresses in Large Horizontal Cylindrical Pressure Vessels on Two Saddle Supports", The Welding Journal Research Supplement, pp. 959-970, 1971

3.0 Acronyms

CA	Corrosion Allowance (minimum design, as specified in the data sheet; not the calculated maximum allowable corrosion allowance [MACA])
DHT	Dehydrogenation Heat Treatment
FEA	Finite Element Analysis
FV	Full Vacuum
GMAW	Gas Metal Arc Welding
GTAW	Gas Tungsten Arc Welding
HAZ	Heat-Affected Zone
ID	Inside Diameter
ISR	Intermediate Stress Relief
MACA	Maximum Allowable Corrosion Allowance (calculated)
MAEP	Maximum Allowable External Pressure
MAP	Maximum Allowable Pressure
MAWP	Maximum Allowable Working Pressure
MDMT	Minimum Design Metal Temperature
MT	Magnetic Particle Testing
N&T	Normalized and Tempered
NDE	Nondestructive Examination
PMI	Positive Material Identification
PQR	Procedure Qualification Record
PT	Liquid Penetrant Testing
PWHT	Post Weld Heat Treatment
Q&T	Quenched and Tempered
RT	Radiographic Testing
SAW	Submerged Arc Welding
UT	Ultrasonic Testing
UTSW	Ultrasonic Shear Wave
WFMT	Wet Fluorescent Magnetic Particle Testing
WPS	Welding Procedure Specification

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4.0 ASME Code Conformance

1. Design and fabrication shall conform to the ASME Boiler and Pressure Vessel Code (ASME Code), Section VIII, Division 1 or Division 2 (hereinafter referred to as 'Division 1' or 'Division 2').
2. All vessels within the scope of the ASME Code shall be Code stamped. Stamping and inspection for other vessels shall be a matter of agreement with Purchaser.
3. The latest edition issued through the date of the agreement (purchase order) shall be used.
4. When the ASME Code, Division 2, is specified, requirements from Division 2 shall be met as a minimum. Applicable corresponding Division 2 paragraphs shall be substituted for Division 1 paragraphs referenced by this specification.
5. Division 1, Appendix 32, and Division 2, paragraphs 4.3.8, 4.4.10, and 4.14.2 of the ASME Code (addressing local thin areas) shall not be used without written authorization from Owner.
6. For a Division 1 vessel, the use of Code Case 2695 shall be permitted.
7. Calculations covered by U-2(g) of Division 1 and Part 5 of Division 2 of the ASME Code shall be supplied to the Owner for review and acceptance. When finite element analysis (FEA) is used, it shall comply with PVM-DU-6158.

5.0 Design

5.1 Design Conditions

5.1.1 Structural Loads

1. Seismic and wind loading shall be in accordance with ASCE 7-16. Unless otherwise stated, the importance factor shall be 1.15.
2. Allowable stresses shall be in accordance with the ASME Code.
3. Local jurisdiction or regulatory standards shall govern if they produce a more stringent design.
4. Other alternative structural codes may be used only with Owner's acceptance.
5. Unless otherwise specified by Owner, transportation design loads shall be based on a lateral acceleration of 0.65 g acting along both the longitudinal axis and the transverse axis, and a vertical acceleration of 1.5 g as a minimum. The allowable stress may be increased by 33 percent for the transportation design condition.

5.1.2 Design Pressure and Temperature

1. Vessel design conditions shall be established by the process engineer.
2. These conditions shall consider startup, normal operation, shutdown, upset, any other conditions reasonably anticipated, and abnormal conditions that may become a governing design per the ASME Code, Division 1, paragraph U-2(a).
3. Multiple pressure/temperature conditions shall be acceptable and are often necessary.

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4. Mechanical design temperature for vessels with internal insulating lining shall not be less than metal temperature under the lining.
   a. Calculation shall be based on maximum operating temperature, atmospheric conditions of no wind, and ambient temperature of 100ºF (40ºC).
   b. Credit shall not be taken for radiative heat loss.
5. Vessels subject to steamout for maintenance shall be designed for 7.5 psi (0.5 bar) external pressure at 350ºF (180°C).

5.1.3 Maximum Allowable Pressures and Corrosion Allowance

1. The vessel data sheet shall state the design pressure, coincident design temperature, and design corrosion allowance (CA). Minimum design CA shall be 0.125 inch (3.0 mm) unless the vessel is clad or overlayed.
2. Pressure vessel shall meet or exceed requirements specified in the vessel data sheet.
3. Design pressure and temperature, maximum allowable corrosion allowance (MACA), and hydrostatic test pressure of a vessel shall not be limited by the mechanical design of minor components such as nozzles (including nozzle flange ratings), manways, opening reinforcement, or bolting, unless Owner's written acceptance is obtained.
4. Maximum allowable working pressure (MAWP) at the coincident temperature shall be set at the process design pressure.
5. MACA methodology shall be used to optimize the design for CA.
   a. Excess component thickness beyond what is required for governing minimum thickness plus design CA shall be used to increase the effective MACA.
   b. The MACA shall be calculated individually for each of the major components first (main shell, head components, and body flanges).
   c. Nozzle reinforcement calculations per UG-37 of the ASME Code, Division 1 or Part 4.5 of Division 2 shall be performed using $A_1 = 0$. Additional thickness that is added to the component for the sole purpose of reinforcement may be used for A_1.

      Note: *The above statement applies when using the area replacement method. Disregard if reinforcement calculations are completed using the Bildy method (pressure area method).*

   d. The limiting MACA shall then be used for design of the minor components (e.g., nozzles).
   e. The CA of minor components shall be equal to or greater than the MACA of their associated major component.
   f. Initial formed head thickness calculations shall be performed before forming with an estimated thickness using $A_1 = 0$ for nozzle reinforcement calculations per UG-37 of the ASME Code, Division 1 or Part 4.5 of Division 2.
   g. Formed head thickness shall be measured after forming. The measured minimum thickness shall be used in the as-built calculations and drawings, and the as-built MACA shall be calculated.
   h. The as-built MACA for each major component shall be included on the vessel drawing and data report.

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i. MACA for each major component shall be determined at least to the nearest 0.01 inch (0.2 mm).

j. MACA shall be calculated regardless of whether the CA is internal or external.

Note: *MACA need not exceed twice the CA.*

6. Manufacturer's calculations shall also include the maximum allowable pressure (MAP) in the new (uncorroded) and cold condition.

7. Vessels designed for internal pressure only shall also be stamped for the calculated maximum allowable external pressure (MAEP) at the design temperature and steamout temperature listed in Section 5.1.2, item 2.

8. Manufacturer's calculations shall include the minimum required thickness for all major components of the vessel.

9. Minimum corroded thickness of shell and heads shall be 0.12 inches (3.0 mm).

5.1.4 Minimum Design Metal Temperature

1. Minimum design metal temperature (MDMT) shall be selected in consideration of the operating cases and of ASME Code rules in Division 1, paragraphs UG-20(b). For Division 2 vessels, paragraph 4.1.5.2(e) shall be referenced.

2. Low-alloy steels covered by this specification are susceptible to temper embrittlement if the maximum design metal temperatures is 650°F (345°C) or greater, and may require special attention by Owner's materials engineer in assigning an MDMT.

   Note: *See Section 6.1, item 17; Section 7.5, item 10; and Section 14.0.*

3. When not specified, the MDMT shall be determined as the lowest of the following:

   a. Coldest 1-day mean ambient temperature at the installation site.

   b. Coldest operating temperature at coincidental pressure minus an additional 25°F (15°C).

   c. 30°F (–1°C).

4. The following MDMT and coincident pressures shall be indicated on both the vessel nameplate and the manufacturer's data report:

   a. MDMT at MAWP.

   b. Lowest allowable MDMT and coincident maximum pressure.

5. All attachments to the pressure boundary shall be considered essential to structural integrity for the purposes of establishing the MDMT of the vessel.

5.1.5 Cyclic Service

1. Vessels designated as subject to cyclic or transient loading shall have their pressure, temperature, or both, screened in accordance with the ASME Code, Division 2, paragraph 5.5.2.

2. If this screening determines that fatigue analysis is required, design rules and details shall be per the ASME Code, Division 2, paragraph 5.5. The use of paragraph 5.5.5 for the fatigue assessment of welds shall not be permitted without Owner acceptance.

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3. Vessels designated as subject to cyclic service or transient loadings shall have all welds blend-ground to form a smooth transition with the parent metal.

   a. Welds need not be ground flat.

   b. Welds shall be sized to meet the ASME Code requirements after grinding.

5.1.6 Load Combinations

Vessels shall be designed with consideration given to the following loads and load combinations:

1. Erect load with full wind—Dead load shall include installed weight of the vessel including internals, platforms, insulation, and other permanent attachments.
2. Operating load with full wind or seismic loads—Operating load shall include all dead and live loads, high-high liquid level and zero process liquid levels, and full and zero pressures or partial or full vacuum (FV) when specified as a design case. Thrust loads from attached piping shall also be considered.
3. Field hydrostatic pressure test load with partial wind—When a field hydrostatic pressure test is required, the vessel shall be assumed to be full of water, and support shall be designed for 50 percent of design wind speed (or 25 percent of wind load).

5.2 Welded Joints

1. Category A and B weld joints shall be double butt welded (Type 1) or an acceptable equivalent agreed to by Owner.
2. Categories C and D joints shall be full penetration welds in accordance with [PVM] GF-C14311 or an acceptable equivalent agreed to by Owner.
3. An acceptable equivalent for Category A, B, C, and D joints in nozzles, elbows, and single-sided closure seams shall be a single-welded butt joint with gas tungsten arc welding (GTAW) root bead.
4. Permanent backup rings shall not be used.
5. Plate and sheet cut by shearing shall be ground to have shear-damaged material removed.
6. On horizontal vessels, no longitudinal seam welds shall be located more than 30 degrees below the horizontal centerline.
7. Weld joints between the support skirt and the vessel shall be as shown on the vessel drawing or data sheet. Other longitudinal and circumferential weld joints shall be in accordance with Section 5.2, item 1.

5.3 Heads and Transitions

1. Ellipsoidal or hemispherical heads shall be preferred. However, torispherical heads may be permitted when designed per the ASME Code, Division 2, Sections 4.3.6 and 4.4.8, and reviewed and accepted by Owner.
2. Toriconical transition sections shall be preferred; however straight conical transitions shall be permitted when reviewed and accepted by Owner.

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